

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2013

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Booklet 1. January 2013

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 4. Calculus—Study and teaching (Secondary)—Manitoba.
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Manitoba Education
School Programs Division
Winnipeg, Manitoba, Canada

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

	Questions	Marks	Total Marks
Booklet 1*	9 Short-Answer	13	31
	6 Long-Answer	18	
Booklet 2	9 Multiple-Choice	9	58
	15 Short-Answer	17	
	10 Long-Answer	32	
Total			89

- * The first 6 questions in *Booklet 1* require a calculator. 
You have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- Read all instructions carefully.
- The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

No marks will be awarded for work done on this page.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

$$A = Pe^{rt}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$\log_a M = \frac{\log_b M}{\log_b a}$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$y = a \sin b(x - c) + d$$

$$y = a \cos b(x - c) + d$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are explained below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

Solve: Give a solution for a problem or determine the value(s) of a variable.

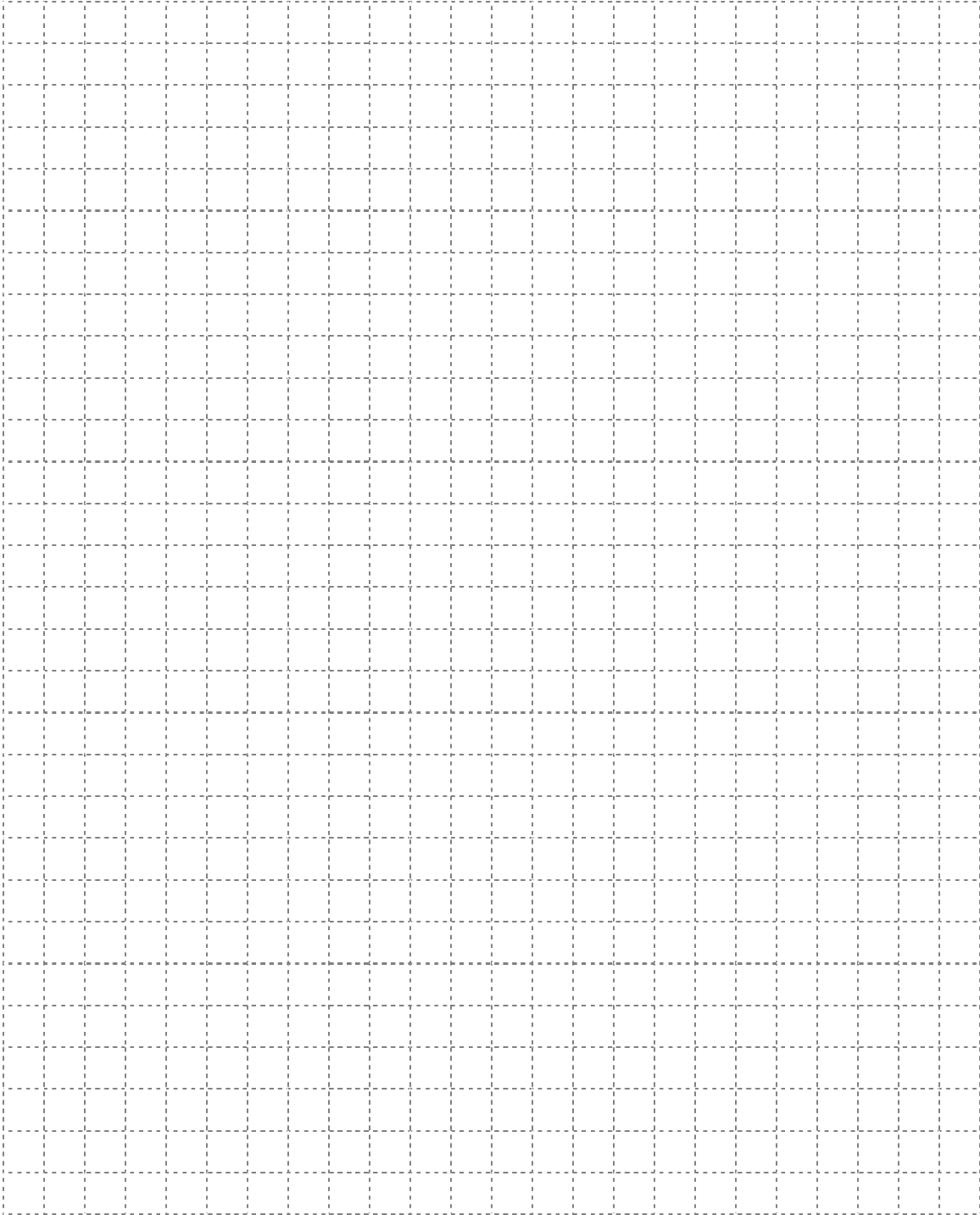
Verify: Establish the truth of a statement by substitution or comparison.

Scrap Paper

No marks will be awarded for work done on this page.

Scrap Paper

No marks will be awarded for work done on this page.



Instructions

- There are 15 questions for a total of 31 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to 3 decimal places unless instructed otherwise.

No marks will be awarded for work done on this page.

Question 1

3 marks

101

Gina correctly started to answer the following question. Complete her solution.

Question: Solve the following equation for all real values of θ .
Express your answer in radians correct to 3 decimal places.

$$3 \sin^2 \theta - 14 \sin \theta - 5 = 0$$

Gina's solution: $3 \sin^2 \theta - 14 \sin \theta - 5 = 0$
 $(3 \sin \theta + 1)(\sin \theta - 5) = 0$

Question 2

3 marks

102

Find and simplify the 6th term in the binomial expansion of $\left(3x^4 - \frac{1}{x^3}\right)^9$.

Question 3

3 marks

103

The number of times a website is visited can be modeled by the function:

$$A = 800(e)^{rt}$$

where A = the total number of visitors at time t

t = the time in days ($t \geq 0$)

r = the rate of growth

After 5 days, 40 000 people have visited the site.

Determine the number of visitors expected after 9 days.

Express your answer as a whole number.

Question 4

3 marks

104

Solve algebraically:

$$10^{3x} = 7^{x+5}$$

Express your answer correct to 3 decimal places.

Question 5

2 marks

105

A word contains two Ms, two Es, two Ns, and no other repeated letters.

Suppose one of the Ns is replaced by an M.

Will this replacement result in greater or fewer permutations?

Justify your reasoning.

Question 6

3 marks

106

There is a group of 16 boys and 12 girls. How many ways can a committee of 3 people be formed if there must be at least 2 girls on the committee?

Express your answer as a whole number.

Note: A calculator is not required for the remaining test questions.

Question 7

2 marks

107

A student is using the formula $s = \theta r$ to find an arc length of a circle. Given a central angle measure of 35° and a radius of 6 cm, the student's solution is as follows:

$$s = (35)(6)$$

$$s = 210 \text{ cm}$$

Explain why this solution is incorrect.

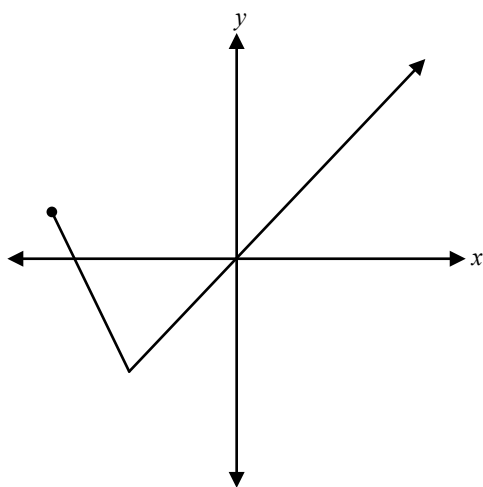
Write the correct solution.

Question 8

1 mark

108

Given the graph of $f(x)$ below, explain how you would sketch the graph of $y = |f(x)|$.



Question 9

1 mark

109

Claire correctly solves the following equation:

$$\log_2(6-x) + \log_2(3-x) = 2$$

She finds two possible values of x : $x = 2$ and $x = 7$.

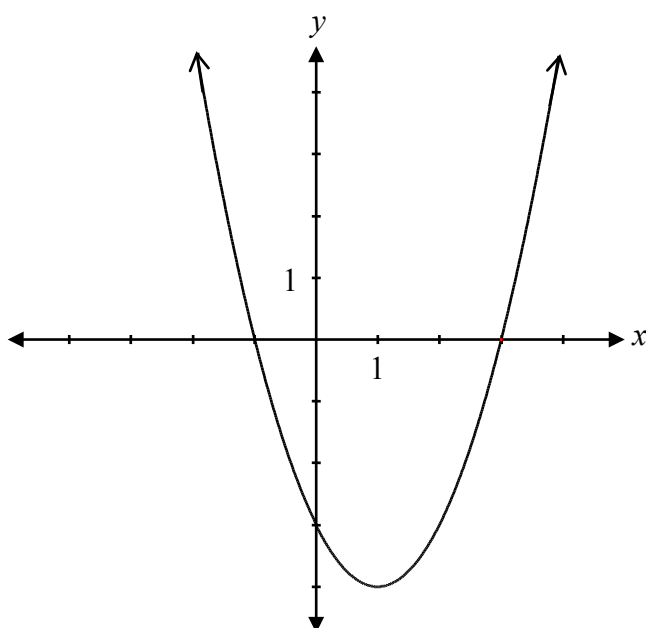
Identify which one of these values is unacceptable and explain why.

Question 10

1 mark

110

Given the graph of the function $f(x)$ below, state the domain of $y = \sqrt{f(x)}$.



Question 11

1 mark

111

A school offers 4 different Science courses, 3 different Mathematics courses, and 2 different English courses.

Julie must select 1 Science course, 1 Mathematics course, and 1 English course. She thinks this creates 9 options for her timetable.

Show why Julie is incorrect.

Question 12

1 mark

112

Explain how Pascal's triangle can be used to determine the coefficients in the binomial expansion of $(x + y)^n$.

Question 13

3 marks

113

Prove the identity below for all permissible values of x :

$$\frac{1 + \cos 2x}{\sin 2x} = \cot x$$

Left-Hand Side	Right-Hand Side

Question 14

2 marks

114

Your classmate, Leo, was absent for one of his math lessons.

Explain to Leo how to determine the cosecant ratio for an angle in standard position given that

$P(-3, -4)$ is a point on the terminal arm of the angle.

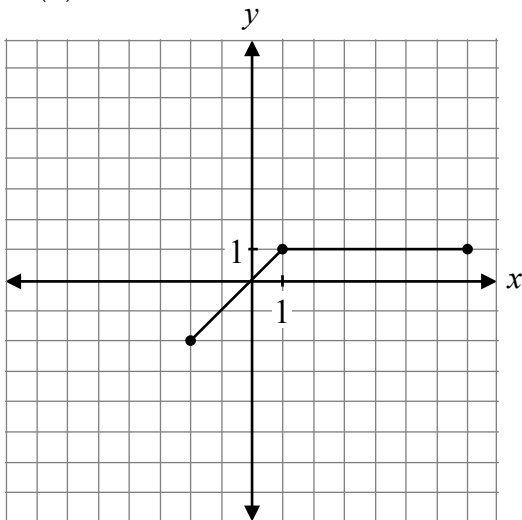
Question 15

2 marks

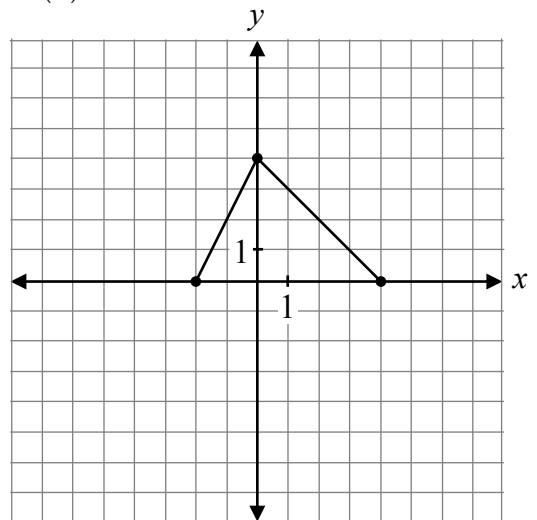
115

Given the following graphs:

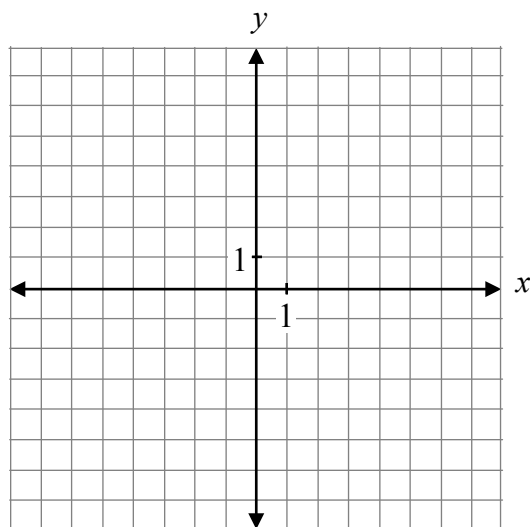
$f(x)$



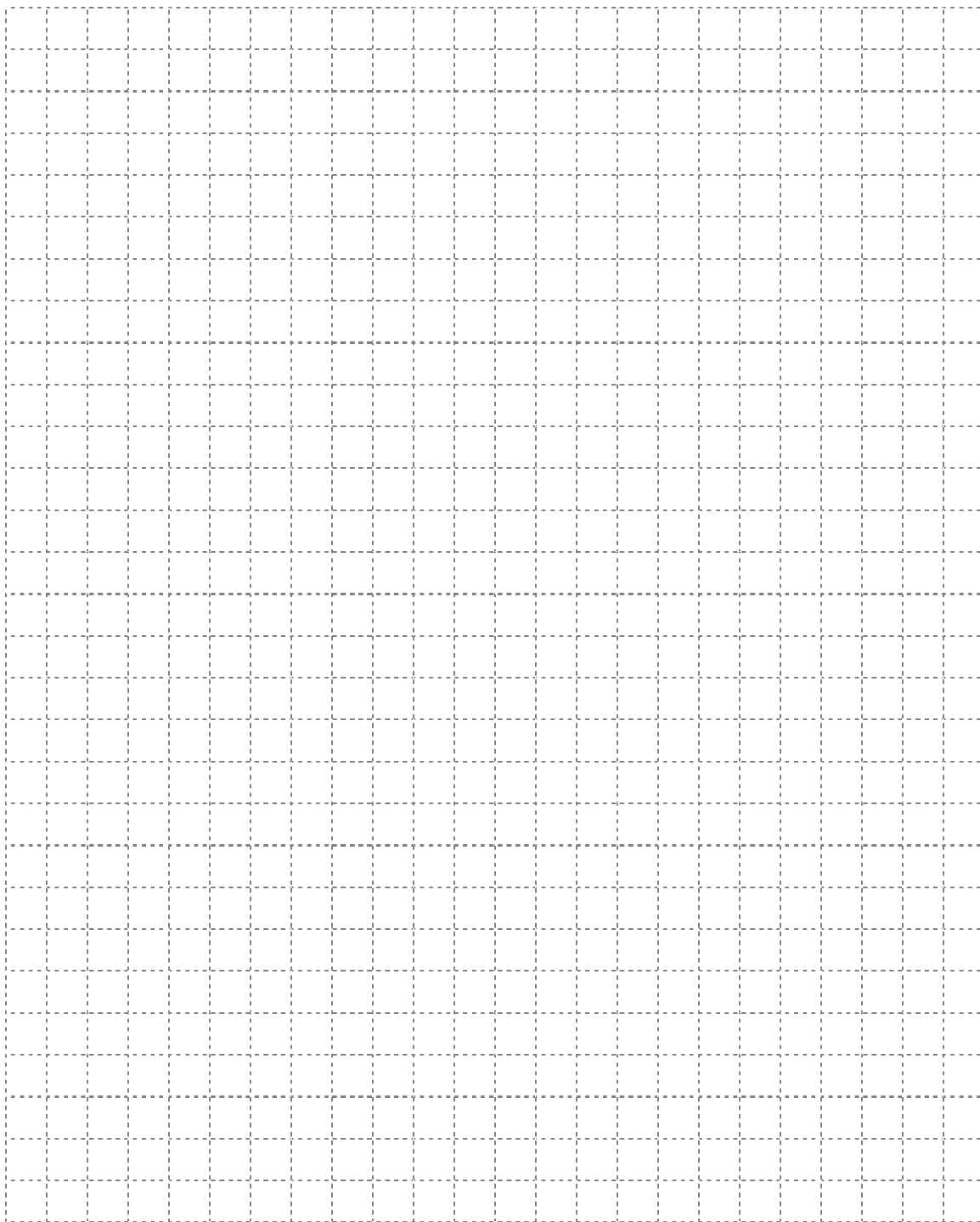
$g(x)$



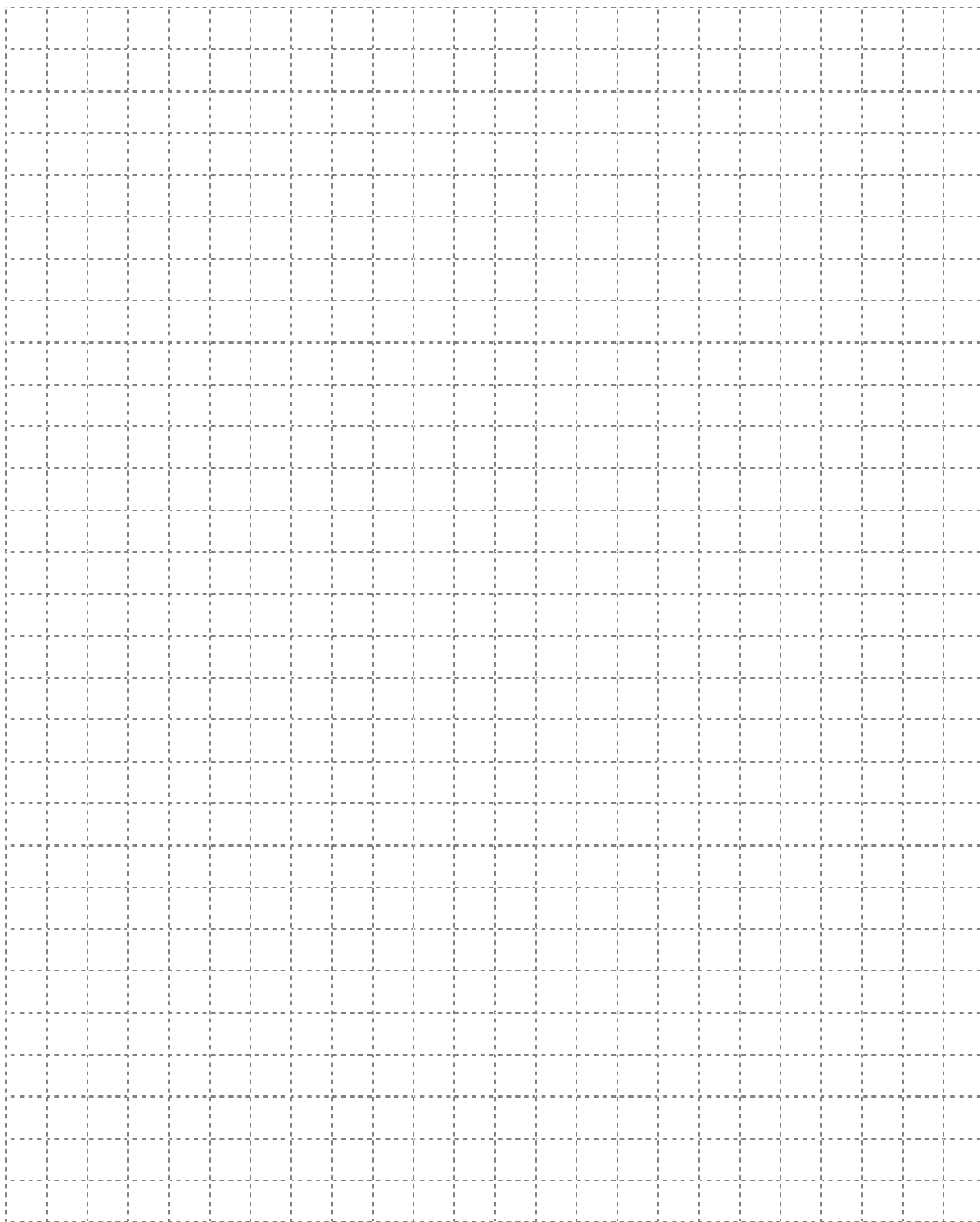
Sketch the graph of $f(x) + g(x)$.



No marks will be awarded for work done on this page.



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Booklet 2

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Instructions

Multiple-Choice Questions

- There are 9 questions each worth one mark.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Short and Long Answer Questions

- There are 25 questions worth a total of 49 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 16

1 mark

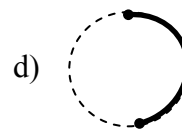
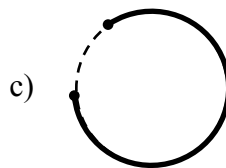
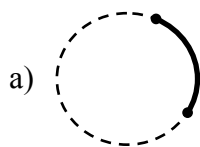
If $(2, 3)$ is a point on the graph of $y = f(x)$, what point must be on the graph of $y = 3f\left(\frac{1}{4}x\right)$?

- a) $\left(\frac{1}{2}, 1\right)$ b) $\left(\frac{1}{2}, 9\right)$ c) $(8, 1)$ d) $(8, 9)$

Question 17

1 mark

Consider the arc drawn on each circle. Which arc measure is closest to 3 radians?



Question 18

1 mark

If $\log_2 x = 4$, then $\log_2(2x)$ is equal to:

- a) 5 b) 8 c) 16 d) 32

Question 19

1 mark

Simplify the following expression:

$$\cos^2 x (1 + \cot^2 x)$$

a) $\sin^2 x$

b) $\cos^2 x$

c) $\cot^2 x$

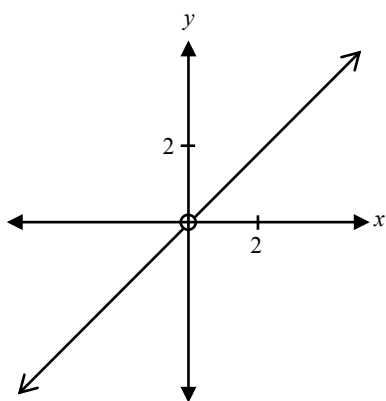
d) $\sec^2 x$

Question 20

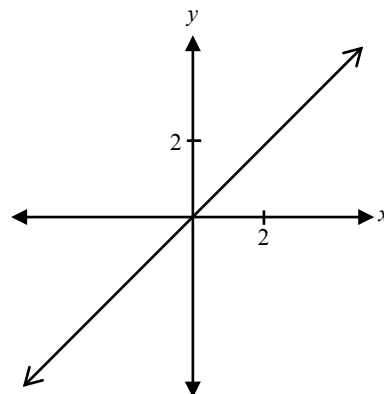
1 mark

Identify the graph of the function $y = \frac{x}{x}$.

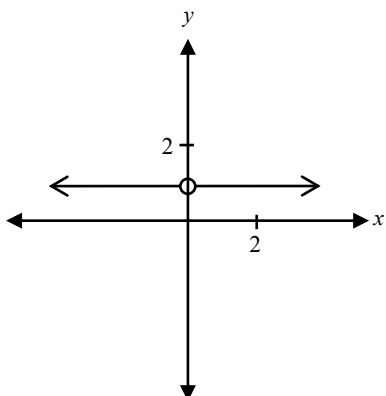
a)



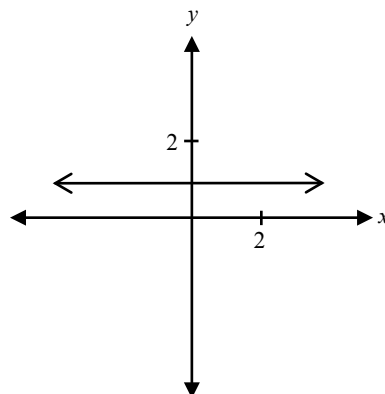
b)



c)



d)



Question 21**1 mark**

How many terms are in the expansion of $(3y^2 - 4z)^7$?

- a) 2 b) 6 c) 7 d) 8

Question 22**1 mark**

Determine one possible restriction for the domain of $y = (x + 3)^2 - 4$ so that its inverse is a function.

- a) $x \leq -3$ b) $x \leq 0$ c) $x \leq 3$ d) $x \leq 4$

Question 23**1 mark**

Find the total possible number of arrangements for 7 adults and 3 children seated in a row if the 3 children must sit together.

- a) $10!$ b) $8!3!$ c) $7!3!$ d) $7!$

Question 24**1 mark**

Identify the value of the x -intercept of the function $y = \ln(x - 2)$.

- a) -1 b) 0 c) 2 d) 3

Question 25

1 mark

116

Given $\log_b a = 3$, give one example of possible values for a and b that make this equation true.

Question 26

1 mark

117

The range of the graph of $y = f(x)$ is $[-3, 2]$.

Explain why there is no effect on the range of the graph that is a result of the transformation $y = f(-x)$.

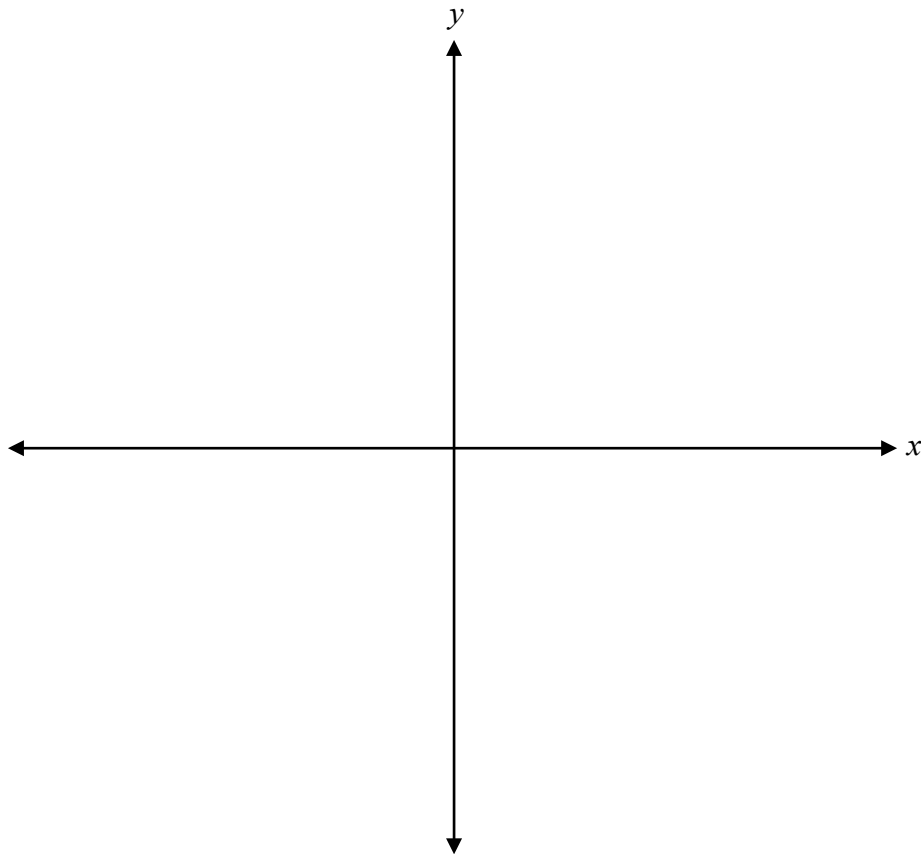
Question 27

3 marks

118

Sketch the graph of $y = (x + 1)(x - 2)^2(x + 5)$.

Identify the x -intercepts and y -intercept.



x -intercepts: _____

y -intercept: _____

Question 28

2 marks

119

The graph of the function $y = \sin x$ has been transformed to create a new graph.

The range of this new graph is $[-4, 4]$ and the zeros are $x = k\frac{\pi}{2}$, where k is an integer.

Write the equation that corresponds to this new graph.

Question 29

1 mark

120

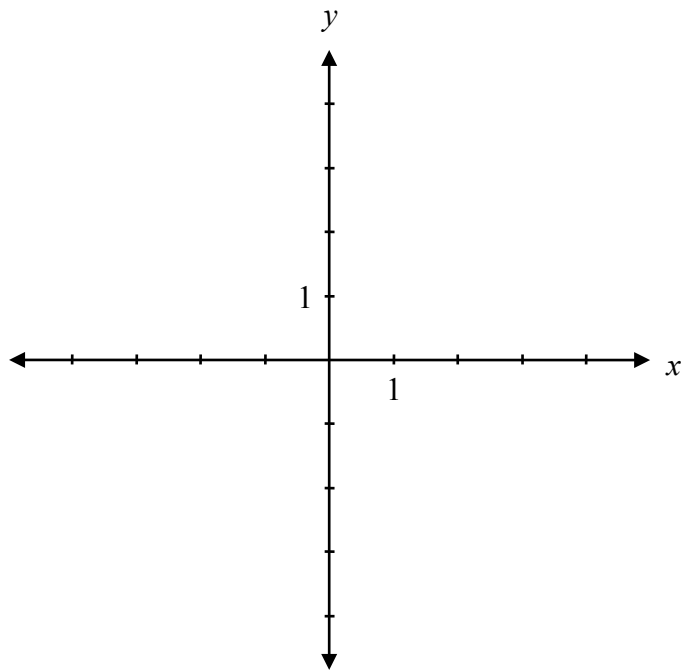
Given the functions $f(x) = x^2 - 1$ and $g(x) = x + 1$, state the domain of $\frac{g(x)}{f(x)}$.

Question 30

a) 2 marks b) 1 mark

121
122

- a) Sketch the graph of $y = 3^x$.



- b) Explain how the graph of $y = 3^x$ can be used to sketch the graph of $y = \log_3 x$.

Question 31

5 marks

123

A box in the shape of a rectangular prism has side lengths x , $x + 2$, and $x + 10$.

Write a function, $V(x)$, to express the volume of the box in terms of x .

Find all possible values of x , given that the volume of the box is 96 cm^3 .

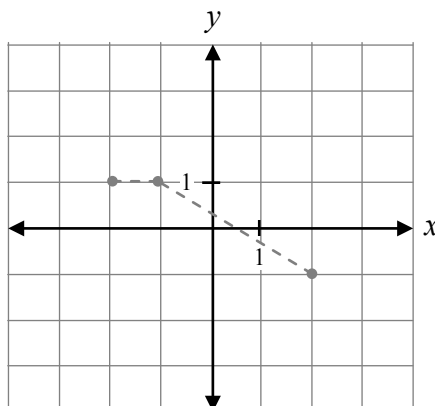
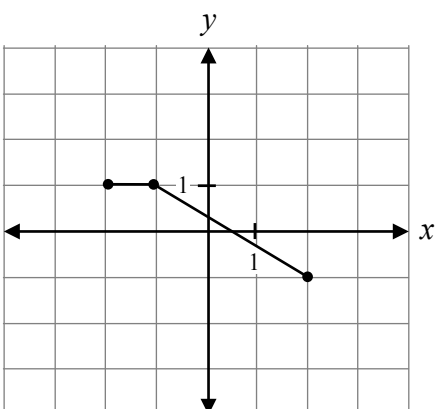
State the dimensions of the box.

Question 32

1 mark

124

Given the graph of $f(x)$ below, sketch the graph of $y = -f(x)$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 33

1 mark

125

Determine the coordinates of a point (x, y) on the unit circle if you are given $\theta = 30^\circ$ where θ is in standard position.

Question 34

3 marks

126

Given the following sinusoidal equation:

$$P(t) = 3000 \sin \left[\frac{\pi}{10}(t - 2010) \right] + 10\,000$$

Determine the maximum value of $P(t)$ and a value of t at which this maximum occurs.

Maximum value of $P(t)$: _____

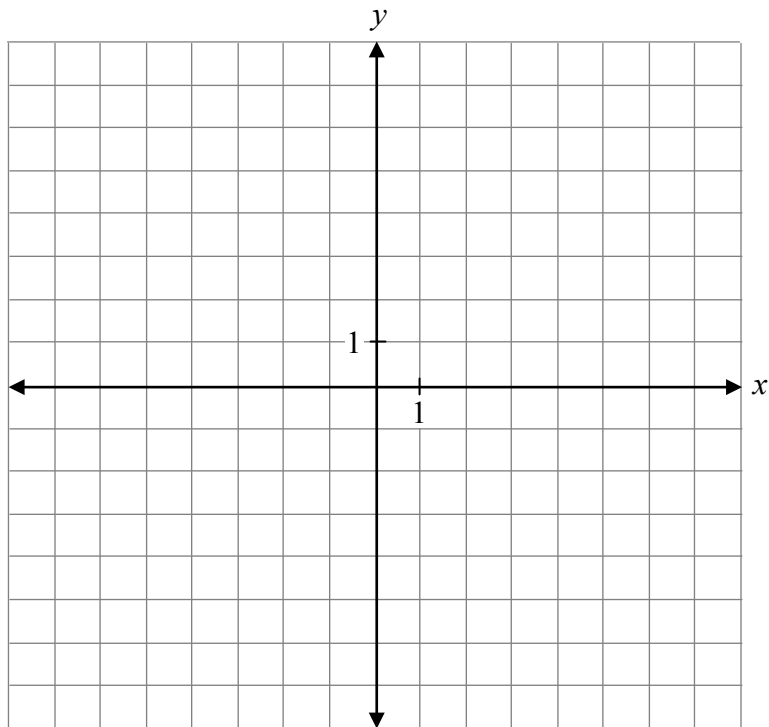
Value of t : _____

Question 35

3 marks

127

Sketch the graph of $y = \sqrt{2x - 2}$.



Question 36

2 marks

128

Given $f(x) = 2x - 6$, write the equation of $f^{-1}(x)$.

Question 37

1 mark

129

Frank tried to expand a logarithmic expression using the laws of logarithms. He made one error.

Frank's solution: $\log_a \frac{(x+2)}{zw} = \log_a x + \log_a 2 - \log_a z - \log_a w$

Write the correct solution.

Question 38

3 marks

130

Determine all non-permissible values of θ over the interval $[0, 2\pi]$.

$$\frac{\sin \theta}{1 + \cos \theta} + \csc \theta + \cot \theta$$

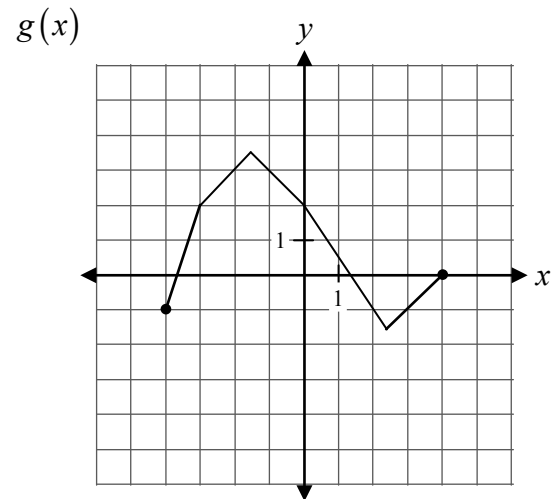
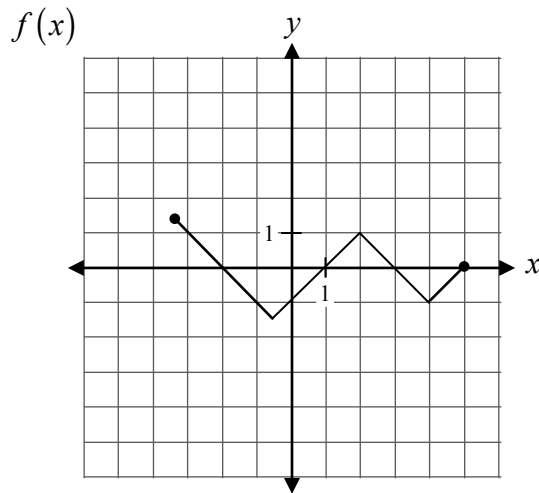
Explain your reasoning.

Question 39

a) 1 mark b) 1 mark c) 1 mark

131
132
133

Given the following graphs:



a) Determine the value of $[f \cdot g](0)$.

b) Determine the value of $g(f(4))$.

c) Determine a value for k where $f(k) = 1$.

Question 40

1 mark

134

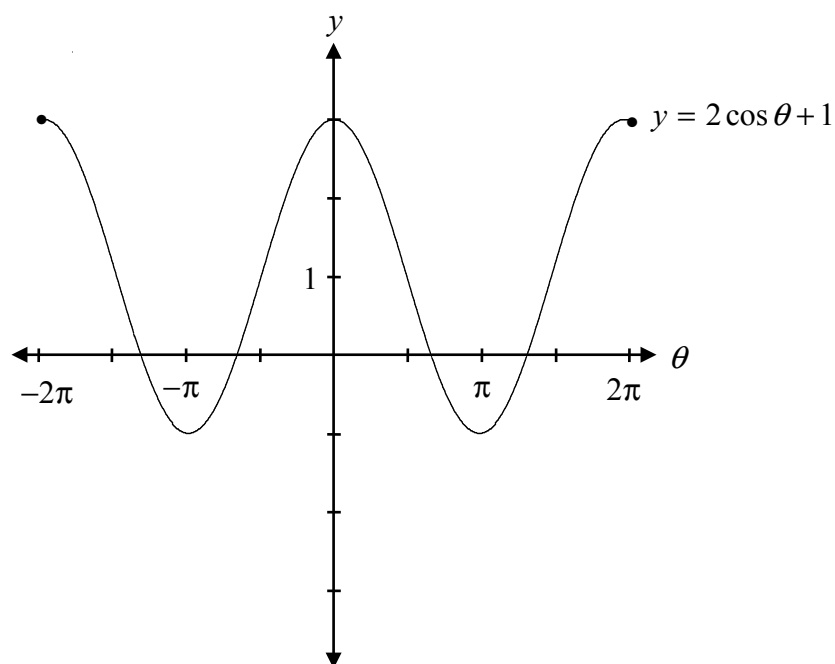
Given that $h(x) = 2x^2 + 5x - 3$ and that $h(x) = f(x) \cdot g(x)$, determine $f(x)$ and $g(x)$.

Question 41

1 mark

135

The graph of $y = 2 \cos \theta + 1$ below can be used to solve the equation $\cos \theta = -\frac{1}{2}$ over the interval $[-2\pi, 2\pi]$. Indicate on the graph where to find the solutions to the equation $\cos \theta = -\frac{1}{2}$.



Question 42

1 mark

136

The function $f(x)$ is transformed.

A new function, $y = \frac{1}{f(x)}$, is created that does not have any vertical asymptotes.

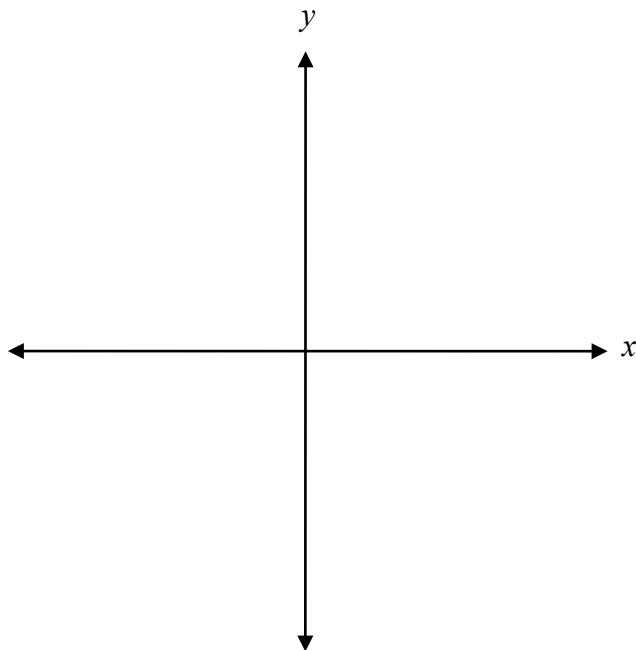
What can you conclude about the original function $f(x)$?

Question 43

1 mark

137

Draw the angle $-\frac{7\pi}{8}$ in standard position.



Question 44

3 marks

138

Determine the exact value of:

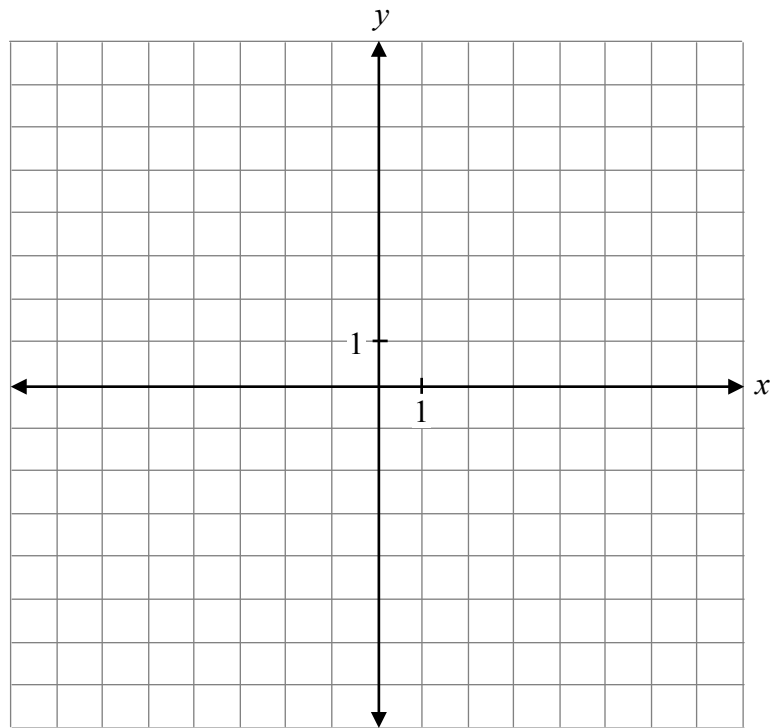
$$4 \cos\left(\frac{11\pi}{12}\right)$$

Question 45

3 marks

139

Sketch the graph of $f(x) = \frac{x-4}{x^2-3x-4}$.



Question 46

1 mark

140

Estimate the value of $\log_5 35$.

Justify your answer.

Question 47

1 mark

141

If $p(x) = x^5 - 12x + 1$, determine the remainder when $p(x)$ is divided by $(x + 2)$.

Question 48**1 mark**

142

Describe the effects on the graph of $y = f(x)$ when asked for the graph of $y = f(x - 3) + 5$.

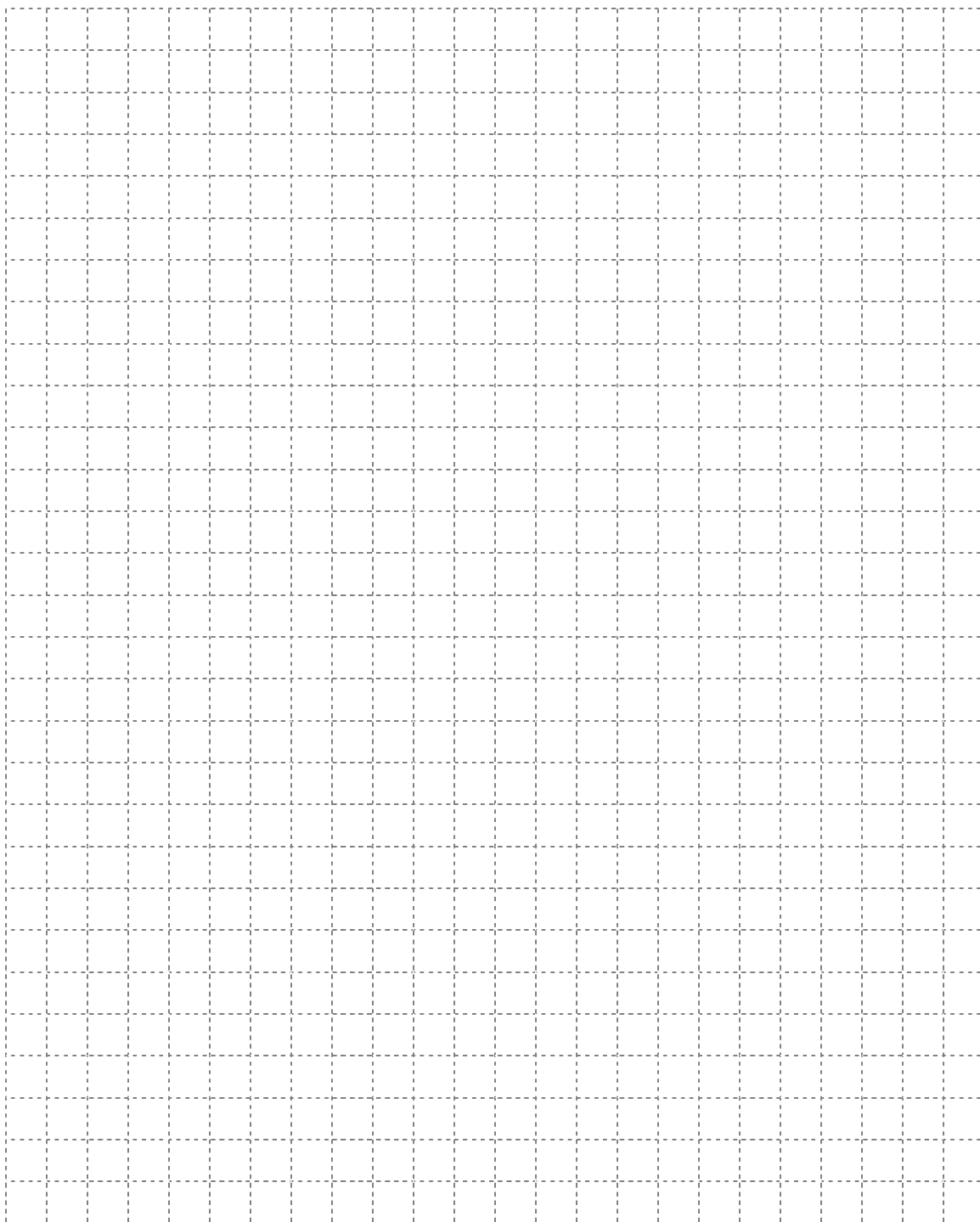
Question 49**3 marks**

143

Find the exact value of the following expression:

$$\sin\left(\frac{11\pi}{3}\right) \cdot \sec\left(\frac{4\pi}{3}\right) \cdot \tan\left(-\frac{5\pi}{6}\right)$$

No marks will be awarded for work done on this page.



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Booklet 1

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$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

$$A = Pe^{rt}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$\log_a M = \frac{\log_b M}{\log_b a}$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$y = a \sin b(x - c) + d$$

$$y = a \cos b(x - c) + d$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are explained below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

Solve: Give a solution for a problem or determine the value(s) of a variable.

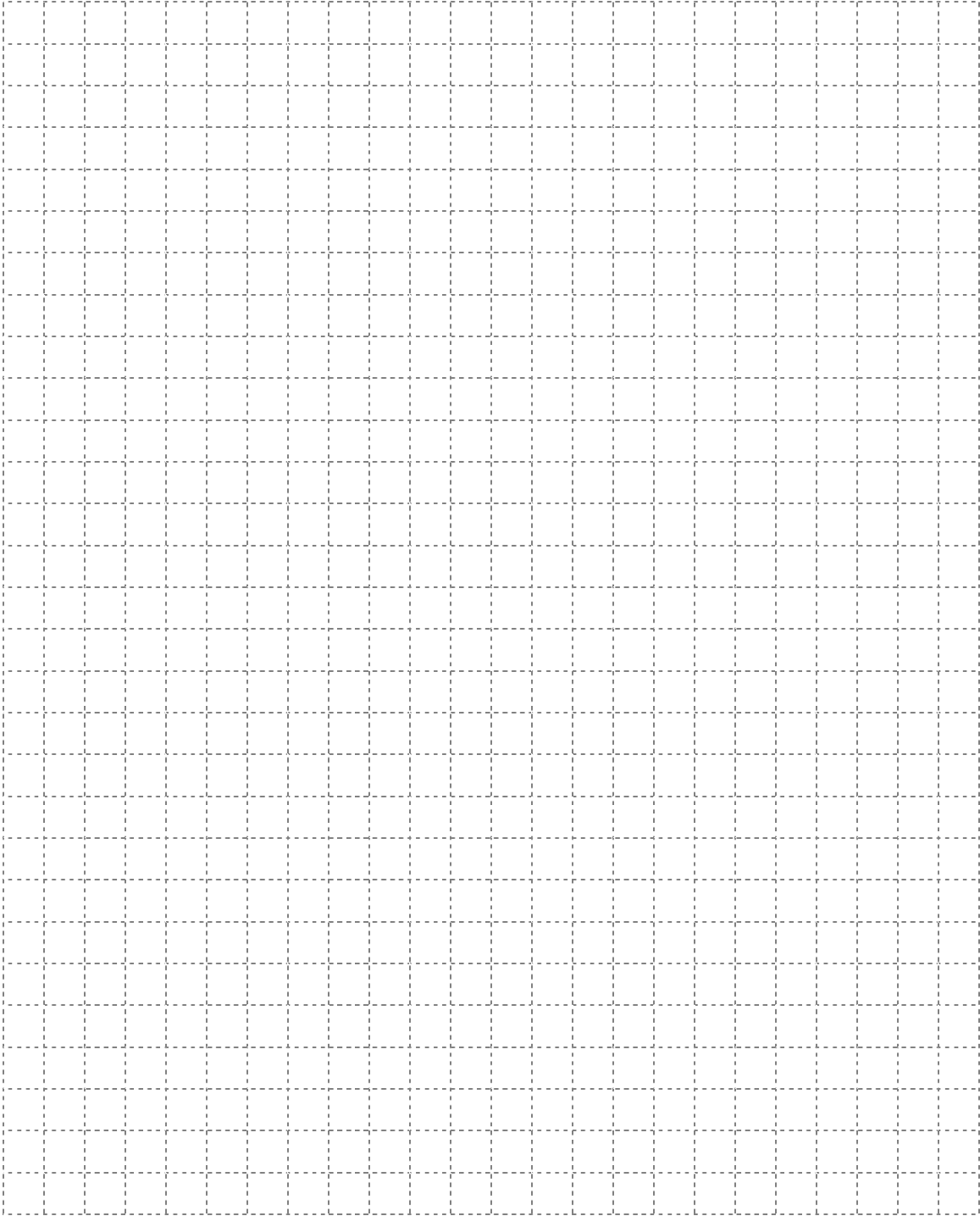
Verify: Establish the truth of a statement by substitution or comparison.

Scrap Paper

No marks will be awarded for work done on this page.

Scrap Paper

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Instructions

- There are 16 questions for a total of 38 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- A calculator icon  appears next to the questions that require a calculator.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to 3 decimal places unless instructed otherwise.

No marks will be awarded for work done on this page.

Question 1

2 marks

101

A central angle of a circle subtends an arc length of 5π cm.
Given the circle has a radius of 9 cm, find the measure of the central angle in degrees.

Question 2

4 marks

102

Solve the equation $\csc^2 \theta + 3 \csc \theta - 4 = 0$ over the interval $[0, 2\pi]$.
Express your answers as exact values or correct to 3 decimal places.

Question 3

3 marks

103

Jess invests \$12 000 at a rate of 4.75% compounded monthly.
How long will it take for Jess to triple her investment?

Express your answer in years, correct to 3 decimal places.

Question 4

3 marks

104

The 4th term in the binomial expansion of $\left(qx^2 - \frac{3}{x}\right)^{10}$ is $414\,720x^{11}$.

Determine the value of q algebraically.

Note: A calculator is not required for the remaining test questions.

Question 5

1 mark

105

Bella has 2 pairs of shoes, 3 pairs of pants, and 10 shirts.

Carey has 4 pairs of shoes, 4 pairs of pants, and 4 shirts.

An outfit is made up of one pair of shoes, one pair of pants, and one shirt.

Who can make more outfits? Justify your answer.

Question 6

2 marks

106

In the binomial expansion of $(x - y)^{10}$, how many terms will be positive?

Justify your answer.

Question 7

4 marks

107

Solve the following equation algebraically where $180^\circ \leq \theta \leq 360^\circ$.

$$2 \sin^2 \theta + 5 \cos \theta + 1 = 0$$

Question 8

3 marks

108

Solve the following equation algebraically:

$$\log_3(x - 4) + \log_3(x - 2) = 1$$

Question 9

1 mark

109

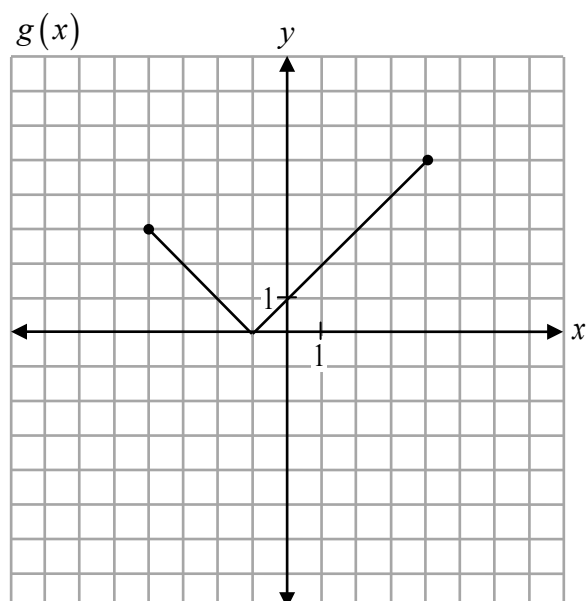
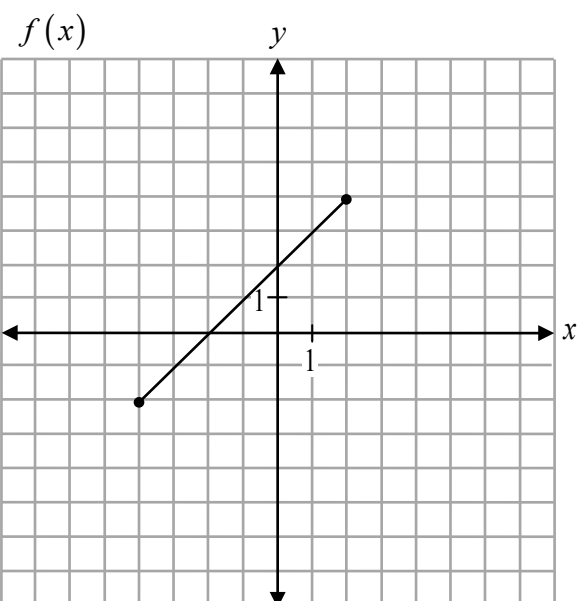
Given that $f(x) = \{(1, 3), (2, 5), (3, 4), (4, 2)\}$, find $f(f(3))$.

Question 10

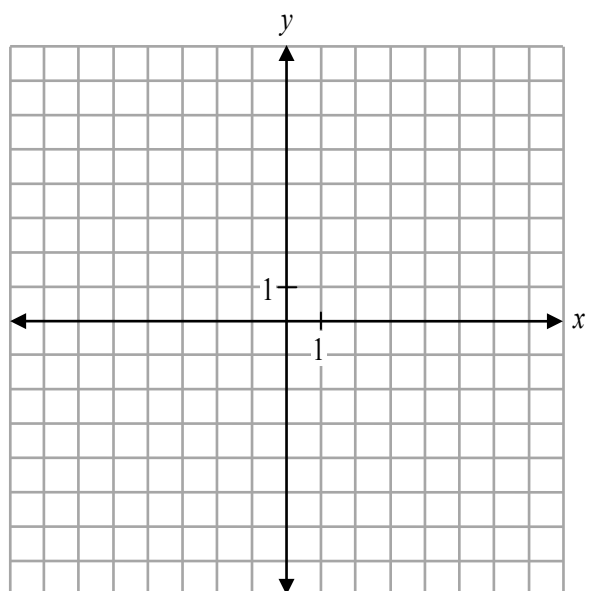
2 marks

110

Given the graphs of $f(x)$ and $g(x)$ below,



sketch the graph of $y = f(x) - g(x)$.



Question 11

2 marks

111

Given the graph of $y = f(x)$, describe the transformations to obtain the graph of the function $y = f(2x - 6)$.

Question 12

1 mark

112

Given $f(x) = \{(-3, 4), (2, 7), (8, 6)\}$, state the domain of the resulting function after $f(x)$ is reflected through the line $y = x$.

Question 13

3 marks

113

Determine the value of y in the following equation:

$$\log_x 27 - \log_x 3 = 2 \log_x y$$

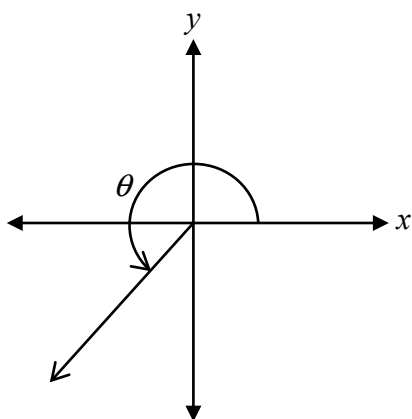
Question 14

1 mark

114

Angle θ , measuring $\frac{5\pi}{4}$, is drawn in standard position as shown below.

Determine the measures of all angles in the interval $[-4\pi, 2\pi]$ that are coterminal with θ .



Question 15

3 marks

115

Prove the identity below for all permissible values of x :

$$\frac{\sin^2 x}{\sec x + 1} = \cos x - \cos^2 x$$

Left-Hand Side	Right-Hand Side

Question 16

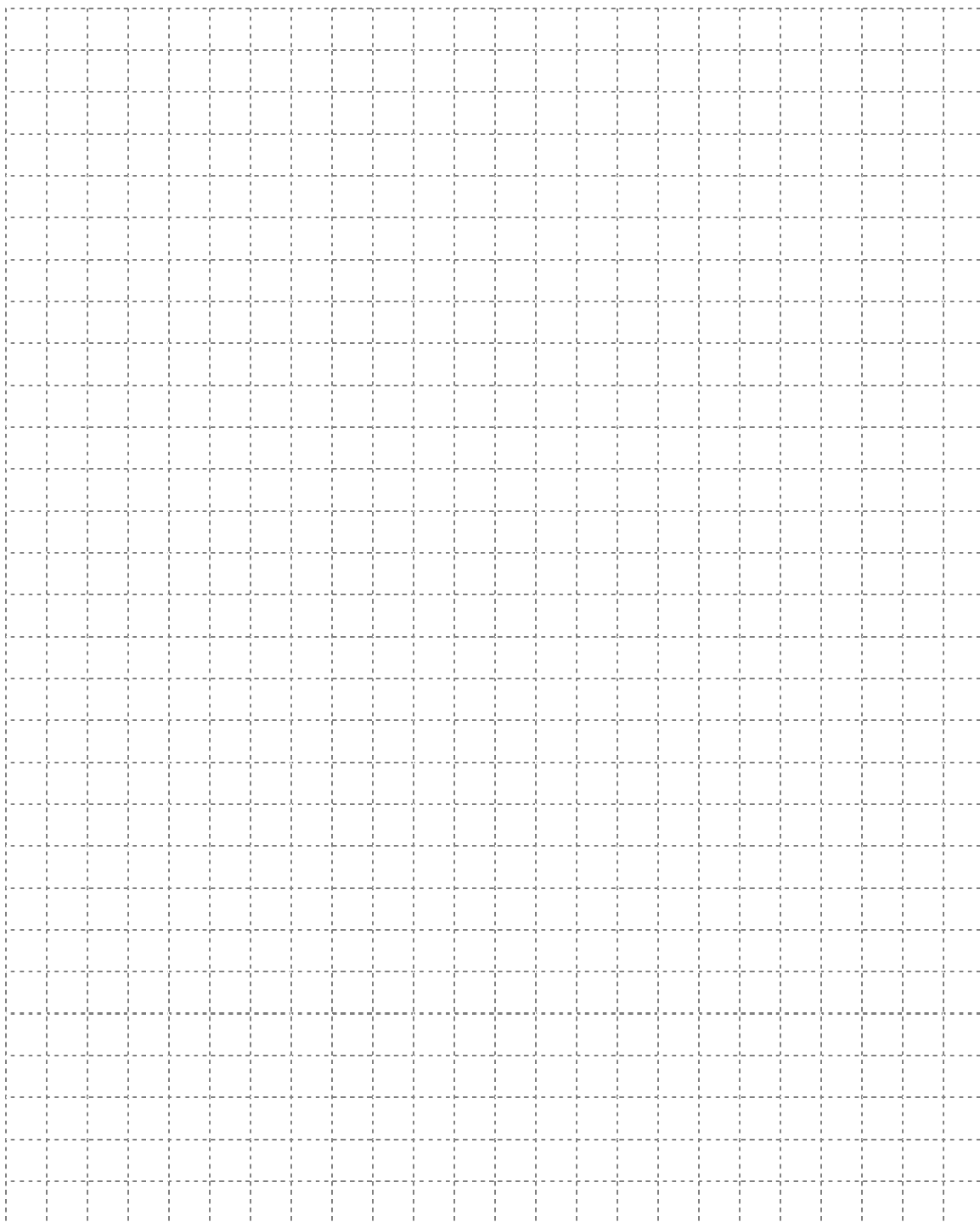
3 marks

116

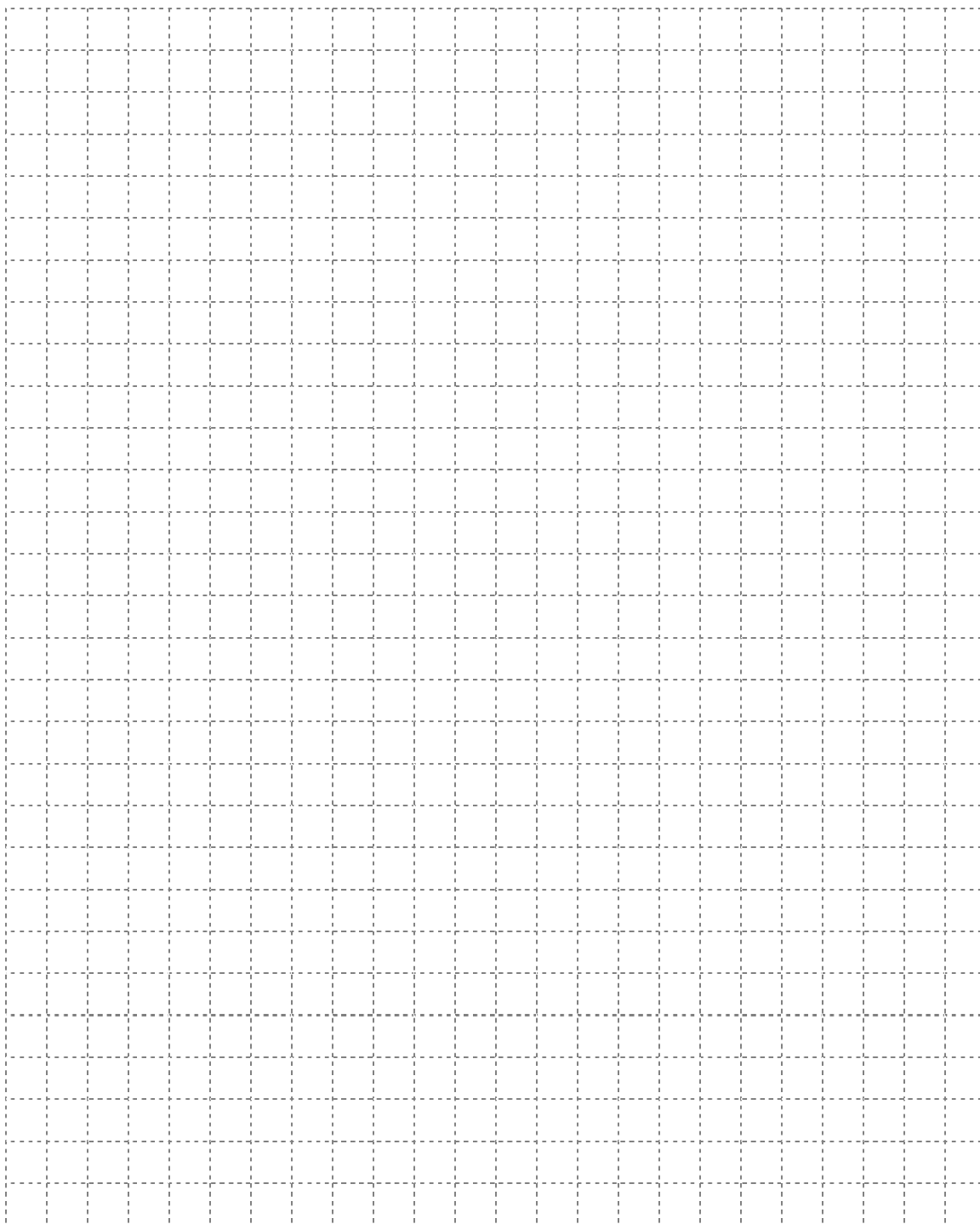
Solve algebraically:

$${}_nC_2 = 4n + 5$$

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2013

Manitoba Education Cataloguing in Publication Data

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Booklet 2. June 2013 [electronic resource]

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 4. Calculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
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Winnipeg, Manitoba, Canada

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Disponible en français.

Available in alternate formats upon request.

Instructions

Multiple-Choice Questions

- There are 8 questions each worth one mark.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Short and Long Answer Questions

- There are 21 questions worth a total of 44 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 17**1 mark**

How many different arrangements are possible when arranging all of the letters of the word SEPTEMBER?

a) $9!$

b) $6!3!$

c) $\frac{9!}{3!}$

d) $\frac{6!}{3!}$

Question 18**1 mark**

Which one of the following angles terminates in Quadrant III?

a) 3 radians

b) $\frac{7\pi}{5}$ radians

c) -210°

d) 500°

Question 19**1 mark**

There are 13 terms in the expansion of $(3x - y)^{2n}$. Determine the value of n .

a) 6

b) 6.5

c) 7

d) 26

Question 20**1 mark**

Which of the following is true about the periods of the three functions below?

$$f(\theta) = 2\sin 3\left(\theta - \frac{\pi}{2}\right)$$

$$g(\theta) = \sin 3\theta + 6$$

$$k(\theta) = 3\sin \theta + 6$$

- a) The graphs of $f(\theta)$ and $g(\theta)$ have the same period.
- b) The graphs of $g(\theta)$ and $k(\theta)$ have the same period.
- c) All of the graphs have the same period.
- d) None of the graphs have the same period.

Question 21**1 mark**

Which of the following represents the general solution to the equation $\tan \theta = -1$?

- a) $\theta = \frac{\pi}{4} + 2k\pi, k \in \mathbb{I}$
- b) $\theta = \frac{\pi}{4} + k\pi, k \in \mathbb{I}$
- c) $\theta = \frac{3\pi}{4} + 2k\pi, k \in \mathbb{I}$
- d) $\theta = \frac{3\pi}{4} + k\pi, k \in \mathbb{I}$

Question 22**1 mark**

If $(3, -2)$ is a point on the graph of $y = f(x)$, what point must be on the graph of $y = 2f(x+1)$?

- a) $(4, -1)$
- b) $(4, -4)$
- c) $(2, 1)$
- d) $(2, -4)$

Question 23

1 mark

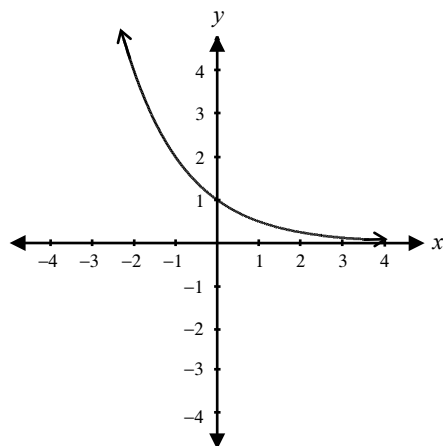
Which equation is represented by the graph sketched below?

a) $y = \left(\frac{1}{2}\right)^{-x}$

b) $y = \left(\frac{1}{2}\right)^x$

c) $y = 2^x$

d) $y = -2^x$



Question 24

1 mark

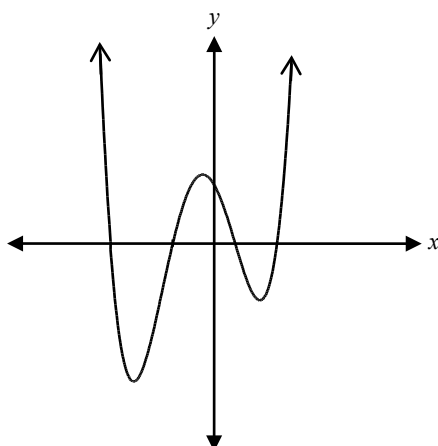
What is the degree of the polynomial represented below?

a) 2

b) 3

c) 4

d) 5

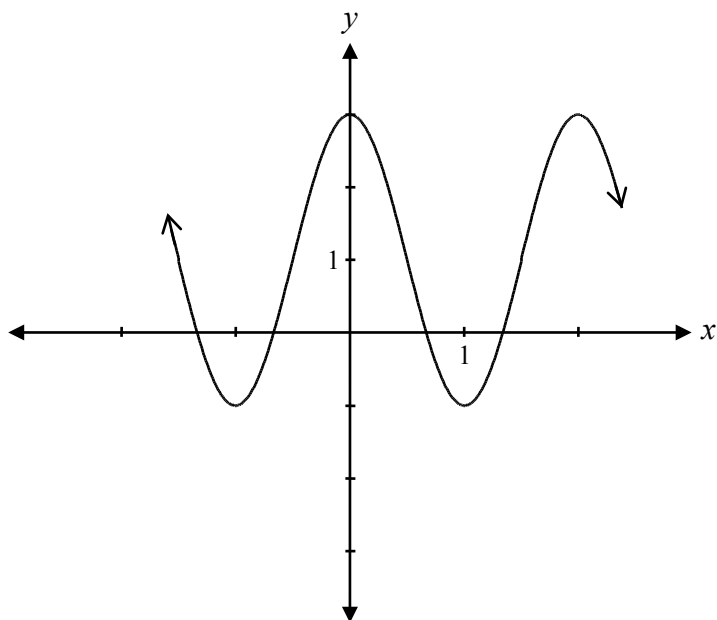


Question 25

1 mark

117

Given the graph of $y = 2 \cos \pi x + 1$ below, determine another equation that will produce the same graph.



$y =$ _____

Question 26

2 marks

118

Given $f(x) = 3$ and $g(x) = x + 2$, determine the domain and range of $h(x) = \frac{f(x)}{g(x)}$.

Domain: _____

Range: _____

Question 27

2 marks

119

Explain how to find the exact value of $\sec\left(\frac{19\pi}{6}\right)$.

Question 28

1 mark

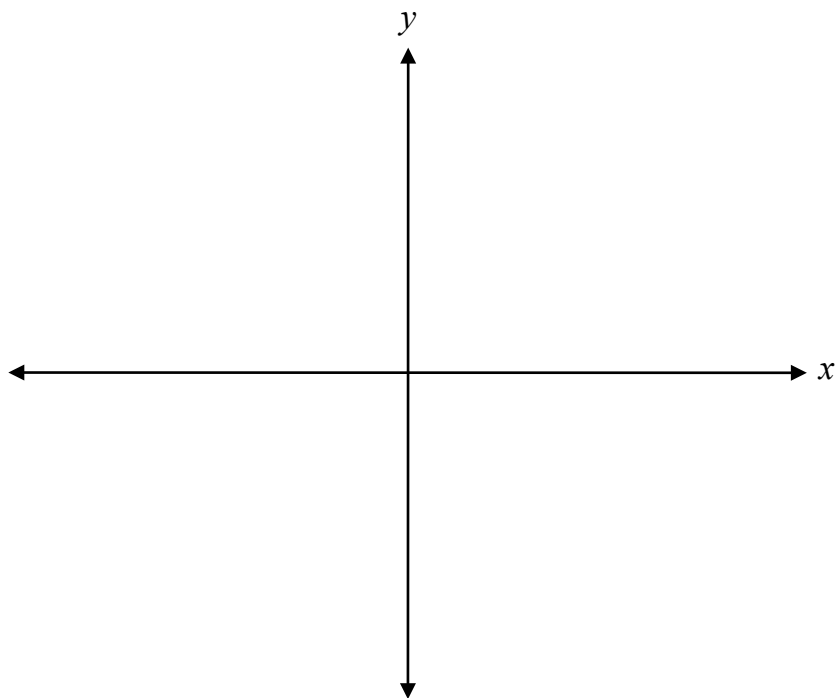
120

Given $f(x) = 4 - x$, verify that $f^{-1}(x) = f(x)$.

Sketch the graph of:

$$f(x) = (2 - x)(x + 3)(x + 1)^2$$

Label the x -intercepts and y -intercept.



Question 30

1 mark

122

Which expression has a larger value?

$$\log_2 36 \text{ or } \log_3 80$$

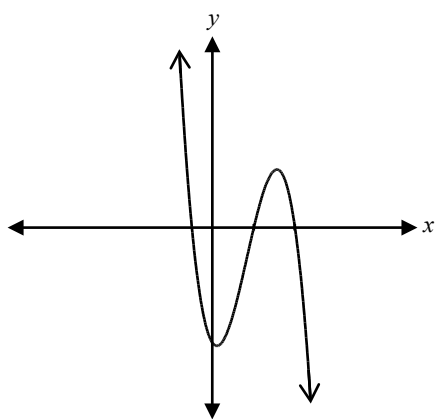
Justify your answer.

Question 31

1 mark

123

The graph below represents the equation $y = ax^3 + 6x^2 + 5x - 10$.



What must be true about the value of a ? Explain your reasoning.

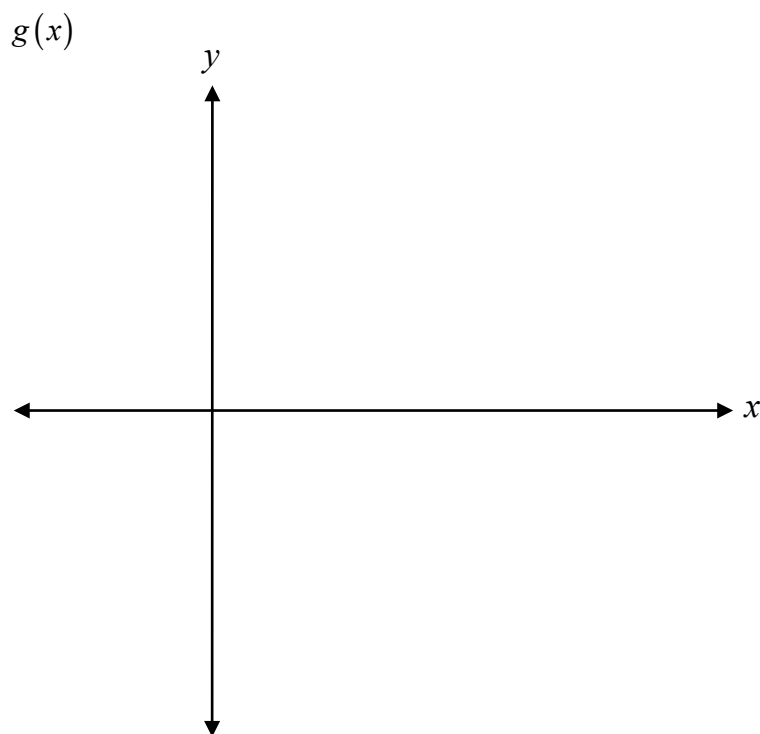
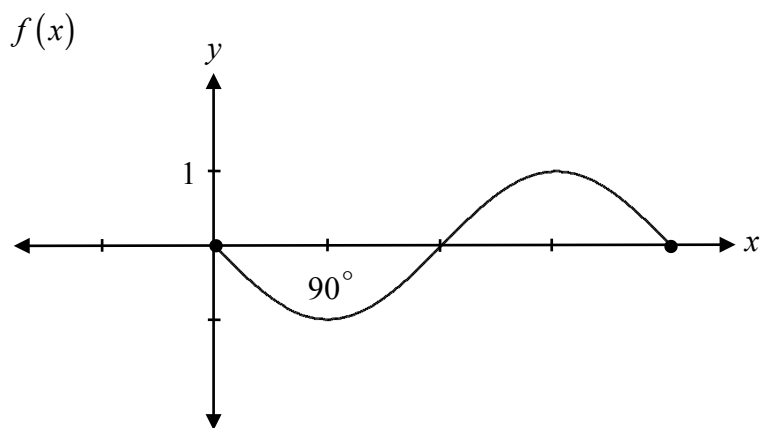
Question 32

2 marks

124

The terminal arm of an angle θ , in standard position, intersects the unit circle in Quadrant IV at a point $P\left(\frac{\sqrt{5}}{4}, y\right)$. Determine the value of $\sin \theta$.

Given the sinusoidal function $f(x)$ below, sketch the graph of $g(x) = |f(x)| - 1$.



Question 34**2 marks**

126

The graph of a rational function, $f(x)$, has a point of discontinuity when $x = 2$ and an asymptote when $x = 4$. Write a possible equation for $f(x)$.

Question 35**2 marks**

127

Given that $(x - 1)$ is one of the factors, express $x^3 - 57x + 56$ as a product of factors.

Question 36

2 marks

128

Give an example using values for A and B , in degrees or radians, to verify that $\cos(A + B) = \cos A + \cos B$ is **not** an identity.

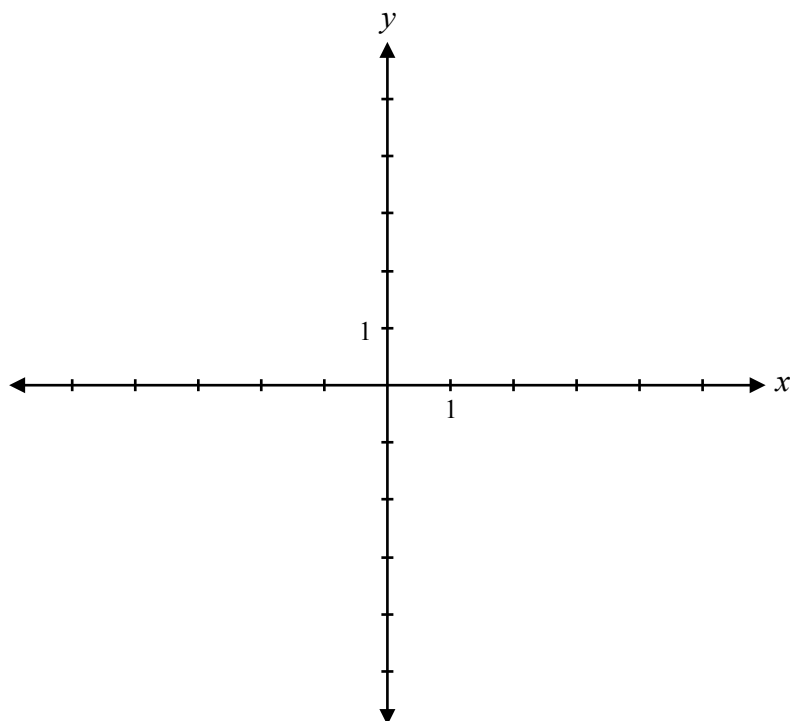
Left-Hand Side	Right-Hand Side

Question 37

3 marks

129

Sketch the graph of $y = \sqrt{x+1} - 2$ and verify that the value of the x -intercept is the same as the solution to the equation $\sqrt{x+1} - 2 = 0$.

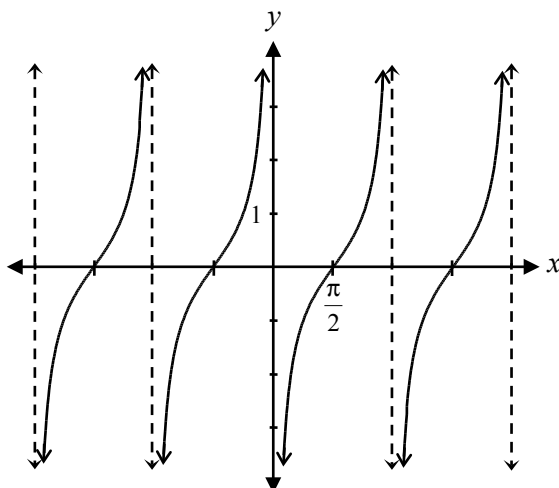


Question 38

1 mark

130

Mohamed is asked to sketch the graph of $y = \tan x$.
His graph is shown below.



Explain why his graph is incorrect.

Question 39

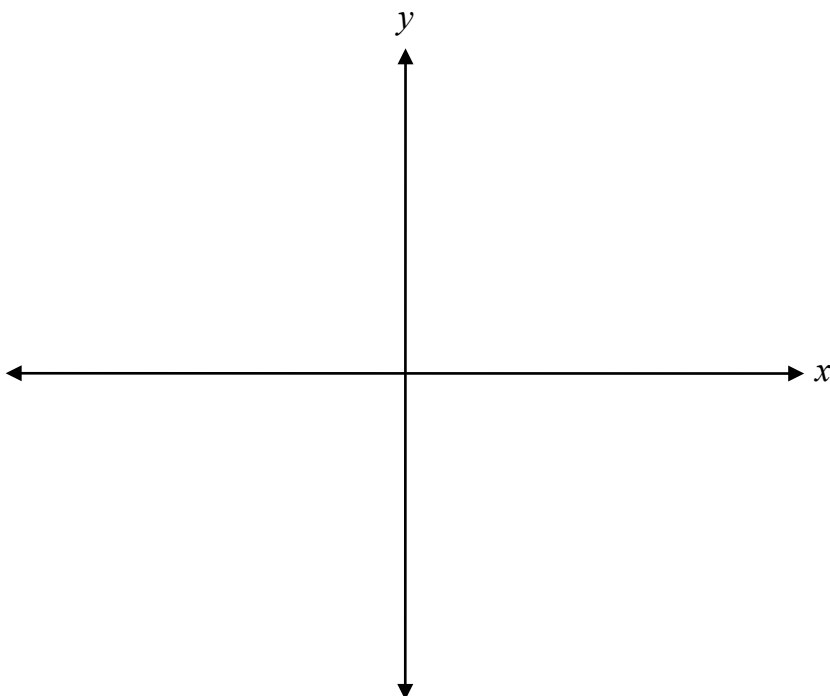
2 marks

131

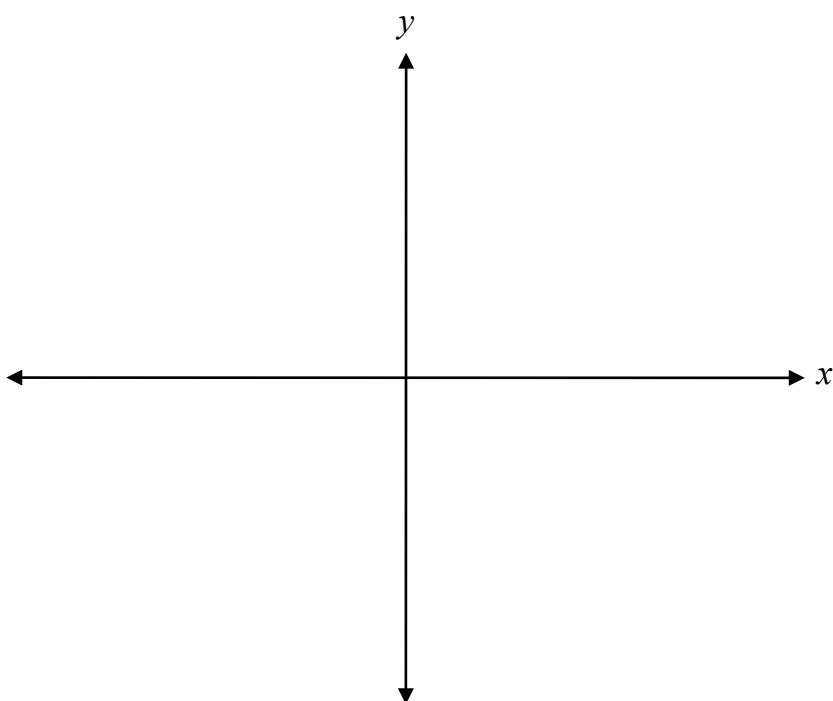
On the interval $0 \leq \theta < 2\pi$, identify the non-permissible values of θ for the trigonometric identity:

$$\tan \theta = \frac{1}{\cot \theta}$$

- a) Sketch the graph of $y = \ln(x)$.



- b) Sketch the graph of $y = -\ln(x - 2)$.



Question 41

2 marks

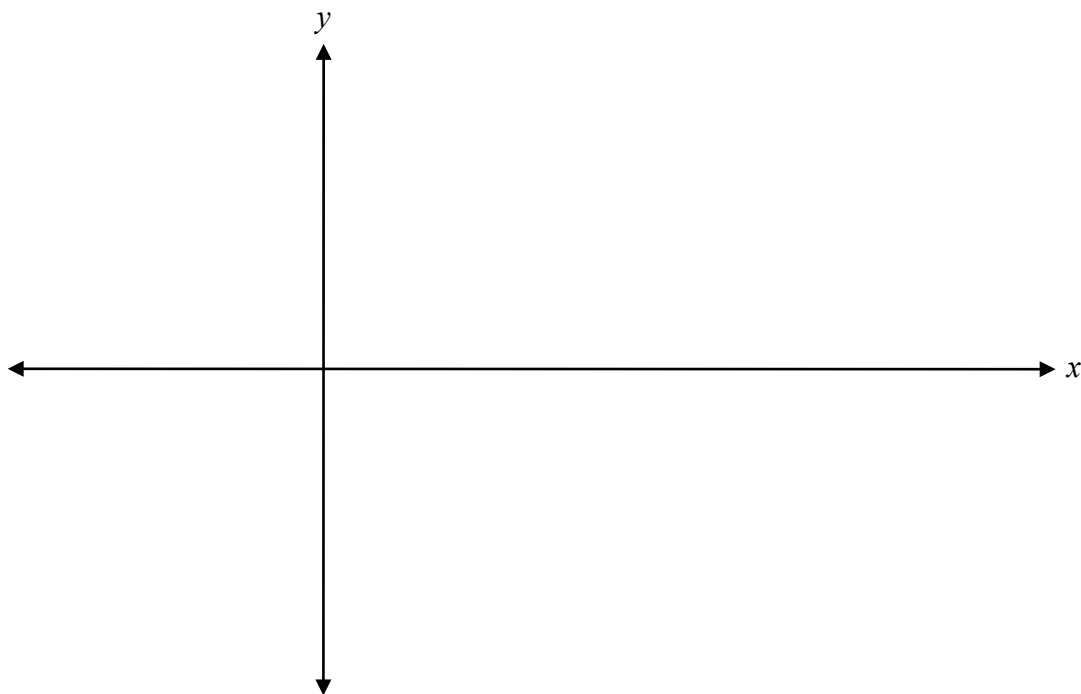
134

Given $f(x) = \sqrt{x - 2}$ and $g(x) = 3x$, write the equation for $h(x) = f(g(x))$.

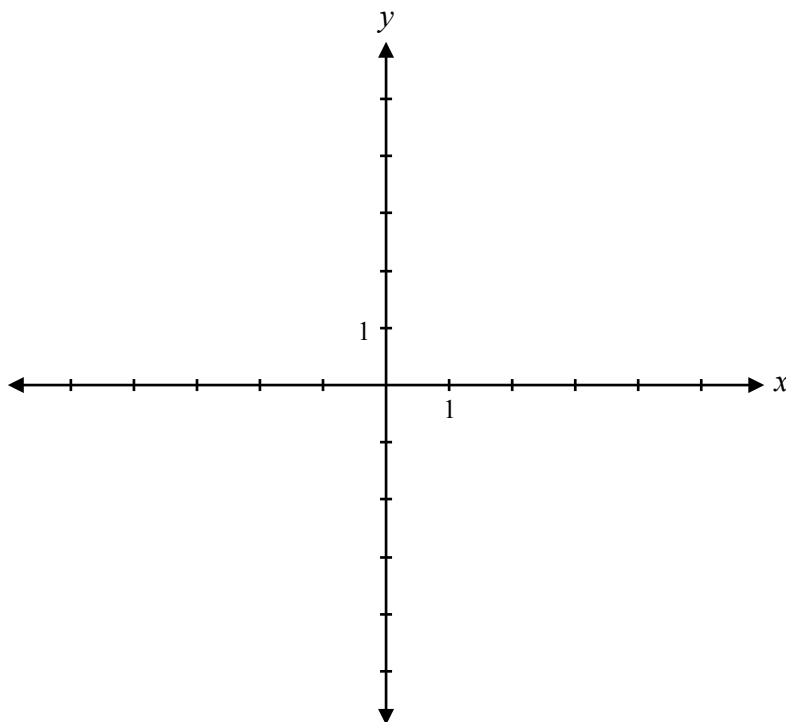
What are the restrictions on the domain of $h(x)$?

Explain your reasoning.

Sketch the graph of $y = 10\cos\left[\frac{\pi}{2}(x - 2)\right]$ over the interval $[0, 6]$.



Sketch the graph of the function $f(x) = \frac{x^2}{x^2 - x}$.



Question 44

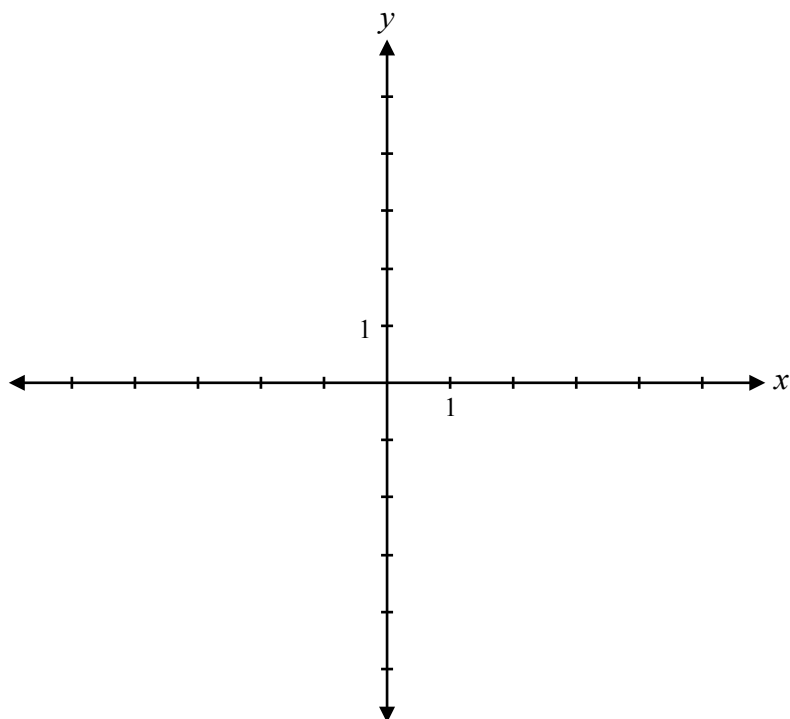
2 marks

137

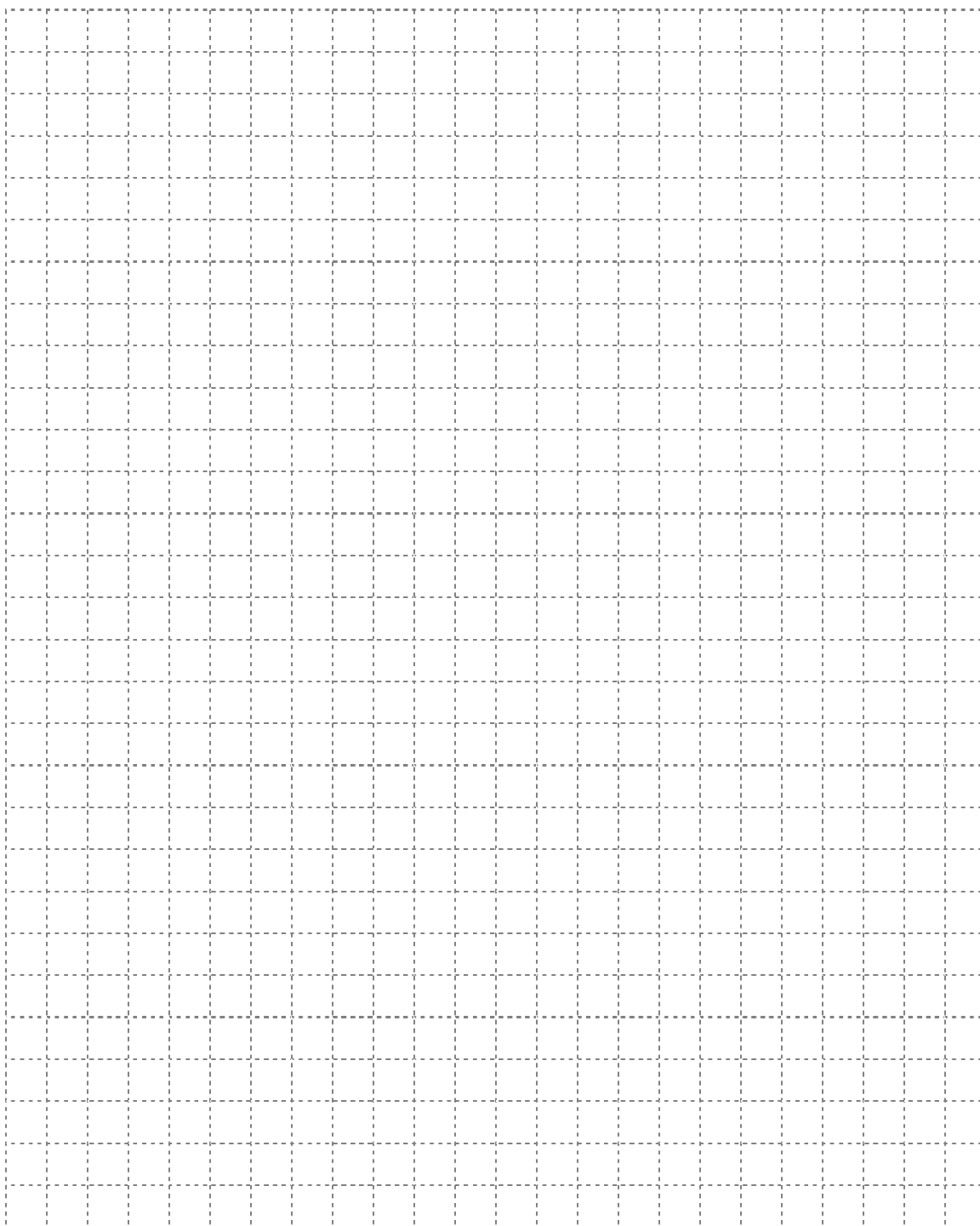
Is $(x - 3)$ a factor of $x^4 - x^3 - 3x^2 + x - 1$?

Justify your answer.

Given $f(x) = x - 1$ and $g(x) = x^2$, write the equation of $y = f(g(x))$ and sketch the graph.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2014

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

	Questions	Marks	Total Marks
Booklet 1*	10 Short-Answer	17	35
	5 Long-Answer	18	
Booklet 2	10 Multiple-Choice	10	57
	10 Short-Answer	15	
	9 Long-Answer	32	
Total			92

* The first 6 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- Read all instructions carefully.
- The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

No marks will be awarded for work done on this page.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are explained below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

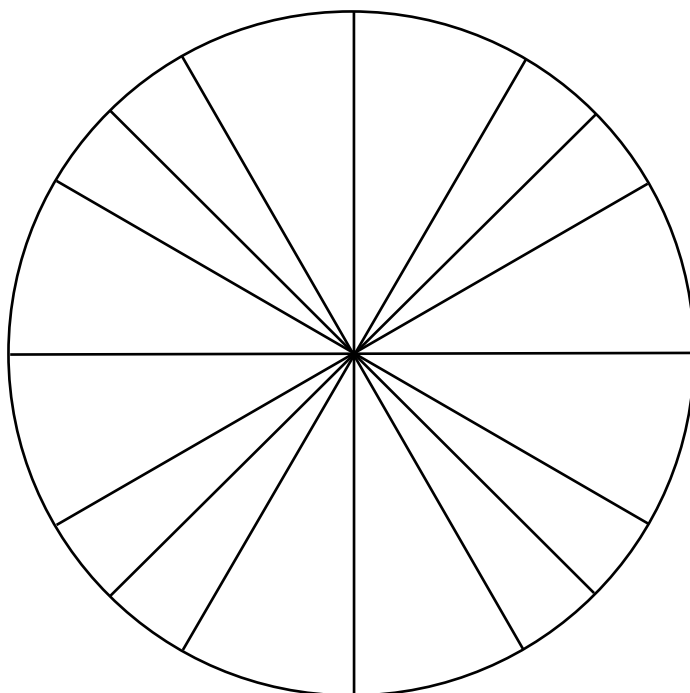
Solve: Give a solution for a problem or determine the value(s) of a variable.

Verify: Establish the truth of a statement by substitution or comparison.

Scrap Paper

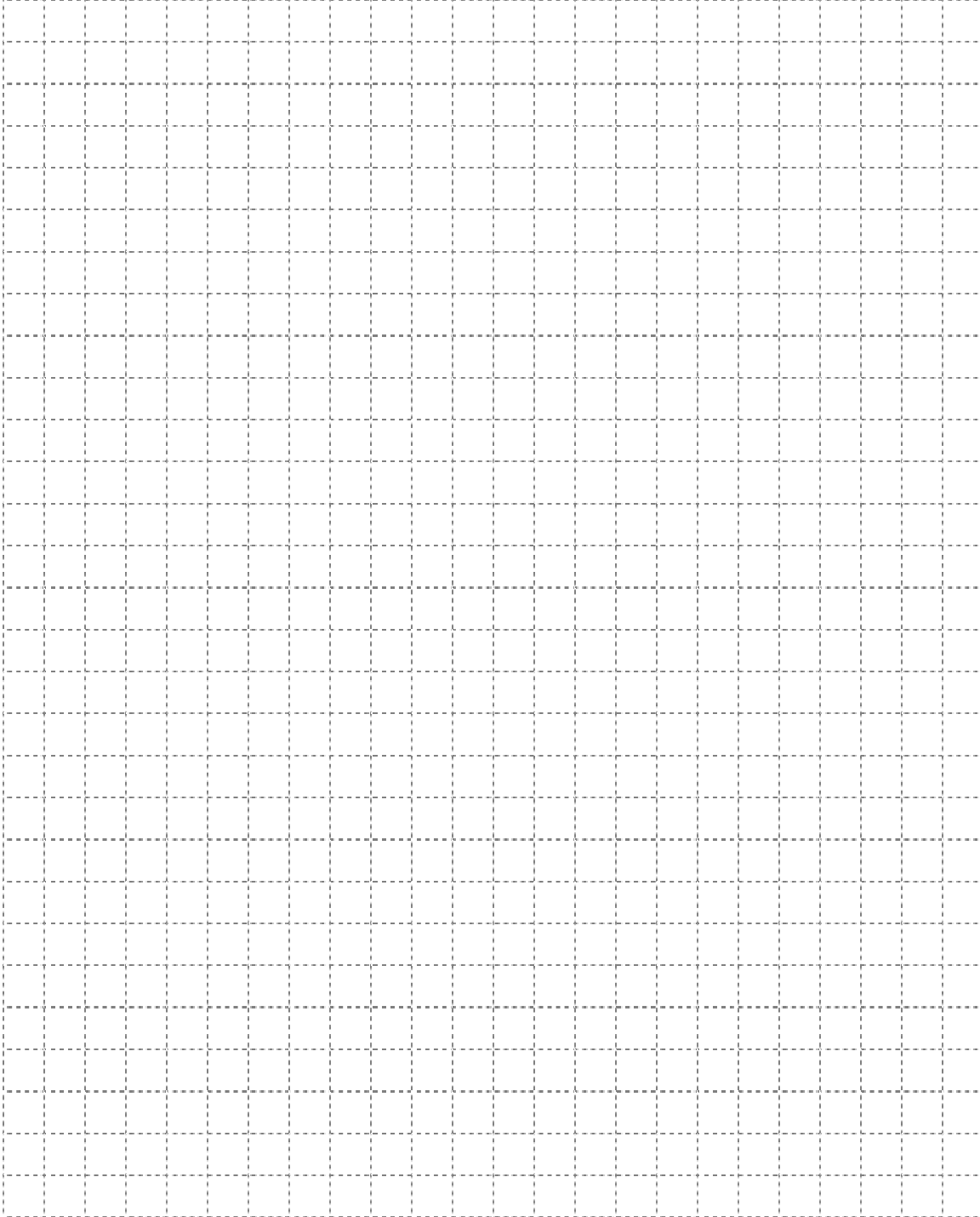
No marks will be awarded for work done on this page.

Unit Circle (can be used if needed)



Scrap Paper

No marks will be awarded for work done on this page.



Instructions

- There are 15 questions for a total of 35 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- A calculator icon  appears next to the questions that require a calculator.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to 3 decimal places unless instructed otherwise.

No marks will be awarded for work done on this page.

Question 1

2 marks

101

Find the coterminal angle to $\frac{27\pi}{5}$ over the interval $[-360^\circ, 0^\circ)$.

Question 2

3 marks

102

Solve the following equation over the interval $0 \leq \theta < 2\pi$.

$$(\tan \theta - 3)(\tan \theta + 1) = 0$$

Question 3

2 marks

103

An earthquake in Vancouver had a magnitude of 6.3 on the Richter scale. An earthquake in Japan had a magnitude of 8.9 on the Richter scale.

How many times more intense was the Japan earthquake than the Vancouver earthquake?

You may use the formula below:

$$M = \log \left(\frac{A}{A_0} \right)$$

where M is the magnitude of the earthquake on the Richter scale

A is the intensity of the earthquake

A_0 is the intensity of a standard earthquake

Express your answer as a whole number.

Question 4

2 marks

104

Find and simplify the last term in the expansion of $(2y - 3x)^7$.

Question 5

3 marks

105

Given $\log_a 9 = 1.129$ and $\log_a 4 = 0.712$, find the value of $\log_a 12$.

Question 6

2 marks

106

How many different ways can 4 girls and 4 boys be arranged in a row if the girls and the boys must alternate?

Note: A calculator is not required for the remaining test questions.

Question 7

4 marks

107

Solve the following equation over the interval $[0, 2\pi]$.

$$2 \cos 2\theta - 1 = 0$$

Question 8

1 mark

108

Alex incorrectly explains to Rashid that the graph of $y = 2f(x) + 5$ means you first move the graph of $y = f(x)$ up 5 units and then multiply the y values by 2.

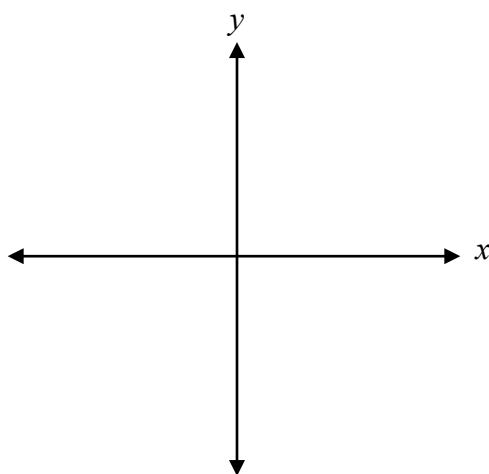
Explain to Rashid the correct way to transform the graph.

Question 9

1 mark

109

Sketch the angle of 5 radians in standard position.

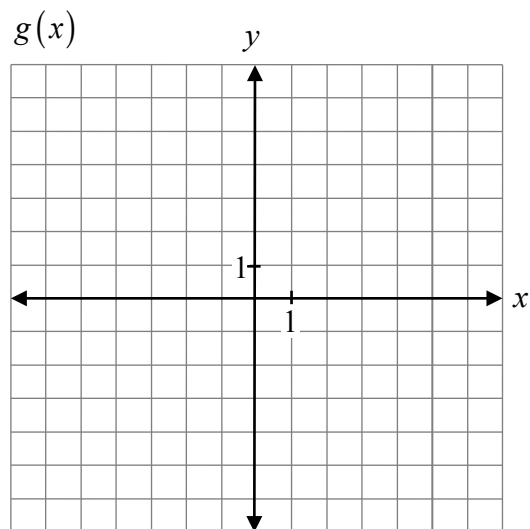
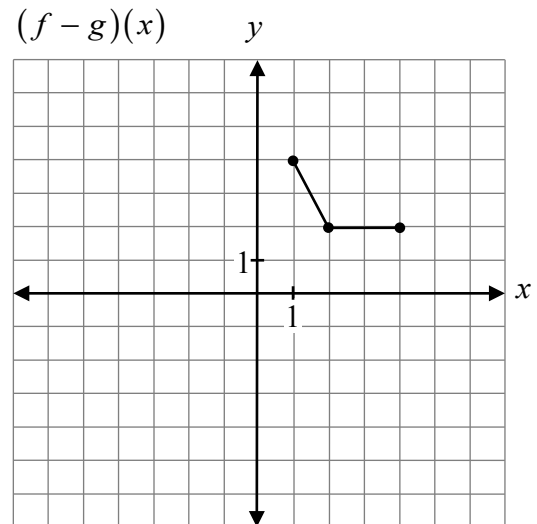
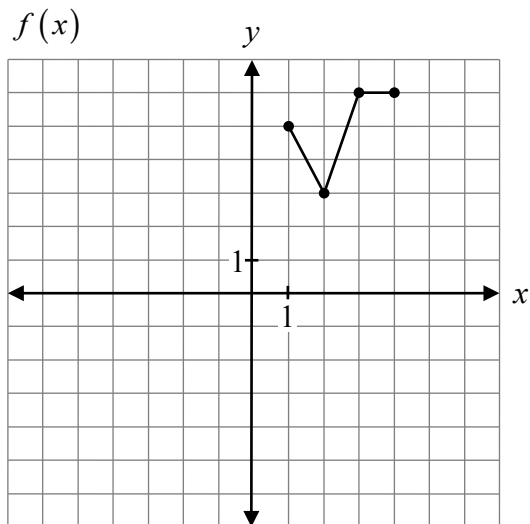


Question 10

2 marks

110

Given the graphs of $f(x)$ and $(f - g)(x)$, sketch the graph of $g(x)$.



Question 11

2 marks

111

A particular math class has a large number of students. From this class, you are to create a committee of 4 students that has at least 1 girl.

Without actually solving the problem, explain the strategy you would use to find the total number of ways to select this committee.

Question 12

a) 2 marks b) 2 marks

112
113

a) Prove the identity below for all permissible values of θ .

$$\frac{1 + 2 \cos^2 \theta}{\cos^2 \theta} = \tan^2 \theta + 3$$

Left-Hand Side	Right-Hand Side

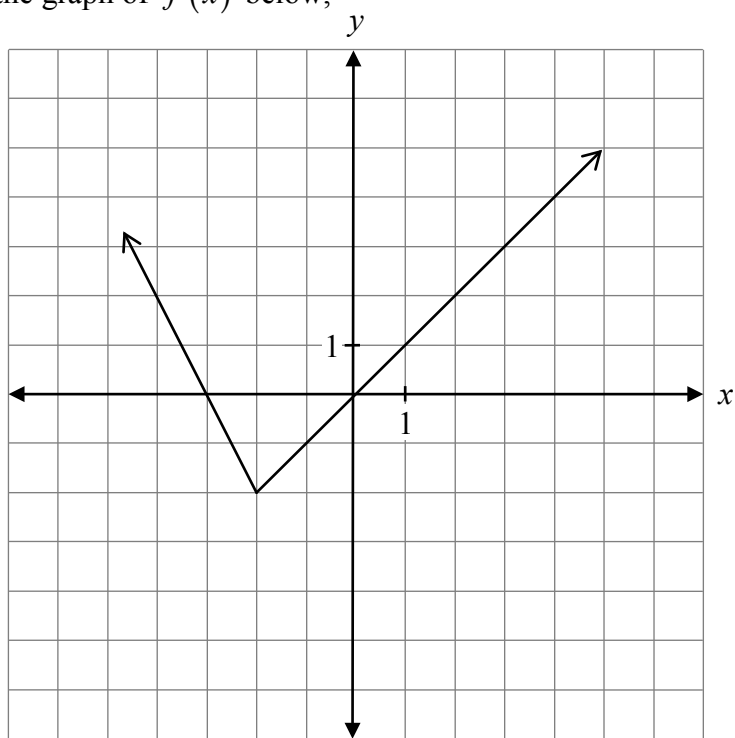
b) Determine all the non-permissible values for θ .

Question 13

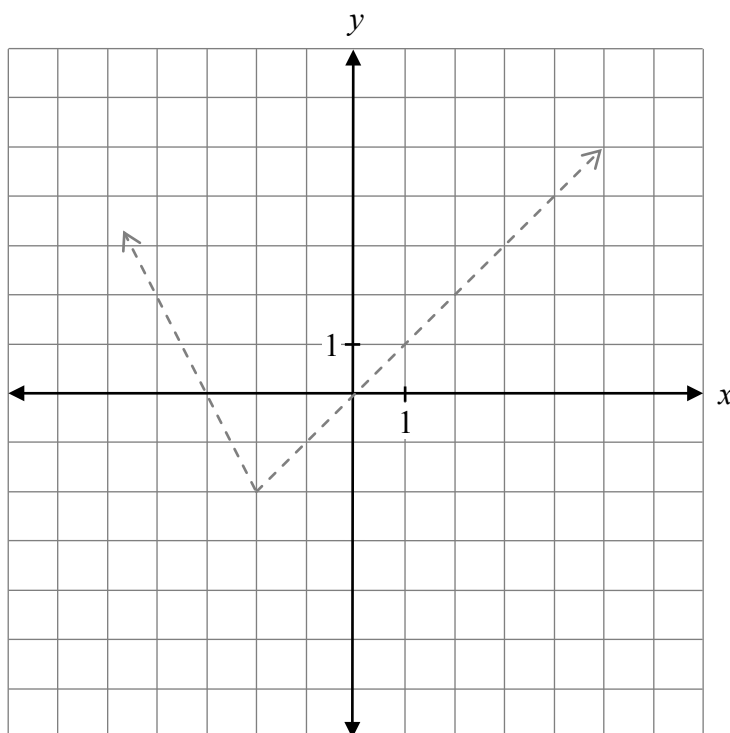
2 marks

114

Given the graph of $f(x)$ below,



sketch the graph of $g(x) = f(x - 2) + 3$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 14

a) 3 marks b) 1 mark

115
116

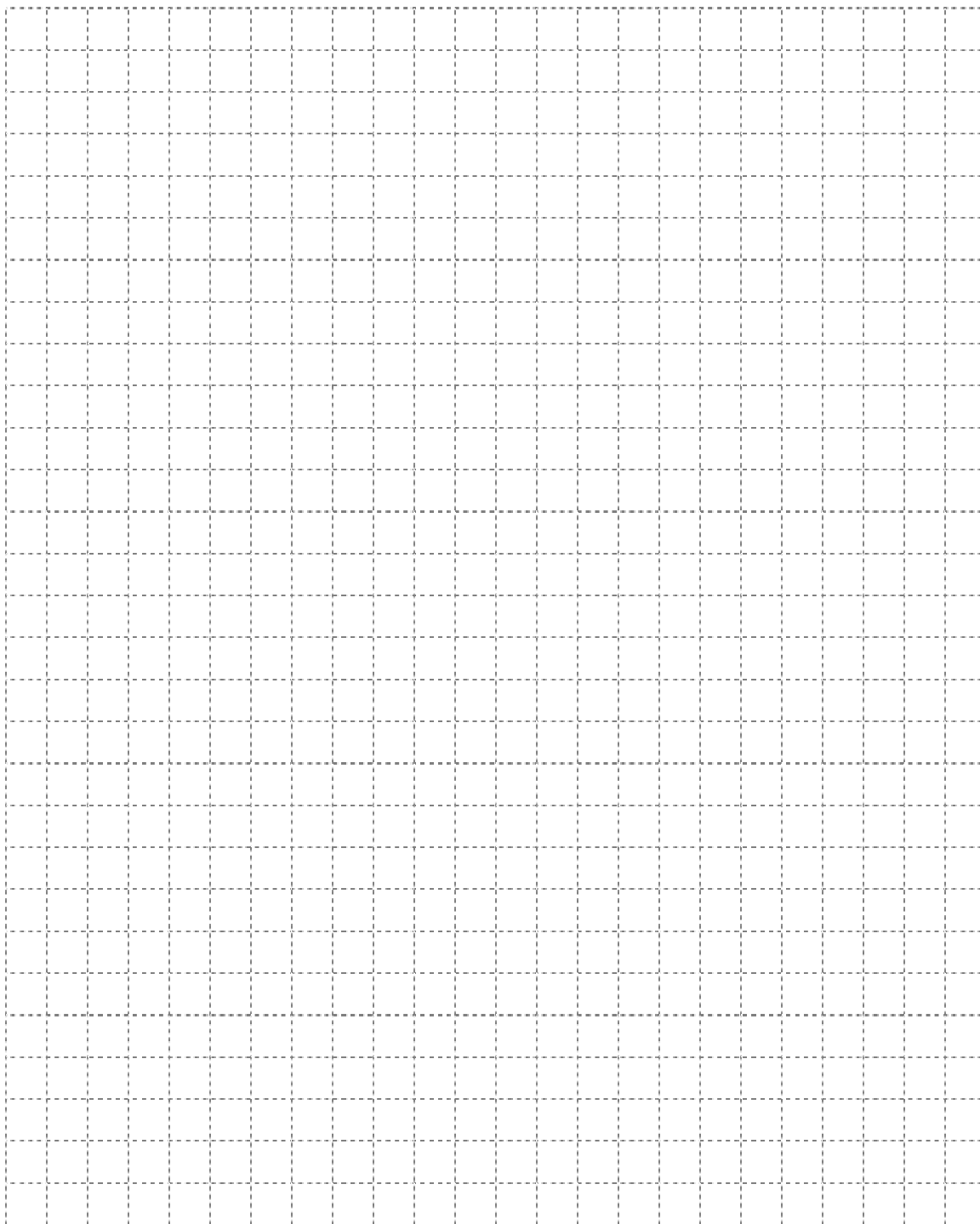
Given that $\sin \alpha = \frac{5}{13}$, where α is in Quadrant II, and $\cos \beta = \frac{2}{5}$, where β is in Quadrant IV, find the exact value of:

a) $\cos(\alpha + \beta)$

b) $\sin 2\alpha$

If $f(x) = x^3$ and $g(x) = 2x - 3$, what is the value of $\left(\frac{f}{g}\right)(-1)$?

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2014

Manitoba Education Cataloguing in Publication Data

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Booklet 2. January 2014 [electronic resource]

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 5. Mathematical ability—Testing.
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Manitoba Education
School Programs Division
Winnipeg, Manitoba, Canada

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Disponible en français.

Available in alternate formats upon request.

Instructions

Multiple-Choice Questions

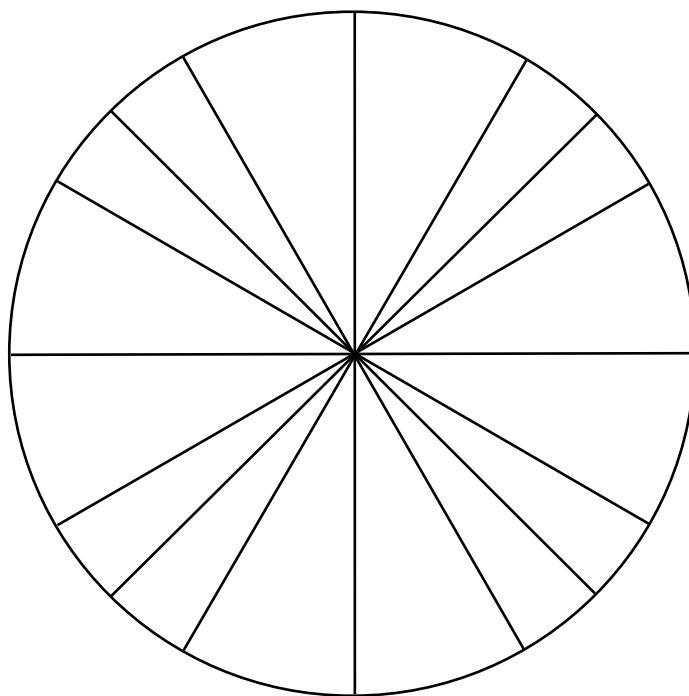
- There are 10 questions each worth one mark.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Short and Long Answer Questions

- There are 19 questions worth a total of 47 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Unit Circle (can be used if needed)



Question 16**1 mark**

If the point $(4, -3)$ lies on the graph of $f(x)$, which point must lie on the graph of $2f(2x)$?

- a) $(8, -6)$ b) $(2, -6)$ c) $\left(8, -\frac{3}{2}\right)$ d) $\left(2, -\frac{3}{2}\right)$

Question 17**1 mark**

The graph of $y = \log_2(2x + 6)$ intersects the graph of $y = 4$ at:

- a) $x = -1$ b) $x = 1$ c) $x = 5$ d) $x = 14$

Question 18**1 mark**

Given the point $A(-3, 5)$ on the terminal arm of an angle θ , identify the value of $\cot \theta$.

- a) $-\frac{3}{5}$ b) $-\frac{5}{3}$ c) $-\frac{4}{5}$ d) $-\frac{5}{4}$

Question 19

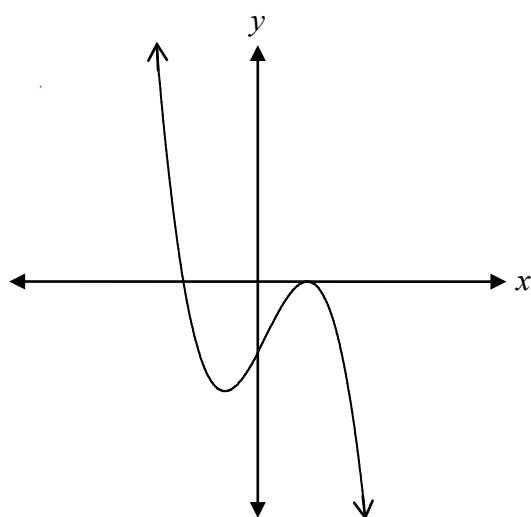
1 mark

The graph of $y = \left(\frac{1}{2}\right)^x$ compared to the graph of $x = \left(\frac{1}{2}\right)^y$ is a:

- a) reflection in the x -axis
- b) reflection in the y -axis
- c) reflection in the line $y = x$
- d) reciprocal function

Question 20

1 mark



Given the above graph of a polynomial function, which one of the following statements can be true?

- a) The function has a degree of 4 with a positive leading coefficient.
- b) The function has a degree of 4 with a negative leading coefficient.
- c) The function has a degree of 3 with a positive leading coefficient.
- d) The function has a degree of 3 with a negative leading coefficient.

Question 21

1 mark

Given that $(x+3)$ is a factor of polynomial $P(x)$, which of the following is true?

- a) $P(-3) = 0$ b) $P(0) = -3$ c) $P(0) = 3$ d) $P(3) = 0$

Question 22

1 mark

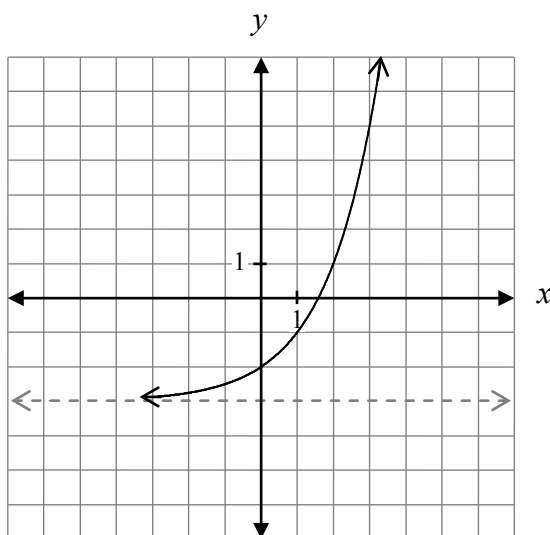
Which of the following is a reasonable estimate for the value of $\log 350$?

- a) 2 b) 2.5 c) 2.8 d) 3

Question 23

1 mark

The graph of the function $f(x)$ shown below is best described by the equation:

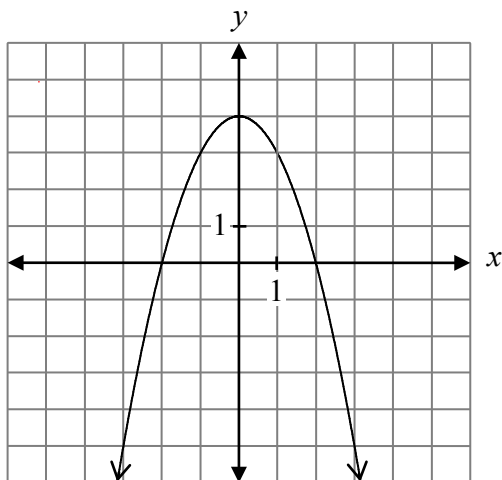


- a) $f(x) = 2^{x+3}$ b) $f(x) = 2^x + 3$ c) $f(x) = 2^{x-3}$ d) $f(x) = 2^x - 3$

Question 24

1 mark

Given the graph of $y = f(x)$, what is the domain of $\sqrt{f(x)}$?



- a) $x \in \mathbb{R}$ b) $-2 \leq x \leq 2$ c) $x \leq -2$ or $x \geq 2$ d) $0 \leq x \leq 4$

Question 25

1 mark

Solve:

$$e^{\ln(5-x)} = 7$$

- a) -2 b) $-\ln 2$ c) $\ln 7 - \ln 5$ d) $\frac{7}{5}$

Question 26

2 marks

118

One of the factors of $P(x) = x^3 - kx^2 - 7x + 10$ is $(x - 2)$.

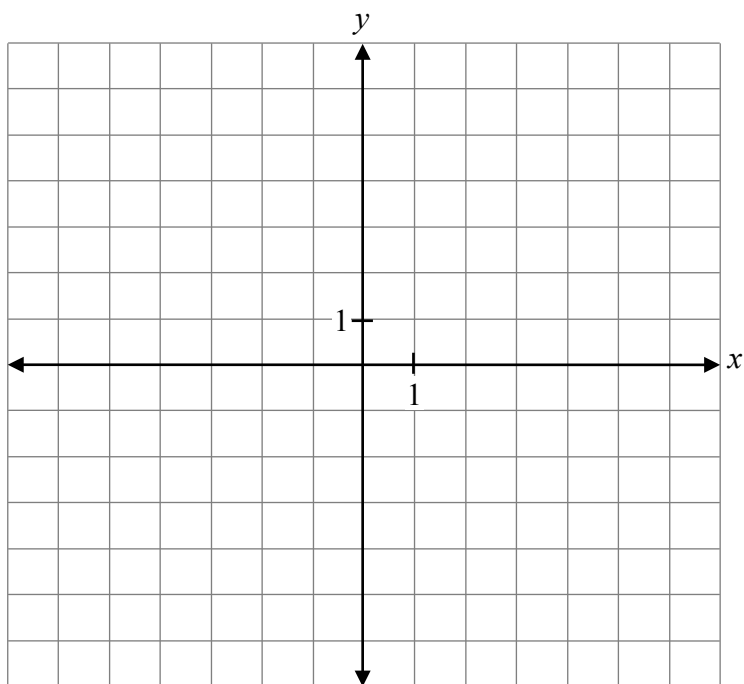
Find the value of k .

Question 27

a) 3 marks b) 1 mark

119
120

- a) Sketch the graph of the function $y = \sqrt{-x} + 1$.



- b) Determine the value of x when $y = 3$.

Question 28

3 marks

121

Solve the following equation:

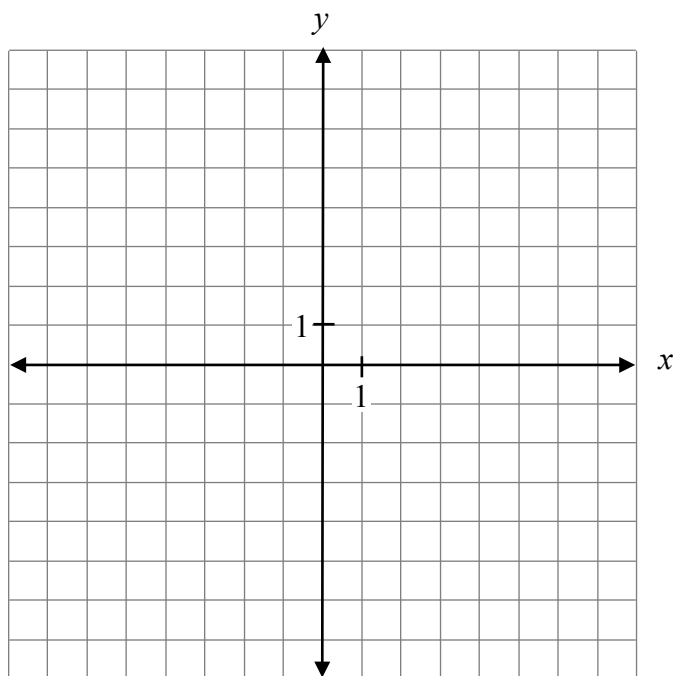
$${}_nP_2 = {}_nC_3$$

Question 29

3 marks

122

Given $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+1}$, sketch the graph of $y = f(g(x))$ and state its domain.



Domain: _____

Question 30

1 mark

123

Write the equation of the horizontal asymptote for the function $f(x) = \frac{x-3}{x-2}$.

Question 31

1 mark

124

The x -intercept of $f(x)$ is 4 and the x -intercept of $g(x)$ is 4.

Benjamin concludes that the x -intercept of $f(x) + g(x)$ is 8.

Do you agree with Benjamin? Justify your answer.

Question 32

4 marks

125

Solve the following equation:

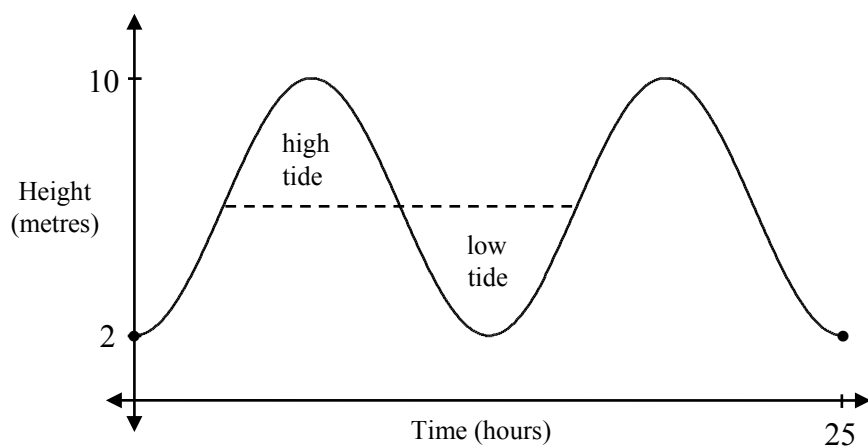
$$2\log_4 x - \log_4 (x + 3) = 1$$

Question 33

a) 1 mark b) 2 marks

126
127

The following graph represents tidal levels in the Bay of Fundy over a 25-hour period.

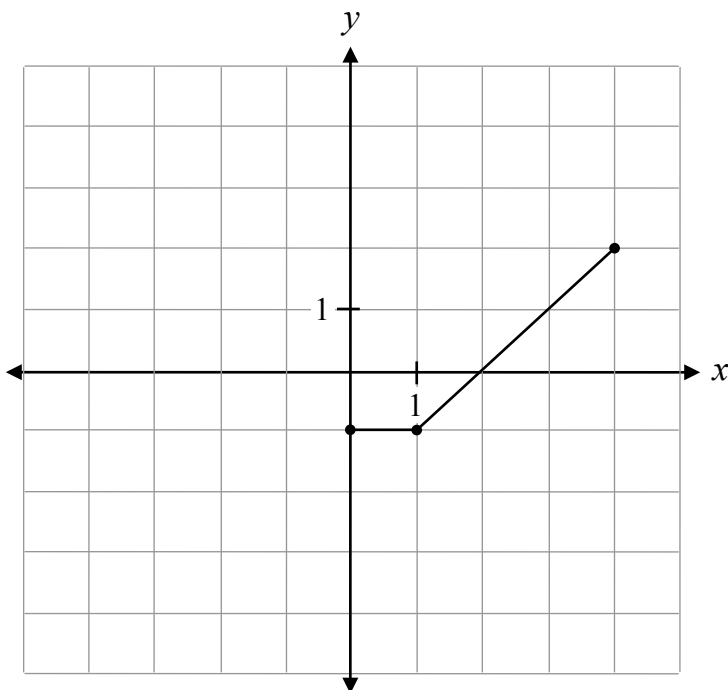


a) What is the average height of the water?

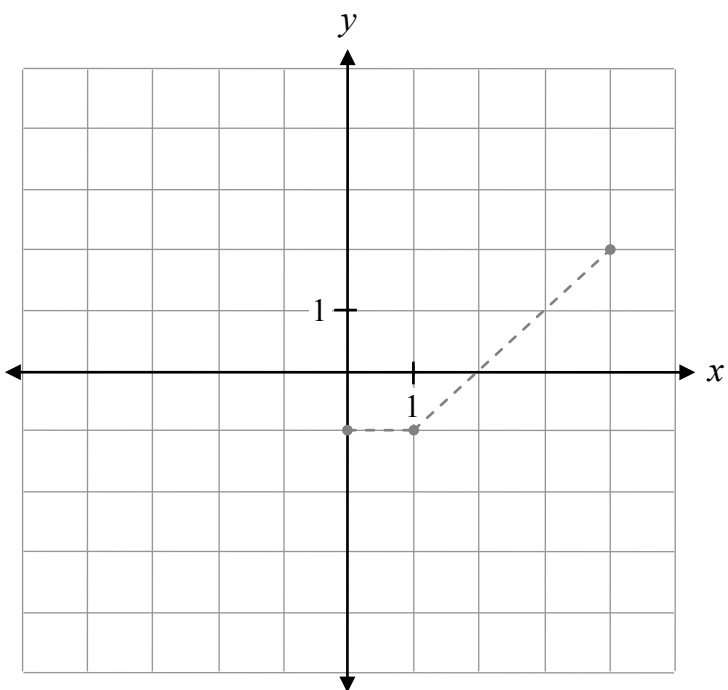
b) What is the period of the graph above?

Explain what the period represents in this situation.

Given the graph of $y = f(x)$ below,



sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 35

1 mark

129

When $P(x)$ is divided by $x - 3$, it has a quotient of $2x^2 + x - 6$ and a remainder of 4.

Determine $P(x)$.

Question 36

2 marks

130

Identify the domain and range of the following function:

$$f(x) = \frac{3}{x^2 + 1}$$

Evaluate:

$$\csc\left(\frac{11\pi}{6}\right) + \sin^2\left(-\frac{3\pi}{4}\right) + \cos\left(\frac{23\pi}{3}\right)$$

Question 38

2 marks

132

Evaluate the coefficient of the term containing x^3 in the expansion of $(1+x)^7$.

Justify your answer.

Question 39

1 mark

133

Which of the following equations could be solved without the use of logarithms?

Without actually solving the problem, explain your choice.

$$4^x = 10^{3x+1} \quad \text{or} \quad \left(\frac{1}{3}\right)^{2x+1} = 27^{4x-1}$$

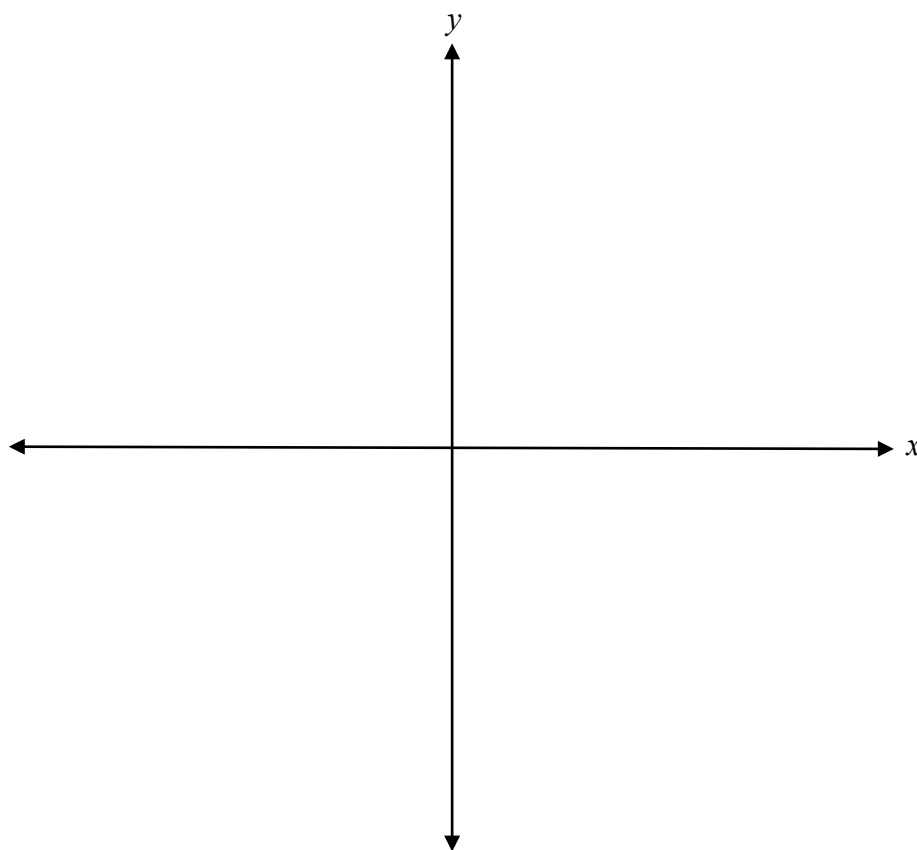
Question 40

4 marks

134

Sketch the graph of $y = x^3 + x^2 - 5x + 3$ given that one of the x -intercepts is 1.

Identify the x -intercepts and y -intercept.



Question 41**1 mark**

135

If $f(x) = \frac{1}{x-2}$ and $g(x) = x - 2$, what is the domain of $f(x) \cdot g(x)$?

Question 42**2 marks**

136

Given $f(x) = (x+1)^2$ for $x \leq -1$, write the equation of $y = f^{-1}(x)$.

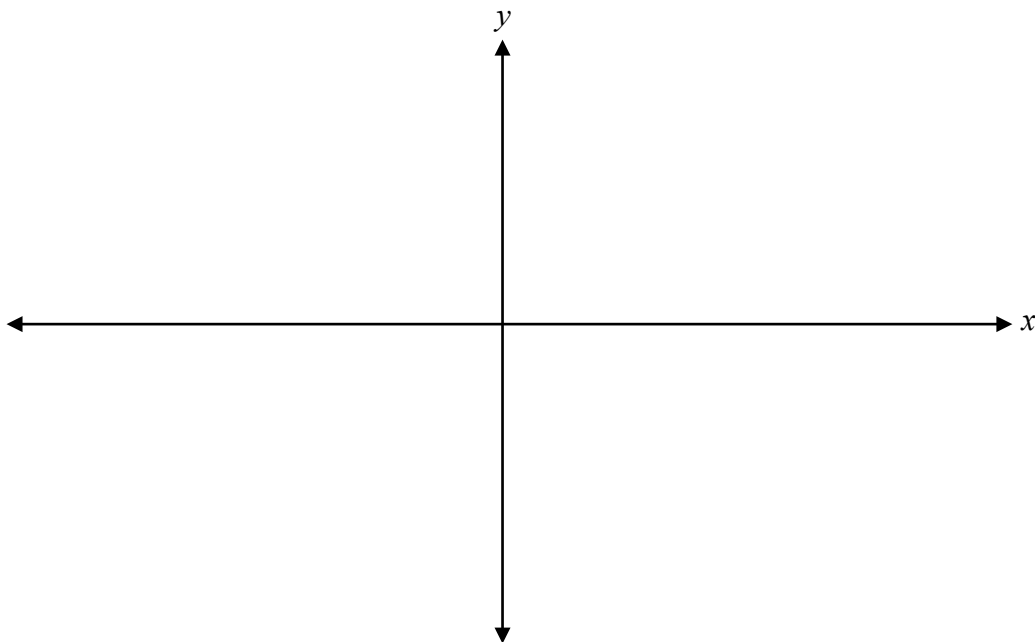
Question 43

4 marks

137

Sketch a graph of at least one period of the function $y = 5 \sin[\pi(x + 1)]$.

Clearly indicate the x -intercepts.



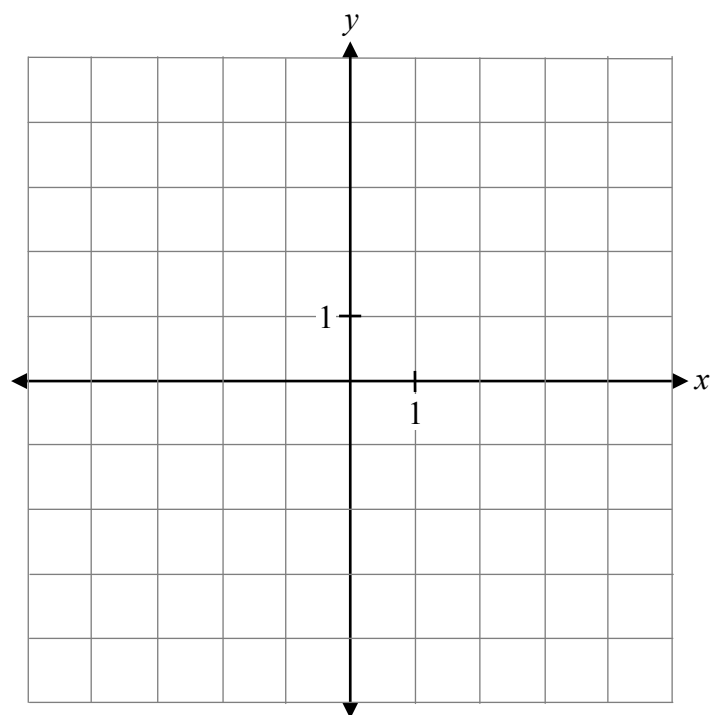
Question 44

4 marks

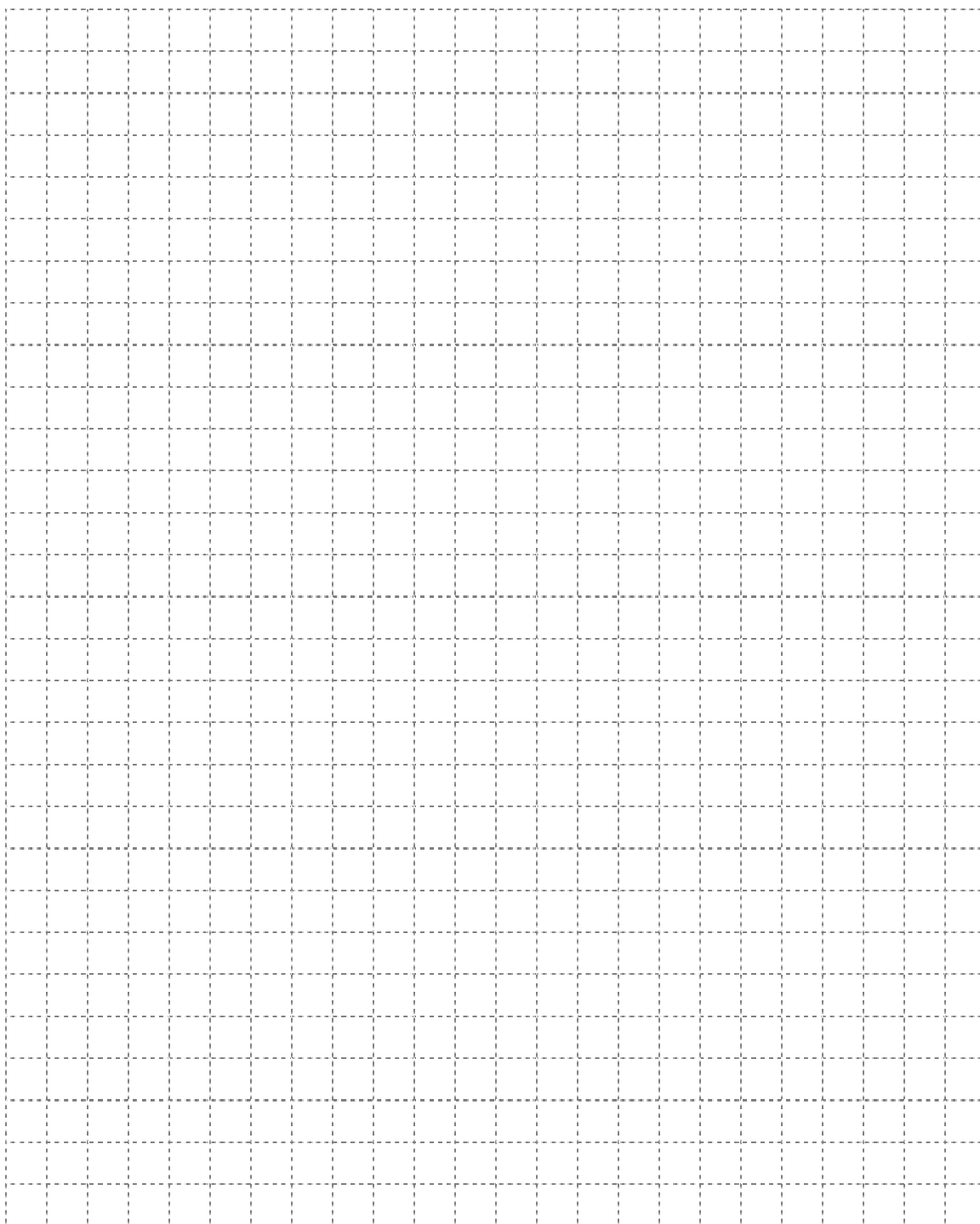
138

Sketch the graph of the following function:

$$f(x) = \frac{x-2}{(2x-3)(x-2)}$$



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2014

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

	Questions	Marks	Total Marks
Booklet 1 *	16 Short-Answer	25	36
	3 Long-Answer	11	
Booklet 2	10 Multiple-Choice	10	54
	12 Short-Answer	18	
	8 Long-Answer	26	
Total			90

- * The first 5 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- § Read all instructions carefully.
- § The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- § Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- § After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

Instructions

- § There are 19 questions for a total of 36 marks.
- § Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- § A calculator icon  appears next to the questions that require a calculator.
- § Write each solution in the space provided.
- § For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- § Graphing calculator solutions must include an explanation of how your final answer is obtained.
- § Your solutions should be neat, organized, and clear.
- § Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- § Express your answers as exact values or correct to 3 decimal places unless instructed otherwise.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are explained below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

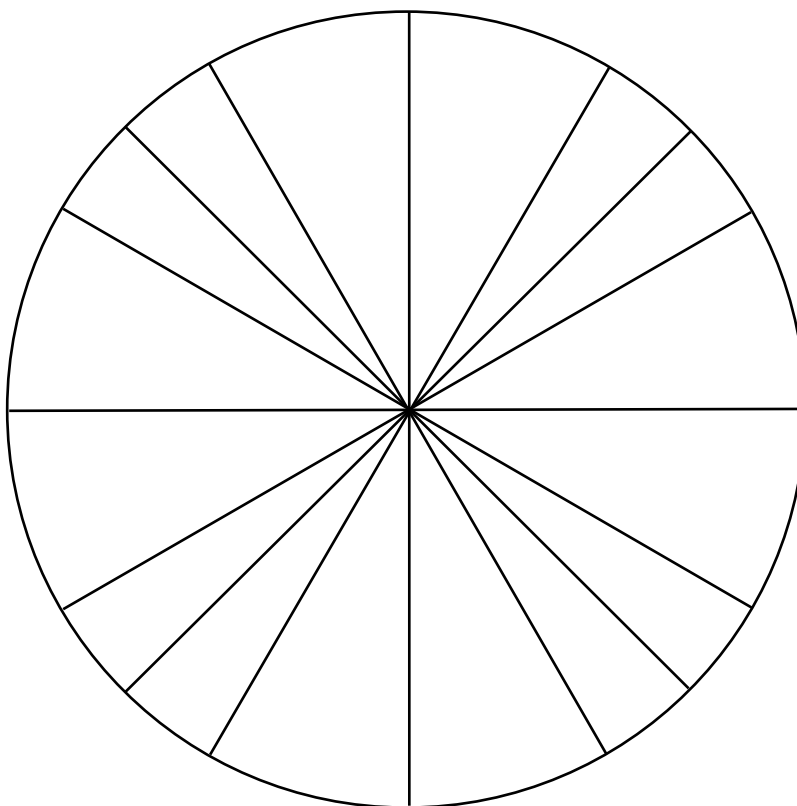
Identify / Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

Solve: Give a solution for a problem or determine the value(s) of a variable.

Verify: Establish the truth of a statement by substitution or comparison.

Unit Circle (can be used if needed)

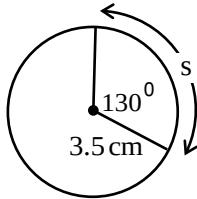


Question 1

2 marks

101

Use the information in the diagram to determine the value of the arc length “s”.



Question 2

2 marks

102

Solve the following equation over the interval $[0, 2\pi)$.

$$\tan^2 \theta + 2.8 \tan \theta + 1.96 = 0$$

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$.

Question 3

3 marks

103

Determine how many monthly investments of \$50 would have to be deposited into a savings account that pays 3% annual interest, compounded monthly, for the account's future value to be \$50,000.

Use the formula:

$$FV = \frac{R \left[(1 + i)^n - 1 \right]}{i}$$

where FV = the future value

R = the investment amount

$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$

$n = \text{the number of investments}$

Express your answer as a whole number.

Question 4

2 marks

104

There are 5 men and 4 women to be seated in a row.

How many arrangements are possible if two men must sit at the beginning of the row and two men must sit at the end of the row?

Question 5

a) 3 marks b) 2 marks

105
106

a) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^8$, determine the 3rd term.

b) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^n$, the 6th term contains x^{25} .
Solve for n .

Note: A calculator is not required for the remaining test questions.

Question 6**1 mark**

107

Given the following two functions, $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$, evaluate $g(f(3))$.

Question 7**2 marks**

108

If θ terminates in quadrant II and $\csc \theta = \frac{3}{2}$, determine the exact value of $\tan \theta$.

Question 8

a) 1 mark b) 1 mark

109
110

a) Determine the remainder when $x^4 - 3x^2 + 1$ is divided by $x + 2$.

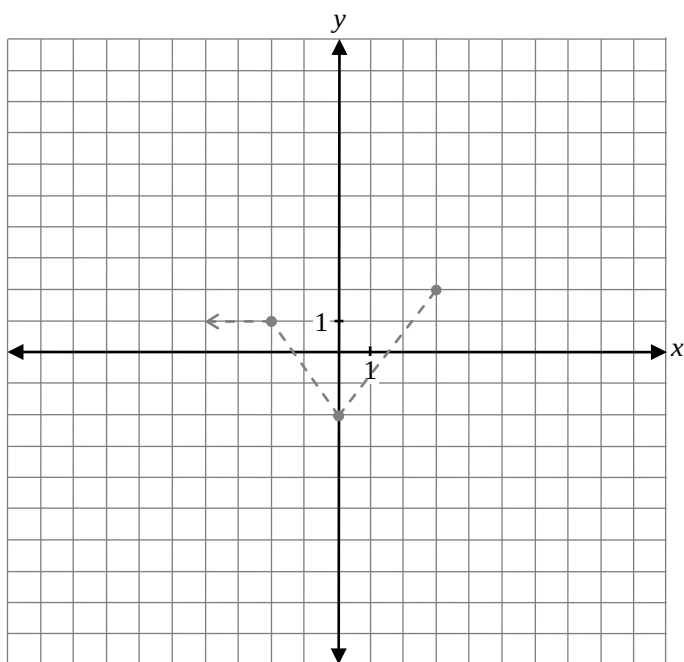
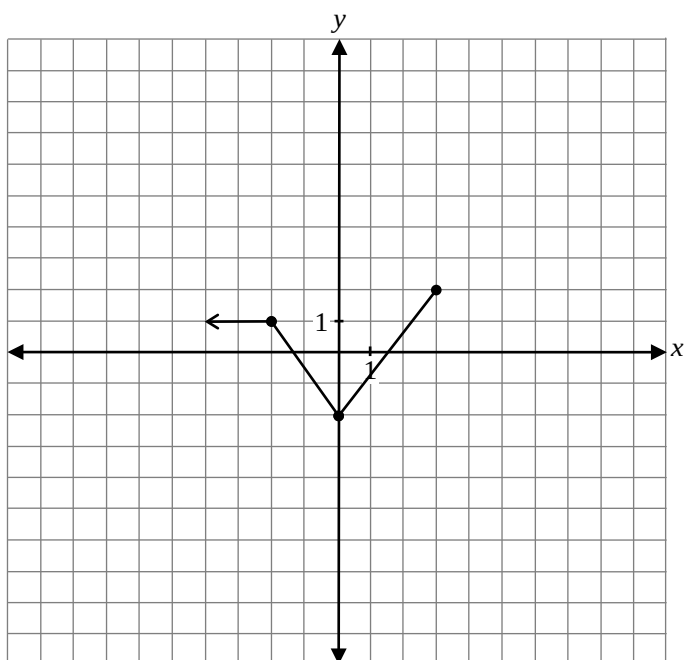
b) Is $x + 2$ a factor of $x^4 - 3x^2 + 1$?
Explain your reasoning.

Question 9

2 marks

111

Given the graph of $y = f(x)$ below, sketch the graph of $y = 2f(x) - 3$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 10

1 mark

112

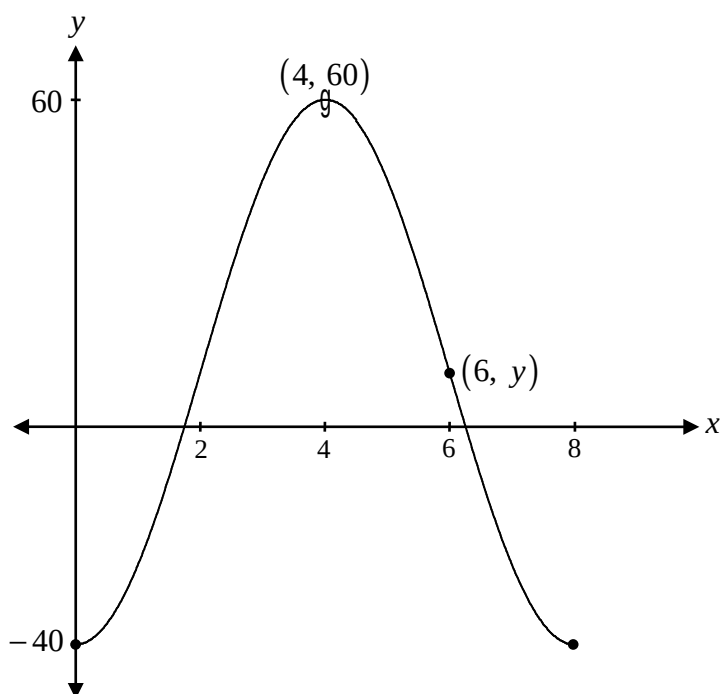
Determine one possible restriction for the domain of $f(x) = (x-1)^2$ so that the inverse of $f(x)$ is a function.

Question 11

1 mark

113

Using the graph of the sinusoidal function below, find the value of y in the point $(6, y)$.



Question 12

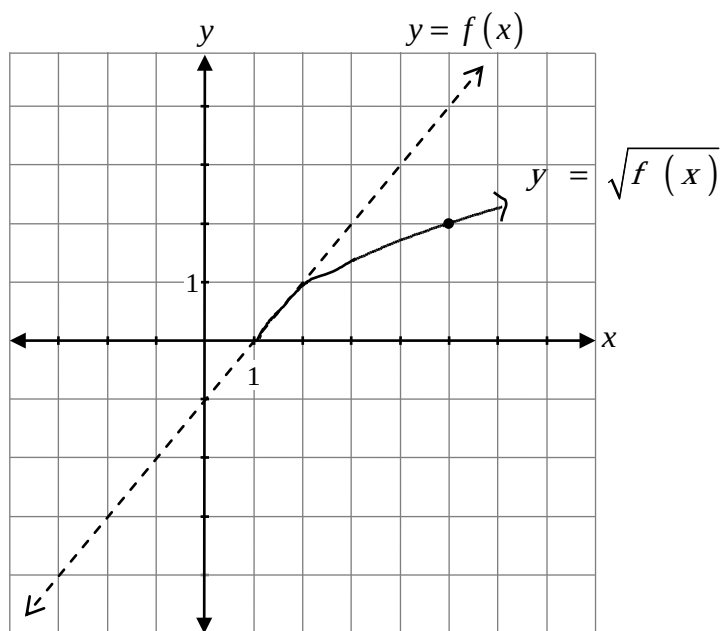
1 mark

114

Billy was given the graph of $y = f(x)$.

He was asked to sketch the graph of $y = \sqrt{f(x)}$.

His answer is given on the graph below.



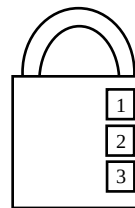
Explain the error Billy made when sketching the graph of $y = \sqrt{f(x)}$.

Question 13

1 mark

115

Explain why a locker combination should really be called a locker permutation.



Question 14

1 mark

116

The graph of $f(x) = x^2 + 4$ is reflected over the x -axis.

Write the equation of the new function.

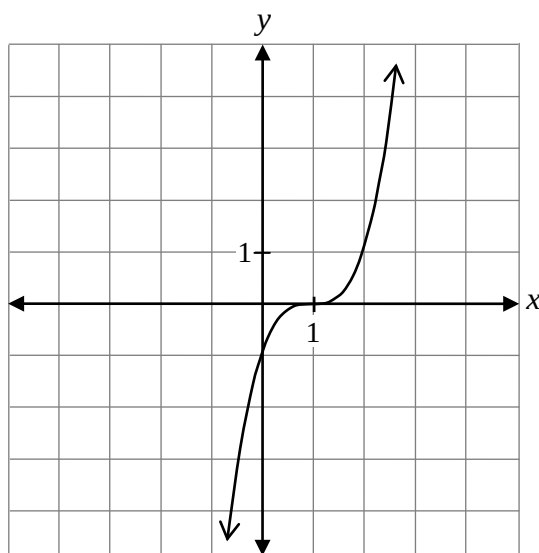
$y =$ _____

Question 15

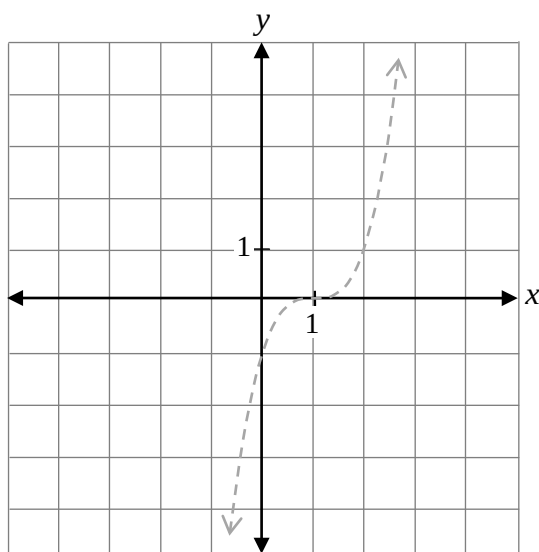
2 marks

117

Given the graph of $y = f(x)$ below,



sketch the graph of $y = \frac{1}{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 16

2 marks

118

Divide $(x^3 - 5x - 4)$ by $(x + 1)$.

Question 17

1 mark

119

You are given the following row of Pascal's Triangle.

1 7 21 35 35 21 7 1

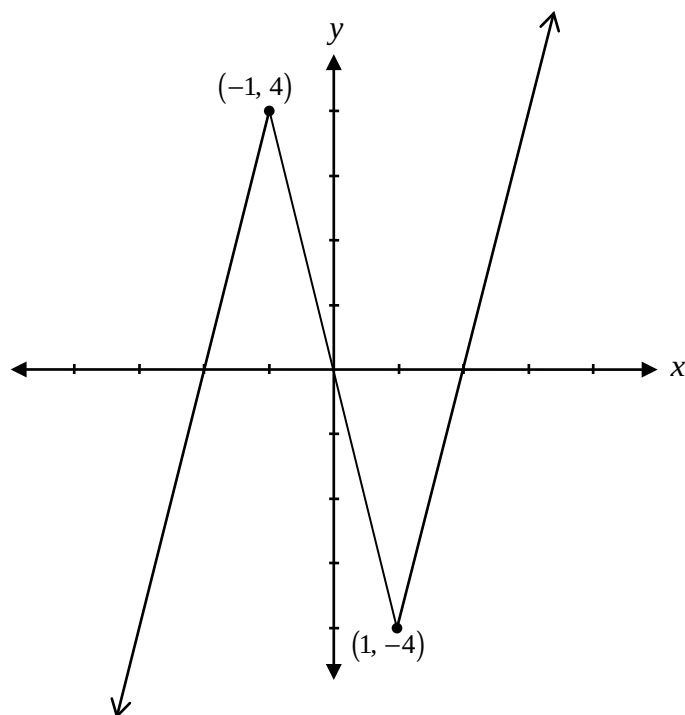
Determine the values of the next row.

Question 18

2 marks

120

Given the graph of $y = f(x)$ below, state the domain and range of $y = \sqrt{f(x)}$.



Domain: _____

Range: _____

Question 19

3 marks

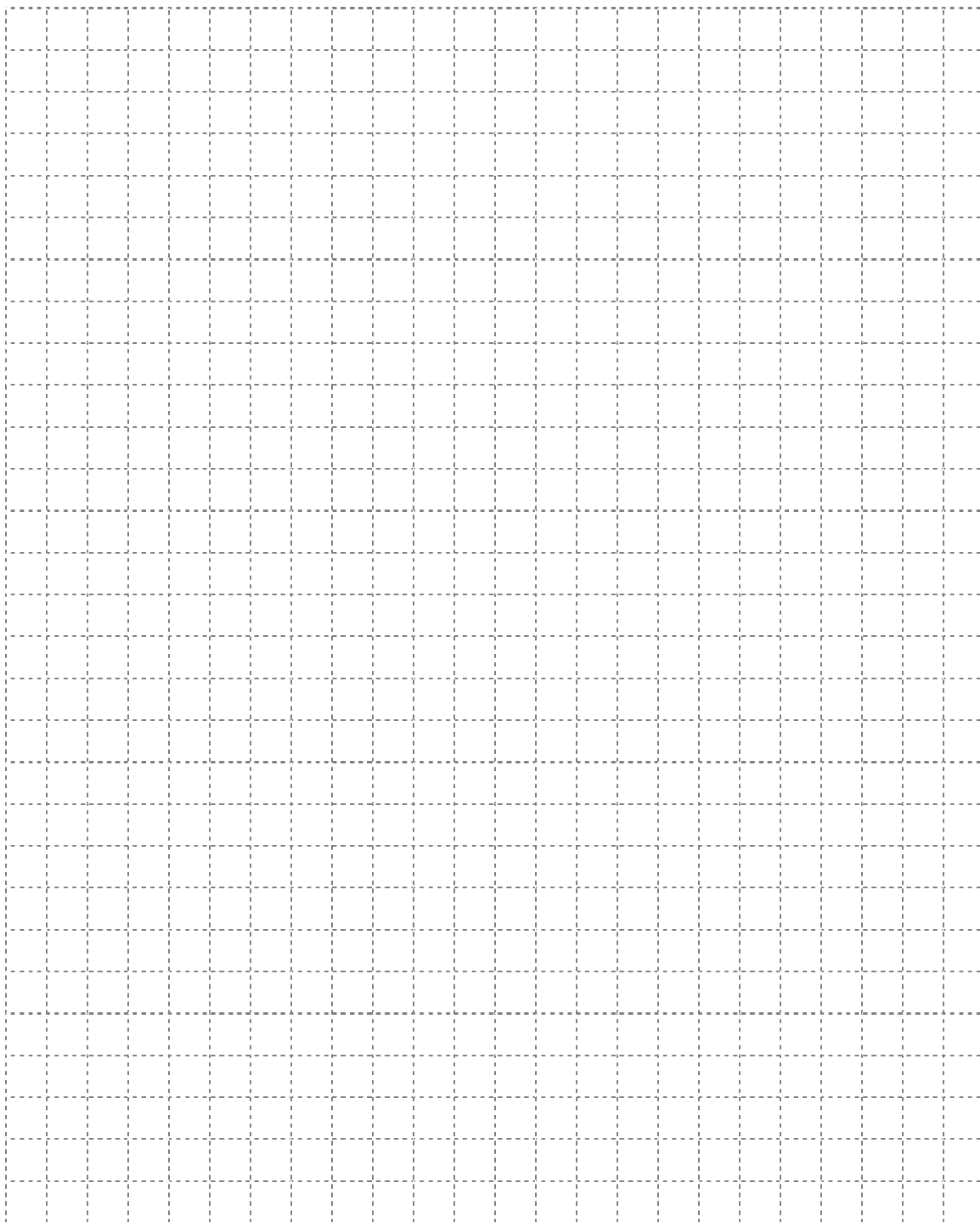
121

Prove the identity below for all permissible values of θ :

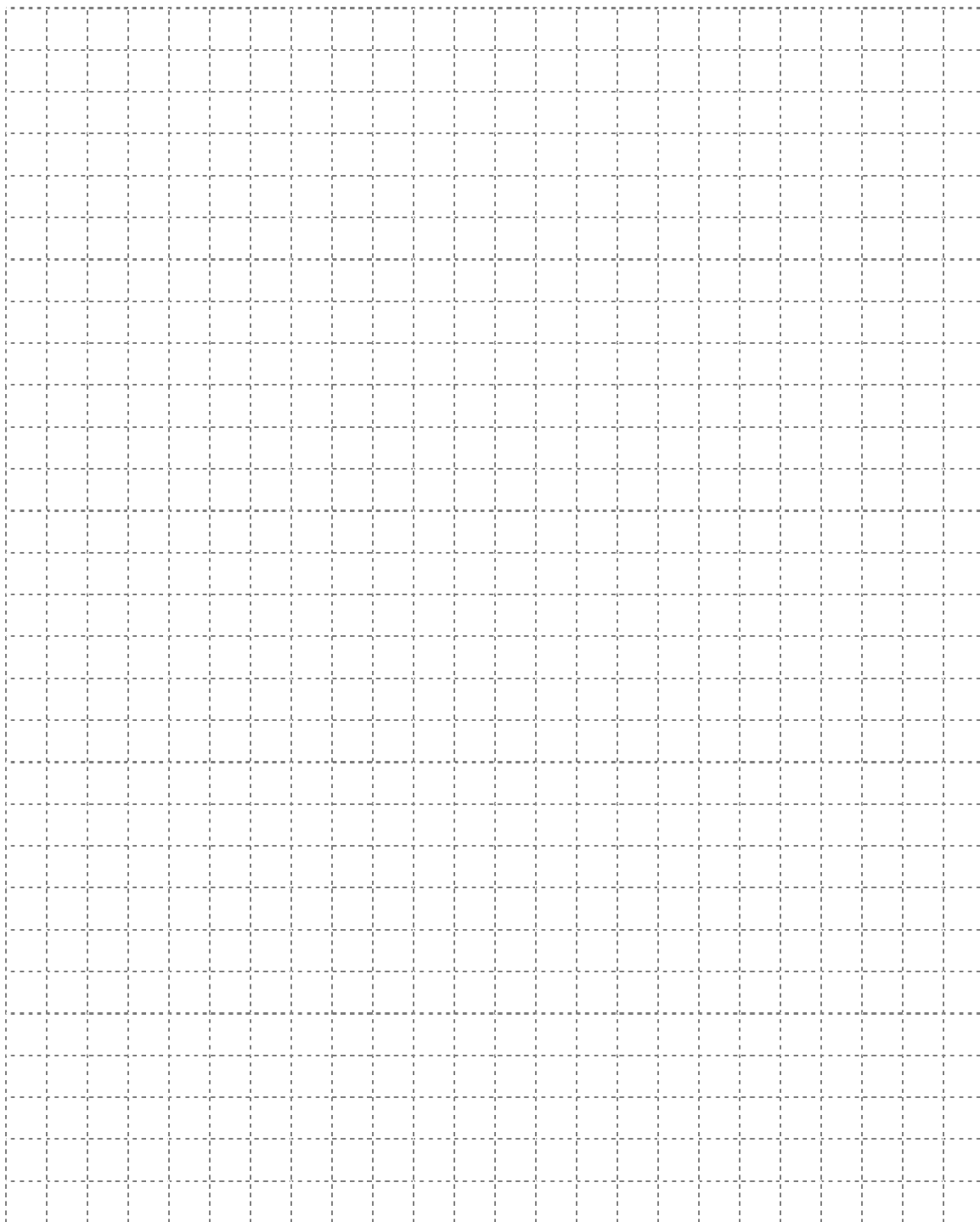
$$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos 2\theta$$

Left-Hand Side	Right-Hand Side

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2014

Manitoba Education and Advanced Learning Cataloguing in Publication Data

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Booklet 2. June 2014 [electronic resource]

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Manitoba Education and Advanced Learning
School Programs Division
Winnipeg, Manitoba, Canada

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Disponible en français.

Available in alternate formats upon request.

Instructions

Multiple-Choice Questions

- § There are 10 questions each worth one mark.
- § Calculators are **not** allowed for this part of the test.
- § You may use the spaces beside each question for rough work.
- § Provide only one answer per question.
- § There is no penalty for guessing.
- § Record your answers on the sheet provided.

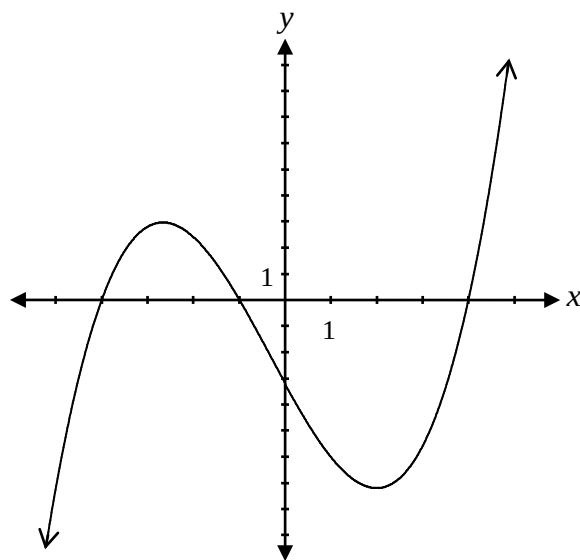
Short and Long Answer Questions

- § There are 20 questions worth a total of 44 marks.
- § Calculators are **not** allowed for this part of the test.
- § For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- § Your solutions should be neat, clear, and well organized.
- § Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 20**1 mark**

Given the graph of the function of $f(x)$ below, what is the range of $y = |f(x)|$?



- a) $y \in \mathbb{I}$ b) $y \geq -7$ c) $y \geq 0$ d) $-4 \leq y \leq -1$ or $y \geq 4$

Question 21**1 mark**

Simplify the following expression:

$$\frac{1}{2} \log_a 36 - \log_a 2$$

- a) $\log_a 3$ b) $\log_a 4$ c) $\log_a 9$ d) $\log_a 12$

Question 22**1 mark**

Given $f(x) = x^2 - x + 2$, an equation that represents the graph of $f(x)$ shifted 3 units to the right is:

a) $y = (x + 3)^2 - (x + 3) - 3$

b) $y = (x - 3)^2 - (x - 3) + 2$

c) $y = (x - 3)^2 - x - 2$

d) $y = x^2 - x + 2 - 3$

Question 23**1 mark**

What is the domain of the function $y = \sqrt{-4x}$?

a) $\{x \in \mathbb{R} \mid x \geq 2\}$

b) $\{x \in \mathbb{R} \mid x \leq 2\}$

c) $\{x \in \mathbb{R} \mid x \geq 0\}$

d) $\{x \in \mathbb{R} \mid x \leq 0\}$

Question 24**1 mark**

Which of the following is true about the two functions below?

$$f(x) = \frac{(x+2)(x-2)}{x-2} \qquad g(x) = \frac{(x-2)(x+1)}{(x+2)(x-2)}$$

a) Both have a point of discontinuity (hole) when $x = 2$.

b) Both have the same vertical asymptote.

c) Both have the same horizontal asymptote.

d) Both have the same y-intercept.

Question 25

1 mark

The general solution to the equation $\cos \theta = -\frac{1}{2}$ is:

a)
$$\left. \begin{aligned} \theta &= \frac{\pi}{3} + 2\pi k \\ \theta &= \frac{5\pi}{3} + 2\pi k \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

b)
$$\left. \begin{aligned} \theta &= \frac{\pi}{3} + \pi k \\ \theta &= \frac{5\pi}{3} + \pi k \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

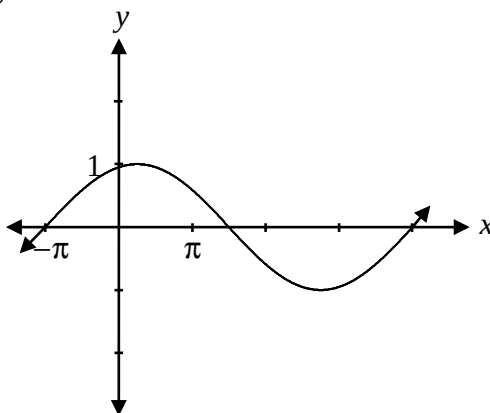
c)
$$\left. \begin{aligned} \theta &= \frac{2\pi}{3} + 2\pi k \\ \theta &= \frac{4\pi}{3} + 2\pi k \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

d)
$$\left. \begin{aligned} \theta &= \frac{2\pi}{3} + \pi k \\ \theta &= \frac{4\pi}{3} + \pi k \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

Question 26

1 mark

If the equation $y = \sin(b(x + \pi))$ is represented by the following graph, what is the value of b ?



a) $\frac{2}{5}$

b) $\frac{5}{2}$

c) $\frac{2\pi}{5}$

d) 5π

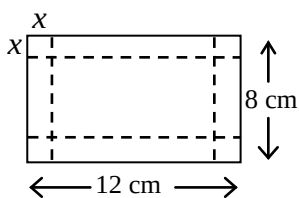
Question 27**1 mark**

Which of the following is closest to the value of $\log_2 40 + \log_5 125$?

- a) 3 b) 8 c) 10 d) 45

Question 28**1 mark**

A sheet of paper 12 cm long and 8 cm wide is used to make a box with no lid. Equal squares of side length x cm are cut from each of the corners and the sides are folded up to make the box.



Which of the following expresses the volume of the box?

- a) $V(x) = x(12 + x)(8 + x)$
b) $V(x) = x(12 - x)(8 - x)$
c) $V(x) = x(12 + 2x)(8 + 2x)$
d) $V(x) = x(12 - 2x)(8 - 2x)$

Question 29**1 mark**

Given that the graph of $f(x)$ contains the point $(-3, 5)$, what point must be on the graph of $f(-x)$?

- a) $(-3, -5)$
b) $(3, 5)$
c) $(3, -5)$
d) $(5, -3)$

Question 30

1 mark

122

Determine one positive and one negative coterminal angle with the angle $\frac{5\pi}{6}$.

Question 31

2 marks

123

Evaluate:

$$\left(\sin \frac{11\pi}{3}\right)\left(\sec \frac{11\pi}{6}\right)$$

Question 32

1 mark

124

Given the equation $2\sin^2\theta - 3\sin\theta + 1 = 0$, verify that $\theta = \frac{\pi}{2}$ is a solution.

Question 33

2 marks

125

Using the laws of logarithms, expand:

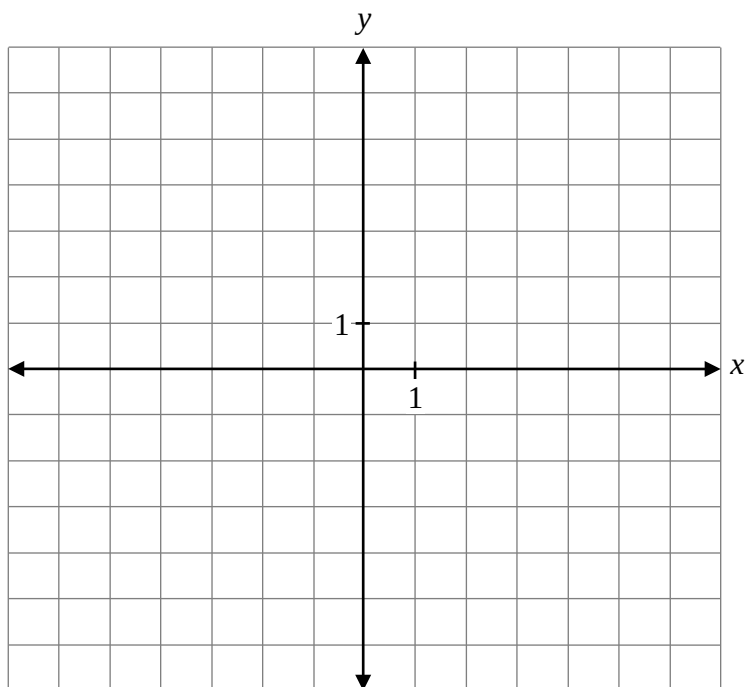
$$\log_a\left(\frac{xy}{z}\right)$$

Question 34

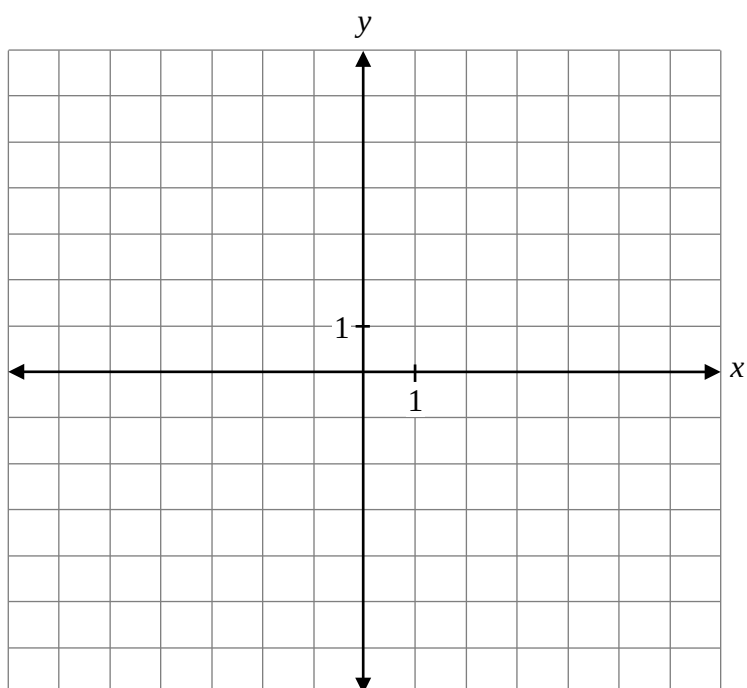
a) 2 marks b) 1 mark

126
127

a) Sketch the graph of $f(x) = 3^x + 1$.



b) Sketch the graph of $f^{-1}(x)$.



Question 35

2 marks

128

Determine the x-intercept and y-intercept of $y = \log_2(x + 4) - 1$.

Question 36

1 mark

129

Explain the error that was made when solving the following equation:

$$\sin 2\theta = \cos \theta, \text{ where } \theta \in \mathbb{R}$$

$$\begin{aligned}\sin 2\theta &= \cos \theta \\ 2\sin \theta \cos \theta &= \cos \theta \\ \frac{2\sin \theta \cos \theta}{\cos \theta} &= \frac{\cos \theta}{\cos \theta} \\ 2\sin \theta &= 1 \\ \sin \theta &= \frac{1}{2} \\ \theta &= \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi, k \in \mathbb{I}\end{aligned}$$

Question 37

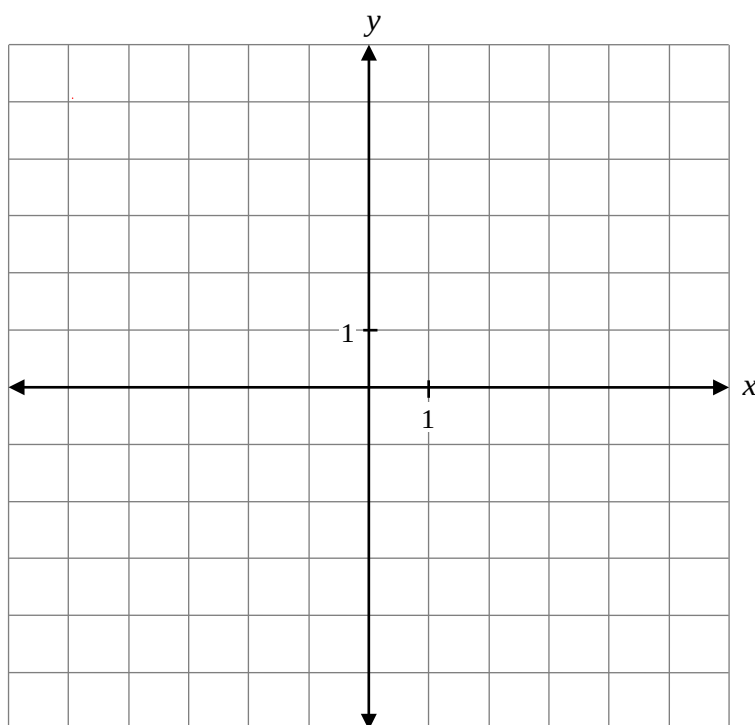
a) 1 mark b) 1 mark

130
131

Given $f(x) = x^2 - 2x - 3$ and $g(x) = x + 1$:

a) Write the equation of $y = f(g(x))$.

b) Sketch the graph of $y = f(g(x))$.



Question 38

1 mark

132

Is the point $\left(\frac{3}{4}, -\frac{\sqrt{3}}{4}\right)$ on the unit circle?

Justify your answer.

Question 39

1 mark

133

Explain why the equation $\sec \theta = \frac{1}{4}$ has no solution.

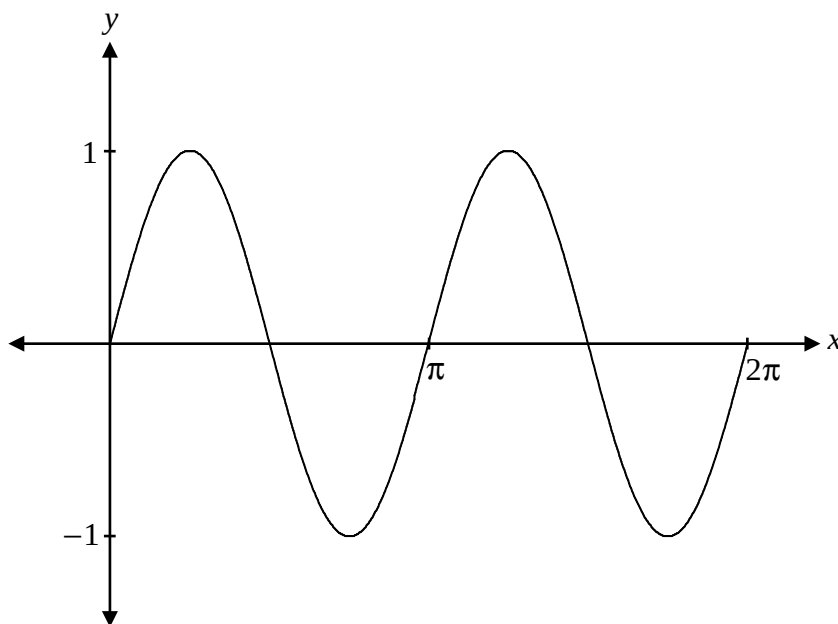
Question 40

1 mark

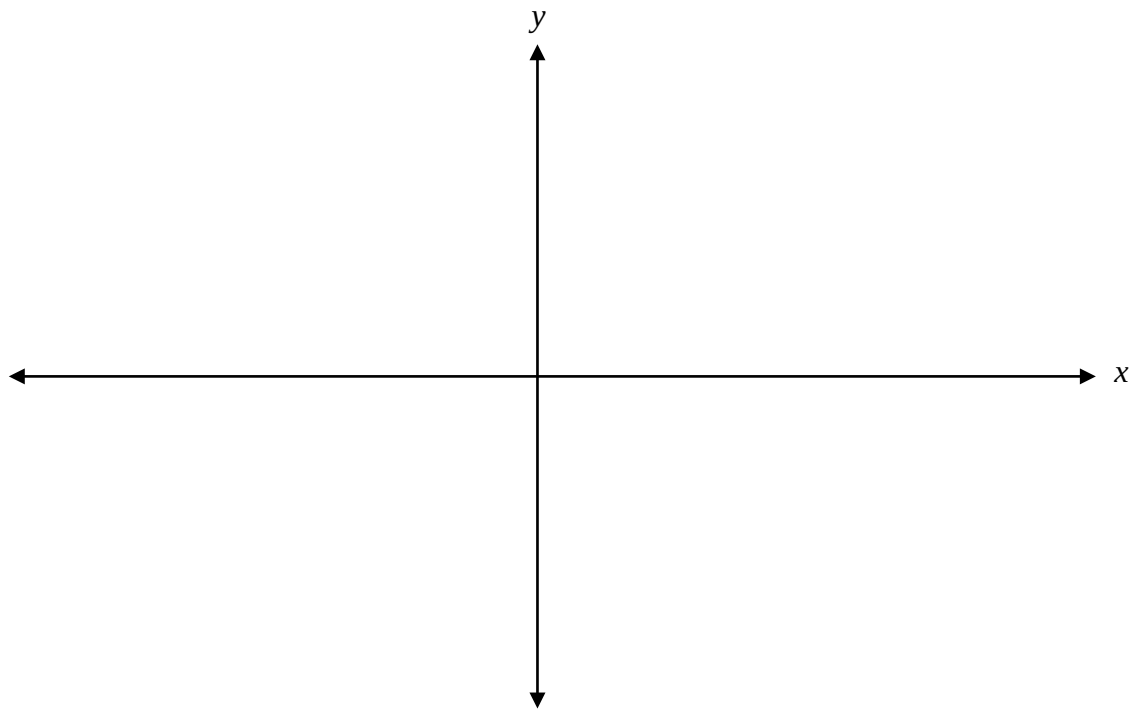
134

The graph of $y = \sin 2x$ is sketched below.

Explain how to use this graph to solve the equation $\sin 2x = \frac{1}{2}$ over the interval $[0, 2\pi]$.



Sketch the graph of $y = -4\cos(2x)$ over the interval $[-\pi, \pi]$.



Question 42

2 marks

136

Write the equation for $f(x)$ that satisfies all of the following conditions:

- $f(x)$ is a polynomial function of degree 4
- $f(x)$ has a zero at 2 with a multiplicity of 3
- $f(x)$ has a zero at -5
- $f(x)$ has a y -intercept of 80

Find the exact value of $\sin\left(\frac{19\pi}{12}\right)$.

Question 44

4 marks

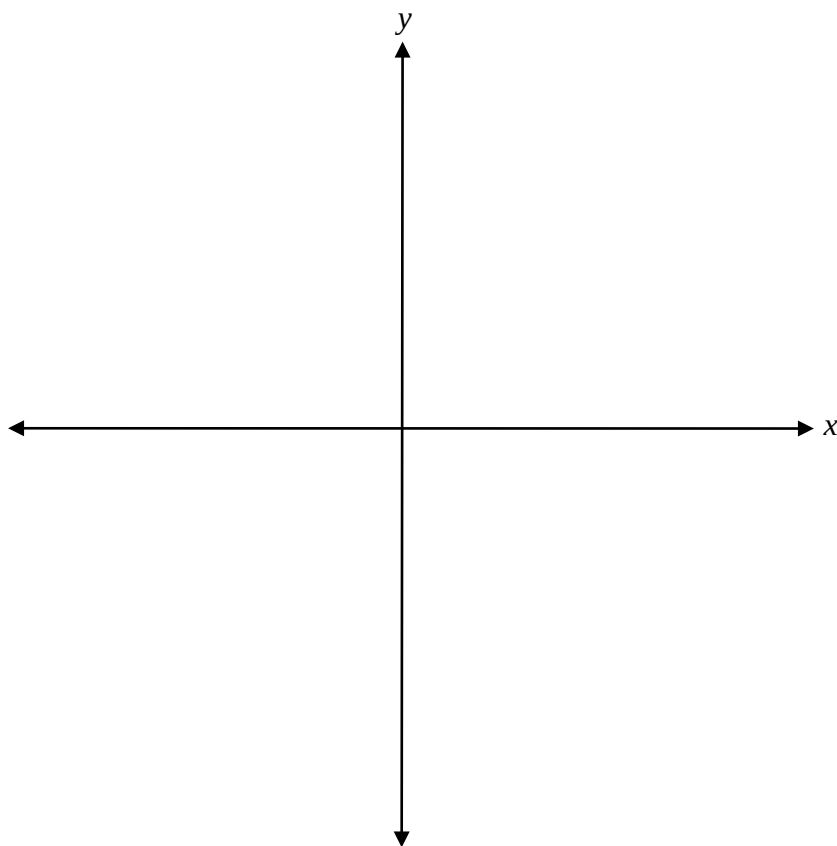
138

Solve the following equation:

$$2\log_2(x-1) - \log_2(x-5) = \log_2(x+1)$$

Sketch the graph of $f(x) = (x-1)^2(x+2)^3$.

Label the x-intercepts and the y-intercept.

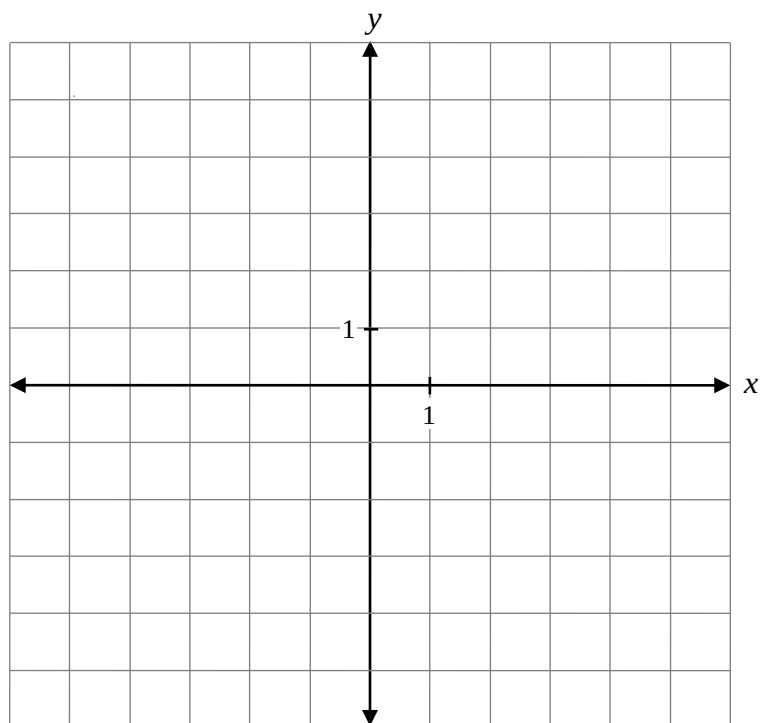


Question 46

4 marks

140

Sketch the graph of $y = -\sqrt{3(x+1)}$.



Question 47

3 marks

141

Solve:

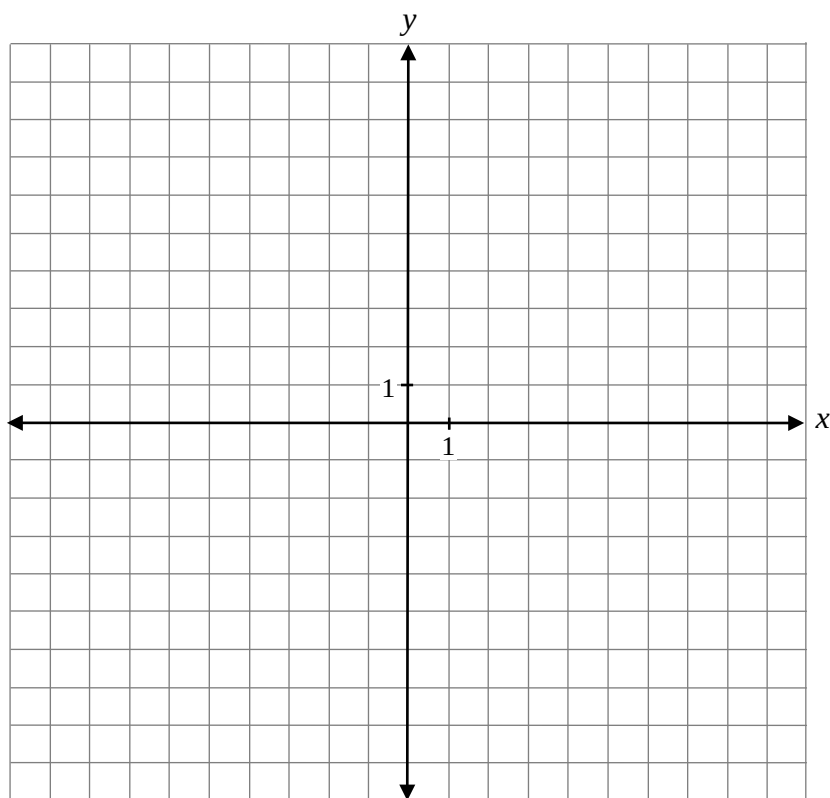
$${}_{n-1}P_2 = 42$$

Question 48

3 marks

142

Sketch the graph of $y = \frac{2x}{x+2}$.

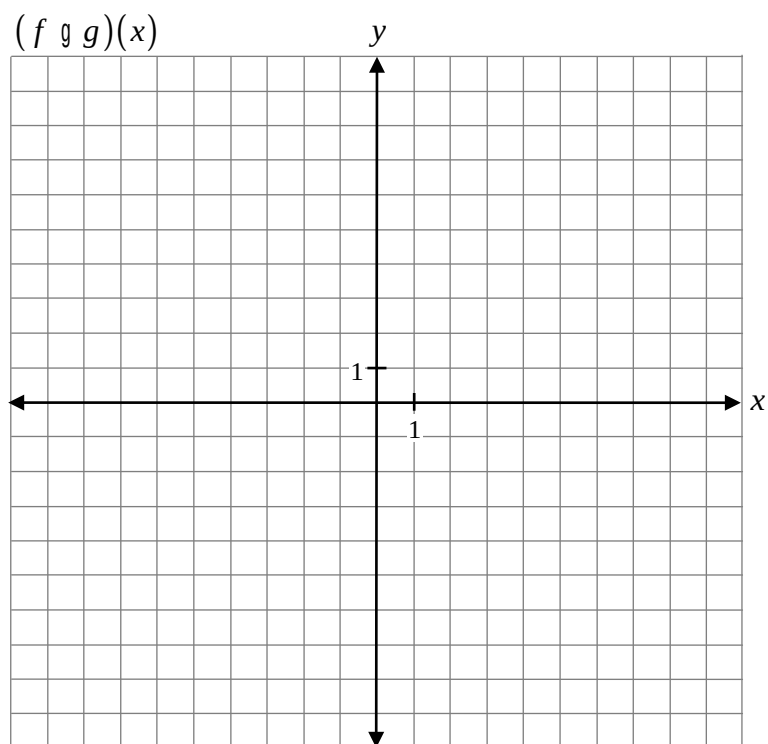
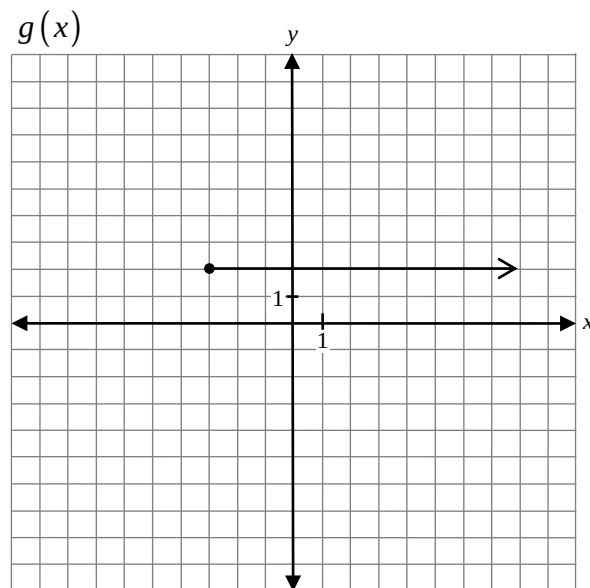
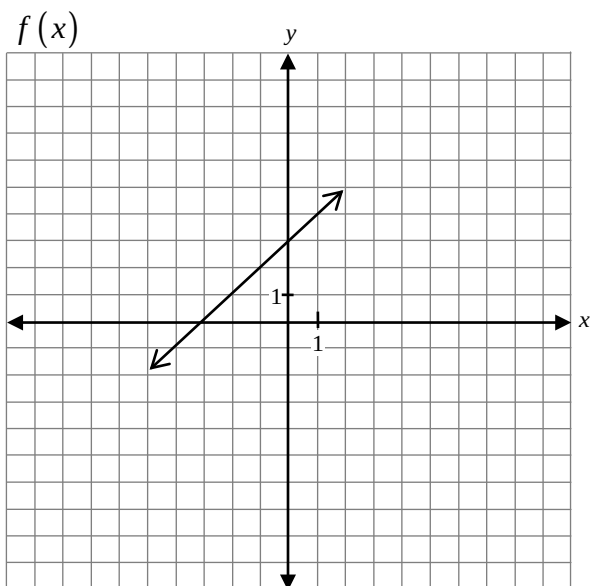


Question 49

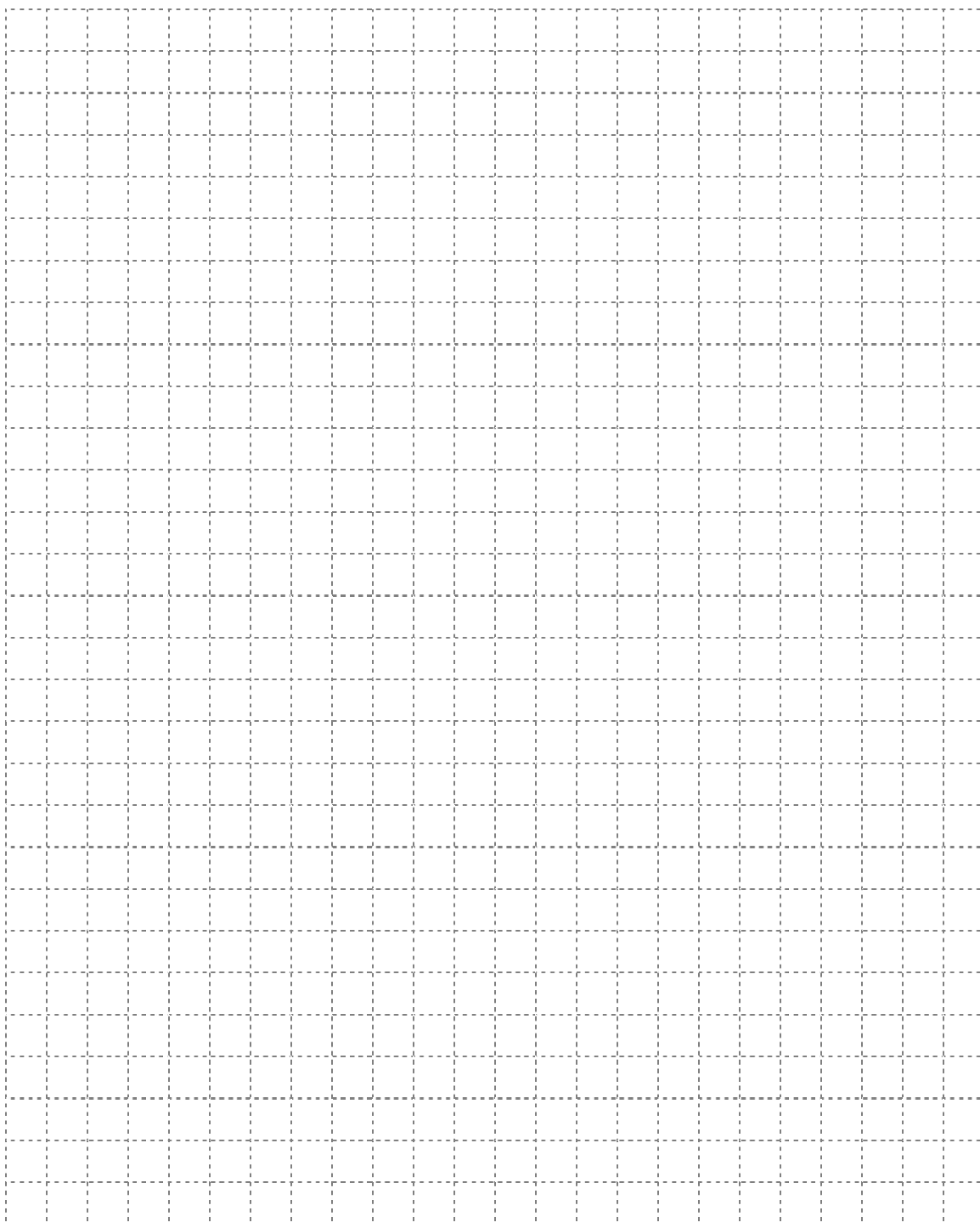
2 marks

143

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f \circ g)(x)$.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2015

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Multiple Choice	Constructed Response	Marks
Booklet 1*	–	15	33
Booklet 2	10	20	57
Total	10	35	90

* The first 6 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- Read all instructions carefully.
- The perforated *Formula Sheet* and the *Terminology Sheet* can be removed from the test booklet. No marks will be given for work done on these pages.
- The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

Instructions

- There are 15 questions for a total of 33 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- A calculator icon  appears next to the questions that require a calculator.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to 3 decimal places unless instructed otherwise.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

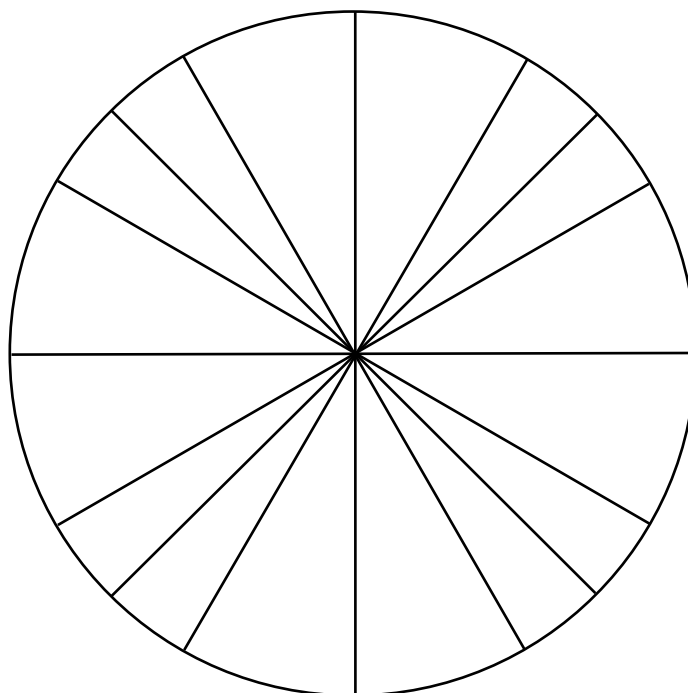
Solve: Give a solution for a problem or determine the value(s) of a variable.

Verify: Establish the truth of a statement by substitution or comparison.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

State: Give an answer without an explanation or justification.

Unit Circle (can be used if needed)



Question 1

1 mark

101

Convert $-\frac{13\pi}{5}$ to degrees.

Question 2

a) 1 mark b) 1 mark c) 1 mark

102
103
104

a) From a group of 9 people, in how many ways can you select a committee of 4 members?

b) From a group of 9 people, in how many ways can you select a president, a vice president, a secretary, and a treasurer?

c) Explain why the answers in a) and b) are different.

Question 3

a) 2 marks b) 2 marks

105
106

A population of 500 bacteria will triple in 20 hours.

Using the formula given below,

$$A = Pe^{rt}$$

A = population after t hours

P = initial population

r = rate of growth

t = time in hours

a) Determine the rate of growth, r .

b) Determine how many hours it will take for the initial population to double with the same rate of growth.

Question 4

a) 1 mark b) 2 marks

107
108

Talla incorrectly solved the following trigonometric equation:

Solve: $2 \sec x - 5 = 0$; $0^\circ \leq x \leq 360^\circ$.

Talla's work:

$$2 \sec x - 5 = 0$$

~~$$\sec x = \frac{5}{2}$$~~

No solution, $\sec x$ cannot be greater than 1.

a) Explain her error.

b) Determine the correct solution.

Question 5

3 marks

109

Simplify the 6th term in the expansion of:

$$\left(2x - \frac{3}{x^2}\right)^{10}$$

Question 6

1 mark

110

Determine the arc length subtended by a central angle if the diameter is 19 cm and the central angle is 1.6 radians.

Note: A calculator is not required for the remaining test questions.

Question 7

4 marks

111

Solve the following equation algebraically for x , where $0 \leq x \leq 2\pi$.

$$2\cos^2 x = -3\sin x$$

Question 8

1 mark

112

In how many different ways can you arrange the letters in the word VOLLEYBALL?

State your answer as a factorial.

Question 9

2 marks

113

Is $(x - 2)$ a factor of the polynomial $p(x) = -x^4 - 3x^3 + 11x^2 + 3x - 10$?

Justify your response.

Question 10

1 mark

114

Determine the period of the sinusoidal function $y = \frac{1}{2} \sin\left(\frac{1}{3}x\right)$.

State your answer in radians.

Question 11

2 marks

115

The domain of $f(x)$ is $x \leq 2$. The domain of $g(x)$ is $x \geq -7$.

State the domain of $f(x) + g(x)$.

Justify your answer.

Question 12

3 marks

116

Prove the identity below for all permissible values of θ .

$$\frac{1}{1 + \cos \theta} = \csc^2 \theta - \frac{\cot \theta}{\sin \theta}$$

Left-Hand Side	Right-Hand Side

Question 13

1 mark

117

Explain how the end behaviours of the graphs of polynomial functions with an even degree and with an odd degree are different.

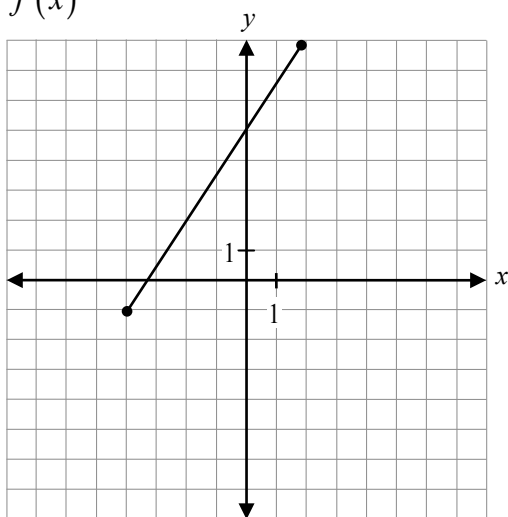
Question 14

2 marks

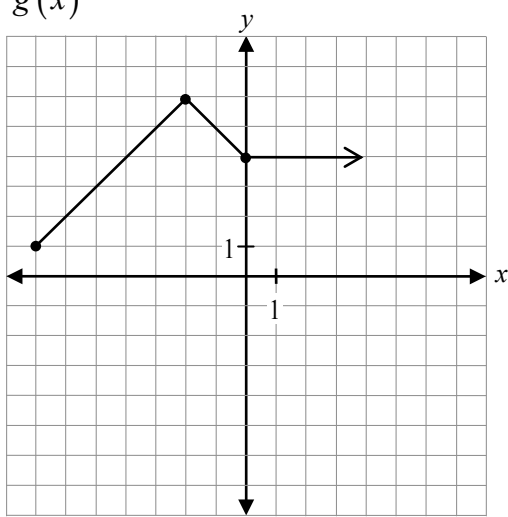
118

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(x) - f(x)$.

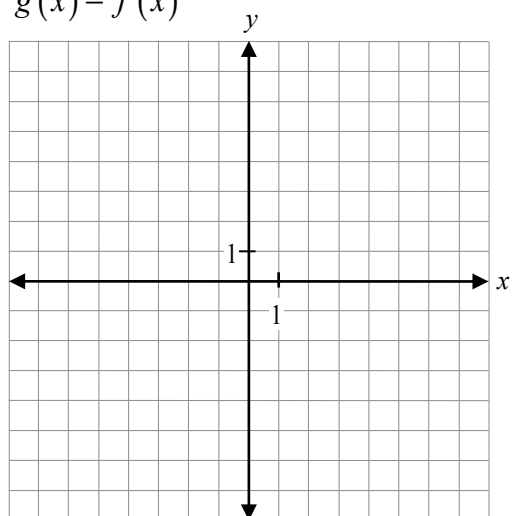
$f(x)$



$g(x)$



$g(x) - f(x)$



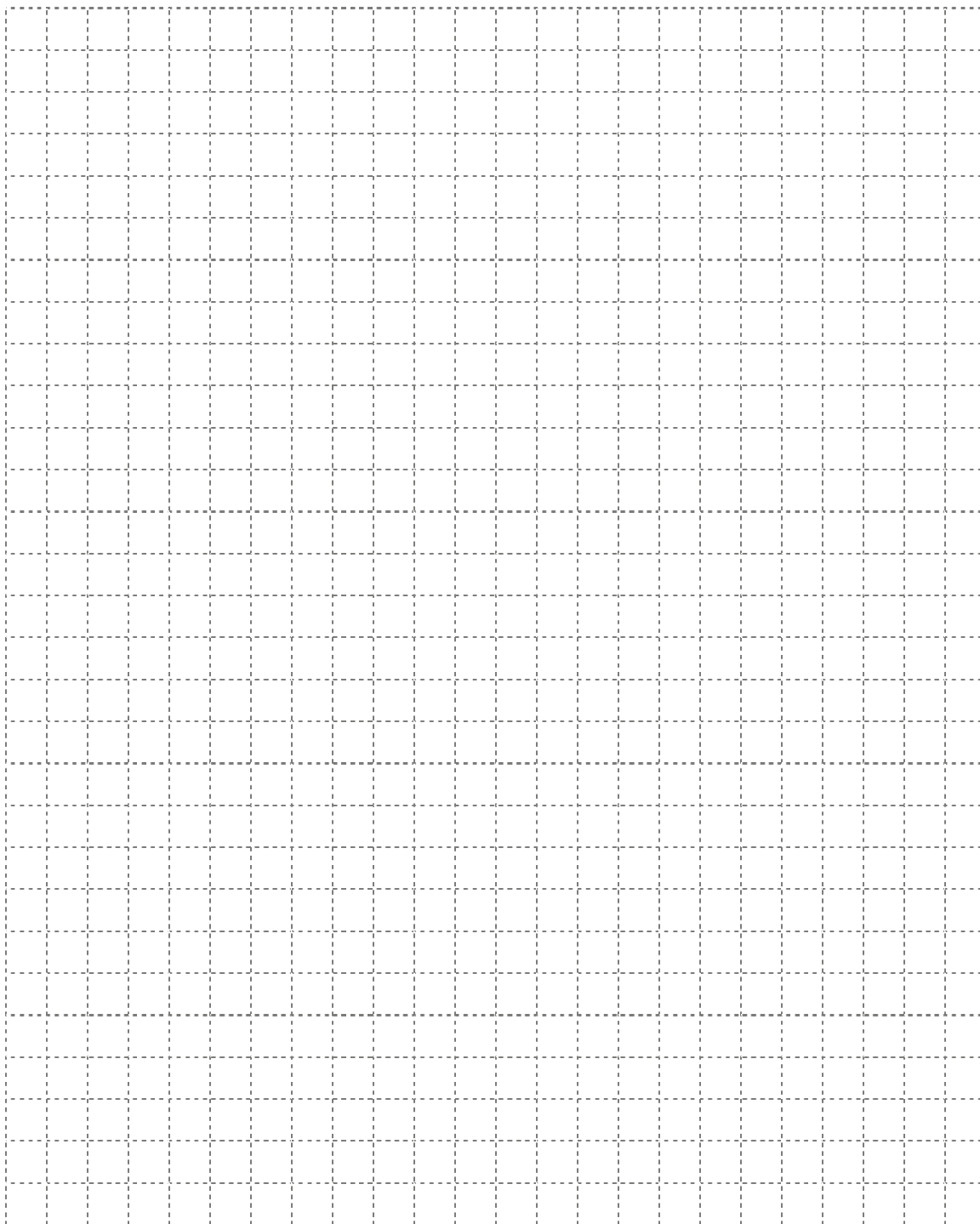
Question 15

2 marks

119

Given $f(x) = -3x + 7$, evaluate $f^{-1}(-2)$.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2015

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Instructions

Multiple-Choice Questions

- There are 10 questions each worth one mark.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- There are 20 questions worth a total of 47 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 16

1 mark

How many terms are there in the expansion of $(x^{12} + 3)^{10}$?

- a) 9
- b) 10
- c) 11
- d) 12

Question 17

1 mark

A co-terminal angle for $\theta = \frac{11\pi}{3}$ in the domain $-2\pi \leq \theta \leq 0$ would be:

- a) $-\frac{5\pi}{3}$
- b) $-\frac{\pi}{3}$
- c) $\frac{\pi}{3}$
- d) $\frac{5\pi}{3}$

Question 18

1 mark

The x-intercept of the graph of $y = 3^x - 1$ is:

- a) -1
- b) 0
- c) 1
- d) 2

Question 19

1 mark

If ${}_nC_5 = {}_nC_3$, the value of n must be:

- a) 3
- b) 5
- c) 8
- d) 15

Question 20

1 mark

What is the domain of the function $f(x) = \sqrt{-(x+1)}$?

- a) $\{x \mid x \in \mathbb{R}, x \neq -1\}$
- b) $\{x \mid x \in \mathbb{R}, x \geq -1\}$
- c) $\{x \mid x \in \mathbb{R}, x \leq -1\}$
- d) $\{x \mid x \in \mathbb{R}\}$

Question 21

1 mark

Identify a non-permissible value of x for the expression $\frac{1}{\cos 2x}$.

- a) 0
- b) $\frac{\pi}{4}$
- c) $\frac{\pi}{2}$
- d) π

Question 22

1 mark

The expression $2 \log x - \frac{1}{3} \log y$ as a single logarithm is:

- a) $\log \frac{x^2}{\sqrt[3]{y}}$
- b) $\log \frac{2x}{3y}$
- c) $-\log x^2 \sqrt[3]{y}$
- d) $\log(x^2 - \sqrt[3]{y})$

Question 23

1 mark

The point $P(\theta)$ lies on the unit circle. What are the coordinates of the point if $\theta = 300^\circ$?

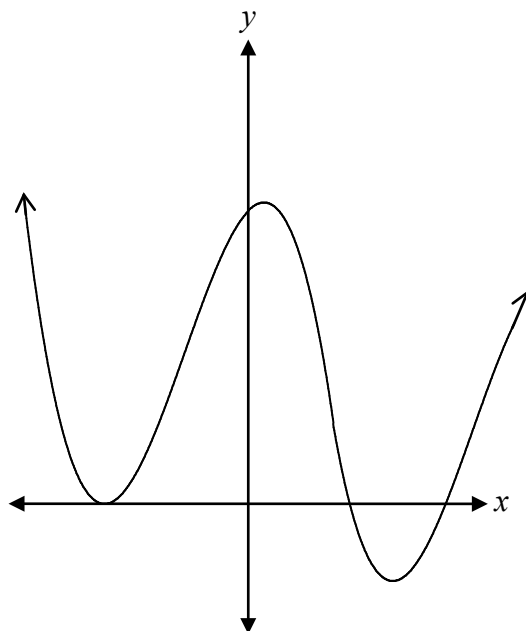
- a) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- b) $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- c) $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- d) $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

Question 24

1 mark

What is the degree of the polynomial function represented by the graph below?

- a) 2
- b) 3
- c) 4
- d) 5



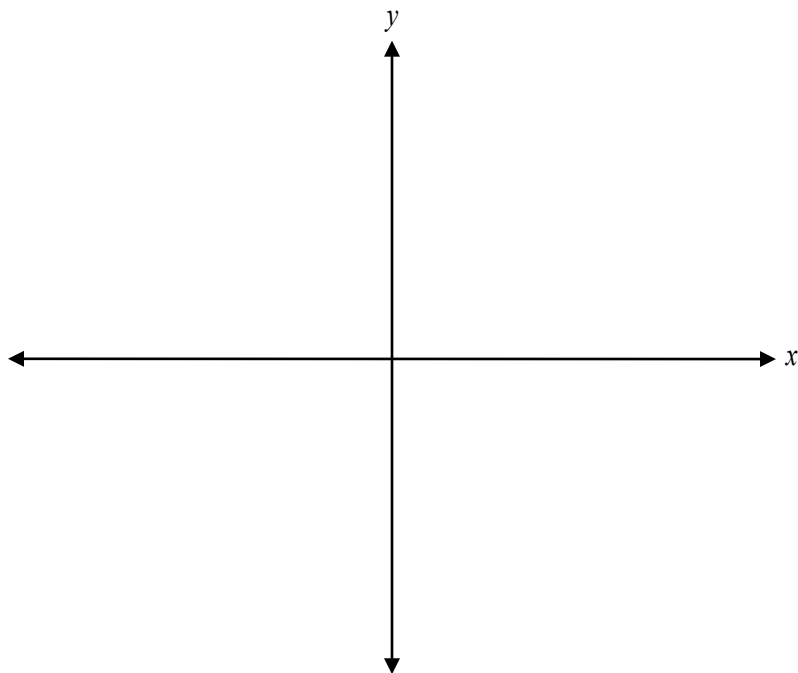
Question 25

1 mark

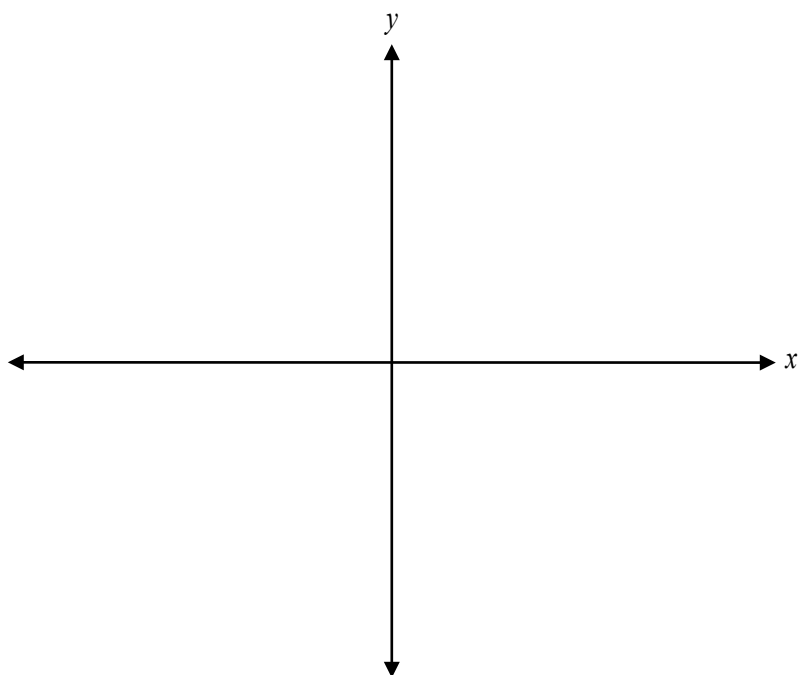
When the point $(-4, -3)$ is reflected in the line $y = x$, the coordinates of the new point are:

- a) $(-3, -4)$
- b) $(3, 4)$
- c) $(4, -3)$
- d) $(-4, 3)$

- a) Sketch the graph of $y = \left(\frac{1}{4}\right)^x$.



- b) Sketch the graph of $y = 2\left(\frac{1}{4}\right)^x$.



Question 27

2 marks

122

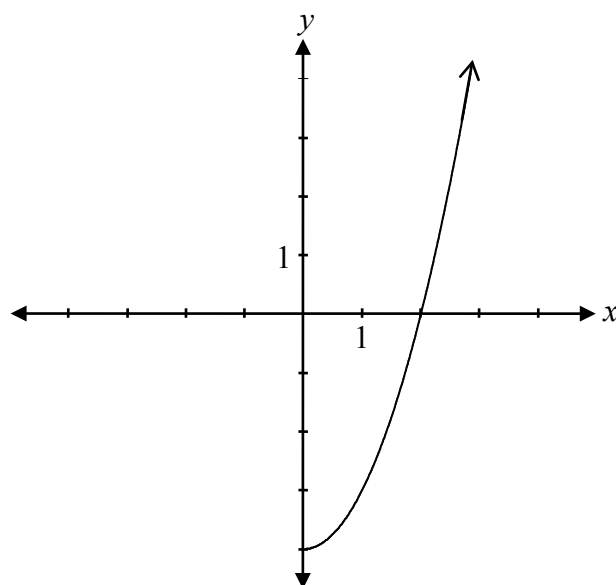
Determine all of the zeroes of the function $p(x) = x^3 - 5x^2 - 2x + 24$, given one of the factors of $p(x)$ is $(x - 3)$.

Question 28

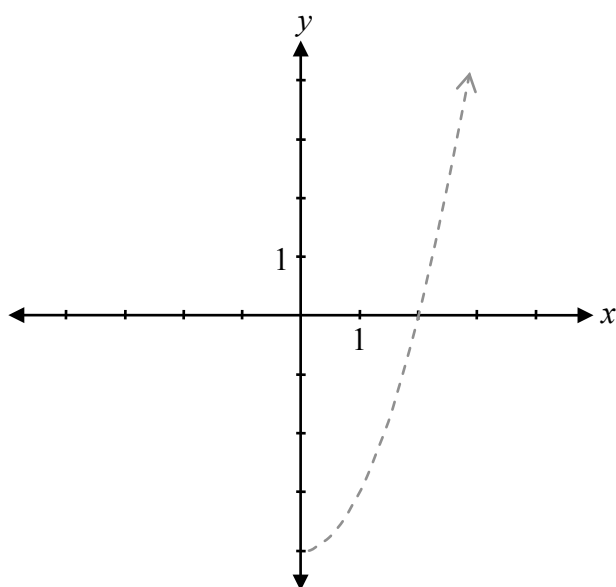
2 marks

123

Given the graph of $f(x)$,

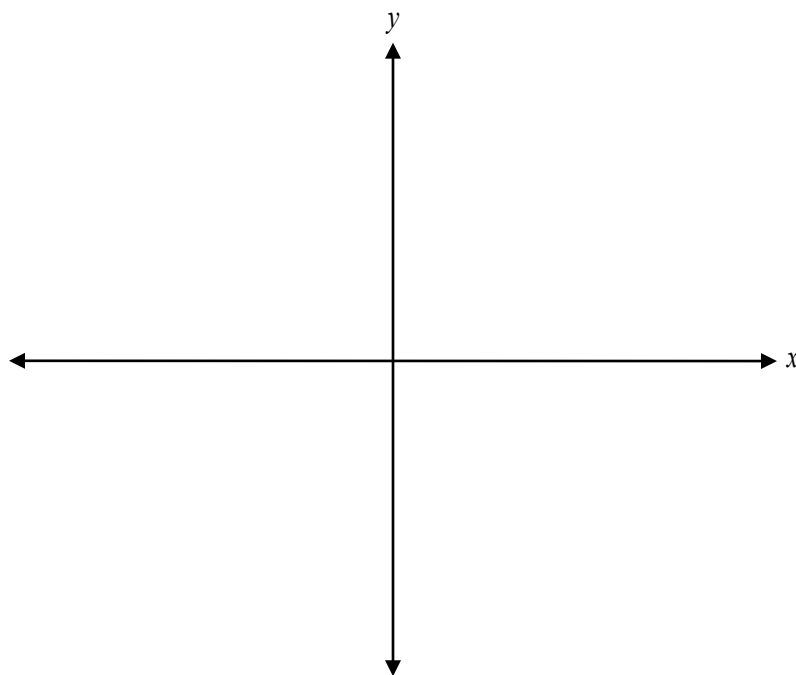


sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.
No marks will be awarded for the graph of $f(x)$.

Sketch the graph of at least one period of the function $y = -2\sin(4x)$.



Question 30

3 marks

125

Evaluate:

$$\frac{1}{2} \log_3 144 - \log_3 4 + 2 \log_3 3$$

Question 31

2 marks

126

Match each function with its correct description.

- a) The graph of this function has a vertical asymptote at $x = -1$.
- b) The graph of this function has a point of discontinuity (hole) at $x = 3$.
- c) The graph of this function has a horizontal asymptote at $y = 4$.
- d) The domain of this function is $x \in \mathbb{R}$.

Place the appropriate letter in this column.

$$f(x) = \frac{4}{x^2 + 1} \quad \underline{\hspace{2cm}}$$

$$g(x) = \frac{4x}{x + 3} \quad \underline{\hspace{2cm}}$$

$$h(x) = \frac{4(x - 3)(x + 2)}{(x - 3)} \quad \underline{\hspace{2cm}}$$

$$k(x) = \frac{4(x - 3)}{(x + 3)(x + 1)} \quad \underline{\hspace{2cm}}$$

Question 32

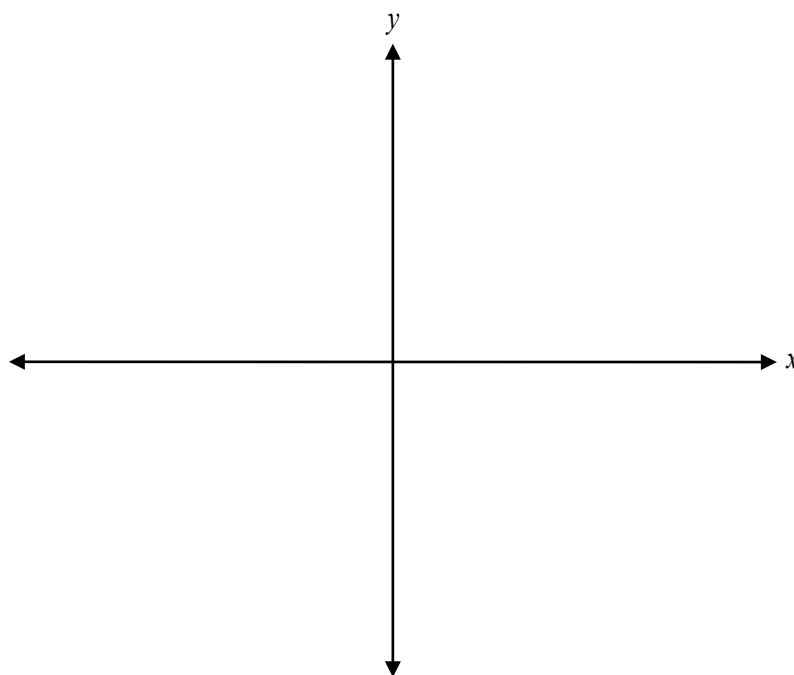
1 mark

127

The point $(-3, 4)$ is on the graph of $y = \frac{1}{2}f(3x)$.

State the coordinates of the corresponding point on the graph of $y = f(x)$.

Sketch the graph of $y = -2(x-1)(x-3)(x+1)$.



Question 34

a) 2 marks b) 1 mark

129
130

- a) Verify that the equation $\frac{1 - \sin^2 x}{\cos x} = \frac{\sin 2x}{2 \sin x}$ is true for $x = \frac{\pi}{3}$.

Left-Hand Side	Right-Hand Side

- b) Explain why verifying the equation for $x = \frac{\pi}{3}$ is insufficient to conclude that the equation is an identity.

Question 35

2 marks

131

Evaluate:

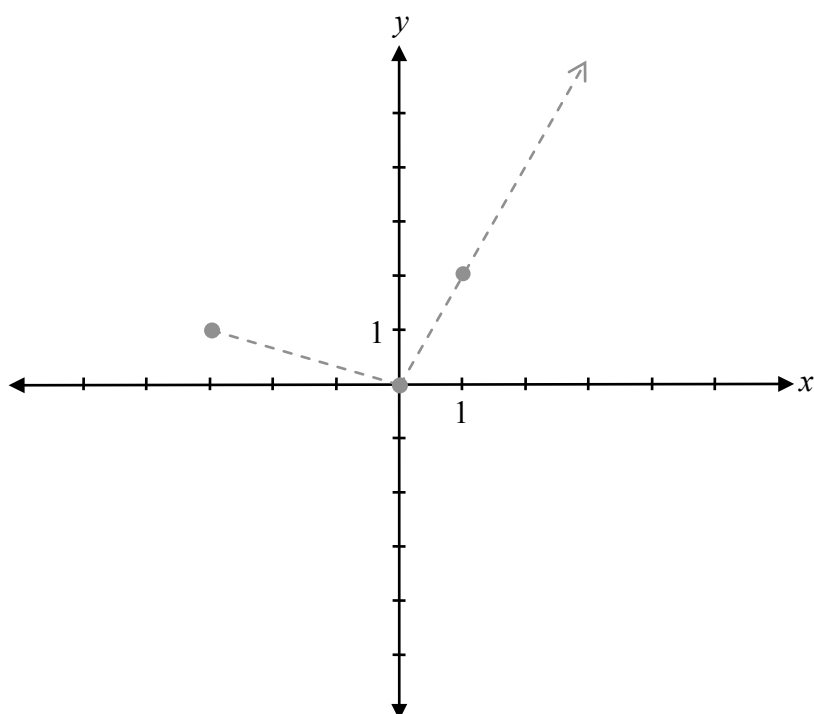
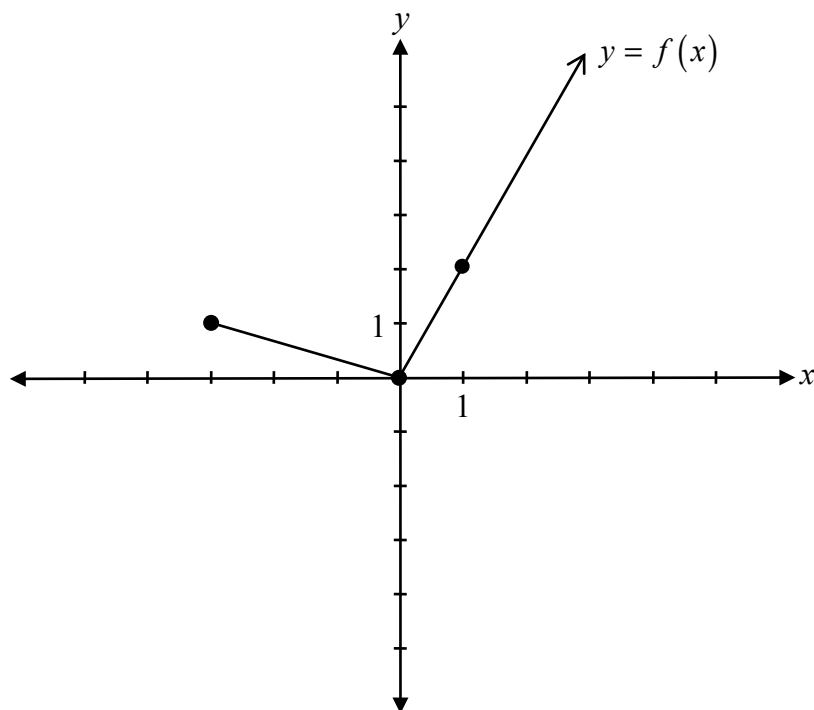
$$\frac{{}_7P_2}{{}_7P_5}$$

Question 36

2 marks

132

Use the graph of $y = f(x)$ to sketch the graph of $y = f(3x) + 1$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 37

3 marks

133

Solve the following equation:

$$\log_4(x+2) + \log_4 3 = \log_4 x$$

Question 38

1 mark

134

Determine the coordinates of the point of discontinuity (hole) for the graph of the function

$$y = \frac{(2-x)(x-3)}{(x-2)}.$$

Evaluate and simplify $\sec\left(\frac{5\pi}{6}\right) \cdot \tan\left(-\frac{\pi}{6}\right)$.

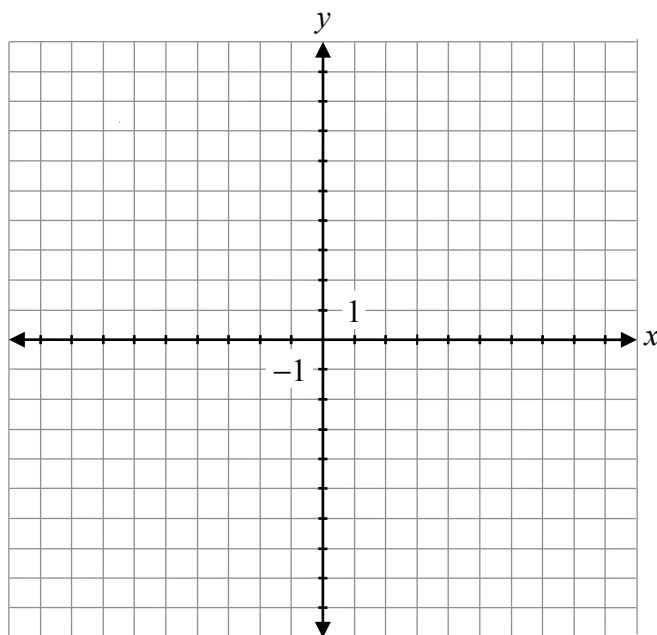
Question 40

4 marks

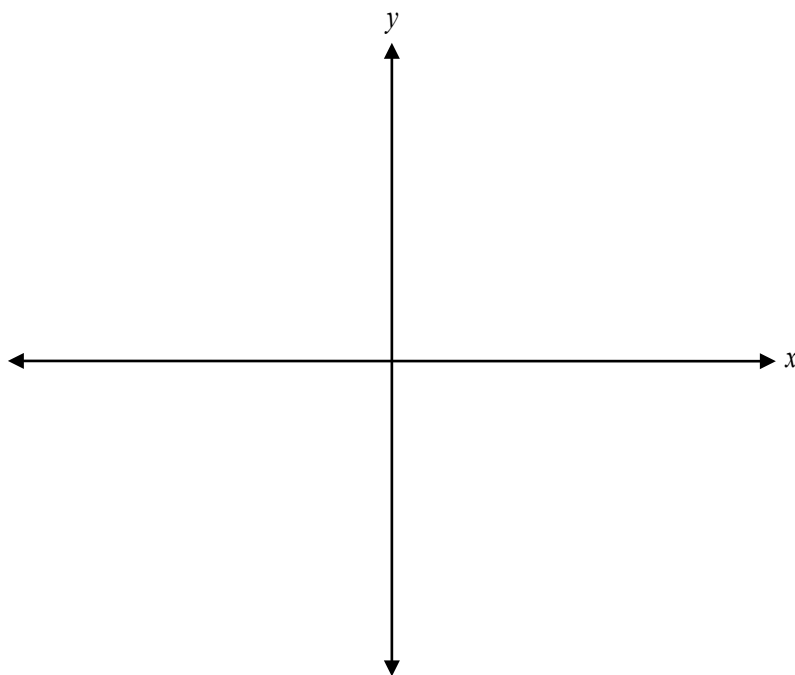
136

Sketch the graph of the following function:

$$y = -2\sqrt{x-3}$$



Sketch the graph of $f(x) = \frac{2x+3}{x+2}$.



Question 42

a) 2 marks b) 1 mark

138
139

a) Given the functions $f(x) = \sqrt{4+x}$ and $g(x) = |3x-6|$, evaluate $f(g(-5))$.

b) Is it possible to evaluate $g(f(-5))$?

Justify your answer.

Question 43

1 mark

140

Identify which of these values is greater. Justify your answer.

$$\log_5 80 \quad \text{or} \quad \log_3 30$$

Question 44

3 marks

141

Given $\cos \alpha = \frac{3}{5}$, where α is in quadrant IV, and $\cos \beta = -\frac{2}{3}$, where β is in quadrant II, determine the exact value of $\sin(\alpha - \beta)$.

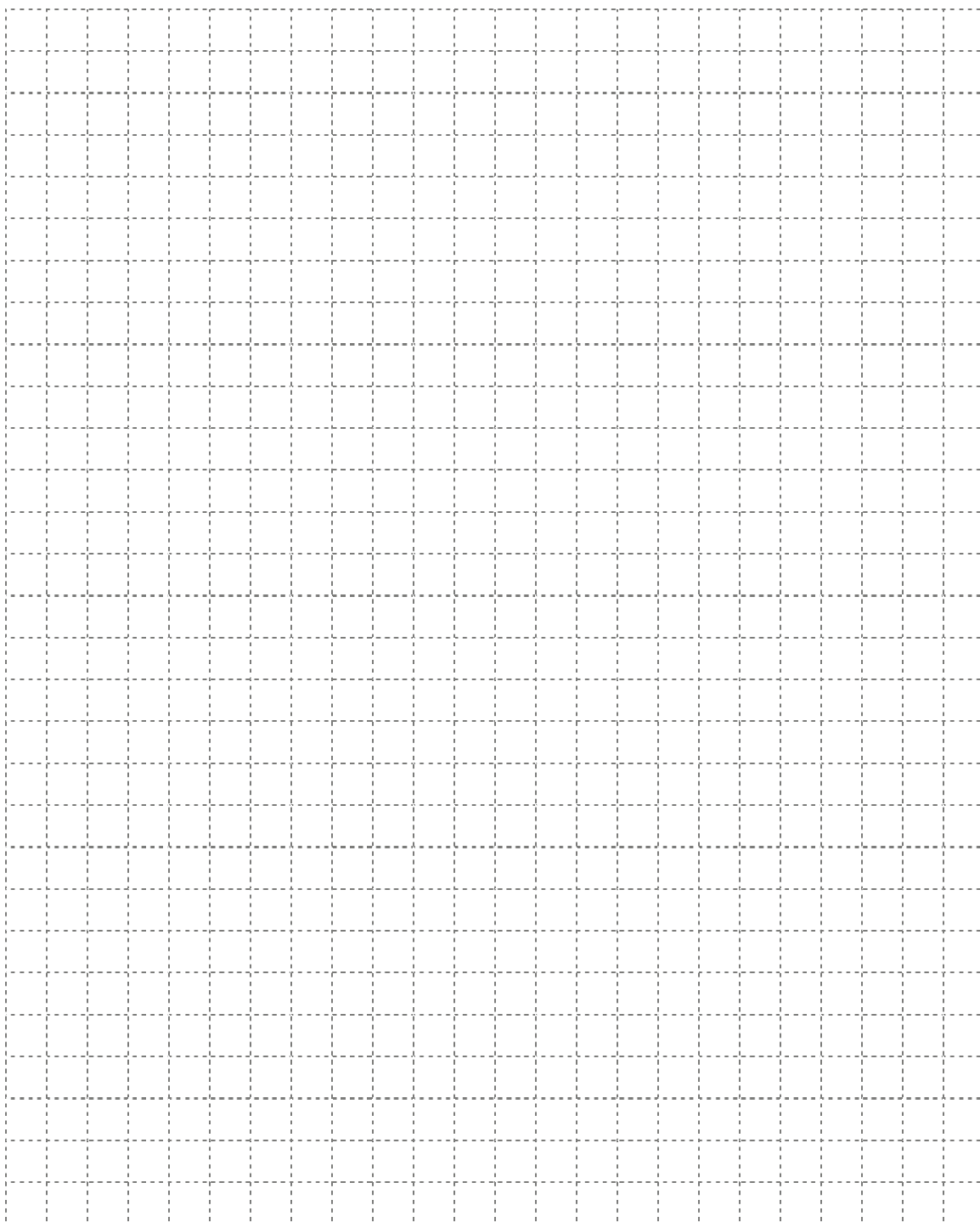
Determine the number of possible sandwiches from the following menu.

MENU

Select one item from each column:

<u>Bread</u>	<u>Sauce</u>	<u>Meat</u>	<u>Vegetable</u>
White	Mayo	Turkey	Tomato
Rye	Mustard	Ham	Onion
Brown		Roast Beef	Lettuce
		Chicken	

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2015

Manitoba Education and Advanced Learning Cataloguing in Publication Data

Grade 12 pre-calculus mathematics achievement test.

Booklet 1. June 2015 [electronic resource]

ISBN: 978-0-7711-5849-0

1. Mathematics—Examinations, questions, etc.
 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Precalculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
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Manitoba Education and Advanced Learning
School Programs Division
Winnipeg, Manitoba, Canada

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	–	13	31
Booklet 2	9	20	59
Total	9	33	90

* The first 6 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- Read all instructions carefully.
- The perforated *Formula Sheet* and the *Terminology Sheet* can be removed from the test booklet. No marks will be given for work done on these pages.
- The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

Instructions

- There are 13 questions for a total of 31 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- A calculator icon  appears next to the questions that require a calculator.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

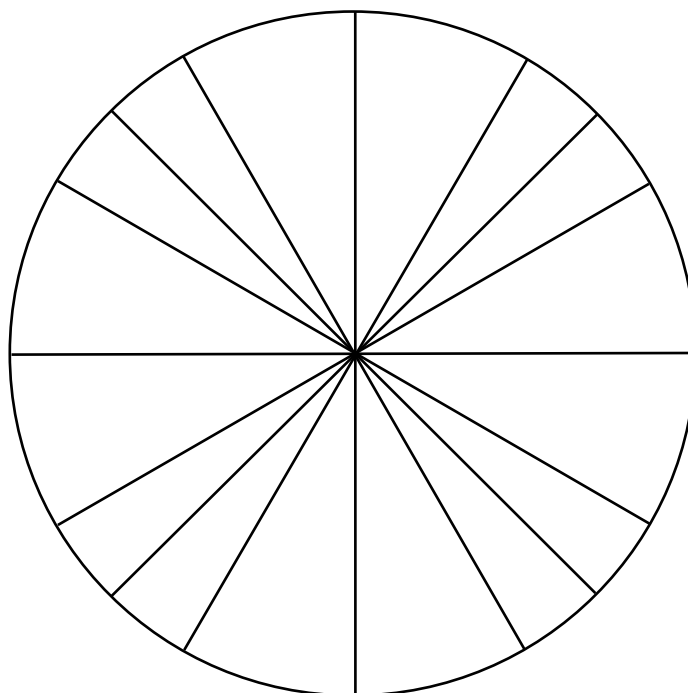
Solve: Give a solution for a problem or determine the value(s) of a variable.

Verify: Establish the truth of a statement by substitution or comparison.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

State: Give an answer without an explanation or justification.

Unit Circle (can be used if needed)

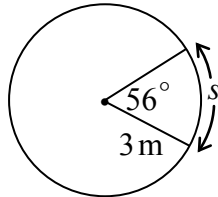


Question 1

2 marks

101

Use the information in the diagram to determine the value of the arc length “ s ”, given the central angle of 56° .



Question 2

4 marks

102

Solve $\tan^2 \theta - 5 \tan \theta + 4 = 0$ where $\theta \in \mathbb{R}$.

Question 3

4 marks

103

Solve:

$$2^{5x} = 3(5)^{x-3}$$

Question 4

3 marks

104

David and Sarah are in a class of 10 boys and 8 girls.

A committee of 3 boys and 2 girls is to be selected from the students in this class.

Determine the number of possible committees if David and Sarah cannot be on the same committee.

Question 5

3 marks

105

In the binomial expansion of $\left(\frac{3}{x^2} - x^5\right)^{10}$, simplify the 7th term.

Question 6

2 marks

106

A lake affected by acid rain has a pH of 4.4.

A person suffering from heartburn has a stomach acid pH of 1.2.

The pH of a solution is defined as $\text{pH} = -\log[H^+]$ where $[H^+]$ is the hydrogen ion concentration.

How many times greater is the hydrogen ion concentration of the stomach than that of the lake?

Express your answer as a whole number.

Note: A calculator is not required for the remaining test questions.

Question 7

4 marks

107

Solve the following equation algebraically over the interval $[0, 2\pi]$.

$$\cos 2\theta - 3 \sin \theta - 2 = 0$$

Question 8

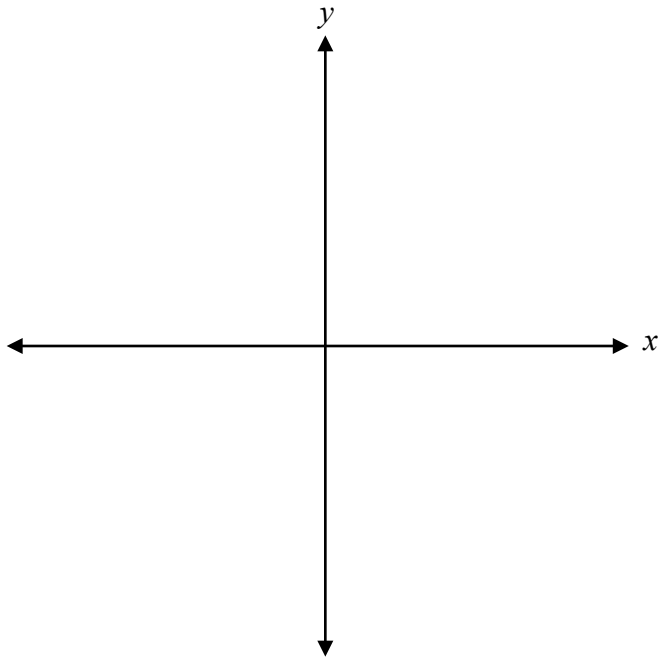
1 mark

108

Explain how the value of n affects the behaviour of the graph of the polynomial function

$p(x) = (x + 3)(x - 1)^n$, as $p(x)$ approaches the x -intercept at $x = 1$.

Sketch the angle -320° in standard position.

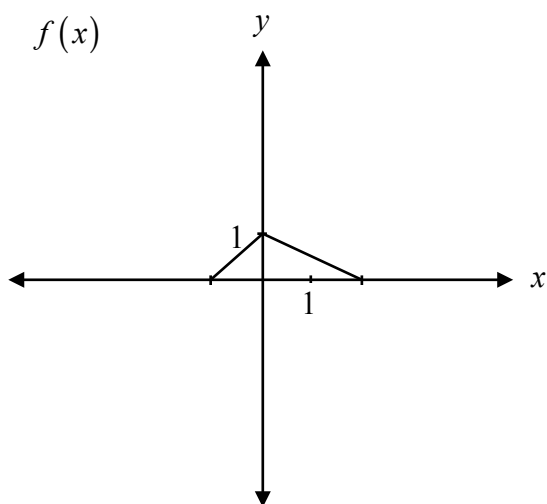


Question 10

1 mark

110

Given the graph of $y = f(x)$, explain how to graph $y = f(-x)$.



Question 11

1 mark

111

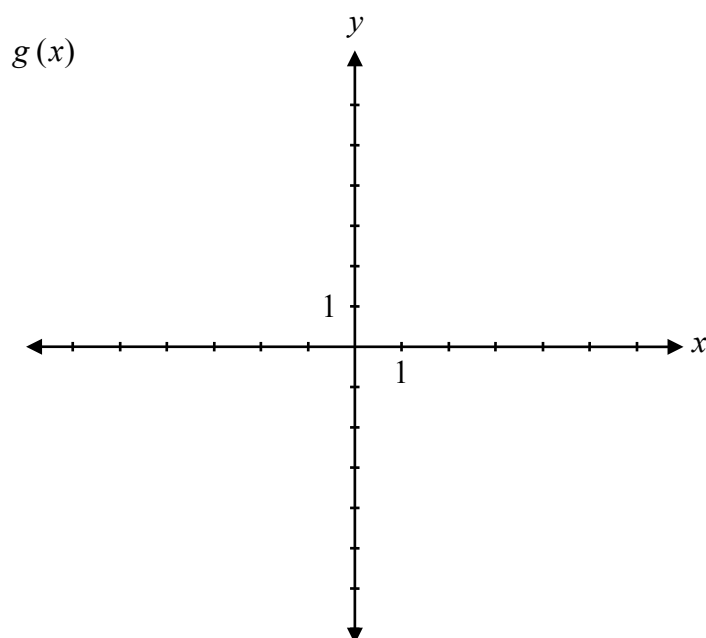
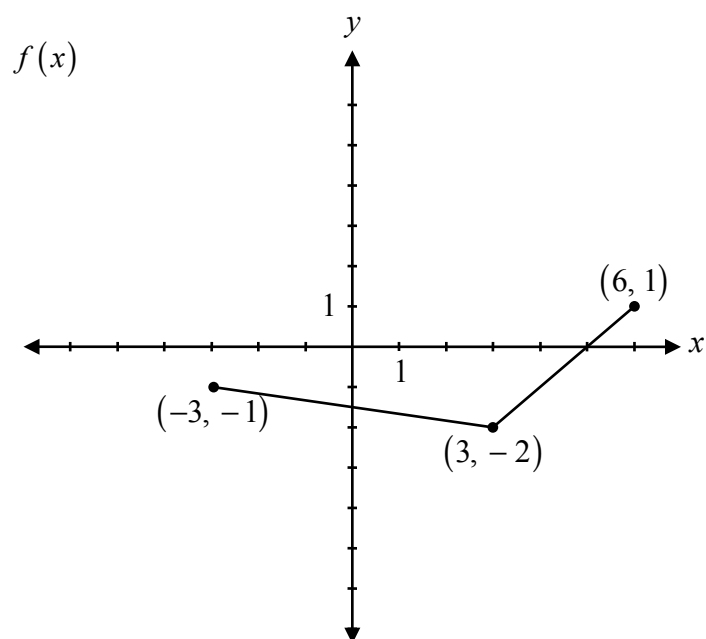
Explain how the graph of $y = \frac{3(x-1)}{(x-1)}$ is different than the graph of $y = 3$.

Question 12

2 marks

112

Given the graph of $f(x)$, sketch the graph of $g(x) = 2f(3x)$.

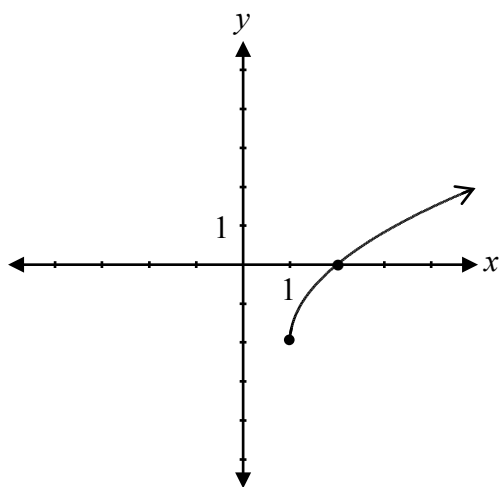


Question 13

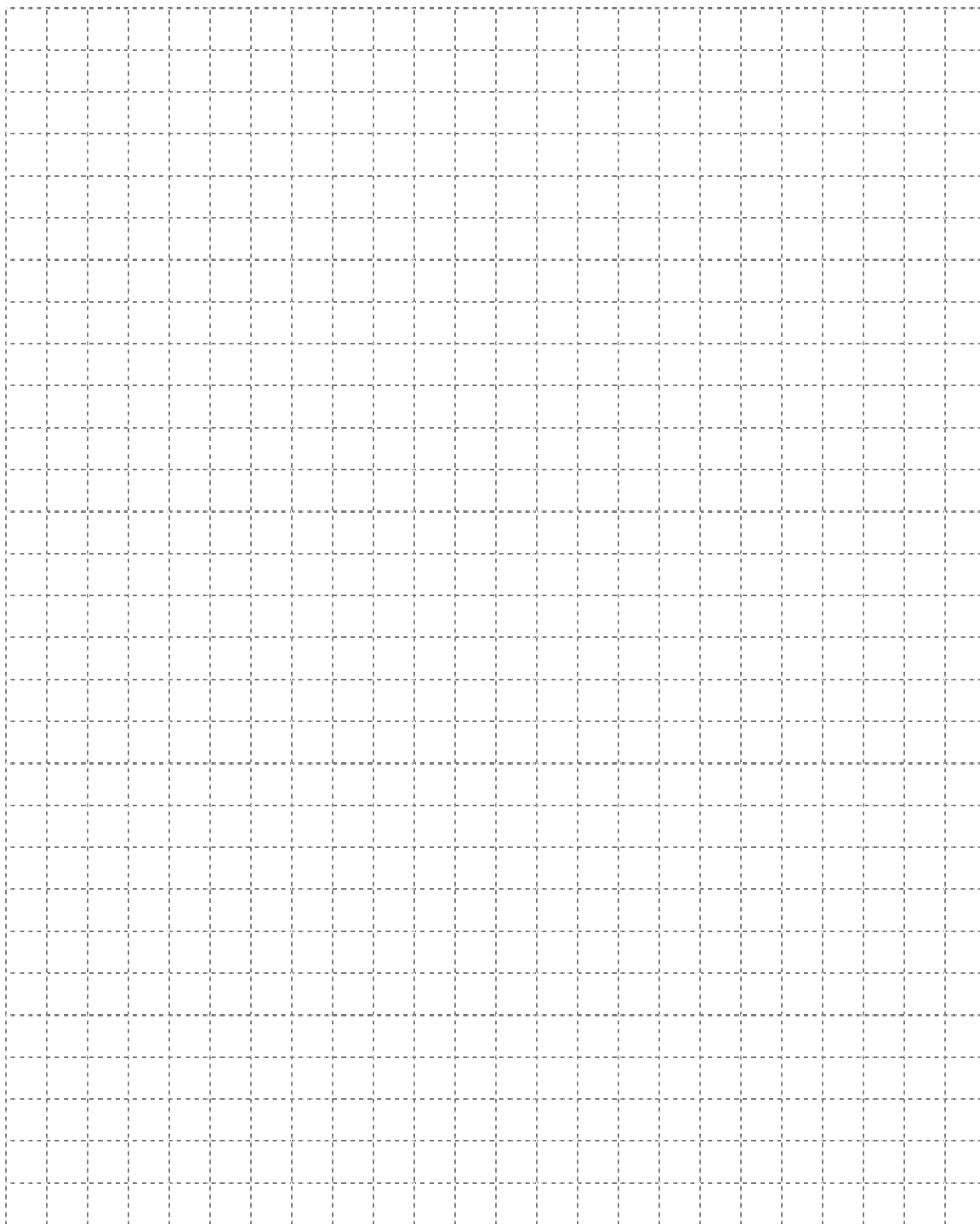
3 marks

113

Determine the equation of the radical function represented by the graph.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2015

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1. Mathematics—Examinations, questions, etc.
 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Precalculus—Study and teaching (Secondary)—Manitoba.
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Instructions

Selected Response Questions

- There are 9 questions each worth one mark.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- There are 20 questions worth a total of 50 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 14

1 mark

There are 2 types of pencils, 3 colours of highlighters, and 5 styles of pens.

If you must select one of each to form a set, how many different sets of writing instruments are possible?

- a) 10
- b) 11
- c) 25
- d) 30

Question 15

1 mark

The point $P(\theta)$ lies on the unit circle. What are the coordinates of the point P if $\theta = 120^\circ$?

- a) $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- b) $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- c) $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
- d) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

Question 16

1 mark

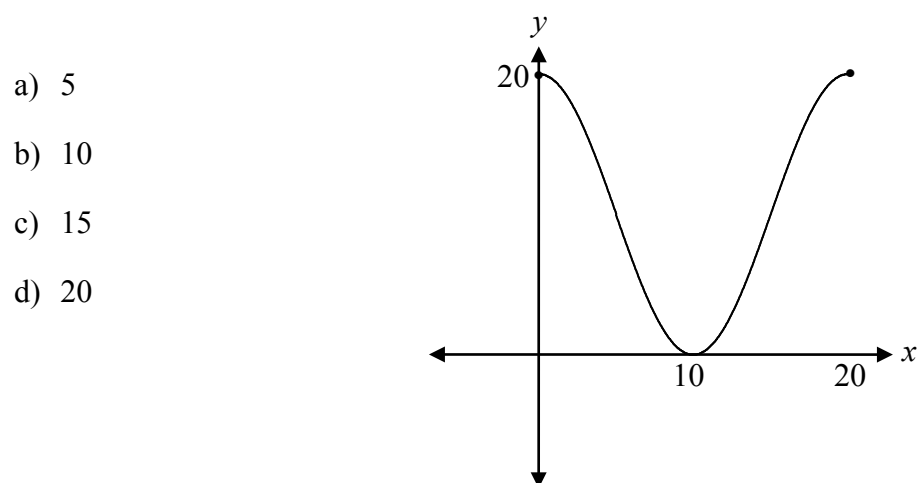
Identify the function that has a domain of $\{x|x \geq 7\}$ and a range of $\{y|y \geq 0\}$.

- a) $f(x) = \sqrt{x} + 7$
- b) $f(x) = \sqrt{x} - 7$
- c) $f(x) = \sqrt{x+7}$
- d) $f(x) = \sqrt{x-7}$

Question 17

1 mark

Using $y = -10 \cos[B(x - C)] + D$, the value of C that corresponds to the following graph is:



Question 18

1 mark

Given the following row of Pascal's Triangle, identify the binomial expansion with these coefficients.

1 5 10 10 5 1

a) $(x + y)^4$

b) $(x + y)^5$

c) $(x + y)^6$

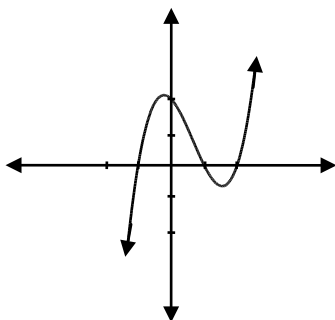
d) $(x + y)^7$

Question 19

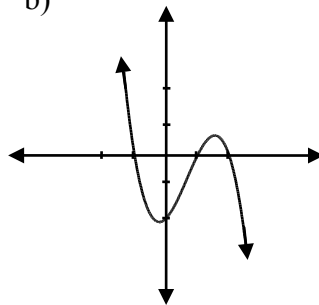
1 mark

Identify the graph of the function $f(x) = -(x - 2)(x - 1)^2(x + 1)$.

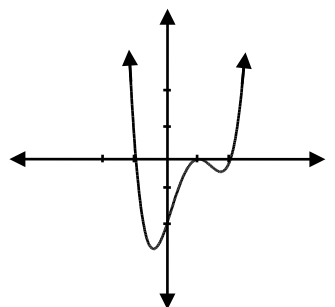
a)



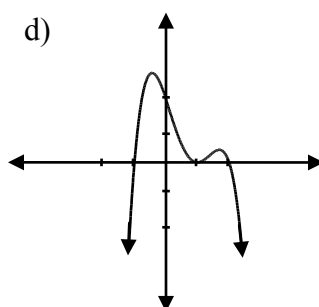
b)



c)



d)



Question 20

1 mark

Determine the value of $\log_9(\log_3 27)$.

a) $\frac{1}{3}$

b) $\frac{1}{2}$

c) 2

d) 3

Question 21

1 mark

If (x, y) is a point on the graph of $y = f(x)$, identify the coordinates of this point on the graph of $g(x) = f(2x) + 5$.

a) $\left(\frac{x}{2}, y + 5\right)$

b) $(2x, y + 5)$

c) $\left(\frac{x}{2}, y - 5\right)$

d) $\left(\frac{x}{2} - 5, y\right)$

Question 22

1 mark

Identify an equivalent expression for $1 + \log_2 5$.

- a) $\log_2 5$
- b) $\log_2 7$
- c) $\log_2 10$
- d) $\log_2 11$

Question 23

4 marks

114

Solve:

$$2\log_4 x - \log_4 (x + 3) = 1$$

Question 24

3 marks 115

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the y -axis
- horizontal translation of 3 units to the right
- vertical translation of 4 units down

Write the equation of $g(x)$ in terms of $f(x)$.

$g(x) =$ _____

Question 25

a) 3 marks b) 1 mark

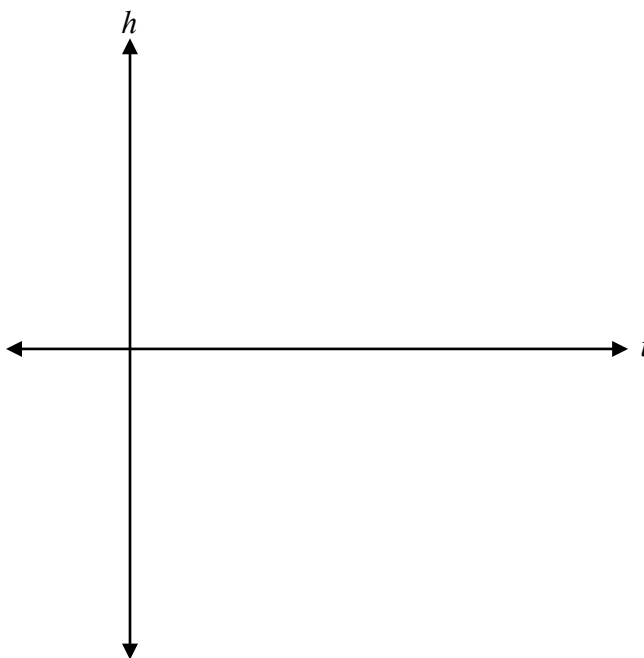
116
117

The height of a bicycle pedal as the bicycle is moving at a constant speed can be represented by the following function:

$$h(t) = 15 \cos \frac{2\pi}{5}t + 30$$

where h is the height of the pedal above the ground, in cm, and t is the time, in seconds.

- a) Sketch a graph of at least one period of this function, where $t \geq 0$.



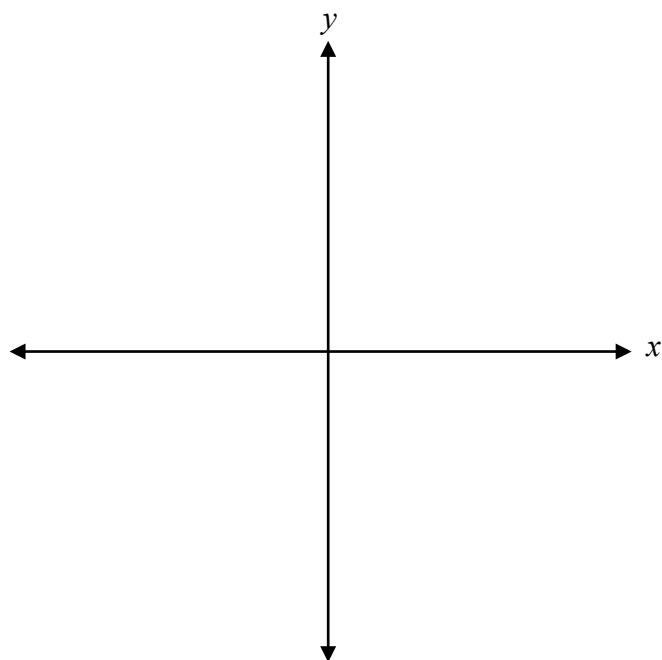
- b) Determine the height of the bicycle pedal at 7.5 seconds.

Question 26

4 marks 118

Sketch the graph of the function $f(x)$ and determine the y -intercept.

$$f(x) = \frac{4}{(x-2)(x+2)}$$



y -intercept: _____

Question 27

1 mark 119

Kim solved the following logarithmic equation:

$$\log_2 \left(-\frac{x}{3} \right) = \log_2 (x - 4)$$

$$-\frac{x}{3} = x - 4$$

$$-x = 3x - 12$$

$$-4x = -12$$

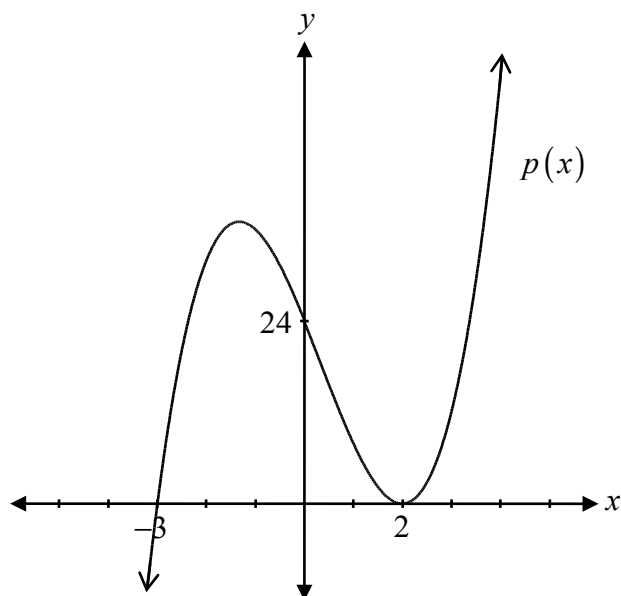
$$\cancel{x = 3}$$

Explain why $x = 3$ is an extraneous solution.

Question 28

3 marks 120

Determine the equation of the polynomial function represented by the graph.



$p(x) =$ _____

Question 29

1 mark 121

Determine the coterminal angles with $\frac{2\pi}{3}$ over the interval $[-2\pi, 4\pi]$.

Question 30

a) 1 mark b) 1 mark

122
123

- a) Solve the following equation:

$$0 = \sqrt{4x - 8} - 2$$

- b) Explain how your answer in a) is related to the graph of $y = \sqrt{4x - 8} - 2$.

Question 31

3 marks 124

Determine the exact value of $\sin \frac{13\pi}{12}$.

Question 32

a) 2 marks b) 1 mark

125
126

Given the functions $f(x) = x + 2$ and $g(x) = \frac{1}{x-5}$:

- a) Determine the equation of the composite function $f(g(x))$ and its domain.

$$f(g(x)) = \underline{\hspace{2cm}}$$

domain: $\underline{\hspace{2cm}}$

- b) Determine the x -intercept and y -intercept of $f(g(x))$.

x -intercept: $\underline{\hspace{2cm}}$

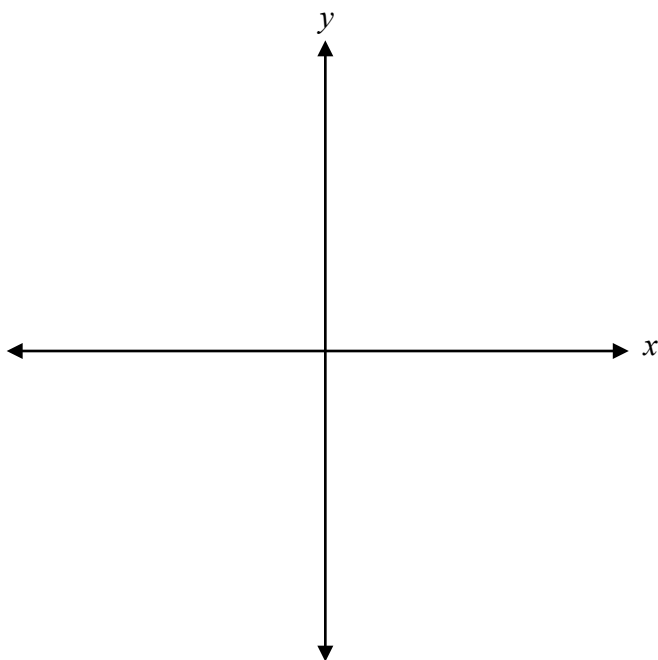
y -intercept: $\underline{\hspace{2cm}}$

Question 33

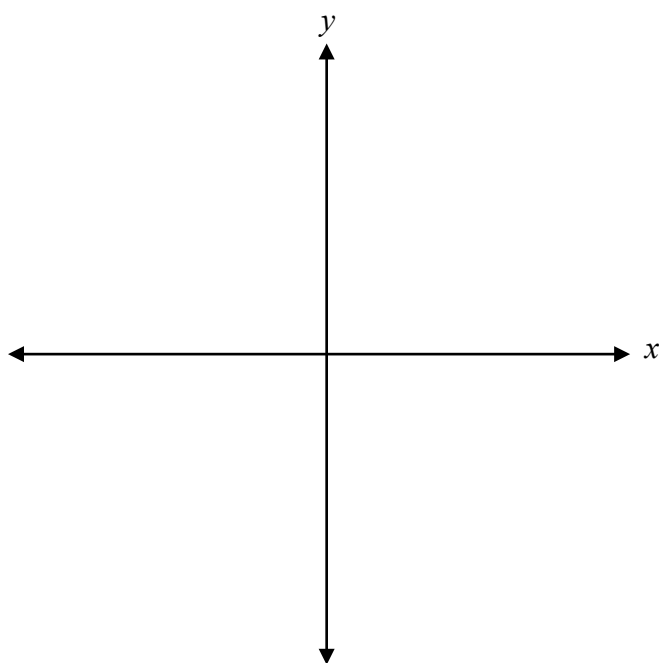
a) 2 marks b) 1 mark

127
128

- a) Sketch the graph of $f(x) = \log_5(x-1)$.



- b) Sketch the graph of $f^{-1}(x)$.



Question 34

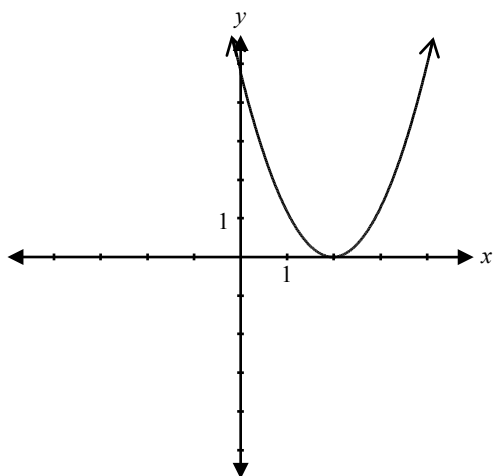
1 mark 129

Explain what the graph of a rational function looks like near a vertical asymptote.

Question 35

1 mark 130

Given the graph of $f(x) = (x - 2)^2$,



determine one possible restriction for the domain of $f(x)$ so that its inverse is a function.

Domain: _____

Question 36

2 marks 131

Over the interval $[0, 2\pi]$, determine the non-permissible values of θ in the expression $\csc \theta (\cos \theta + 1)$.

Question 37

1 mark 132

Explain why ${}_3C_8$ is undefined.

Question 38

3 marks 133

Solve:

$${}_nP_3 = 48(n-1)$$

Question 39

a) 2 marks b) 1 mark c) 1 mark

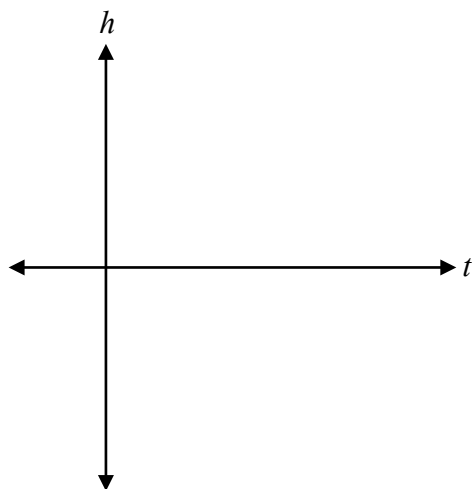
134
135
136

Christine dives off a diving board.

Her dive is modelled by the function $h(t) = t^3 - 3t^2 - t + 3$, where h is her height in metres, relative to the water surface and t is the time in seconds after diving off the diving board.

a) Given that $(t + 1)$ is a factor for the function $h(t)$, determine the other factors.

b) Sketch the graph of the function $h(t)$ for the time interval $t = 0$ to $t = 3$.



c) Determine how many seconds Christine is underwater.

Question 40

3 marks 137

Prove the identity for all permissible values of x .

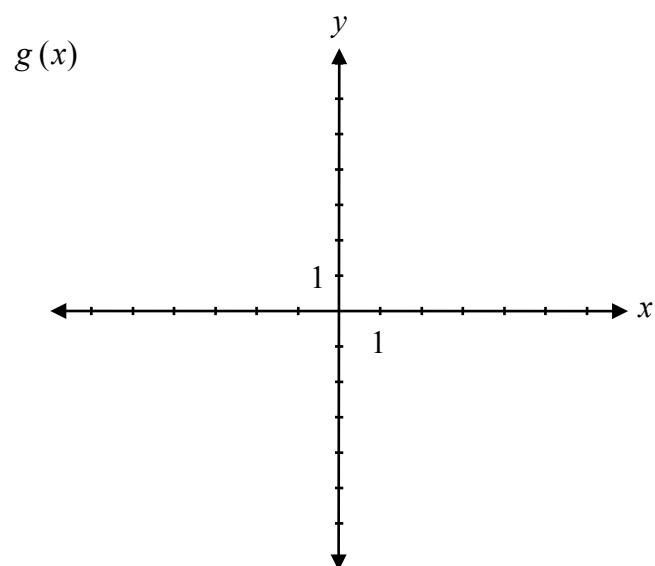
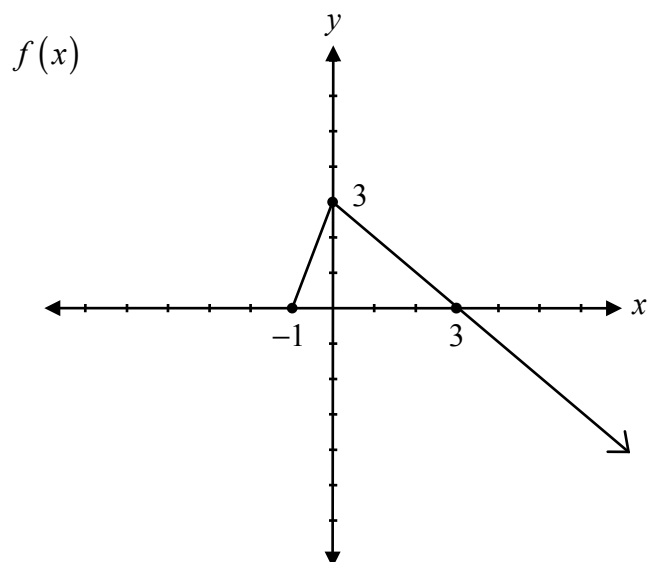
$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

Left-Hand Side	Right-Hand Side

Question 41

2 marks 138

Given the graph of $f(x)$, sketch the graph of the function $g(x) = -|f(x)|$.

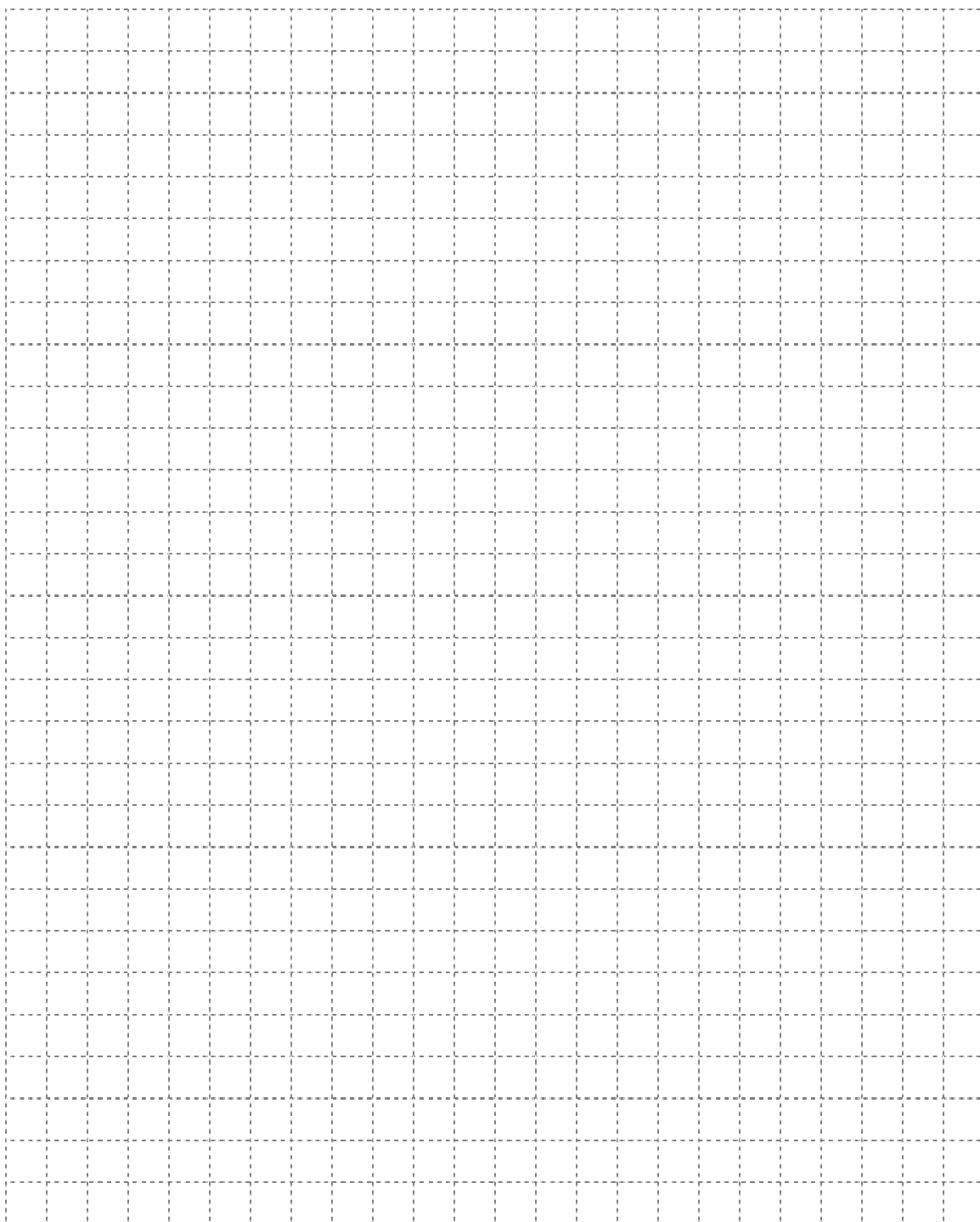


Question 42

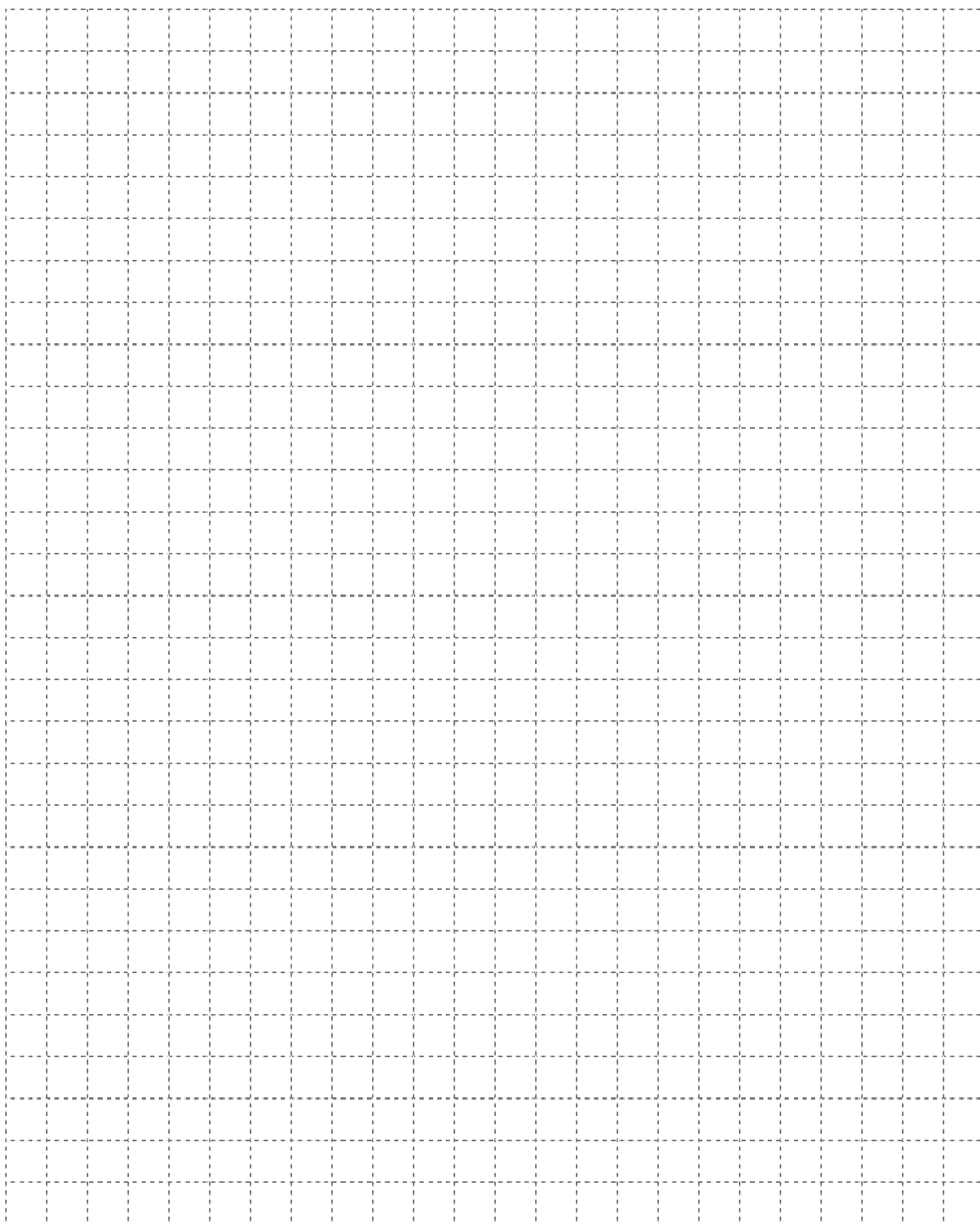
2 marks 139

Given that $\cot \theta = -\frac{2}{5}$, where θ is in Quadrant IV, determine the exact value of $\sin \theta$.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2016

Manitoba Education and Advanced Learning Cataloguing in Publication Data

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Booklet 1. January 2016

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1. Mathematics—Examinations, questions, etc.
 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Precalculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
- I. Manitoba. Manitoba Education and Advanced Learning.
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Manitoba Education and Advanced Learning
School Programs Division
Winnipeg, Manitoba, Canada

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	–	13	29
Booklet 2	8	24	63
Total	8	37	92

* The first 4 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- Read all instructions carefully.
- The perforated *Formula Sheet* and the *Terminology Sheet* can be removed from the test booklet. No marks will be given for work done on these pages.
- The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

Instructions

- There are 13 questions worth a total of 29 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- A calculator icon  appears next to the questions that require a calculator.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

Solve: Give a solution for a problem or determine the value(s) of a variable.

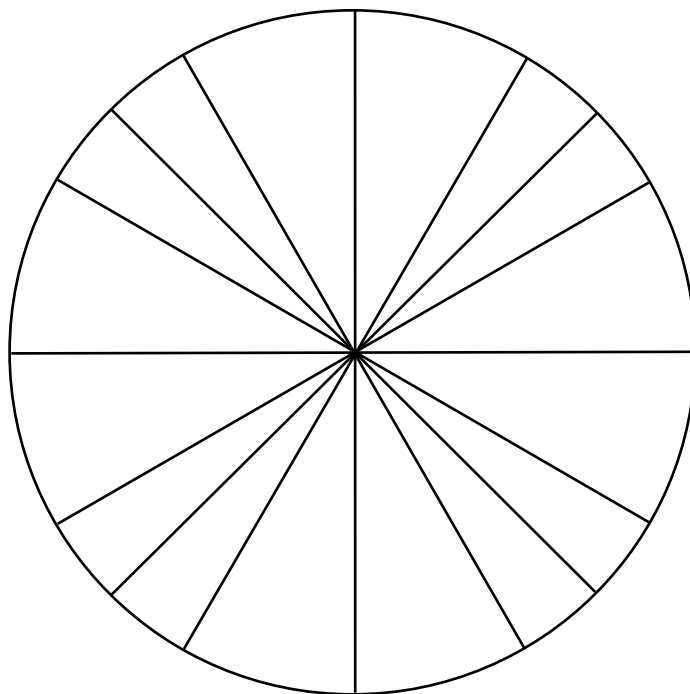
Verify: Establish the truth of a statement by substitution or comparison.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

State: Give an answer without an explanation or justification.

Describe: Use words to provide the process or to report details of the response.

Unit Circle (can be used if needed)



Question 1

2 marks

101

A pizza with a diameter of 15 inches is cut into equal slices, each with a central angle of 36° .
Determine the length of the crust on the outer edge of one slice of pizza.

Question 2

a) 1 mark b) 2 marks

102
103

There are 9 girls and 7 boys in a math class from which a committee of 5 is to be chosen.

- a) How many different committees of 5 can be formed if one of the boys, William, must be on the committee?

- b) How many different committees of 5 can be formed if there must be 2 girls and 3 boys on the committee?

Question 3

3 marks

104

Solve the following equation over the interval $[0, 2\pi]$:

$$\sin^2 \theta + 6 \sin \theta - 2 = 0$$

Question 4

4 marks

105

Solve:

$$6(5)^{3x+2} = 9^{2-x}$$

Note: A calculator is not required for the remaining test questions.

Question 5

4 marks

106

Solve $(2 \sin \theta - 1)(\sin \theta + 1) = 0$ where $\theta \in \mathbb{R}$.

Question 6

1 mark

107

The roots of the polynomial equation $3(x - 2)(x + 1)^2 = 0$ are $x = 2$ and $x = -1$.

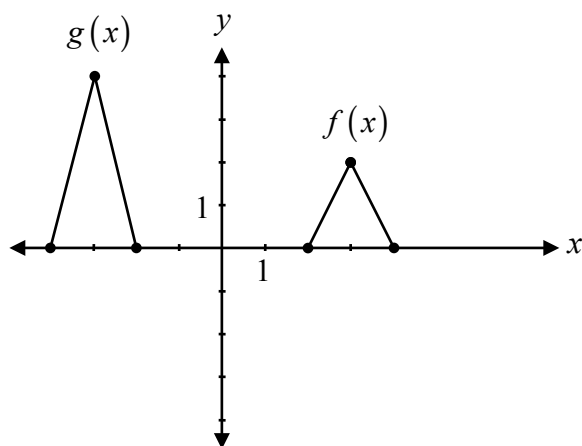
Explain what these roots represent on the graph of $p(x) = 3(x - 2)(x + 1)^2$.

Question 7

2 marks

108

Determine an equation for $g(x)$ as a transformation of $f(x)$.



$g(x) =$ _____

Question 8

1 mark

109

A student must determine the factors of $5x^4 - 2x^3 + 4x - 1$. He used 5, -2 , 4, and -1 as the coefficients of the polynomial when using synthetic division.

Explain the student's error.

Question 9

2 marks

110

Describe the transformations of $y = f(x)$ when asked to sketch the graph of $y = -f(x - 4)$.

Question 10

3 marks

111

Prove the identity below for all permissible values of θ :

$$\sin \theta + \frac{\cos \theta}{\tan \theta} = \frac{1}{\cos \theta \tan \theta}$$

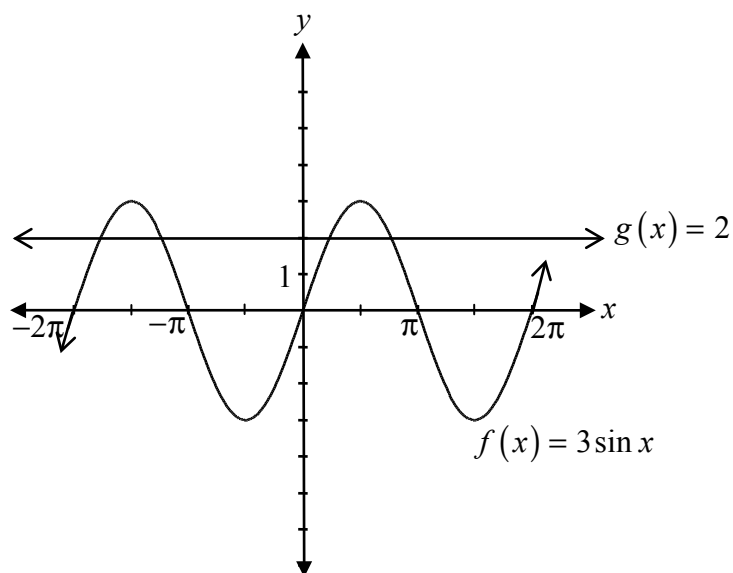
Left-Hand Side	Right-Hand Side

Question 11

1 mark

112

Describe how to use the graphs of $f(x) = 3 \sin x$ and $g(x) = 2$ to solve the equation $3 \sin x = 2$.



Question 12

1 mark

113

A hockey arena has 5 doors.

Determine the number of ways that you can enter through one door but exit through a different door.

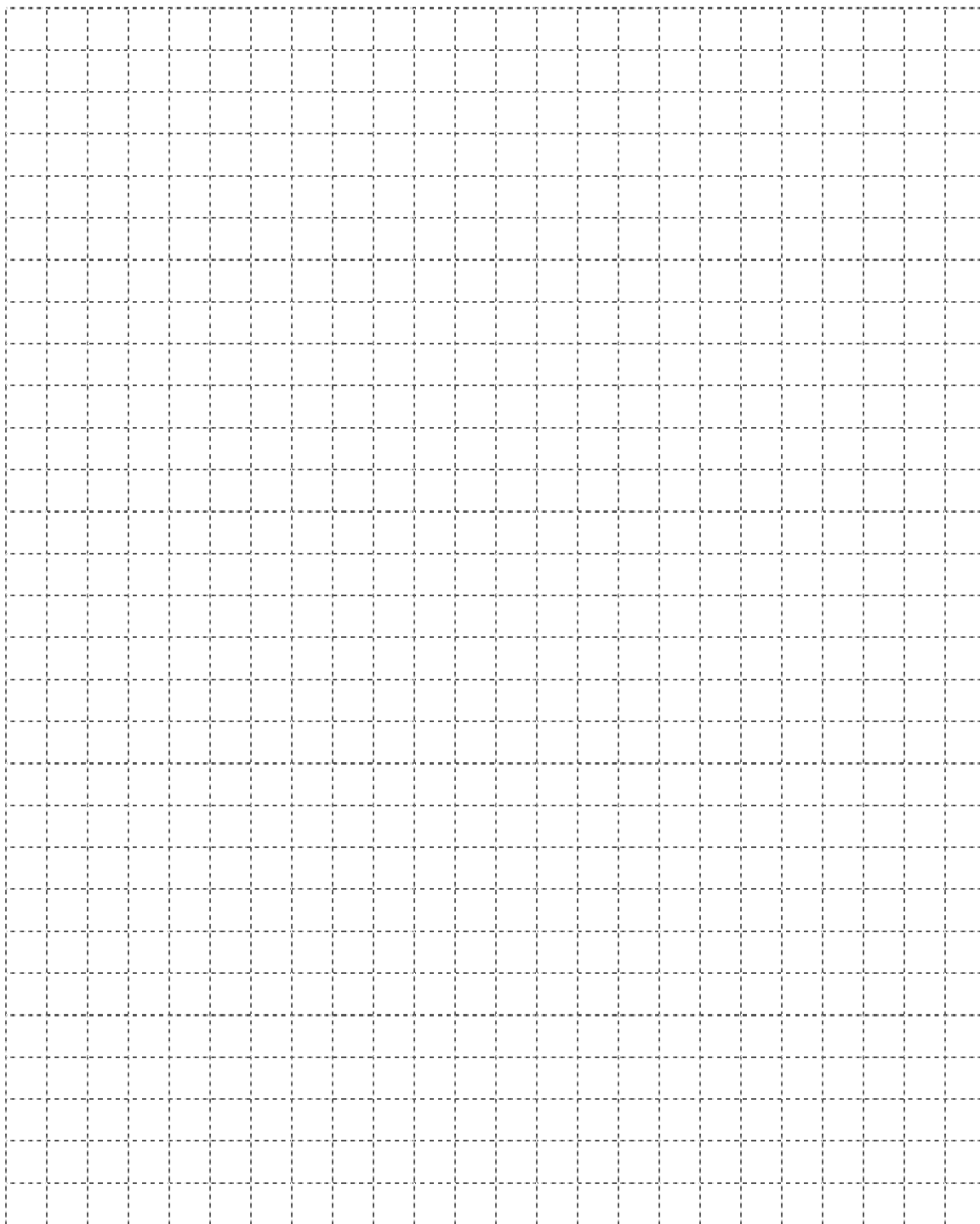
Question 13

2 marks

114

Given that $(x + 3)$ is one of the factors, express $2x^3 + 7x^2 + 2x - 3$ as a product of factors.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2016

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 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Precalculus—Study and teaching (Secondary)—Manitoba.
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Instructions

Selected Response Questions

- There are 8 questions worth a total of 8 marks.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- There are 24 questions worth a total of 55 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 14

1 mark

Identify the maximum number of x -intercepts for a polynomial function of degree 3.

- a) 1
- b) 2
- c) 3
- d) 4

Question 15

1 mark

The graph of $y = f(x)$ contains the point (a, b) . The graph of $g(x)$ is a transformation of the graph of $f(x)$ and contains the point $(3a, b)$.

Identify the function that represents $g(x)$.

- a) $g(x) = f(3x)$
- b) $g(x) = 3f(x)$
- c) $g(x) = f\left(\frac{x}{3}\right)$
- d) $g(x) = \frac{1}{3}f(x)$

Question 16

1 mark

The angle 2.95 radians, in standard position, terminates in quadrant:

- a) I
- b) II
- c) III
- d) IV

Question 17

1 mark

Evaluate:

$$2 \sin \frac{\pi}{8} \cos \frac{\pi}{8}$$

a) $\frac{1}{2}$

b) $\frac{\sqrt{2}}{2}$

c) 1

d) $\sqrt{2}$

Question 18

1 mark

Identify which of the following represents the 5th term in the expansion of $(4x^2 - 2y^3)^{15}$.

a) ${}_{15}C_5 (4x^2)^{10} (-2y^3)^5$

b) ${}_{15}C_5 (4x^2)^{11} (-2y^3)^4$

c) ${}_{15}C_4 (4x^2)^{10} (-2y^3)^5$

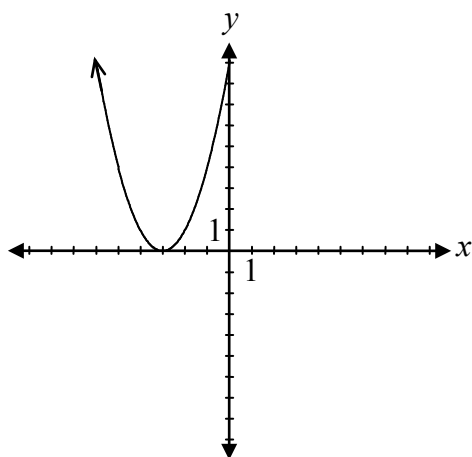
d) ${}_{15}C_4 (4x^2)^{11} (-2y^3)^4$

Question 19

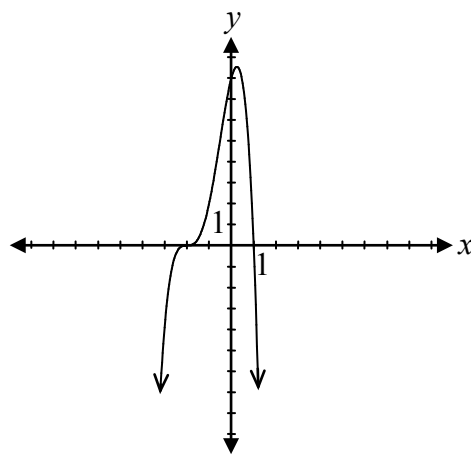
1 mark

Identify which of the following graphs of polynomial functions has a zero with a multiplicity of 3.

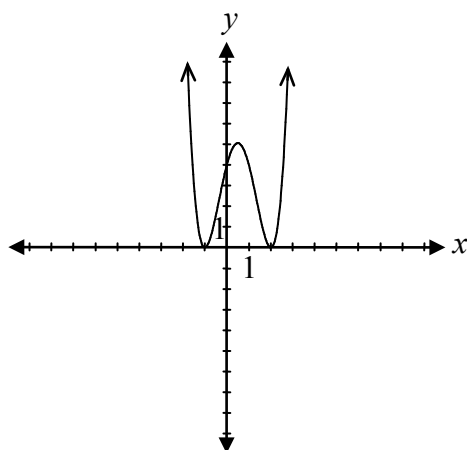
a)



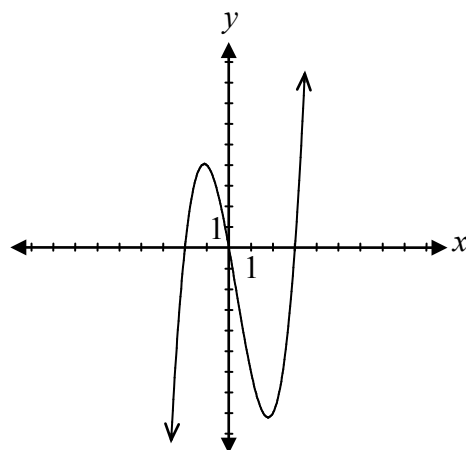
b)



c)



d)



Question 20

1 mark

A non-permissible value of x for the function $f(x) = \frac{1}{\cos x + 1}$ is:

- a) -1
- b) 0
- c) π
- d) $\frac{3\pi}{2}$

Question 21

1 mark

Identify which of the following statements is true for the rational function $f(x) = \frac{4(x-1)(x-2)}{(x-1)(x+3)}$.

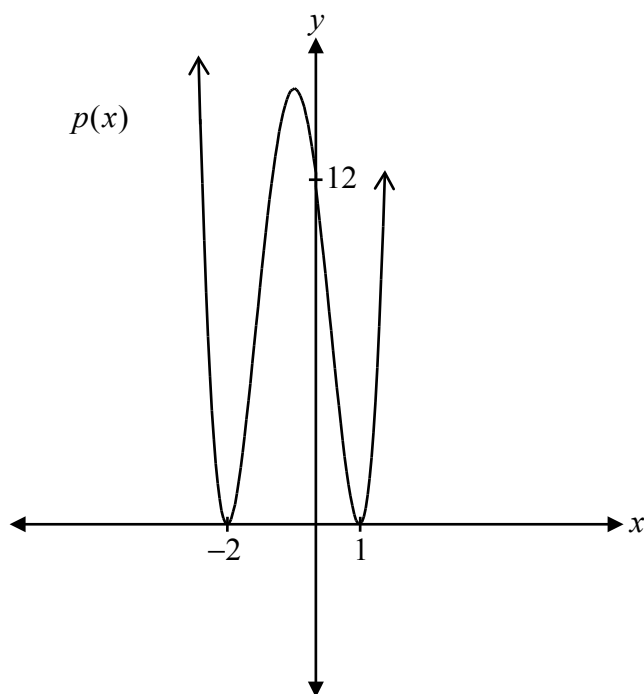
- a) The equation of the horizontal asymptote is $y = 4$.
- b) The equation of the vertical asymptote is $x = 1$.
- c) The y -intercept is 0 .
- d) There is a point of discontinuity (hole) when $x = 2$.

Question 22

3 marks

115

Determine the equation of the polynomial function, $p(x)$, represented by the graph.



$p(x) =$ _____

Question 23

1 mark

116

Evaluate:

$$\log_4 2$$

Question 24

2 marks

117

Evaluate:

$$\left(\cos \frac{11\pi}{3}\right)\left(\csc \frac{11\pi}{6}\right)$$

Question 25

1 mark

118

Estimate the value of $\log_2 5$.

Justify your answer.

Question 26

2 marks

119

If θ terminates in quadrant III and $\cos \theta = -\frac{6}{7}$, determine the exact value of $\tan \theta$.

Question 27

1 mark

120

Given $f(x) = x^2 + x - 4$ and $g(x) = \sqrt{x+5}$, Taz was asked to find $f(g(x))$.

Taz's solution:

$$\begin{aligned} f(g(x)) &= (\sqrt{x+5})^2 + x - 4 \\ &= x + 5 + x - 4 \\ &= 2x + 1, \quad x \geq -5 \end{aligned}$$

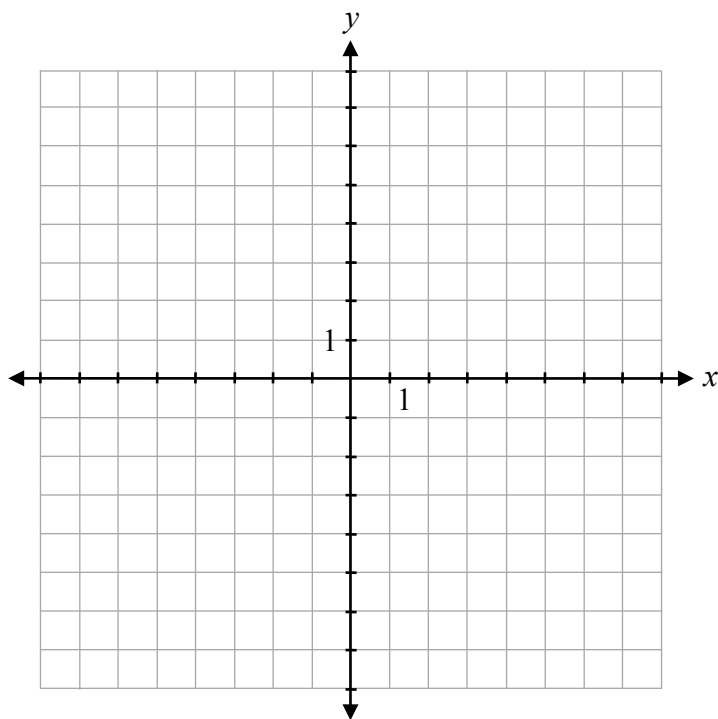
Describe the error in Taz's solution.

Question 28

3 marks

121

Sketch the graph of the function $f(x) = 3 \log_2(x+1)$.



Question 29

1 mark

122

Write an equation of a rational function that would not have any vertical asymptotes.

Question 30

2 marks

123

Determine the exact value of $\tan 75^\circ$.

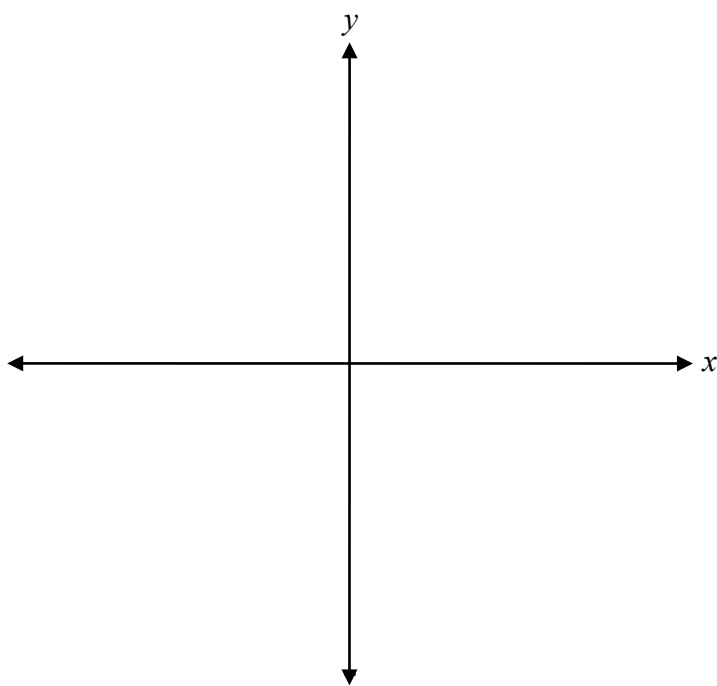
Question 31

4 marks

124

Sketch the graph of the following function:

$$f(x) = \frac{(x+3)(x-3)}{x(x-3)}$$



In the binomial expansion of $\left(\frac{1}{x^3} - 2x^2\right)^9$, determine which term contains x^3 .

Question 33

4 marks

126

José and Dana get on a Ferris wheel, which is 1 metre off the ground. The diameter of the Ferris wheel is 16 metres. Their ride lasts for 4 minutes, in which time the Ferris wheel makes one revolution.

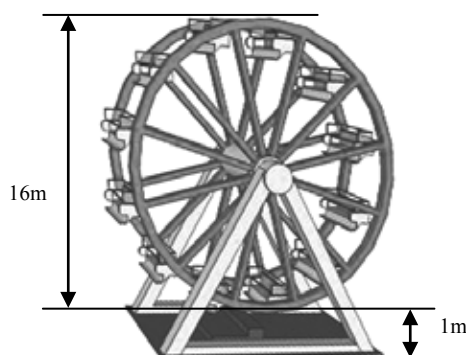
Determine the values of A , B , C , and D , if the sinusoidal function that models the situation is $h(t) = A \cos[B(t - C)] + D$, where h is the height at which José and Dana are located on the Ferris wheel, from the ground, in metres, and t is the time, in minutes.

$A =$ _____

$B =$ _____

$C =$ _____

$D =$ _____



Question 34

3 marks

127

Solve algebraically:

$${}_nP_3 = 4!(n-1)$$

Question 35

2 marks

128

Given $f(x) = \frac{2}{x-1}$, determine the equation of the inverse, $f^{-1}(x)$.

Question 36

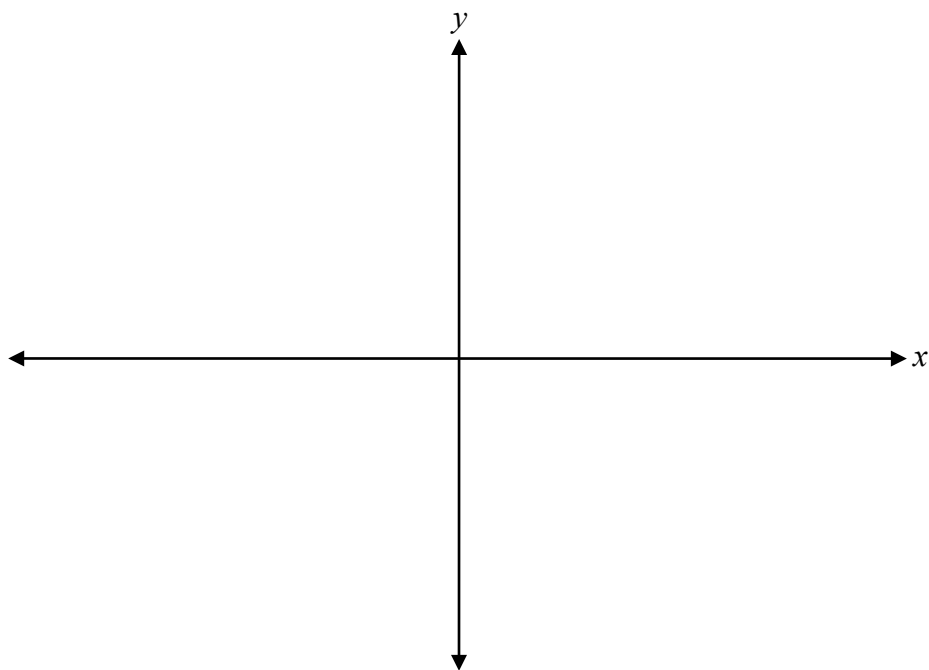
3 marks

129

Solve:

$$4\log_3 2 - \frac{1}{3}\log_3 8 = \log_3 a$$

Sketch the graph of at least one period of the function $y = 3 \cos(\pi x) - 1$.



Question 38

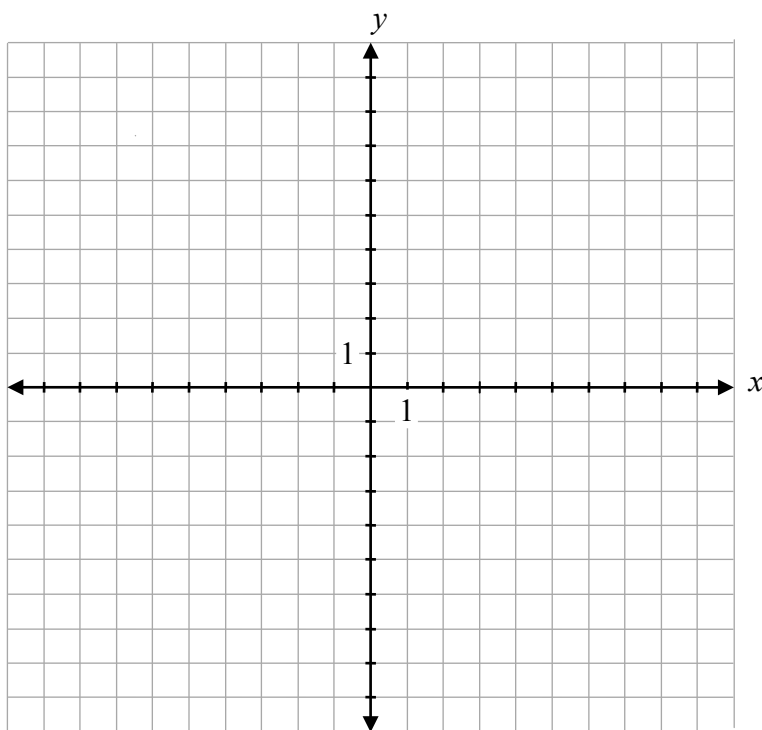
3 marks

131

Using the laws of logarithms, fully expand the expression:

$$\log_a \left(\frac{x^3}{y\sqrt{z}} \right)$$

Sketch the graph of $f(x) = 3\sqrt{x-2} + 1$.



Question 40

a) 1 mark b) 1 mark

133
134

a) Determine the domain of the graph of the function $f(x) = \sqrt{x^2 - 4}$.

b) Explain why the domain of $f(x) = \sqrt{x^2 - 4}$ is restricted.

Question 41

a) 1 mark b) 1 mark

135
136

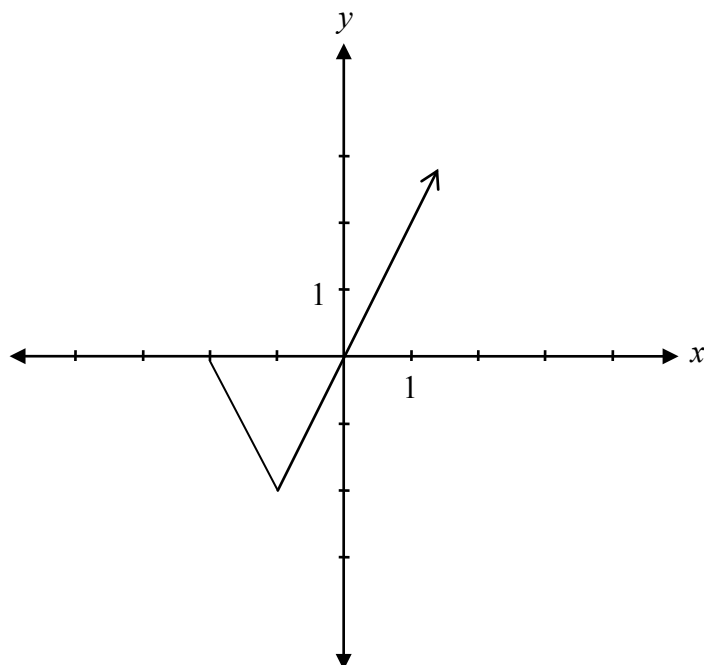
Given the point $(-12, -18)$ on the graph of $f(x)$, determine the new points after the following transformations of $f(x)$.

a) $\frac{1}{f(x)}$

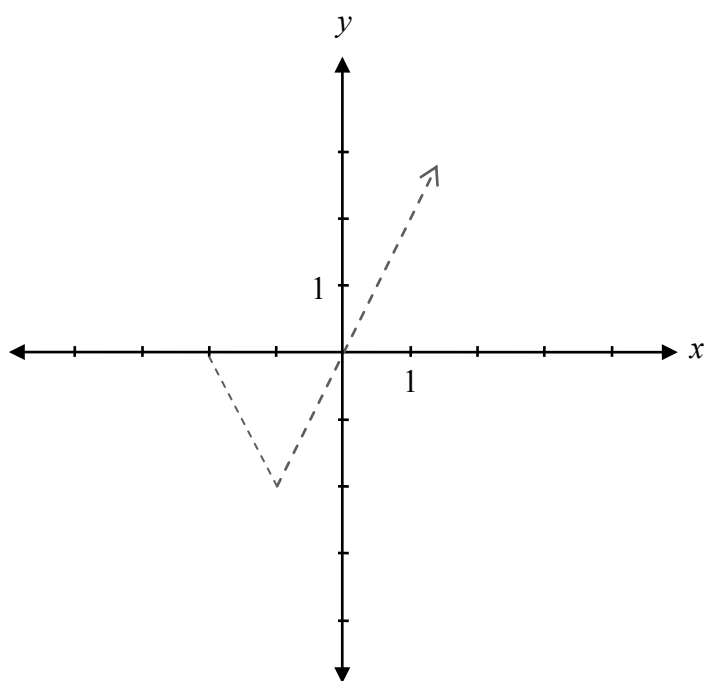
b) $f(-x) + 10$

Explain why there is no solution for the equation $\csc \theta = -\frac{1}{2}$.

Given the graph of $y = f(x)$,



sketch the graph of $y = |f(2x)| + 1$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 44

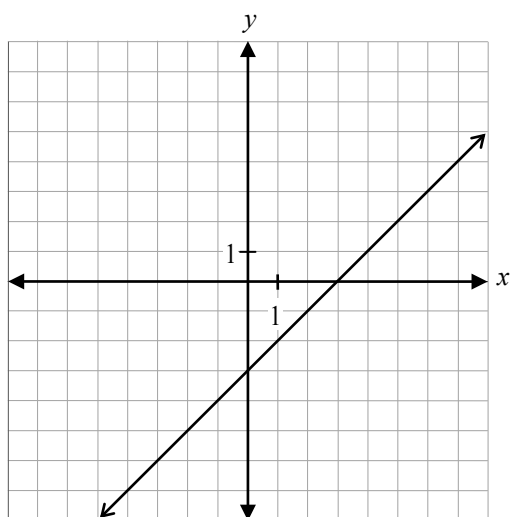
1 mark

139

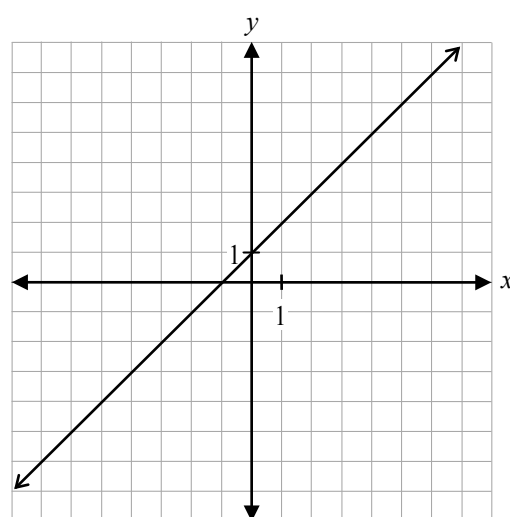
Given $f(x) = 2^x + 1$, state the equation of the horizontal asymptote.

Given the following graphs of $f(x) = x - 3$ and $g(x) = x + 1$,

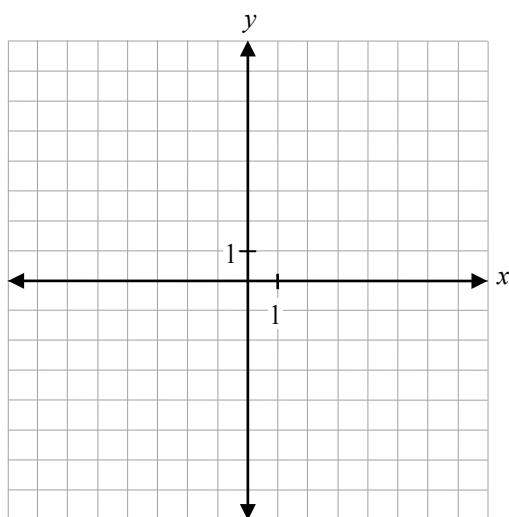
$f(x)$



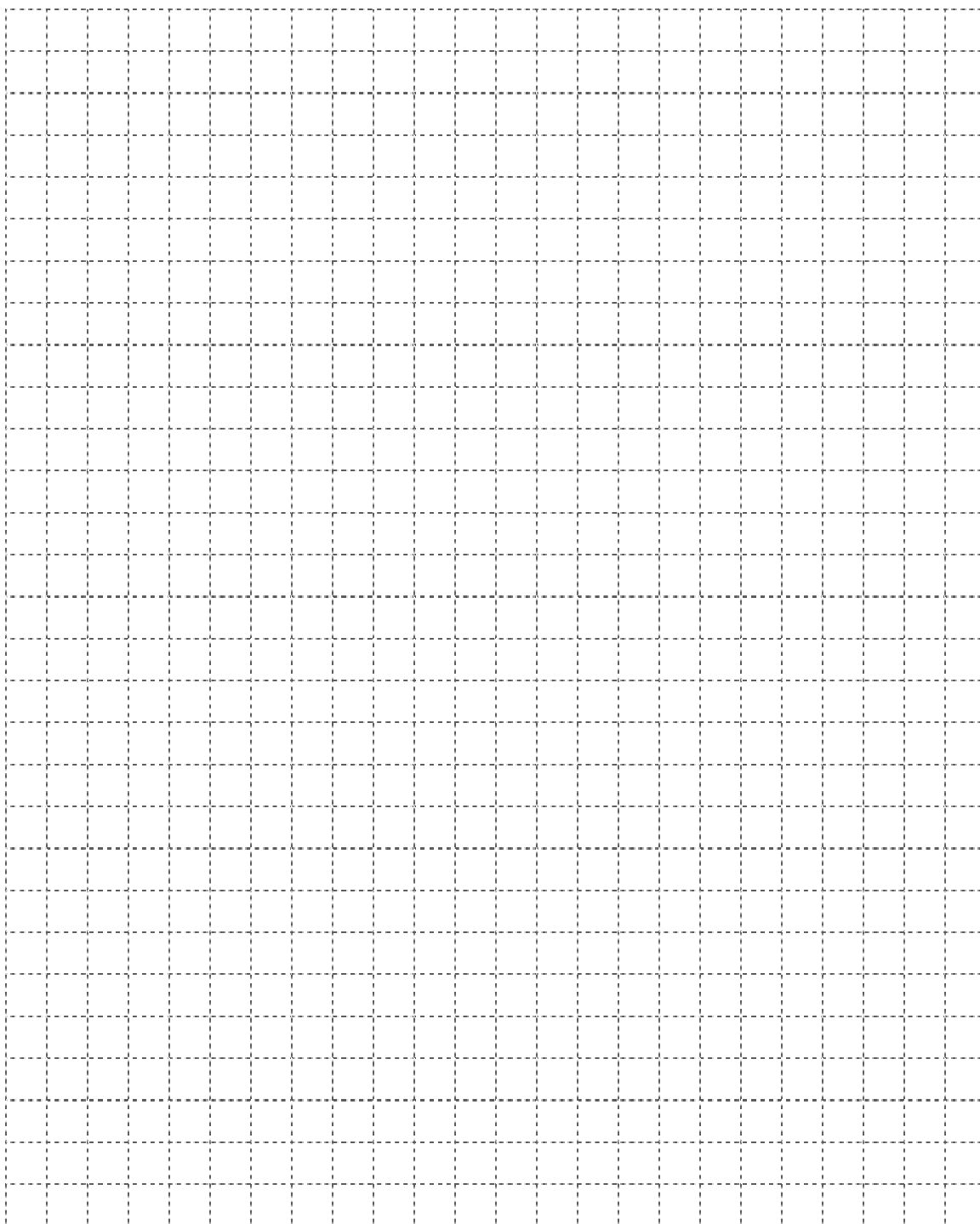
$g(x)$



sketch the graph of $h(x) = (f \cdot g)(x)$.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2016

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Booklet 1. June 2016

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1. Mathematics—Examinations, questions, etc.
 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Precalculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
 - I. Manitoba. Manitoba Education and Advanced Learning.
- 510.76

Manitoba Education and Advanced Learning
School Programs Division
Winnipeg, Manitoba, Canada

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	–	14	34
Booklet 2	8	20	56
Total	8	34	90

* The first 5 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- Read all instructions carefully.
- The perforated pages can be removed from the test booklet. No marks will be given for work done on these pages.
- The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

Instructions

- There are 14 questions worth a total of 34 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- A calculator icon  appears next to the questions that require a calculator.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a (MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a (M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

Solve: Give a solution for a problem or determine the value(s) of a variable.

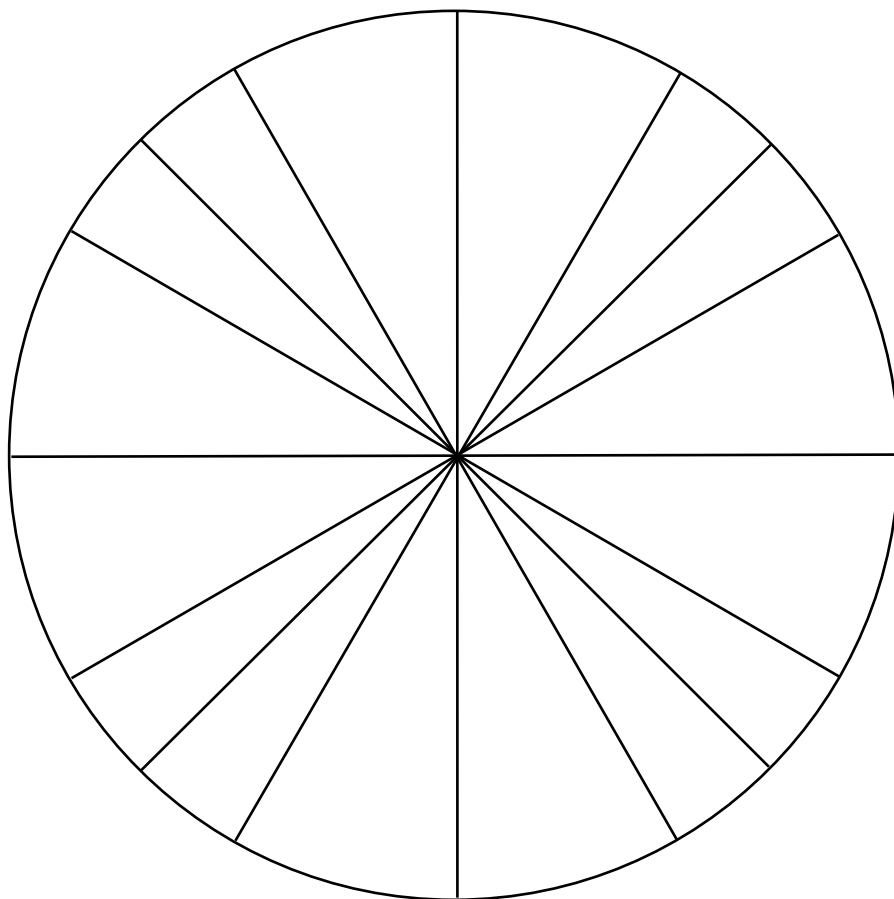
Verify: Establish the truth of a statement by substitution or comparison.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

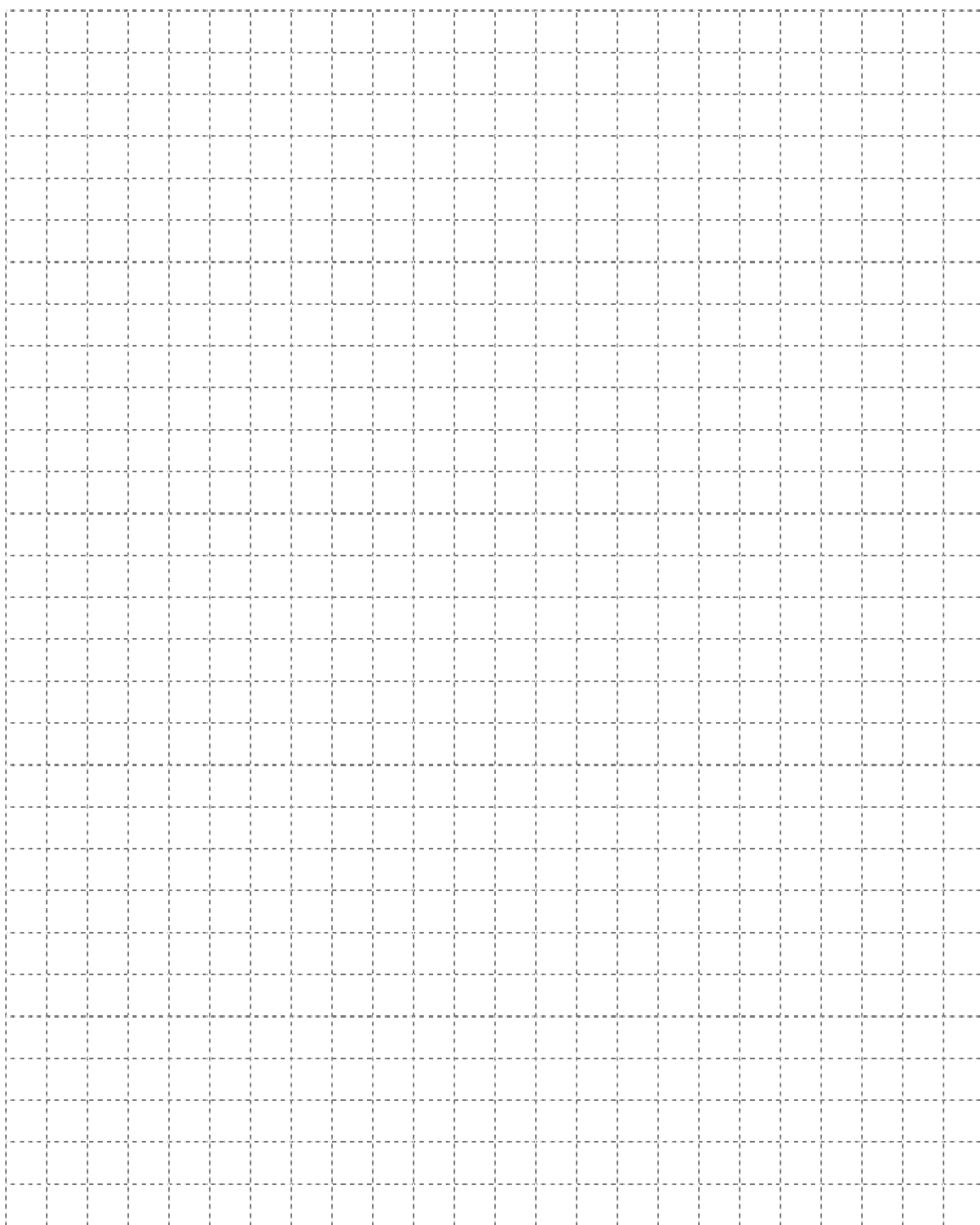
State: Give an answer without an explanation or justification.

Describe: Use words to provide the process or to report details of the response.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



Question 1

2 marks

101

A wheel has a diameter of 20 cm and rotates through a central angle of 252° .
Determine how far the wheel rolled.

Question 2

3 marks

102

Solve the following equation over the interval $[0, 2\pi]$:

$$3 \sin^2 \theta - 10 \sin \theta - 8 = 0$$

Question 3

3 marks

103

Determine and simplify the fourth term in the expansion of $(2x^4 - 3y)^8$.

Question 4

3 marks

104

Sheeva's bank is lending her \$50 000 at an annual interest rate of 6%, compounded monthly, to purchase a car.

Given that the last payment will be a partial payment, determine how many full monthly payments of \$800 Sheeva will have to make.

The formula below may be used.

$$PV = \frac{R \left[1 - (1 + i)^{-n} \right]}{i}$$

where PV = the present value of the amount borrowed

R = the amount of each periodic payment

$i = \frac{\text{annual interest rate (as a decimal)}}{\text{the number of compounding periods per year}}$

n = the number of equal periodic payments

Express your answer as a whole number.

Question 5

2 marks

105

An employee asked 10 people in an ice cream shop to wait in line.

Determine the number of different arrangements possible if two of the people, Jamie and John, refused to stand next to each other in the line.

Note: A calculator is not required for the remaining test questions.

Question 6

1 mark

106

The point $(-2, 4)$ is on the graph of $f(x)$.

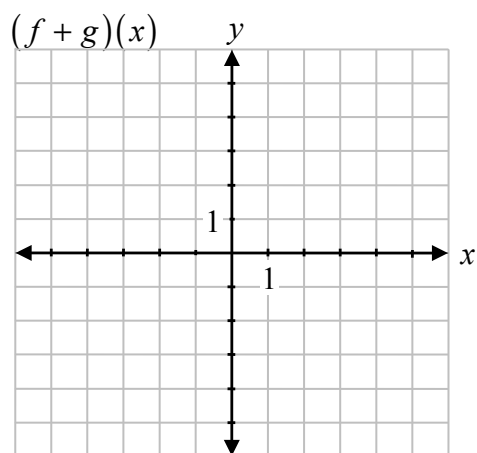
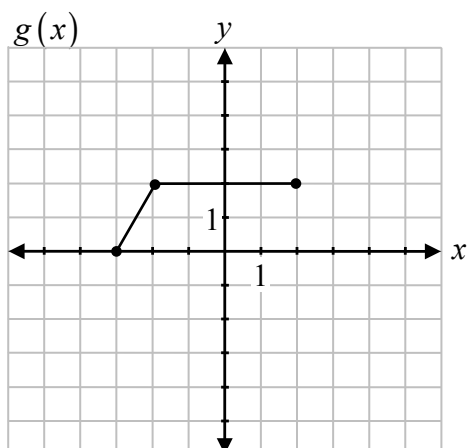
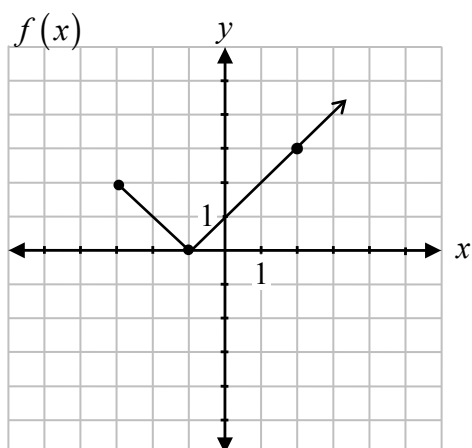
State the coordinates of the corresponding point when $f(x)$ is reflected over the y -axis.

Question 7

2 marks

107

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f + g)(x)$.



Question 8

3 marks

108

Using the laws of logarithms, fully expand the expression:

$$\log_2 \left(\frac{w^3 x}{y-1} \right)$$

Question 9

4 marks

109

Solve the following equation algebraically for θ , where $0 \leq \theta \leq 2\pi$:

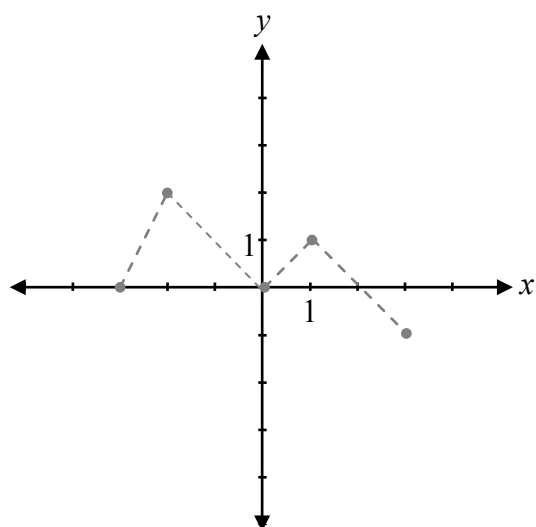
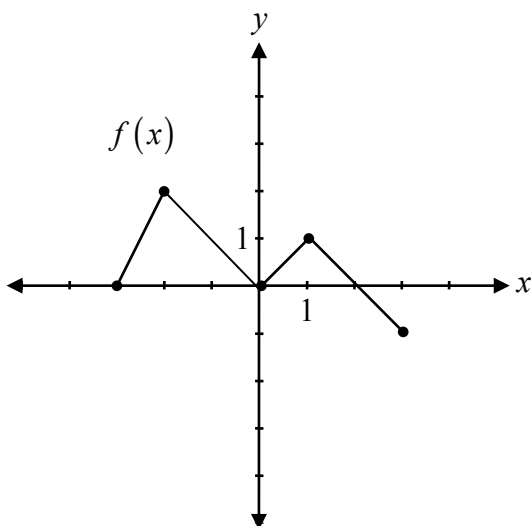
$$2 \cos 2\theta = 1$$

Question 10

3 marks

110

Given the graph of $y = f(x)$, sketch the graph of $y = 2|f(x-1)|$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 11

3 marks

111

Prove the identity for all permissible values of θ :

$$\cos \theta + \tan \theta \sin \theta = \frac{\tan \theta \sin \theta}{1 - \cos^2 \theta}$$

Left-Hand Side	Right-Hand Side

Question 12

1 mark

112

Raoul has 8 shirts, 5 pairs of pants, and 3 hats. He adds the options together and determines that he has 16 different outfits to wear.

Raoul made an error in calculating the number of different outfits. Describe how to determine the correct number of outfits.

Question 13

a) 1 mark b) 1 mark

113
114

Given $f(x) = 2x - 1$ and $g(x) = x^2 + 1$:

a) Determine $f(x) \cdot g(x)$.

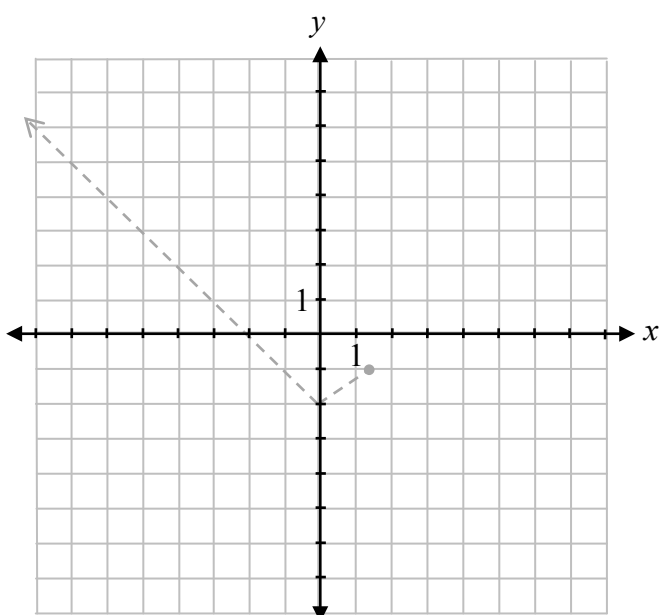
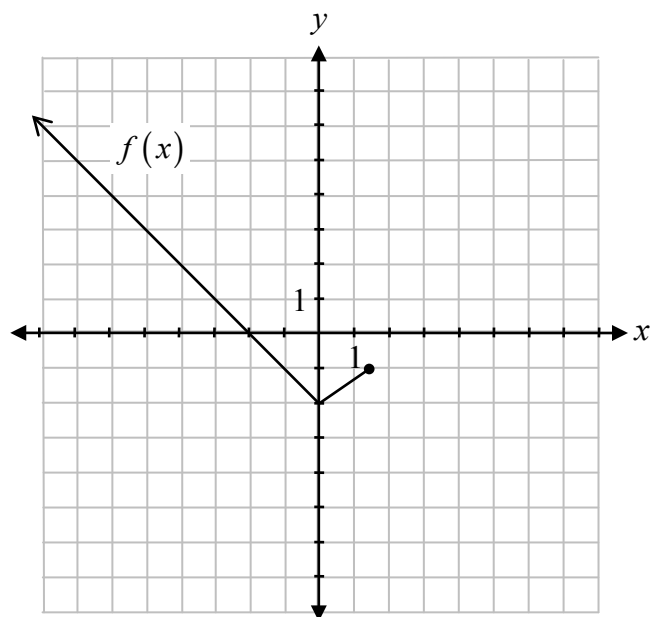
b) Determine $g(g(x))$.

Question 14

2 marks

115

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2016

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Booklet 2. June 2016

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Instructions

Selected Response Questions

- There are 8 questions worth a total of 8 marks.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- There are 20 questions worth a total of 48 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 15

1 mark

Given the polynomial function $P(x) = x^4 - 5x^2 - 2x + 6$, if $P(1) = 0$, identify which statement is true.

- a) The y -intercept is 1.
- b) $P(x)$ has a factor of $(x + 1)$.
- c) The graph has a zero at 1.
- d) The graph has a zero at -1 .

Question 16

1 mark

There are 6 different books that are being distributed evenly amongst three people. Identify which expression represents the number of possible combinations.

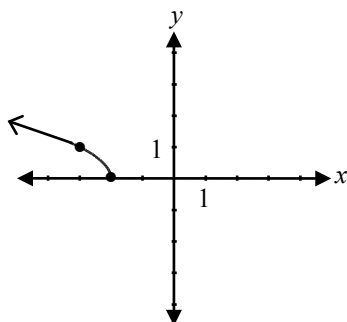
- a) ${}_6C_2 \cdot {}_6C_2 \cdot {}_6C_2$
- b) ${}_6C_2 \cdot {}_4C_2 \cdot {}_2C_2$
- c) ${}_2C_2 \cdot {}_2C_2 \cdot {}_2C_2$
- d) $3 \cdot {}_6C_2$

Question 17

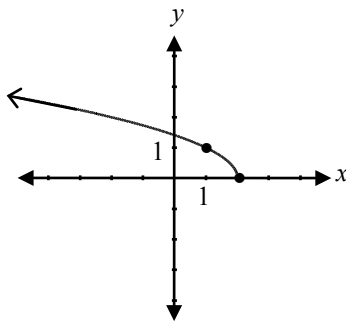
1 mark

Identify the graph that corresponds to the function $f(x) = -\sqrt{(x-2)}$.

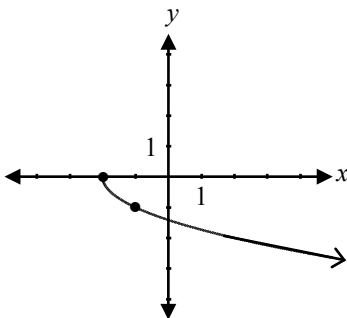
a)



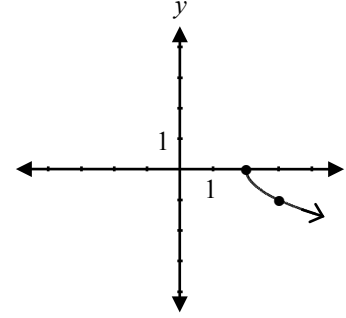
b)



c)



d)



Question 18

1 mark

Solve:

$$7^{\log_7 2} = x$$

a) $x = 1$

b) $x = 2$

c) $x = 7$

d) $x = 49$

Question 19

1 mark

Identify the equation that has a general solution of $\left. \begin{array}{l} \theta = \frac{\pi}{6} + 2\pi k \\ \theta = \frac{5\pi}{6} + 2\pi k \end{array} \right\}$ where $k \in \mathbb{Z}$.

a) $\sin \theta = \frac{1}{2}$

b) $\cos \theta = \frac{1}{2}$

c) $\sin \theta = \frac{\sqrt{3}}{2}$

d) $\cos \theta = \frac{\sqrt{3}}{2}$

Question 20

1 mark

Identify the function that has a domain of $x \leq -2$ and a range of $y \geq 3$.

a) $y = \sqrt{x+2} + 3$

b) $y = \sqrt{-(x+2)} + 3$

c) $y = -\sqrt{x-2} - 3$

d) $y = -\sqrt{-(x-2)} - 3$

Question 21

1 mark

Given $f(x) = 3x + 2$, identify $f^{-1}(x)$.

a) $f^{-1}(x) = -3x - 2$

b) $f^{-1}(x) = 2x + 3$

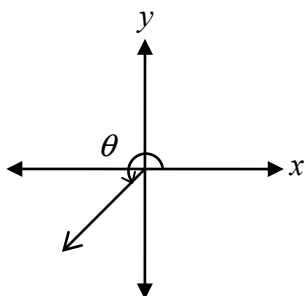
c) $f^{-1}(x) = \frac{x}{3} - 2$

d) $f^{-1}(x) = \frac{x-2}{3}$

Question 22

1 mark

Identify a possible value for the angle θ sketched in standard position.



a) 2

b) 3

c) 4

d) 5

Question 23

3 marks

116

Solve the following equation:

$$\log_3(x+3) + \log_3(x-5) = 2$$

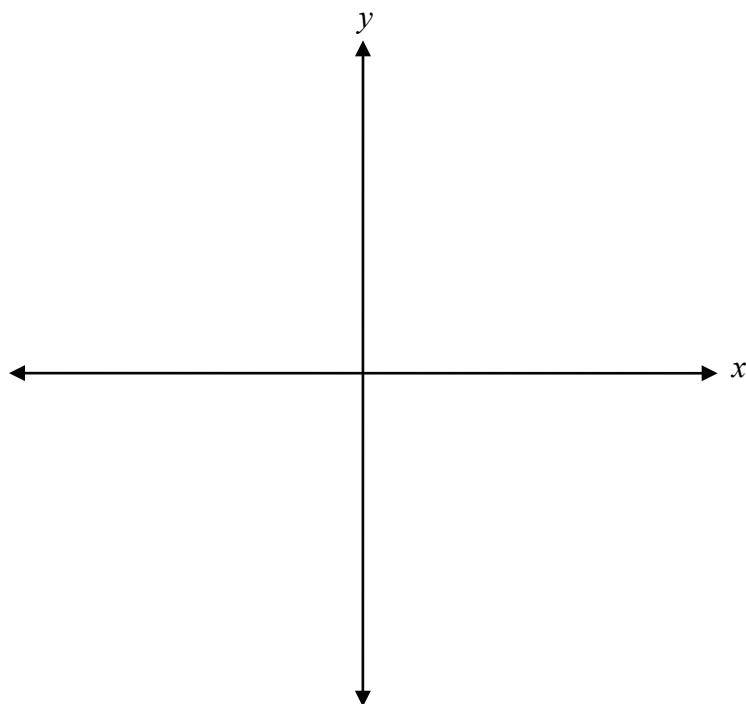
Question 24

1 mark

117

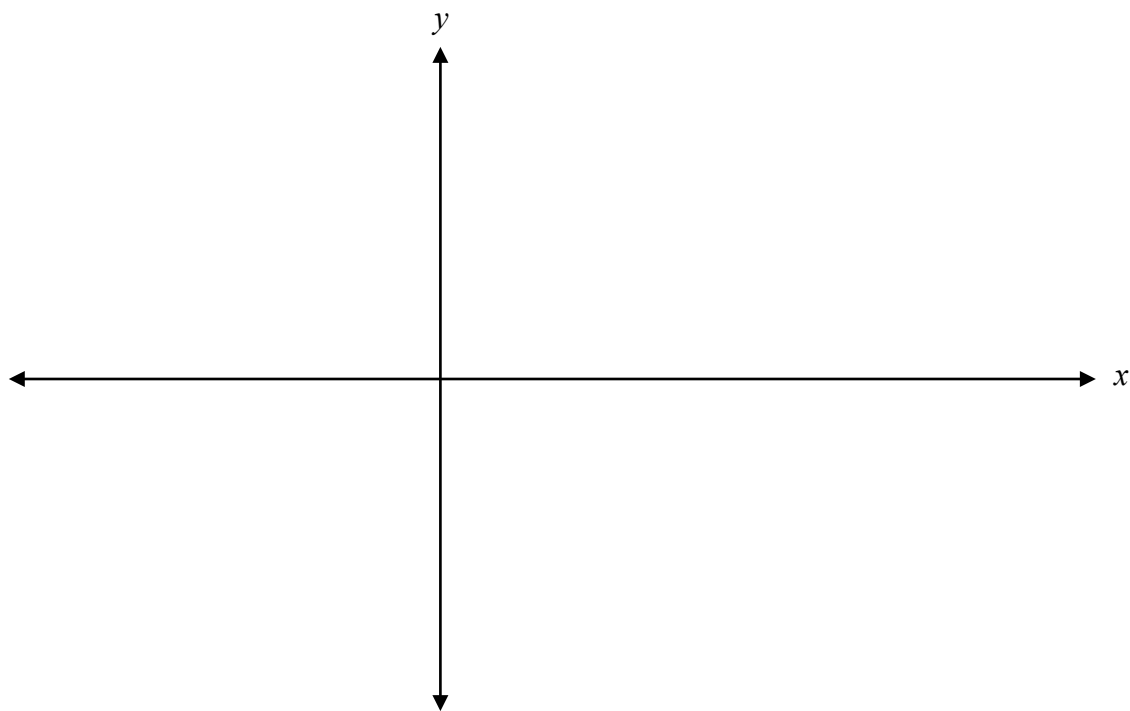
State a coterminal angle for $\theta = \frac{9\pi}{4}$.

Sketch the graph of the function $f(x) = \frac{2x+2}{x^2-1}$.



Justify why the binomial expansion of $(x + x^3)^7$ does not have a term containing x^{10} .

Sketch the graph of $y = -\sin\left(\frac{\pi}{2}(x-1)\right) + 3$ over the domain $[0, 6]$.



Question 28

2 marks

121

When $P(x) = 3x^4 - kx^3 + 5x - 14$ is divided by $(x + 2)$, the remainder is -8 .

Determine the value of k .

Question 29

a) 3 marks b) 1 mark

122
123

Given that $\cos \alpha = \frac{7}{12}$ where α is in quadrant IV, and $\sin \beta = \frac{3}{5}$ where β is in quadrant I, determine the exact value of:

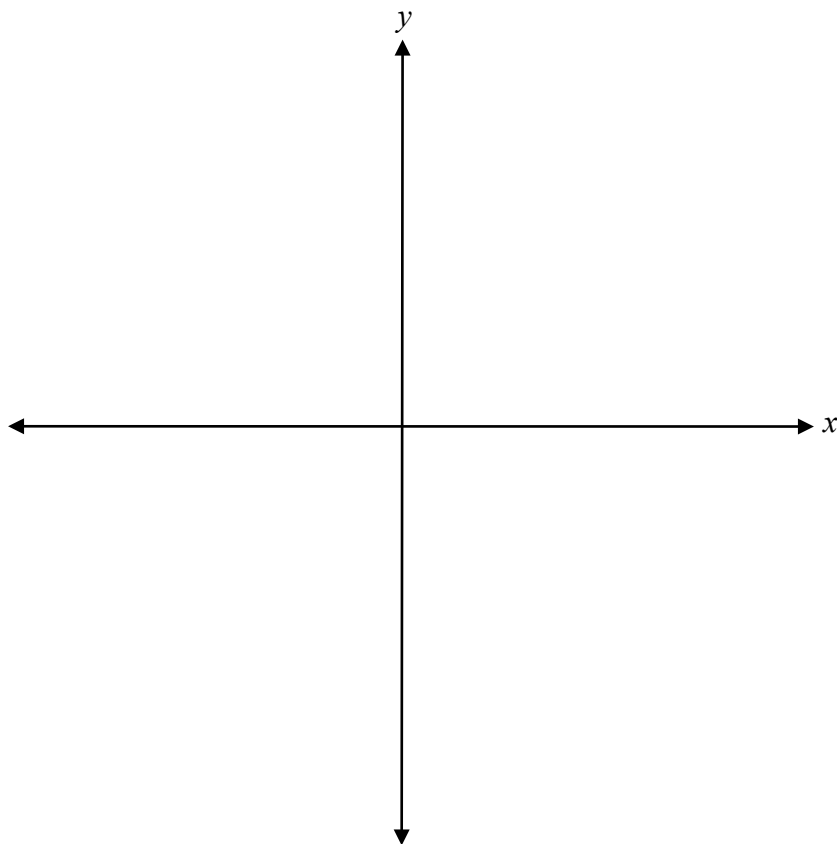
a) $\sin(\alpha - \beta)$

b) $\csc(\alpha - \beta)$

Describe the difference between the graph of $f(x) = \frac{7(x+2)}{x+2}$ and the graph of

$$g(x) = \frac{7(x-2)}{x+2} \text{ at } x = -2.$$

Sketch the graph of $f(x) = 3^x + 2$.



Question 32

3 marks

126

Solve algebraically:

$${}_nC_3 = n - 2$$

Describe the error that was made when solving the following equation:

$$\sin^2 \theta + \sin \theta - 2 = 1$$

$$\sin^2 \theta + \sin \theta = 3$$

$$\sin \theta (\sin \theta + 1) = 3$$

$$\sin \theta = 3 \quad \sin \theta + 1 = 3$$

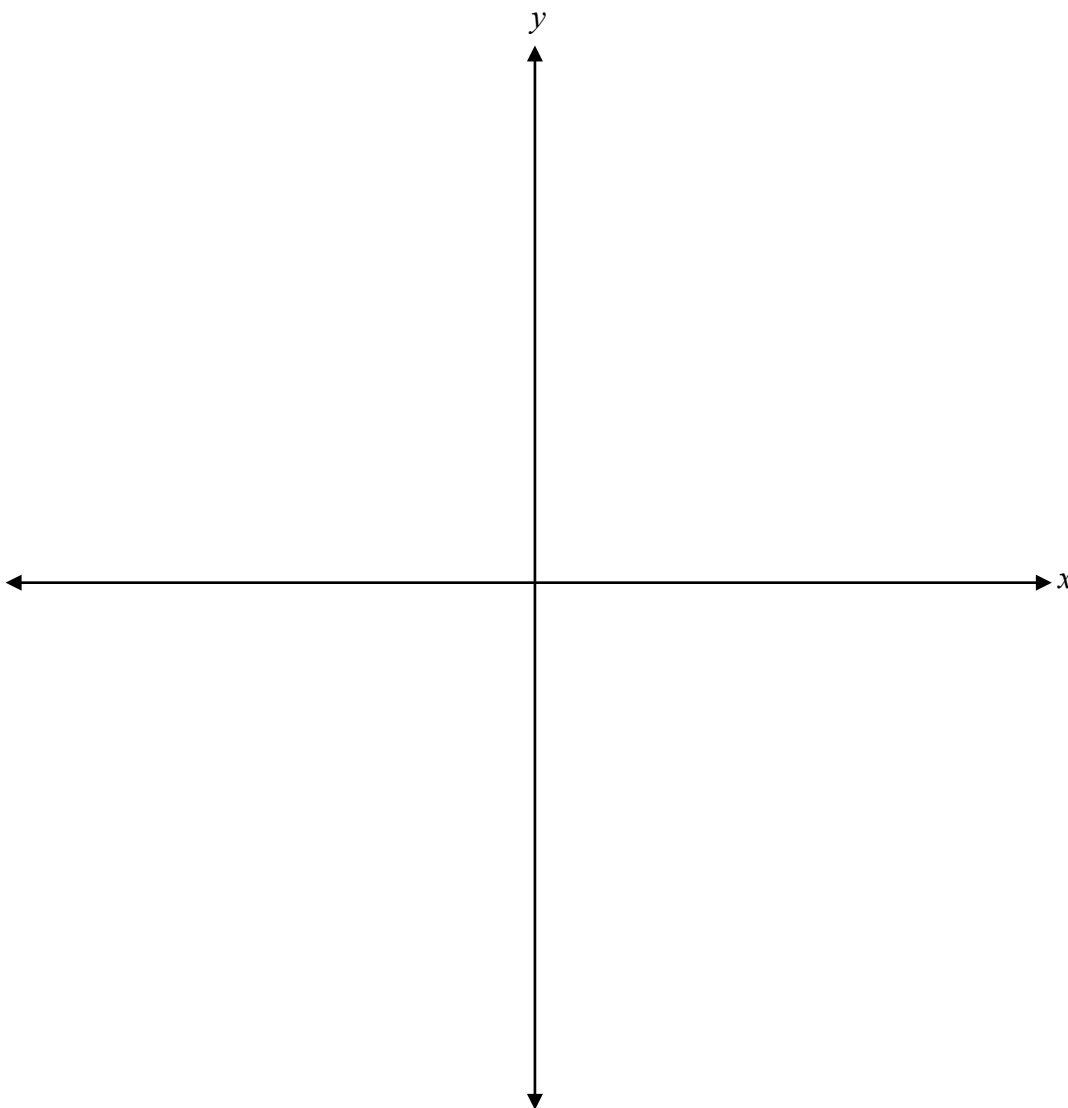
$$\sin \theta = 2$$

$\therefore$ No solution

$\therefore$ No solution

Sketch the graph of the polynomial function with the following characteristics.

- a y -intercept of -9
- zeroes at -1 and 3
- the zero at -1 has a multiplicity of 1 and the zero at 3 has a multiplicity of 2



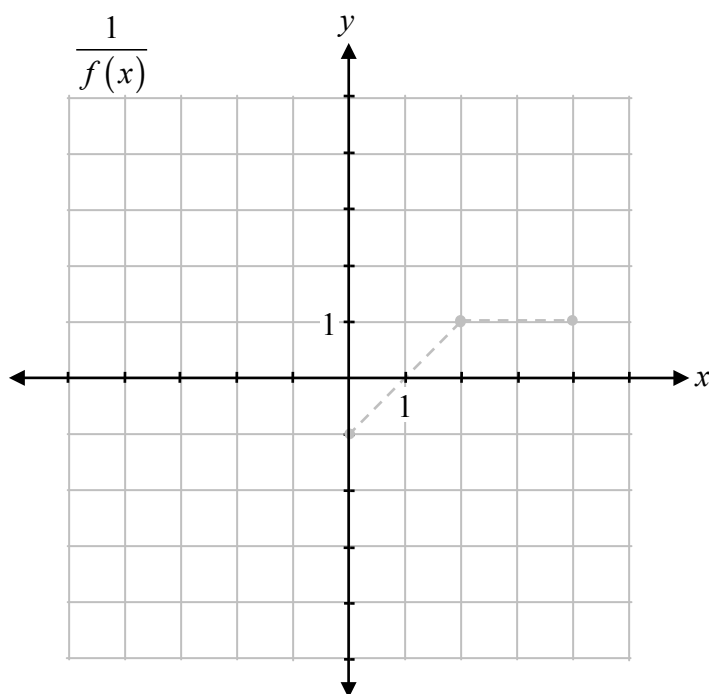
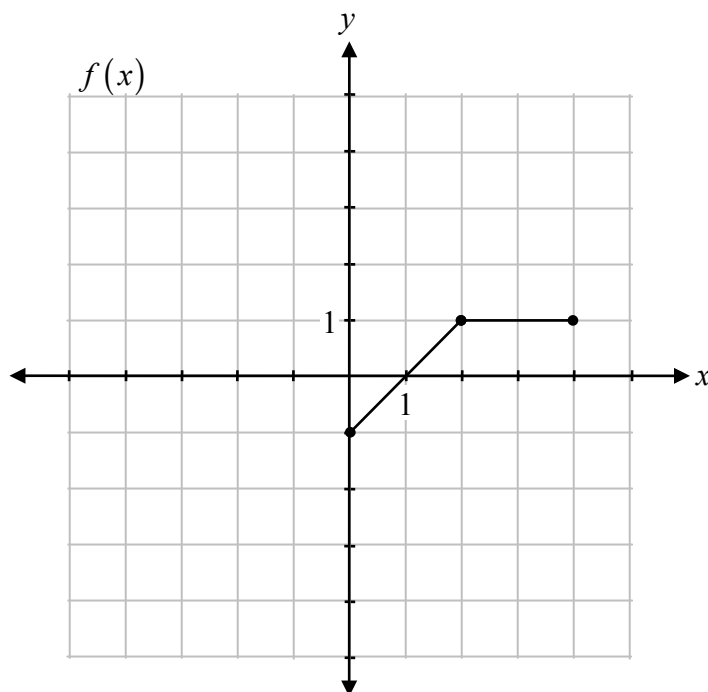
Question 35

2 marks

129

Given $\cot \theta = -\frac{1}{3}$, where θ is in quadrant II, determine the exact value of $\sin \theta$.

Given the function $f(x)$, sketch the graph of the reciprocal, $\frac{1}{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

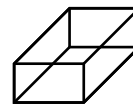
No marks will be awarded for the graph of $f(x)$.

Question 37

3 marks

131

The volume of a planter, in the shape of a rectangular prism, can be modelled by the polynomial function $V(x) = x^3 + 3x^2 - 34x + 48$.



Determine the factors of the function, $V(x)$, which represent possible dimensions of this planter.

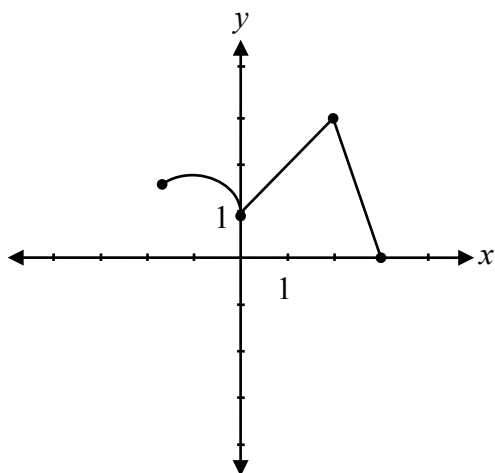
$V(x) =$ _____

Question 38

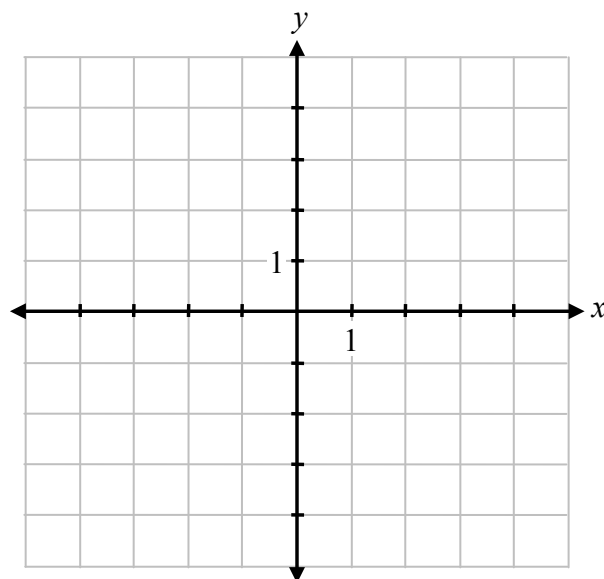
1 mark

132

Describe how to determine the range of the inverse of the following graph.



Sketch the graph of the function $y = \sqrt{2x} + 1$.



Question 40

a) 2 marks b) 1 mark

134
135

Given the following characteristics of a sinusoidal function:

- an amplitude of 2
- a vertical translation down 3 units
- a period of $\frac{\pi}{4}$

a) Determine an equation of this sinusoidal function in the form $y = a \sin b(x - c) + d$.

b) Determine the range of this function.

Range:

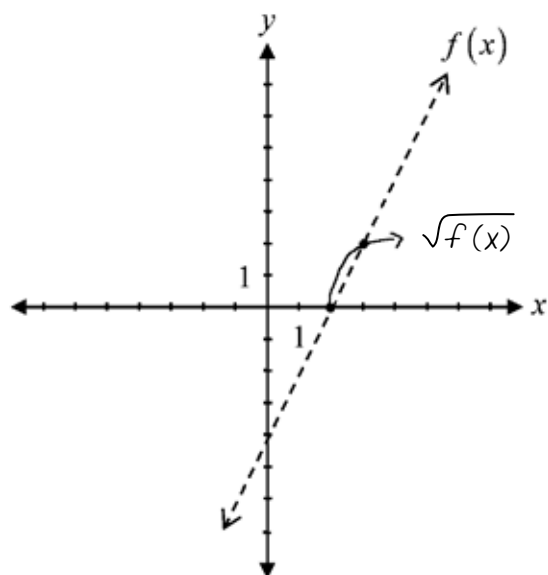
Question 41

1 mark

136

Suah was given the graph of $f(x)$ and asked to graph $y = \sqrt{f(x)}$.

Her solution is given on the graph below.



Describe the error Suah made when sketching the graph of $y = \sqrt{f(x)}$.

Question 42

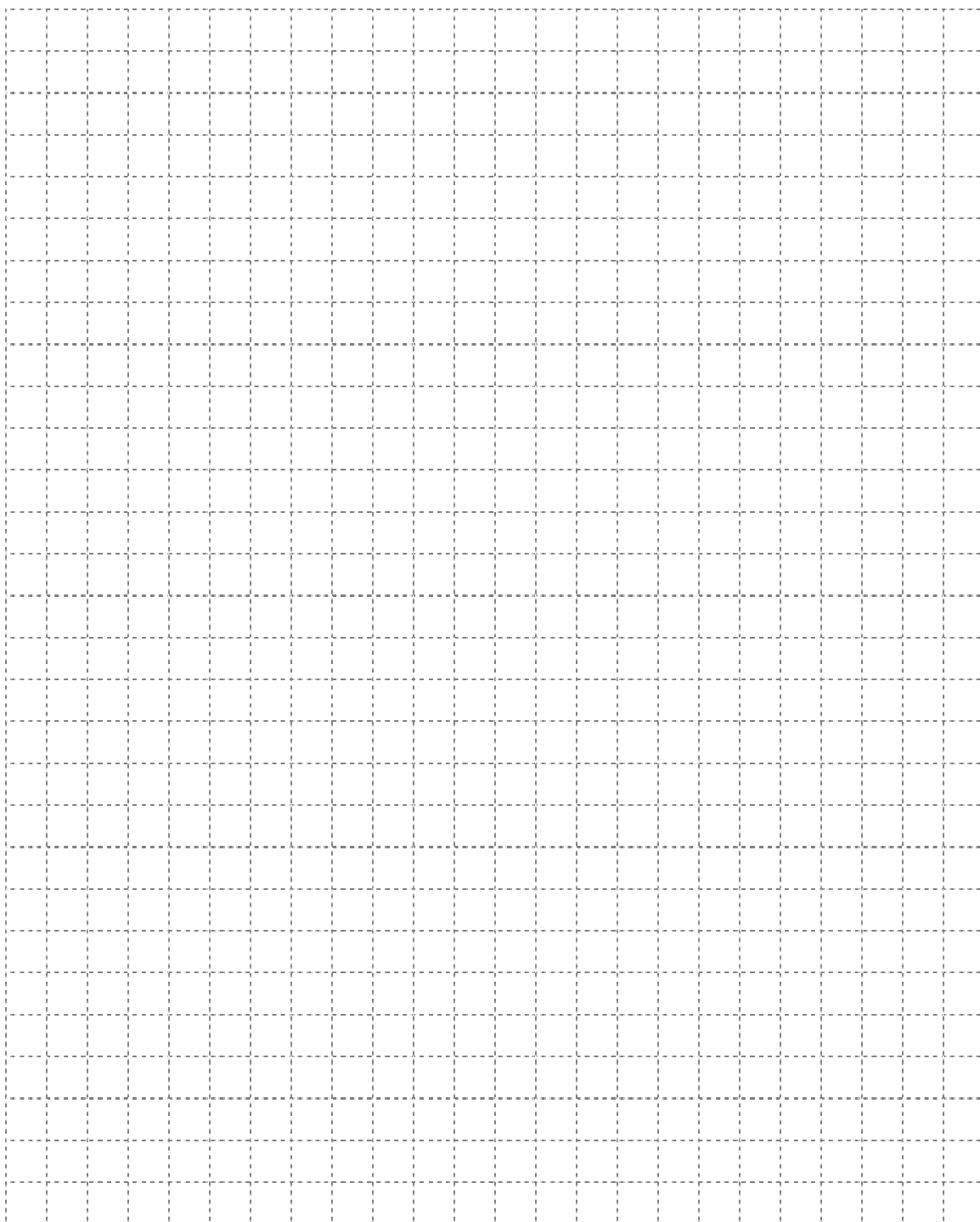
3 marks

137

Solve:

$$9^{2x+1} = 27^x$$

No marks will be awarded for work done on this page.



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Grade 12
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Achievement Test

Booklet 1

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Formula Sheet

$$s = \theta r$$

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$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

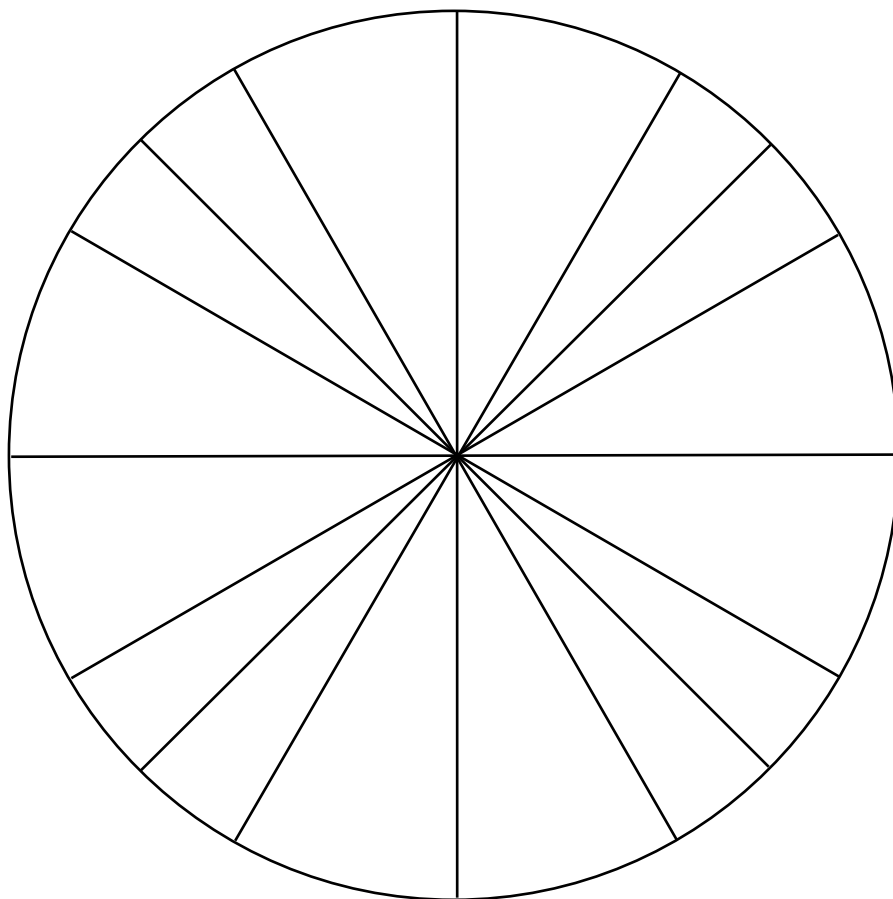
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

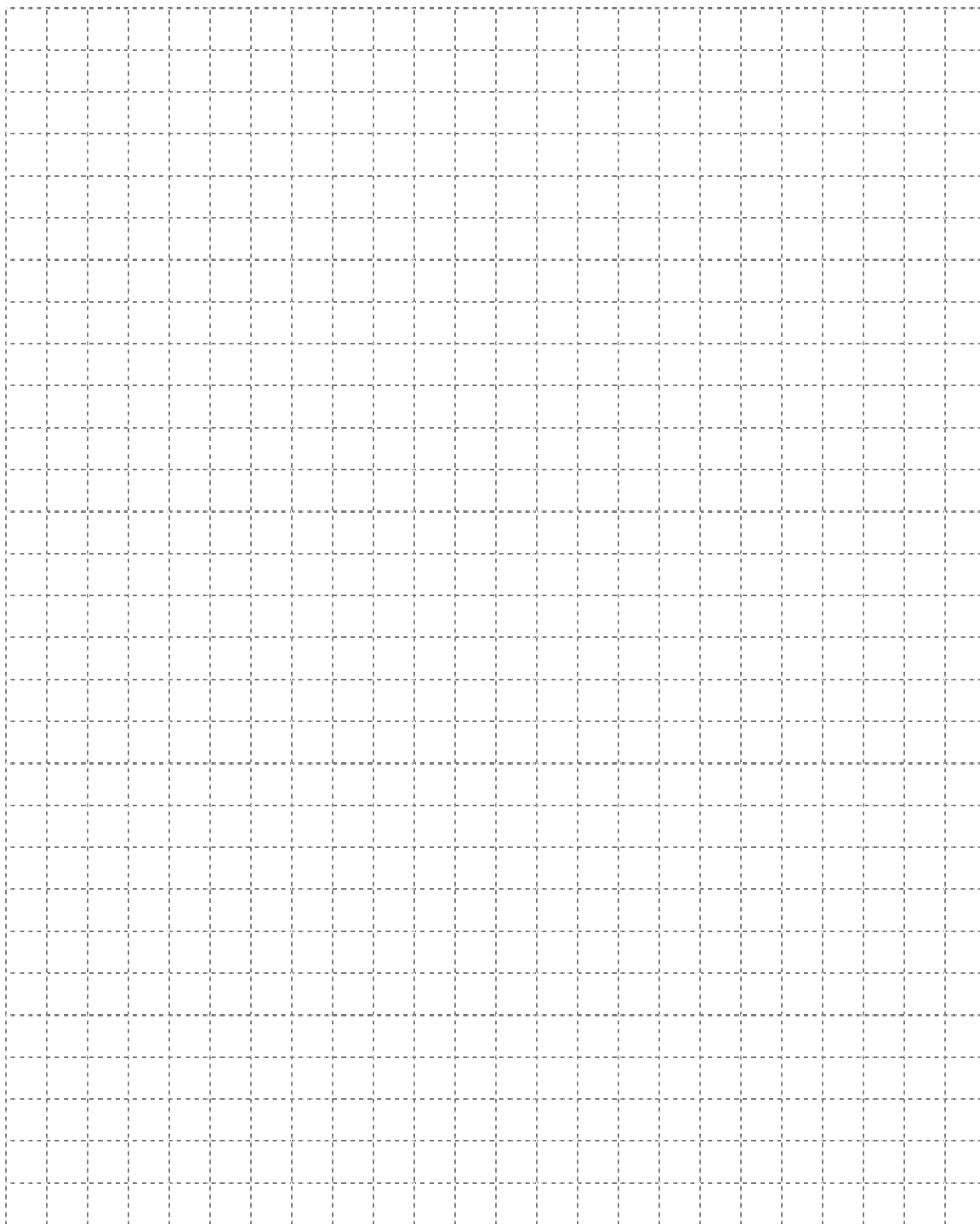
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



Question 1

1 mark 101

There are 24 different movies Kiandra can download to her computer. Determine the number of ways she can select 15 movies.

Question 2

a) 1 mark b) 1 mark

102
103

Given $\theta = 40^\circ$,

a) convert θ to radians.

b) determine the coterminal angles of θ where $\theta \in \mathbb{R}$.

Question 3

3 marks 104

Peter invests \$560 per month at an annual interest rate of 4.2%, compounded monthly. Determine how many monthly investments he will need to make to obtain at least \$500 000.

Express your answer as a whole number.

Use the formula:

$$FV = \frac{R[(1+i)^n - 1]}{i}$$

where FV = the future value

R = the investment amount each period

$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$

n = the number of investments

Question 4

2 marks 105

Ishmael has 4 dogs, 5 cats, and 3 horses.

If he arranges all of them in a row, determine how many ways they can be arranged if each type of animal must be grouped together.

Note: A calculator is not required for the remaining test questions.

Question 5

3 marks 106

Solve the following equation algebraically over the interval $0 \leq \theta \leq 2\pi$.

$$2 \cos^2 \theta + 9 \cos \theta - 5 = 0$$

Question 6

2 marks 107

Determine which term contains $\frac{1}{x^6}$ in the binomial expansion of $\left(\frac{2}{x^3} + 3x^2\right)^7$.

Question 7

1 mark 108

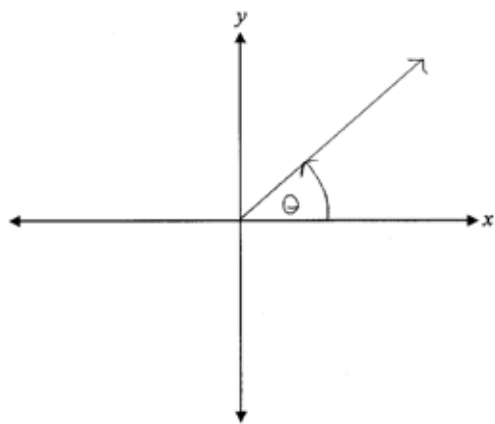
Determine the radius of a circle which has an arc length of 5 cm with a central angle of 3 radians.

Question 8

1 mark 109

Tyler incorrectly sketched the angle $\theta = -\frac{7\pi}{4}$ in standard position.

Describe his error.



Question 9

a) 1 mark b) 3 marks

110
111

Given the identity $\sec \theta + \cos \theta = \frac{2 - \sin^2 \theta}{\cos \theta}$,

a) determine the non-permissible values of θ , over the interval $0 \leq \theta \leq 2\pi$.

b) prove the identity for all permissible values of θ .

Left-Hand Side	Right-Hand Side

Question 10

2 marks ¹¹²

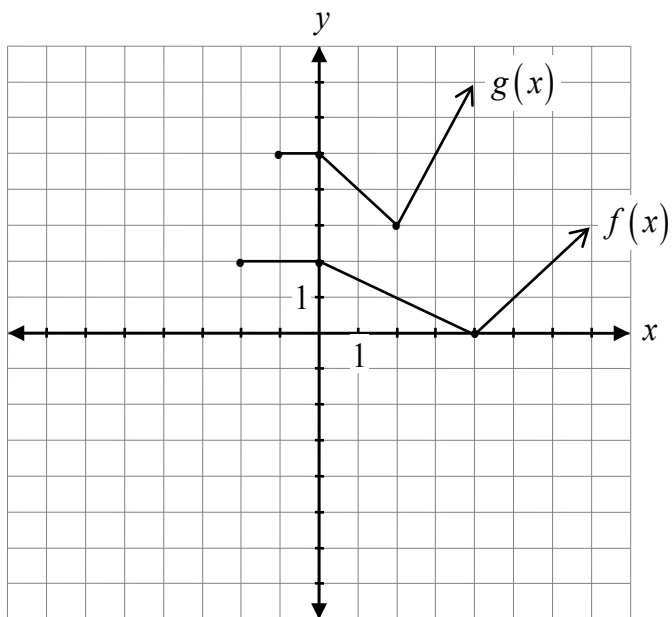
Expand using the laws of logarithms.

$$\log\left(\frac{a}{b^4}\right)$$

Question 11

2 marks 113

State the equation of $g(x)$ in terms of $f(x)$.

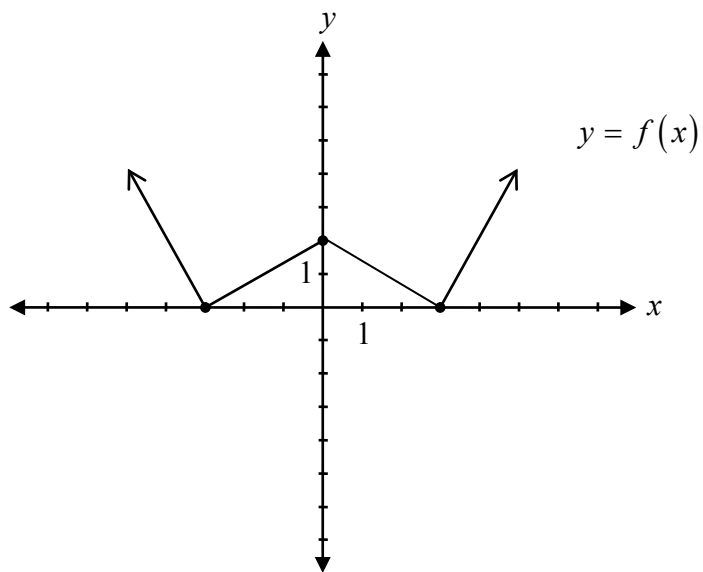


$g(x) =$ _____

Question 12

1 mark 114

Explain why the inverse of the graph of $y = f(x)$ is not a function.



Question 13

2 marks ¹¹⁵

Solve the following equation algebraically:

$$\log(x^2 + 5) - \log(x^2 + 1) = \log 3$$

Question 14

2 marks 116

Describe the transformations used to obtain the graph of the function $y = 5f(x + 1)$ from the graph of $y = f(x)$.

Question 15

3 marks ¹¹⁷

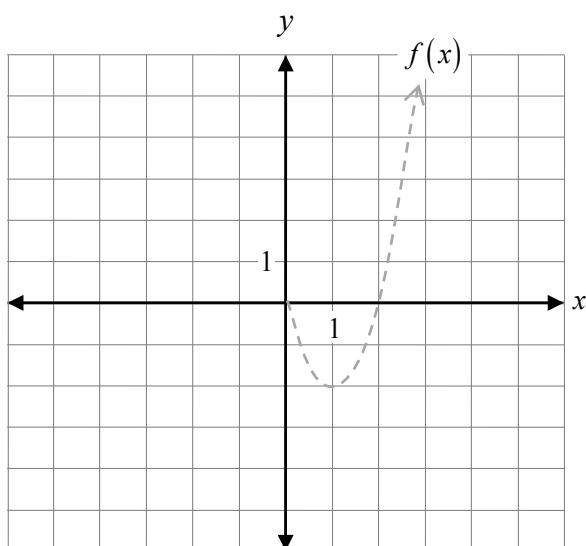
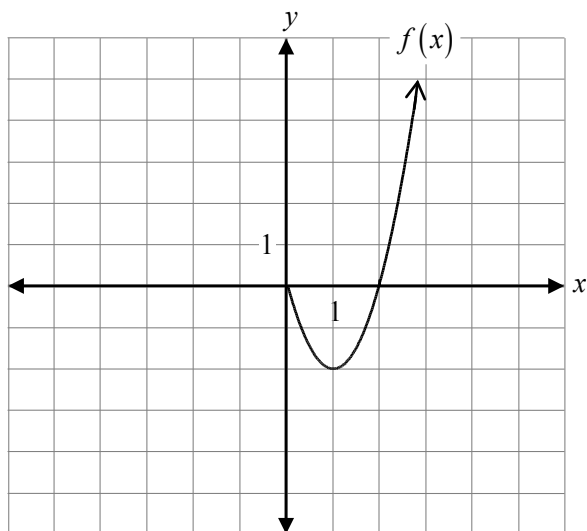
Solve algebraically:

$${}_nC_2 = 3n + 4$$

Question 16

3 marks 118

Given the graph of $f(x)$, sketch the graph of $y = \left| \frac{1}{2} f(x-1) \right|$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 17

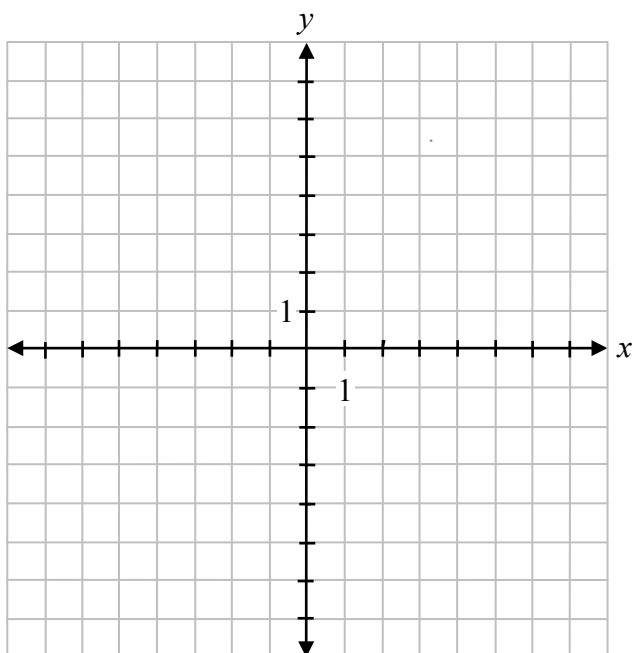
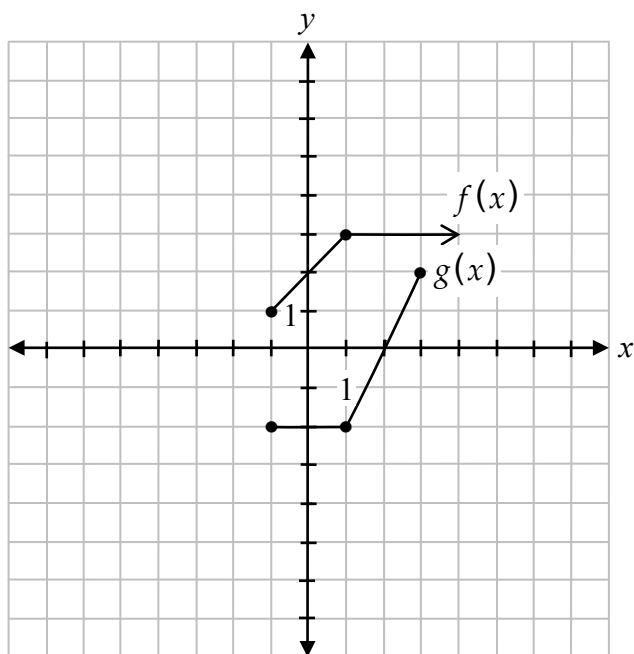
1 mark 119

Explain why $f(x) = (x+2)^3(x-1)^{\frac{1}{2}}$ is not a polynomial function.

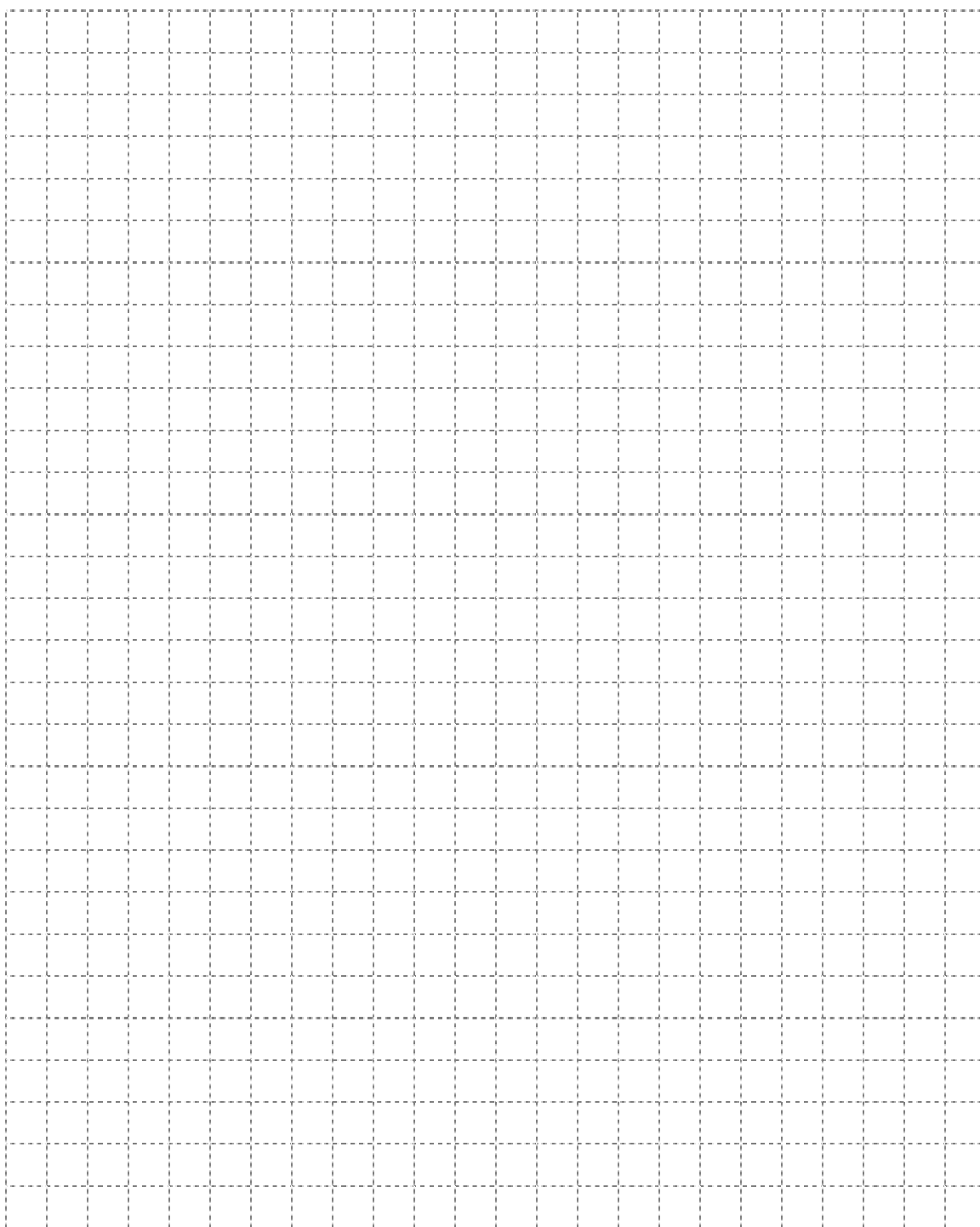
Question 18

2 marks 120

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = (f \circ g)(x)$.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2017

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 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Precalculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
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Websites are subject to change without notice.

Disponible en français.

Available in alternate formats upon request.

Instructions

Selected Response Questions

- There are 9 questions worth a total of 10 marks.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- There are 19 questions worth a total of 43 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 19

1 mark

Identify the trigonometric function that is equivalent to $\sin \frac{\pi}{4} \cos \frac{\pi}{3} + \cos \frac{\pi}{4} \sin \frac{\pi}{3}$.

a) $\sin \frac{2\pi}{7}$

b) $\sin \frac{7\pi}{12}$

c) $\cos \frac{2\pi}{7}$

d) $\cos \frac{7\pi}{12}$

Question 20

1 mark

Identify the logarithmic form of $5^x = 6$.

a) $\log_5 x = 6$

b) $\log_5 6 = x$

c) $\log_6 x = 5$

d) $\log_6 5 = x$

Question 21

1 mark

Given $f(\theta) = 3 \cos 2\theta - 1$ and $g(\theta) = \sin \theta + 1$, identify which statement is true.

- a) Both functions have the same period.
- b) Both functions have the same amplitude.
- c) Both functions have the same minimum value.
- d) Both functions have the same maximum value.

Question 22

1 mark

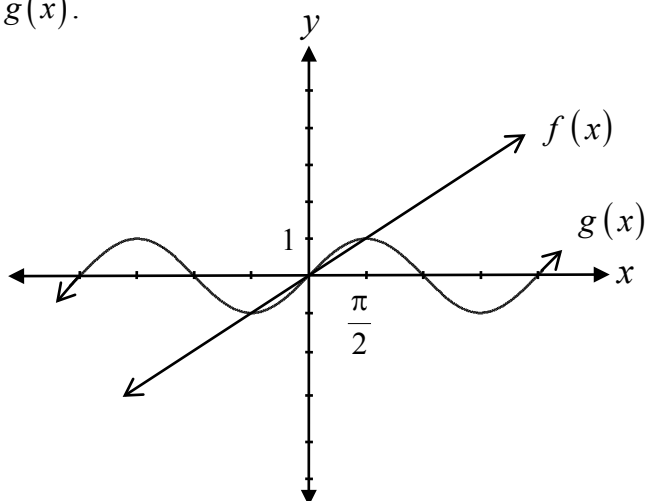
Identify the fourth term in the expansion of $(x + y)^5$.

- a) $10x^4y$
- b) $10x^3y^2$
- c) $10x^2y^3$
- d) $10xy^4$

Question 23

1 mark

Given the graphs of $f(x)$ and $g(x)$, identify the choice with all of the solutions of the equation $f(x) = g(x)$.



a) $x = -2\pi, -\pi, 0, \pi, 2\pi$

b) $x = -\frac{\pi}{2}, 0, \frac{\pi}{2}$

c) $x = \frac{\pi}{2}$

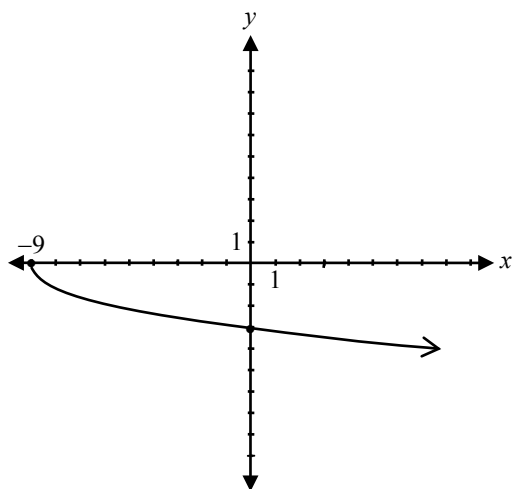
d) $x = -1, 0, 1$

Question 24

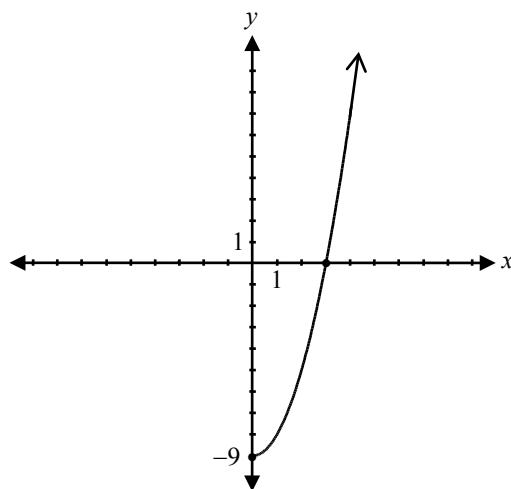
1 mark

Identify the graph of $f^{-1}(x)$ if $f(x) = x^2 - 9, x \geq 0$.

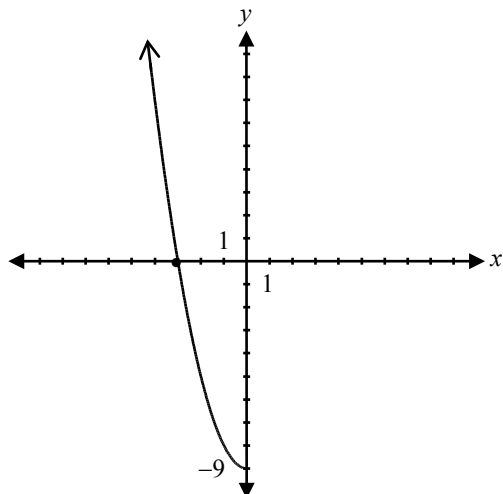
a)



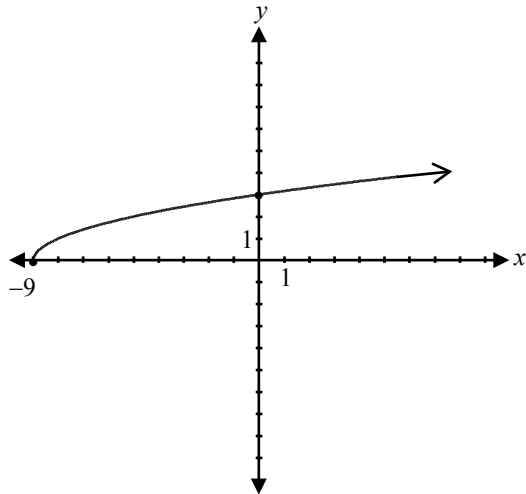
b)



c)



d)



Question 25

1 mark

Using the remainder theorem, identify which value of x results in a remainder of zero given

$$p(x) = x^3 + 7x^2 + 14x + 8.$$

- a) 1
- b) 0
- c) -1
- d) -3

Question 26

1 mark

Evaluate $\cos\left(\cos\left(\frac{3\pi}{2}\right)\right)$.

- a) 1
- b) $\frac{1}{2}$
- c) 0
- d) -1

Question 27

2 marks 121

Match the following equations with their graphs:

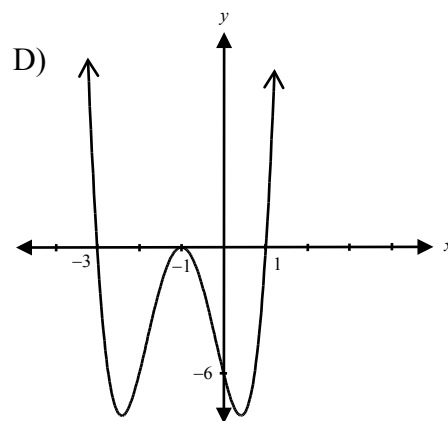
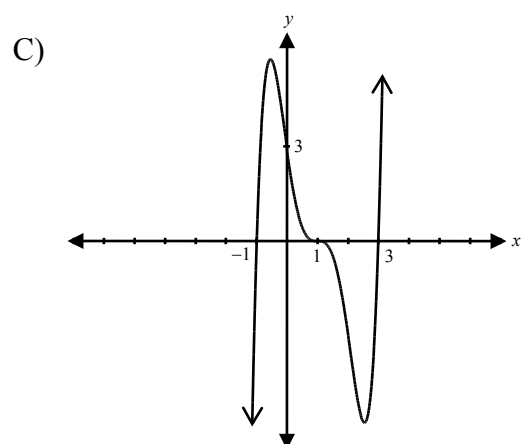
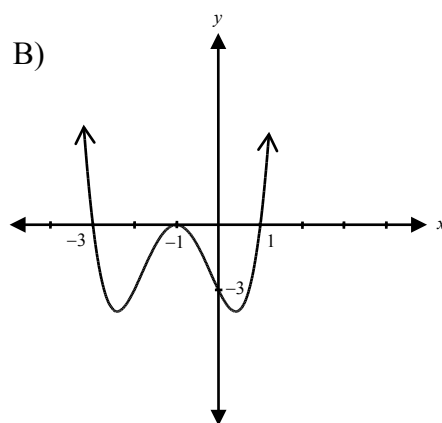
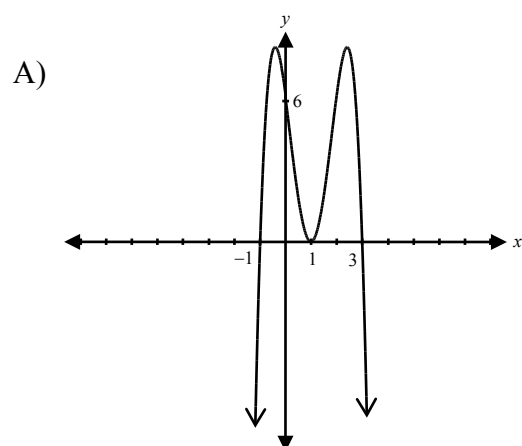
Place the appropriate letter
in this column.

$$f(x) = (x-1)^3(x+1)(x-3) \quad \underline{\hspace{2cm}}$$

$$g(x) = (x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm}}$$

$$h(x) = -2(x-1)^2(x+1)(x-3) \quad \underline{\hspace{2cm}}$$

$$k(x) = 2(x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm}}$$



Question 28

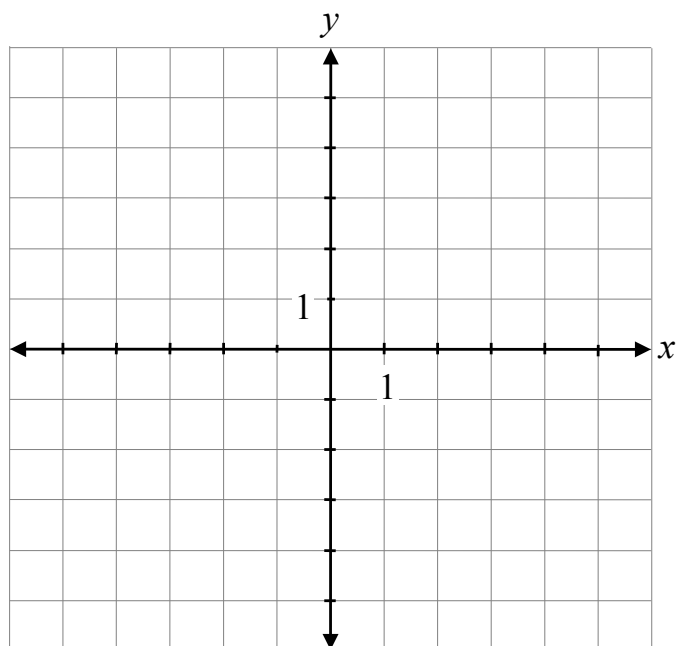
2 marks 122

If $\log 6 = p$, $\log 5 = r$ and $\log 2 = q$, express $\log 60$ in terms of p , q and r .

Question 29

4 marks 123

Sketch the graph of $y = \sqrt{-2x} + 1$.



Question 30

1 mark 124

Justify why the letters of the word FRANCE have a greater number of possible arrangements than the letters of the word CANADA.

Question 31

a) 1 mark b) 3 marks

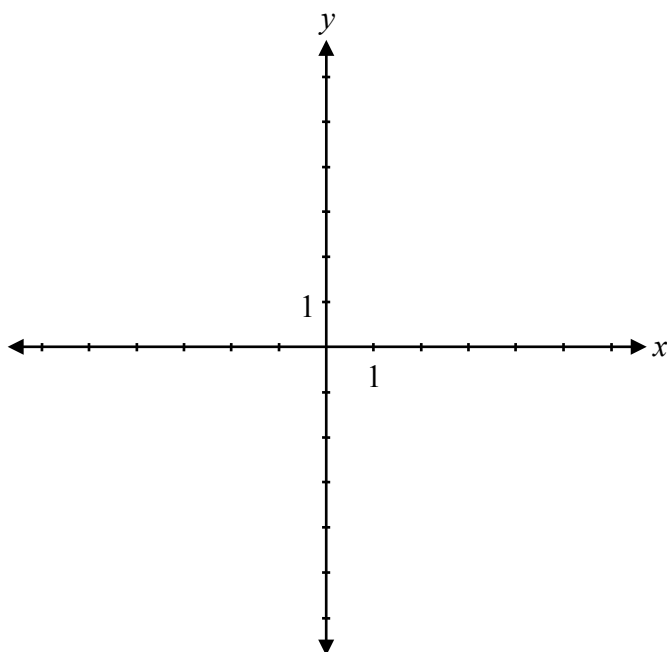
125
126

Given $f(x) = \frac{1}{x-2}$ and $g(x) = x+5$,

a) determine the equation for $f(g(x))$.

$f(g(x)) =$ _____

b) sketch the graph of $f(g(x))$.



Question 32

a) 3 marks b) 1 mark

127
128

Given that $\sin \alpha = \frac{3}{7}$, where α is in Quadrant II, and $\cos \beta = \frac{4}{5}$, where β is in Quadrant IV, determine the exact value of:

a) $\sin(\alpha - \beta)$

b) $\cos 2\alpha$

Question 33

2 marks 129

Determine the domain and range of $f(x) = \sqrt{x-5} - 1$.

Domain: _____

Range: _____

Question 34

1 mark 130

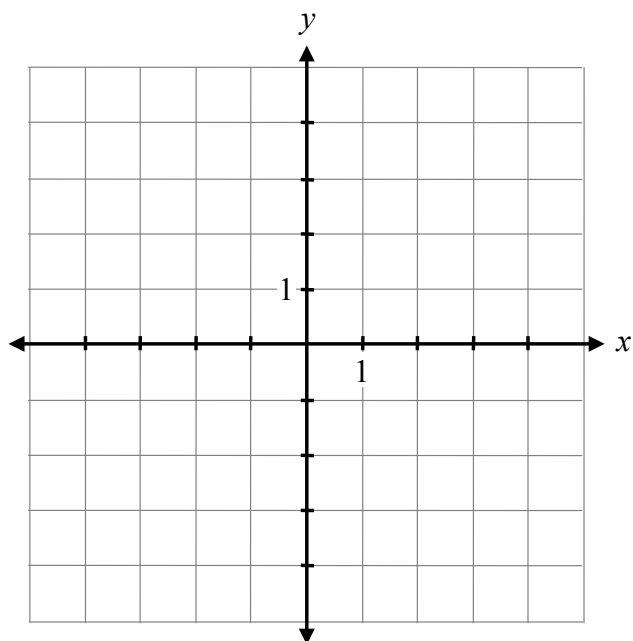
Justify why 4.7 is a better estimate than 4.3 for the value of $\log_2 26$.

Question 35

2 marks 131

Sketch the graph of the function:

$$f(x) = \frac{x(x-2)(x-4)}{(x-2)}$$



Question 36

3 marks 132

Evaluate:

$$\sec^2\left(\frac{\pi}{6}\right) + \tan\left(\frac{7\pi}{6}\right) \csc\left(-\frac{2\pi}{3}\right)$$

Question 37

1 mark 133

The graph of $f(x) = 3x + 7$ is reflected over the y -axis.

Determine the equation of the new function.

$y =$ _____

Question 38

2 marks 134

Determine the equations of all of the asymptotes of the function:

$$y = \frac{2x + 1}{x - 3}$$

Question 39

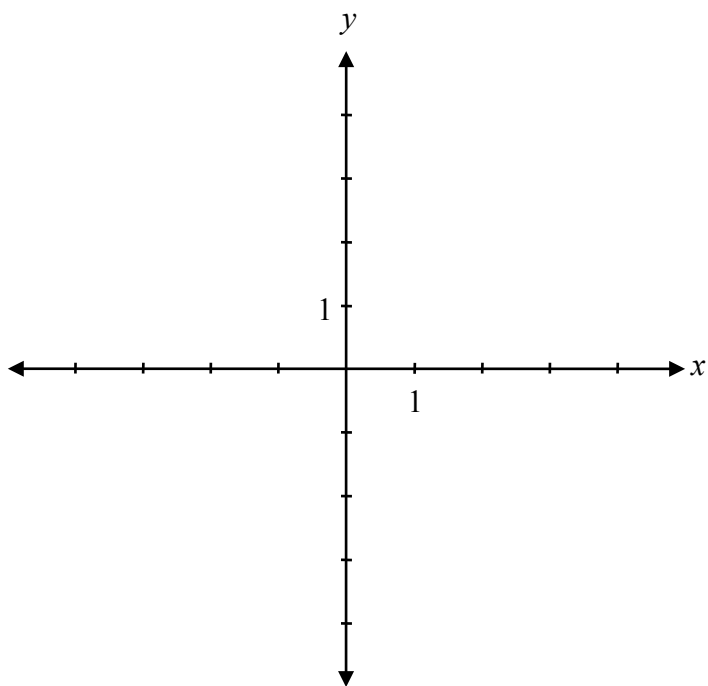
2 marks 135

One of the zeros of $p(x) = x^3 + 6x^2 - 32$ is $x = 2$. Determine all of the other zeros of $p(x)$.

Question 40

3 marks 136

Sketch the graph of $y = -2^x + 2$.



Question 41

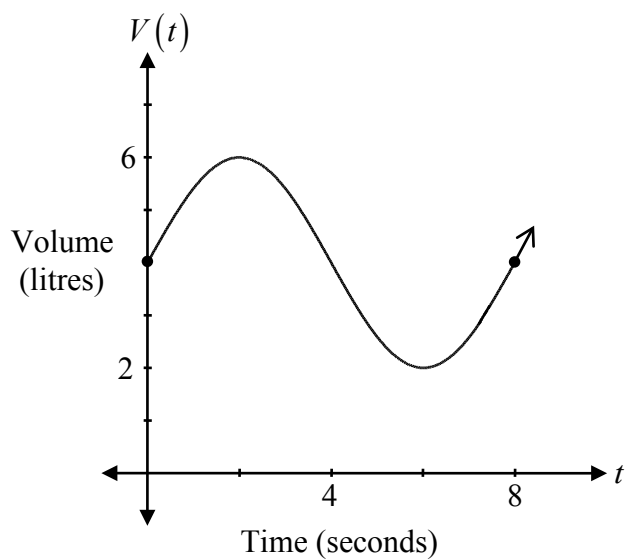
1 mark 137

Given the function $f(x) = \frac{2}{x} - 1$, justify why $f(f(2))$ is undefined.

Question 42

3 marks 138

The following graph represents the volume of air in an adult's lungs. If $V(t)$ is the volume of air in litres and t is the time in seconds, determine an equation that represents this sinusoidal function.



$V(t) =$ _____

Question 43

1 mark 139

Explain why the domain of $y = \log_2(x - 1)$ is $x > 1$.

Question 44

2 marks 140

The point $(-2, 7)$ is on the terminal arm of an angle in standard position.

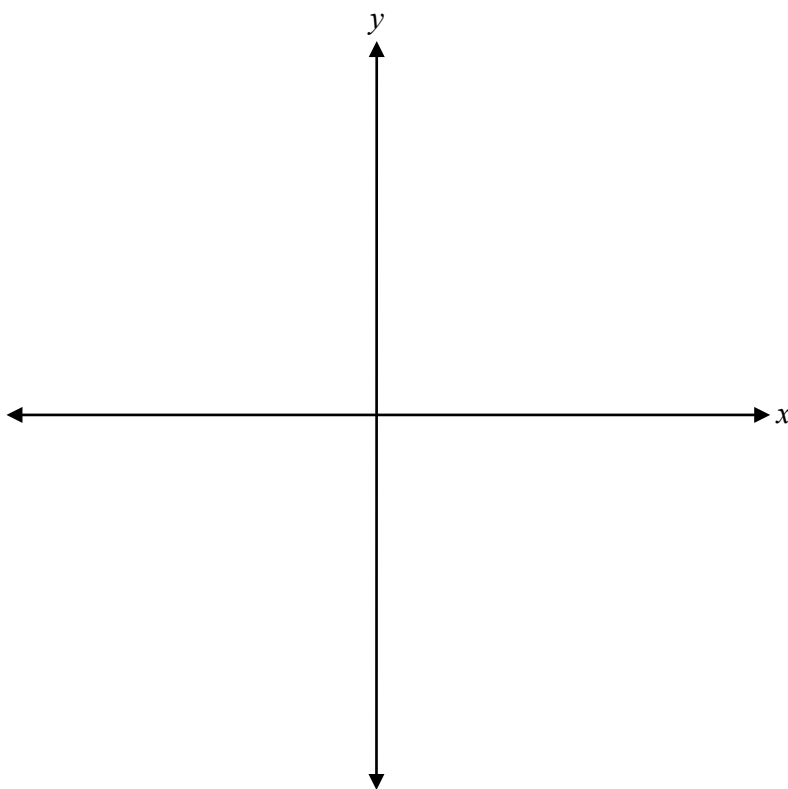
Determine the coordinates of the corresponding point, $P(\theta)$, on the unit circle.

Question 45

3 marks 141

Sketch a graph of $P(x)$ that satisfies all of the following conditions:

- $P(x)$ is a polynomial function of degree 3.
- $P(x)$ has a zero at -3 with a multiplicity of 2.
- $P(x)$ has a zero at 1.
- $P(x)$ has a leading coefficient of -3 .

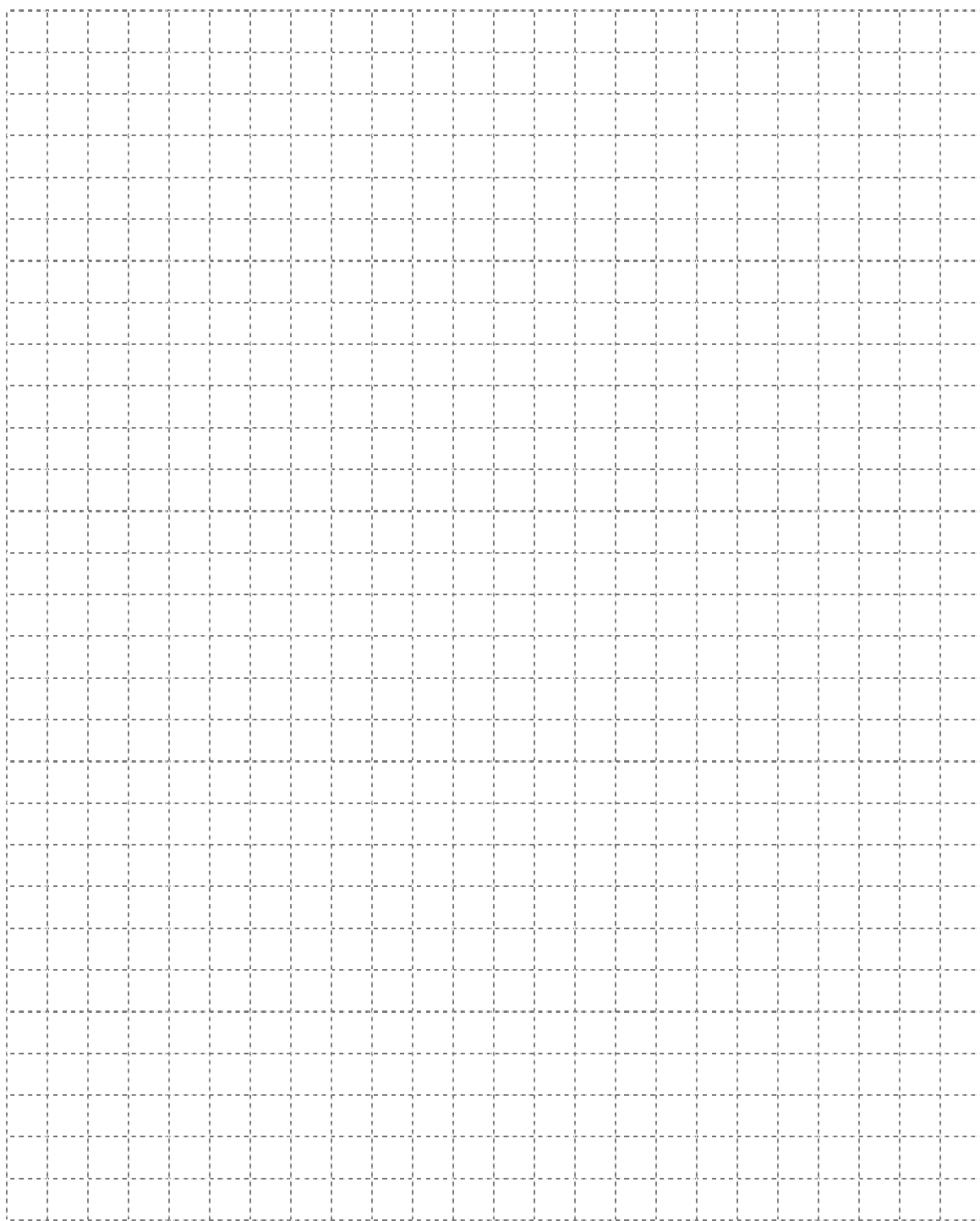


Question 46

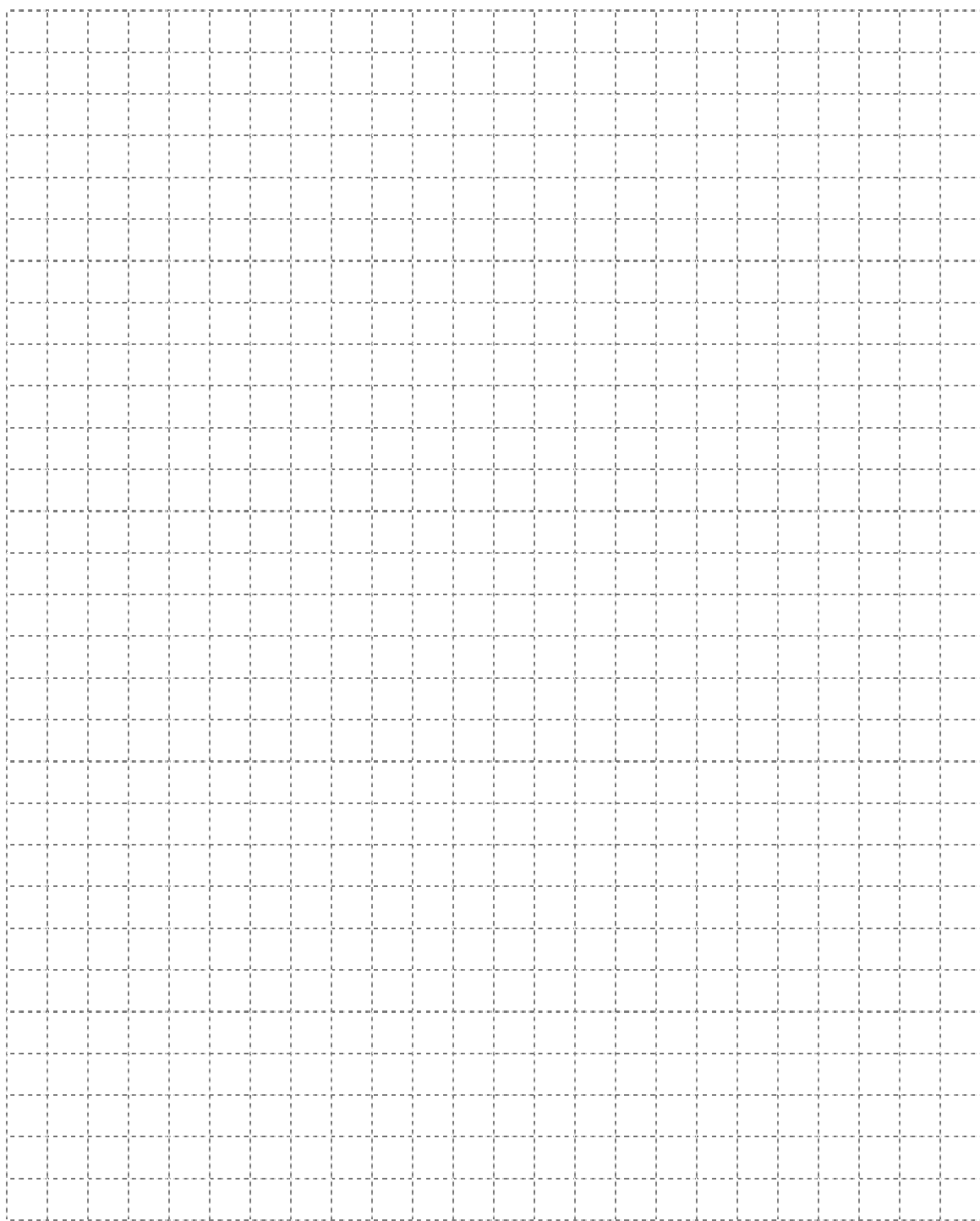
2 marks 142

Determine how many 4-digit numbers greater than 4000 can be made using the digits 2, 3, 4, 5, and 6 if repetitions are not allowed.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2017

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Booklet 1. June 2017

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 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Pre-calculus—Study and teaching (Secondary)—Manitoba.
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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	–	17	33
Booklet 2	10	22	57
Total	10	39	90

* The first 5 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

GENERAL DIRECTIONS

- Read all instructions carefully.
- The perforated pages can be removed from the test booklet. No marks will be given for work done on these pages.
- The blank pages at the back of each booklet may be used as scrap paper, but must **not** be removed from the test booklet. No marks will be given for work done on these pages.
- Note that diagrams and graphs provided in the test booklets may not be drawn to scale.
- After 45 minutes, put away your calculator. Even though you may not have finished *Booklet 1*, *Booklet 2* will be distributed at this time. You may choose to continue working on *Booklet 1* or start working on *Booklet 2*, but you will no longer have access to your calculator.

Instructions

- There are 17 questions worth a total of 33 marks.
- Calculators (scientific or graphing) are allowed for the first 45 minutes of the test.
- A calculator icon  appears next to the questions that require a calculator.
- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

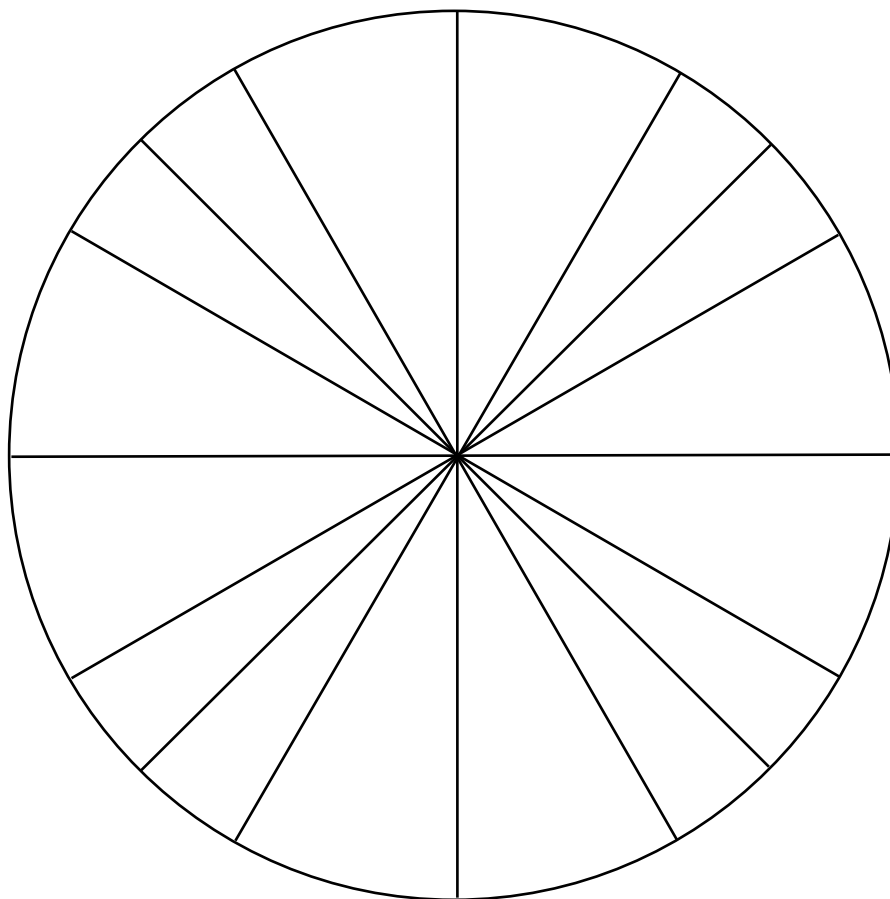
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

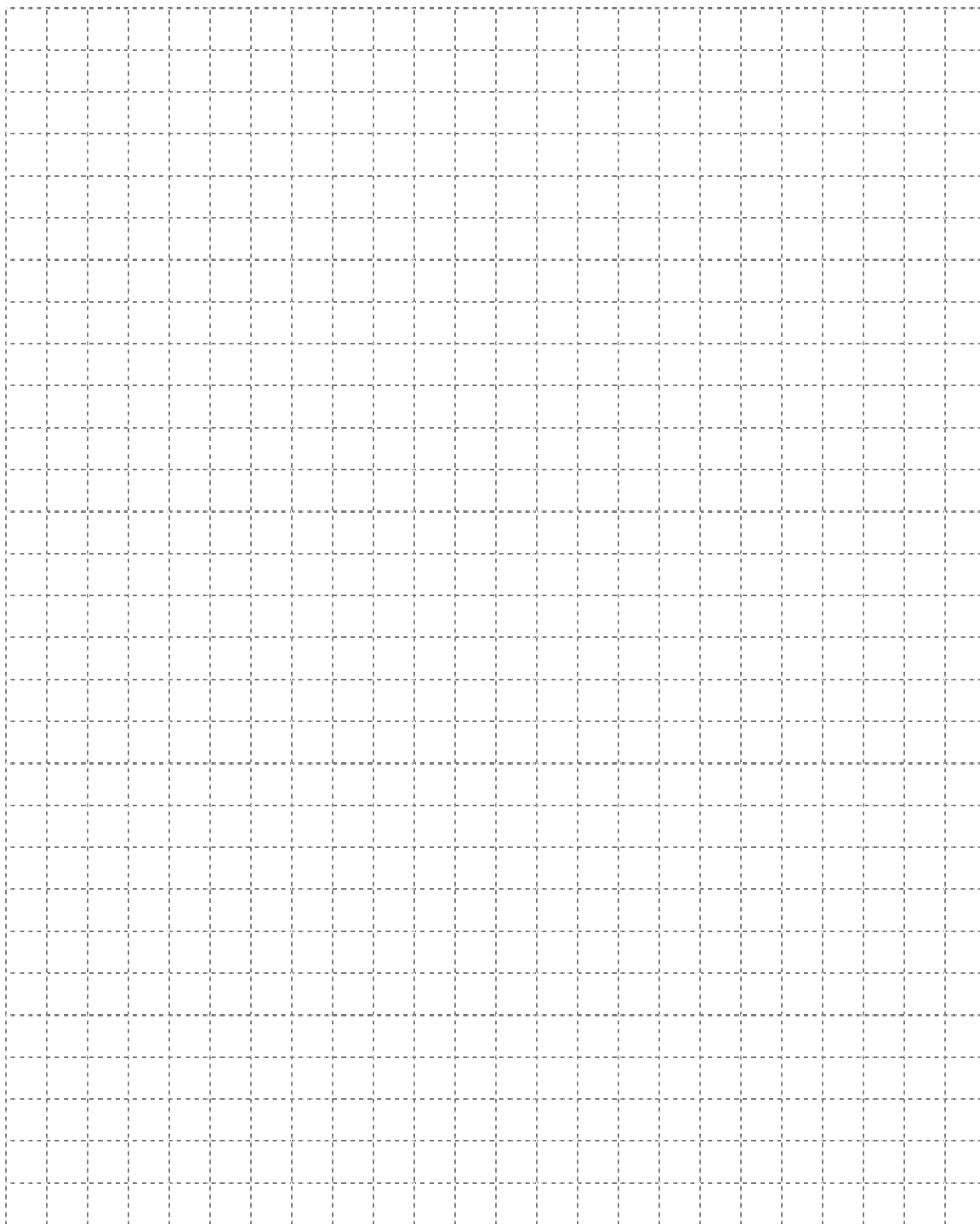
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



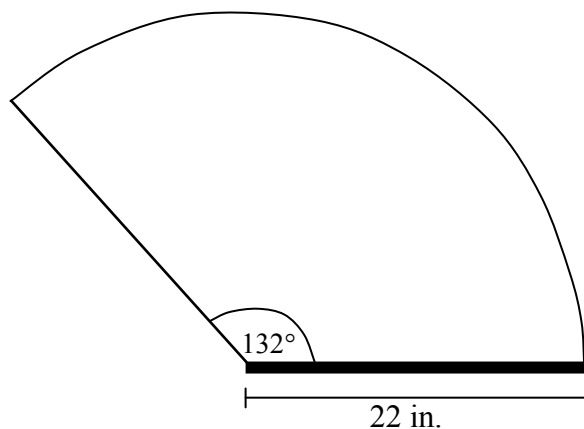
No marks will be awarded for work done on this page.



Question 1

2 marks 101

A section of a car windshield is cleaned by a wiper as shown in the diagram below. The arm of the wiper is 22 inches, and it rotates through a central angle of 132° . Determine the length of the arc that is created by the tip of the wiper.



Question 2

2 marks 102

There are 20 boys and 11 girls who can be selected to be on a team.

Determine the number of ways that 7 boys and 5 girls can be selected for this team.

Question 3

2 marks 103

A water filtration system which removes impurities from a sample of water can be modelled by

$$P = 0.25(0.55)^n, \text{ where:}$$

P = the percentage of impurities remaining, in decimal form

n = the number of filters

Determine, algebraically, how many filters are required so that less than 1% of the impurities remain in the water sample. Express your answer as a whole number.

Question 4

3 marks 104

In the binomial expansion of $\left(x^2 - \frac{2}{y}\right)^8$, determine the middle term in simplified form.

Question 5

3 marks 105

Solve the following equation algebraically over the interval $[0, 2\pi]$.

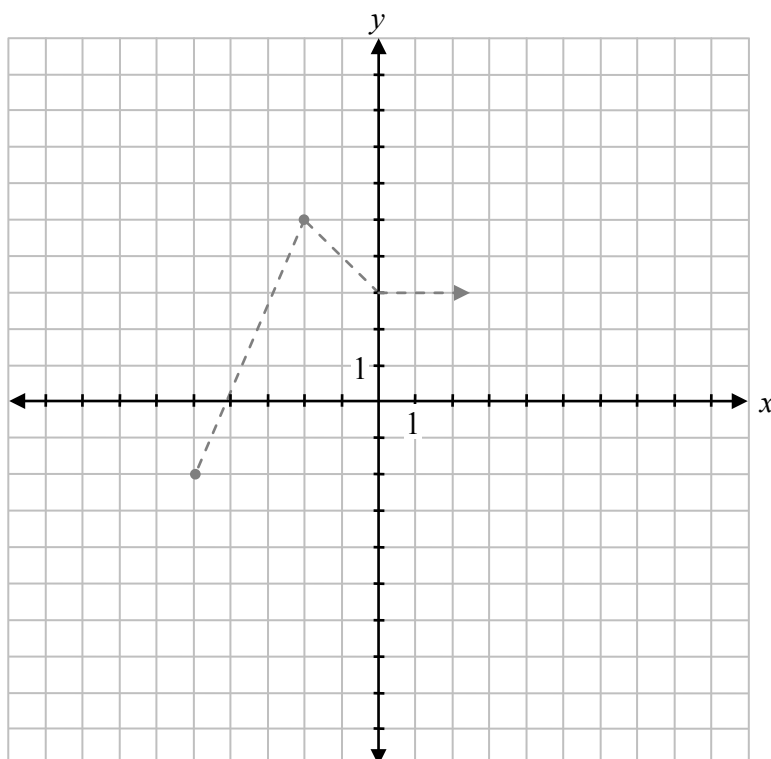
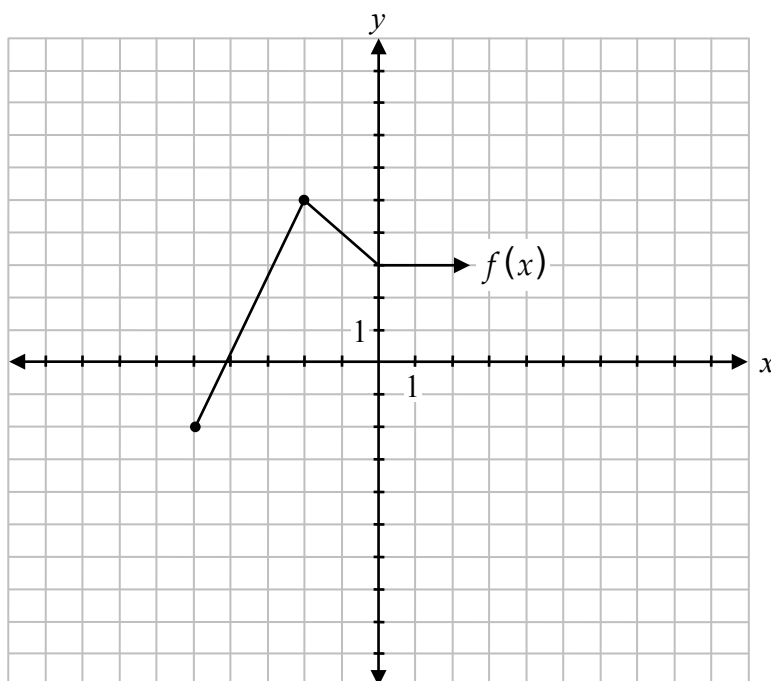
$$6 \sin^2 \theta + \sin \theta - 1 = 0$$

Note: A calculator is not required for the remaining test questions.

Question 6

2 marks 106

Given the graph of $y = f(x)$, sketch the graph of $y = f(-x + 4)$.

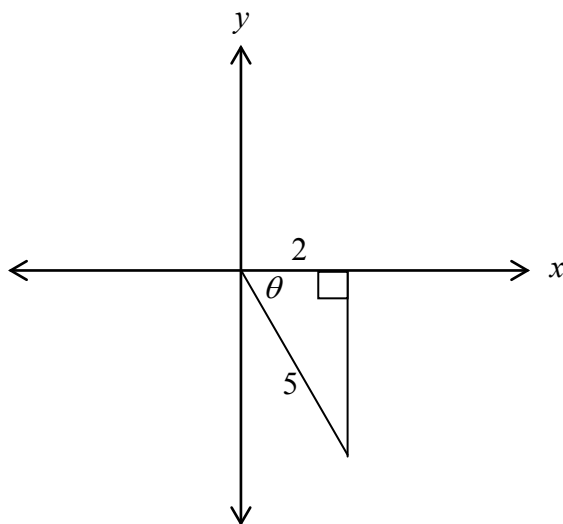


The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 7

2 marks 107

Given the following triangle, determine $\csc \theta$.



Question 8

3 marks 108

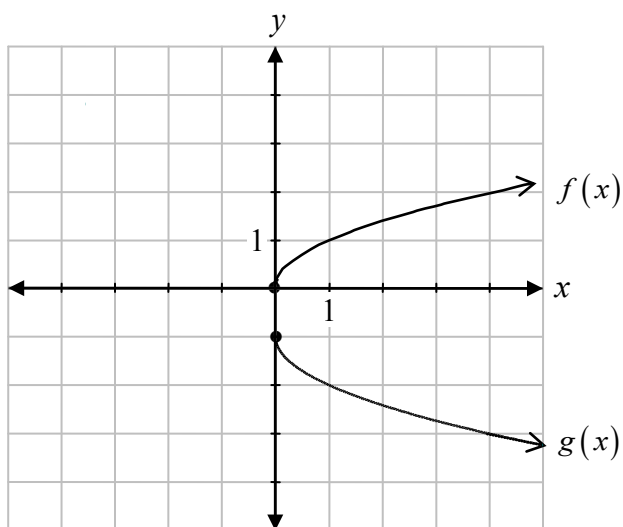
Solve algebraically.

$${}_nP_2 = 9n$$

Question 9

2 marks 109

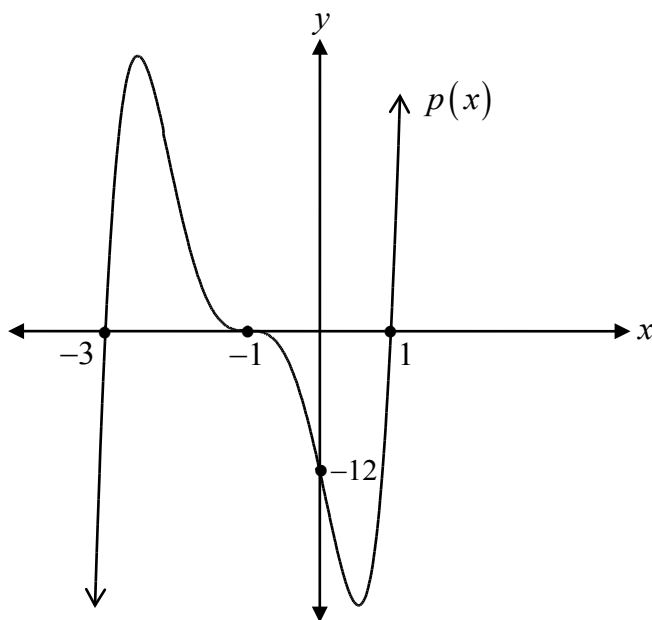
Describe the transformations applied to the graph of $f(x)$ to obtain the graph of $g(x)$.



Question 10

2 marks 110

Determine, algebraically, the value of the leading coefficient of the graph of the polynomial function, $p(x)$.



Question 11

1 mark ¹¹¹

Frank, Liam, Chan, and Thao are going to a movie.

Determine the number of ways they can sit in a row of four chairs, if Frank and Chan must sit beside each other.

Question 12

2 marks 112

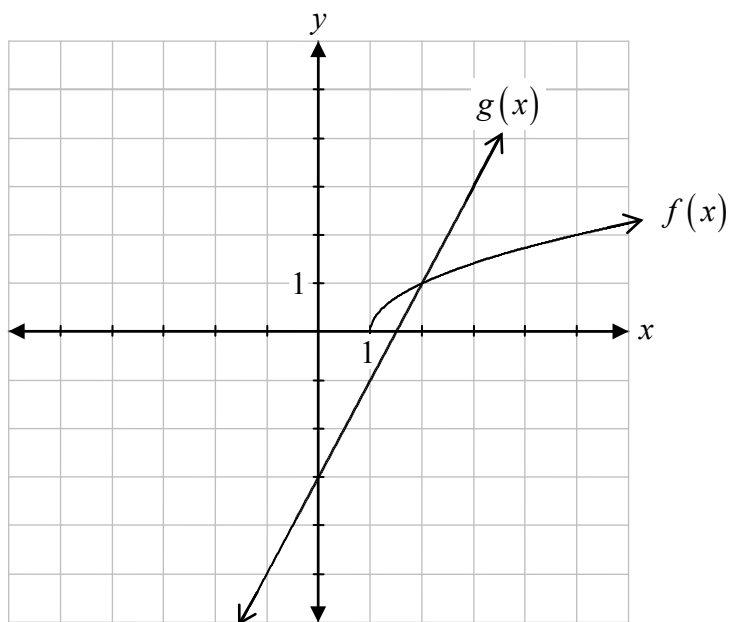
Determine, algebraically, if $f(x) = \frac{1}{x+5}$ and $g(x) = \frac{1}{x-5}$ are inverses of each other.

Justify your answer.

Question 13

1 mark 113

Using the graphs of $y = f(x)$ and $y = g(x)$, solve $f(x) = g(x)$.



Question 14

1 mark 114

An angle in standard position measures $\frac{3\pi}{4}$.

Determine in which quadrant the terminal arm of this angle is located after a rotation of 3 radians.

Justify your answer.

Question 15

3 marks 115

Prove the following identity for all permissible values of θ .

$$\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$$

Left-Hand Side	Right-Hand Side

Question 16

1 mark 116

If the range of $y = f(x)$ is $-3 \leq y \leq 6$, determine the range of $y = 2f(3x)$.

Question 17

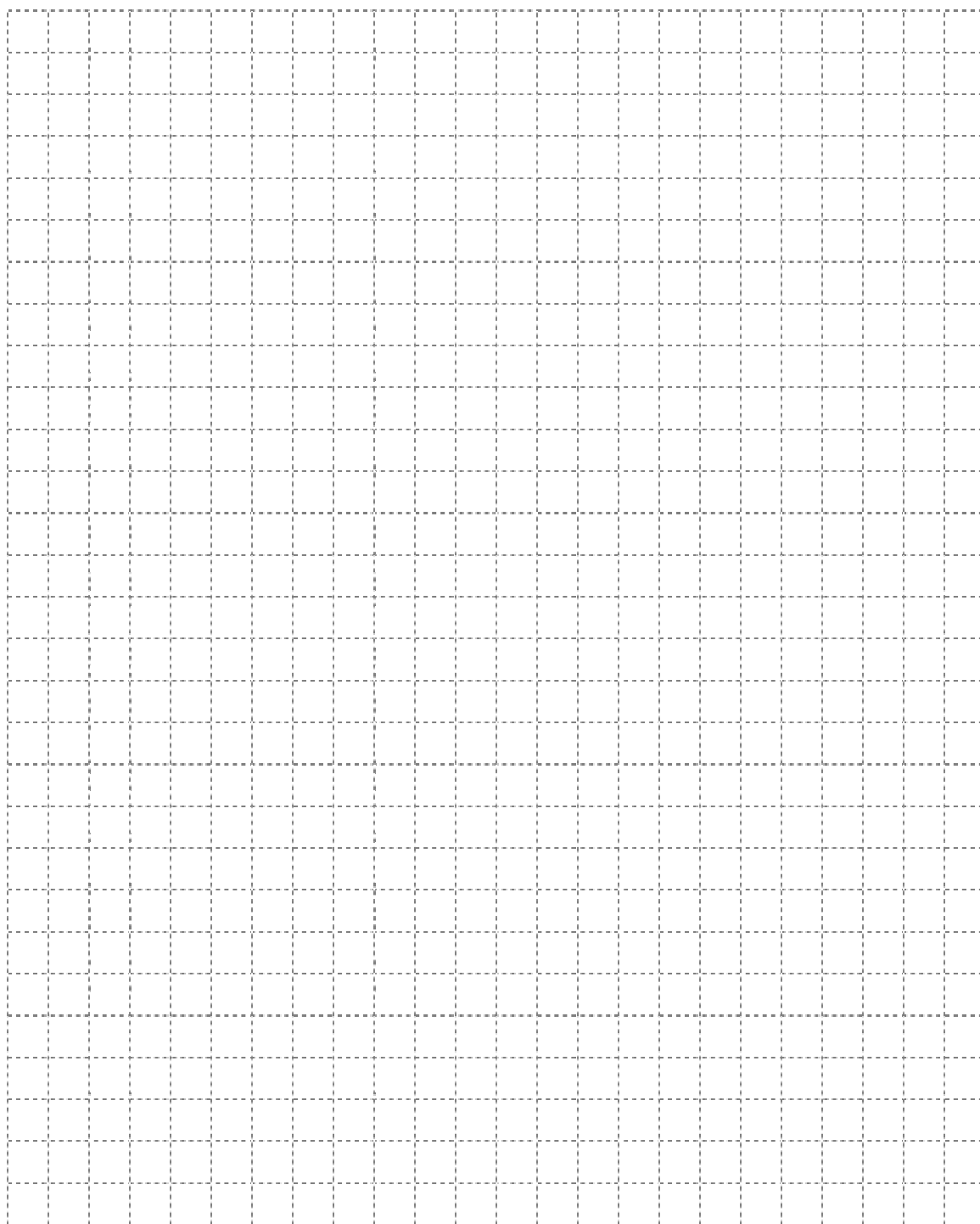
1 mark 117

Maurice incorrectly solved the equation, $\sin \theta + 1 = 0$, over the interval $[0^\circ, 360^\circ]$.

$$\begin{aligned}\sin \theta + 1 &= 0 \\ \sin \theta &= -1 \\ \sin \theta &= 270^\circ\end{aligned}$$

Describe his error.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2017

Manitoba Education and Training Cataloguing in Publication Data

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Booklet 2. June 2017

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1. Mathematics—Examinations, questions, etc.
 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Precalculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
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Winnipeg, Manitoba, Canada

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Instructions

Selected Response Questions

- There are 10 questions worth a total of 11 marks.
- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- There are 22 questions worth a total of 46 marks.
- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

No marks will be awarded for work done on this page.

Question 18**1 mark**

If $P(3, 5)$ is a point on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = f(x - 1) + 7$.

- a) $(2, 12)$
- b) $(4, -2)$
- c) $(2, -2)$
- d) $(4, 12)$

Question 19**1 mark**

Identify how the graph of $y = 3^x$ is transformed to the graph of $y = 3^{-x}$.

- a) reflected over the x -axis
- b) reflected over the y -axis
- c) reflected over both the x -axis and the y -axis
- d) reflected over the line $y = x$

Question 20**1 mark**

Identify the equation $\log_a b = c$ in exponential form.

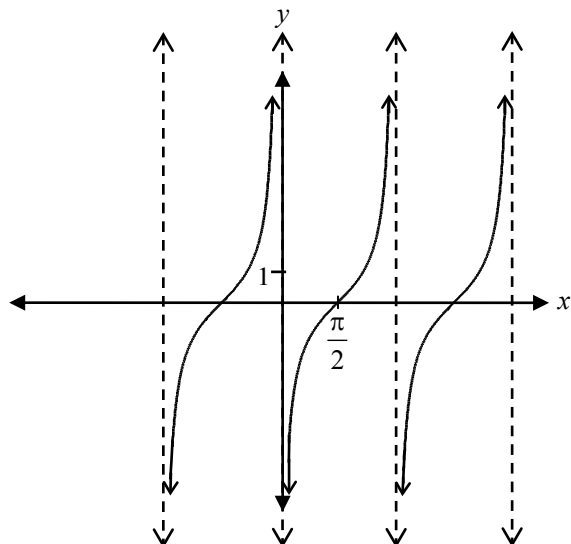
- a) $b^c = a$
- b) $a^c = b$
- c) $a^b = c$
- d) $c^a = b$

Question 21

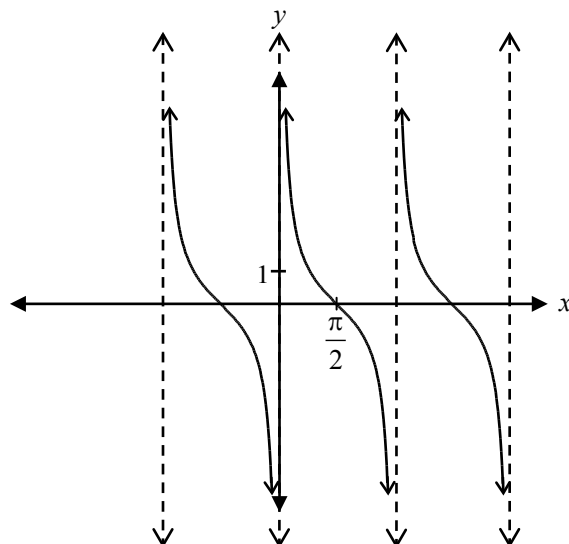
1 mark

Identify the graph of $y = \tan x$.

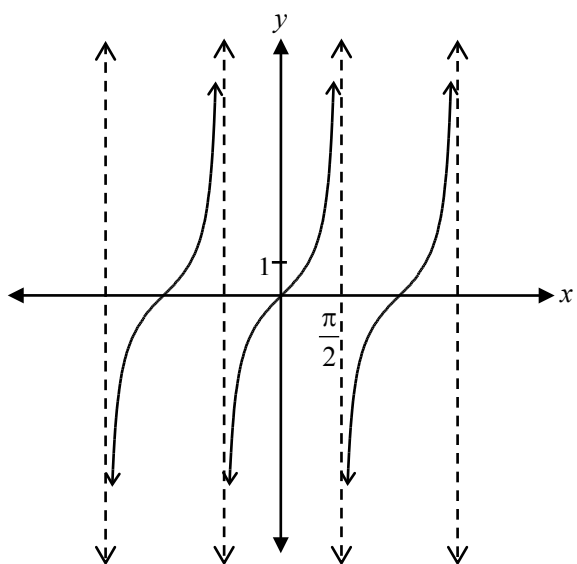
a)



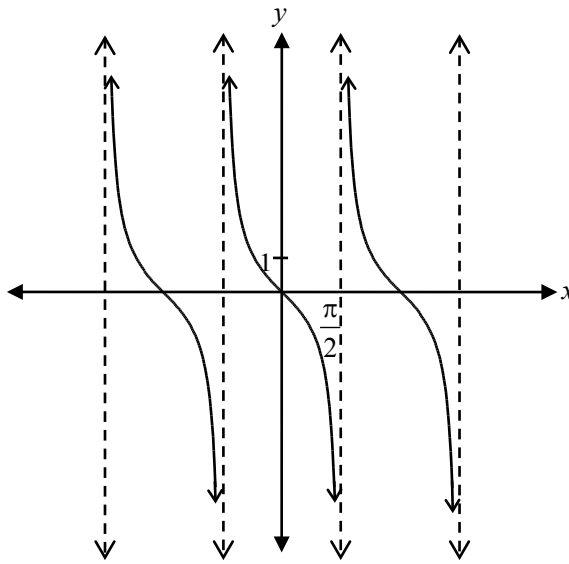
b)



c)



d)

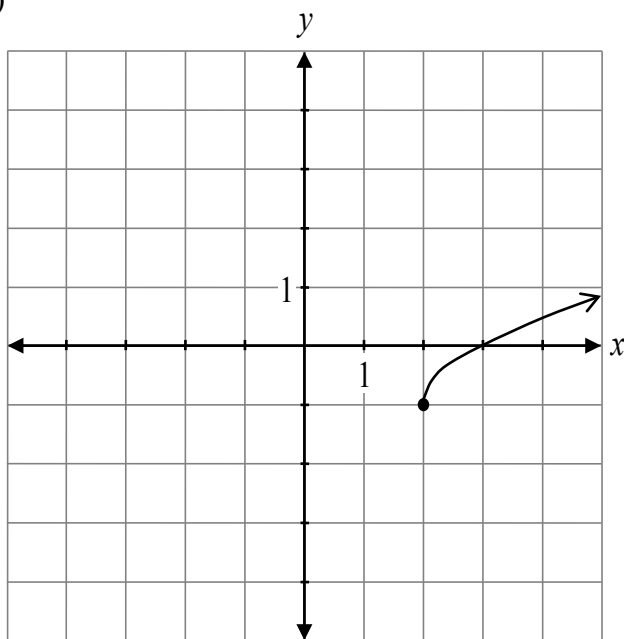


Question 22

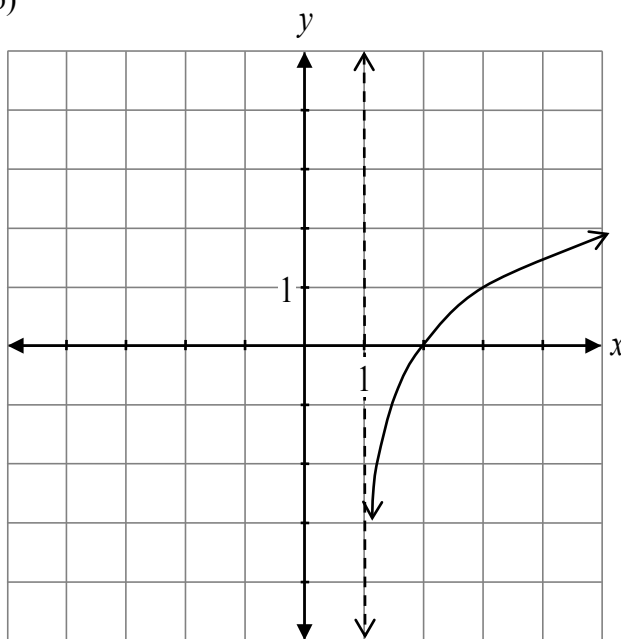
1 mark

Identify which of the following graphs represents a logarithmic function.

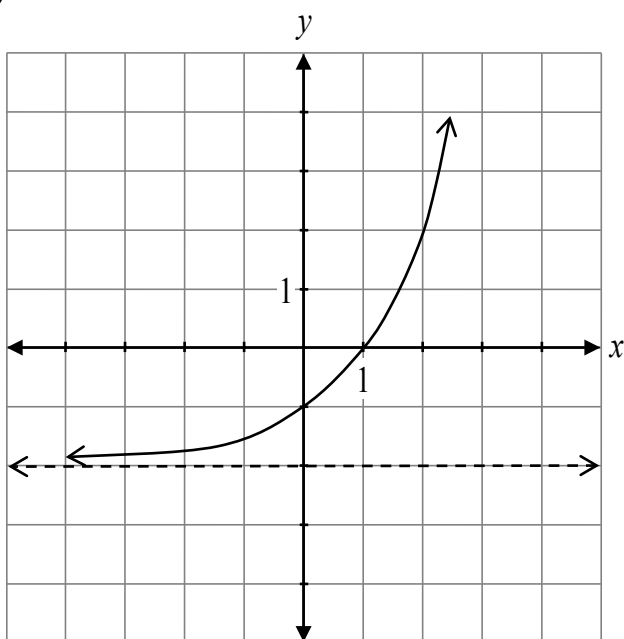
a)



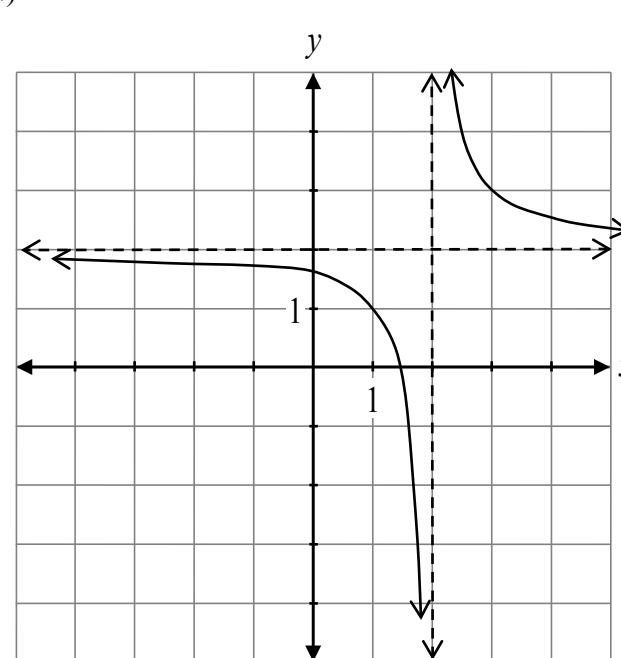
b)



c)



d)



Question 23**1 mark**

If the volume of a box is represented by $V(x) = (x + 4)(x + 2)(x - 1)$, identify a possible value of x .

- a) -4
- b) -1
- c) 1
- d) 4

Question 24**1 mark**

Identify a coterminal angle for $\theta = -\frac{\pi}{3}$.

- a) $\frac{\pi}{3}$
- b) $\frac{4\pi}{3}$
- c) $\frac{7\pi}{3}$
- d) $\frac{11\pi}{3}$

Question 25**1 mark**

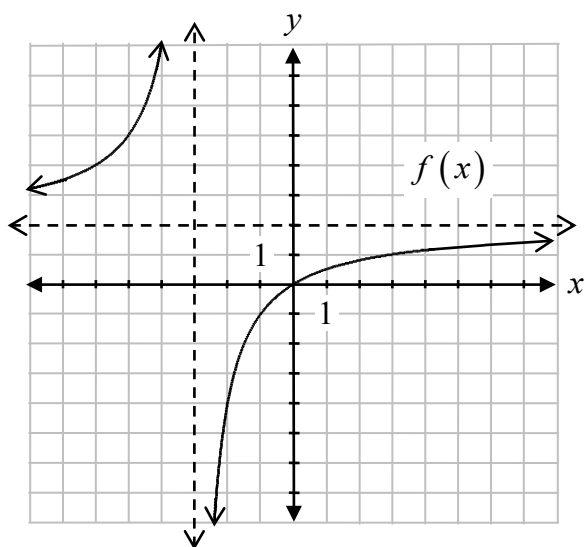
Identify the value of n in the equation ${}_nC_3 = {}_nC_6$.

- a) 3
- b) 6
- c) 9
- d) 18

Question 26

1 mark

Identify the equation of the function, $f(x)$, for the following graph.



- a) $f(x) = \frac{2x}{x+3}$
- b) $f(x) = \frac{2}{x+3}$
- c) $f(x) = \frac{2x^2}{x(x+3)}$
- d) $f(x) = \frac{3x^2}{x(x+2)}$

Question 27

2 marks 118

Match the following radical functions with their graphs.

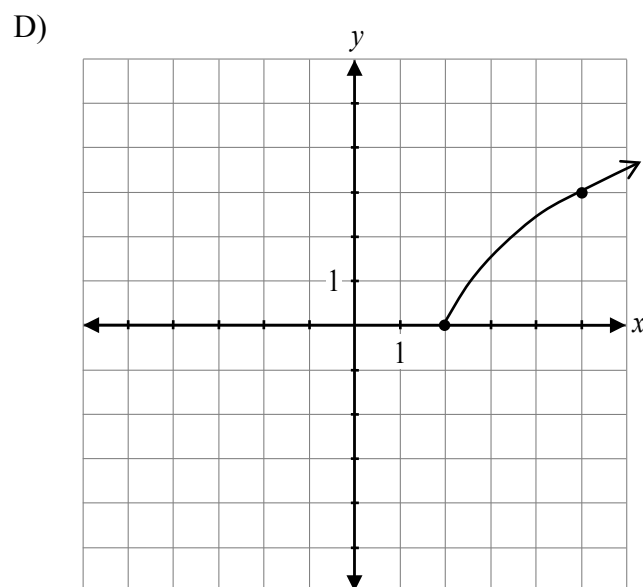
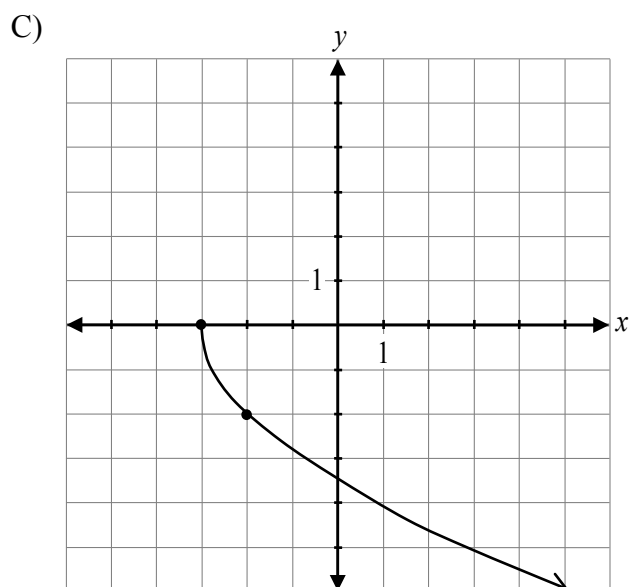
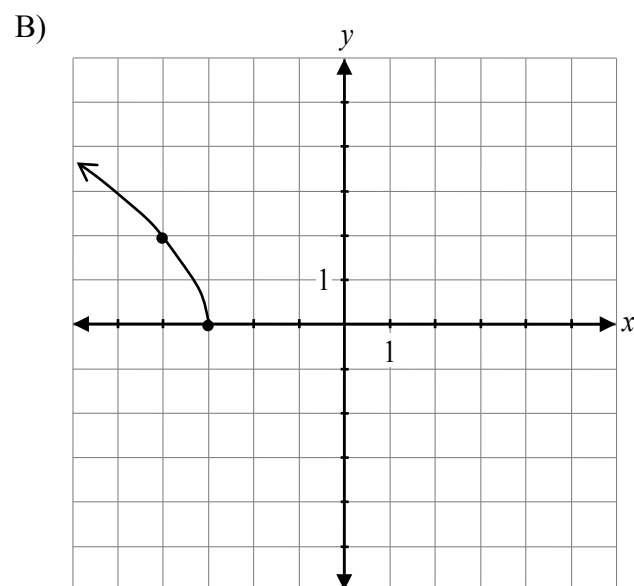
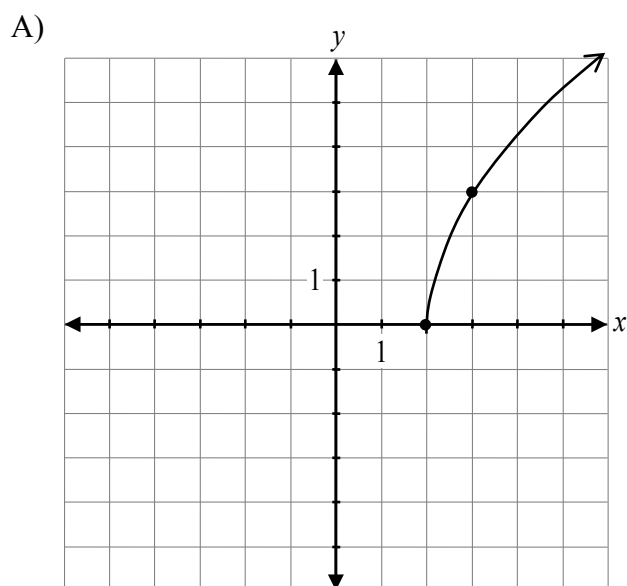
Place the appropriate letter in this column.

$$f(x) = 2\sqrt{-(x+3)} \quad \underline{\hspace{2cm}}$$

$$g(x) = -2\sqrt{(x+3)} \quad \underline{\hspace{2cm}}$$

$$h(x) = 3\sqrt{(x-2)} \quad \underline{\hspace{2cm}}$$

$$k(x) = \sqrt{3(x-2)} \quad \underline{\hspace{2cm}}$$



Question 28

3 marks 119

Express $p(x) = x^3 - 2x^2 - 4x + 8$ as a product of factors.

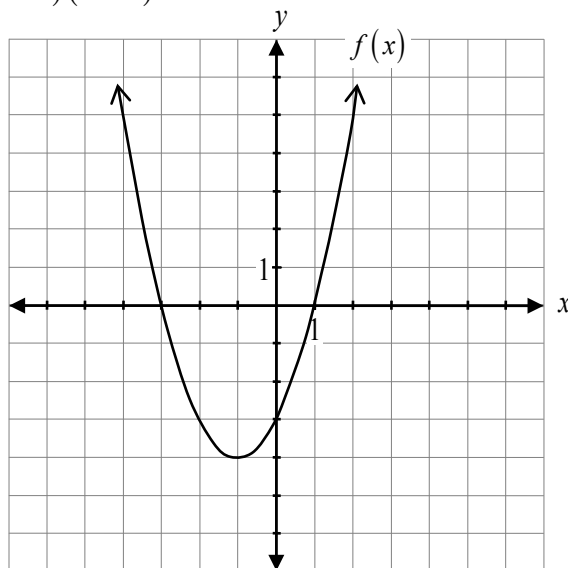
$p(x) =$ _____

Question 29

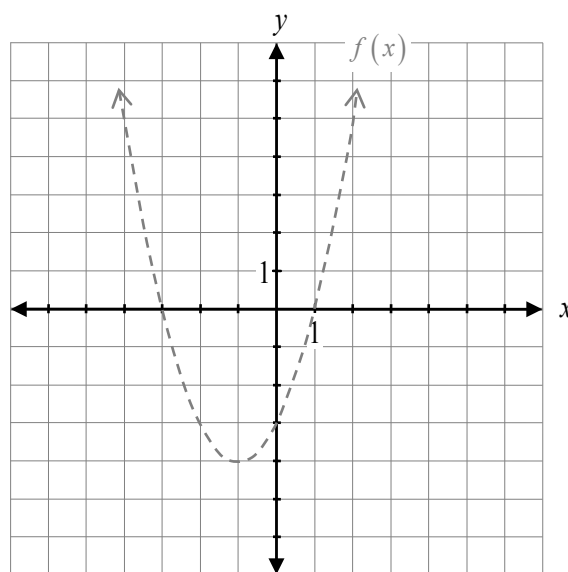
a) 3 marks b) 1 mark

120
121

Given the graph of $f(x) = (x+3)(x-1)$,



a) sketch the graph of $g(x) = \frac{1}{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

b) describe how to sketch the graph of $h(x) = |f(x)|$.

Question 30

1 mark 122

Describe how the value of m in the equation $y = \log_3(x - m)$, $m \in \mathbb{R}$, affects the asymptote on the graph of $y = \log_3 x$.

Question 31

2 marks 123

Solve algebraically.

$$25^x = \left(\frac{1}{5}\right)^{-3x+1}$$

Question 32

4 marks 124

Solve $\cos 2\theta = 0$, where $\theta \in \mathbb{R}$.

Question 33

1 mark 125

Describe a difference between the graphs of $y = f(x)$ and $y = g(x)$.

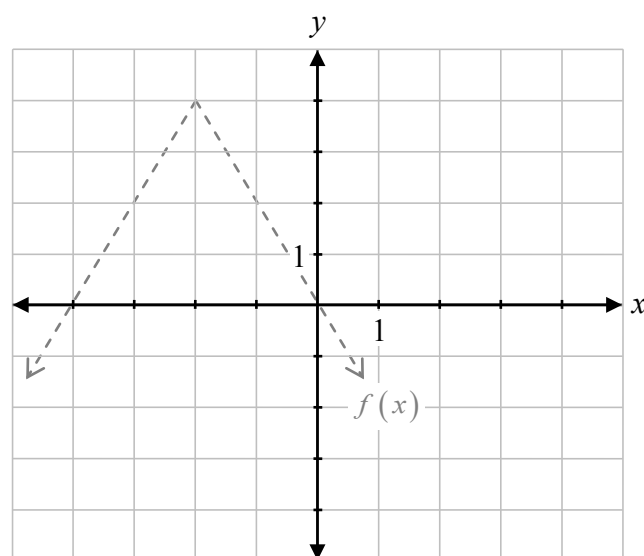
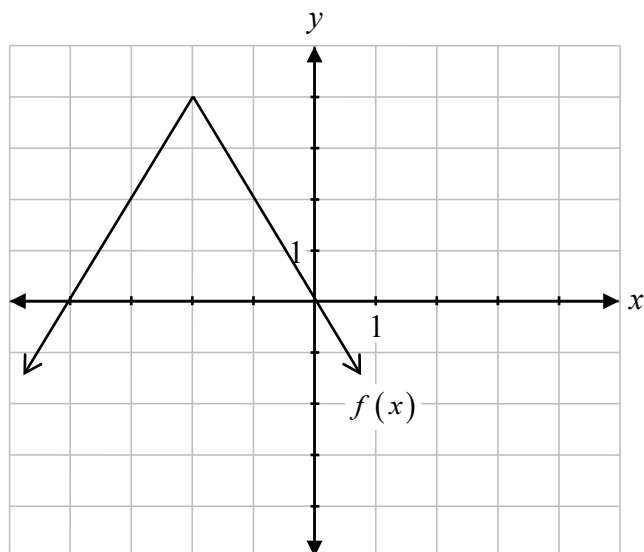
$$f(x) = -2(x+1)^2(x+3)$$

$$g(x) = 2(x+1)^2(x+3)$$

Question 34

2 marks 126

Given the graph of $y = f(x)$, sketch the graph of $\sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.
No marks will be awarded for the graph of $f(x)$.

Question 35

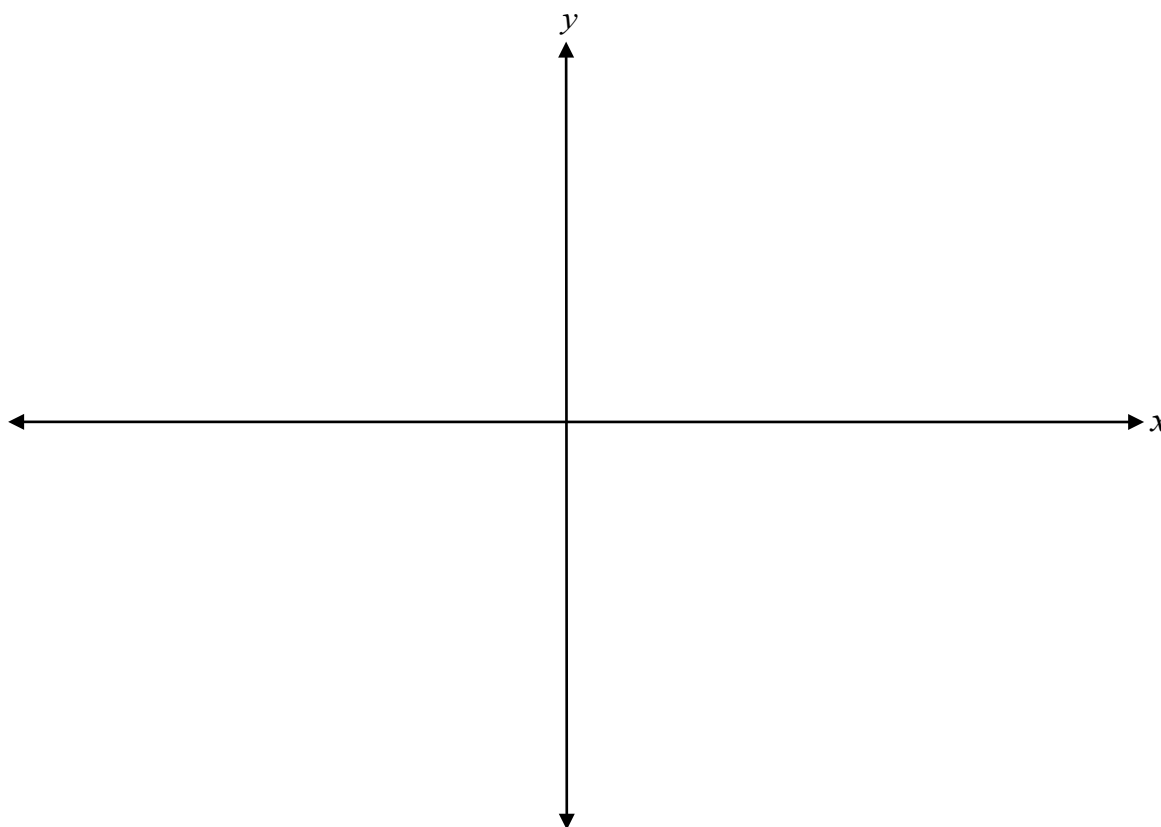
1 mark 127

Describe the relationship between the zeros of the function $f(x) = (2x - 1)(x + 3)^2$, the roots of the equation $(2x - 1)(x + 3)^2 = 0$, and the x -intercepts of the graph of $y = f(x)$.

Question 36

3 marks 128

Sketch a graph of at least one period of the function $f(x) = \cos\left[\frac{1}{2}\left(x + \frac{\pi}{2}\right)\right] - 3$.



Question 37

1 mark 129

Verify that $\theta = \frac{4\pi}{3}$ is a solution of the equation $4\cos^2 \theta - 1 = 0$.

Question 38

1 mark 130

Describe how to determine the equation of the horizontal asymptote of a rational function when the degree of the polynomial in the numerator and the degree of the polynomial in the denominator are equal.

Question 39

2 marks 131

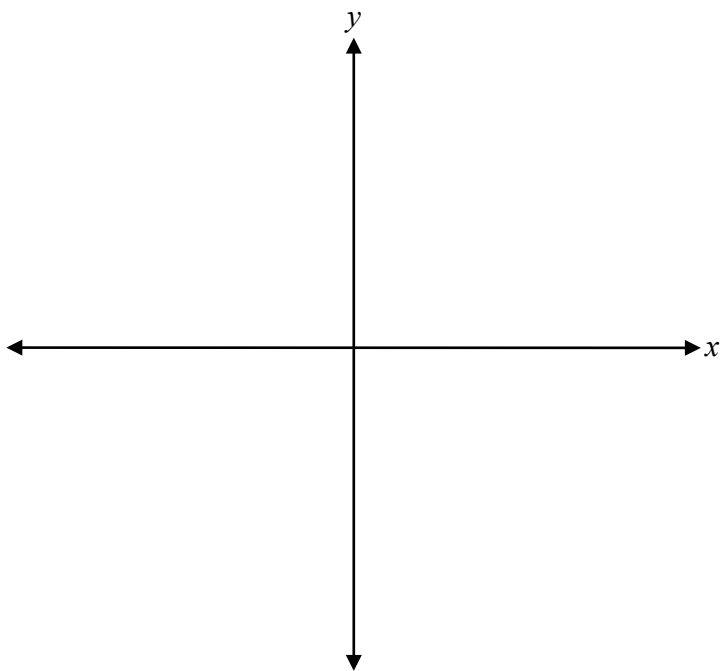
Evaluate.

$$\frac{\cot\left(-\frac{5\pi}{6}\right)}{\sin\left(\frac{17\pi}{3}\right)}$$

Question 40

4 marks 132

Sketch the graph of the function $f(x) = \frac{-1}{(x-1)^2}$ and determine the range.



Range: _____

Question 41

a) 1 mark b) 1 mark

133
134

Given $f(x) = \sqrt{x-2}$ and $g(x) = x^2 + 1$,

a) determine $g(f(x))$.

$g(f(x)) =$ _____

b) explain why the domain of $g(f(x))$ is restricted.

Question 42

3 marks 135

Solve algebraically.

$$2 \log_a 3 + \log_a 4 = 2, \text{ where } a > 0$$

Question 43

2 marks 136

Solve $\sec \theta + 2 = 0$ over the interval $[0, 2\pi]$.

Question 44

1 mark 137

Determine the x -intercept of the graph of $f(x) = e^x - 1$.

Question 45

1 mark 138

Given the 5th row of Pascal's triangle, determine the values of the next row.

1 4 6 4 1

Question 46

2 marks 139

Evaluate.

$$\log_2 80 - \log_2 10$$

Question 47

1 mark 140

State the amplitude of $f(x) = -2\sin(x - \pi) - 1$.

Question 48

3 marks 141

Determine the exact value of $\cos 15^\circ$.

Question 49

a) 1 mark b) 1 mark

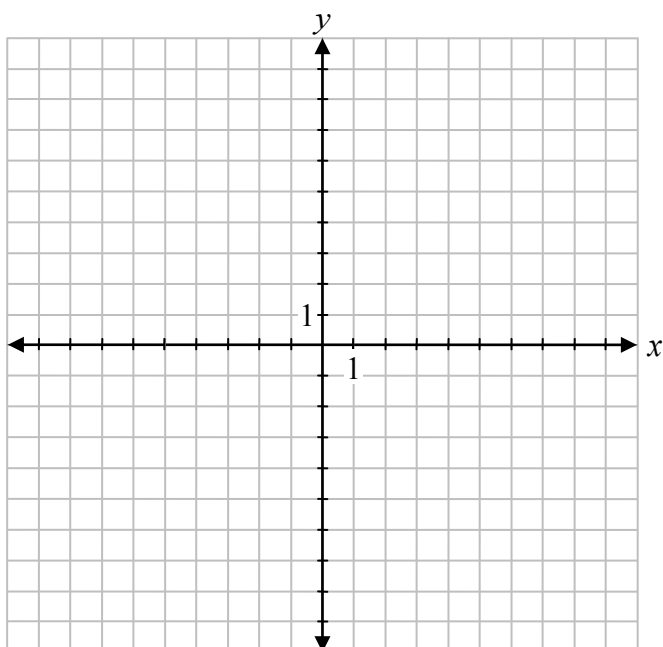
142
143

Given $f(x) = x^2 + 5x + 6$, $g(x) = x + 3$, and $h(x) = f(x) - g(x)$,

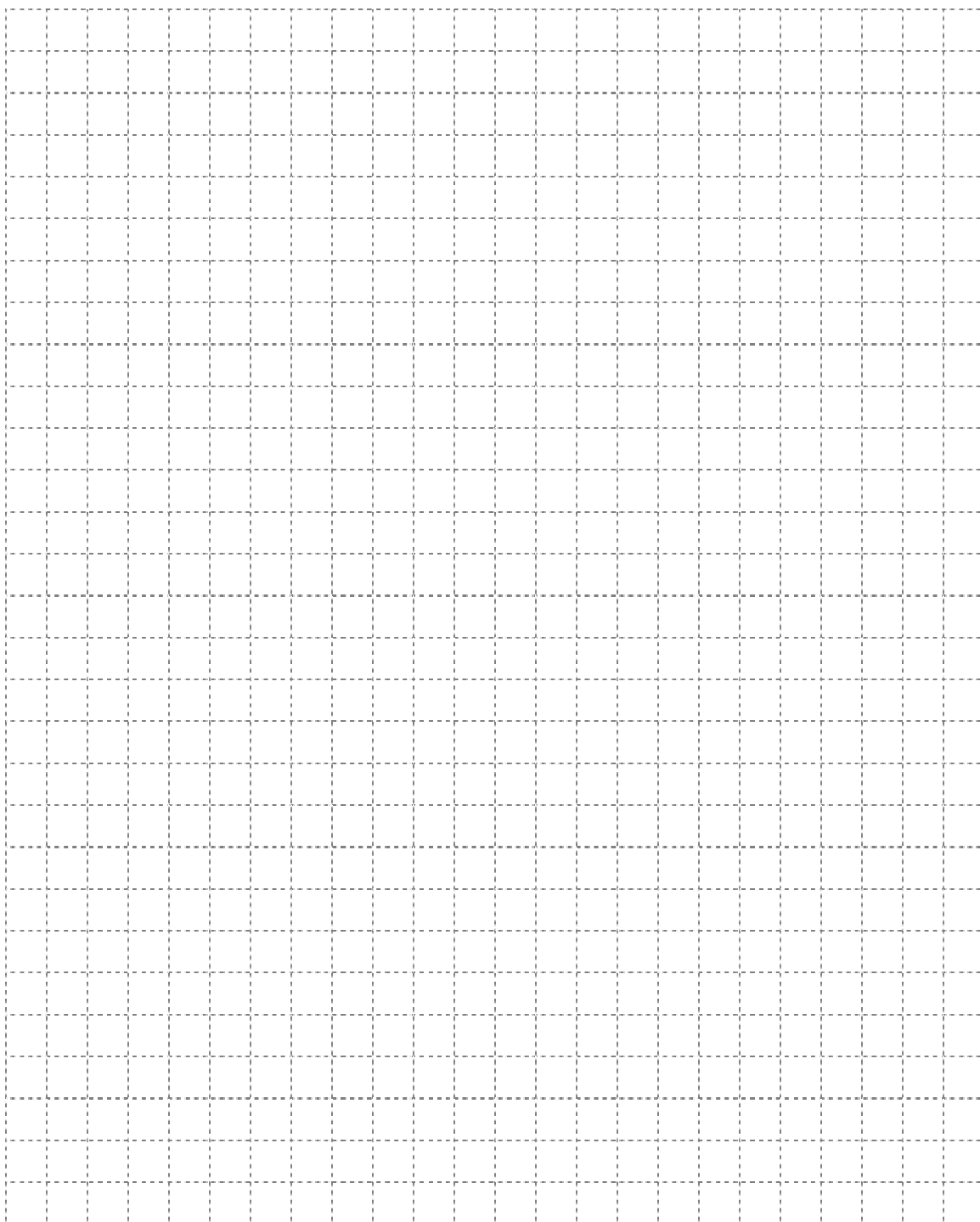
a) determine $h(x)$.

$h(x) =$ _____

b) sketch the graph of $y = h(x)$.



No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2018

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Booklet 1. January 2018

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1. Mathematics—Examinations, questions, etc.
 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Pre-calculus—Study and teaching (Secondary)—Manitoba.
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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	15	32
Booklet 2	9	25	57
Total	9	40	89

* The first 5 questions in *Booklet 1* require a calculator.  You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited.

No marks will be awarded for work done on this page.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a (MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a (M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

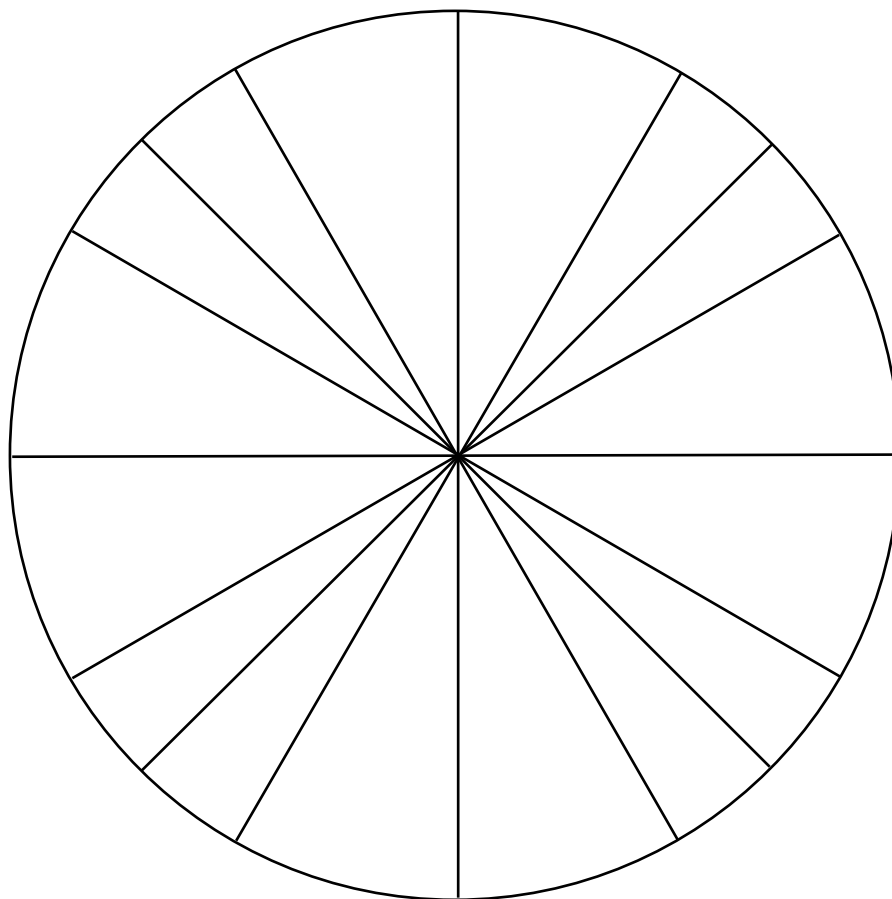
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

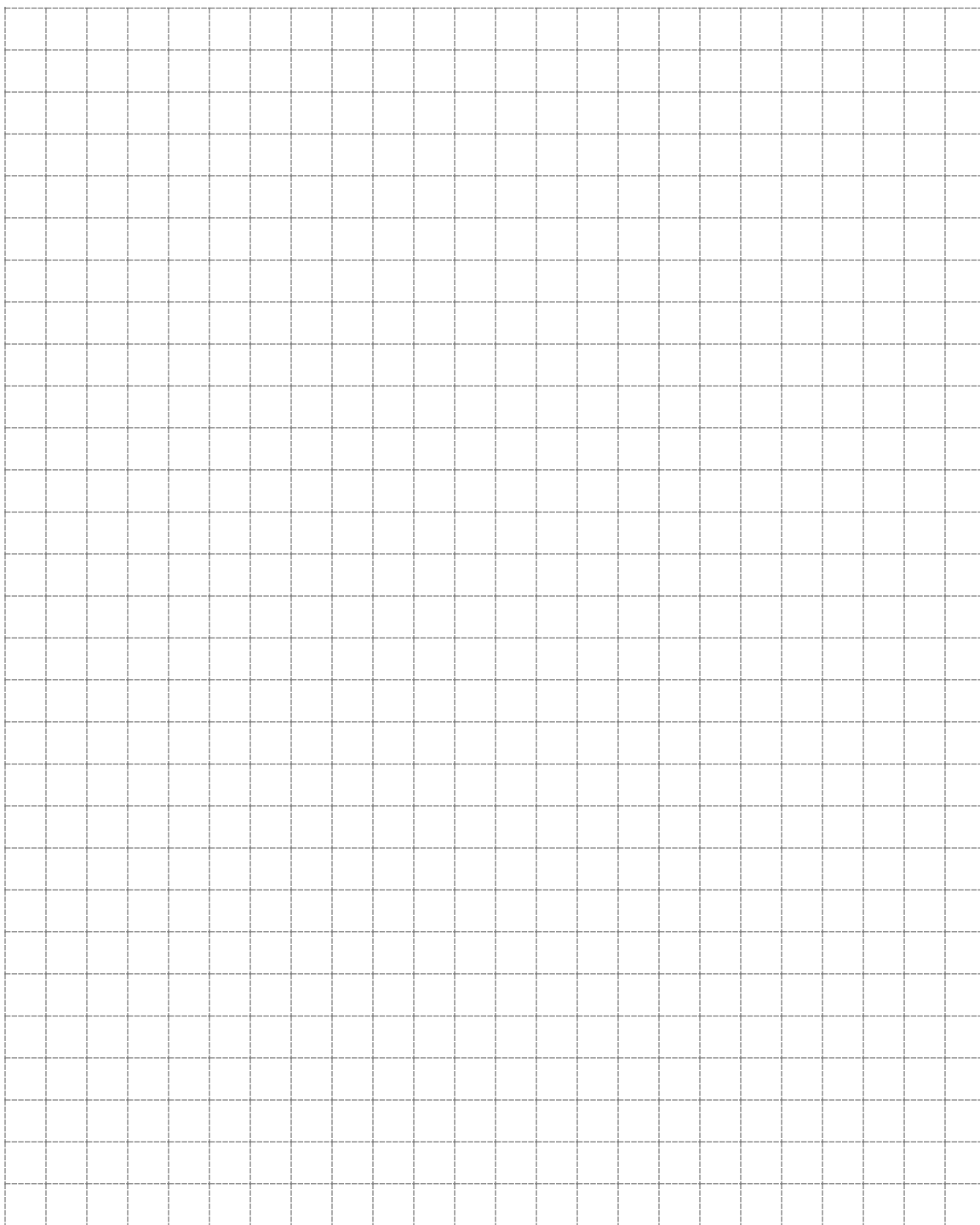
State: Give an answer without an explanation or a justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



Question 1

2 marks 101

A group of 7 friends decide to go to a movie.

Determine how many ways the friends can sit in a row if two of the friends refuse to sit next to each other.

Question 2

2 marks 102

Gabrielle listens to her radio at a sound level of 80 dB. She attended a music concert that had a sound level of 115 dB. Determine how many times more intense the music concert was than the radio.

You may use the formula:

$$\beta = 10 \log \left(\frac{I}{I_0} \right)$$

where β is the intensity level of sound, measured in dB

I is the intensity of sound

I_0 is the standard minimum intensity that a person can hear

Question 3

4 marks 103

Solve, algebraically.

$$2(7)^x = 3^{2x-3}$$

Question 4

4 marks 104

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta + 2 \csc \theta - 8 = 0$$

Question 5

1 mark 105

You have forgotten the code to unlock your cell phone. You know the code is made up of four numbers from 0 to 9.

Determine the number of possible codes, if repetition is allowed.

Note: A calculator is not required for the remaining test questions.

Question 6

2 marks 106

In the binomial expansion of $\left(\frac{7}{x^3} - 3x^7\right)^n$, the 5th term contains x^7 .

Determine the value of n .

Question 7

1 mark

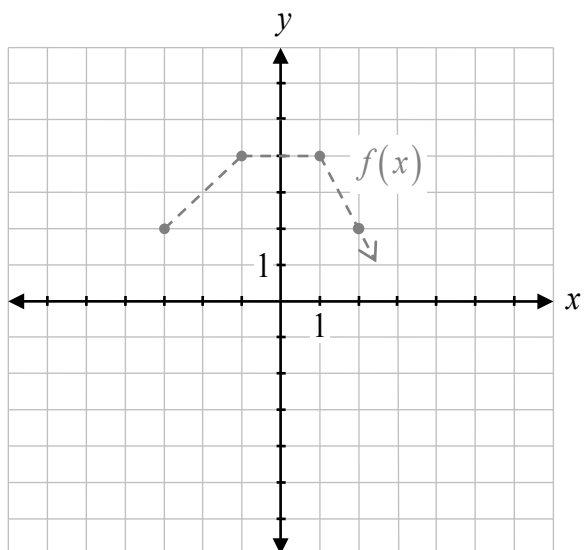
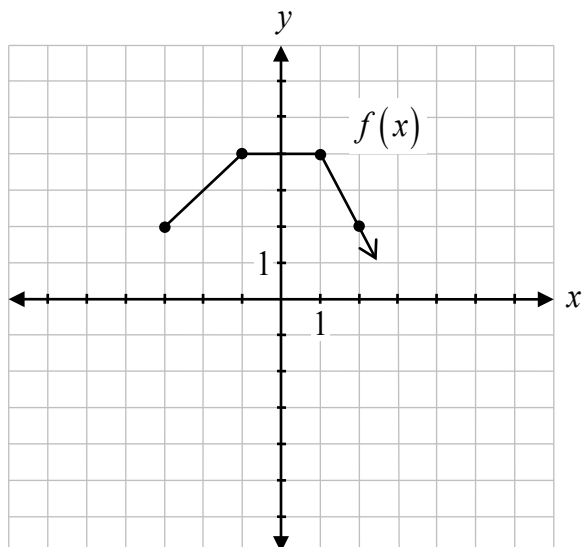
107

Given the domain of $f(x)$ is $\{-6, 1, 3, 4\}$ and the range of $f(x)$ is $\{-4, 7, 10, 15\}$, state the domain of $f^{-1}(x)$.

Question 8

1 mark 108

Given the graph of $y = f(x)$, sketch the graph of its inverse.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 9

3 marks

109

Prove the following identity for all permissible values of θ .

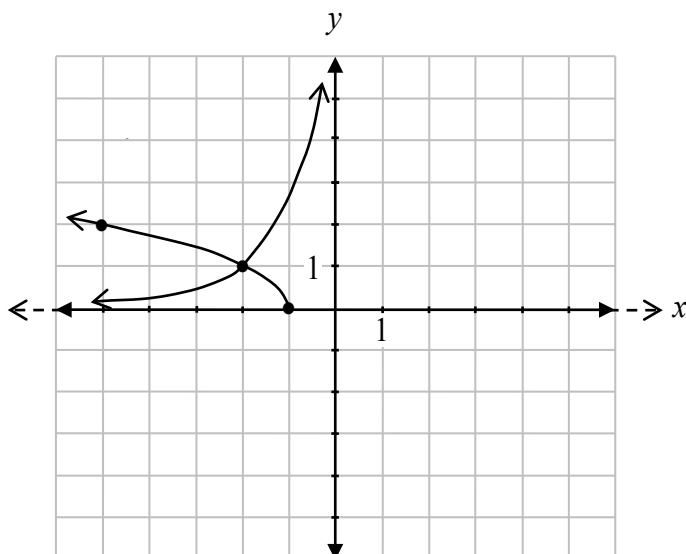
$$\frac{1 + \cos \theta}{1 - \sin^2 \theta} = \sec \theta + \tan^2 \theta + 1$$

Left-Hand Side	Right-Hand Side

Question 10

1 mark 110

Thomas used graphs to solve the equation $e^{x+2} = \sqrt{-(x+1)}$.



He incorrectly states the solution as $(-2, 1)$.

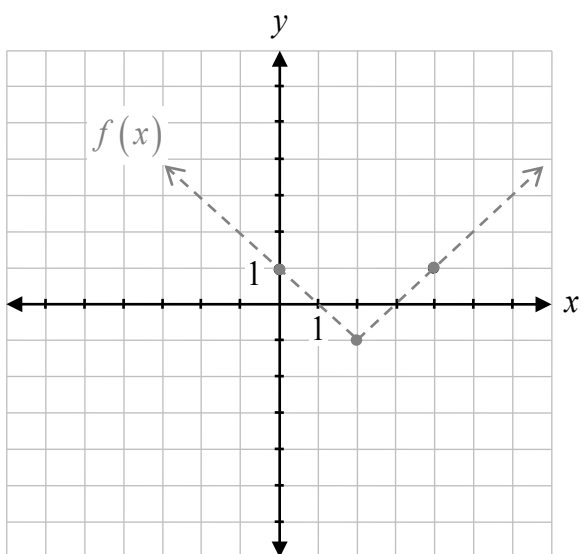
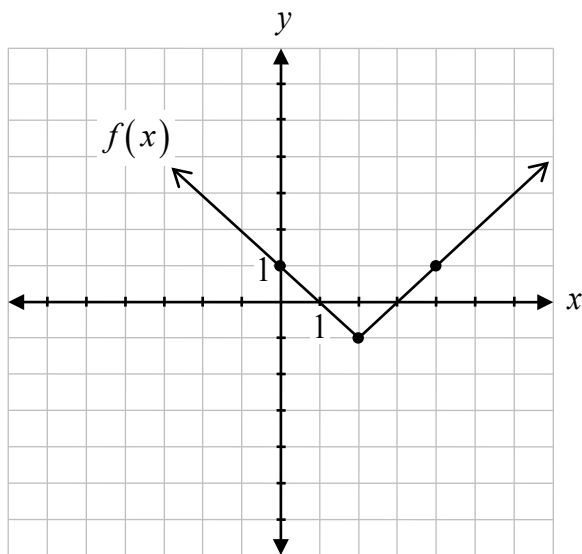
Describe how Thomas should have stated the solution.

Question 11

2 marks

111

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 12

a) 1 mark b) 1 mark

112
113

When a polynomial, $P(x)$, is divided by $(x - 2)$ the resulting equation is

$$\frac{P(x)}{x - 2} = x^2 - x + 1 + \frac{3}{x - 2}.$$

a) Explain why $x - 2$ is not a factor of $P(x)$.

b) Determine the equation for the polynomial function $P(x)$.

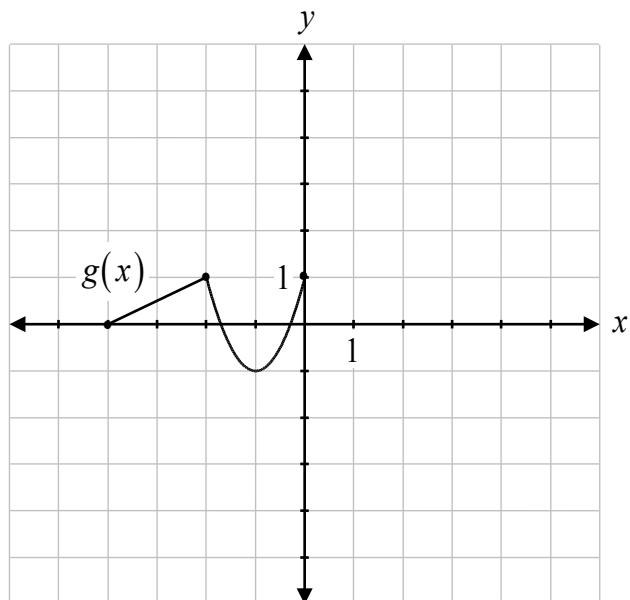
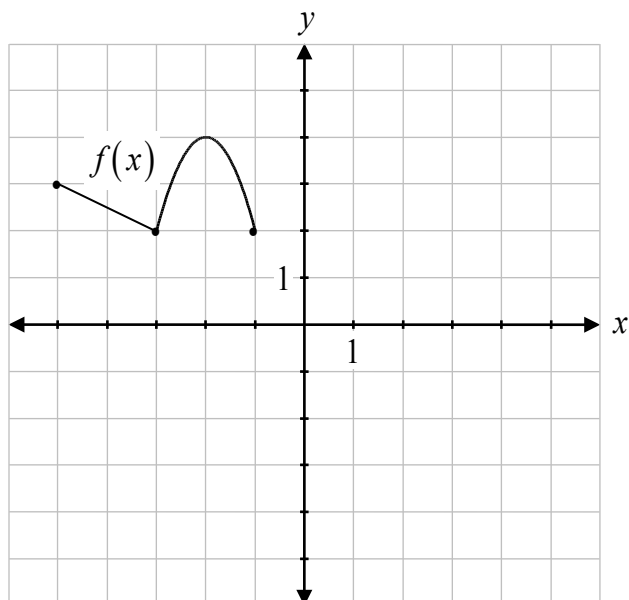
$$P(x) = \underline{\hspace{10cm}}$$

Question 13

3 marks

114

Determine the equation for $g(x)$ in terms of $f(x)$.



$g(x) =$ _____

Question 14

1 mark

115

Explain why the binomial expansion of $(2x + y)^9$ does not have a middle term.

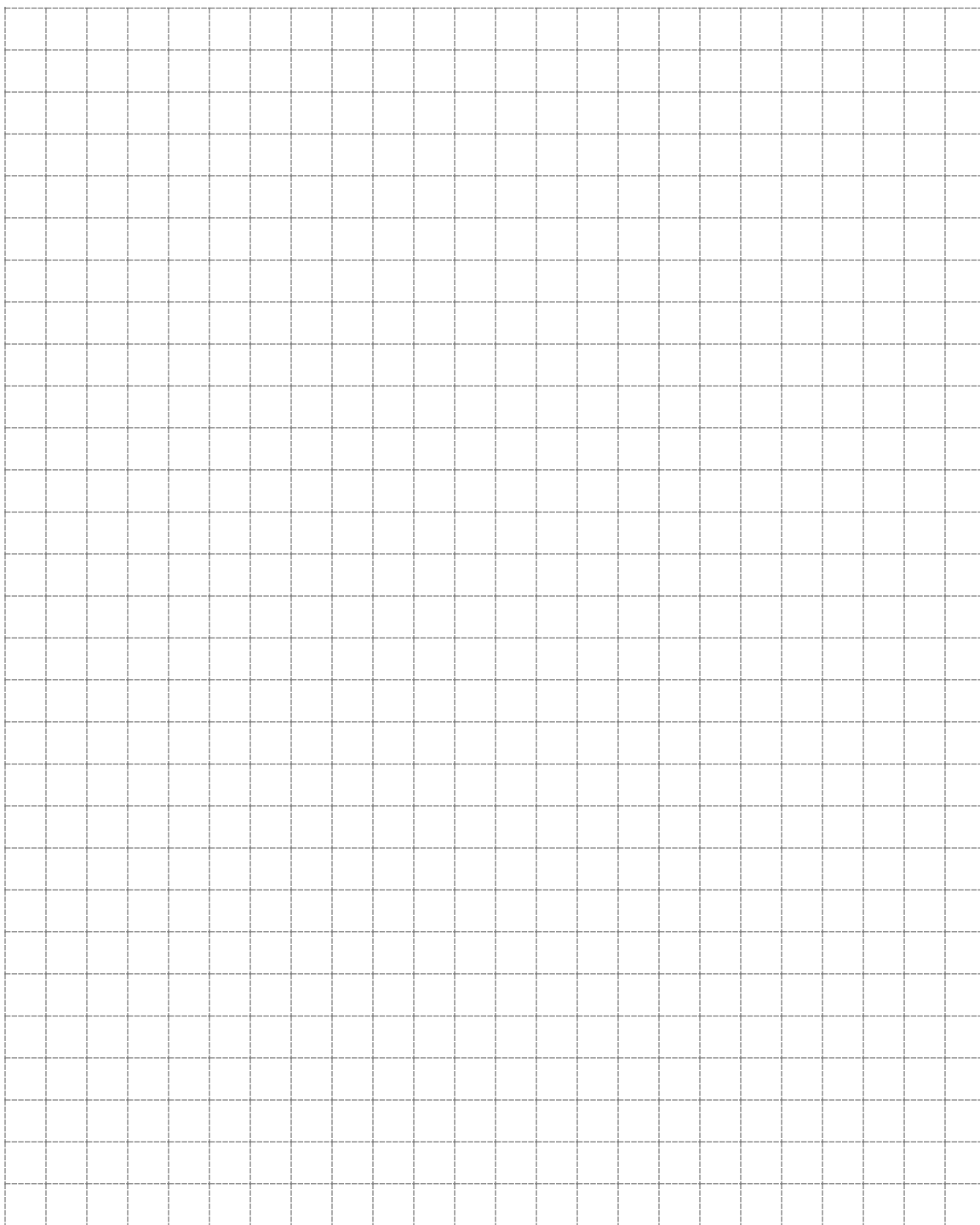
Question 15

3 marks

116

Using the laws of logarithms, completely expand the expression $\log\left(\frac{5\sqrt{a}}{b^3}\right)$.

No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2018

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Booklet 2. January 2018

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1. Mathematics—Examinations, questions, etc.
 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	15	32
Booklet 2	9	25	57
Total	9	40	89

* The first 5 questions in *Booklet 1* require a calculator.  You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited.

No marks will be awarded for work done on this page.

Question 16**1 mark**

Identify 10° in radians.

a) $\frac{1800}{\pi}$

b) $\frac{\pi}{1800}$

c) $\frac{18}{\pi}$

d) $\frac{\pi}{18}$

Question 17**1 mark**

The polynomial function, $P(x) = a(x-1)^2(x+4)^2$, has a y -intercept of -8 .

Identify the value of a .

a) -2

b) $-\frac{1}{2}$

c) $\frac{1}{2}$

d) 2

Question 18**1 mark**

Identify the value of $\log_4\left(\frac{1}{16}\right)$.

- a) -2
- b) $-\frac{1}{2}$
- c) $\frac{1}{2}$
- d) 2

Question 19**1 mark**

Given the angle $\frac{25\pi}{7}$, identify the coterminal angle on the interval $[-2\pi, 0]$.

- a) $\frac{18\pi}{7}$
- b) $\frac{11\pi}{7}$
- c) $-\frac{3\pi}{7}$
- d) $-\frac{10\pi}{7}$

Question 20**1 mark**

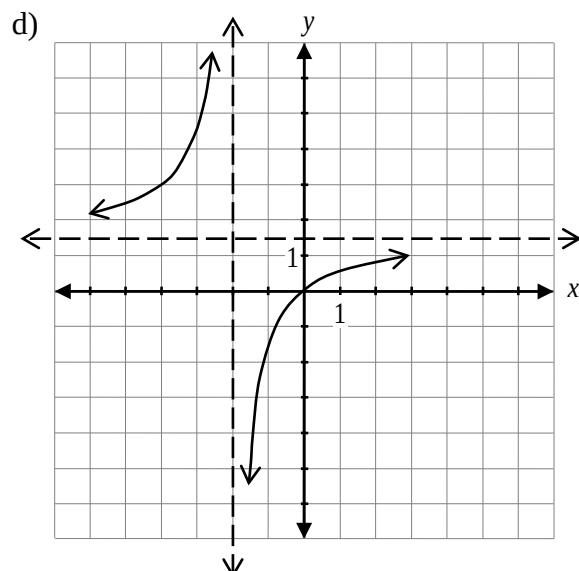
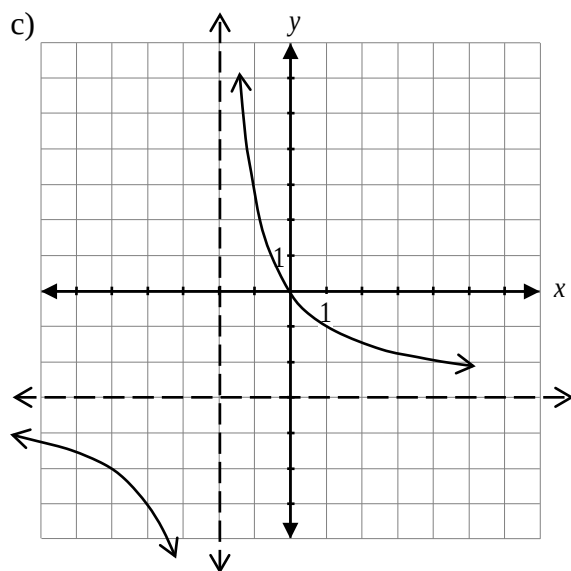
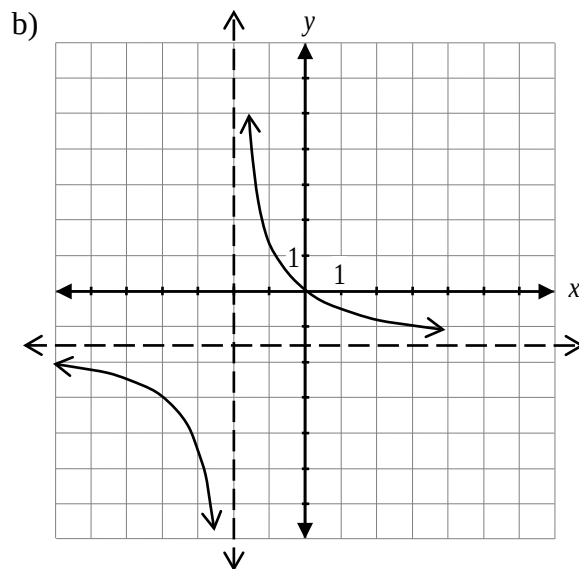
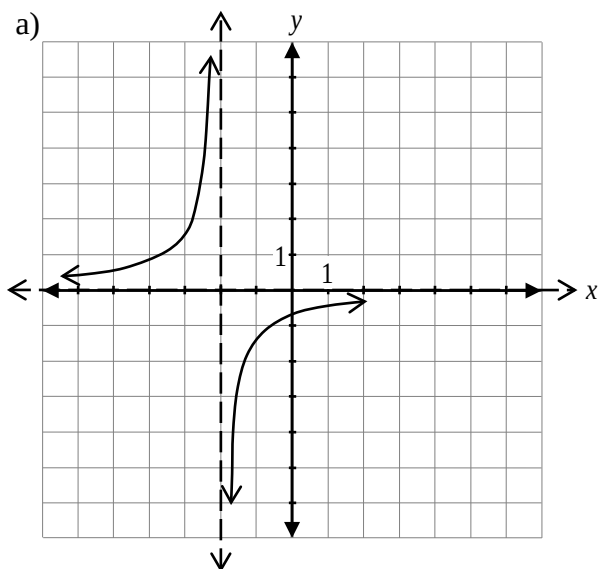
Identify which expression cannot be evaluated.

- a) ${}_7P_0$
- b) ${}_7P_6$
- c) ${}_7P_7$
- d) ${}_7P_8$

Question 21

1 mark

Identify the graph of $f(x) = \frac{-3x}{2x+4}$.



Question 22**1 mark**

Given a point $(-2, 0)$ on the graph of $y = f(x)$, identify the coordinates of the corresponding point on the graph of $y = 4f\left(\frac{1}{2}x\right)$.

- a) $(-8, 0)$
- b) $(-4, 0)$
- c) $(-2, 0)$
- d) $(-1, 0)$

Question 23**1 mark**

Identify the non-permissible value of θ for the expression $\frac{\cos \theta}{1 + \sin \theta}$.

- a) $\frac{\pi}{2}$
- b) π
- c) $\frac{3\pi}{2}$
- d) 2π

Question 24**1 mark**

Identify the function with an asymptote at $x = -3$.

- a) $y = \log(x + 3)$
- b) $y = \log x + 3$
- c) $y = \log(x - 3)$
- d) $y = \log x - 3$

Question 25

3 marks 117

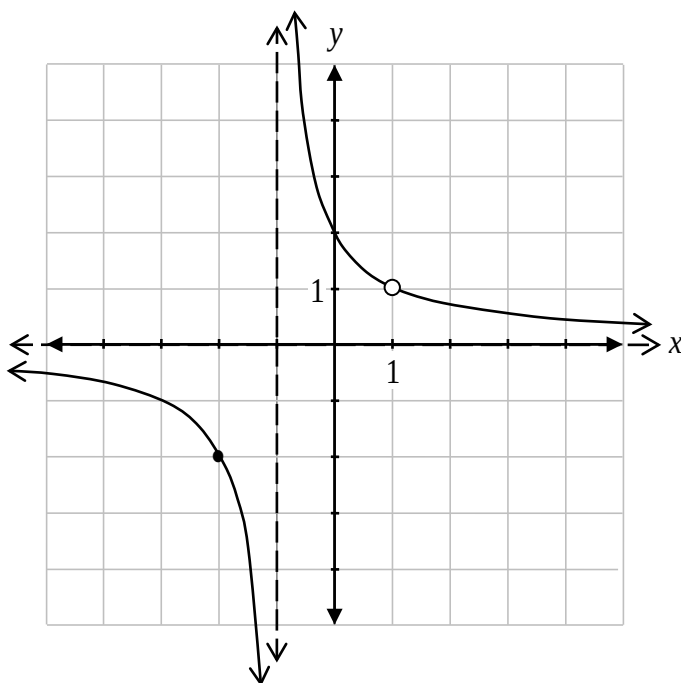
Evaluate the following expression.

$$\tan\left(\frac{2\pi}{3}\right)\csc\left(\frac{-2\pi}{3}\right) + \cos(3\pi)$$

Question 26

1 mark 118

State the range of the graph below.

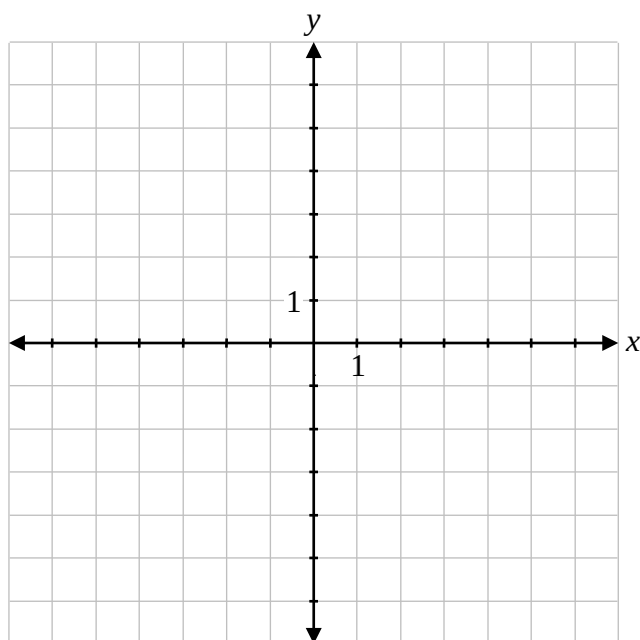


Range: _____

Question 27

2 marks 119

Sketch the graph of the function $f(x) = \frac{2x^2 - 5x}{x}$.



Question 28

1 mark 120

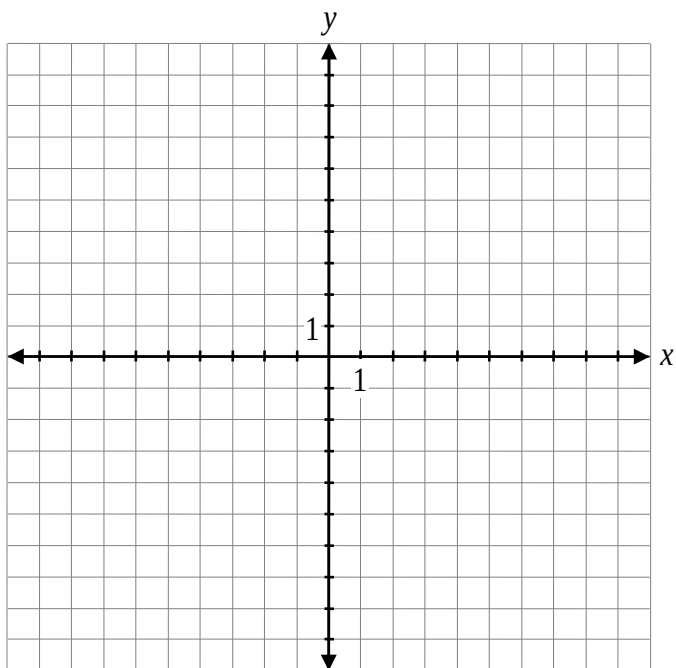
State a possible value of n if the polynomial function $P(x) = (x - 1)^2(x + 2)^n$ has a range of $[0, \infty)$.

Question 29

2 marks

121

Sketch the graph of $y = \left(\frac{1}{2}\right)^{x-1}$.



Question 30**1 mark**

122

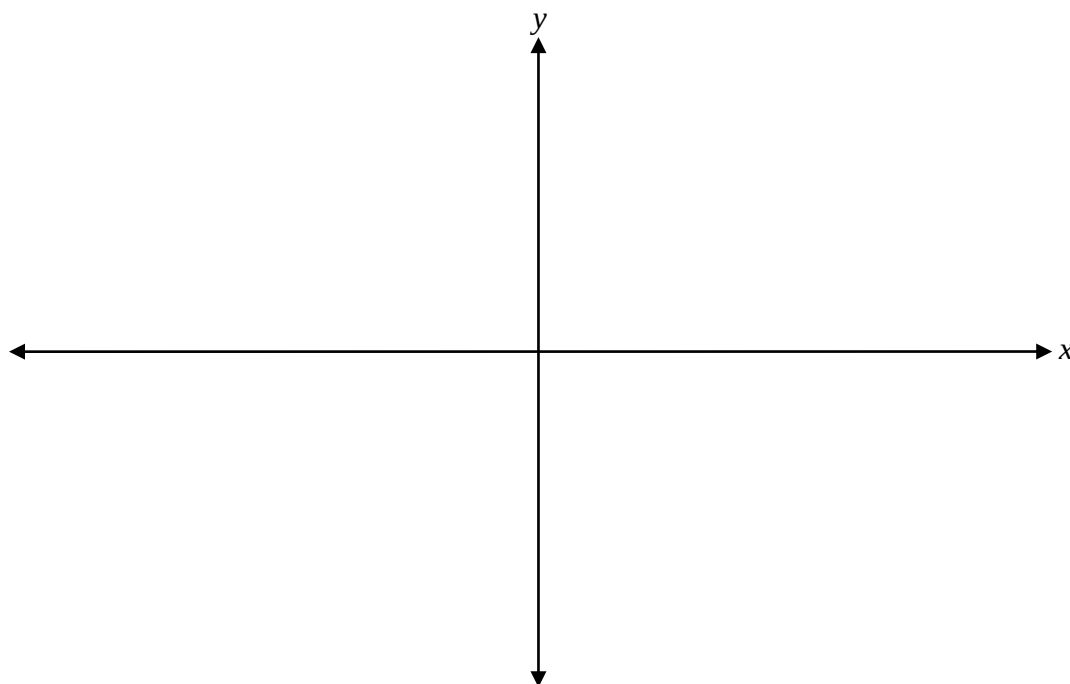
Solve.

$$\log_x 27 = 3$$

Question 31**2 marks**

123

Sketch at least two periods of the graph $y = \tan x$.

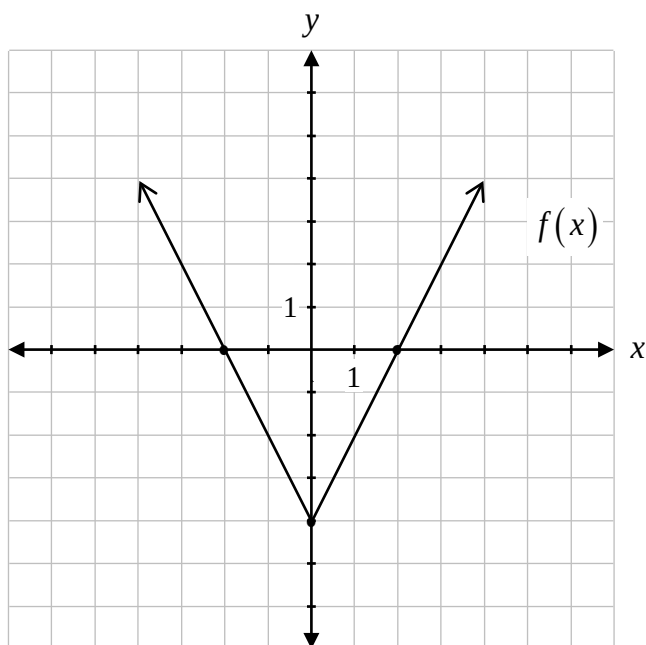


Question 32

1 mark

124

Given the graph of $f(x)$, determine the domain of $\frac{1}{f(x)}$.



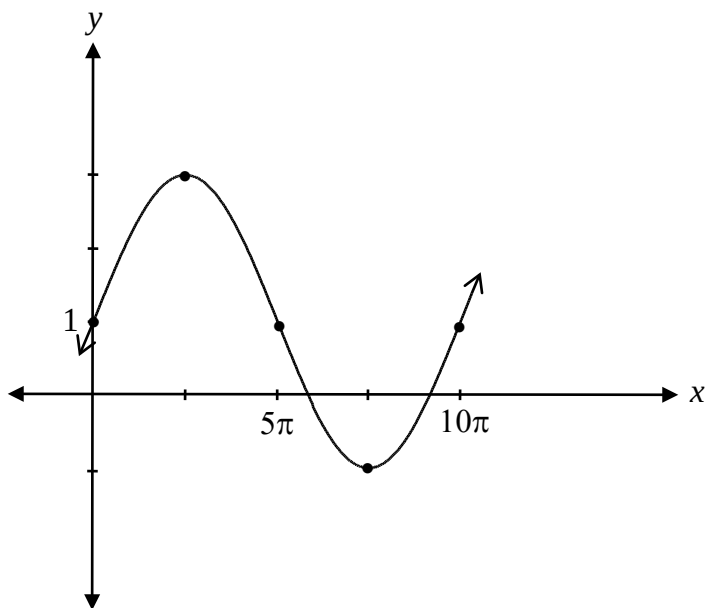
Domain: _____

Question 33

3 marks

125

Determine the values of A, B, and D of the sinusoidal function in the form $y = A \sin(Bx) + D$.



A = _____

B = _____

D = _____

Question 34

1 mark 126

Determine if the point $\left(-\frac{\sqrt{7}}{5}, \frac{2}{5}\right)$ is on the unit circle.

Justify your answer.

Question 35

3 marks 127

Solve, algebraically.

$$\frac{{}_nC_5}{{}_nC_4} = 6$$

Question 36

2 marks

128

Given $\sin \alpha = \frac{4}{5}$, where α is in quadrant II, determine the exact value of $\sin 2\alpha$.

Question 37

a) 1 mark b) 2 marks

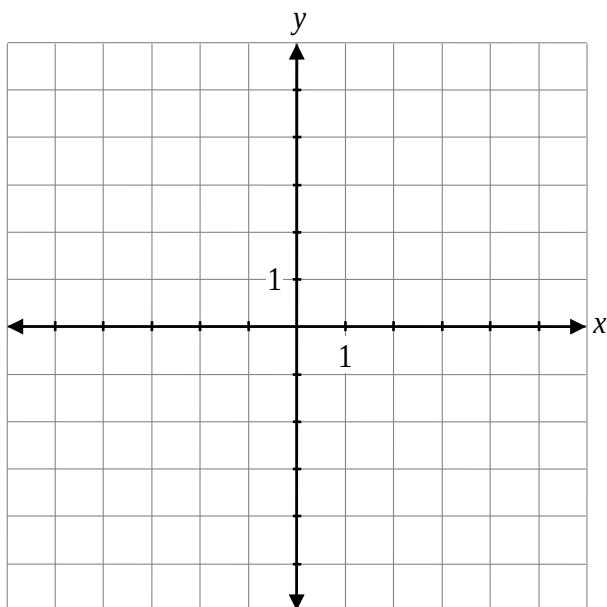
129
130

Given the functions $f(x) = x + 1$ and $g(x) = \sqrt{x}$,

a) determine the equation of $g(f(x))$.

$$g(f(x)) = \underline{\hspace{4cm}}$$

b) sketch the graph of $g(f(x))$.



Question 38

1 mark 131

Steve is asked to determine an equation with a larger period than the period of the graph of $y = \cos(2x)$.

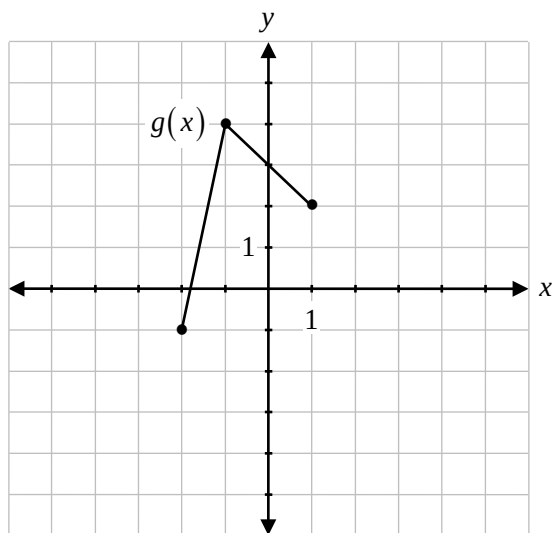
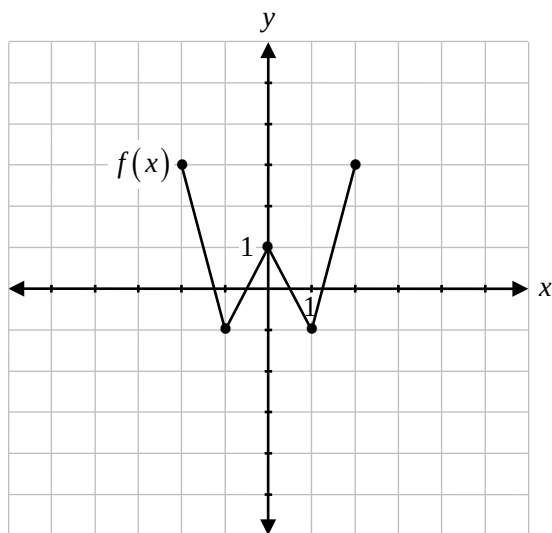
Justify why Steve's answer of $y = \cos(6x)$ is incorrect.

Question 39

a) 1 mark b) 1 mark

132
133

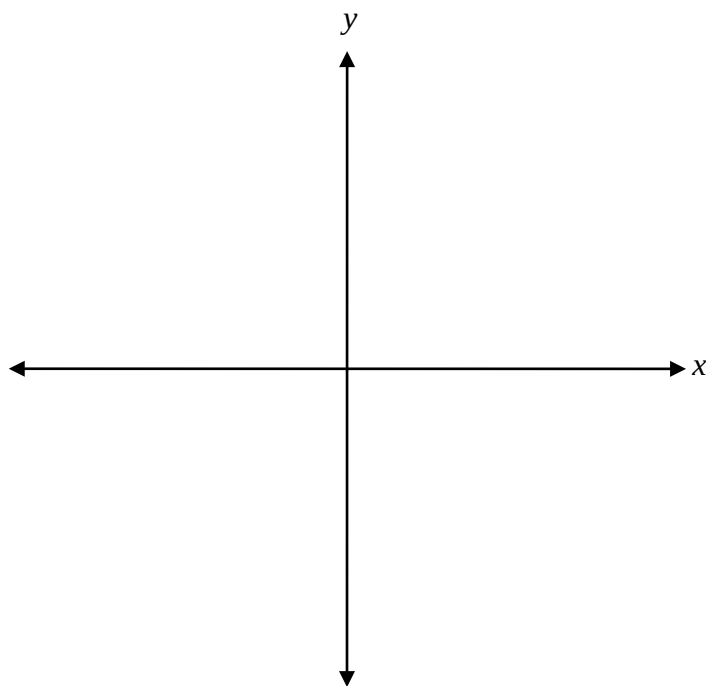
Given the graphs of $f(x)$ and $g(x)$,



a) determine the value of $(f \cdot g)(-1)$.

b) determine the value of $g(f(0))$.

Sketch the graph of $P(x) = -(x-1)^3(x-3)(x+1)$.



Question 41

a) 1 mark b) 1 mark

135
136

The point $(-\sqrt{3}, 1)$ is on the terminal arm of an angle θ , in standard position.

a) Determine $\tan \theta$.

b) Determine a possible value of θ , in radians.

Question 42**1 mark**

137

Describe the transformation used to obtain the graph of $y = \log_5 x$ given the graph of $y = 5^x$.

Question 43**2 marks**

138

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$, where $\theta \in \mathbb{R}$.

Question 44**1 mark** 139

Given that the point (a, b) is on the graph of $f(x)$, describe how you would determine the corresponding point on the graph of $y = \sqrt{f(x)}$.

Question 45**1 mark** 140

Evaluate.

$$\cos\left(\frac{\pi}{20}\right)\cos\left(\frac{\pi}{5}\right) - \sin\left(\frac{\pi}{20}\right)\sin\left(\frac{\pi}{5}\right)$$

Question 46**3 marks**

141

Describe the transformations used to obtain the graph of the function $y = f(-x + 6) - 8$ from the graph of $y = f(x)$.

Question 47**2 marks**

142

State the equations of all the asymptotes of the function, $y = \frac{1}{3x+1}$.

Question 48

3 marks

143

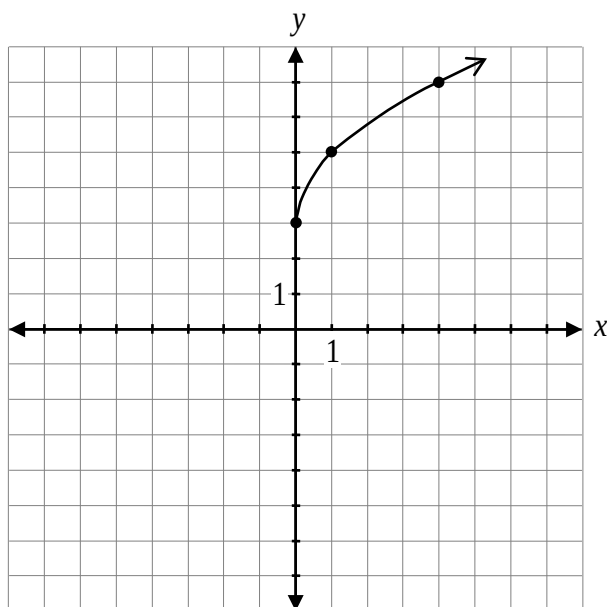
Determine the zeros of the polynomial function $P(x) = 2x^3 + 5x^2 - 4x - 3$.

Question 49

2 marks

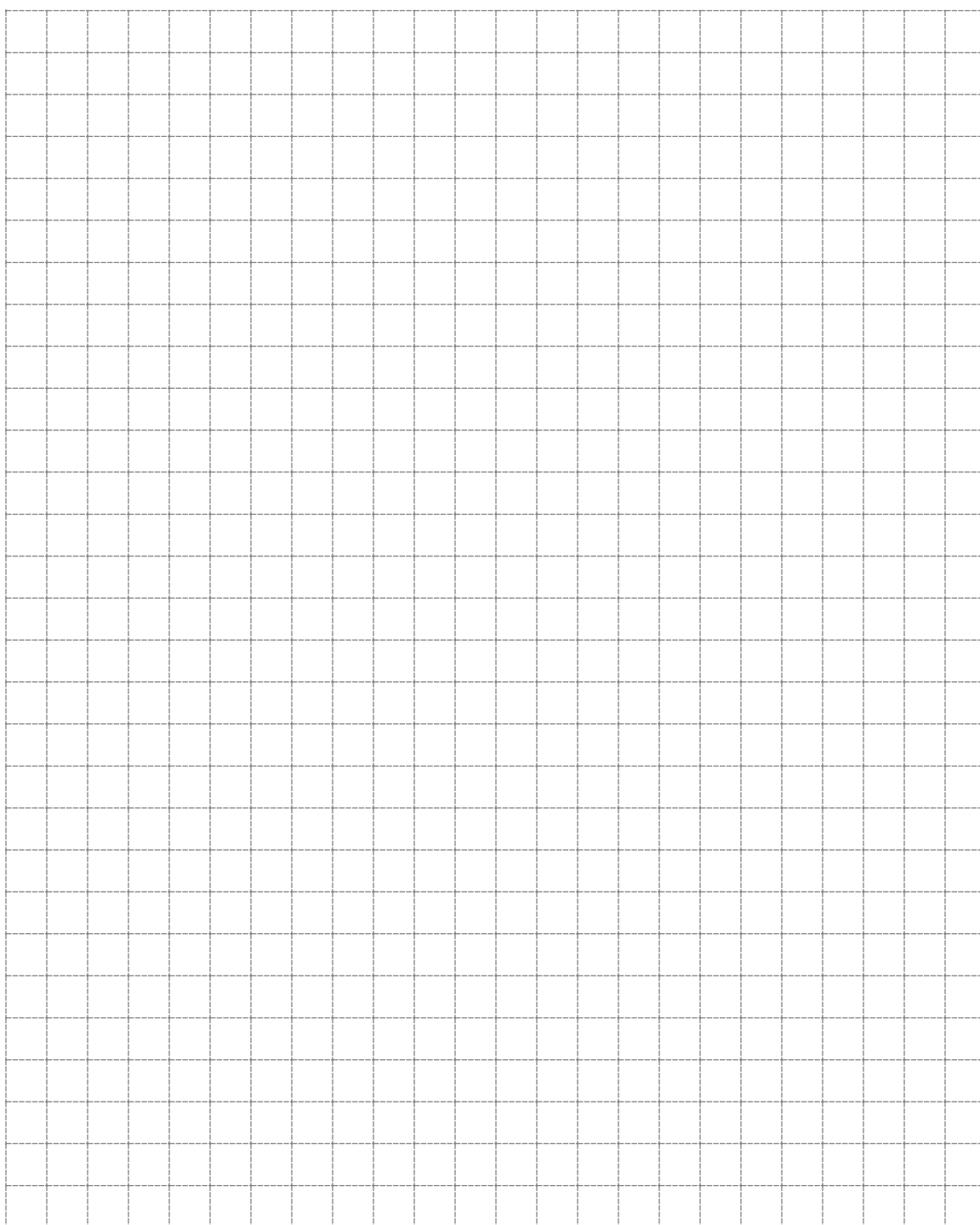
144

Determine the equation of the radical function represented by the graph.



$y =$ _____

No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2018

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Booklet 1. June 2018

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Grade 12 Pre-Calculus Mathematics Achievement Test

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* The first 6 questions in *Booklet 1* require a calculator. 
You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited.

No marks will be awarded for work done on this page.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a (MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a (M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

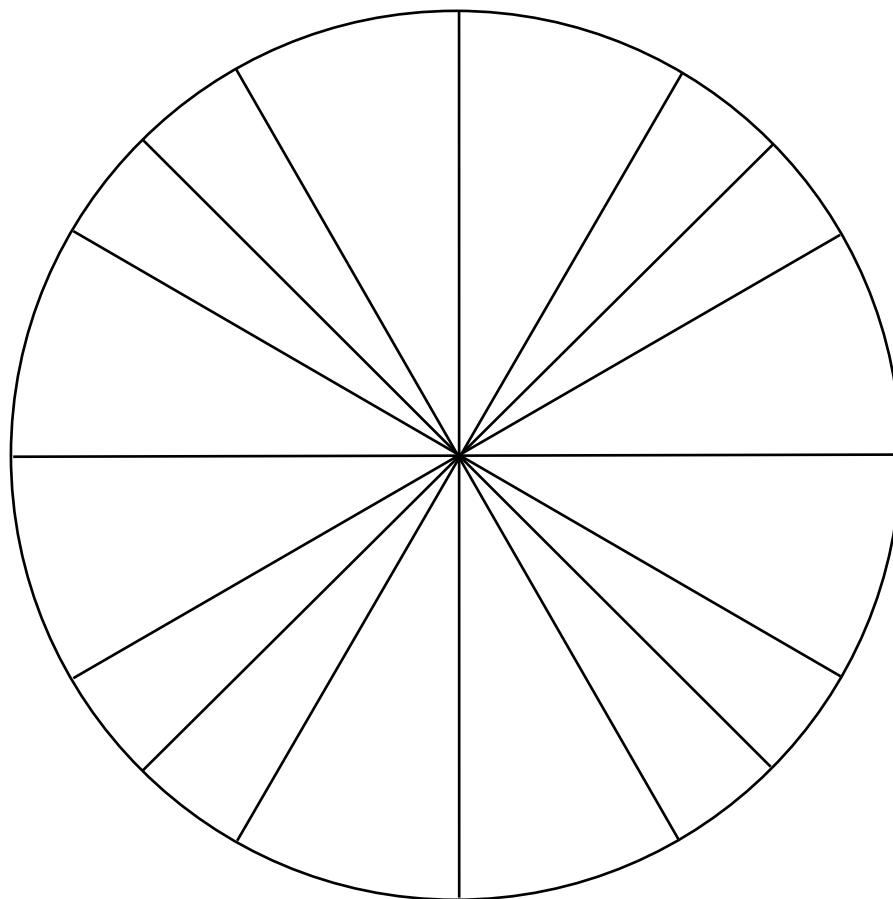
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

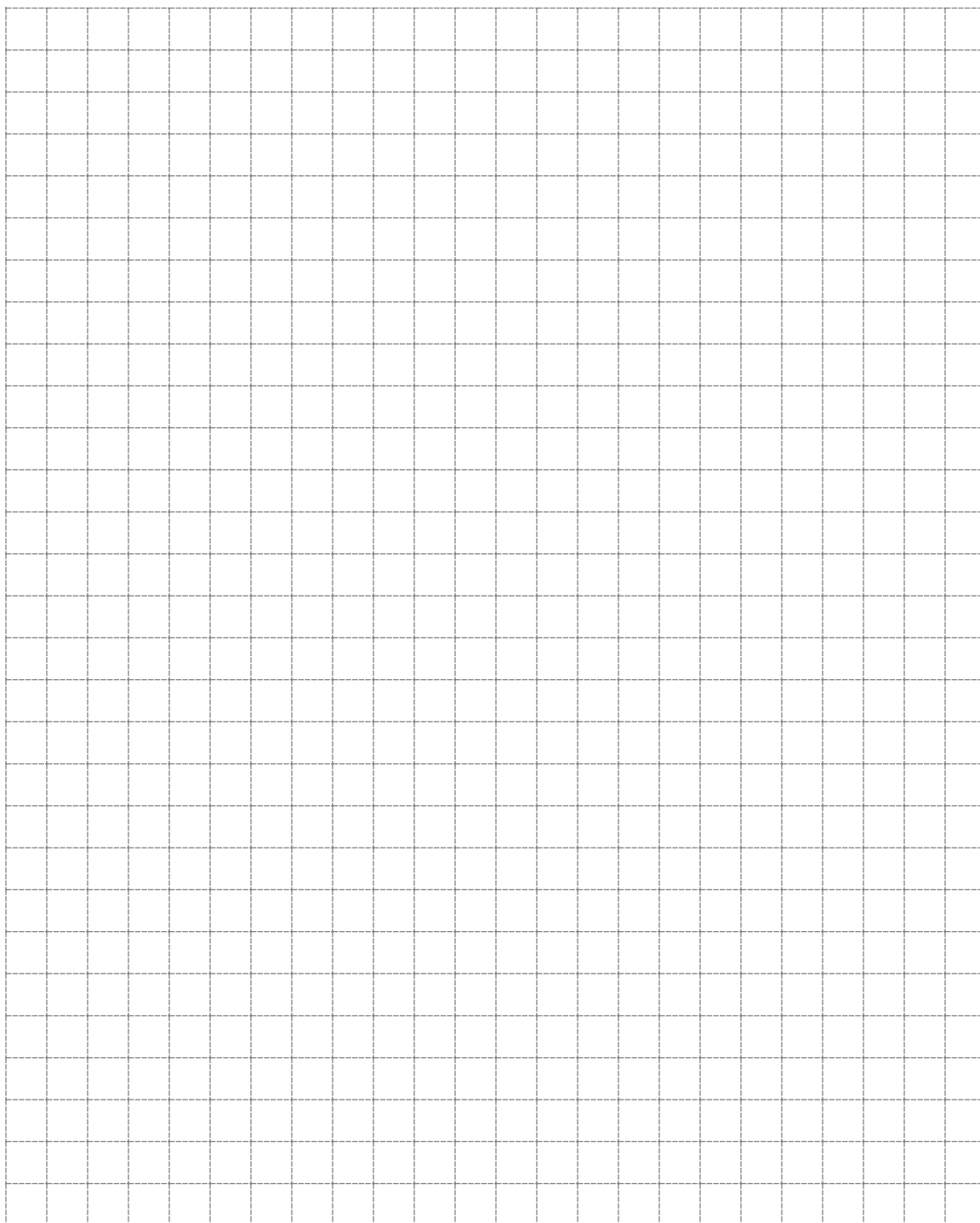
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



Question 1

2 marks 101

Pierre pushes his car into a garage. The radius of a tire on his car is 22 cm. Determine the distance travelled by his car if the tire rotated a total of 1000° .

Question 2

2 marks 102

Solve, algebraically.

$$7^{\frac{x}{2}} = 85$$

Question 3

4 marks 103

Solve, algebraically, over the interval $[0, 2\pi)$.

$$\sin x(\sec x + 3) = 0$$

Question 4

3 marks 104

Brahim invests \$2500 at an annual interest rate of 6.75% compounded monthly. Determine, algebraically, how many years it will take for his investment to reach an amount of \$10 500.

Use the formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A = the amount of the investment after t years

P = the principal of the investment

r = the annual interest rate (as a decimal)

n = the number of compounding periods per year

t = the length of the investment in years

Question 5

2 marks 105

There are 13 adults and 18 children who can be selected to go on a trip. Determine the number of ways 4 adults and 7 children can be selected if Sandra, one of the adults, must be selected.

In the binomial expansion of $\left(\frac{2}{x^2} - x^3\right)^9$, determine and simplify the 6th term.

Note: A calculator is not required for the remaining test questions.

Question 7

1 mark 107

Given that $f(x) = \{(-1, 0), (0, 2), (1, -3), (2, 4)\}$, evaluate $f(f(0))$.

Question 8

1 mark 108

The point $\left(-\frac{5}{6}, b\right)$ is on the unit circle and is in quadrant III.

Determine the exact value of b .

Question 9

1 mark 109

Given the following row of Pascal's Triangle, determine the values of the next row.

1 6 15 20 15 6 1

Question 10

3 marks

110

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the x -axis
- vertical stretch by a factor of 3
- horizontal stretch by a factor of 4

State the equation of $g(x)$ in terms of $f(x)$.

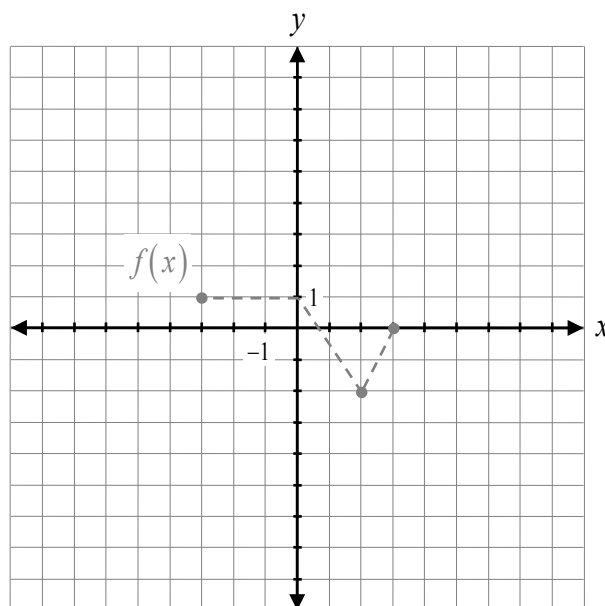
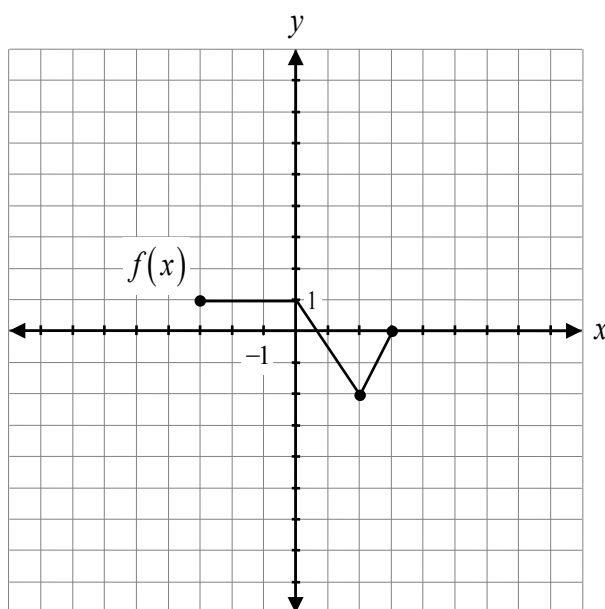
$g(x) =$ _____

Question 11

3 marks

111

Given the graph of $f(x)$, sketch the graph of $y + 1 = 2f(x - 3)$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$

Question 12

1 mark 112

State the equation of the horizontal asymptote of $f(x) = \frac{2x^2 - 3x + 5}{4x^2 + 2x - 7}$.

Question 13

2 marks

113

Given that $(x + 4)$ is one of the factors of $p(x) = x^3 + 6x^2 - 32$, express $p(x)$ in completely factored form.

$p(x) =$ _____

Question 14

3 marks

114

Prove the identity for all permissible values of θ .

$$\frac{2\cos^2\theta}{1-\cot\theta} = \frac{\sin 2\theta}{\tan\theta - 1}$$

Left-Hand Side	Right-Hand Side

Question 15

1 mark 115

Restaurant A has 5 types of hamburgers, 2 types of french fries, and 10 types of drinks.

Restaurant B has 4 types of hamburgers, 5 types of french fries, and 6 types of drinks.

If a meal is made up of a hamburger, french fries, and a drink, justify which restaurant offers a greater variety of meals.

Question 16

2 marks

116

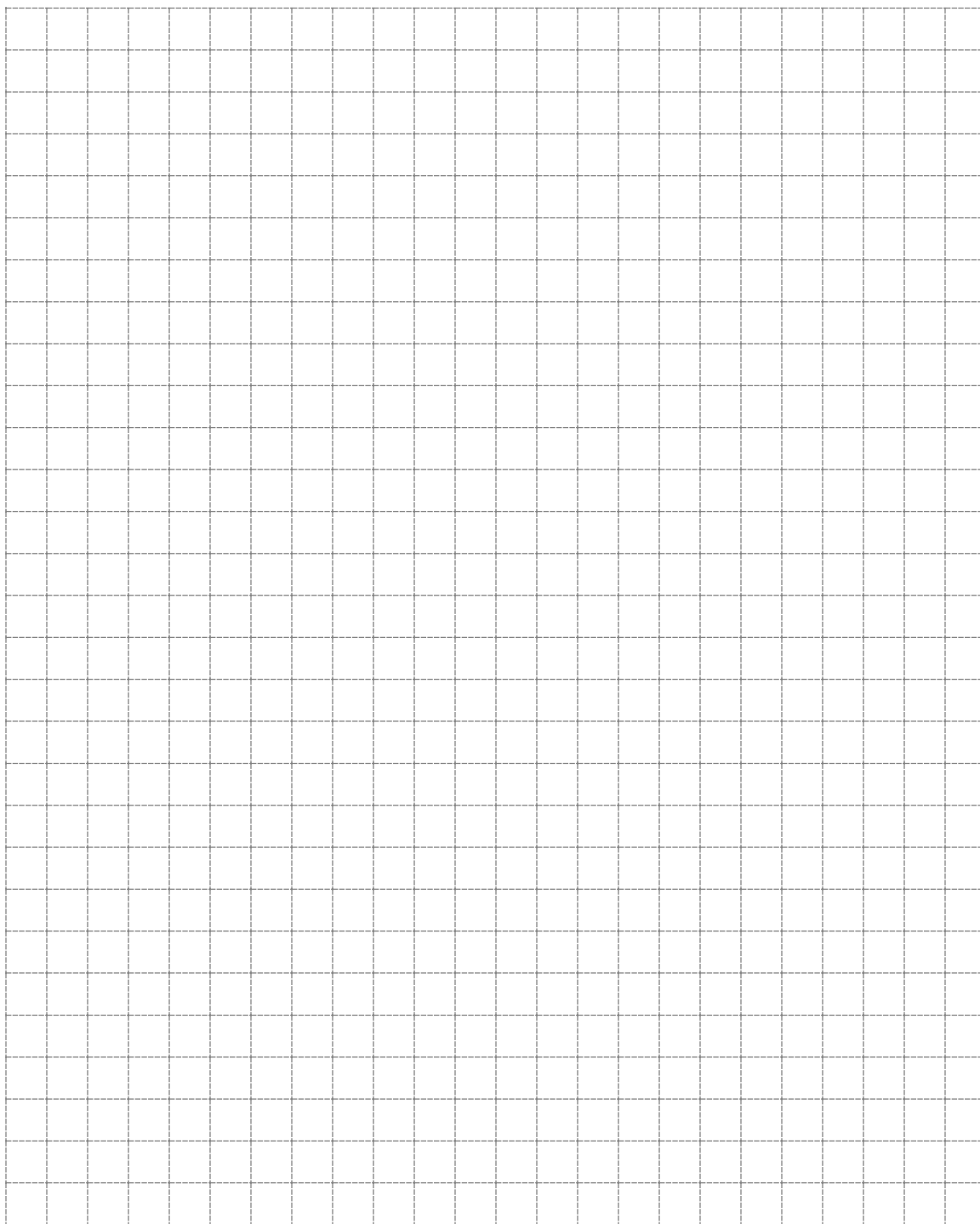
Express $\log_7(2x - 5) + 2\log_7 3$ as a single logarithm.

Question 17

1 mark 117

Explain why $11!$ is not the total number of 11-letter arrangements that can be made from the word CELEBRATION.

No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2018

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Manitoba Education and Training
Winnipeg, Manitoba, Canada

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time: 3 hours

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	17	35
Booklet 2	9	22	54
Total	9	39	89

* The first 6 questions in *Booklet 1* require a calculator.  You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
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Constructed Response Questions

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Question 18**1 mark**

Given $f(x) = x - 1$, identify a point on the graph of $y = \sqrt{f(x)}$.

- a) $(0, -1)$
- b) $(3, 2)$
- c) $(1, 0)$
- d) $(0, 1)$

Question 19**1 mark**

Identify the total number of possible arrangements of 6 adults and 4 children seated in a row if the children must sit together.

- a) $6!4!$
- b) $7!4!$
- c) $10!$
- d) $6!$

Question 20**1 mark**

Identify the exact value of $\sec\left(-\frac{7\pi}{3}\right)$.

- a) -2
- b) $-\frac{2}{\sqrt{3}}$
- c) $\frac{2}{\sqrt{3}}$
- d) 2

Question 21**1 mark**

Given $\log_x\left(\frac{1}{25}\right) = -2$, identify the value of x .

- a) -5
- b) $-\frac{1}{5}$
- c) $\frac{1}{5}$
- d) 5

Question 22**1 mark**

Identify the equation for all of the asymptotes on the graph of $y = \tan x$.

- a) $x = k\pi, k \in \mathbb{Z}$
- b) $x = 2k\pi, k \in \mathbb{Z}$
- c) $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$
- d) $x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

Question 23**1 mark**

If $p(x) = 3(m)(x+1)^2$ is a cubic function with a y -intercept of -12 , identify the missing factor, m .

- a) $m = x - 4$
- b) $m = x + 4$
- c) $m = x + 12$
- d) $m = x - 12$

Question 24**1 mark**

Identify the number of negative terms in the binomial expansion of $(x - y)^5$.

- a) 2
- b) 3
- c) 5
- d) 6

Question 25**1 mark**

Given $f(x) = x^2$, identify which equation represents the graph of $y = f(x)$ after a translation of 5 units to the left.

- a) $y = (x + 5)^2$
- b) $y = (x - 5)^2$
- c) $y = x^2 - 5$
- d) $y = x^2 + 5$

Question 26**1 mark**

When a polynomial, $p(x)$, is divided by $(x - 7)$, the remainder is 24. Identify the only statement that must be true.

- a) $x = 7$ is a zero of $p(x)$
- b) $p(7) = 24$
- c) $x = 24$ is a zero of $p(x)$
- d) the y -intercept is 24

Question 27

1 mark 118

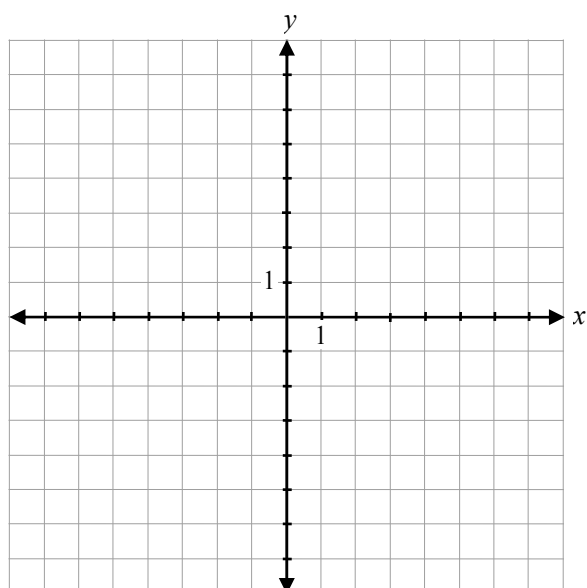
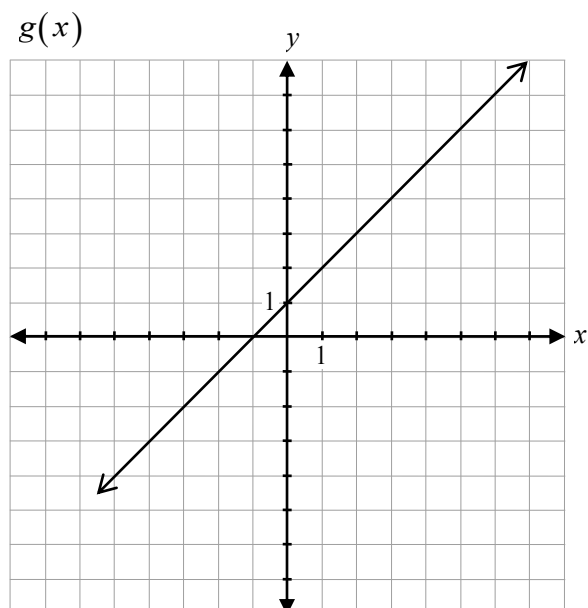
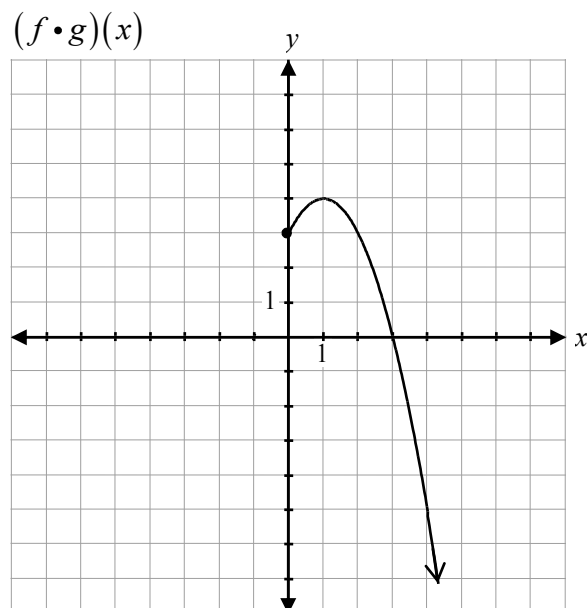
Given $f(x) = \frac{(2x+1)(x-8)}{(x-8)(x+4)}$, state the equation(s) of the vertical asymptote(s).

Question 28

2 marks

119

Given the graph of $(f \cdot g)(x)$ and $g(x)$, sketch the graph of $f(x)$.



Question 29

1 mark 120

Brian was asked to state the zeros of the polynomial $p(x) = (x + 2)(x - 5)(x - 1)$.

Brian's response:

$$\text{Zeros: } (x+2), (x-5), (x-1)$$

Explain why his response is incorrect.

Question 30

2 marks 121

Simplify ${}_{n+3}C_2$.

Question 31

2 marks

122

Verify that the equation $2\cos^2 x = \sin x + 1$ is true for $x = \frac{\pi}{6}$.

Question 32

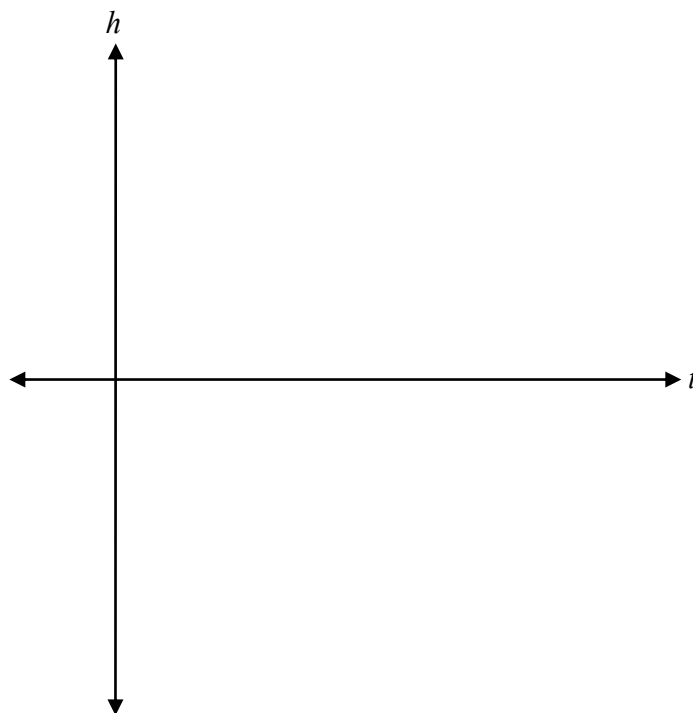
a) 2 marks b) 1 mark

123
124

The height of a fish jumping out of the water can be modelled by the function

$h(t) = -t(t-1)(t-4)(t-5)$ where $h(t)$ is the height of the fish above or below the water in cm, and t is the time in seconds, $t \geq 0$.

a) Sketch a graph representing the height of the fish with respect to time over the interval $[0, 5]$.



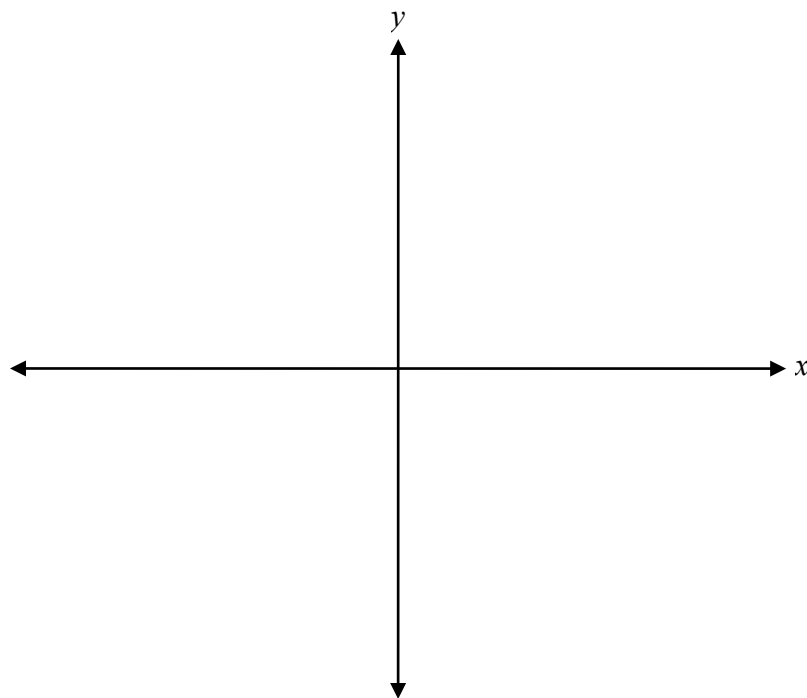
b) State, from the graph in a), the total number of seconds that the fish is above the water.

Question 33

1 mark 125

Describe the behaviour of the graph of $y = 5^x + 4$ as it approaches $y = 4$.

Sketch the angle of 6 radians in standard position.

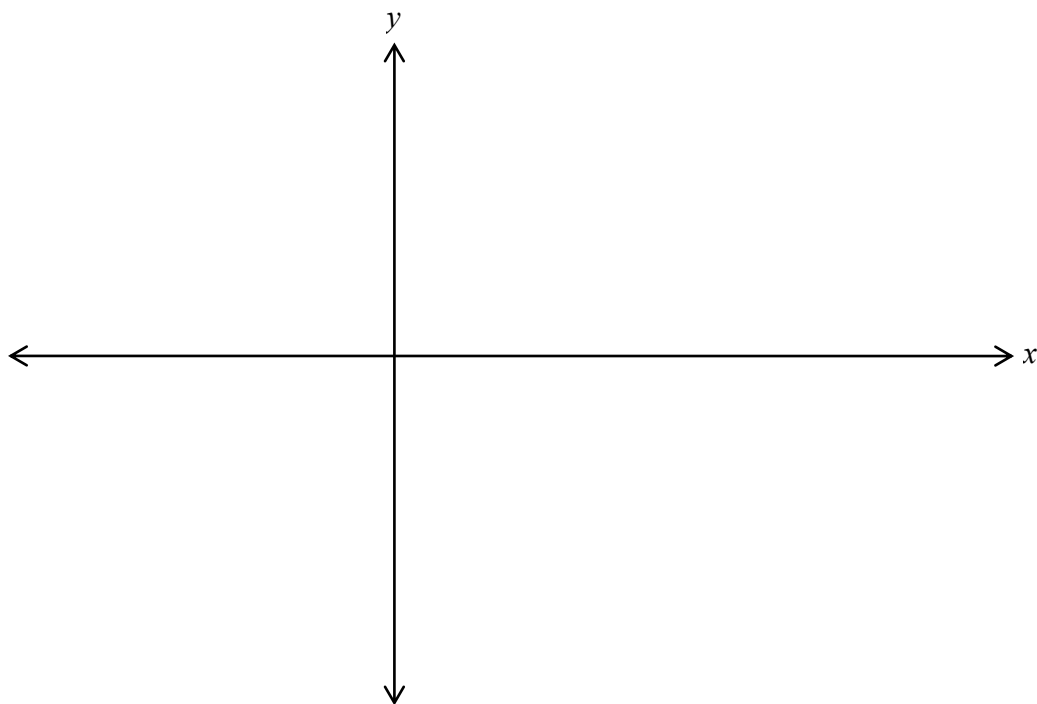


Question 35

4 marks

127

Sketch the graph of the function $y = 4 \sin\left(\frac{\pi}{3}x\right) - 2$ over the domain $[-3, 6]$.



Question 36

a) 1 mark b) 1 mark

128
129

Given $f(x) = 3x - 12$ and $g(x) = x - 4$,

a) determine the equation of $h(x) = \left(\frac{f}{g}\right)(x)$.

$h(x) =$ _____

b) describe what the non-permissible value represents on the graph of $h(x)$.

Question 37

2 marks 130

Determine an equation of a radical function, $f(x)$, with a domain of $x \geq 5$ and a range of $y \geq -2$.

$$f(x) = \underline{\hspace{10cm}}$$

Question 38

3 marks 131

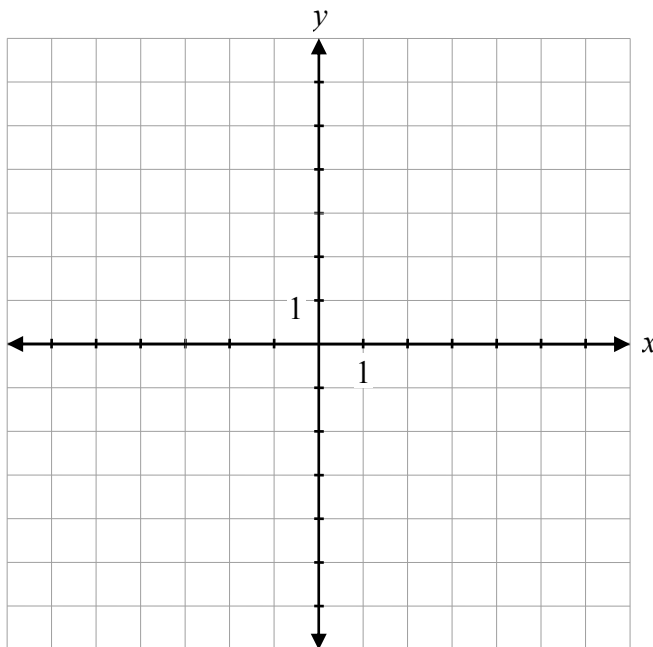
Determine the exact value of $\cos\left(\frac{17\pi}{12}\right)$.

Question 39

4 marks

132

Sketch the graph of $y = \frac{1}{2}\sqrt{-x} + 1$.



Question 40

2 marks 133

Given $f(x) = \frac{3x}{4} + 9$, determine the equation of $f^{-1}(x)$.

Question 41

1 mark 134

Describe the end behaviour of the polynomial function $p(x) = -(x - 2)(x + 3)^2$.

Question 42

a) 2 marks b) 1 mark

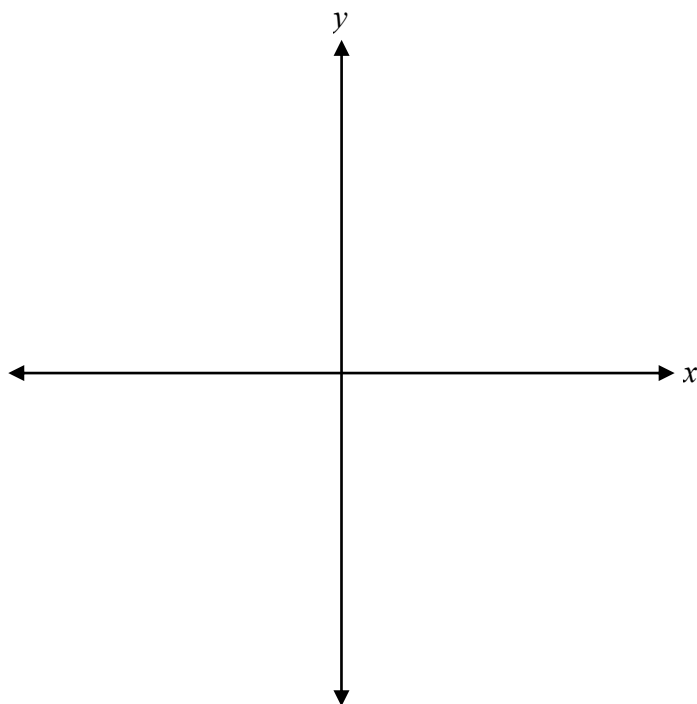
135
136

Given $\csc \theta = -4$ and θ is in quadrant IV,

a) determine the exact value of $\cos \theta$.

b) determine the exact value of $\cot \theta$.

Sketch the graph of the function $f(x) = \frac{10}{x^2 + 2}$.



Question 44

1 mark 138

Explain why only one of the following equations could be solved algebraically without using logarithms.

$$3^{5x} = 6^{2x-1} \quad \text{or} \quad 16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$$

Question 45

1 mark 139

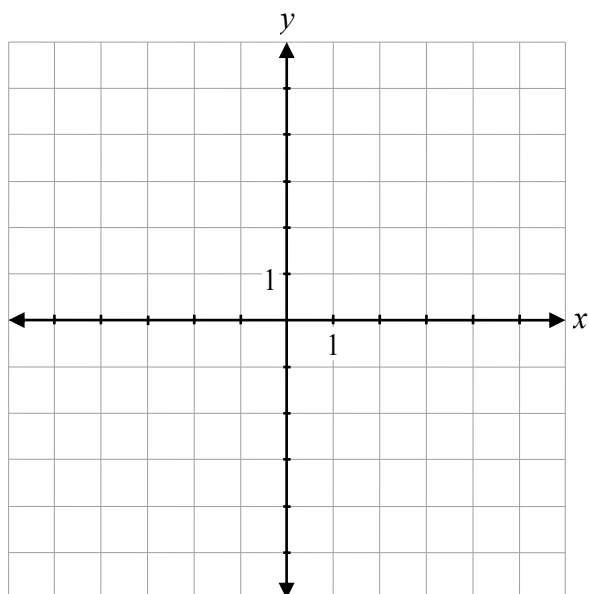
Given a graph of $y = f(x)$, describe how to sketch the graph of $y = |f(x)|$.

Question 46

2 marks

140

Sketch the graph of $y = \log_3(x + 4)$.



Question 47

3 marks 141

Solve, algebraically.

$$\log x + \log 4 - \log(x - 2) = \log 5$$

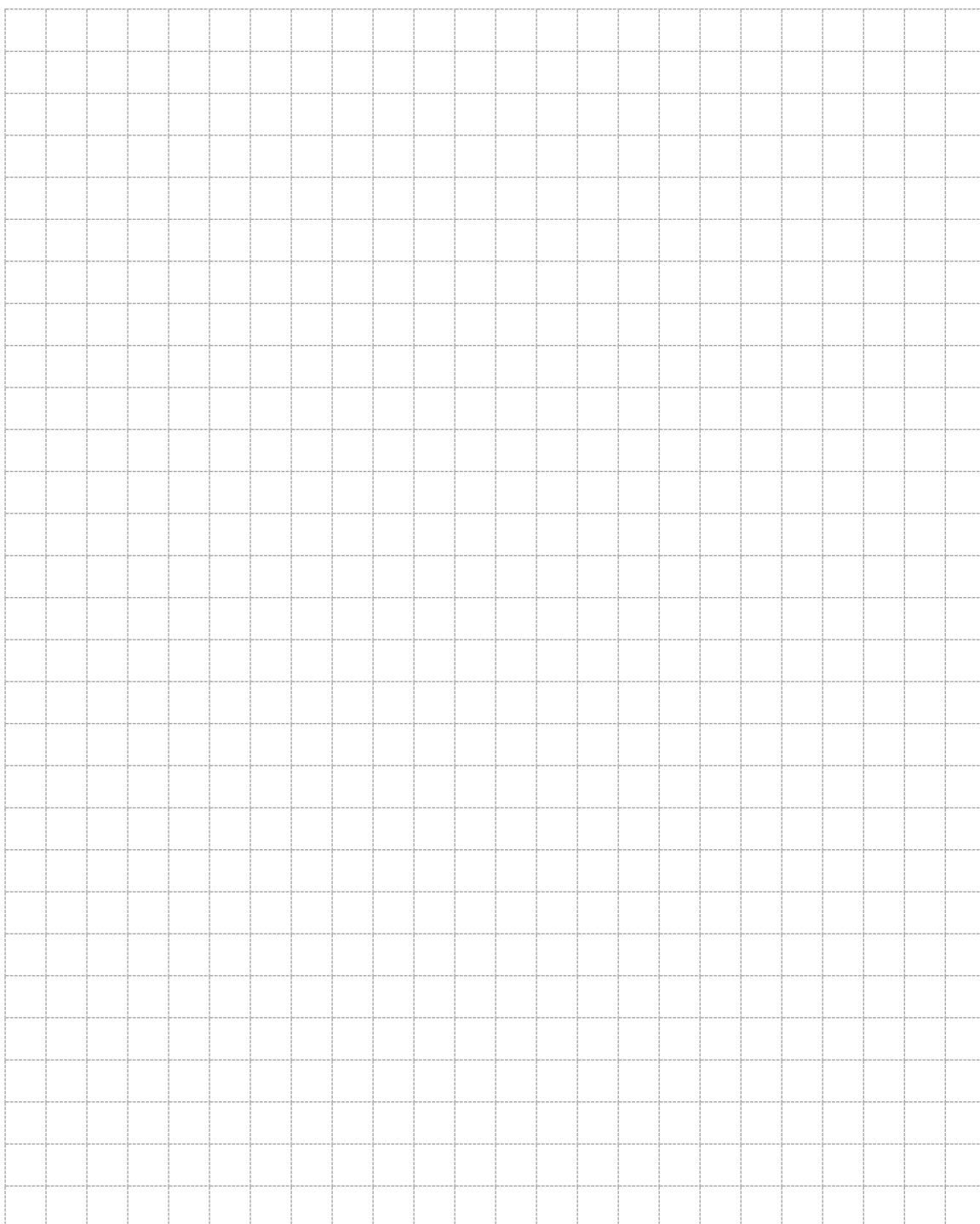
Question 48

2 marks

142

Given $\sin \theta = \frac{1}{2}$, determine all possible values of θ over the interval $[-2\pi, 2\pi]$.

No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2019

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 2. Educational tests and measurements—Manitoba.
 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Pre-calculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
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DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Graphing calculator solutions must include an explanation of how your final answer is obtained.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

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Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

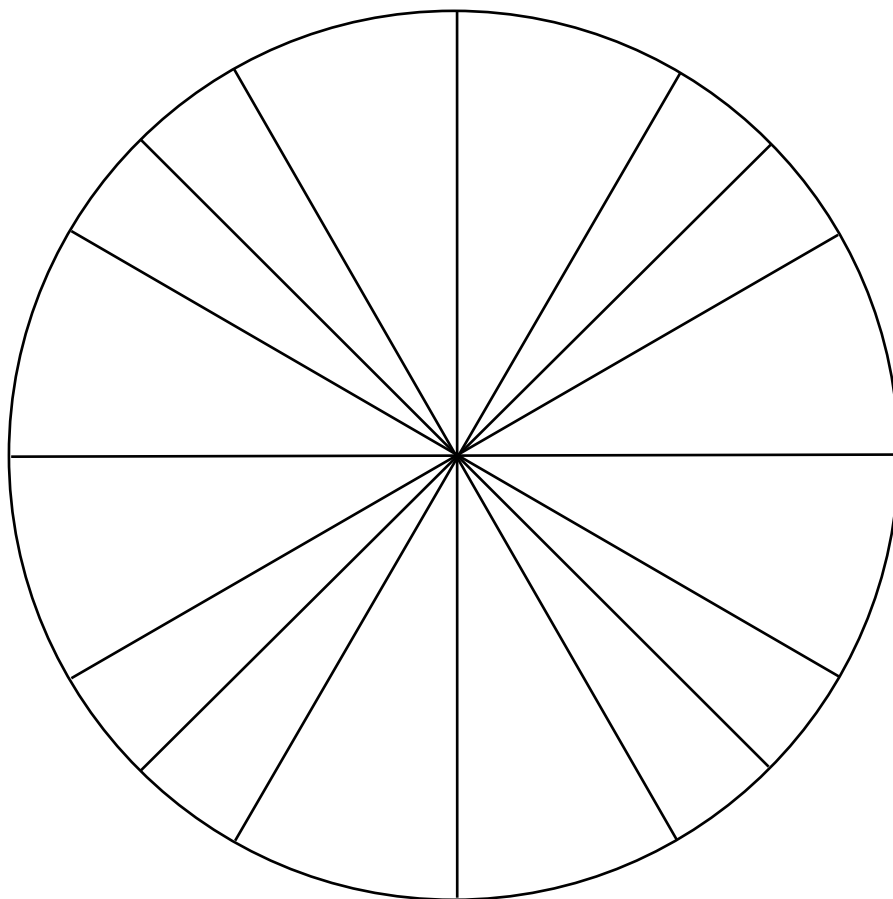
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

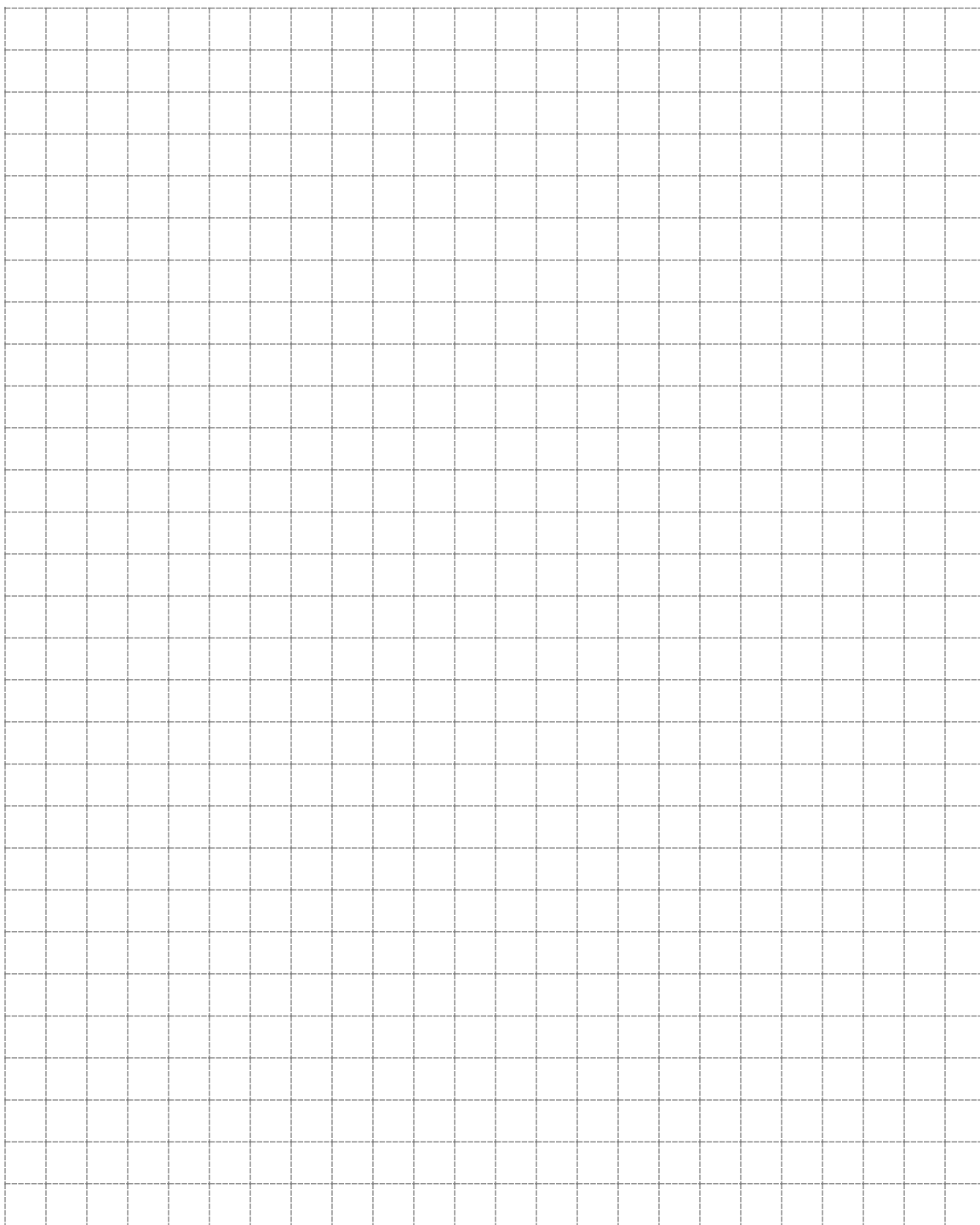
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



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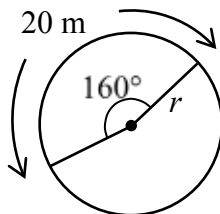


Question 1



2 marks 101

Determine the length of the radius, r , given an arc length of 20 metres and a central angle of 160° .



Question 2

1 mark 102

There are eight cars parked in a row. Determine the number of possible arrangements of these eight cars if Mrs. Jones must always park in the third spot and Mr. Rodriguez must always park in the last spot.

Question 3



3 marks 103

Bill wins \$1 300 000 in a lottery and invests the entire amount at an annual interest rate of 2.5% compounded quarterly. He will withdraw \$10 000 at the end of every three months.

Determine, algebraically, the total number of withdrawals, including the partial amount, that Bill can make until there is no money left. Express your answer as a whole number.

Use the formula:

$$PV = \frac{R[1 - (1 + i)^{-n}]}{i}$$

where PV = the present value deposited

R = the amount of each withdrawal

n = the number of equal withdrawals

$i = \frac{\text{the annual interest rate (in decimal form)}}{\text{the number of compounding periods}}$

Question 4

3 marks 104

Determine and simplify the 12th term in the binomial expansion of $\left(x^3 - \frac{1}{2x^2}\right)^{12}$.

Question 5

3 marks 105

Solve, algebraically.

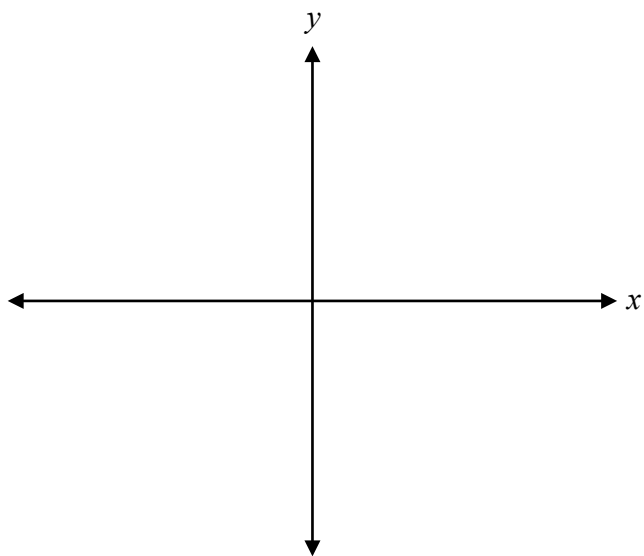
$$e^{2x-3} = 7^{x+1}$$

Note: A calculator is not required for the remaining test questions.

Question 6

1 mark 106

Sketch the angle $\frac{7\pi}{3}$ in standard position.



Question 7

3 marks 107

Determine, algebraically, all of the zeros of the polynomial function $P(x) = x^4 - 5x^3 - 4x^2 + 20x$.

Question 8

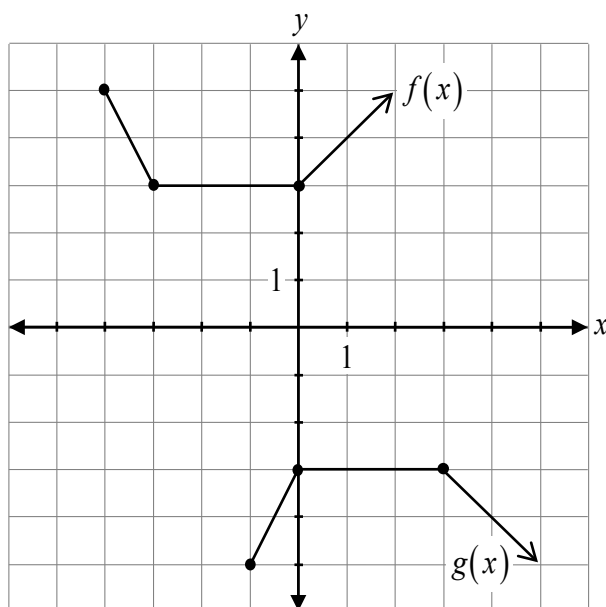
1 mark 108

Justify why four of the terms in the binomial expansion of $(-x + y)^6$ are positive.

Question 9

2 marks 109

Determine the equation of $g(x)$ in terms of $f(x)$.



$g(x) =$ _____

Question 10

3 marks 110

Prove the identity for all permissible values of x .

$$\frac{\sin x + \tan x}{\cot x + \csc x} = \frac{\sin^2 x}{\cos x}$$

Left-Hand Side	Right-Hand Side

Question 11

1 mark 111

Given that the point $(-2, 1)$ is on the graph of $y = f(x)$, describe how the coordinates of the corresponding point on the graph of $y = f(4x)$ are different.

Question 12

3 marks 112

Using the laws of logarithms, completely expand the expression:

$$\log \left(\frac{x^2 \sqrt{y}}{w-1} \right)$$

Question 13

1 mark 113

Given $\sec \theta = -\frac{5}{4}$ and $\tan \theta > 0$, state the quadrant in which θ terminates.

Justify your answer.

Question 14

2 marks 114

State an equation of a rational function that has a vertical asymptote at $x = -8$ and a horizontal asymptote at $y = 9$.

Question 15

1 mark 115

Given the point $(5, 1)$, state the coordinates of the corresponding point after a reflection across the line $y = x$.

Question 16

1 mark 116

Simplify.

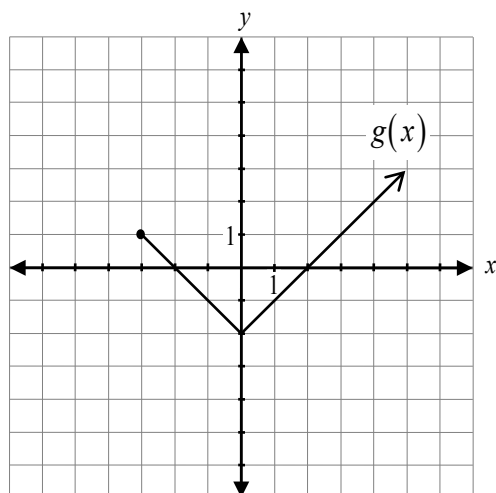
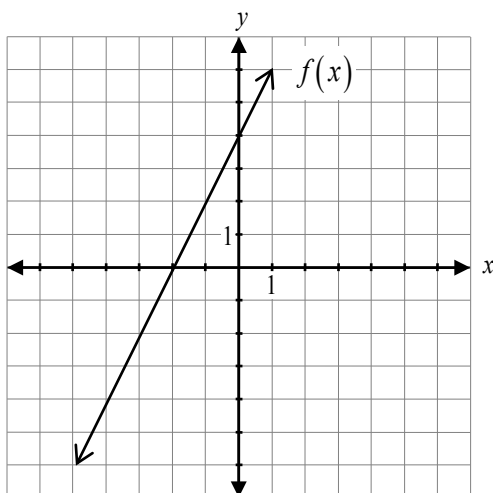
$$\frac{(n-13)!}{(n-12)!}$$

Question 17

a) 1 mark b) 1 mark

117
118

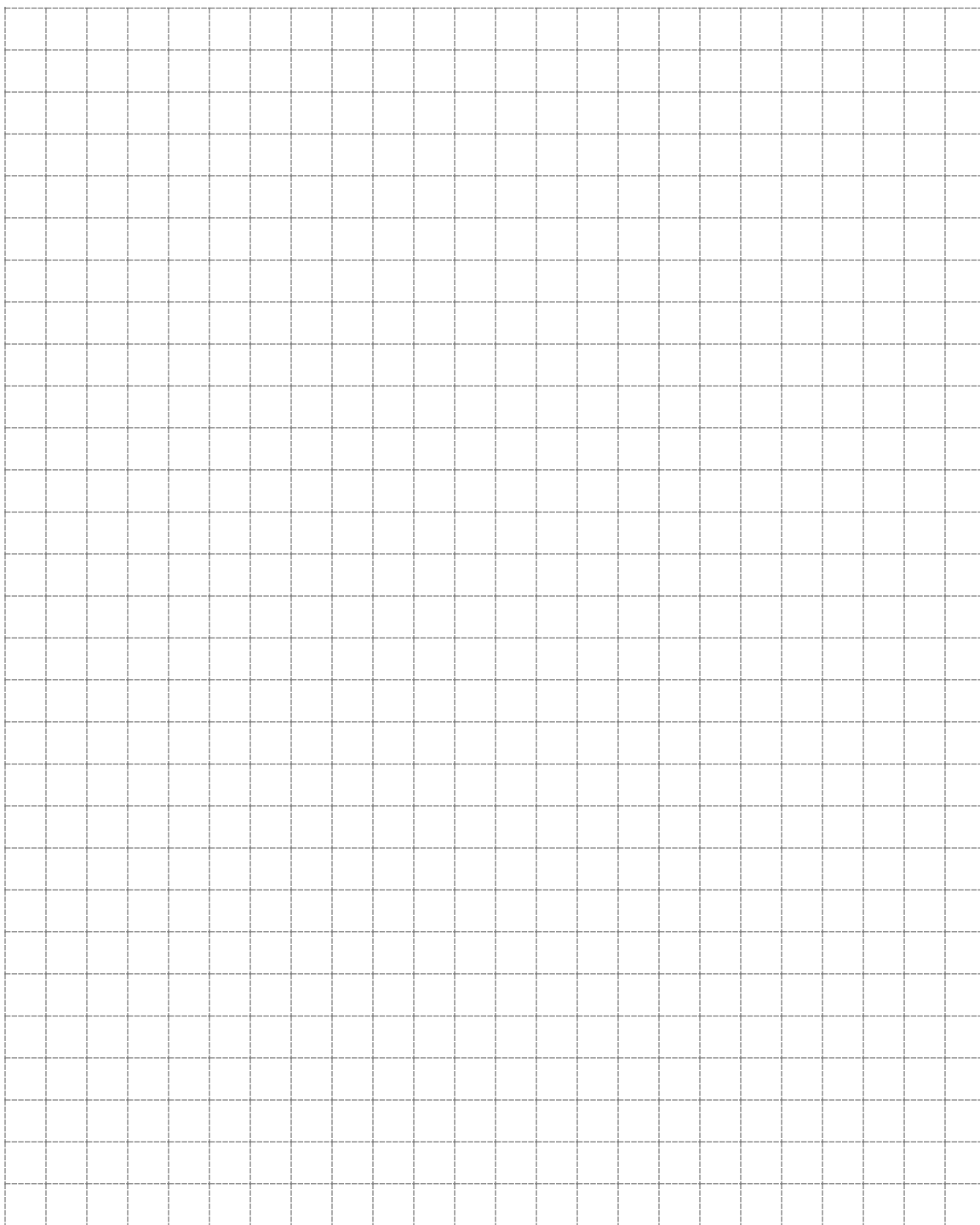
Given the graphs of $y = f(x)$ and $y = g(x)$,



a) determine the value of $f(g(2))$.

b) determine the value of $(g - f)(-3)$.

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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2019

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Question 18**1 mark**

Identify the remainder when $P(x) = 3x^3 - x^2 + 1$ is divided by $(x - 2)$.

- a) -27
- b) -19
- c) 11
- d) 21

Question 19**1 mark**

Identify the logarithmic form of $2^x = \frac{1}{4}$.

- a) $\log_2 x = \frac{1}{4}$
- b) $\log_x 2 = \frac{1}{4}$
- c) $\log_2 \left(\frac{1}{4} \right) = x$
- d) $\log_x \left(\frac{1}{4} \right) = 2$

Question 20**1 mark**

Leah's Pizzeria offers 9 different pizza toppings. Identify the expression that represents the number of different types of pizzas, with 3 different toppings, that can be made.

- a) ${}_9C_3$
- b) ${}_9P_3$
- c) $\frac{9!}{3!}$
- d) $9!3!$

Question 21**1 mark**

Given $(5, -4)$ is a point on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = \frac{1}{f(x)}$.

a) $\left(\frac{1}{5}, -4\right)$

b) $\left(5, -\frac{1}{4}\right)$

c) $\left(\frac{1}{5}, -\frac{1}{4}\right)$

d) $(-4, 5)$

Question 22**1 mark**

Identify the non-permissible value of x for $1 + \sec x$ over $[0, \pi]$.

a) 0

b) $\frac{\pi}{4}$

c) $\frac{\pi}{2}$

d) π

Question 23**1 mark**

Indicate the combination that represents the circled term in the given row of Pascal's triangle.

$$1 \quad 4 \quad 6 \quad \textcircled{4} \quad 1$$

a) ${}_4C_3$

b) ${}_4C_4$

c) ${}_5C_3$

d) ${}_5C_4$

Question 24**1 mark**

Identify the x -intercept on the graph of $f(x) = \sqrt{2(x+5)}$.

- a) -5
- b) 0
- c) $\sqrt{10}$
- d) 5

Question 25**1 mark**

Identify the coterminal angle of $\frac{\pi}{5}$ over the interval $-\pi \leq \theta \leq 4\pi$.

- a) $-\frac{9\pi}{5}$
- b) $-\frac{\pi}{5}$
- c) $\frac{3\pi}{5}$
- d) $\frac{11\pi}{5}$

Question 26**1 mark**

Given $f(x) = \{(2, 6), (3, 2), (3, 4), (6, 5)\}$, identify the value of $f(f(2))$.

- a) 3
- b) 4
- c) 5
- d) 6

Question 27

1 mark

The graph of $f(x) = (x - 1)^2$ is translated 2 units to the left and 3 units up. Identify the equation of the transformed graph, $g(x)$.

a) $g(x) = (x + 1)^2 + 3$

b) $g(x) = (x - 3)^2 + 3$

c) $g(x) = (x + 2)^2 + 3$

d) $g(x) = (x - 2)^2 + 3$

Question 28

2 marks 119

Given $\csc \theta = -\frac{8}{5}$, determine the exact value of $\cos 2\theta$.

Question 29

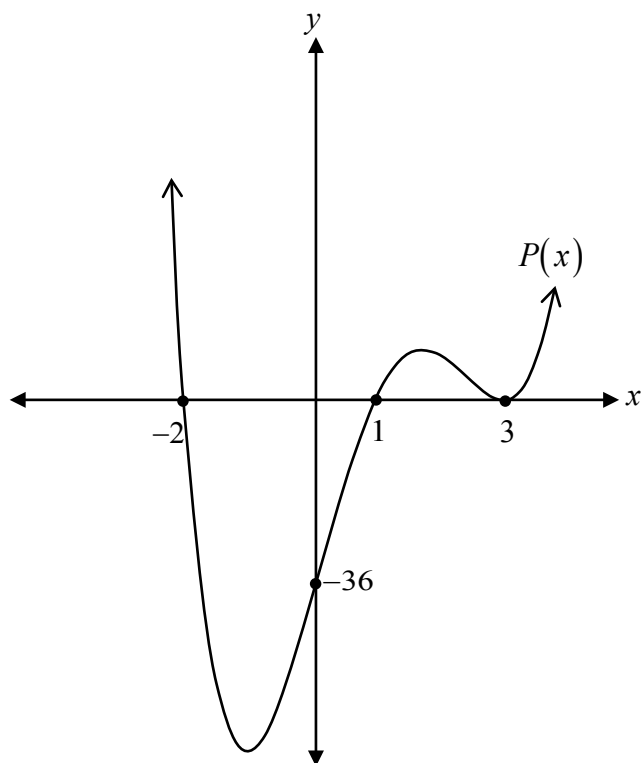
1 mark 120

Determine the period of the sinusoidal function, $f(x) = -6 \cos\left(\frac{\pi}{6}(x+1)\right) + 5$.

Question 30

2 marks 121

Determine, algebraically, the equation of $P(x)$, given the graph of the polynomial function $P(x)$.



$P(x) =$ _____

Question 31

4 marks 122

Solve $2\sin^2\theta - 7\sin\theta - 4 = 0$ where $\theta \in \mathbb{R}$.

Question 32

1 mark 123

Justify that the shapes of the graphs of $f(x) = (x+1)^2(x-1)$ and $g(x) = (x+1)^2(x-1)^3$ are different as they approach the x -intercept at $x=1$.

Question 33

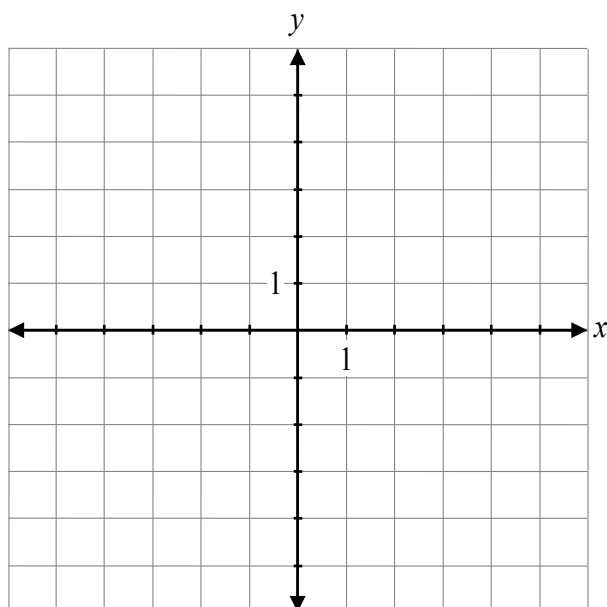
2 marks 124

Determine the exact value of $\cot \theta$ if $\cos \theta = -\frac{4}{7}$ and $\sin \theta$ is positive.

Question 34

3 marks 125

Sketch the graph of $f(x) = -\log_2(x) + 2$.



Question 35

1 mark 126

State the range of $f(x) = \sqrt{x+4}$.

Range: _____

Question 36

1 mark 127

Sophie correctly solved the logarithmic equation, $\log_7(x-1) = \log_7(2x-2)$.

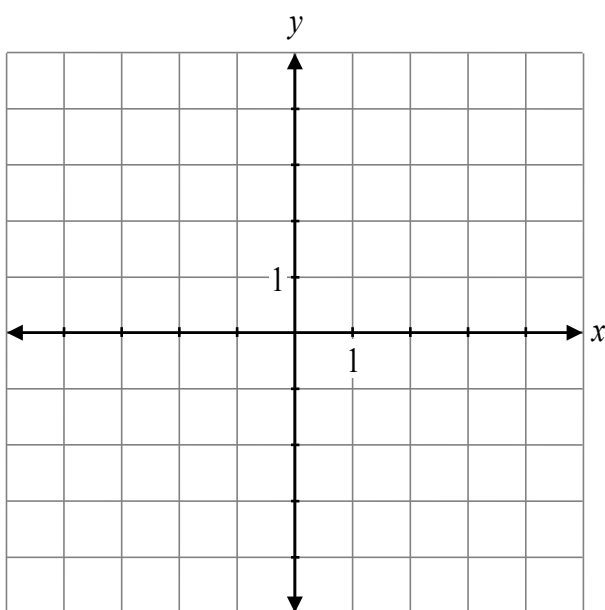
$$x-1 = 2x-2$$

$$-1+2 = 2x-x$$

$$~~1 = x~~$$

Explain why $x=1$ is an extraneous root.

Sketch the graph of $f(x) = \sqrt{4x} - 1$.



Question 38

3 marks 129

Solve, algebraically.

$${}_nC_2 = 2n + 7$$

Question 39

1 mark 130

Given $f(x) = x^2 - 1$ and $g(x) = x - 3$, explain why the domain of $h(x) = \frac{f(x)}{g(x)}$ has a restriction when $x = 3$.

Evaluate.

$$\frac{\cot\left(\frac{11\pi}{6}\right)\sin\left(-\frac{4\pi}{3}\right)}{\cos\left(\frac{2\pi}{3}\right)}$$

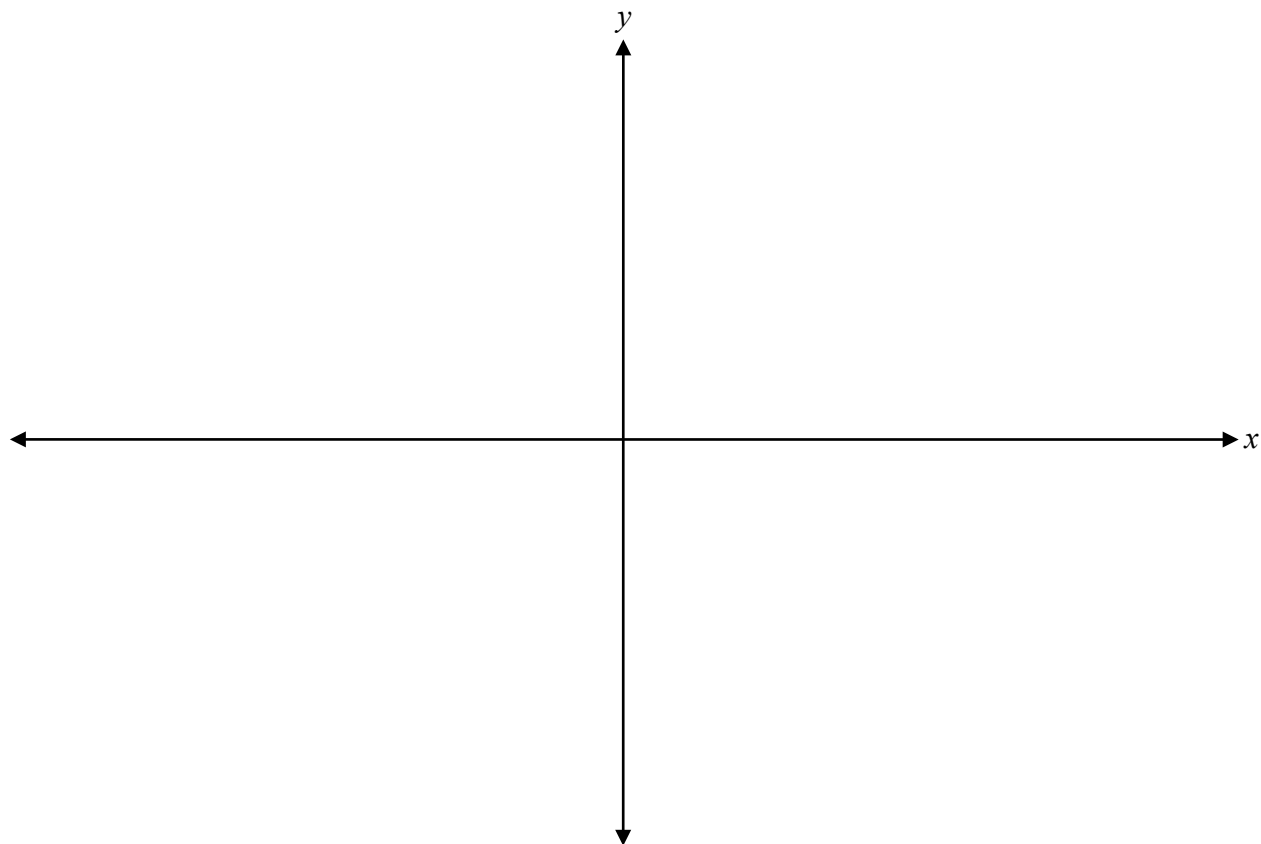
Question 41

1 mark 132

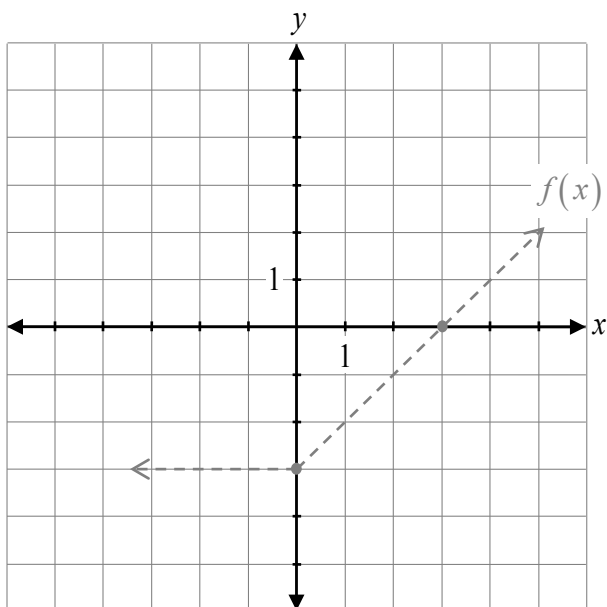
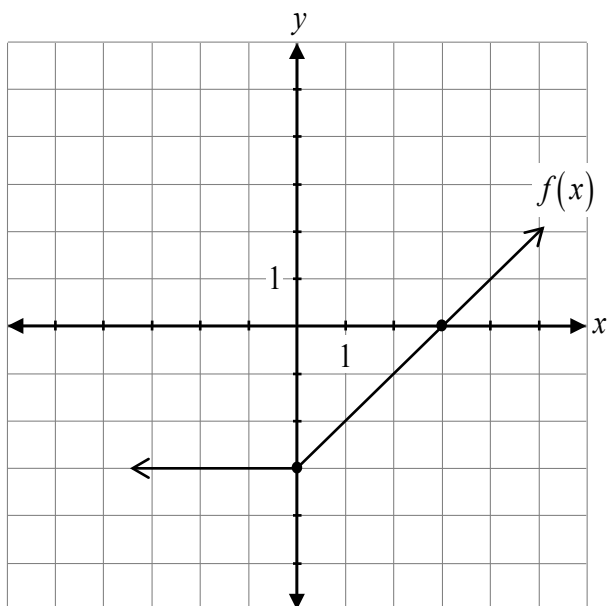
Solve, algebraically.

$$\log_2(\log_3 x) = 2$$

Sketch the graph of the function $y = 5 \sin\left(\frac{\pi}{4}x\right) + 1$ over the domain $[-4, 8]$.



Given the graph of $y = f(x)$, sketch the graph of $y = |f(-x)|$.

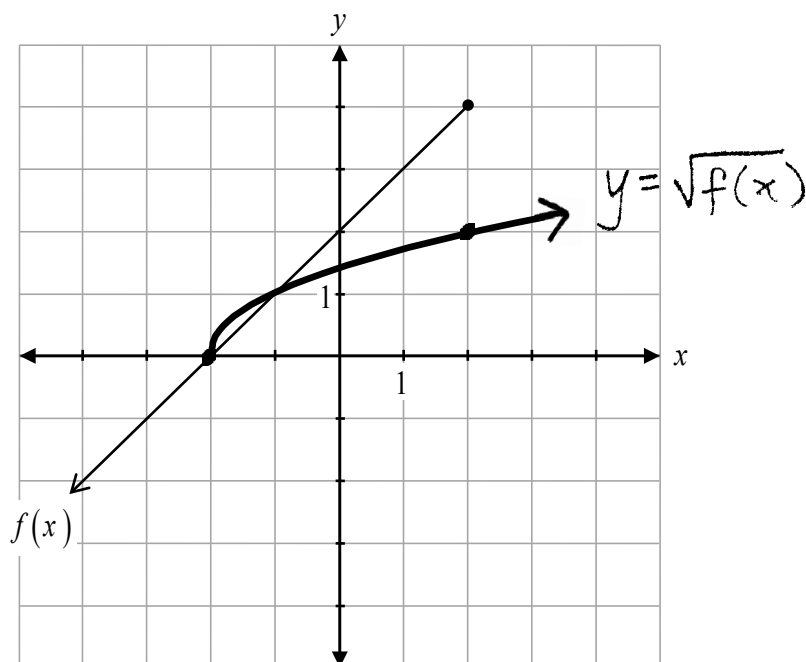


The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 44

1 mark 135

Savannah used the graph of $y = f(x)$ to sketch the graph of $y = \sqrt{f(x)}$. Her solution is given below. Describe her error.



Determine the exact value of $\sin\left(\frac{13\pi}{12}\right)$.

Question 46

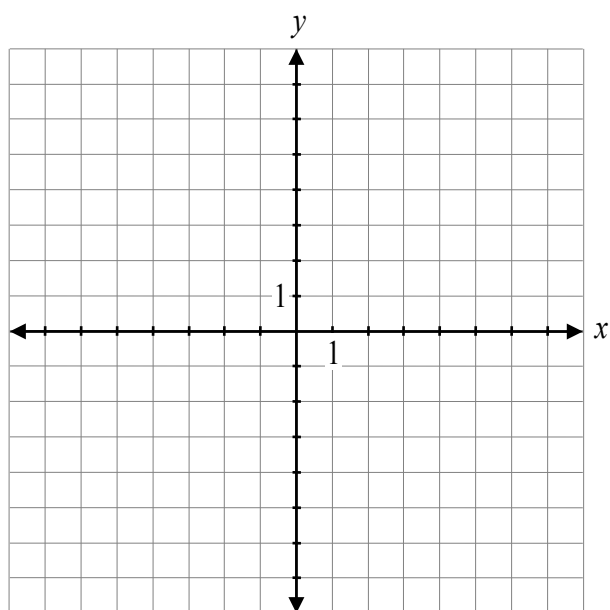
1 mark 137

State the equation of the horizontal asymptote of $f(x) = \frac{3x}{x-1}$.

Question 47

4 marks 138

Sketch the graph of $f(x) = \frac{5x - 10}{x^2 + x - 6}$.

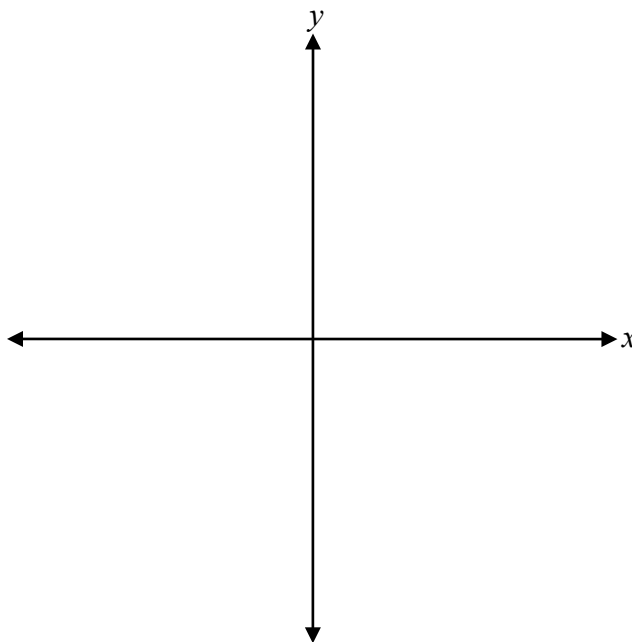


Question 48

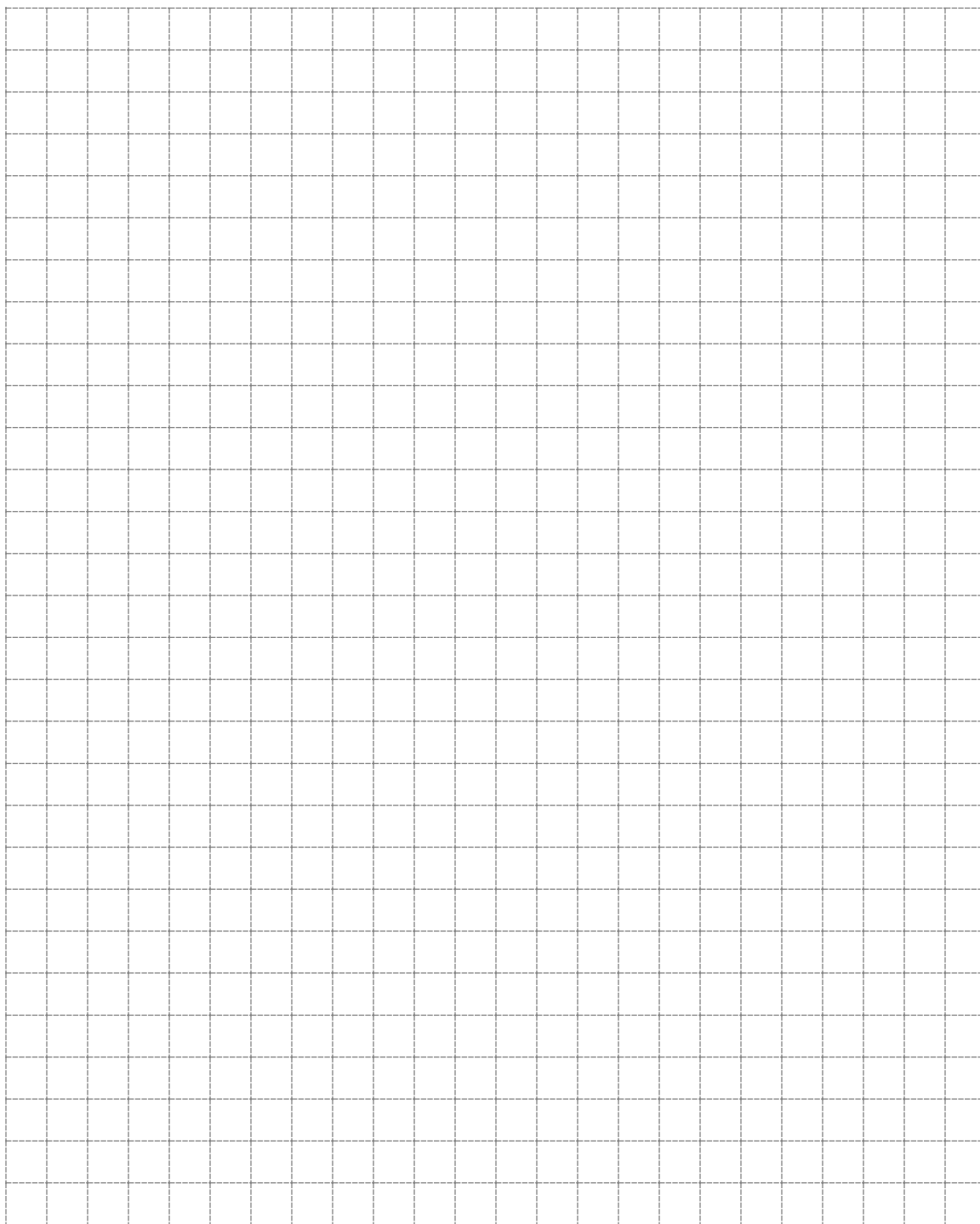
2 marks 139

Determine, algebraically, the inverse of $f(x) = 3x + 4$.

Sketch the graph of $P(x) = -(x+1)(x-2)(x+3)$.



No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2019

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Booklet 1. June 2019

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 3. Mathematics—Study and teaching (Secondary)—Manitoba.
 4. Pre-calculus—Study and teaching (Secondary)—Manitoba.
 5. Mathematical ability—Testing.
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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	17	34
Booklet 2	9	23	56
Total	9	40	90

* The first 4 questions with the symbol  in *Booklet 1* require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited. Please turn off your cell phone and all other such devices.

No marks will be awarded for work done on this page.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify/Indicate: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

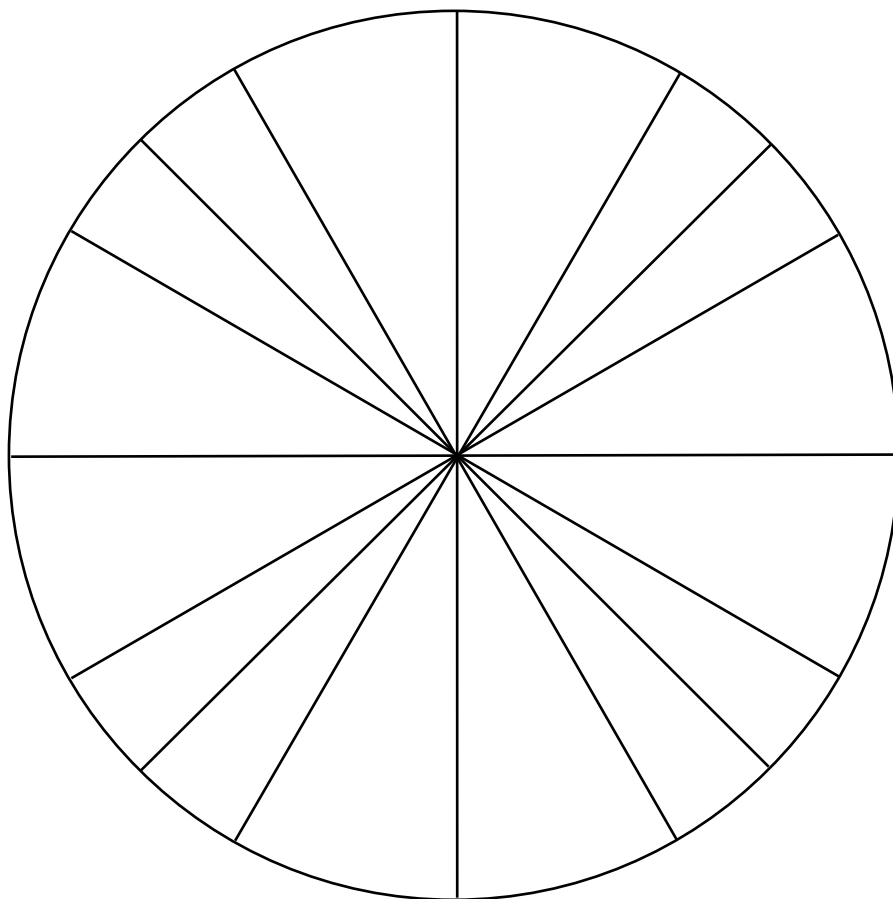
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

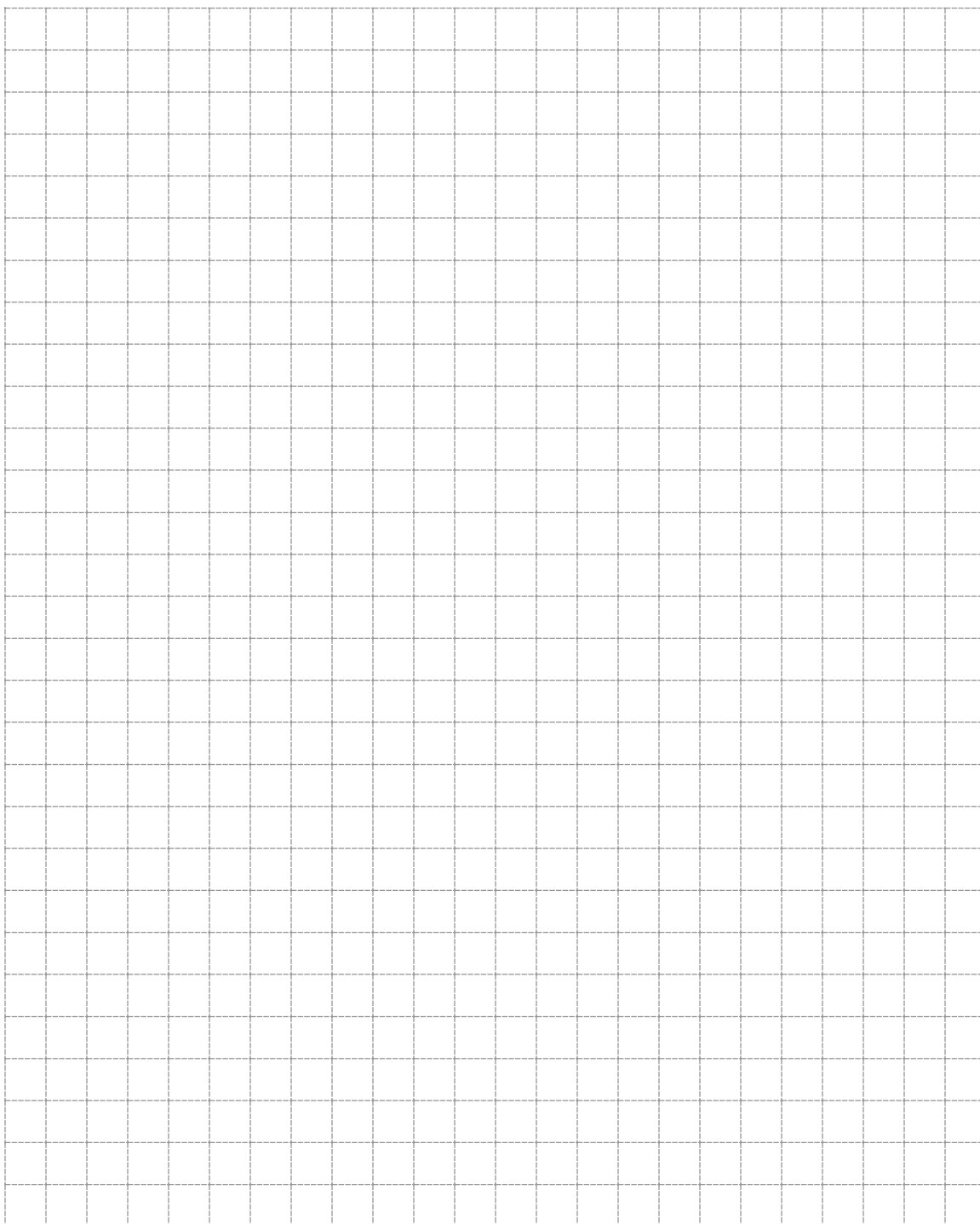
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



Question 1

2 marks 101

Avery has 4 adventure books, 5 mystery books, and 1 comic book.

Determine the number of ways he can arrange all of the books on his shelf if each type of book must be grouped together.

Question 2

4 marks 102

Solve the following equation, algebraically, over the interval $[0, 2\pi]$.

$$5 \cos^2 \theta - \cos \theta - \sin^2 \theta = 0$$

Question 3

3 marks 103

Given that $-6048x^2y^5$ is the sixth term in the expansion of $(3x - m)^7$, determine m .

Question 4

a) 2 marks b) 1 mark

104
105

A series of blood tests measures the concentration of a prescribed drug. This concentration decreases according to the formula $A = Pe^{rt}$ where:

A is the concentration at time t

P is the initial concentration

r is the rate of change

t is the time, in hours, after the first blood test

The initial concentration is 3.8900 units/mL. Three hours later, the concentration is 1.7505 units/mL.

a) Determine the rate of change, r , algebraically.

b) Determine the concentration of the prescribed drug four hours after the initial concentration of 3.8900 units/mL was measured. Express the answer correct to 4 decimal places.

Note: A calculator is not required for the remaining test questions.

Question 5

1 mark 106

Ariane uses the formula $s = \theta r$ to determine the arc length of a circle that has a central angle of 20° and a radius of 15 cm.

Below is Ariane's work:

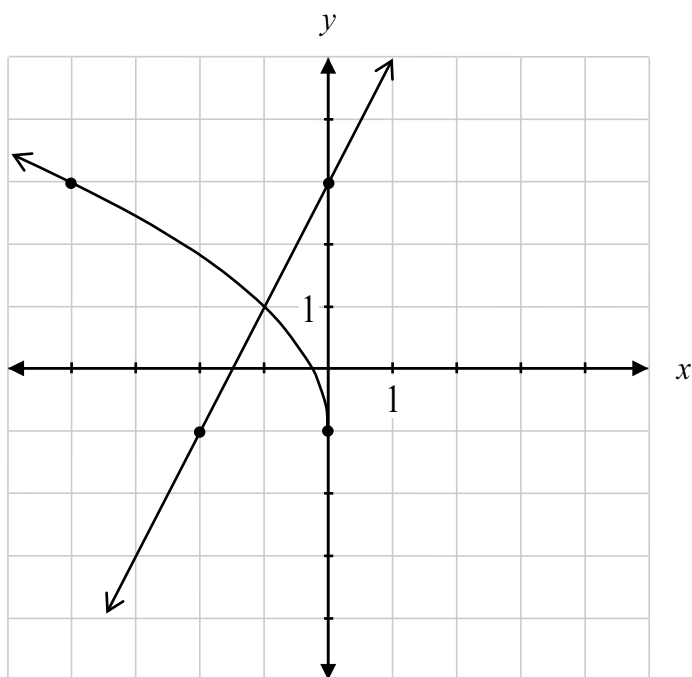
$$\begin{aligned} s &= \theta r \\ s &= (20)(15) \\ s &= 300 \text{ cm} \end{aligned}$$

Describe her error.

Question 6

1 mark 107

Using the graphs below, state the solution of the equation $2x + 3 = 2\sqrt{-x} - 1$.



Question 7

3 marks 108

Solve, algebraically.

$$\log_2(x+3) = 5 - \log_2(x-1)$$

Question 8

1 mark 109

Explain why the value of n must be greater than or equal to the value of r , when using ${}_nC_r$.

Question 9

3 marks 110

Given that $y = |x|$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection in the x -axis
- vertical translation 5 units down
- horizontal stretch by a factor of 3

$g(x) =$ _____

Question 10

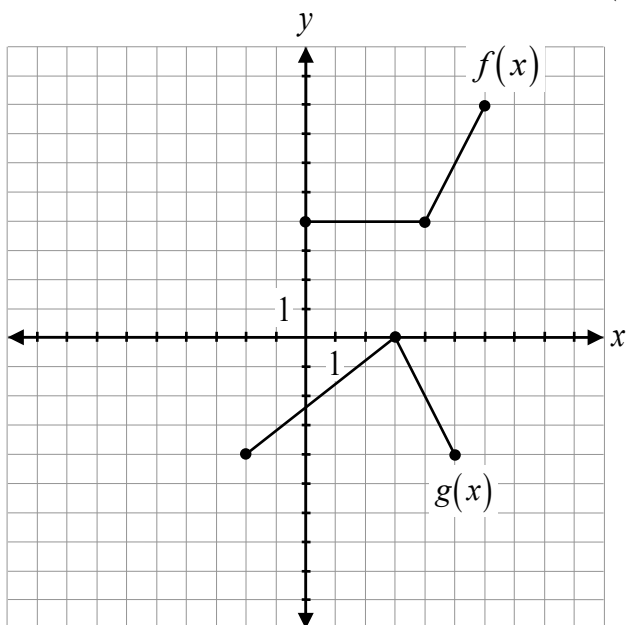
1 mark 111

Explain why the graph of $y = \frac{x-1}{x^2+x-6}$ has a horizontal asymptote at $y = 0$.

Question 11

1 mark 112

Given the graphs of $f(x)$ and $g(x)$, evaluate $g(f(2))$.



Question 12

1 mark 113

Kennedy was asked to solve the equation $\tan \theta = 1$ over all real numbers.

Below is Kennedy's solution:

$$\tan \theta = 1$$
$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Describe her error.

Question 13

3 marks 114

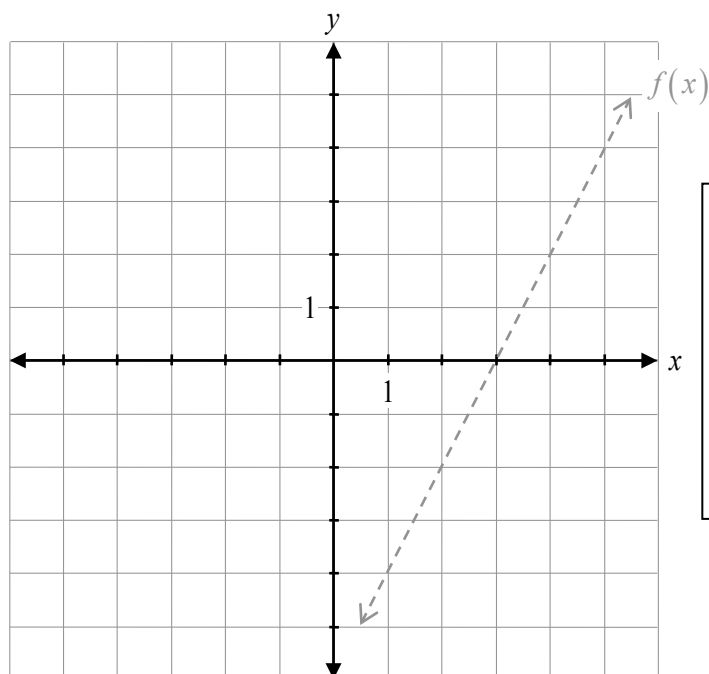
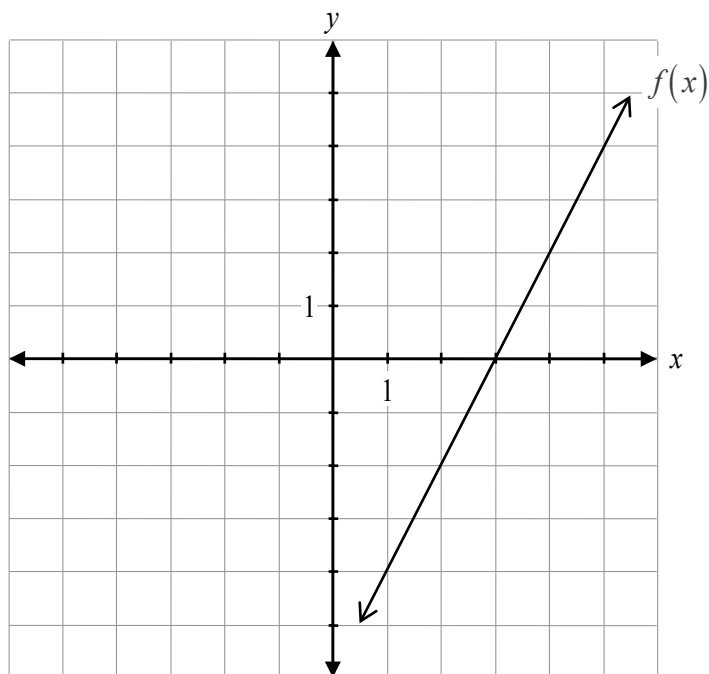
Solve, algebraically.

$${}_nC_3 = 3({}_nP_2)$$

Question 14

2 marks 115

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 15

3 marks 116

Prove the identity for all permissible values of θ .

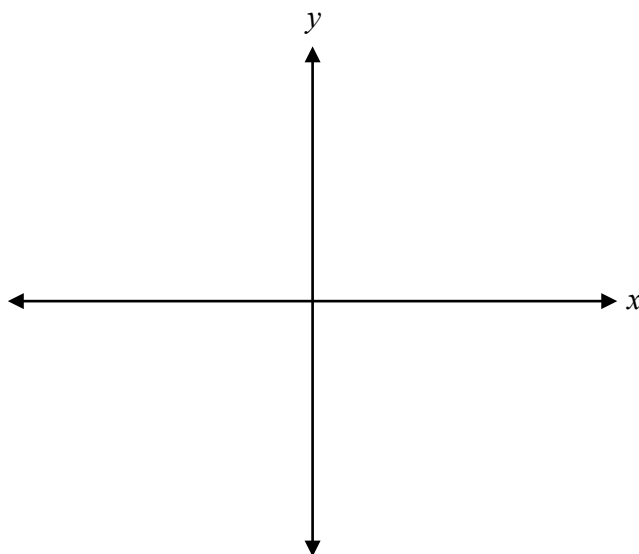
$$\frac{\sec \theta - \tan \theta \sin \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Left-Hand Side	Right-Hand Side

Question 16

1 mark 117

Sketch the angle of $-\frac{\pi}{12}$ radians in standard position.



Question 17

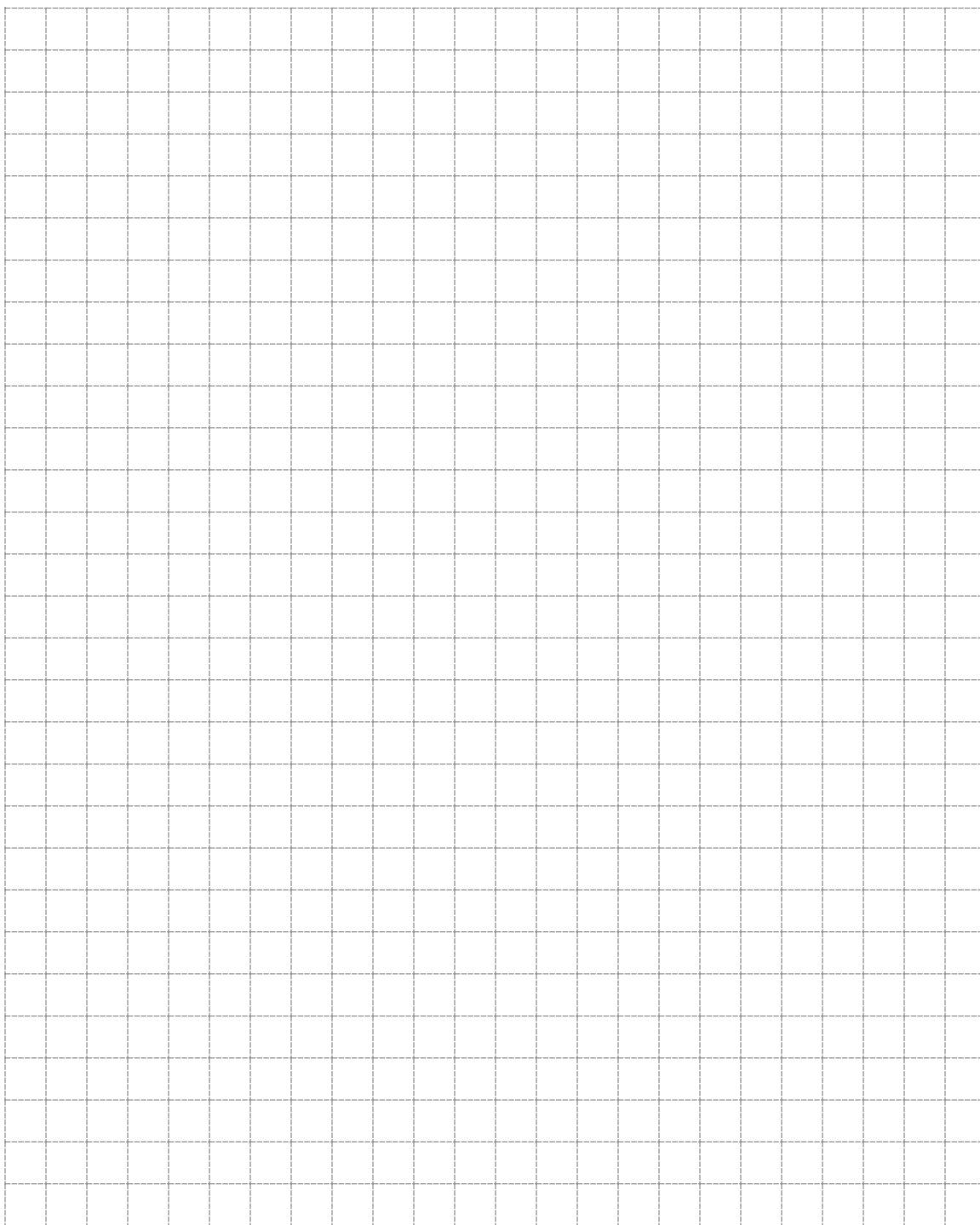
1 mark 118

Given that $h(x) = 2x^2 - 7x - 15$, determine possible equations of the functions $f(x)$ and $g(x)$ if $h(x) = f(x) \cdot g(x)$.

$f(x) =$ _____

$g(x) =$ _____

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2019

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Booklet 2	9	23	56
Total	9	40	90

* The first 4 questions with the symbol  in *Booklet 1* require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.
- Write each solution in the space provided.

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No marks will be awarded for work done on this page.

Question 18**1 mark**

The range of $y = f(x)$ is $-6 \leq y \leq 12$. The range of the transformed function $y = af(x)$ is $-2 \leq y \leq 4$. Identify the value of a .

- a) -3
- b) $-\frac{1}{3}$
- c) $\frac{1}{3}$
- d) 3

Question 19**1 mark**

Identify the expression which is equivalent to $3 \log y - \frac{1}{2} \log x + \log z$.

- a) $\log \left(\frac{y^3}{\sqrt{xz}} \right)$
- b) $\log \left(\frac{y^3 z}{\sqrt{x}} \right)$
- c) $\log \left(\frac{y^3}{x^2 z} \right)$
- d) $\log \left(\frac{y^3 z}{x^2} \right)$

Question 20**1 mark**

Identify the measure of the angle $-\frac{2\pi}{9}$ in degrees.

- a) -400°
- b) -40°
- c) 40°
- d) 320°

Question 21**1 mark**

If $y = f(x)$ has a domain of $[2, 5]$ and a range of $[6, 10]$, identify the domain of $y = f^{-1}(x)$.

a) $\left[\frac{1}{2}, \frac{1}{5}\right]$

b) $[-5, -2]$

c) $[-10, -6]$

d) $[6, 10]$

Question 22**1 mark**

Identify which of the following is a polynomial function.

a) $p(x) = -\frac{1}{2}(x+2)^3(x-3)$

b) $p(x) = 2x^{\frac{1}{2}} + x - 3$

c) $p(x) = 3x^{-4} + x^2 - 6$

d) $p(x) = 2^x + 3$

Question 23**1 mark**

Identify the total number of terms in the expansion of $(x - y)^9$.

a) 8

b) 9

c) 10

d) 11

Question 24**1 mark**

Identify the exact value of $2\cos^2(15^\circ) - 1$.

- a) 1
- b) $\frac{1}{2}$
- c) $\frac{\sqrt{3}}{2}$
- d) $\sqrt{3}$

Question 25**1 mark**

The zeros of the function $y = f(x)$ are $x = -2$ and $x = 3$. Identify the zeros of the function $g(x) = 2f(x - 4)$.

- a) $x = -6$ and $x = -1$
- b) $x = 2$ and $x = 7$
- c) $x = -4$ and $x = 6$
- d) $x = 0$ and $x = 10$

Question 26**1 mark**

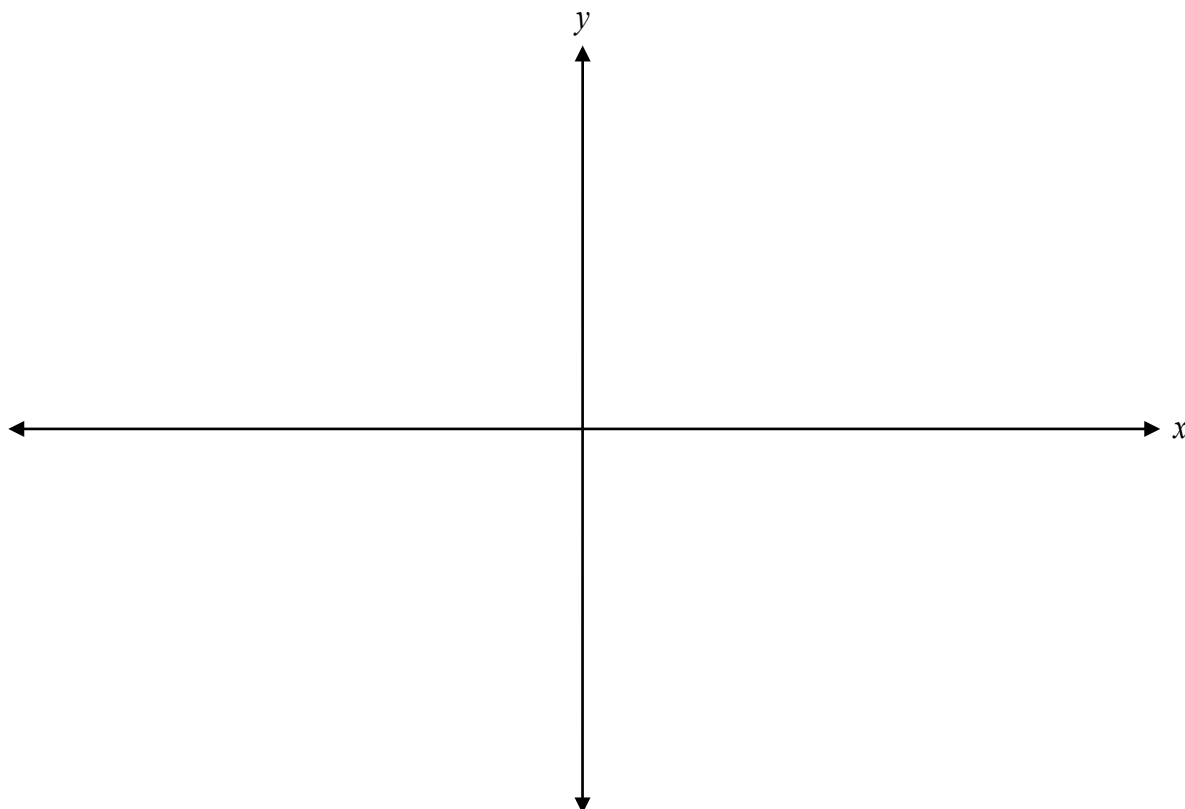
Identify the value of $\log_4\left(\frac{1}{4}\right)$.

- a) -16
- b) -1
- c) 1
- d) 16

Question 27

4 marks 119

Sketch the graph of at least one period of the function $y = -\cos\left(x + \frac{\pi}{4}\right) + 3$.



Question 28

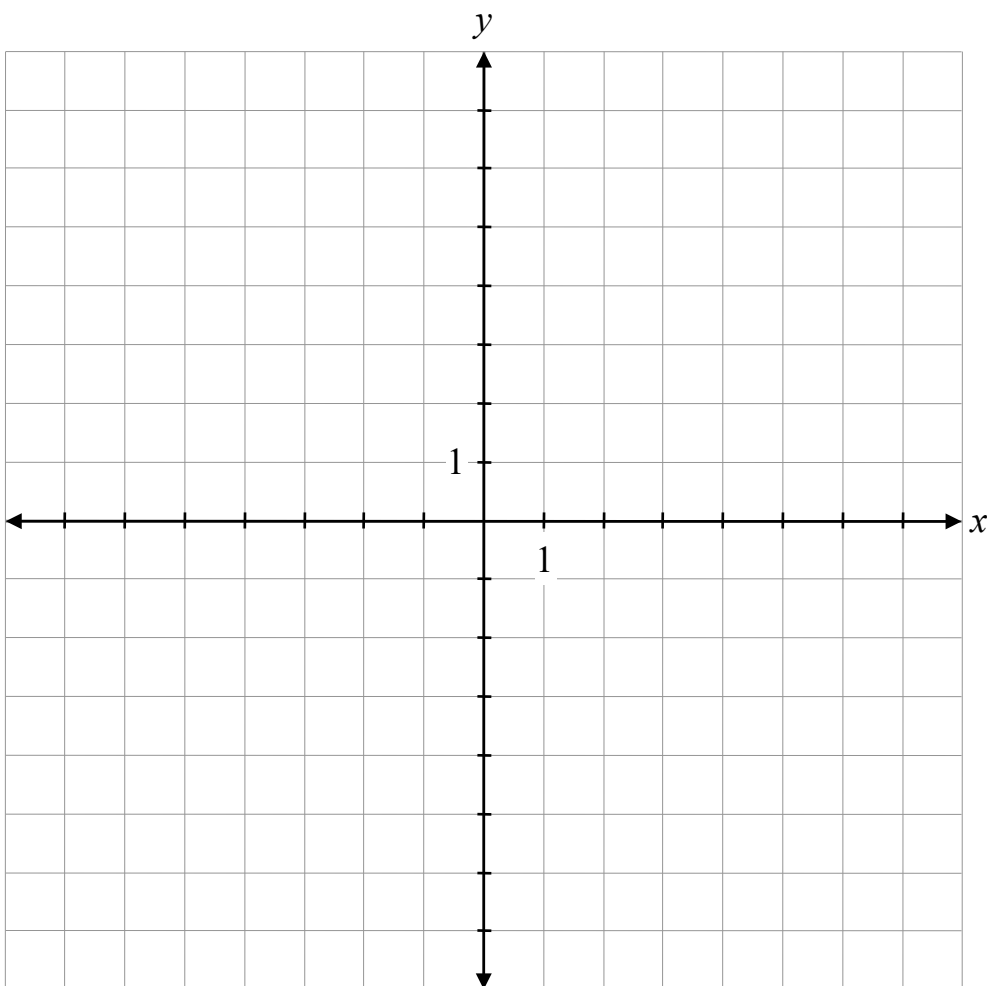
1 mark 120

Justify that $(x - 5)$ is not a possible factor of the function $P(x) = x^3 - 3x^2 - 4x + 12$.

Question 29

4 marks 121

Sketch the graph of $f(x) = \frac{6}{(x+2)(x-3)}$ and state the y-intercept.



y-intercept: _____

Question 30

2 marks 122

Determine how many 3-digit odd numbers less than 300 are possible using the digits 1, 2, 3, 4, 5, 6 if repetition is not allowed.

Question 31

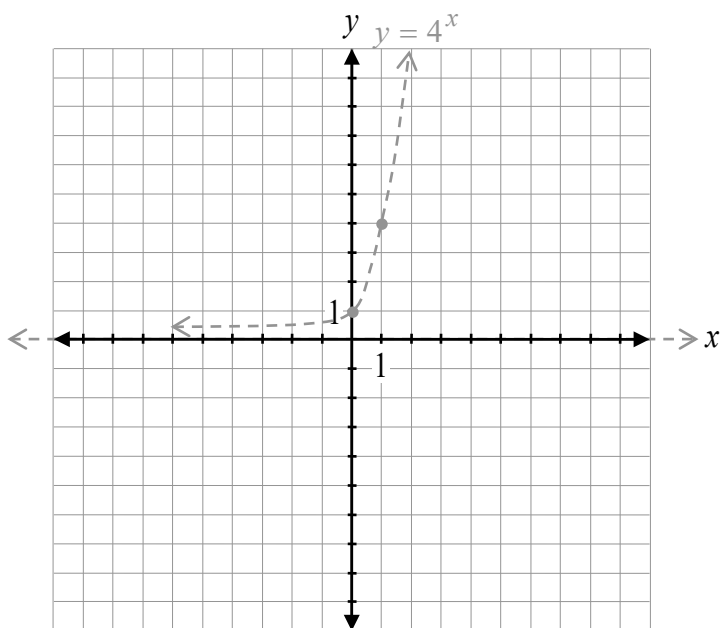
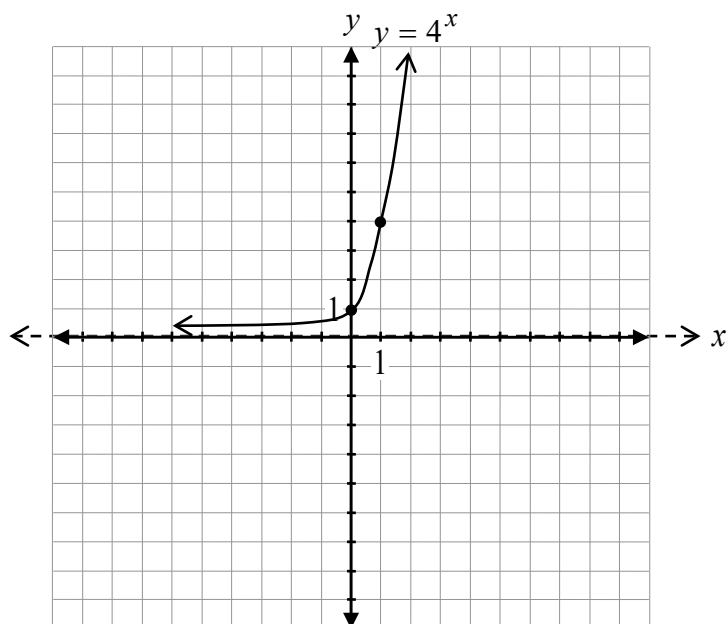
3 marks 123

Given that $\cos \alpha = -\frac{5}{13}$ and $\sin \beta = \frac{2}{3}$, where α and β terminate in the same quadrant, determine the exact value of $\cos(\alpha - \beta)$.

Question 32

3 marks 124

Given the graph of $y = 4^x$, sketch the graph of $y = 2(4)^{x-3} + 1$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 33

1 mark 125

Determine the coterminal angle of $\frac{\pi}{5}$ over the interval $[-2\pi, 0]$.

Question 34

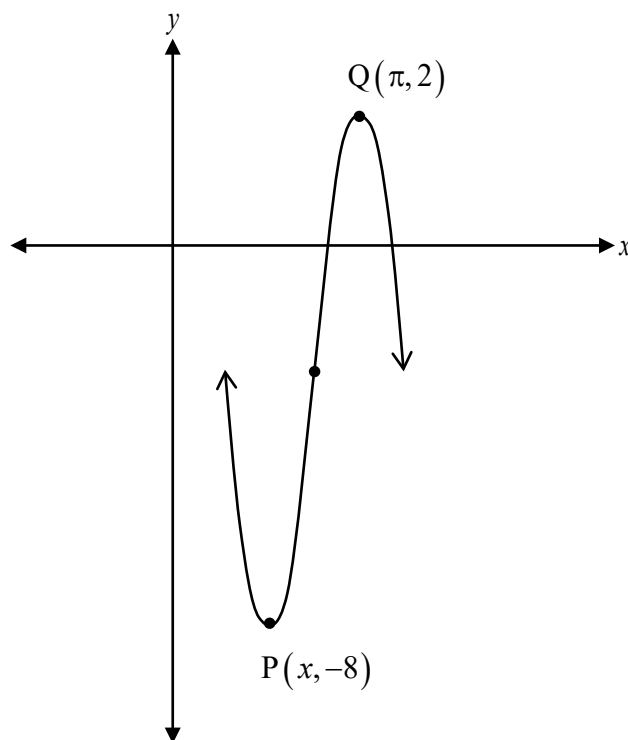
1 mark 126

State the domain of the graph of $y = \log(x - 4) - 8$.

Question 35

1 mark 127

Given the graph of $y = 5 \sin \left[2 \left(x + \frac{\pi}{4} \right) \right] - 3$, determine the exact value of the x -coordinate in the point P.



Question 36

2 marks 128

Verify that the following equation is true for $x = \frac{5\pi}{6}$.

$$\frac{\cos x}{1 - \sin x} = \frac{1 + \sin x}{\cos x}$$

Left-Hand Side	Right-Hand Side

Question 37

2 marks 129

Given that $(x + 1)$ is one of the factors of $P(x) = x^3 - x^2 + kx - 8$, determine the value of k .

Question 38

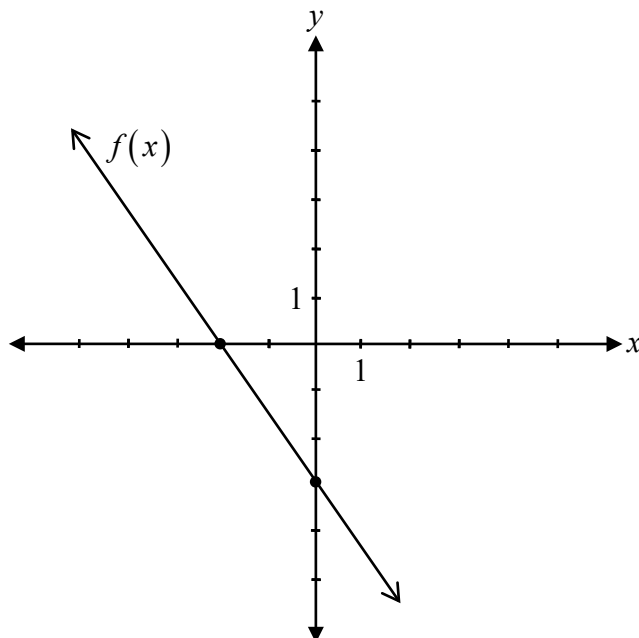
1 mark 130

Given the function $f(x) = \sqrt{x}$, describe how to use transformations to determine the domain of the function $g(x) = f(x + 2) + 1$.

Question 39

1 mark 131

Given the graph of $y = f(x)$, state the equation of the vertical asymptote of $y = \frac{1}{f(x)}$.



Question 40

2 marks 132

Solve, algebraically.

$$16^x = 64^{2x-1}$$

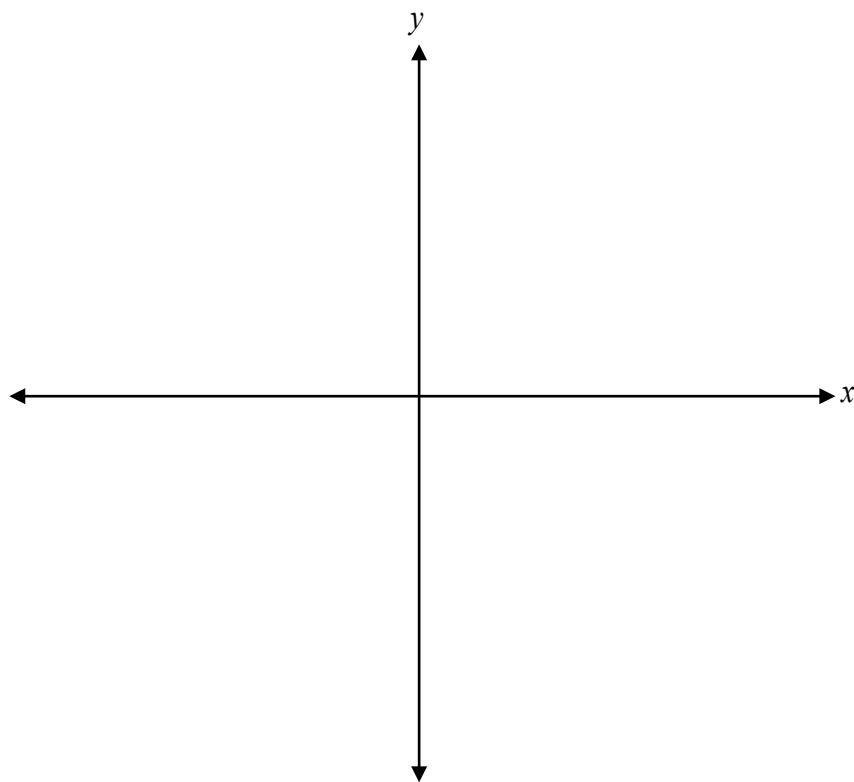
Question 41

2 marks 133

Given one of the factors of $P(x) = x^3 + 2x^2 - 5x - 6$ is $(x + 3)$, express $P(x)$ in completely factored form.

$P(x) =$ _____

Sketch the graph of $p(x) = 3(x+1)^2(x-2)^2$.



Question 43

2 marks 135

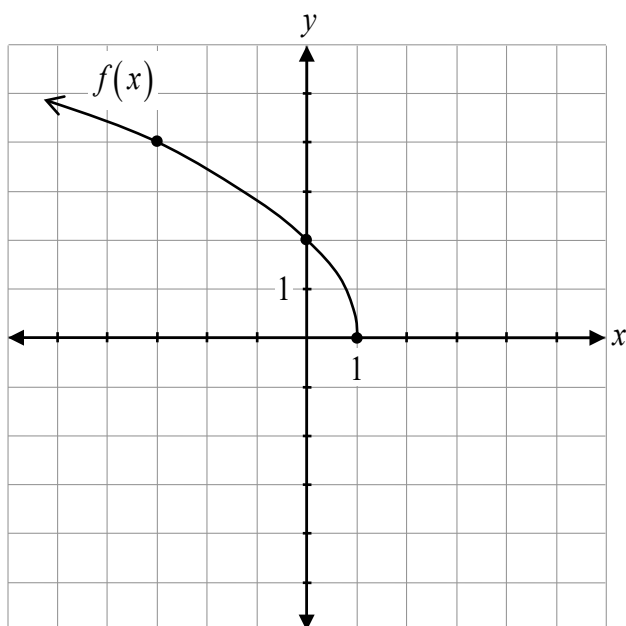
Given that $f(x) = x^2 - 4$ and $g(x) = \sqrt{x}$, determine $f(g(x))$ and state its domain.

$$f(g(x)) = \underline{\hspace{10cm}}$$

Question 44

3 marks 136

Determine a possible equation of the function $f(x)$.



$f(x) =$ _____

Question 45

1 mark 137

Explain why the graph of $y = \log_2 x$ does not have a y -intercept.

Question 46

3 marks 138

Evaluate.

$$\sin^2\left(-\frac{\pi}{3}\right) + \cos\left(\frac{17\pi}{6}\right)\sec\left(\frac{\pi}{6}\right)$$

Question 47

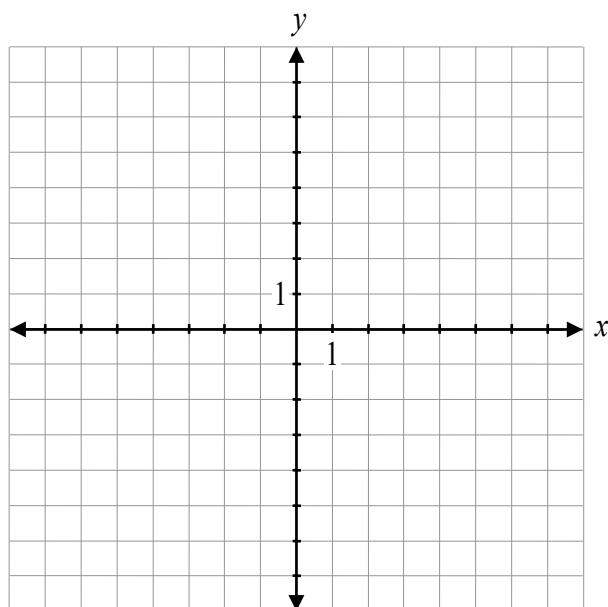
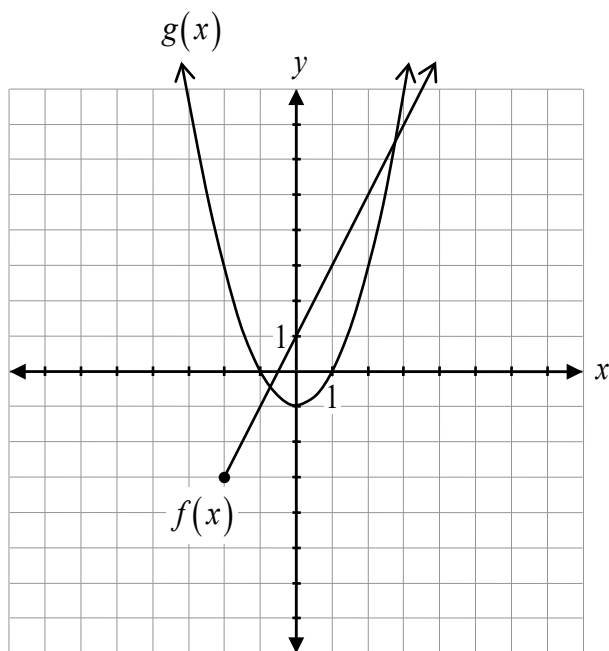
1 mark 139

Determine the coordinates of the point of discontinuity (hole) on the graph of $y = \frac{x^2 - 3x}{x}$.

Question 48

2 marks 140

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) + g(x)$.

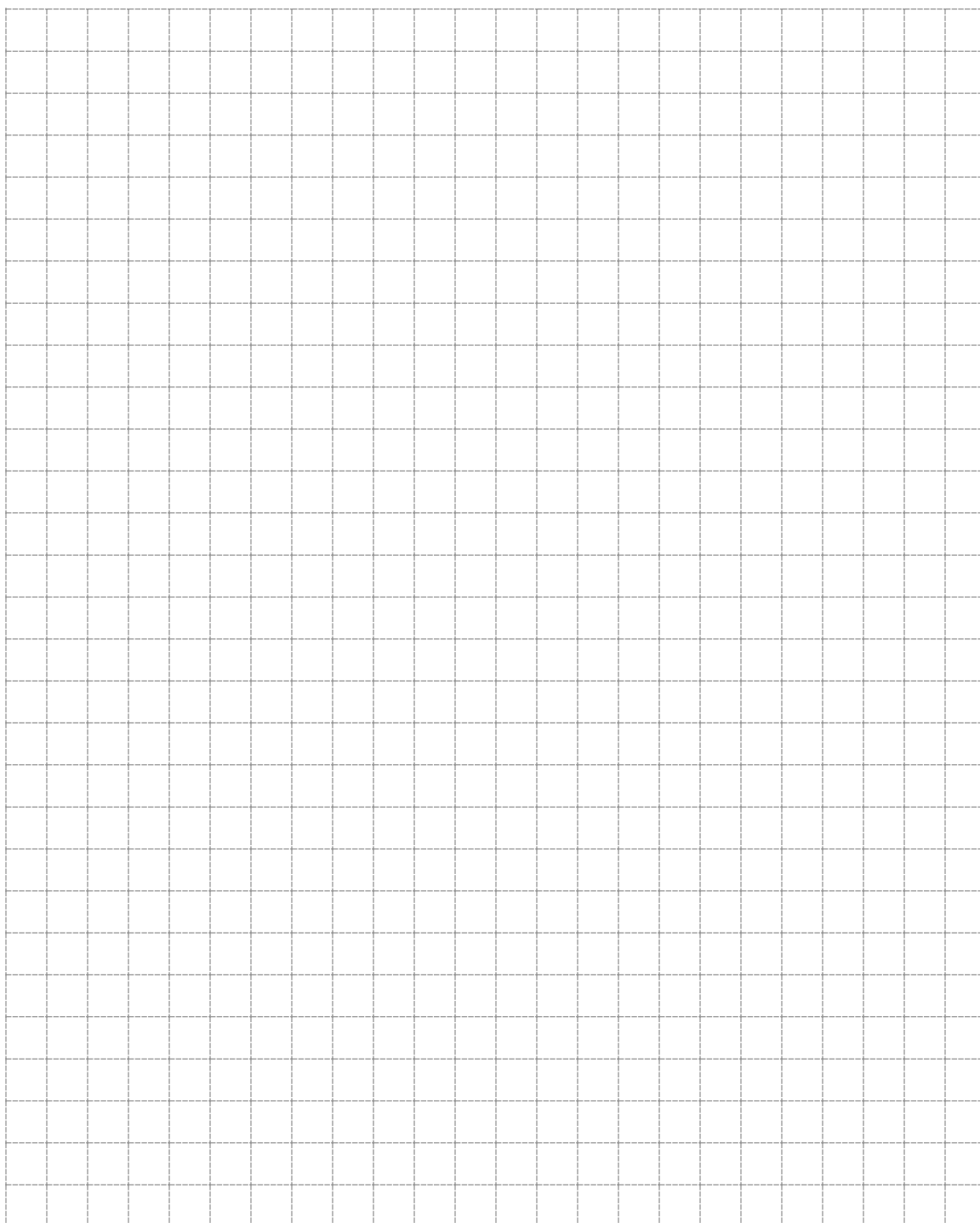


Question 49

2 marks 141

Given that $\csc \theta = -\frac{4}{\sqrt{7}}$ and $\cos \theta > 0$, determine the exact value of $\tan \theta$.

No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2020

Grade 12 pre-calculus mathematics achievement test.
Booklet 1. January 2020

This resource is available in print and electronic formats.

ISBN: 978-0-7711-6351-7 (print)

ISBN: 978-0-7711-6353-1 (pdf)

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DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

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	Selected Response	Constructed Response	Marks
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Booklet 2	8	21	56
Total	8	35	90

* The first 4 questions with the symbol  in *Booklet 1* require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

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- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
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Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{For } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

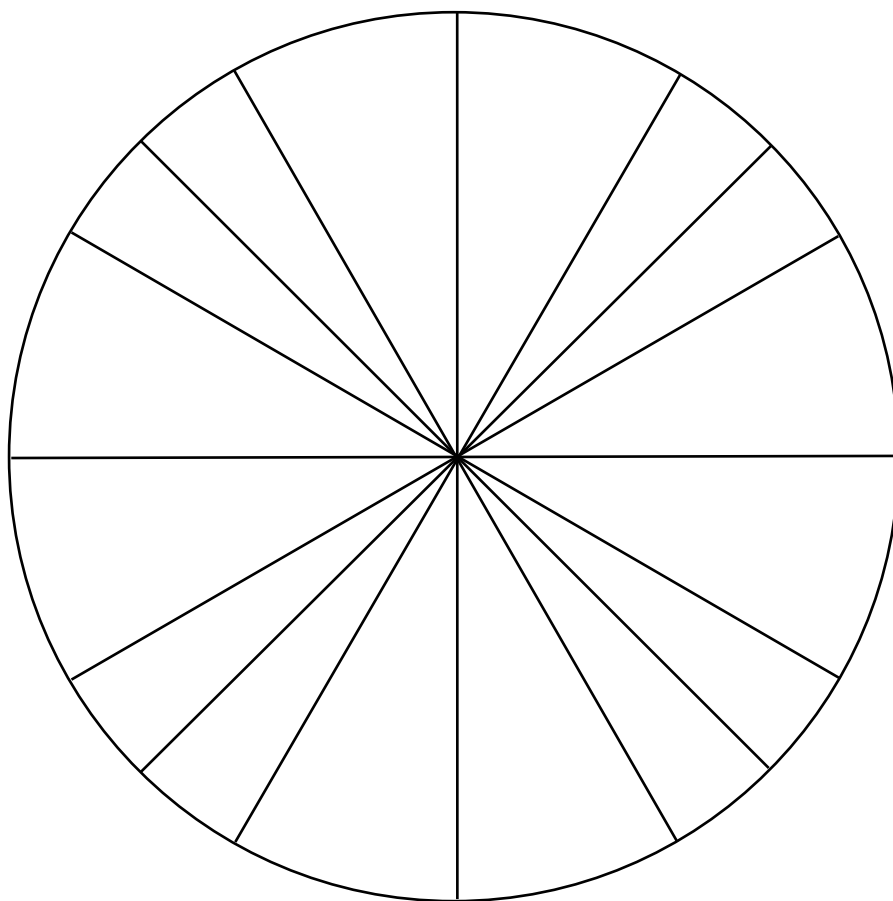
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

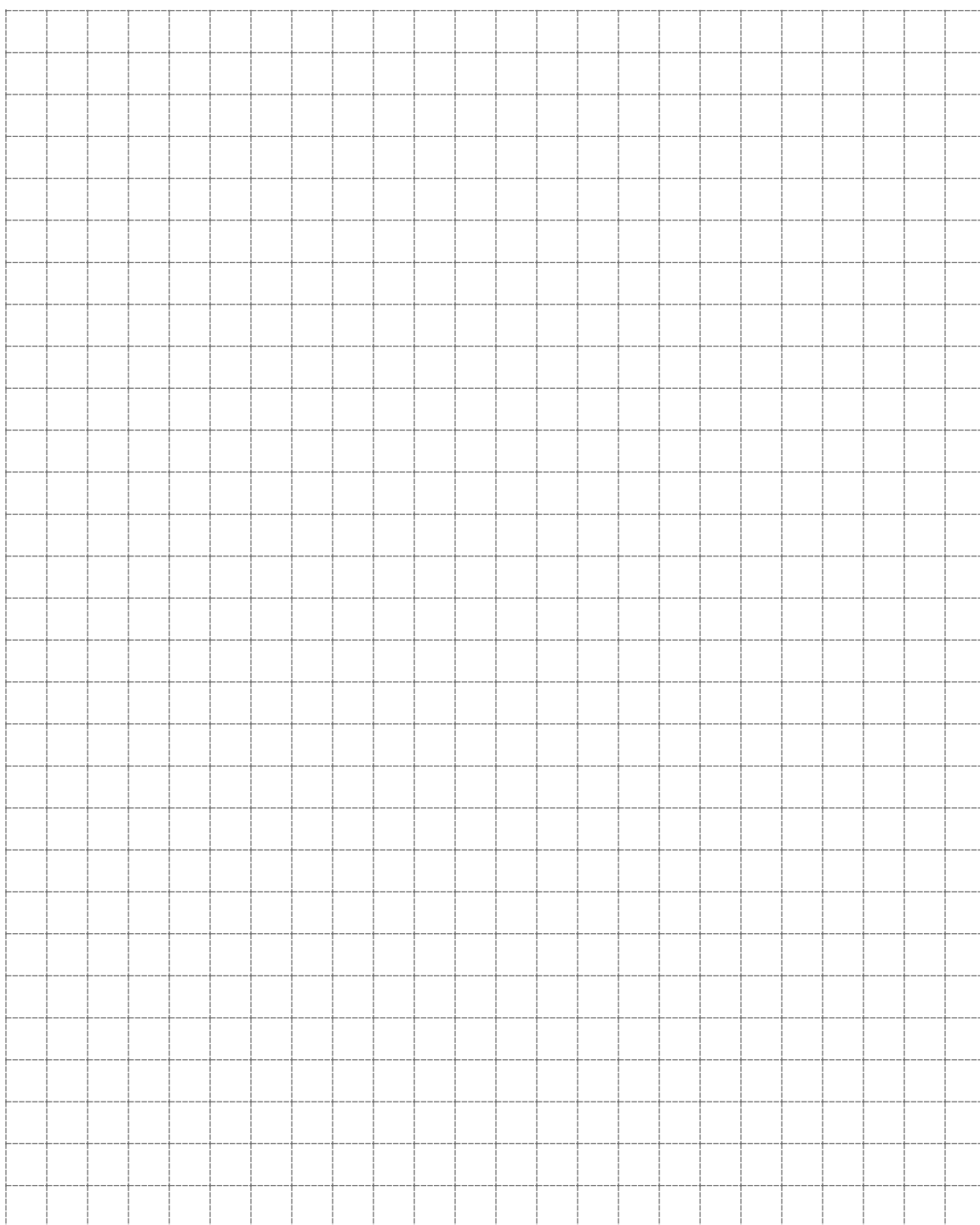
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



Question 1

3 marks 101

Determine and simplify the 8th term in the binomial expansion of $\left(x - \frac{2}{x^3}\right)^{10}$.

Question 2

2 marks

102

The temperature of hot chocolate is recorded as it cools down. The data shows that the temperature cools according to the equation:

$$T = 82(0.87)^t + 16$$

where T is the temperature of the hot chocolate, in degrees Celsius after t minutes and

t is the time, in minutes, after the hot chocolate is made.

In order to avoid burns, hot chocolate should not be served at a temperature higher than 71°C . Determine, algebraically, the amount of time it will take the hot chocolate to reach this temperature.

Question 3

3 marks 103

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$3\sin^2\theta + 6\sin\theta + 2 = 0$$

Question 4



3 marks 104

A committee of 4 people is to be selected from 6 high school students and 5 middle years students.

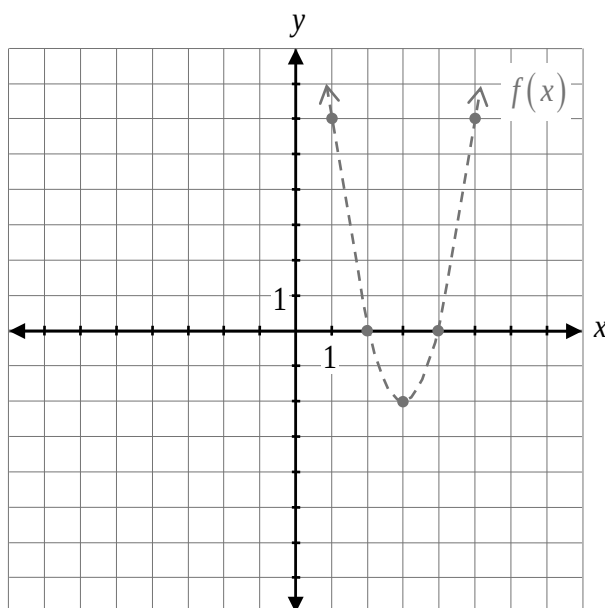
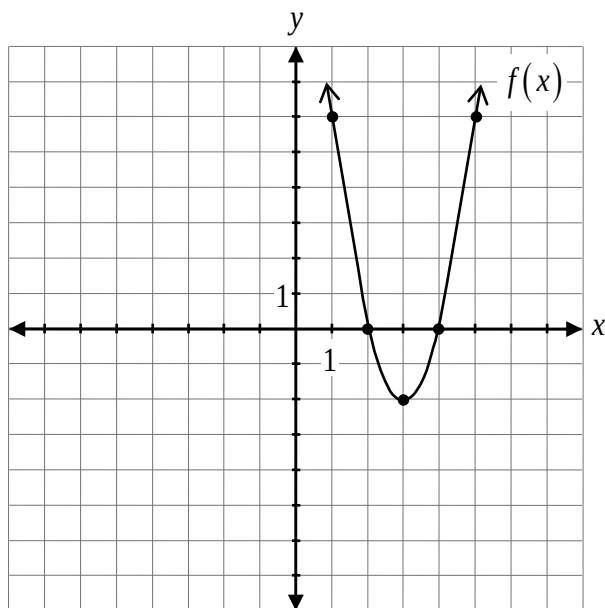
Determine the number of possible committees if the committee must have at least 3 high school students.

Note: A calculator is not required for the remaining test questions.

Question 5

2 marks 105

Given the graph of $y = f(x)$, sketch the graph of $y = -|f(x)|$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 6

3 marks 106

Describe the transformations used to obtain the graph of the function $y = 4f\left(\frac{x}{2}\right) - 3$ from the graph of $y = f(x)$.

Question 7

2 marks 107

If $\log 4 = m$ and $\log 3 = n$, express $\log 48$ in terms of m and n .

Question 8

2 marks 108

State the domain and range of the radical function, $f(x) = -3\sqrt{x-8} + 1$.

Domain: _____

Range: _____

Question 9

1 mark

109

Explain why the graph of $y = \tan x$ does not have an amplitude.

Question 10

3 marks 110

Prove the following identity for all permissible values of θ .

$$\frac{\sec \theta - \sin^2 \theta \sec \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Left-Hand Side

Right-Hand Side

Question 11**2 marks** 111

Given that $x = 3$ is one of the zeros of $p(x) = x^3 - 7x - 6$, express $p(x)$ in completely factored form.

$p(x) =$ _____

Question 12

3 marks 112

Solve, algebraically.

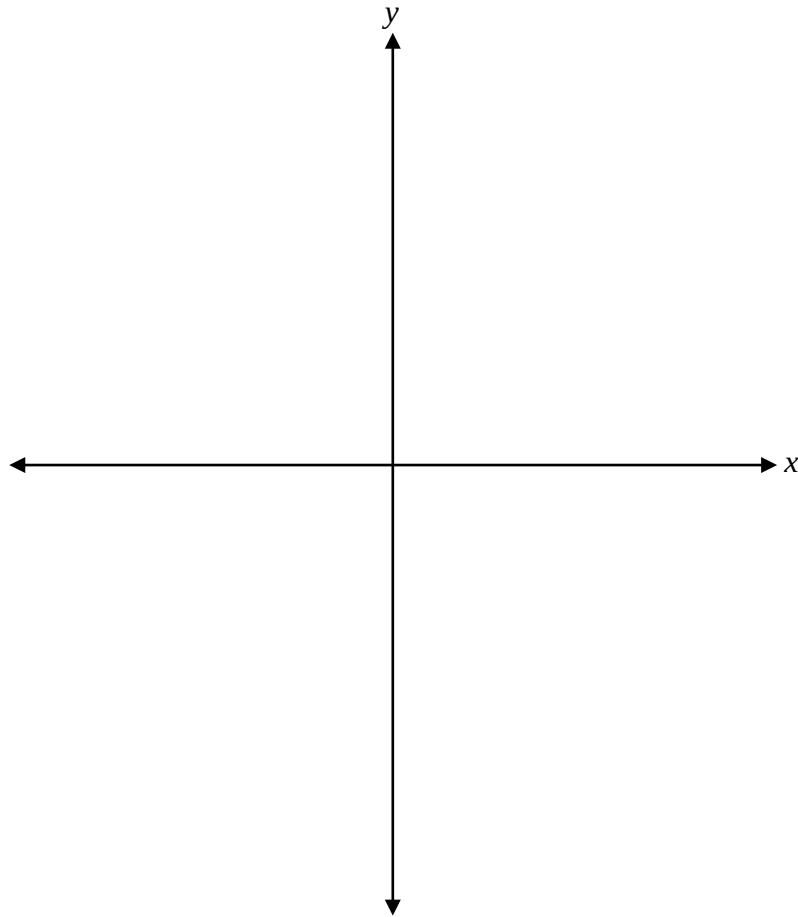
$${}_{n+1}P_2 = 6$$

Question 13

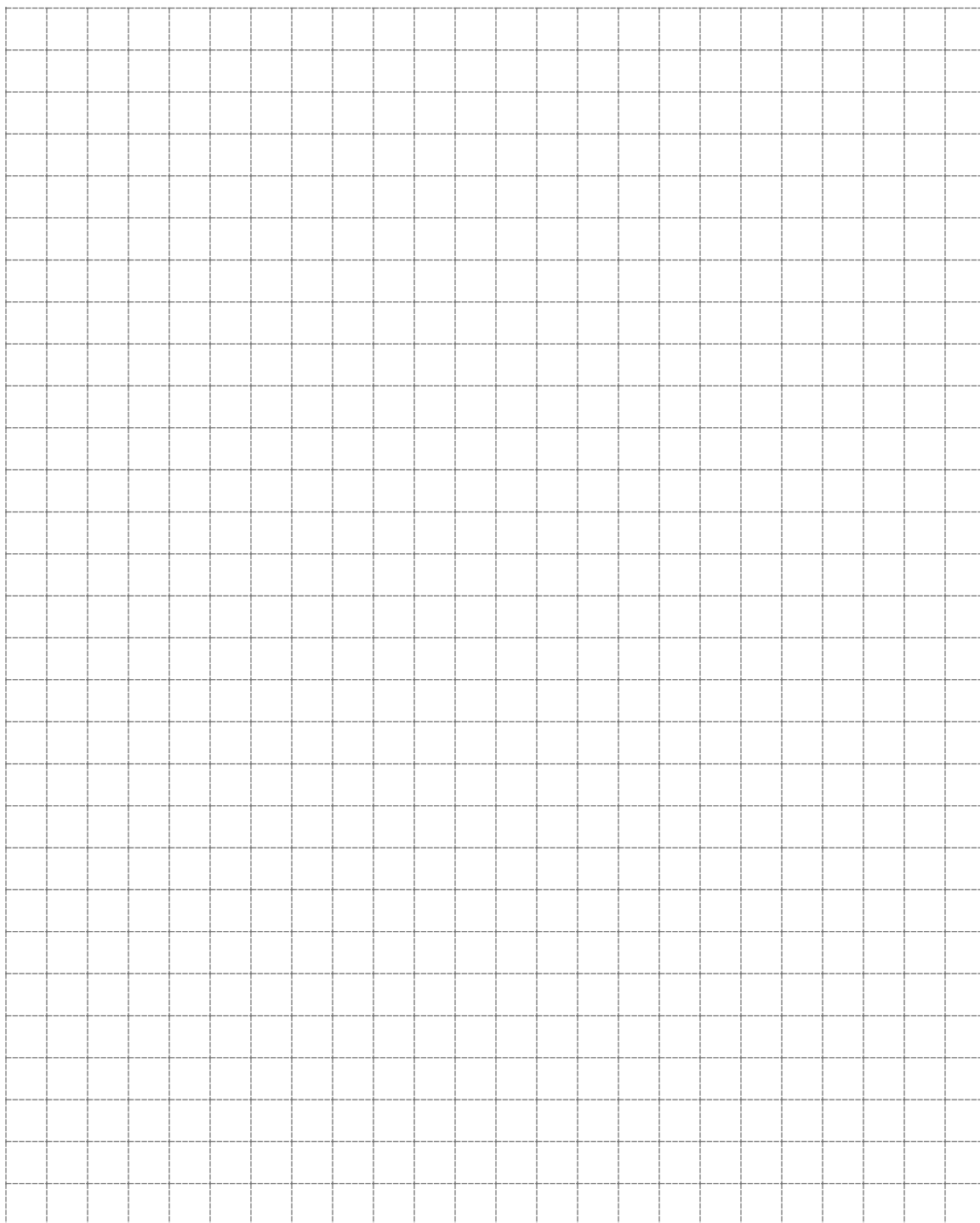
2 marks 113

Express $\log(3x - 1) - \log x + \log 9$ as a single logarithm.

Sketch the graph of $P(x) = -(x-2)^2(x+3)$.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2020

Grade 12 pre-calculus mathematics achievement test.
Booklet 2. January 2020

This resource is available in print and electronic formats.

ISBN: 978-0-7711-6359-3 (print)

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Total	8	35	90

* The first 4 questions with the symbol  in *Booklet 1* require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- Write each solution in the space provided.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.

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Question 15**1 mark**

Given $f(x) = -2x - 5$ and $g(x) = x + 7$, identify the equation of $h(x) = f(g(x))$.

- a) $h(x) = -2x^2 - 35$
- b) $h(x) = -2x - 19$
- c) $h(x) = -2x^2 - 19x - 35$
- d) $h(x) = -2x + 2$

Question 16**1 mark**

The function $y = f(x)$ has a domain of $[2, 5]$ and a range of $[-6, 3]$. Identify the range of the function $y = f^{-1}(x)$.

- a) $[-6, 3]$
- b) $[-5, 2]$
- c) $[-3, 6]$
- d) $[2, 5]$

Question 17**1 mark**

Identify the function that has a graph with a point of discontinuity (hole) at $x = -3$.

- a) $y = \frac{x+3}{x^2-9}$
- b) $y = \frac{x-3}{x^2-9}$
- c) $y = \frac{x^2-9}{x-3}$
- d) $y = \frac{x^2+9}{x+3}$

Question 18**1 mark**

Identify the value of $\ln e$.

- a) 0
- b) $\log e$
- c) 1
- d) e

Question 19**1 mark**

When the polynomial function, $p(x)$, is divided by $(x-4)$ the remainder is 17. Identify which statement is true.

- a) $p(4) = 17$
- b) $p(-4) = 17$
- c) $p(4) = 0$
- d) $p(-4) = 0$

Question 20**1 mark**

Identify the expression equivalent to $\frac{(n-6)!}{(n-4)!}$.

- a) $(n-4)(n-5)$
- b) $\frac{1}{(n-4)(n-5)}$
- c) $(n-6)(n-5)$
- d) $\frac{1}{(n-6)(n-5)}$

Question 21**1 mark**

Identify the quadrant in which θ terminates if $\sec \theta = -\frac{4}{3}$ and $\sin \theta > 0$.

- a) I
- b) II
- c) III
- d) IV

Question 22**1 mark**

Identify the expression that represents all angles that are coterminal with $\frac{\pi}{3}$.

- a) $\frac{\pi}{3} + \pi k, k \in \mathbb{Z}$
- b) $\frac{\pi}{3} + \pi k, k \in \mathbb{R}$
- c) $\frac{\pi}{3} + 2\pi k, k \in \mathbb{Z}$
- d) $\frac{\pi}{3} + 2\pi k, k \in \mathbb{R}$

Question 23

1 mark 115

Given $f(x) = \frac{1}{2}x - 3$, state the coordinates of an invariant (unchanged) point when sketching the graph of $y = \sqrt{f(x)}$.

Question 24

2 marks 116

Solve, algebraically.

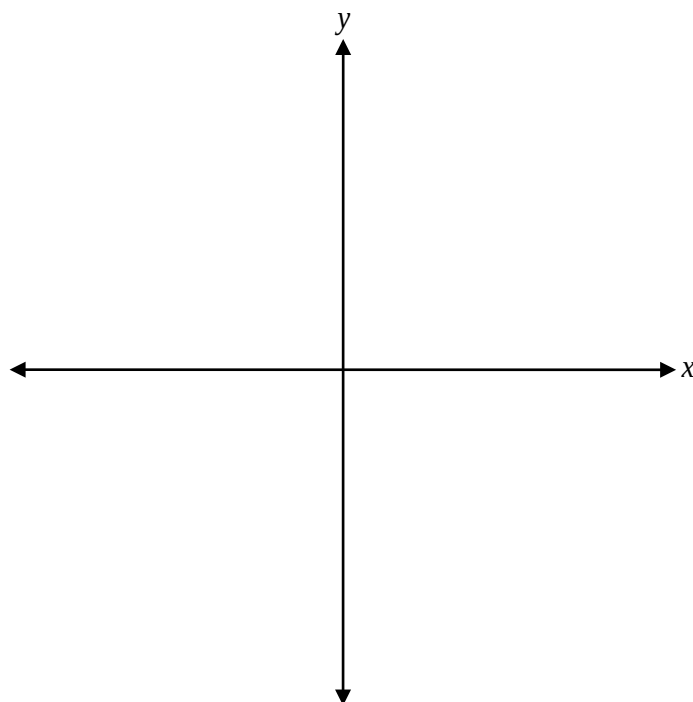
$$125^{3x+4} = \left(\frac{1}{5}\right)^x$$

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ and $\cos \theta = -\frac{\sqrt{2}}{2}$, justify that the value of $\tan(2\theta)$ is undefined.

Question 26

3 marks 118

Sketch the graph of $y = -2\log_3 x$.



Evaluate.

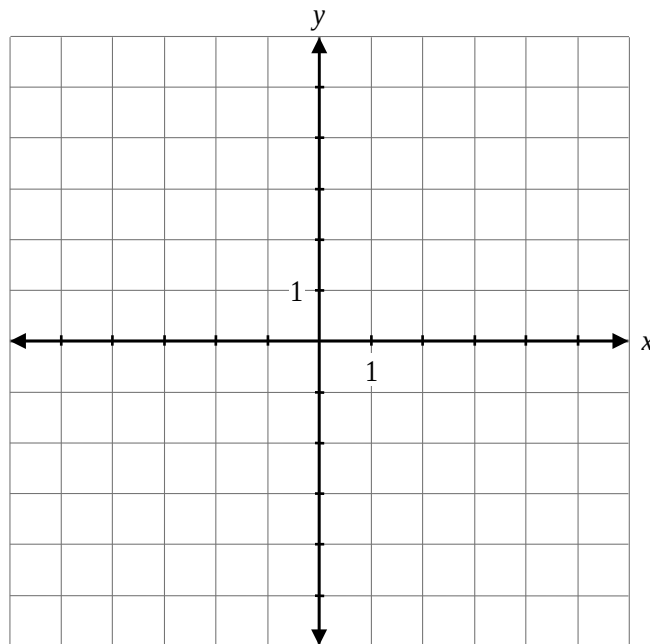
$$\cos\left(\pi \cdot \sin\left(-\frac{\pi}{6}\right)\right)$$

Question 28

a) 1 mark b) 1 mark

120
121

a) Given $f(x) = -x + 2$, sketch the graph of $h(x) = f(f(x))$.



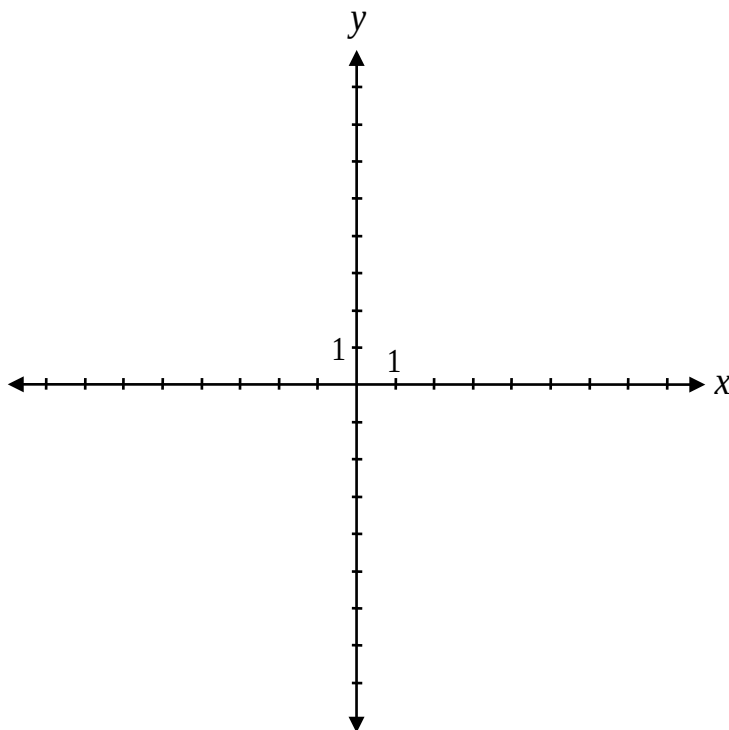
b) Explain why the domain of $h(x) = f(f(x))$ does not have any restrictions.

Question 29

a) 3 marks b) 1 mark

122
123

a) Sketch the graph of $f(x) = \frac{3x-5}{2x+4}$.



b) State the range of $f(x)$.

Range: _____

Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has a zero at 3 with a multiplicity of 2
- $p(x)$ has zeroes at -1 and -2
- $p(x)$ passes through the point $(2, 24)$.

$p(x) =$ _____

Question 31

3 marks 125

Verify, by substitution, that the equation $\frac{\cos \theta + \cot \theta}{\cot \theta} = 1 + \sin \theta$ is true for $\theta = \frac{2\pi}{3}$.

Left-Hand Side	Right-Hand Side

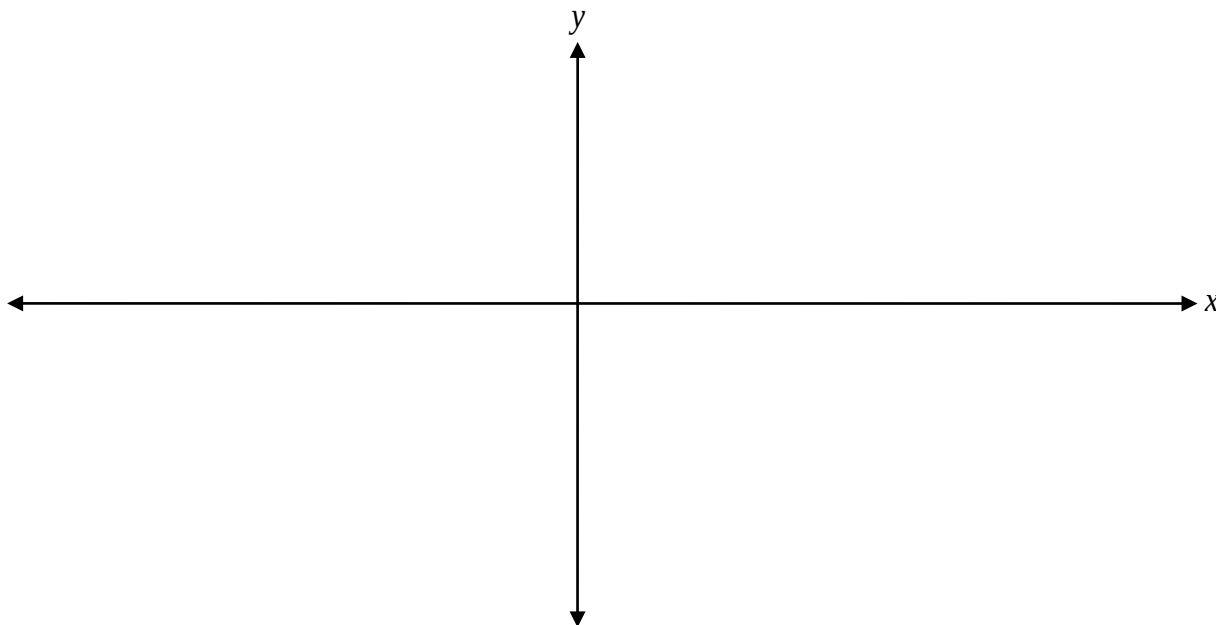
Jyugo was asked to state the equation of the vertical asymptote(s) on the graph of $f(x) = \frac{3x-15}{x^2-5x}$.

His solution:

$$f(x) = \frac{3(x-5)}{x(x-5)}$$
$$x=0 \quad x=5$$

Explain why his solution is incorrect.

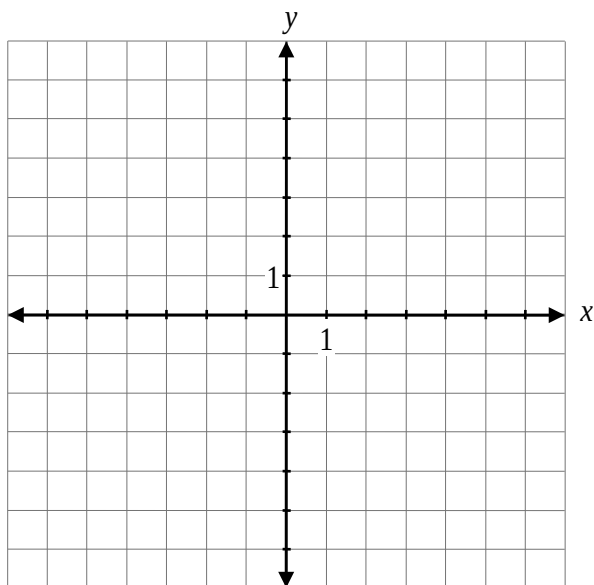
Sketch the graph of $y = \sin\left(\frac{1}{2}x\right) - 1$ over the interval $[-2\pi, 2\pi]$.



Question 34

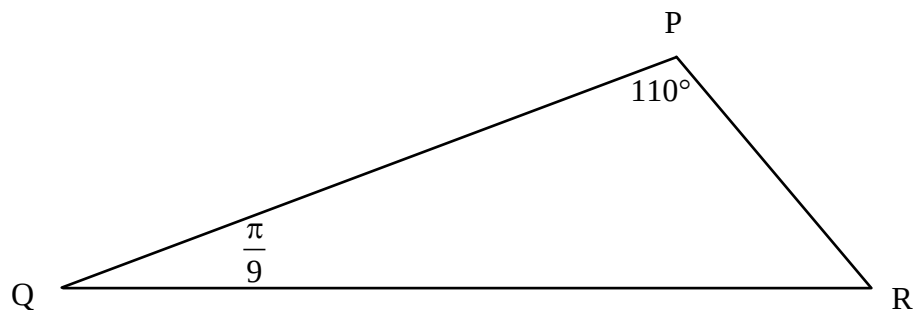
4 marks 128

Sketch the graph of $y = \sqrt{-2x + 6}$.



Question 35**2 marks** 129

Determine the measure of angle R, in radians.

**Question 36****1 mark** 130

Given $f(x) = \sqrt{x-4}$, state the equation of the resulting function, $g(x)$, after a reflection over the x -axis.

$g(x) =$ _____

Question 37

a) 1 mark b) 3 marks

131
132

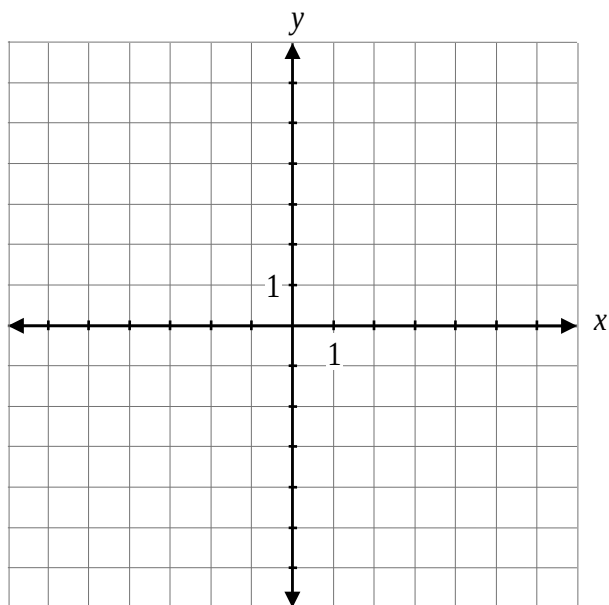
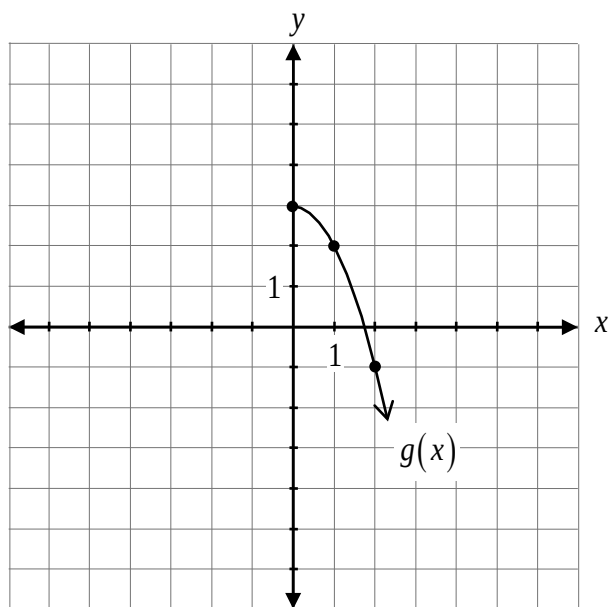
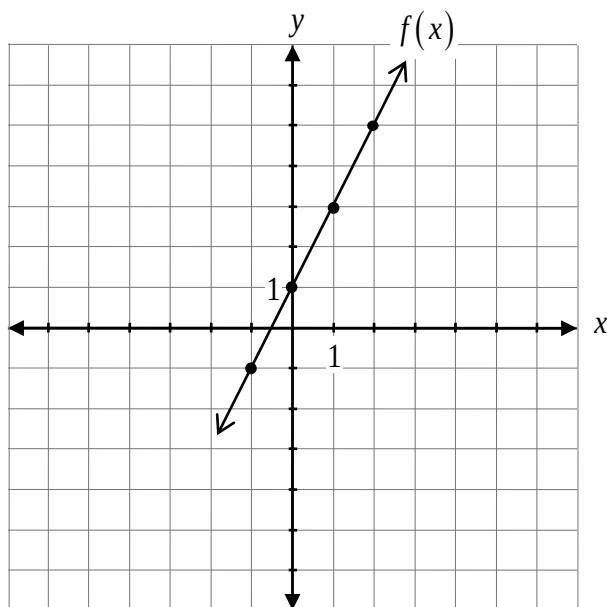
a) Determine the coterminal angle of $\frac{29\pi}{12}$ over the interval $[0, 2\pi]$.

b) Determine the exact value of $\sin\left(\frac{29\pi}{12}\right)$.

Question 38

2 marks 133

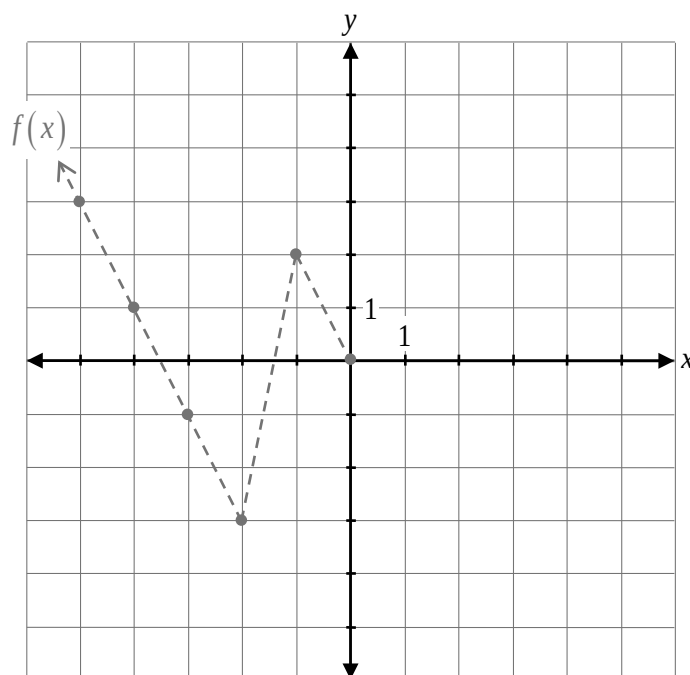
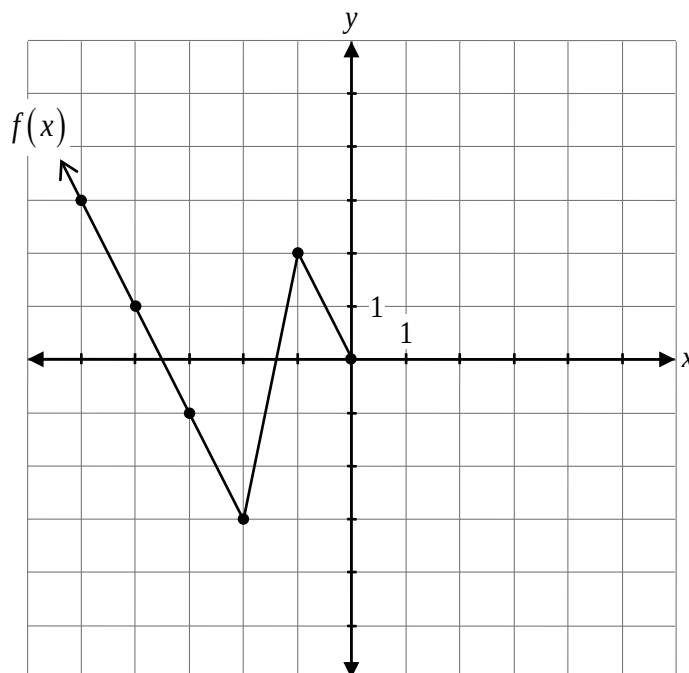
Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) - g(x)$.



Determine an equation of a sinusoidal function that has the following characteristics:

- an amplitude of 3
- a period of 6
- a minimum value of -5

Given the graph of $y = f(x)$, sketch the graph after a reflection over the line $y = x$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 41

a) 1 mark b) 1 mark

136
137

a) Determine the remainder when $3x^3 + 5x^2 - 13x - 3$ is divided by $(x + 3)$.

b) Is $(x + 3)$ a factor of $3x^3 + 5x^2 - 13x - 3$? Explain your reasoning.

Question 42**3 marks**

138

State the range, the y-intercept, and the equation of the asymptote of the exponential function,

$$f(x) = 3^{x-1} + 2.$$

Range: _____

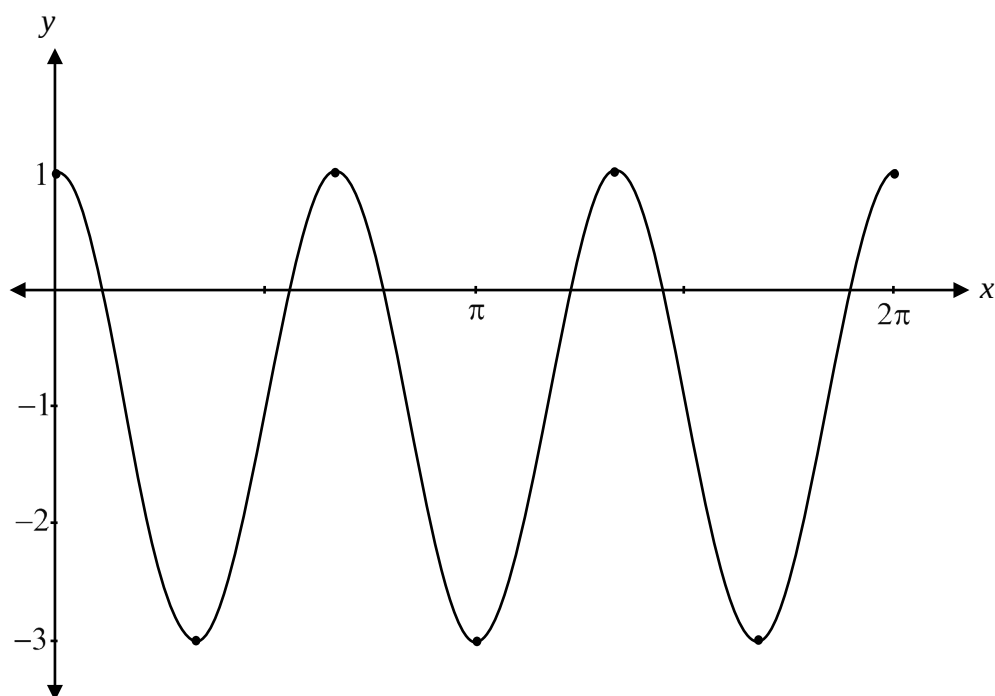
y-intercept: _____

Equation of the asymptote: _____

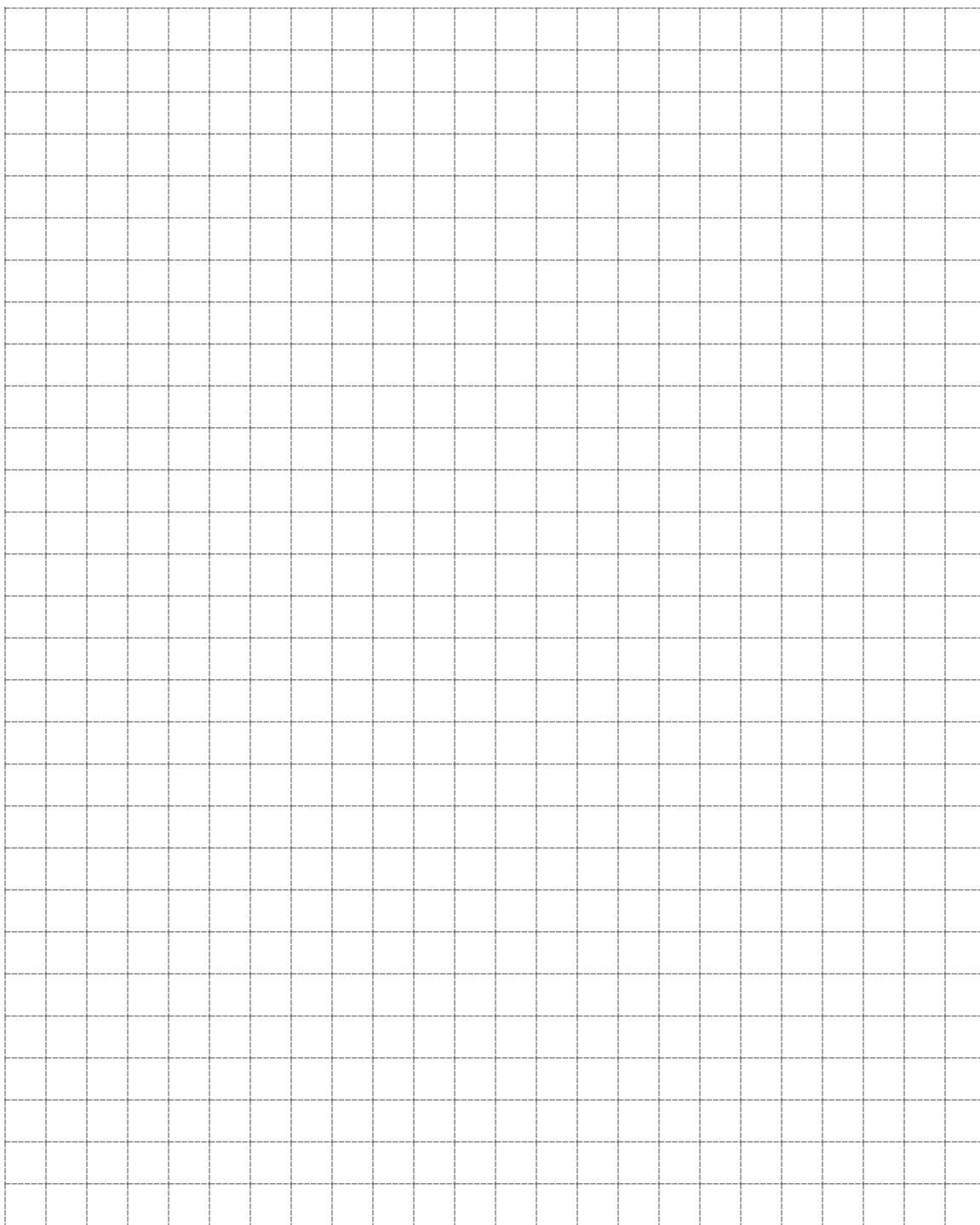
Question 43**1 mark**

139

The graph of $y = 2 \cos(3x) - 1$ below can be used to solve the equation $0 = 2 \cos(3x) - 1$ over the interval $[0, 2\pi]$. Indicate on the graph where to find at least one solution to the equation $0 = 2 \cos(3x) - 1$.



No marks will be awarded for work done on this page.



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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2024

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Booklet 1. January 2024

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Winnipeg, Manitoba, Canada

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$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

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$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a (MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

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$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{Given } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

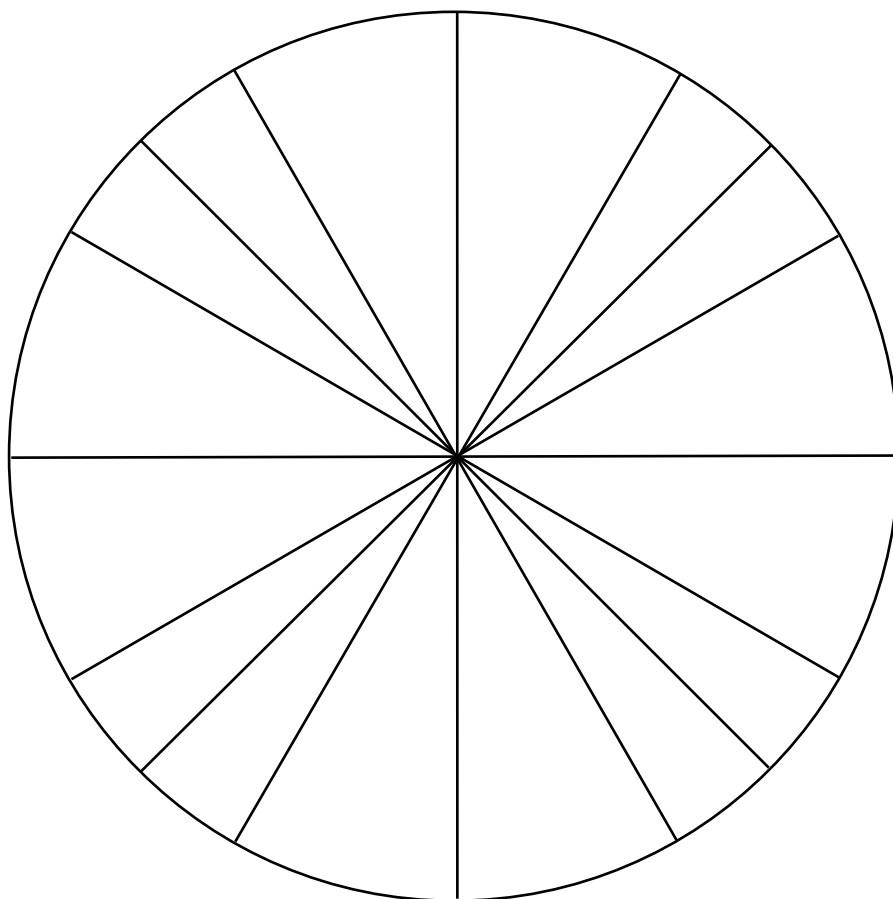
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

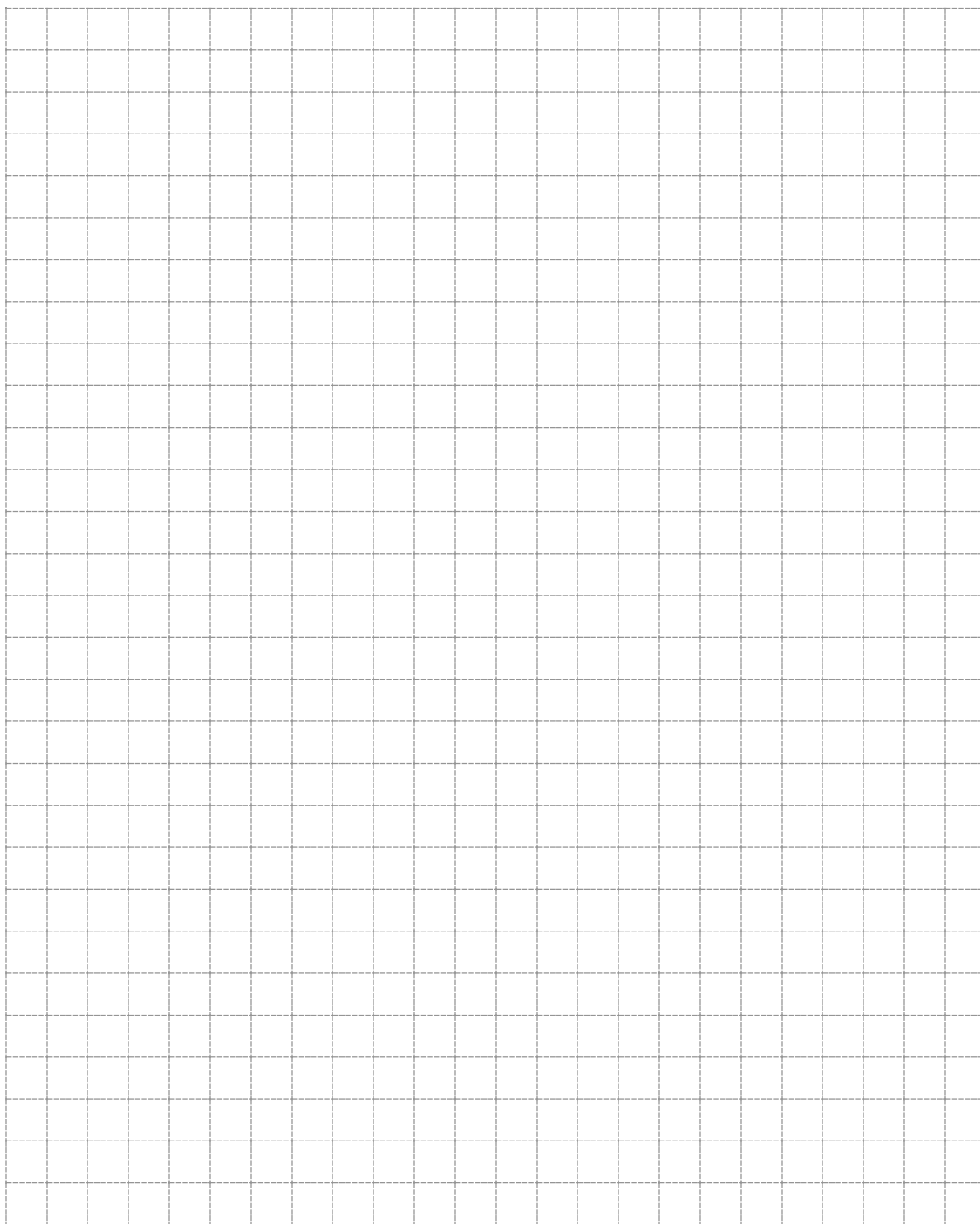
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



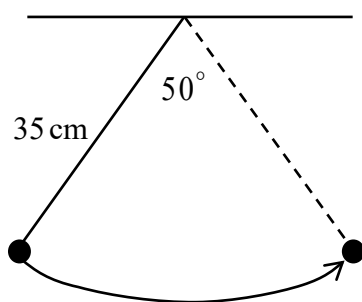
No marks will be awarded for work done on this page.



Question 1

2 marks 101

A pendulum that is 35 cm long swings through an angle of 50° . Determine the length of the arc through which the pendulum swings.



Question 2

4 marks 102

Solve algebraically, where $0 \leq \theta \leq 2\pi$.

$$2\cos^2\theta = \sin^2\theta - 2\cos\theta$$

Question 3

2 marks 103

Determine the number of arrangements of the letters in the word ATTENTION which begin with the letter A.

Question 4

3 marks 104

Solve for x , algebraically.

$$e^{2x+1} = 5^x$$

Question 5

2 marks 105

There are 10 teachers and 17 students who would like to attend a field trip.

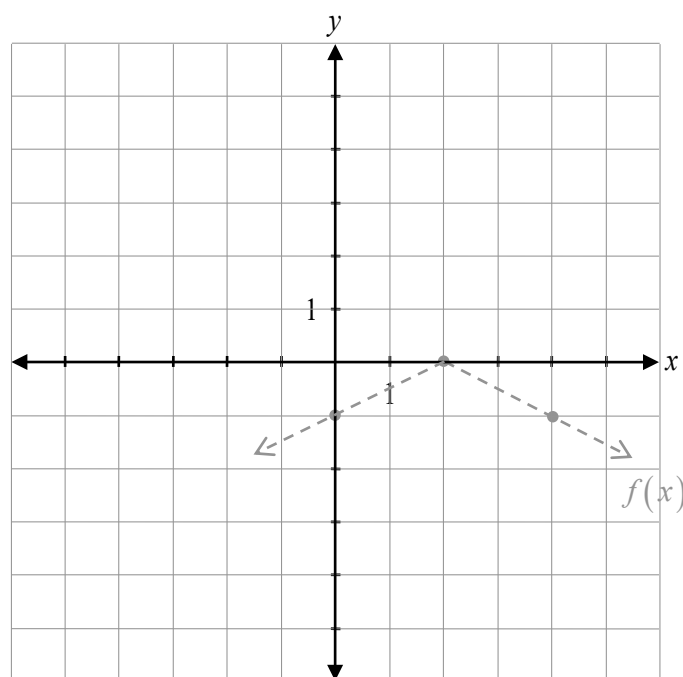
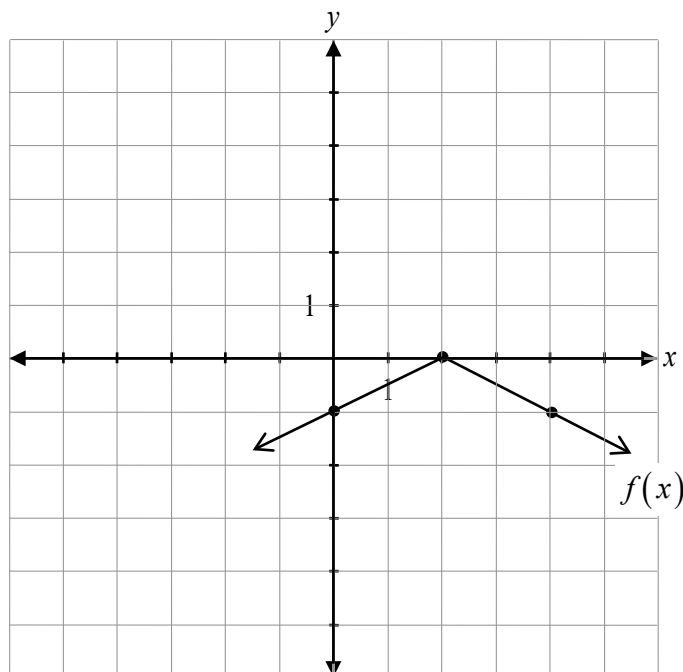
Determine the number of ways that 3 teachers and 9 students can be selected given that Mr. Jones and Mrs. Carol, two of the teachers, must be selected to attend the field trip.

Note: A calculator is not required for the remaining test questions.

Question 6

2 marks 106

Given the graph of $f(x)$, sketch the graph of $y = -f(2x)$.



The graph of $f(x)$ has already been drawn for your reference.
No marks will be awarded for the graph of $f(x)$

Question 7

1 mark

107

There are 5 roads between Anneville and Berrybourg, and 2 roads between Berrybourg and Carriton.

Determine how many ways Blake can travel from Anneville to Carriton and back to Anneville, given the following conditions:

- he must travel through Berrybourg in both directions
- he cannot use the same road twice

Determine which term contains x^0 in the binomial expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

Question 9

2 marks 109

Given $f(x) = x^3 + 1$, determine the equation of $f^{-1}(x)$.

Question 10

1 mark

110

Guillermo was asked to determine the number of ways to select a president, a vice president, and a treasurer from a group of 11 people.

His solution: ${}_{11}C_3$.

Explain why he should have used a permutation instead of a combination.

Question 11

3 marks 111

Prove the following identity for all permissible values of x .

$$\frac{\csc^2 x \sec x}{\tan x + \cot x} = \csc x$$

Left-Hand Side	Right-Hand Side

Question 12

3 marks 112

Determine the value of x , algebraically.

$$5\log_a 2 - \frac{1}{4}\log_a 16 = \log_a x$$

Question 13

1 mark 113

Tamara must determine the factors of $x^4 - 13x + 2x^3 - 14x^2 + 24$.

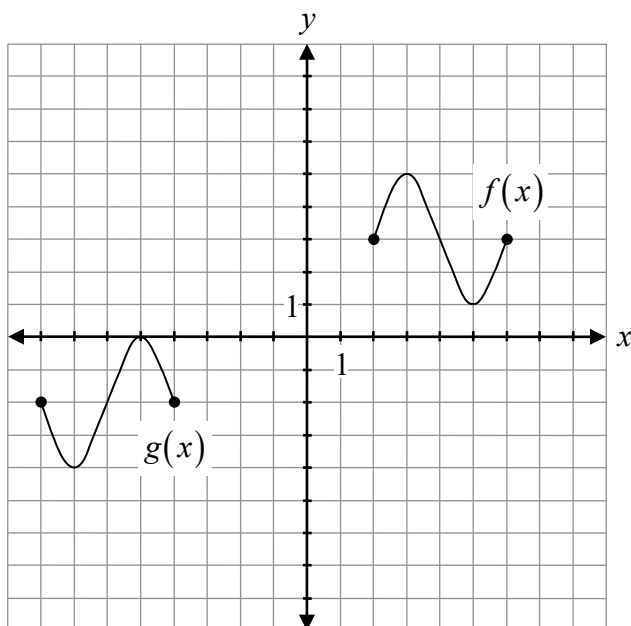
Explain why the coefficients Tamara used to set up her synthetic division are not written correctly.

$$\begin{array}{r|rrrrr} & 1 & -13 & 2 & -14 & 24 \end{array}$$

Question 14

3 marks 114

Determine the equation of the graph of $g(x)$ in terms of $f(x)$.



$g(x) =$ _____

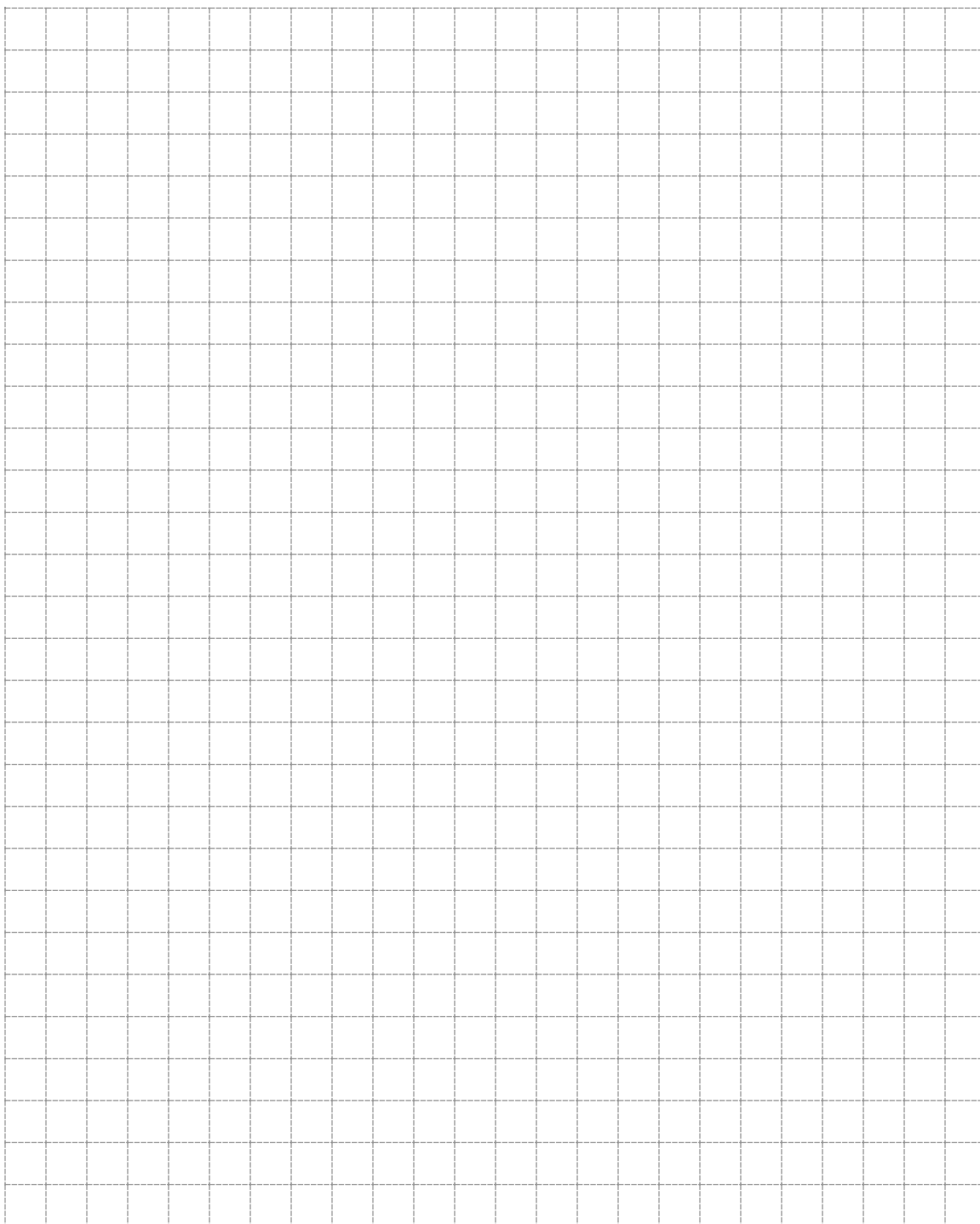
Question 15

2 marks 115

Expand, using the laws of logarithms.

$$\log_2 \left[\frac{(x-1)(x-2)}{x} \right]$$

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2024

Grade 12 pre-calculus mathematics achievement test.
Booklet 2. January 2024

This resource is available in print and electronic formats.

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Manitoba Education and Early Childhood Learning
Winnipeg, Manitoba, Canada

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1	-	15	33
Booklet 2	8	24	57
Total	8	39	90

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- Write each solution in the space provided.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.

No marks will be awarded for work done on this page.

Question 16**1 mark**

Identify the range of the function $g(x) = \frac{1}{2}f(x+1)$, given that the range of the function $y = f(x)$ is $[-6, 4]$.

- a) $[-12, 8]$
- b) $[-7, 3]$
- c) $[-5, 5]$
- d) $[-3, 2]$

Question 17**1 mark**

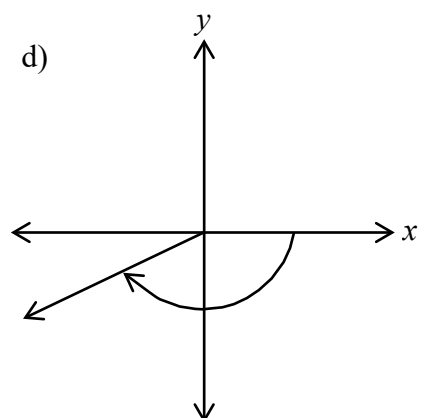
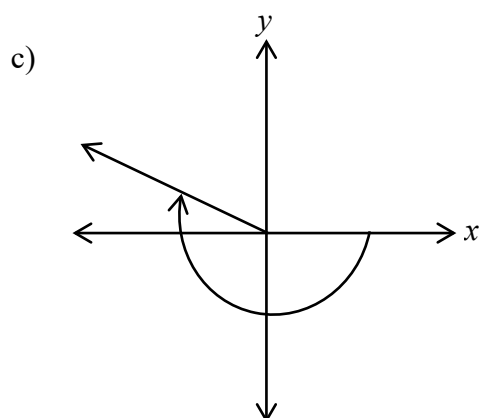
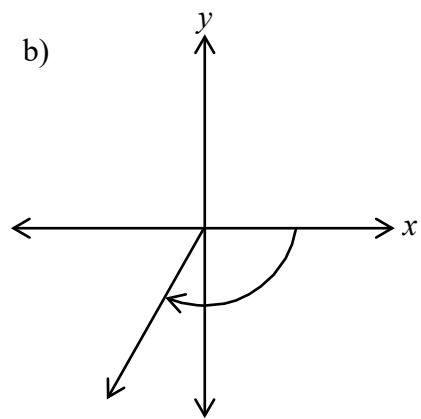
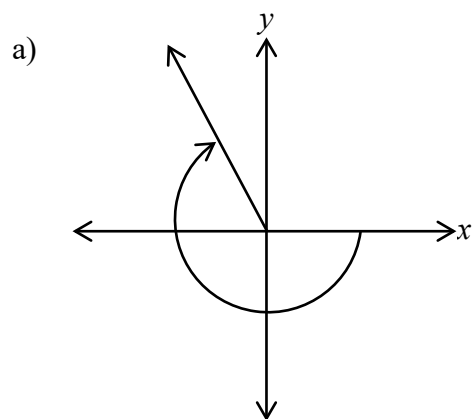
Identify the value of a , given that there are 11 terms in the expansion of $(3x^4 - y)^{2a}$.

- a) 5
- b) 6
- c) 10
- d) 11

Question 18

1 mark

Identify the angle that best represents $\theta = -\frac{6\pi}{5}$.

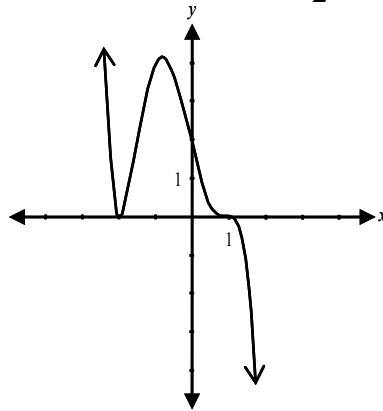


Question 19

1 mark

Identify a possible value for n , given the graph of $y = -\frac{1}{2}(x+2)^2(x-1)^n$.

- a) 1
- b) 2
- c) 3
- d) 4



Question 20

1 mark

Identify the statement that is false, given $g(x) = \frac{8x^2}{x^2 - 16}$.

- a) the graph of $g(x)$ has one x -intercept.
- b) the graph of $g(x)$ has a point of discontinuity (hole) at $x = 0$.
- c) the graph of $g(x)$ has two vertical asymptotes.
- d) the graph of $g(x)$ has a horizontal asymptote at $y = 8$.

Question 21

1 mark

Identify the equivalent form of $\log_a \left(\frac{1}{x^2} \right)$.

- a) $-2\log_a x$
- b) $1 - 2\log_a x$
- c) $2\log_a x$
- d) $-2\log_a \left(\frac{1}{x} \right)$

Question 22

1 mark

Identify which one of the following expressions is equivalent to ${}_{13}C_6$.

a) ${}_{13}P_6$

b) ${}_{13}C_7$

c) ${}_{12}P_7$

d) ${}_{12}C_6$

Question 23

1 mark

Identify the equation of $h(x) = f(x) - g(x)$, given $f(x) = x + 5$ and $g(x) = 4x + 1$.

a) $h(x) = -3x + 6$

b) $h(x) = -3x + 4$

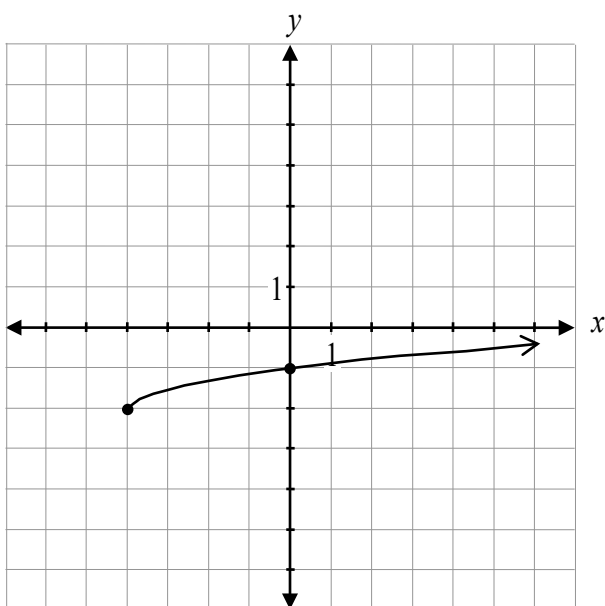
c) $h(x) = 3x + 6$

d) $h(x) = 3x - 4$

Question 24

3 marks 116

Determine the equation of the radical function represented by the graph.



$y =$ _____

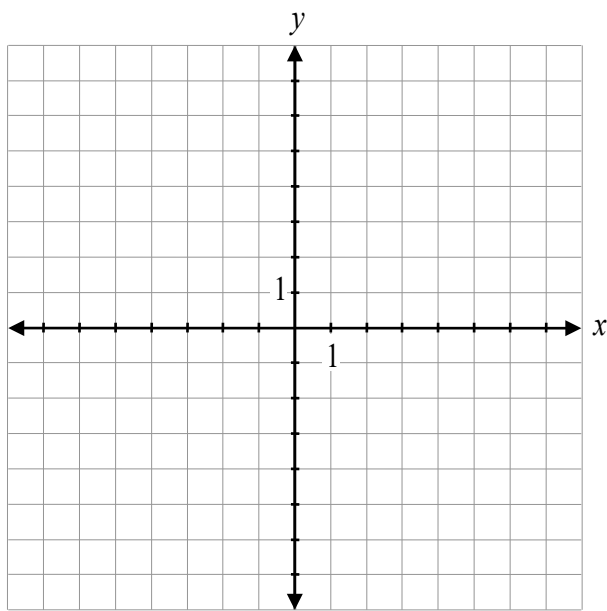
Determine the exact value of x .

$$\sec\left(\frac{2\pi}{3}\right)\left(\sin\left(-\frac{5\pi}{3}\right)\right)(x) = 3$$

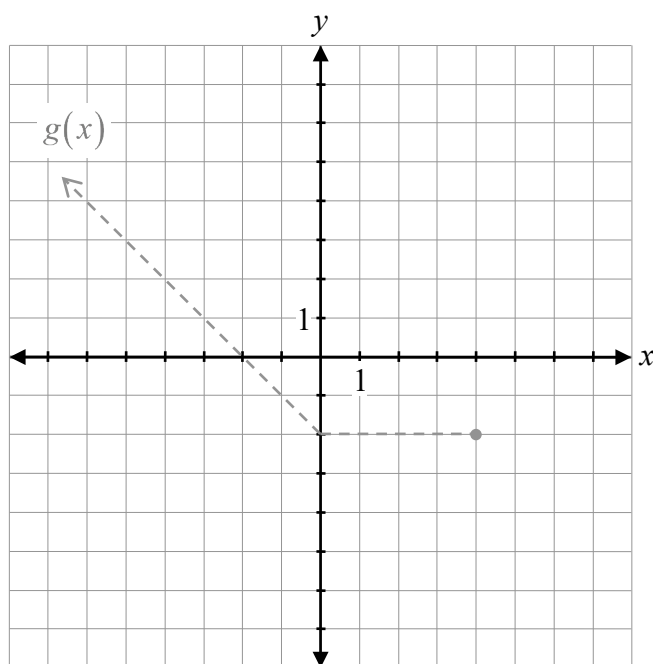
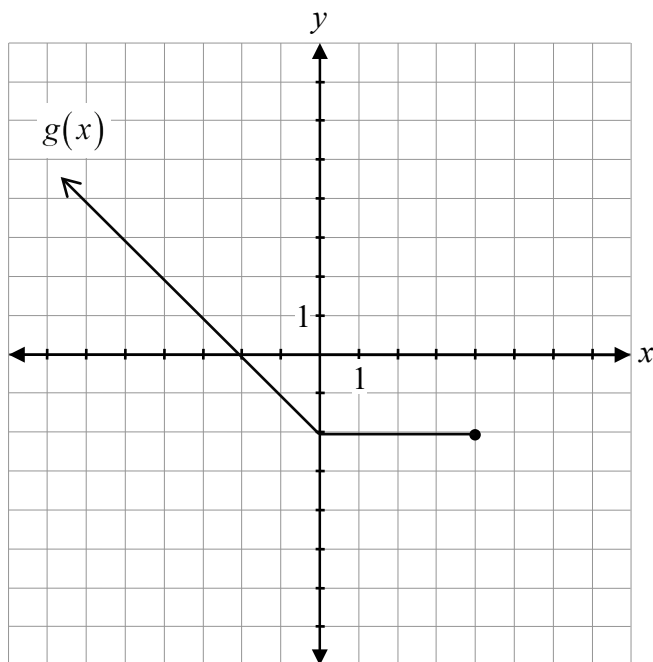
Question 26

3 marks 118

Sketch the graph of $y = 2^{-x} - 3$.



Given the graph of $y = g(x)$, sketch the graph of $y = \frac{1}{g(x)}$.



The graph of $g(x)$ has already been drawn for your reference.
No marks will be awarded for the graph of $g(x)$.

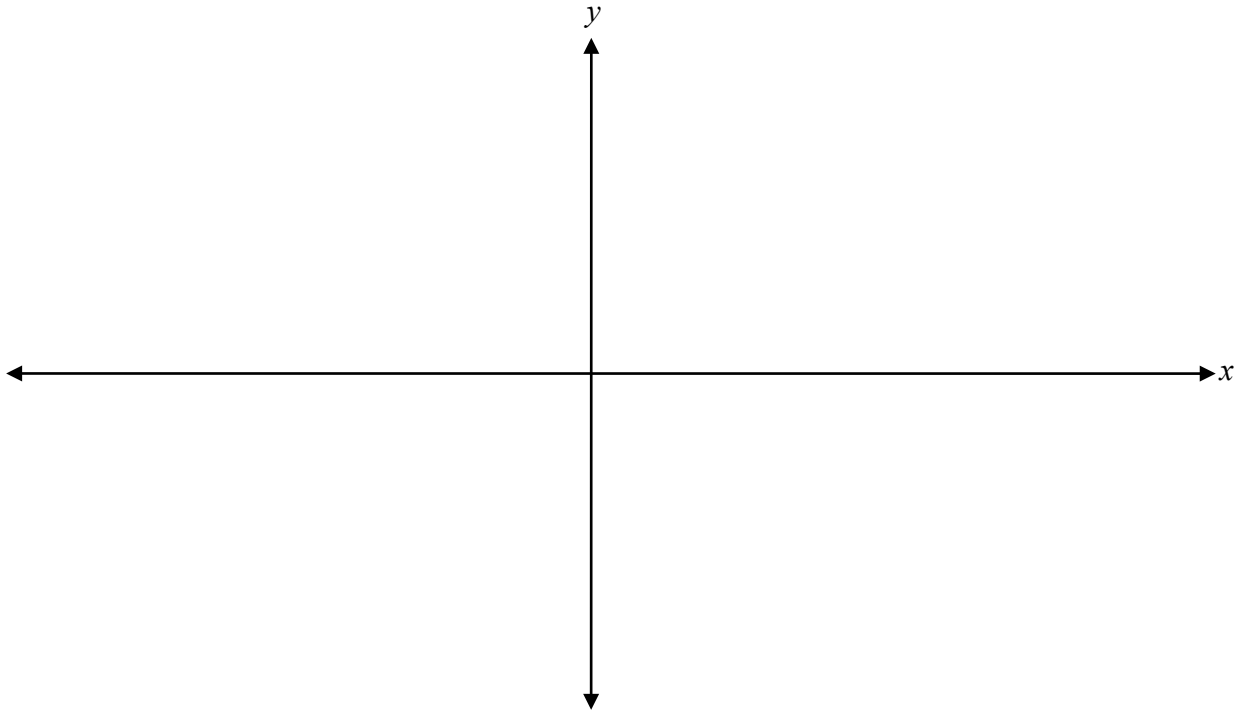
Determine the exact value of $\tan\left(\frac{\pi}{12}\right)$.

Explain why the graph of $g(x) = \frac{3}{x^2 + 4}$ does not have a vertical asymptote.

Solve, algebraically.

$$\log_3 x + \log_3 (x + 8) = 2$$

Sketch at least one period of the graph of the function $y = \sin(3(x + 30^\circ)) - 1$.

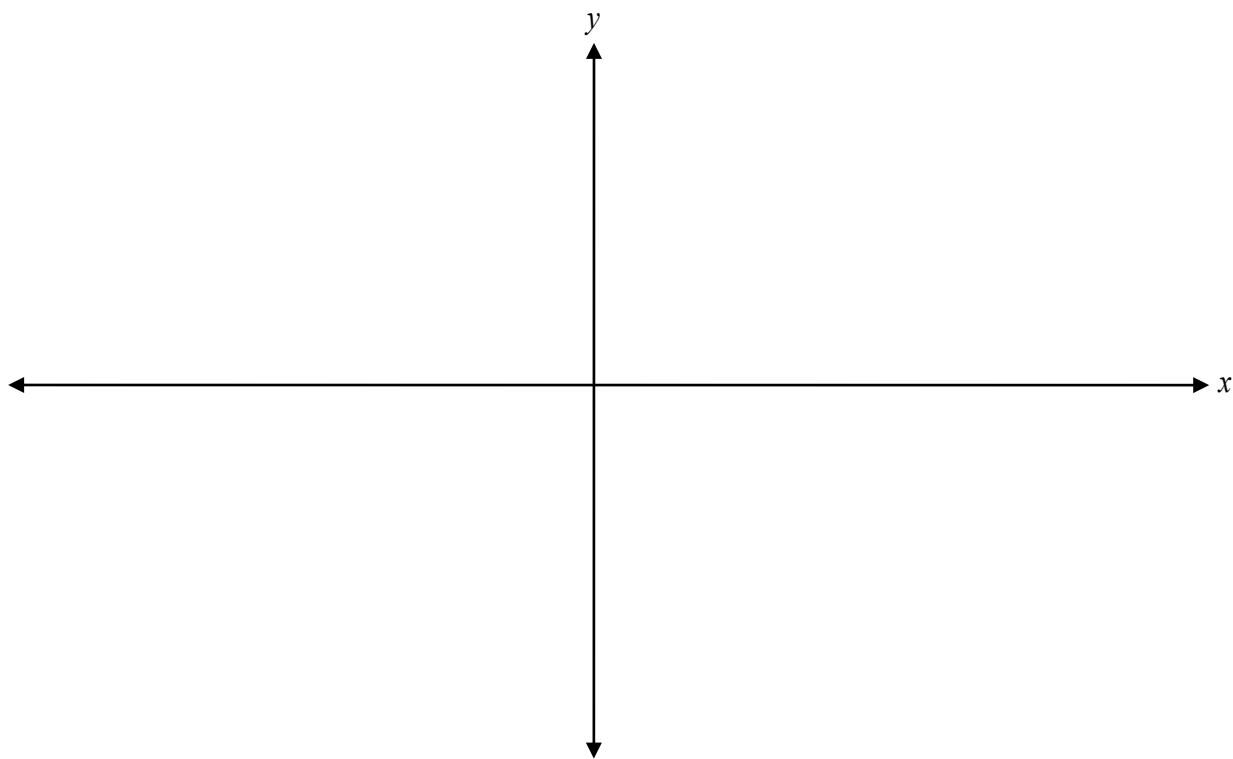


Question 32

1 mark 124

Explain why the domain of the function, $f(x) = \log(x - 3)$, is $x > 3$.

Sketch the graph of $p(x) = -(x-3)(x+1)^2(x-5)$.



Question 34

2 marks 126

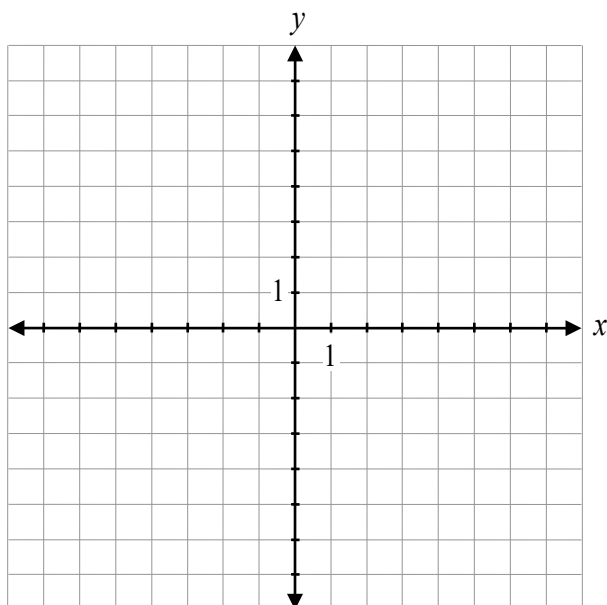
Given that $\sin \theta = -\frac{2}{3}$ and $\tan \theta > 0$, determine the exact value of $\sin 2\theta$.

Question 35

1 mark 127

Justify whether $\frac{5\pi}{8}$ and $-\frac{11\pi}{4}$ are coterminal angles.

Sketch the graph of $f(x) = \frac{-2x(x+1)(x-3)}{2x}$.



Given $\frac{\sin \theta + \cos \theta \csc \theta}{\sin \theta}$, determine the non-permissible values of θ , where $\theta \in \mathbb{R}$.

Question 38

2 marks 130

Write an equation of a rational function that has a horizontal asymptote at $y = 0$ and a vertical asymptote at $x = 6$.

Question 39

a) 1 mark b) 2 marks

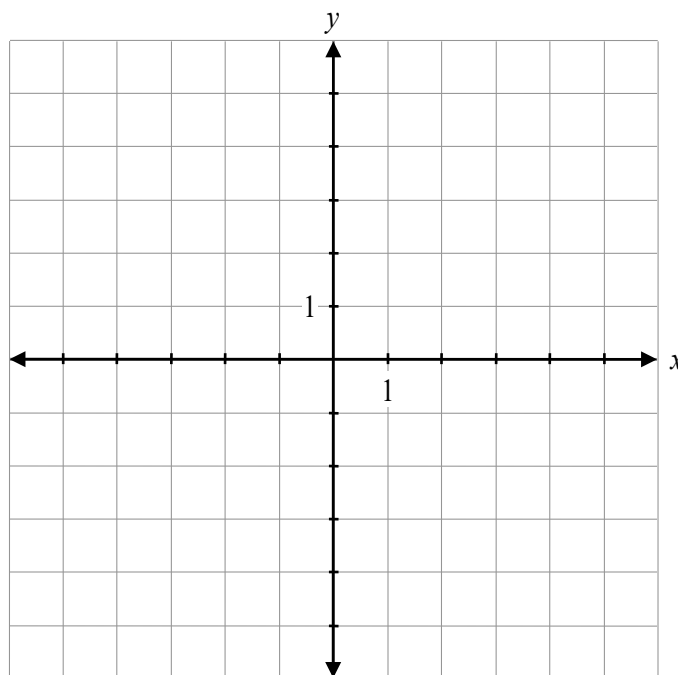
131
132

Given the functions $f(x) = \sqrt{x-1}$ and $g(x) = x^2$,

a) state the equation of $g(f(x))$.

$g(f(x)) =$ _____

b) sketch the graph of $g(f(x))$.



Question 40

1 mark 133

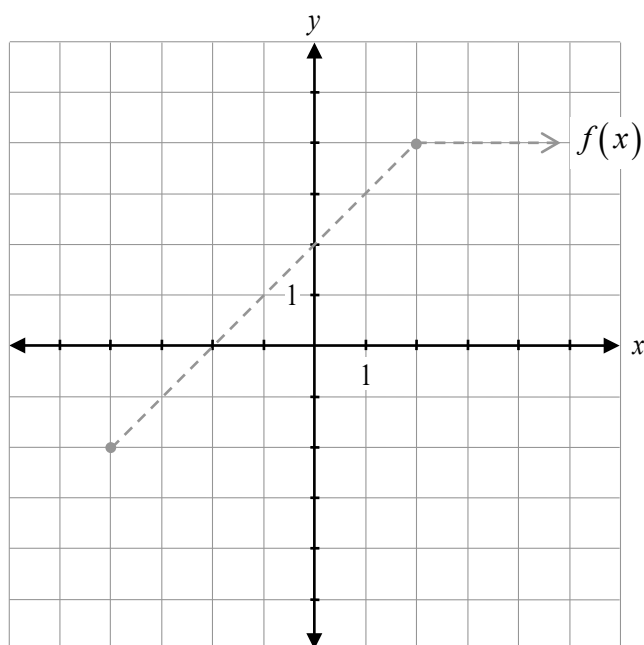
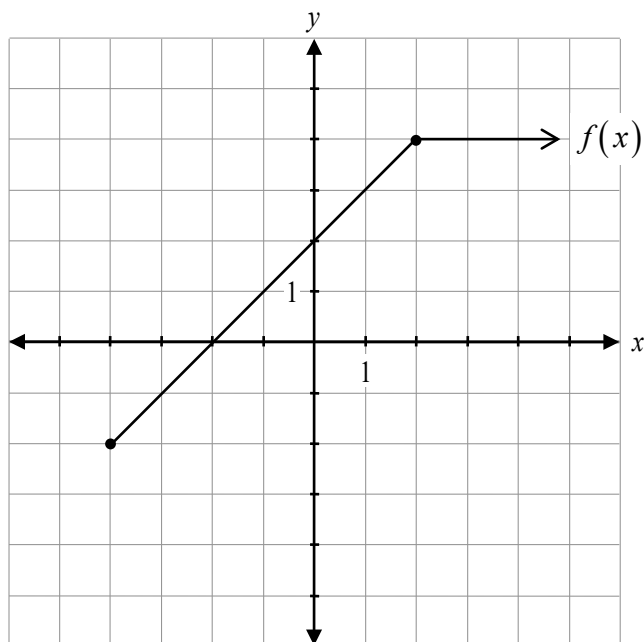
Suzanne was asked to determine the value of $\tan \theta$, given that $\sec \theta = -\frac{8}{3}$ and θ terminates in quadrant II.

Her solution:

$$\begin{aligned}(-3)^2 + y^2 &= (8)^2 \\ y^2 &= 55 \\ y &= \sqrt{55} \\ \tan \theta &= \frac{\sqrt{55}}{3}\end{aligned}$$

Describe her error.

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

The point $P(\theta) = (0, -1)$ lies on the unit circle. State the angle θ , over the interval $[2\pi, 4\pi]$.

Question 43

1 mark 136

Describe how the transformations of $f(x)$ on the graphs of $g(x) = f(3x - 6)$ and $h(x) = f(3(x - 6))$ are different.

Question 44

a) 1 mark b) 1 mark

137
138

a) Solve.

$$\sqrt{2x+5}-3=0$$

b) Describe how the solution in a) relates to the graph of $y = \sqrt{2x+5} - 3$.

Determine all of the zeros of the function $p(x) = x^3 - 2x^2 - 9x + 18$.

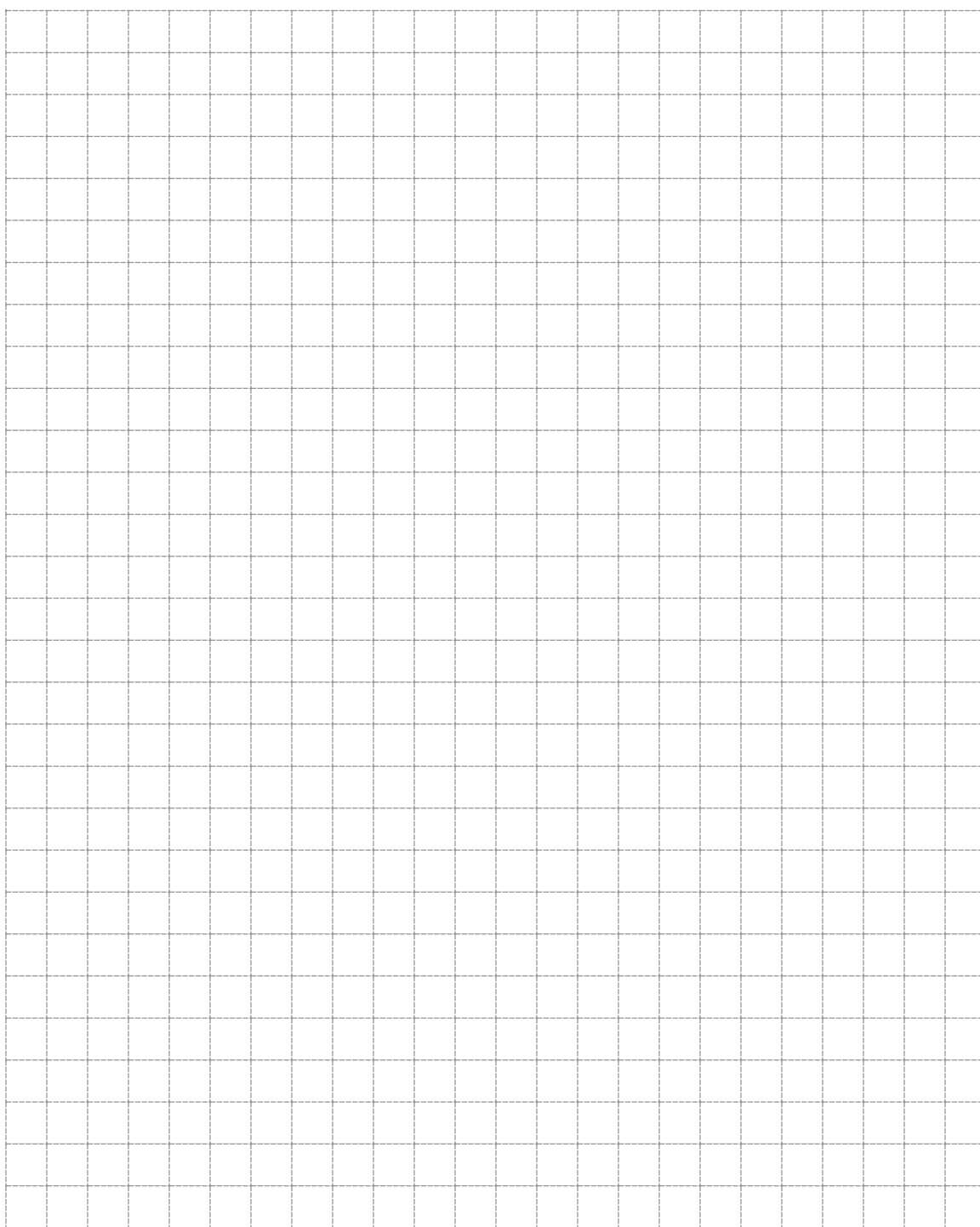
Given that the point $\left(\frac{\sqrt{23}}{6}, y\right)$ is on the unit circle, determine the exact value(s) of y .

Question 47

1 mark 141

State one zero of the function $y = \tan x$.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2024

Grade 12 pre-calculus mathematics achievement test.
Booklet 1. June 2024

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

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Additional Time Allowed: 30 minutes

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	16	34
Booklet 2	10	21	56
Total	10	37	90

* The first 5 questions with the symbol  in *Booklet 1* require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited. Please turn off your cell phone and all other such devices.

No marks will be awarded for work done on this page.

Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_n P_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_n C_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_n C_k a^{n-k} b^k$$

$$\text{Given } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain*, *identify*, and *justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

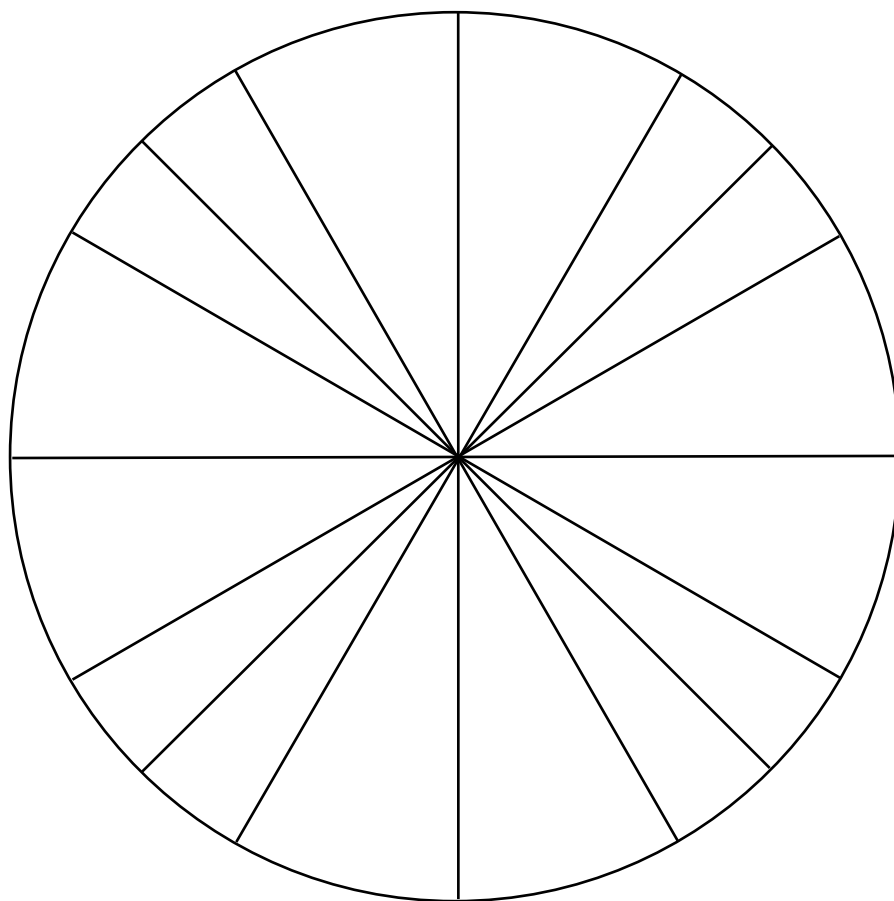
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

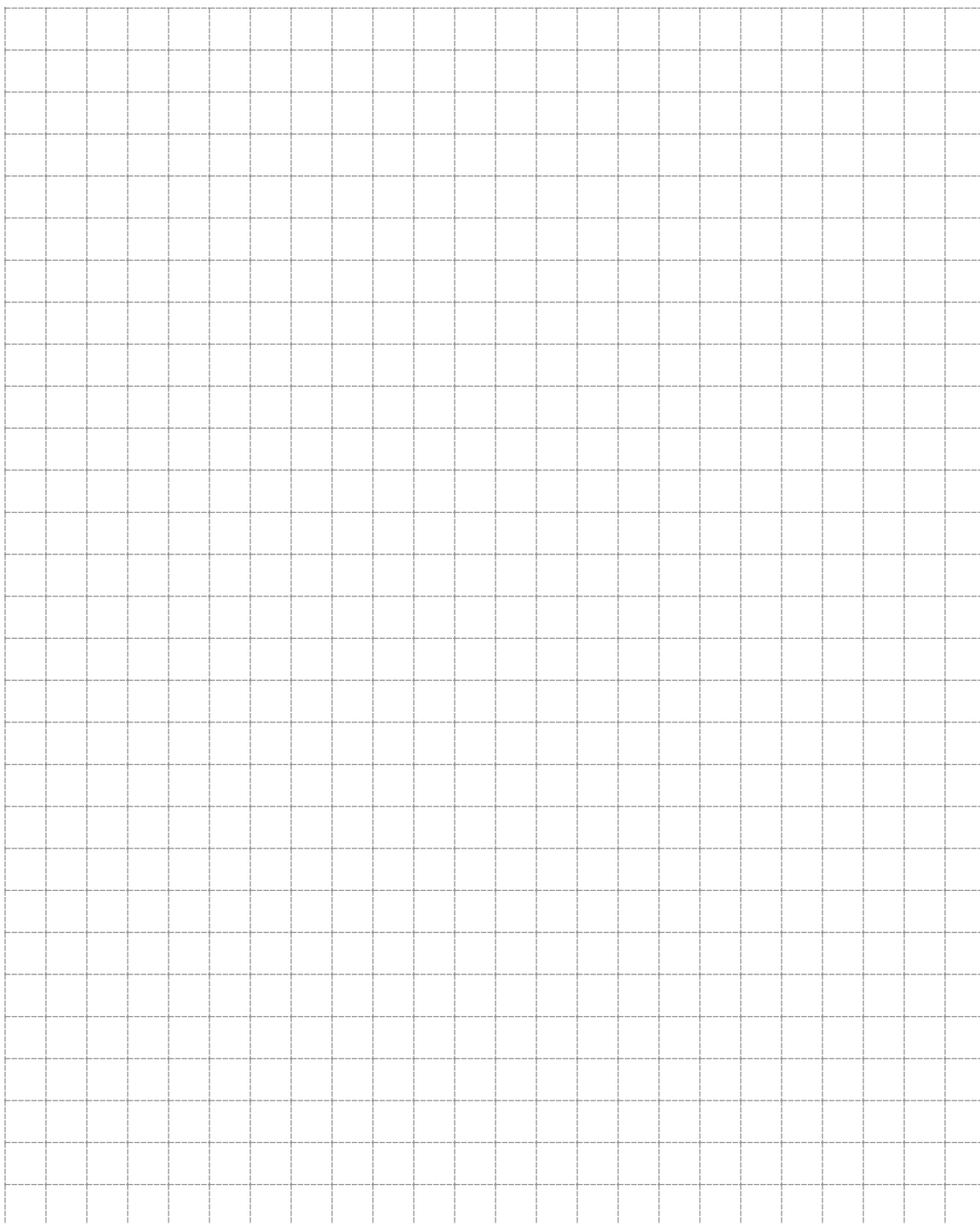
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.



Question 1



1 mark 101

A soccer team consists of 14 players. Determine the number of ways a captain and an assistant captain can be selected.

Question 2



2 marks 102

The pH of a person's blood can be found using the formula

$$\text{pH} = 6.1 + \log\left(\frac{B}{C}\right)$$

where B represents the concentration of bicarbonate in the blood, in mEq/L

C represents the concentration of carbonic acid in the blood, in mEq/L

Kansas has a blood pH of 7.41. In her blood sample, the concentration of carbonic acid is 1.41 mEq/L. Determine the concentration of bicarbonate in her blood.

Express the answer correct to one decimal place.

Question 3



3 marks 103

Solve $(2 \tan x - 1)(\tan x + 1) = 0$ where $x \in \mathbb{R}$.

Question 4



3 marks 104

Determine and simplify the 7th term in the binomial expansion of $\left(\frac{2}{x^2} + 3x\right)^9$.

Question 5



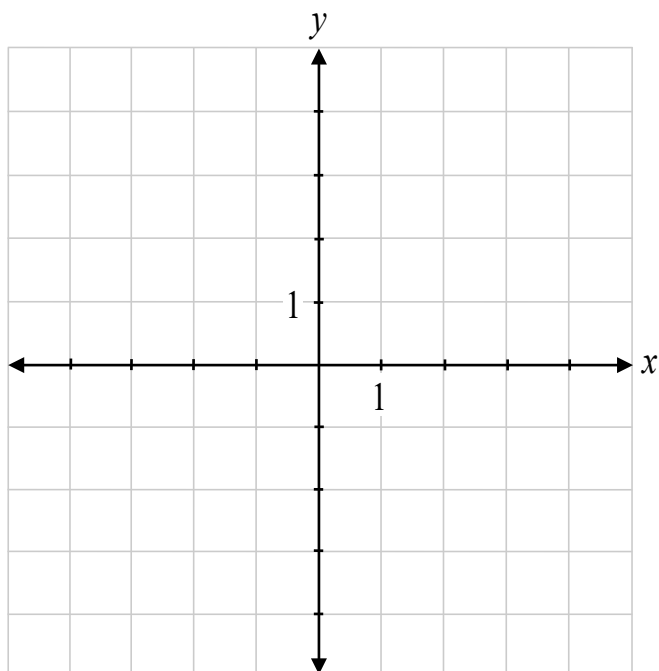
2 marks 105

A winch, with a diameter of 14 cm, is used to pull a boat out of the water. As the winch rotates, it pulls in a cable. Determine the length of cable that is pulled in if the winch rotates 526° .



Note: A calculator is not required for the remaining test questions.

Sketch the graph of $f(x) = \sqrt{\frac{1}{3}x} - 1$.



Question 7

3 marks 107

Given $y = f(x)$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection over the y -axis
- vertical stretch by a factor of 3
- horizontal translation 2 units left

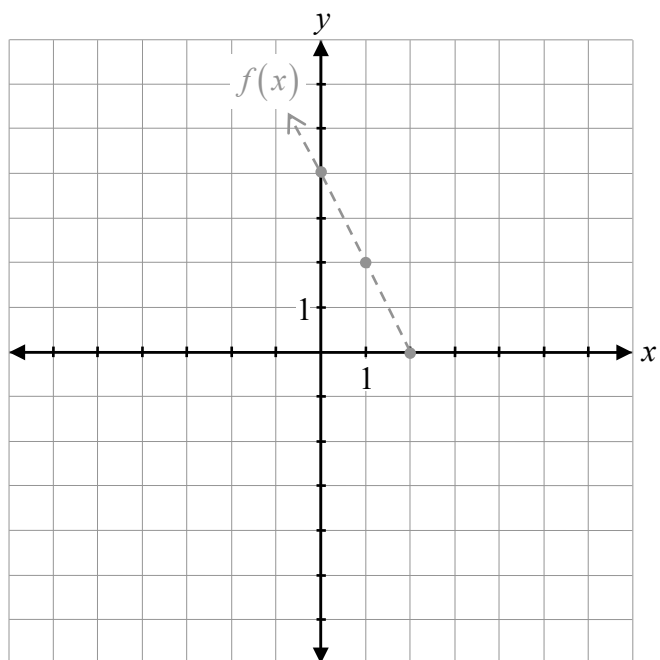
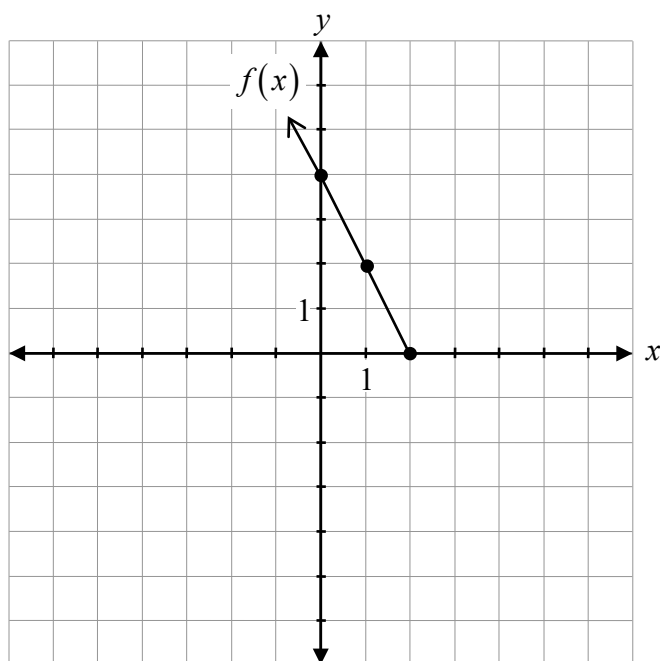
$g(x) =$ _____

Explain why the graph of $y = \frac{6x+7}{3x+5}$ has a horizontal asymptote at $y = 2$.

Question 9

3 marks 109

Given the graph of $y = f(x)$, sketch the graph of $y = -\sqrt{f(x)} + 2$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 10

3 marks 110

Using the laws of logarithms, completely expand the given expression.

$$\log\left(\frac{A}{\sqrt[3]{B} \cdot C^4}\right)$$

Question 11

3 marks 111

Prove the following identity for all permissible values of θ .

$$\cot \theta - \tan \theta = \frac{2 \cos 2\theta}{\sin 2\theta}$$

Left-Hand Side	Right-Hand Side

Question 12

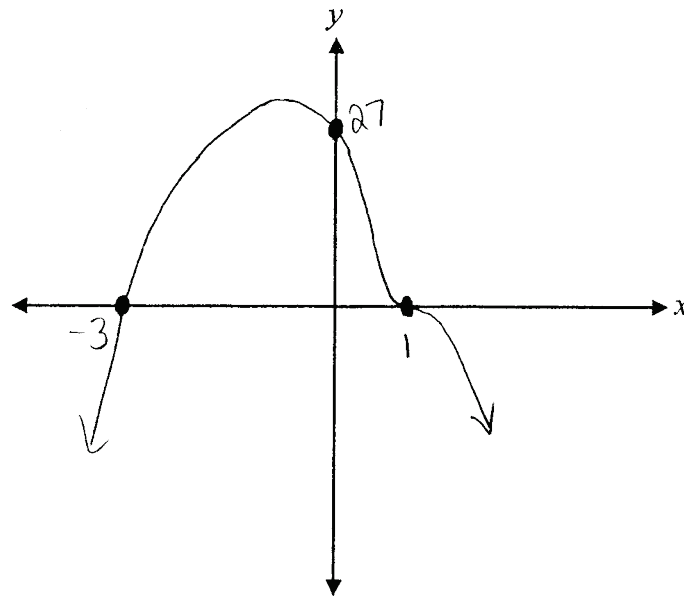
3 marks 112

Express $p(x) = -2x^3 + x^2 + 13x + 6$ in completely factored form.

$p(x) =$ _____

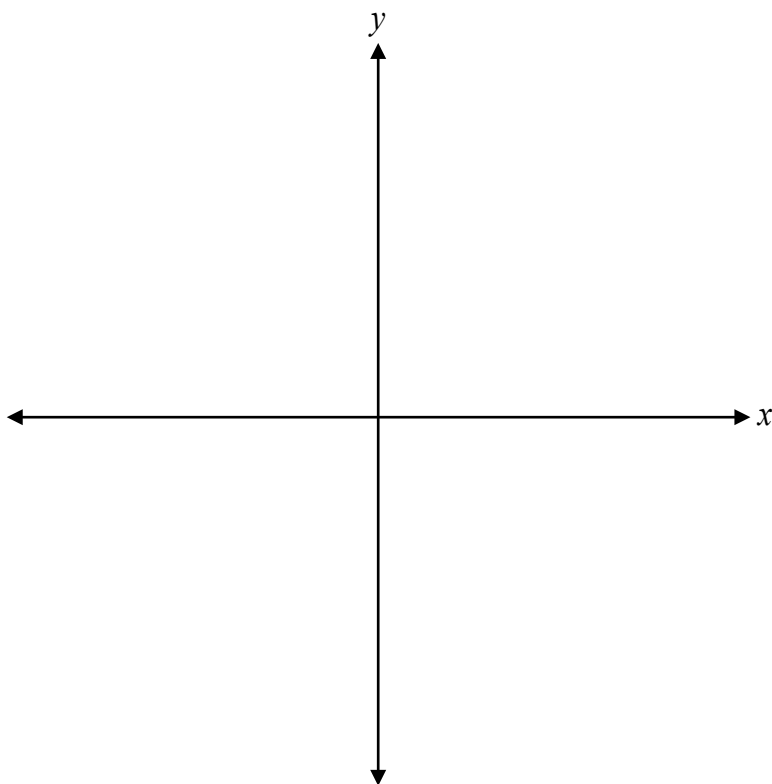
Kaitlyn was asked to sketch the graph of the polynomial function, $p(x) = -(x-1)(x+3)^3$.

Kaitlyn's graph:

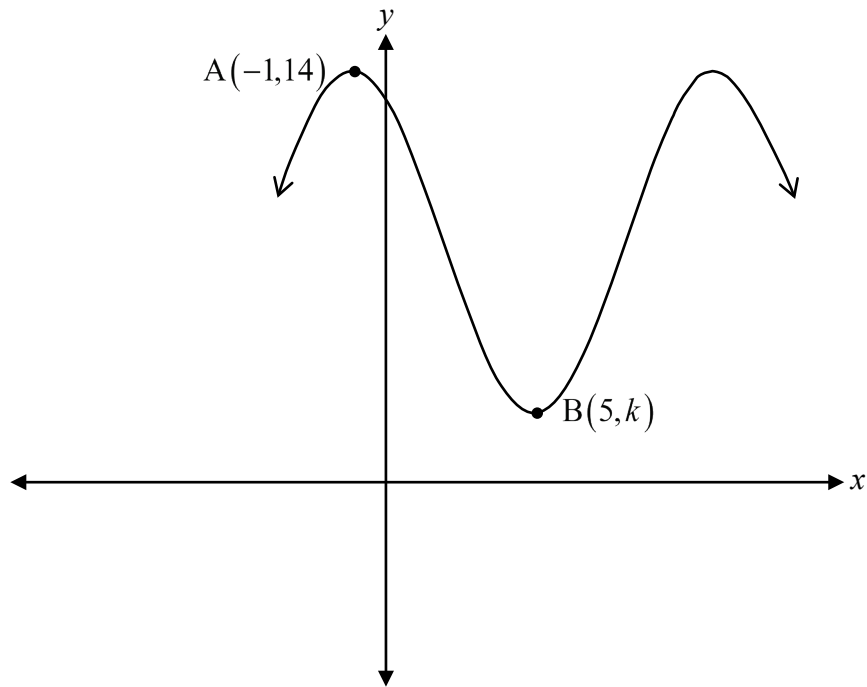


Describe an error on her graph.

Sketch the angle of -5 radians in standard position.



The graph of a sinusoidal function is sketched below. The point A is a maximum and the point B is a minimum. State the value of k if the amplitude of the function is 6.

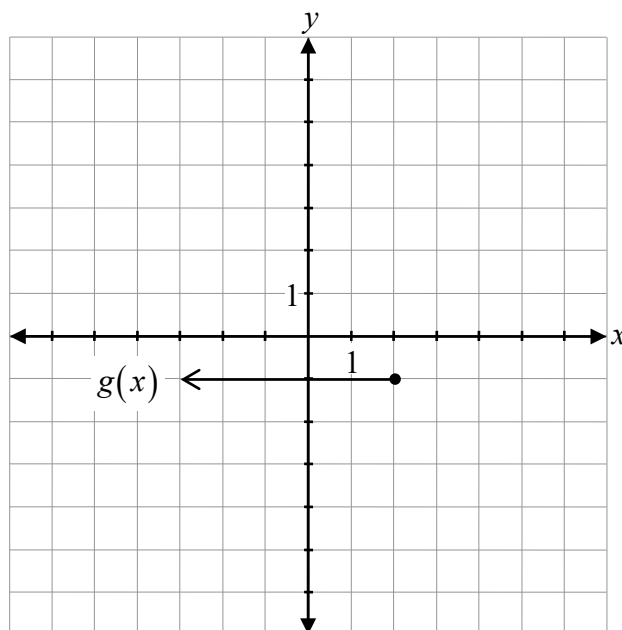
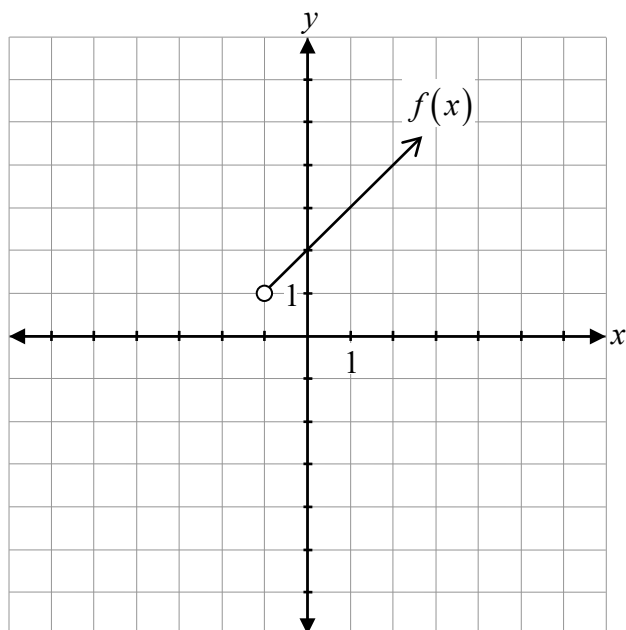


Question 16

1 mark

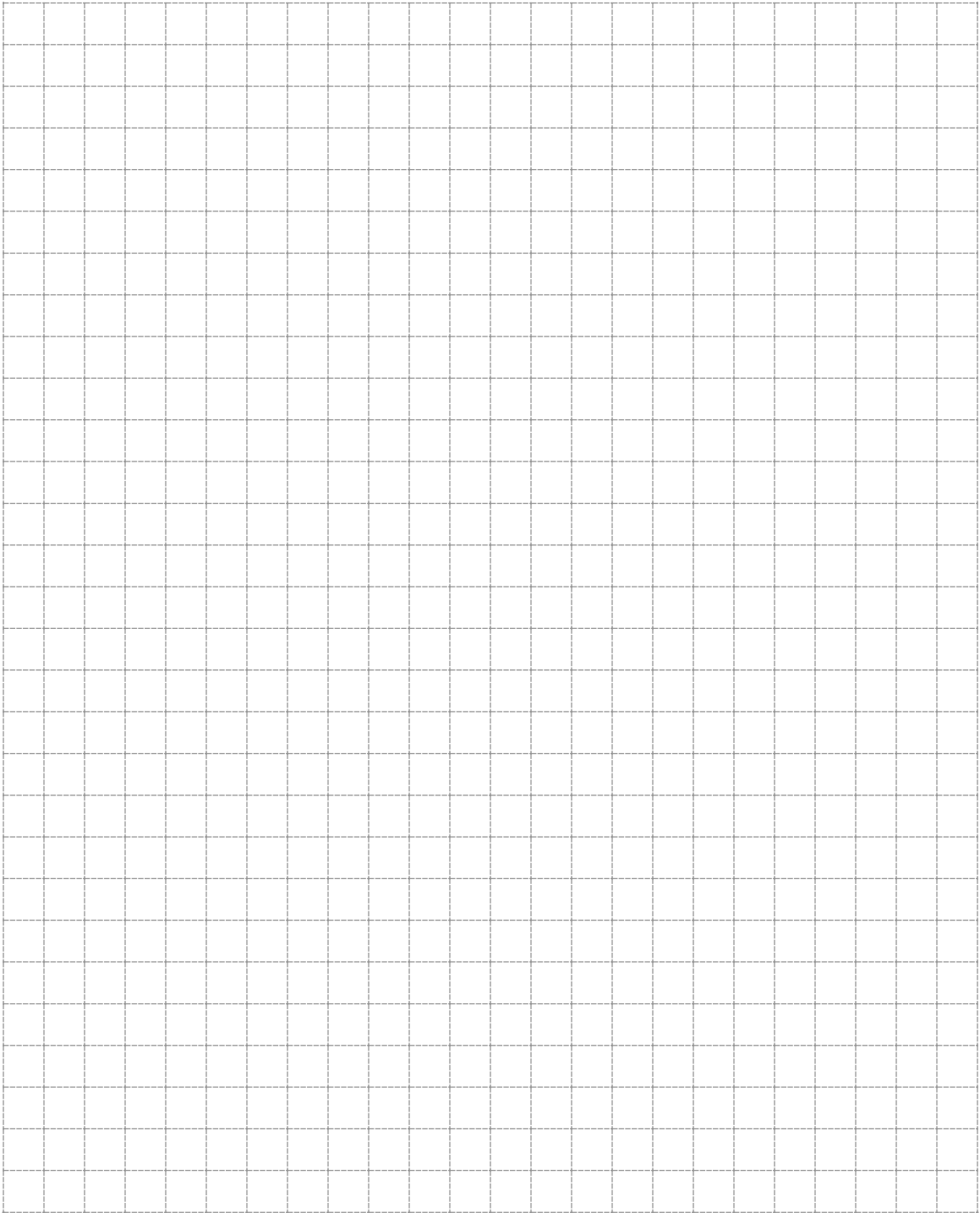
116

Given the graphs of $f(x)$ and $g(x)$, state the domain of $(f + g)(x)$.



Domain: _____

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

No marks will be awarded for work done on this page.

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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2024

Grade 12 pre-calculus mathematics achievement test.
Booklet 2. June 2024

This resource is available in print and electronic formats.

ISBN: 978-0-7711-6628-8 (print)

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Numbers and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1	—	16	34
Booklet 2	10	21	56
Total	10	37	90

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- Write each solution in the space provided.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.

No marks will be awarded for work done on this page.

Question 17**1 mark**

Given that the point $(-13, 7)$ is on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = f^{-1}(x+2)$.

- a) $(7, -11)$
- b) $(5, -13)$
- c) $(7, -15)$
- d) $(9, -13)$

Question 18**1 mark**

Given that $x = 0$, $x = 1$, $x = 2$ and $x = -3$ are zeros of the polynomial function $p(x)$, identify a possible equation for $p(x)$.

- a) $p(x) = -3x(x+1)(x-2)$
- b) $p(x) = (x+1)(x+2)(x-3)^2$
- c) $p(x) = x(x+3)^2(x-1)(x-2)$
- d) $p(x) = (x-2)(x-1)^3(x+3)$

Question 19**1 mark**

Identify the interval where the angle, θ , could exist if $\cos \theta = -\frac{4}{5}$ and $\csc \theta = \frac{5}{3}$.

- a) $\left(0, \frac{\pi}{2}\right)$
- b) $\left(\frac{\pi}{2}, \pi\right)$
- c) $\left(\pi, \frac{3\pi}{2}\right)$
- d) $\left(\frac{3\pi}{2}, 2\pi\right)$

Question 20**1 mark**

Identify a non-permissible value of θ in the identity, $\frac{\cot \theta}{\csc \theta} = \cos \theta$.

- a) π
- b) $\frac{\pi}{4}$
- c) $\frac{3\pi}{2}$
- d) $\frac{\pi}{2}$

Question 21**1 mark**

Identify the solution of the equation, $e^{2x} = 16$.

- a) $x = \ln 8$
- b) $x = 8$
- c) $x = \frac{\ln 16}{2}$
- d) $x = \frac{\log 16}{2}$

Question 22**1 mark**

A committee of 6 people is to be selected from 90 students. Identify an expression that represents the number of possible committees if two of the students, Evelyn and Wayne, must be on the committee.

- a) ${}_{90}C_4$
- b) ${}_{90}C_6$
- c) ${}_{88}C_6$
- d) ${}_{88}C_4$

Question 23**1 mark**

Identify the expression for the number of arrangements of the letters in the word RUSSELL.

- a) $\frac{7!}{4!}$
- b) $7!$
- c) $\frac{7!}{2!}$
- d) $\frac{7!}{2!2!}$

Question 24**1 mark**

Given the row of Pascal's triangle below, identify the third value of the next row.

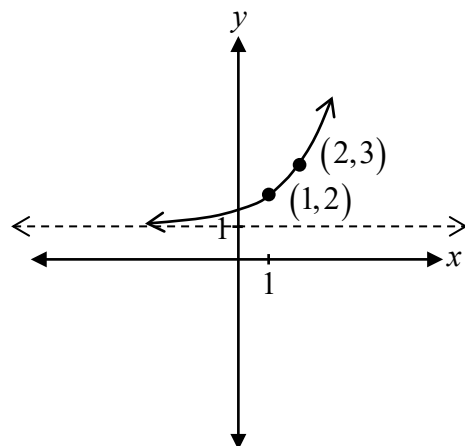
1 7 21 35 35 21 7 1

- a) 8
- b) 28
- c) 56
- d) 70

Question 25**1 mark**

Identify the equation represented by the graph sketched below.

- a) $y = \left(\frac{1}{2}\right)^x + 1$
- b) $y = 2^{x-1} + 1$
- c) $y = 2^{x+1} + 1$
- d) $y = \left(\frac{1}{2}\right)^{x+1} + 1$



Match the following functions with their correct graphs.

Place the appropriate letter in this column.

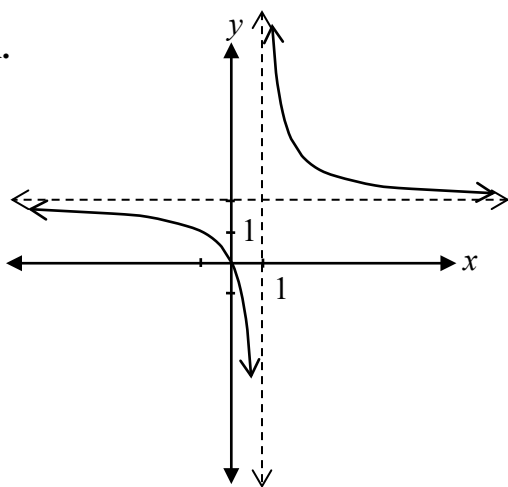
$$f(x) = \frac{1}{x-2}$$

$$g(x) = \frac{-x}{x-2}$$

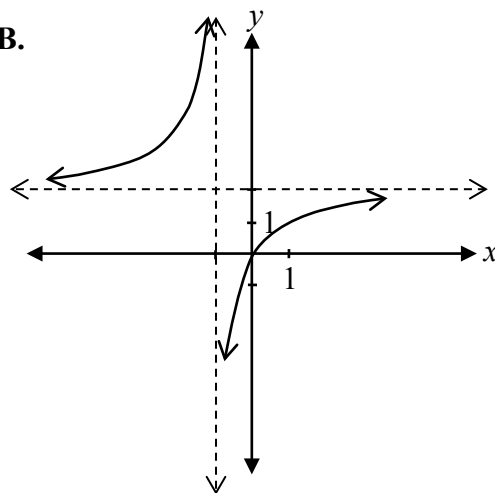
$$h(x) = \frac{2x}{x-1}$$

$$k(x) = \frac{2x}{x+1}$$

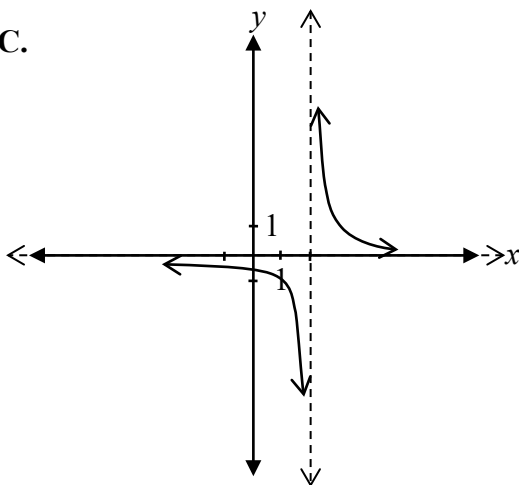
A.



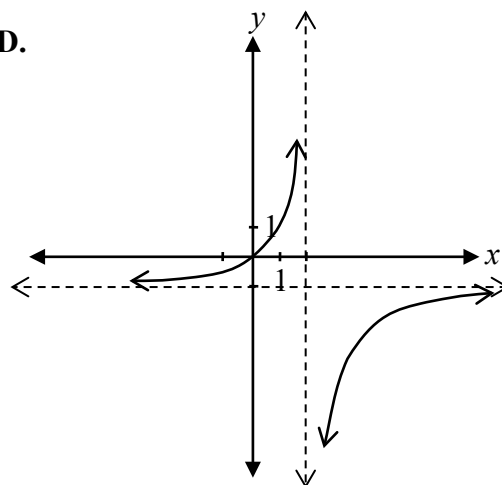
B.



C.



D.



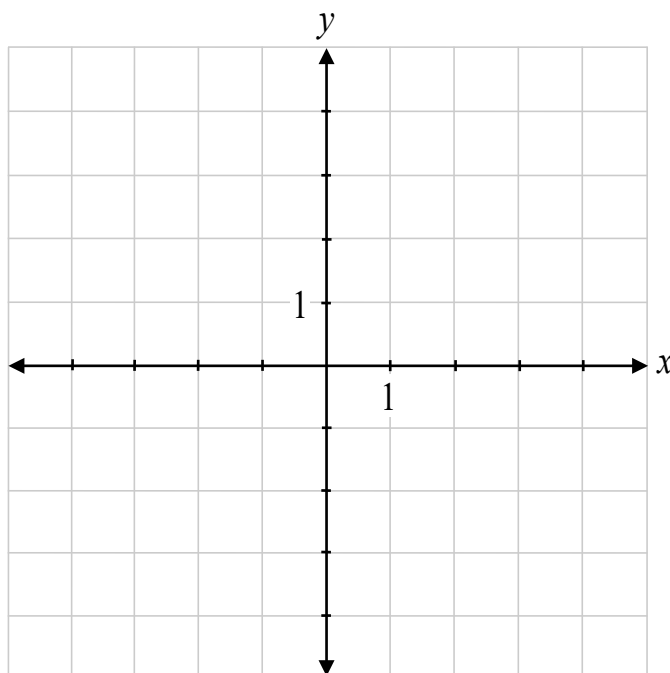
Evaluate.

$$\sin^2\left(\frac{11\pi}{3}\right)\cot\left(\frac{\pi}{4}\right) + \csc\left(\frac{11\pi}{6}\right)$$

Question 28

3 marks 119

Sketch the graph of $y = \log_2[-(x - 2)]$.



Question 29

1 mark 120

Justify that ${}_{12}C_7 = {}_{12}C_5$.

Solve, algebraically.

$$\left(\frac{1}{4}\right)^{3x} = 8^{x-3}$$

Question 31

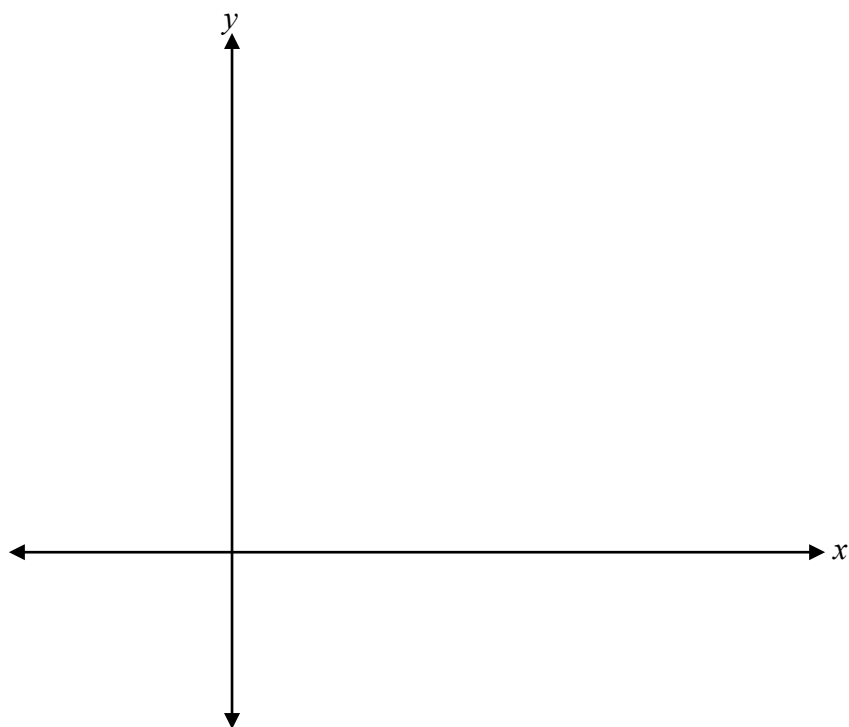
2 marks 122

State the domain and range of the function $y = -\sqrt{x-5} + 4$.

Domain: _____

Range: _____

Sketch at least one period of the graph of $y = 4 \cos \left[3 \left(x + \frac{\pi}{6} \right) \right] + 2$.



Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has zeros at -5 and 1
- all zeros of $p(x)$ have a multiplicity of 2
- the graph of $p(x)$ has a y -intercept at 75

$p(x) =$ _____

Vitaly was asked to solve the exponential equation $5^{2x+7} = 3 \cdot 7^x$.

Vitaly's work:

$$\begin{aligned}\log(5^{2x+7}) &= \log(3 \cdot 7^x) \\ (2x+7)\log 5 &= \log(21^x) \\ 2x\log 5 + 7\log 5 &= x\log 21 \\ 7\log 5 &= x\log 21 - 2x\log 5 \\ 7\log 5 &= x(\log 21 - 2\log 5) \\ x &= \frac{7\log 5}{\log 21 - 2\log 5}\end{aligned}$$

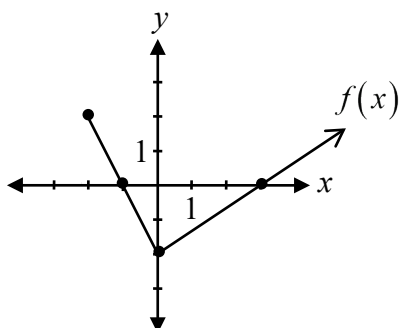
Describe his error.

Question 35

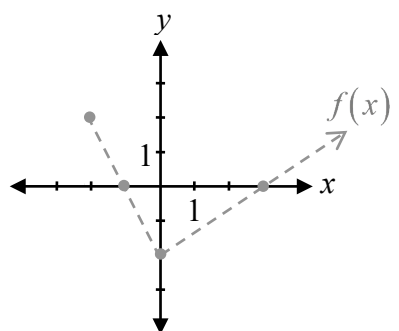
a) 1 mark b) 1 mark c) 1 mark

126
127
128

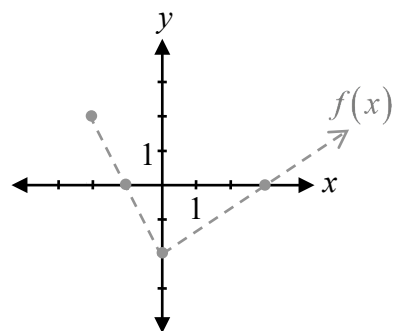
Given the graph of $f(x)$,



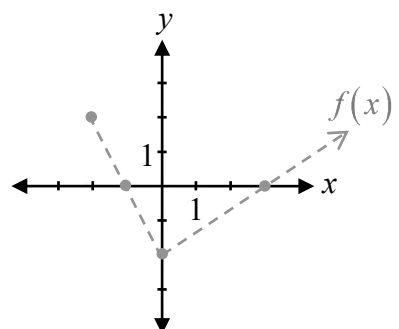
a) sketch the graph of $g(x) = -f(x)$.



b) sketch the graph of $k(x) = |f(x)|$.



c) sketch the graph of $r(x) = f(2x)$.



Determine the exact value of $\sin\left(\frac{17\pi}{12}\right)$.

Question 37

a) 1 mark b) 1 mark

130
131

Given $f(x) = \sqrt{x+3}$ and $g(x) = x-8$,

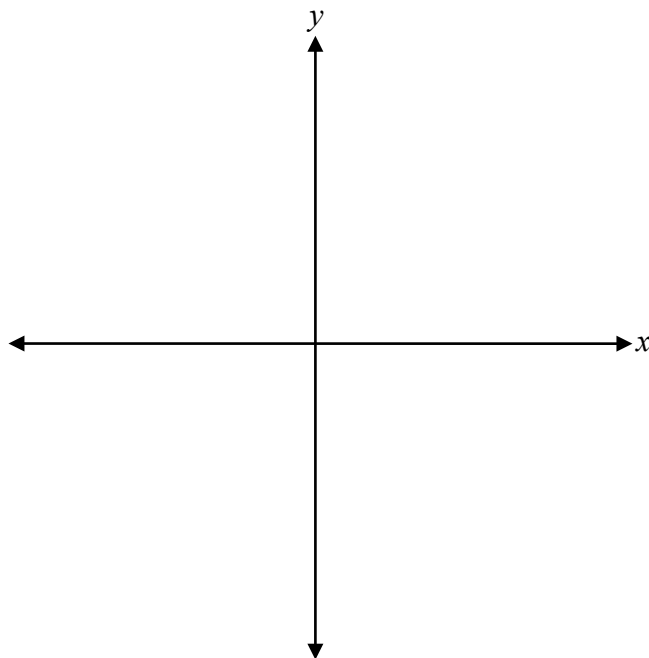
a) determine the value of $g(f(1))$.

b) justify that the value of $(f-g)(-5)$ is undefined.

Solve, algebraically.

$$48\binom{n}{4} = {}_{(n+1)}P_4$$

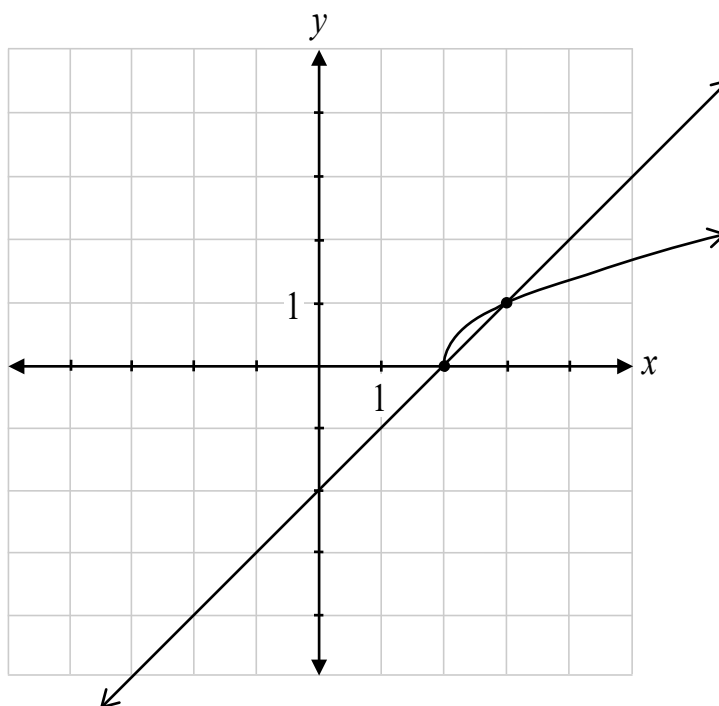
Sketch the graph of $p(x) = (x+1)(x+2)(x-4)$.



Evaluate.

$$\log_5(\log_2 32)$$

Describe how to use the graph to solve the equation $\sqrt{x-2} = x-2$.



Question 42

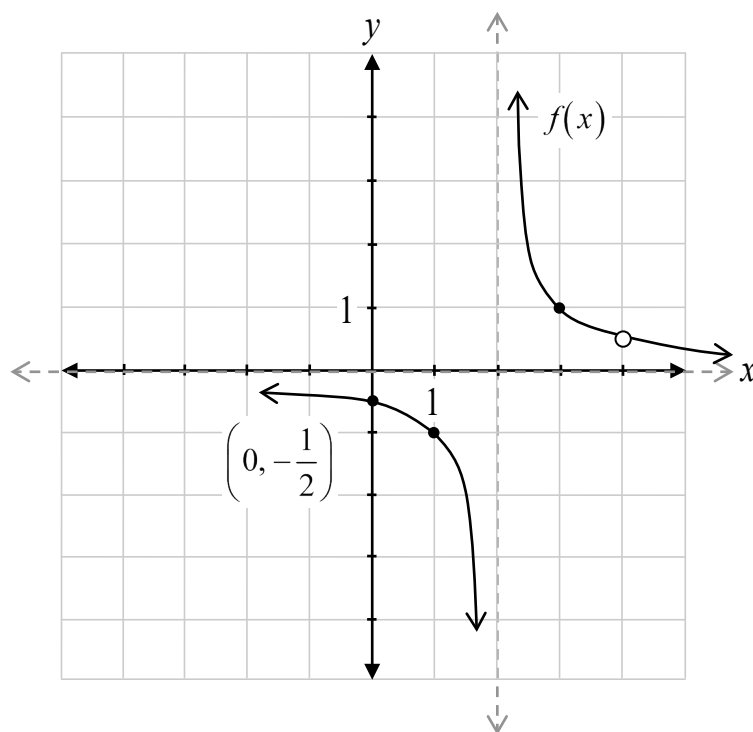
3 marks 136

Verify, by substitution, that the equation $\frac{\sec \theta}{\tan \theta + \cot \theta} = \sin 2\theta$ is true for $\theta = \frac{\pi}{3}$.

Left-Hand Side	Right-Hand Side

Describe the transformation(s) to the graph of $y = 3^x$ which results in a graph with a y -intercept of -2 .

Determine the equation of the rational function represented by the graph.



$$f(x) = \underline{\hspace{2cm}}$$

Justify that $f(x) = 3x + 2$ and $g(x) = \frac{x-3}{2}$ are not inverses of each other.

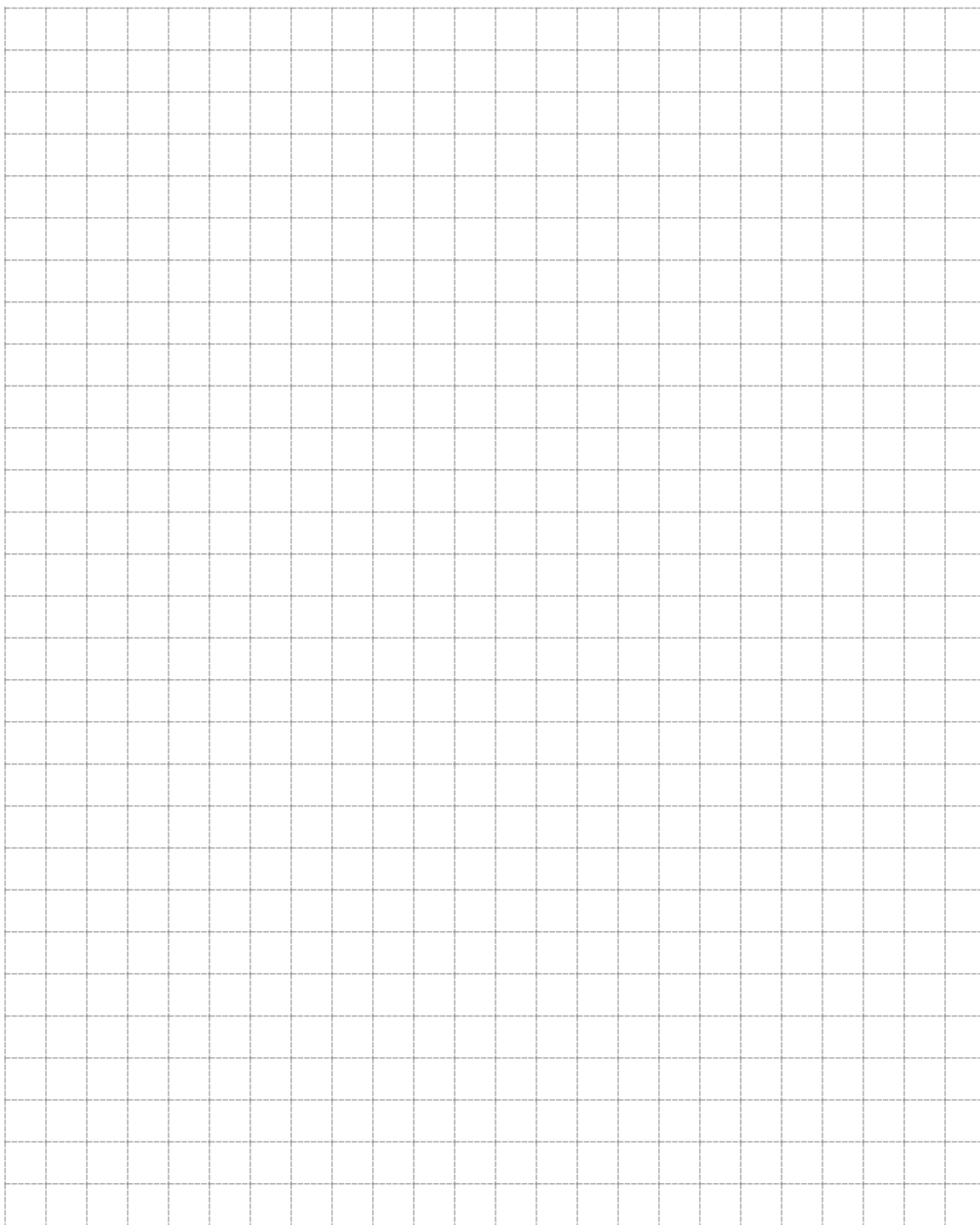
The terminal arm of an angle, θ , in standard position passes through the point $(-3, 6)$ and intersects the unit circle at the point $P(\theta)$. Determine the coordinates of the point $P(\theta)$.

Question 47

2 marks 141

Determine the exact value of $2\sin\theta\cos\theta$ if $\theta = \frac{\pi}{12}$.

No marks will be awarded for work done on this page.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet I

January 2025

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 1 (January 2025)

This resource is available in print and electronic formats.

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Number and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	16	40
Booklet 2	9	19	50
Total	9	35	90

* The first 5 questions with the symbol  in Booklet 1 require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

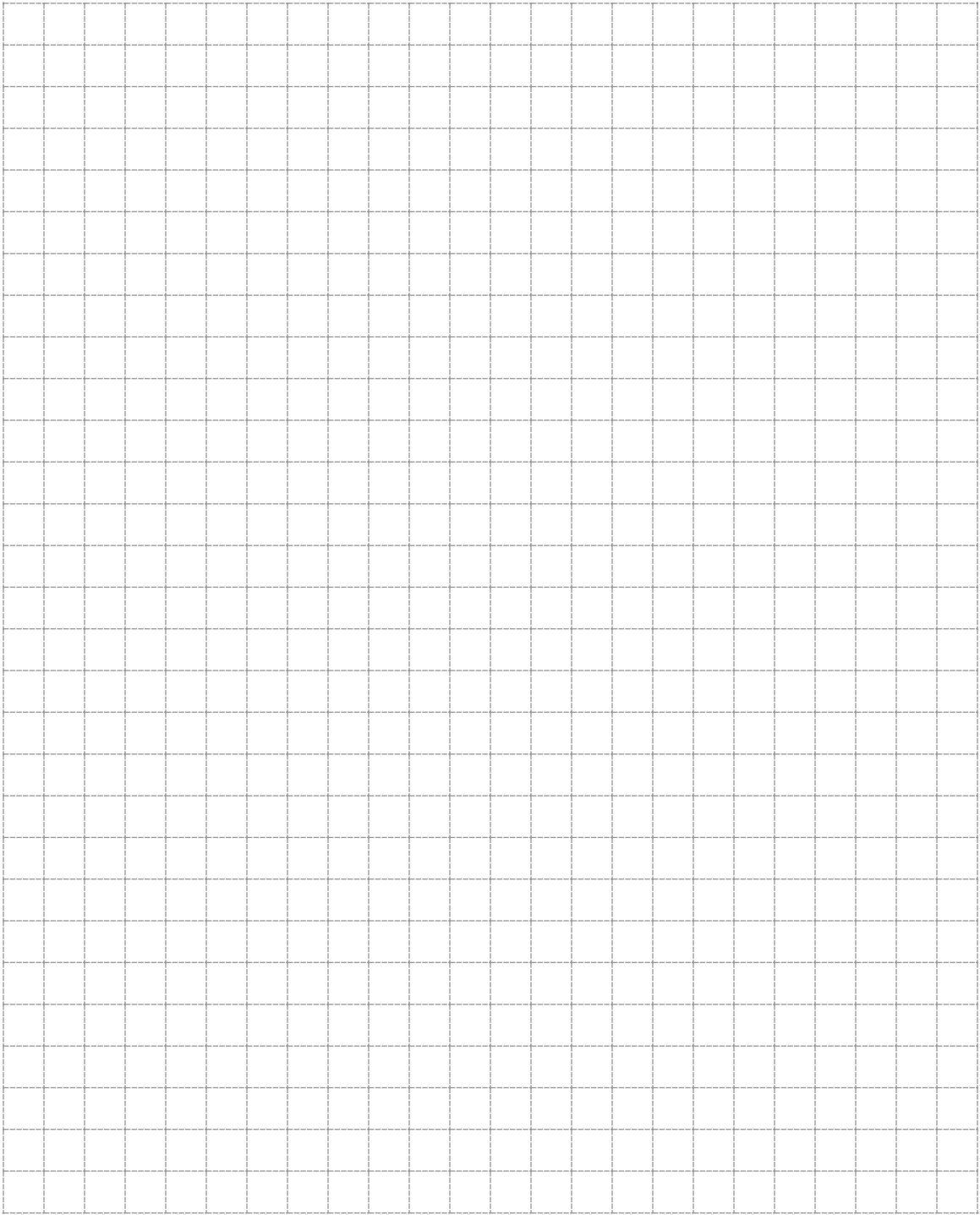
Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited. Please turn off your cell phone and all other such devices.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_nP_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_nC_k a^{n-k} b^k$$

$$\text{Given } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain, identify, and justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

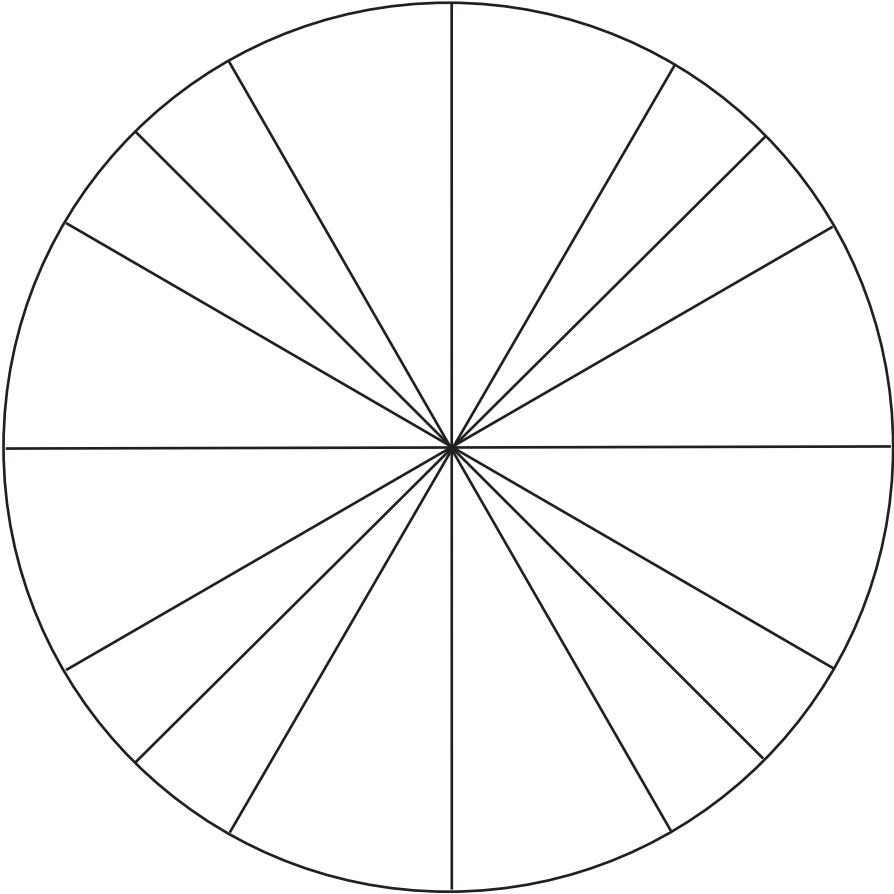
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

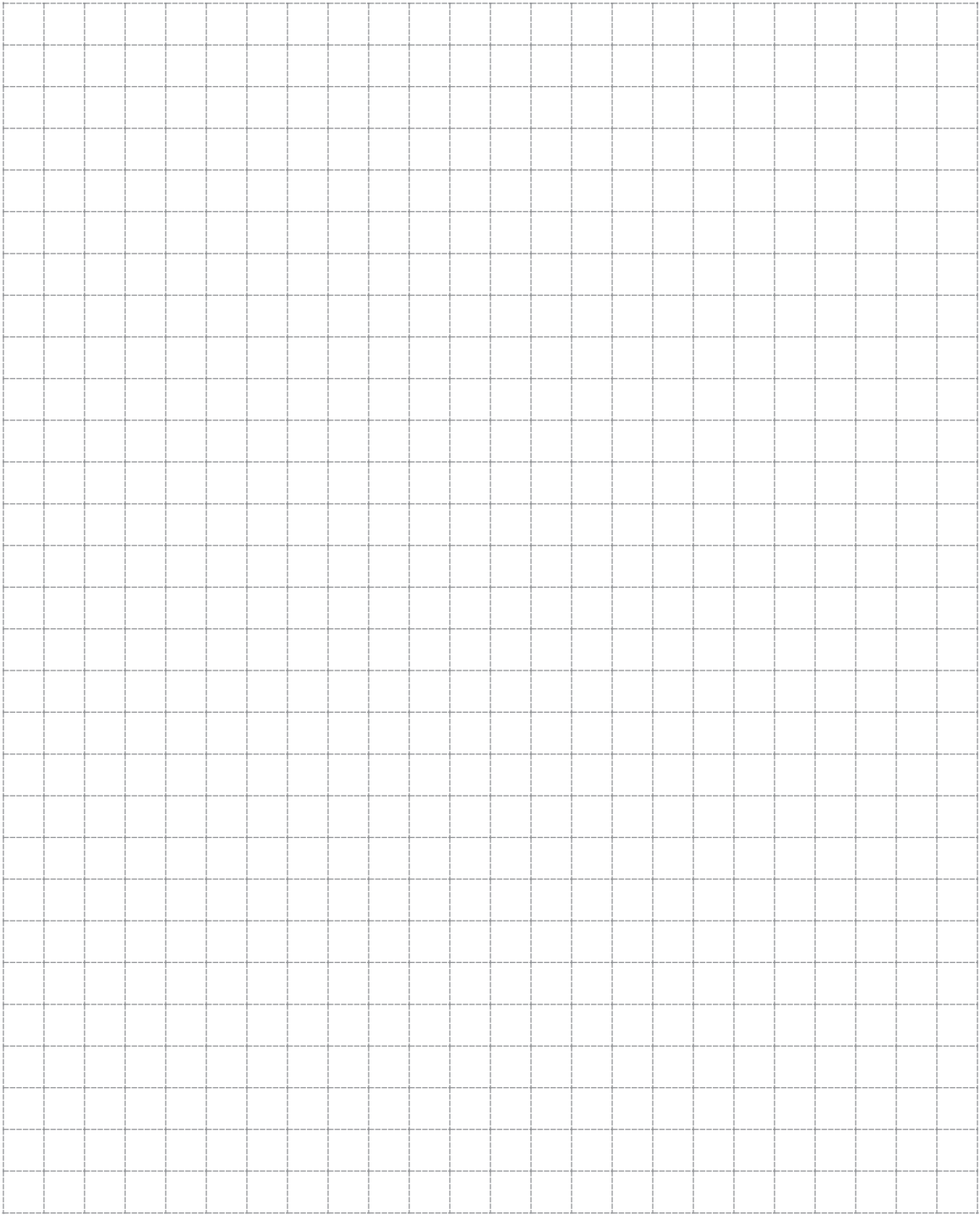
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 1 **1 mark**

101

Youyi wants to buy a new cell phone from a company that offers 33 different types of cell phones. Each cell phone is available in 5 different colours. Determine the total number of available options.

Question 2 **2 marks**

102

Determine the central angle subtended by an arc length of 30 cm on a circle with a diameter of 20.6 cm. Express your answer in degrees.

Question 3 **3 marks**

103

A choir has 14 sopranos. Karin and Ellen are two of them. Determine the number of possible ensembles of 3 sopranos that could be formed if either Karin or Ellen, or both, must be included.

Question 4 **3 marks**

104

Loïc invests \$5000 at an annual interest rate of 6.75% compounded semi-annually. The value of his investment grows according to the following formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A is the value of the investment after t years

P is the amount of the initial deposit

r is the annual interest rate (as a decimal)

n is the number of compounding periods per year

t is the time in years

Determine, algebraically, the length of time it will take Loïc's investment to grow to \$9500.

Question 5 **4 marks**

105

Solve, algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta - \csc \theta - 6 = 0$$

Note: A calculator is not required for the remaining test questions.

Question 6

2 marks

106

Determine how many 4-digit numbers can be made using the digits 0, 1, 2, 3, 4, and 5, if the third digit must be 3, and repetition is not allowed.

Question 7

3 marks

107

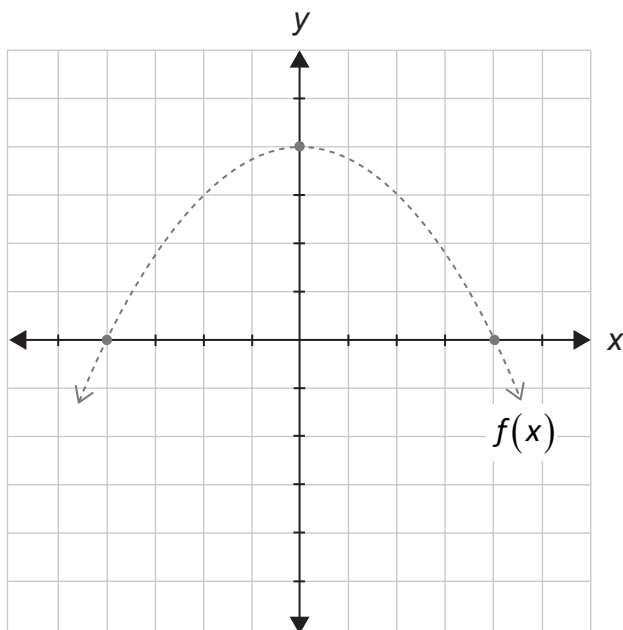
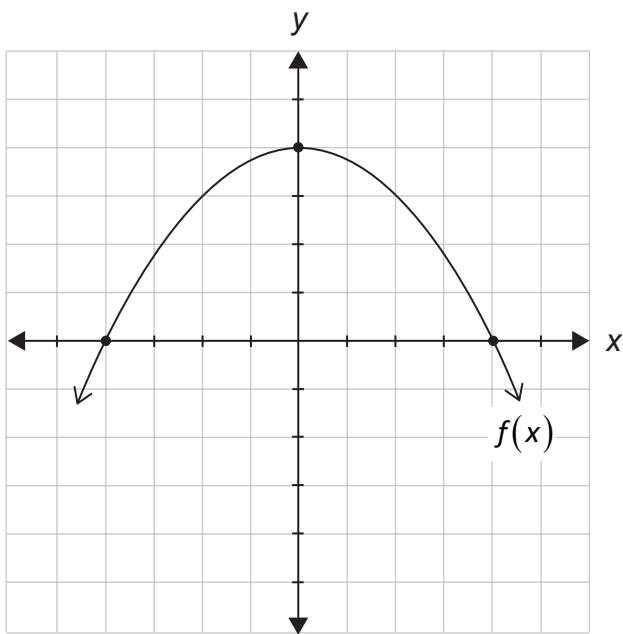
Describe the transformations required to obtain the graph of the function $y = -4f\left(\frac{1}{3}x\right)$ from the graph of $y = f(x)$.

Question 8

2 marks

108

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 9

1 mark

109

Describe how to determine if $(x - a)$ is a factor of the polynomial function, $p(x)$.

Question 10

3 marks

110

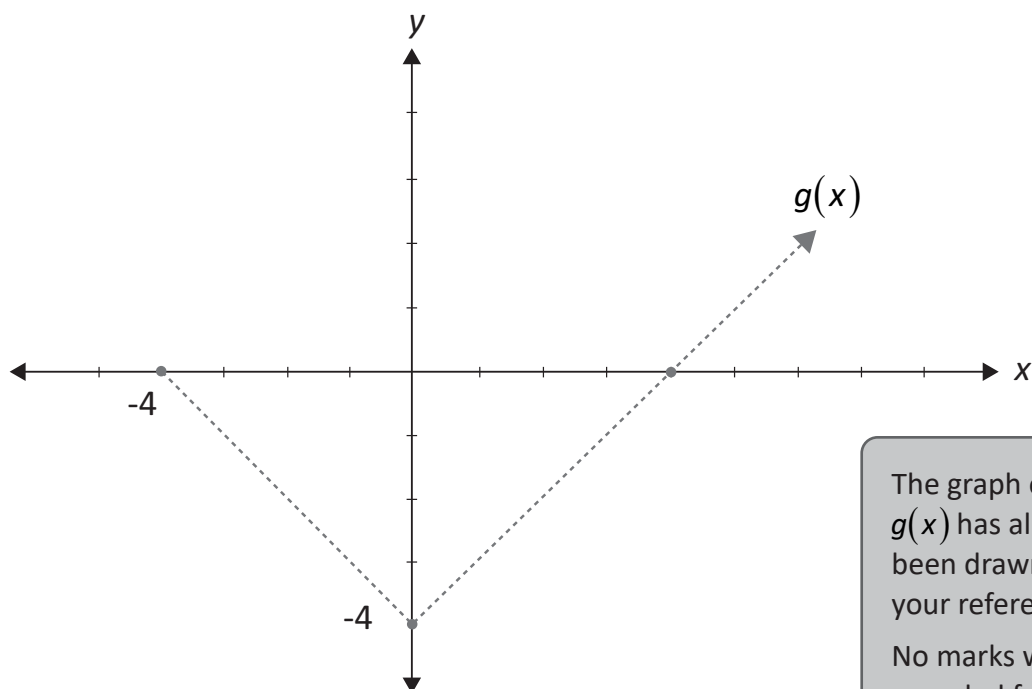
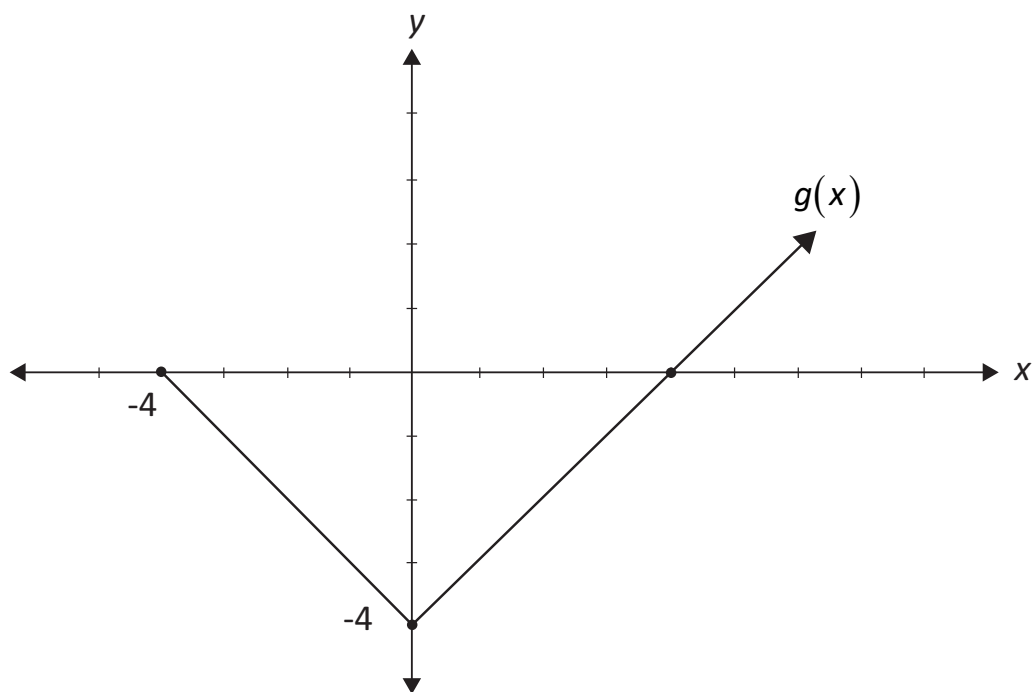
Express $\frac{1}{2}\log x - 2\log y + \log z$ as a single logarithm.

Question 11

3 marks

111

Given the graph of $g(x)$, sketch the graph of $f(x) = \frac{1}{g(x)}$.



The graph of $g(x)$ has already been drawn for your reference.
No marks will be awarded for the graph of $g(x)$.

Question 12

a) 1 mark b) 3 marks

112
113

Given the identity $\frac{1 - \tan^2 \theta}{\sec^2 \theta} = \cos 2\theta$,

a) state a non-permissible value of θ .

b) prove the identity for all permissible values of θ .

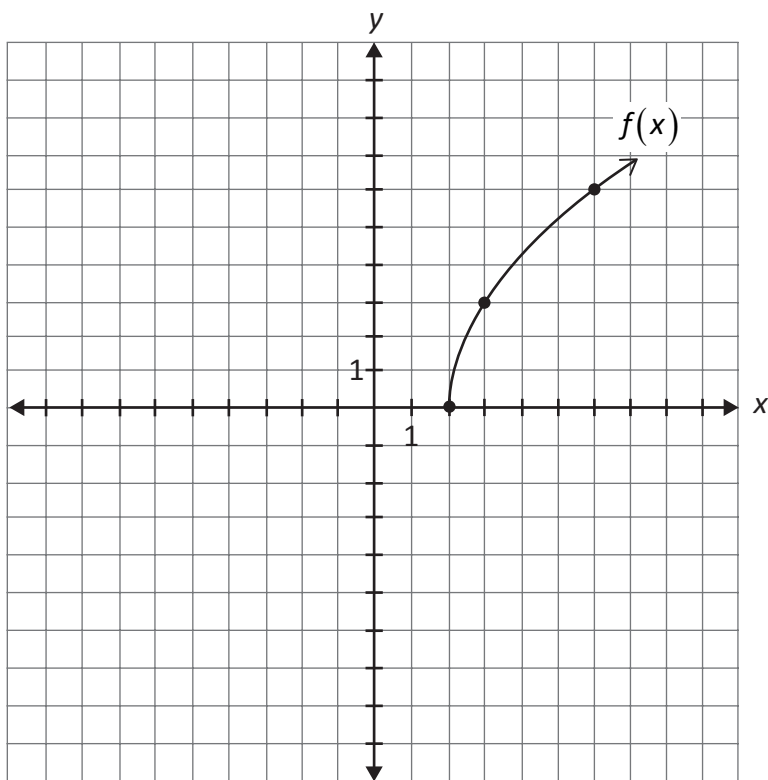
Left-Hand Side	Right-Hand Side

Question 13

2 marks

114

Determine an equation of the radical function represented by the graph of $f(x)$.



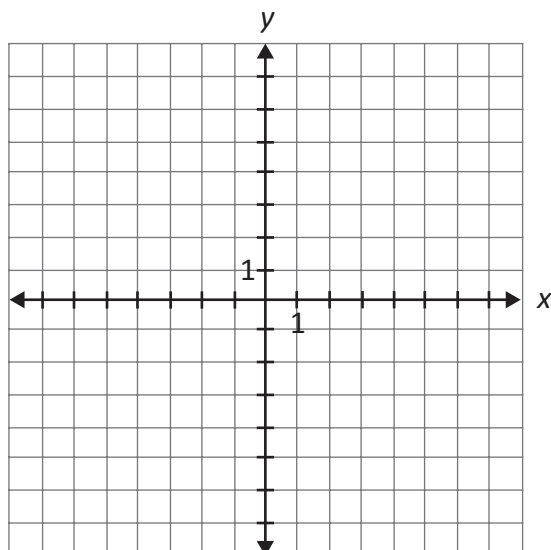
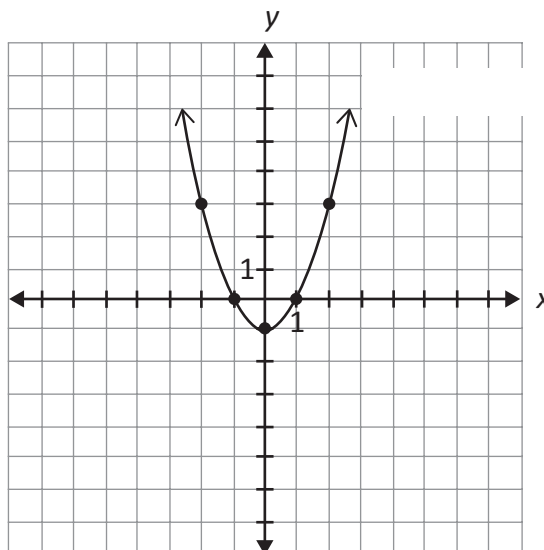
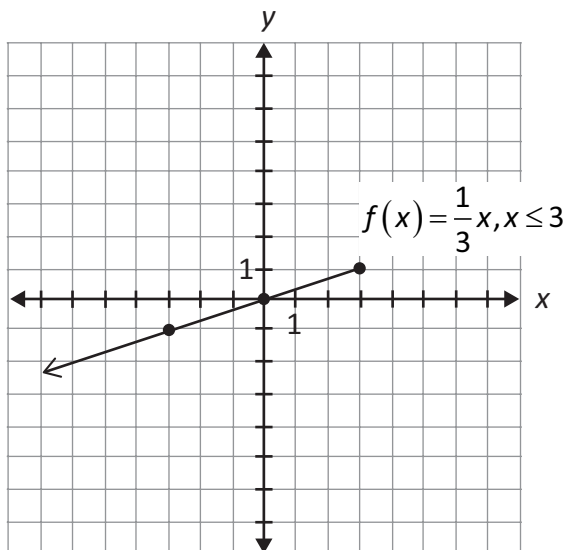
$f(x) =$ _____

Question 14

2 marks

115

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(f(x))$.



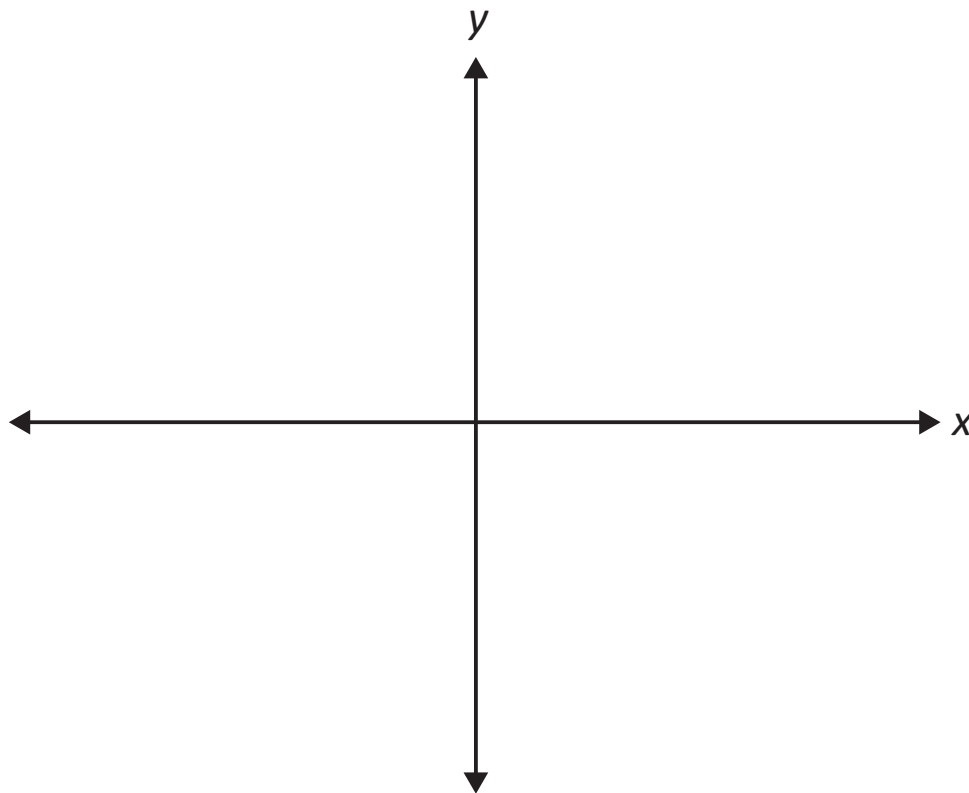
Question 15

3 marks

116

Sketch a graph of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 3
- $p(x)$ has a zero at -5 with a multiplicity of 2
- $p(x)$ has a zero at 1
- $p(x)$ has a leading coefficient of -2

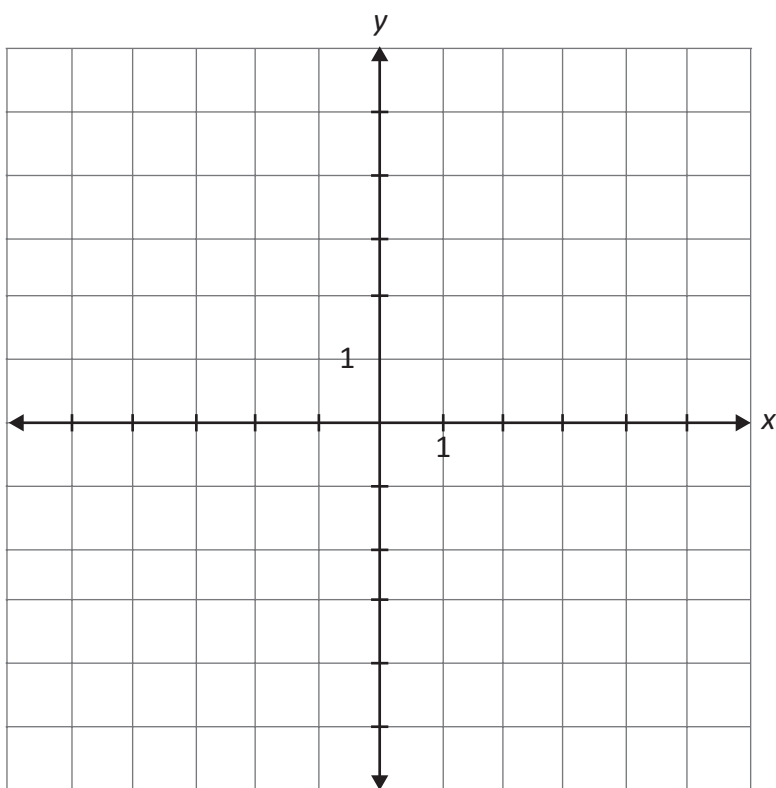


Question 16

2 marks

117

Sketch the graph of $f(x) = \frac{x^2 + x}{x}$.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2025

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 2 (January 2025)

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Number and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1	—	16	40
Booklet 2	9	19	50
Total	9	35	90

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

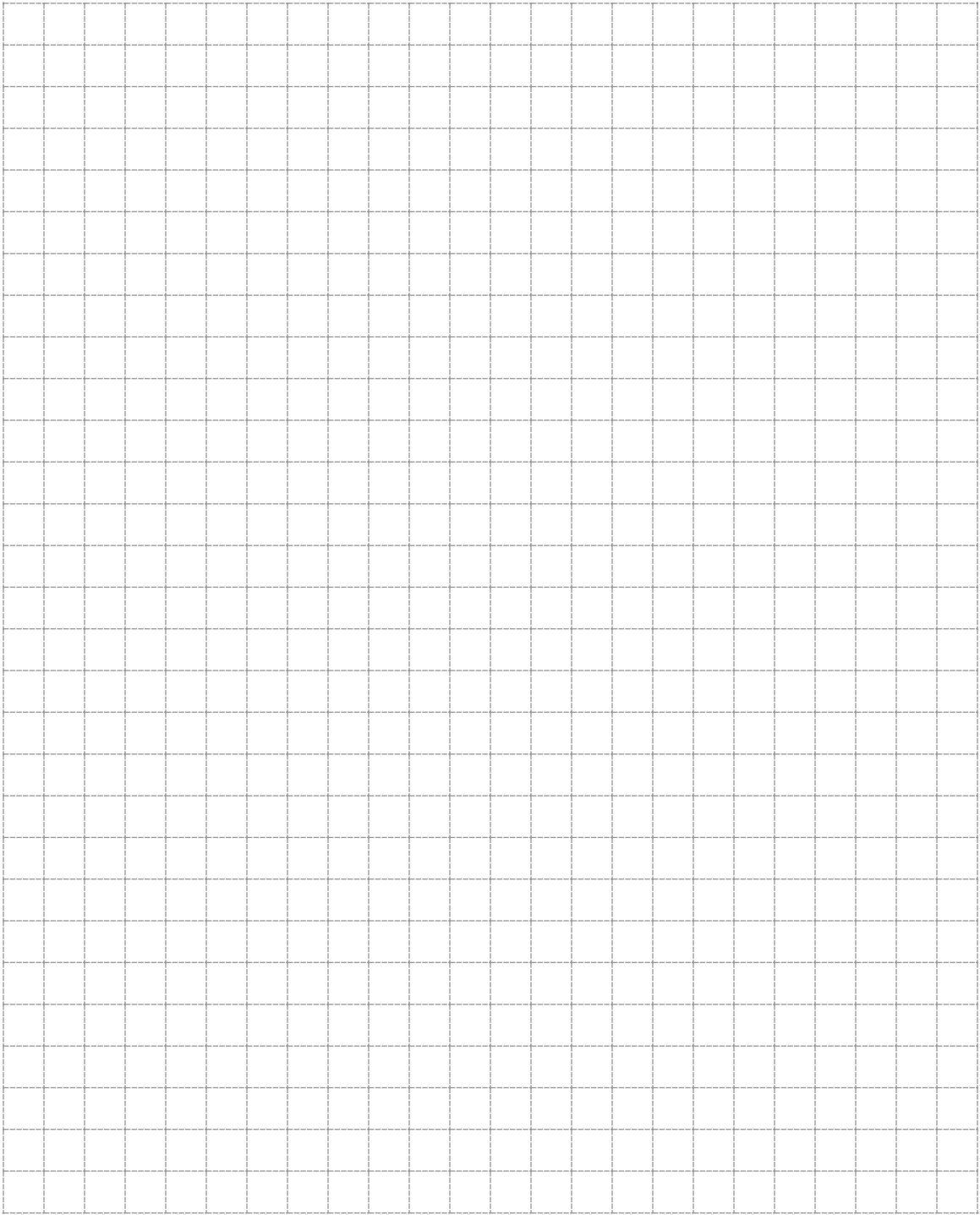
Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- Write each solution in the space provided.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 17**1 mark**

Identify the function that has a different range than $f(x) = x^2$.

- a. $y = 4f(x)$
- b. $y = f(x) + 4$
- c. $y = f(x + 4)$
- d. $y = f(-x)$

Question 18**1 mark**

Identify the exact value of $\sin\left(-\frac{7\pi}{3}\right)$.

- a. $-\frac{\sqrt{3}}{2}$
- b. $-\frac{1}{2}$
- c. $\frac{1}{2}$
- d. $\frac{\sqrt{3}}{2}$

Question 19**1 mark**

Identify the value of a in the equation, $\log_a 3 = \frac{1}{4}$.

- a. $\frac{1}{64}$
- b. $3^{\frac{1}{4}}$
- c. 12
- d. 81

Question 20**1 mark**

Identify a value of x where the graphs of $y = \cos x$ and $y = \sin x$ intersect.

a. $-\frac{3\pi}{4}$

b. $-\frac{\pi}{4}$

c. 0

d. $\frac{3\pi}{4}$

Question 21**1 mark**

Identify the number of negative terms in the binomial expansion of $(2x - y)^6$.

a. 0

b. 3

c. 4

d. 6

Question 22**1 mark**

Identify the general solution of the equation, $\sin x = 1$.

a. $x = k\pi, k \in \mathbb{Z}$

b. $x = 2k\pi, k \in \mathbb{Z}$

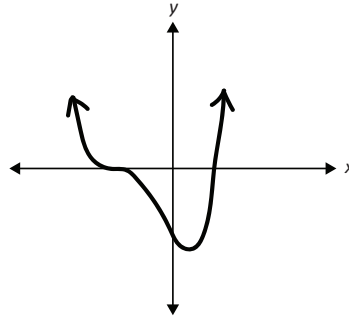
c. $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

d. $x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

Question 23**1 mark**

Identify a possible degree of the polynomial function represented below.

- a. 2
- b. 3
- c. 4
- d. 5

**Question 24****1 mark**

Identify the expression that is equivalent to $\cos x \cot x$.

- a. $\frac{1 - \sin x}{\sin x}$
- b. $\frac{\sin^2 x - 1}{\sin x}$
- c. $\sin x$
- d. $\frac{1 - \sin^2 x}{\sin x}$

Question 25**1 mark**

Given that the point $(-7, -5)$ is on the graph of $y = g(x)$, identify the corresponding point on the graph of $y = |g(x)|$.

- a. $(7, 5)$
- b. $(-7, 5)$
- c. $(7, -5)$
- d. $(-5, -7)$

Question 26**2 marks**

118

Given that $x = 1$ is one of the zeros of $p(x) = x^3 + x^2 - 10x + 8$, express $p(x)$ in completely factored form.

$p(x) =$ _____

Question 27**2 marks**

119

Determine the exact value of $\frac{\csc^2 \theta - 1}{1 - \cos^2 \theta}$ if $\theta = \frac{5\pi}{6}$.

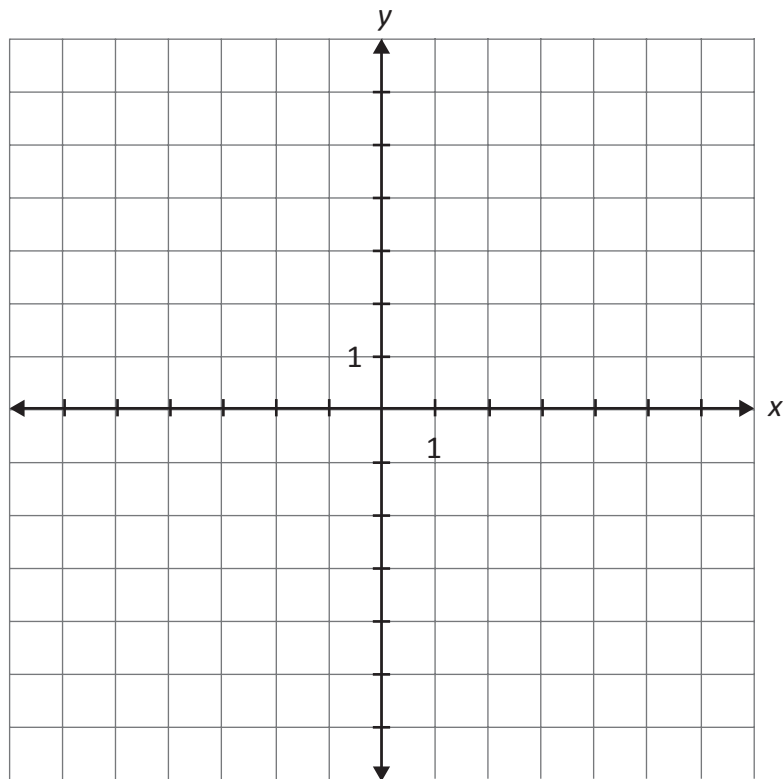
Question 28

3 marks

120

Solve, graphically.

$$-2x + 5 = \sqrt{x - 1}$$



Question 29

2 marks

121

If $\log_a 2 = p$ and $\log_a 5 = q$, express $\log_a 40$ in terms of p and q .

Question 30**1 mark**

122

Describe how to sketch the graph of $y=f^{-1}(x)$, given the graph of a linear function, $y=f(x)$.

Question 31**3 marks**

123

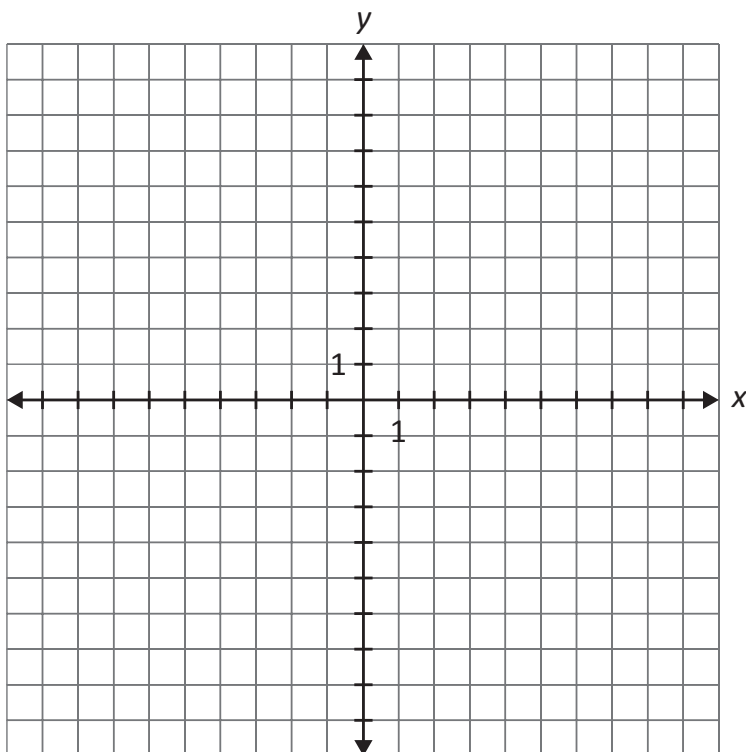
Given that $\sin \alpha = -\frac{4}{5}$ where α is in quadrant III, and $\cos \beta = \frac{12}{13}$ where β is in quadrant IV, determine the exact value of $\cos(\alpha - \beta)$.

Question 32

3 marks

124

Sketch the graph of $f(x) = \frac{2x-4}{x+1}$.



Question 33

a) 1 mark b) 1 mark c) 1 mark

125
126
127

Due to the tidal variations, the depth of the water in a bay is represented by the function,

$$d(t) = -2\cos\left[\frac{\pi}{6}(t+4)\right] + 16$$

where d is the depth of the water, in metres and
 t is the time, in hours.

a) Determine the depth of the water when $t = 2$.

b) State the minimum depth of the water.

c) Determine the length of time between two consecutive low tides (minimum depths).

Question 34

3 marks

128

Evaluate.

$$\sin^2\left(\frac{5\pi}{4}\right)\sec\left(\frac{13\pi}{3}\right) - \cot\left(\frac{7\pi}{4}\right)$$

Question 35

a) 1 mark b) 1 mark

129
130

a) State the non-permissible value(s) of the function, $f(x) = \frac{x-3}{x^2-9}$.

b) Describe what each non-permissible value represents on the graph of $f(x) = \frac{x-3}{x^2-9}$.

Question 36

1 mark

131

Determine which of the given values is greater. Justify your answer.

$$\log_3 38 \text{ or } \log_4 50$$

Question 37**1 mark**

132

Determine if the point $\left(\frac{3}{4}, \frac{\sqrt{5}}{4}\right)$ is on the unit circle. Justify your answer.

Question 38**2 marks**

133

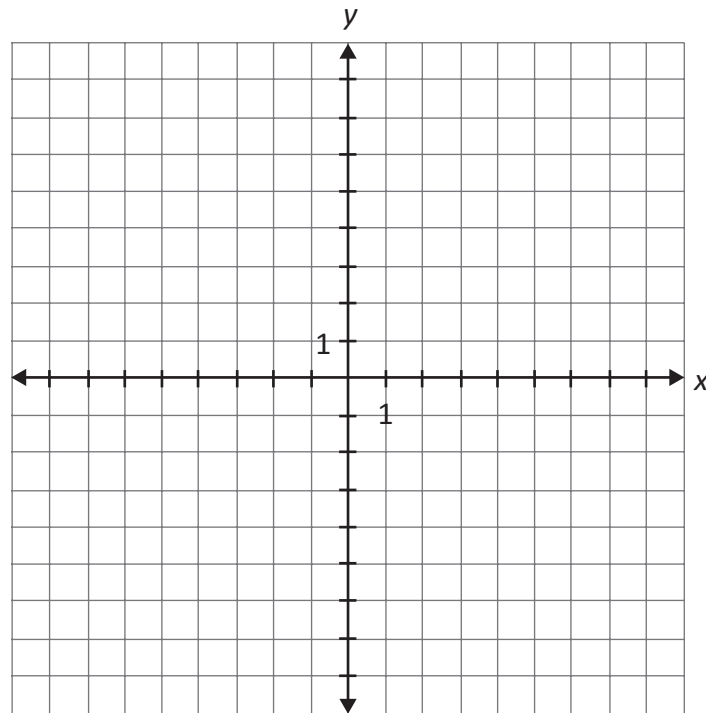
Determine which term contains x^5 in the binomial expansion of $\left(2x + \frac{3}{x}\right)^{11}$.

Question 39

3 marks

134

Sketch the graph of $y = 2(3^x) + 1$.



Question 40**2 marks**

135

Solve, algebraically.

$$\log(2x + 1) = \log(x - 2) + 1$$

Question 41

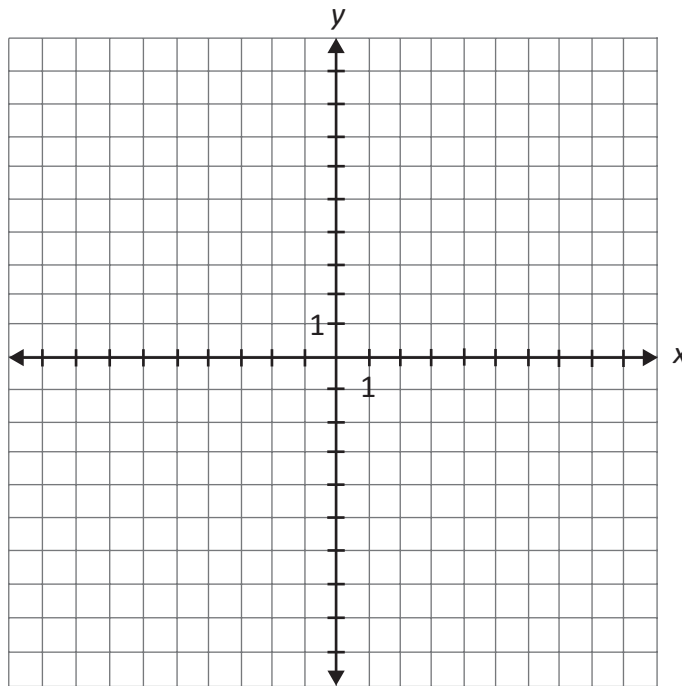
a) 1 mark b) 1 mark

136
137

Given the functions, $f(x) = x + 2$ and $g(x) = x - 2$,

a) state the equation of $h(x) = f(x) \cdot g(x)$.

b) sketch the graph of $h(x) = f(x) \cdot g(x)$.



Question 42**1 mark**

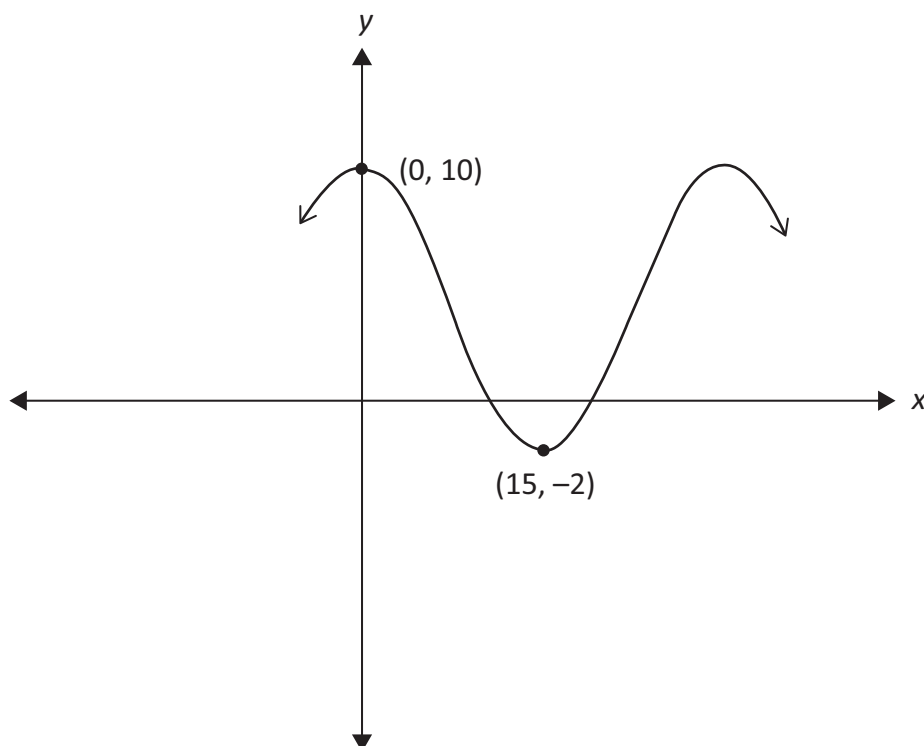
138

Given $p(x) = -2x^3 + 3x + 1$, determine the remainder when $p(x)$ is divided by $(x - 2)$.

Question 43**3 marks**

139

Determine the equation of the sinusoidal function in the form $y = A\cos(Bx) + D$.



$y =$ _____

Question 44**2 marks**

140

Determine the total number of arrangements of the letters in the word TORONTO.

State your answer as a whole number.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet I

June 2025

Grade 12 pre-calculus mathematics achievement test.
Booklet 1. June 2025

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Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Number and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	15	33
Booklet 2	9	22	57
Total	9	37	90

* The first 4 questions with the symbol  in Booklet 1 require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

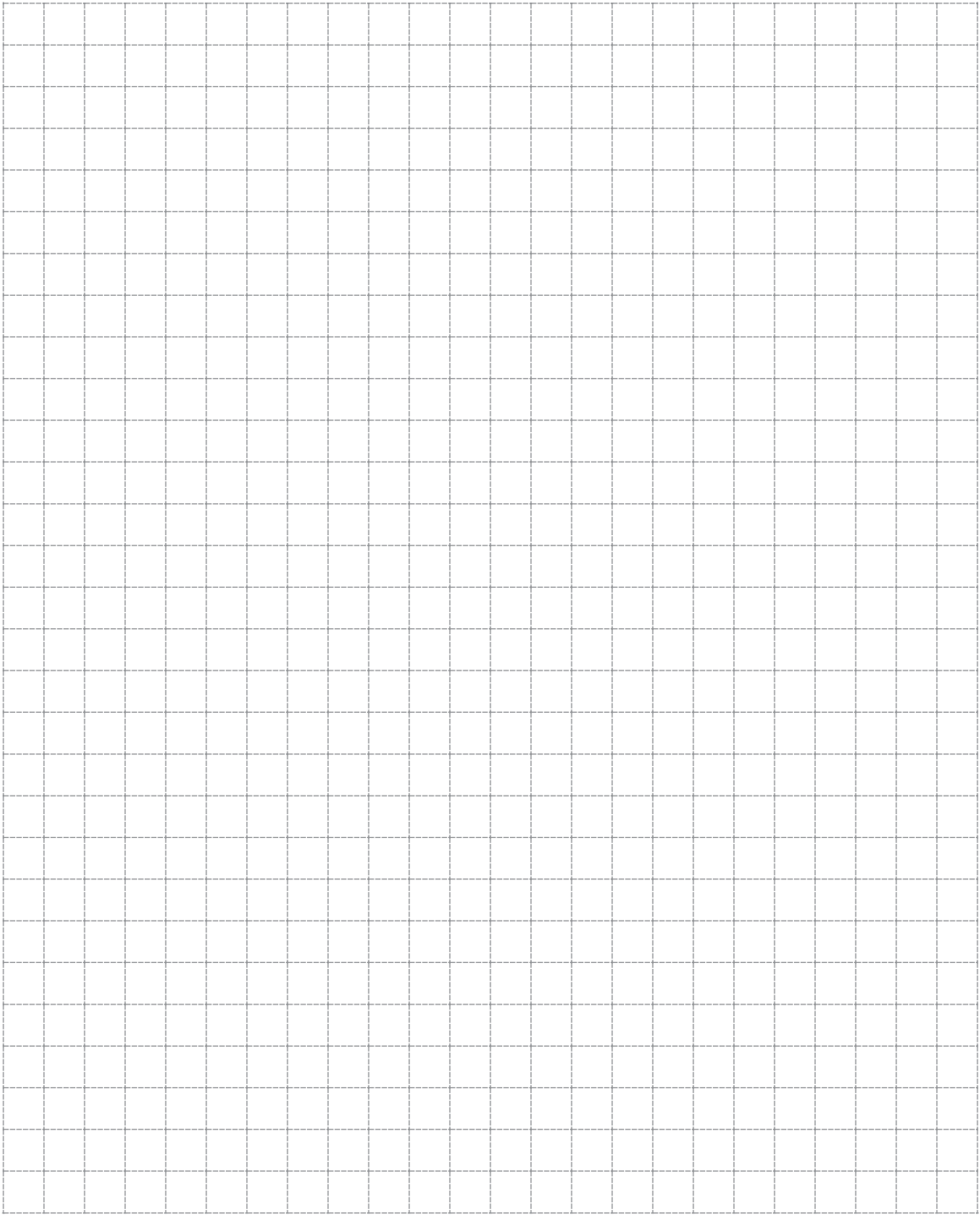
Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited. Please turn off your cell phone and all other such devices.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_nP_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_nC_k a^{n-k} b^k$$

$$\text{Given } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain, identify, and justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

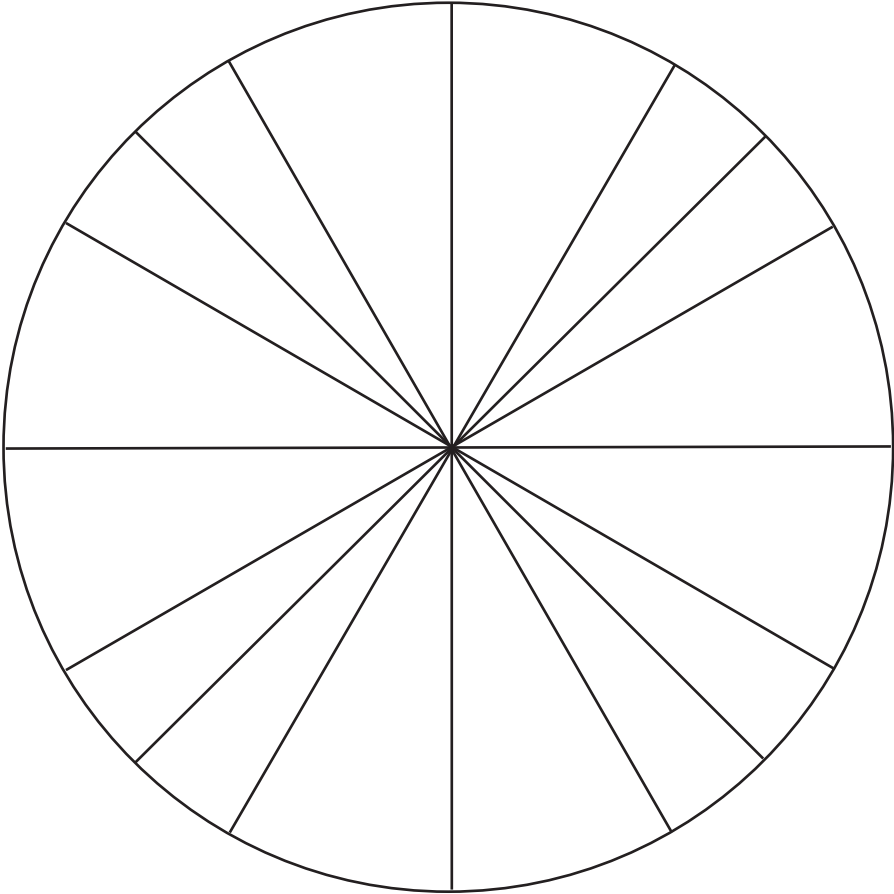
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of two coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

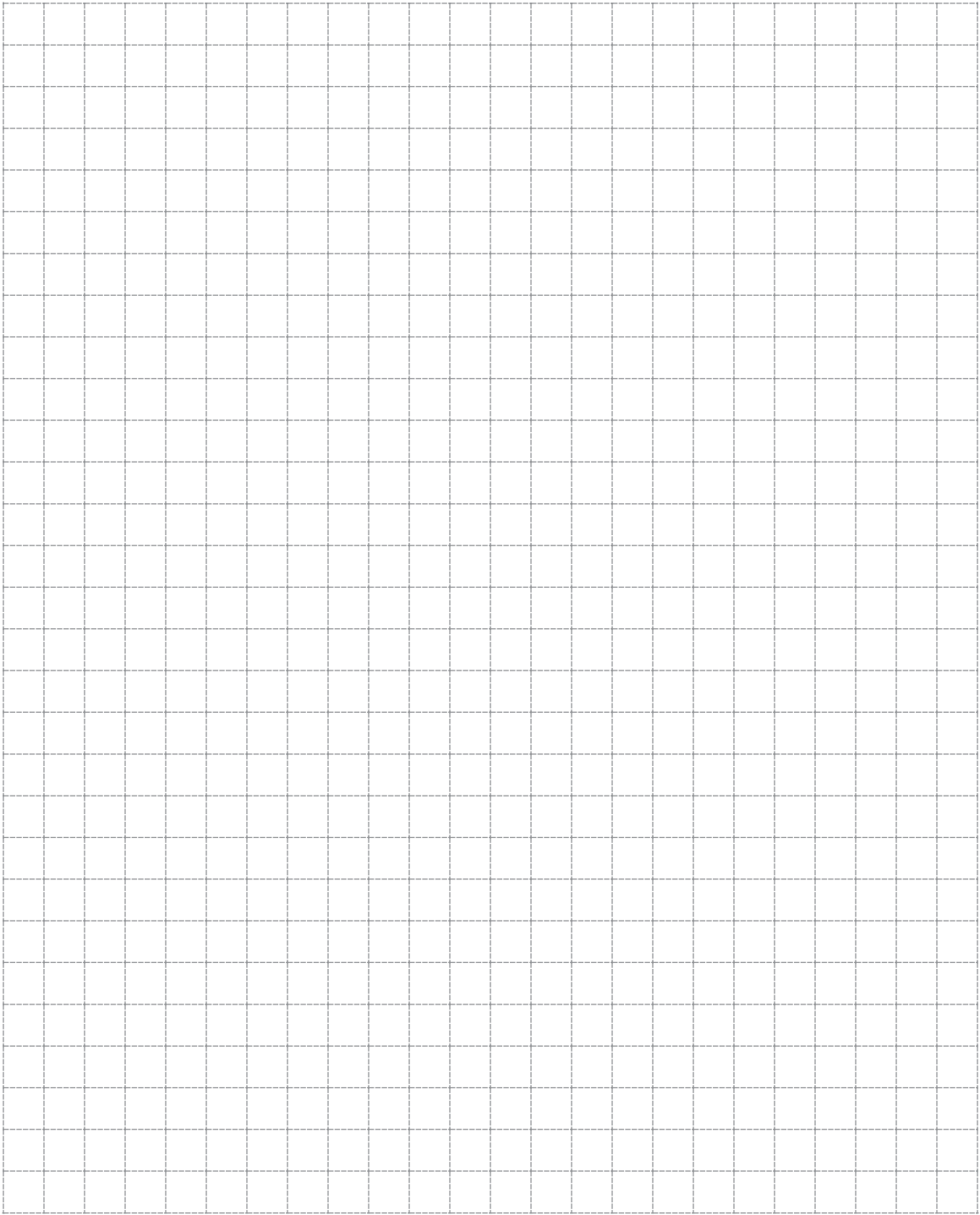
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 1 **1 mark**

101

Manitoba postal codes consist of three letters and three digits. Determine the total number of possible postal codes in Manitoba if the following characteristics must be met:

- The postal code must begin with the letter R.
- Only 18 letters of the alphabet can be used.
- The letters and digits must alternate.
- The letters and digits can be repeated.

Question 2 **2 marks**

102

The percent of people who click on a link, in response to seeing an advertisement on YouTube, can be modeled by the formula:

$$R(t) = -(0.8)^{0.2t} + 0.6$$

where $R(t)$ is the percentage, in decimal form, of people who click on the link

t is the time in days.

Determine, algebraically, the number of days it will take for 45% of people who see the advertisement to click on the link.

Question 3 **3 marks**

103

Determine and simplify the middle term in the binomial expansion of $\left(\frac{5}{x} + 4x^3\right)^6$.

Question 4 **3 marks**

104

Solve $3\tan^2x + 5\tan x - 6 = 0$, over the interval $[0, 2\pi]$.

Note: A calculator is not required for the remaining test questions.

Question 5

a) 1 mark b) 1 mark

105
106

Given $\theta = -80^\circ$,

a) determine the measure of θ , in radians.

b) state all the coterminal angles of θ .

Question 6

1 mark

107

State a possible value of n if the polynomial function $p(x) = (x+1)^n(x-4)^2$ has a range of $(-\infty, \infty)$.

Question 7

4 marks

108

Describe the transformations used to obtain the graph of the function $y = -2f(x + 8) - 5$ from the graph of $y = f(x)$.

Question 8

2 marks

109

State the domain and range of the function $f(x) = \frac{-2}{x^2}$.

Domain: _____

Range: _____

Question 9

1 mark

110

Given the following row of Pascal's Triangle, state the values in the next row.

1 5 10 10 5 1

Question 10

2 marks

111

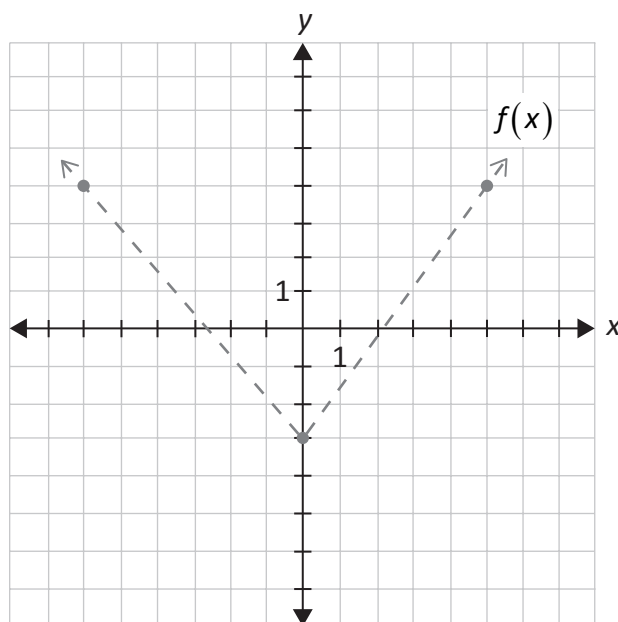
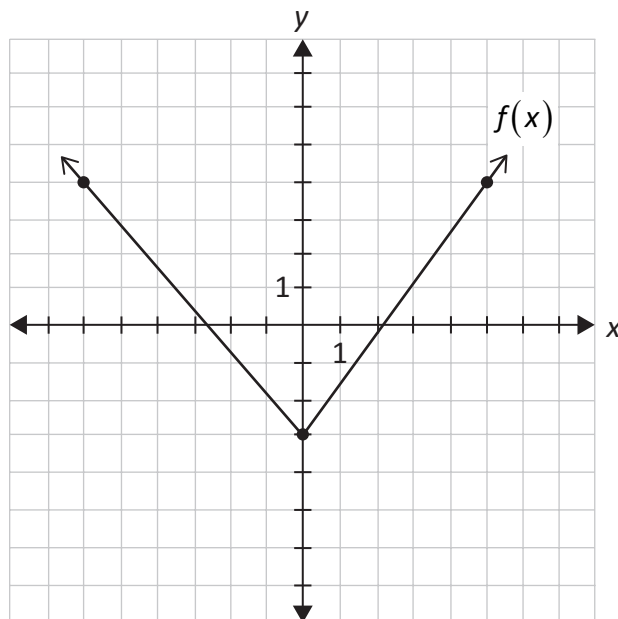
Given $f(x) = 3 - 2x$, determine the equation of $f^{-1}(x)$.

Question 11

2 marks

112

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 12

3 marks

113

Prove the identity for all permissible values of x .

$$\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x} = \csc x \sec x$$

Left-Hand Side	Right-Hand Side

Question 13

1 mark

114

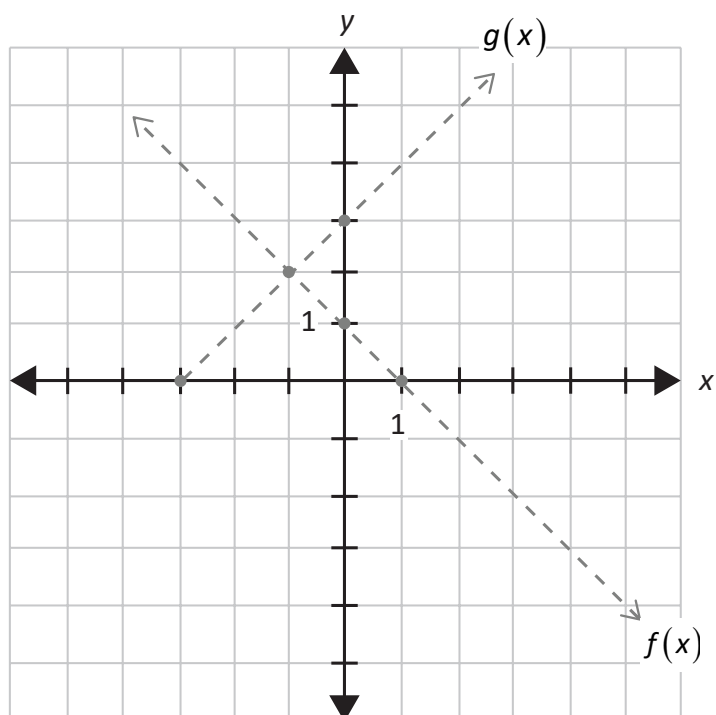
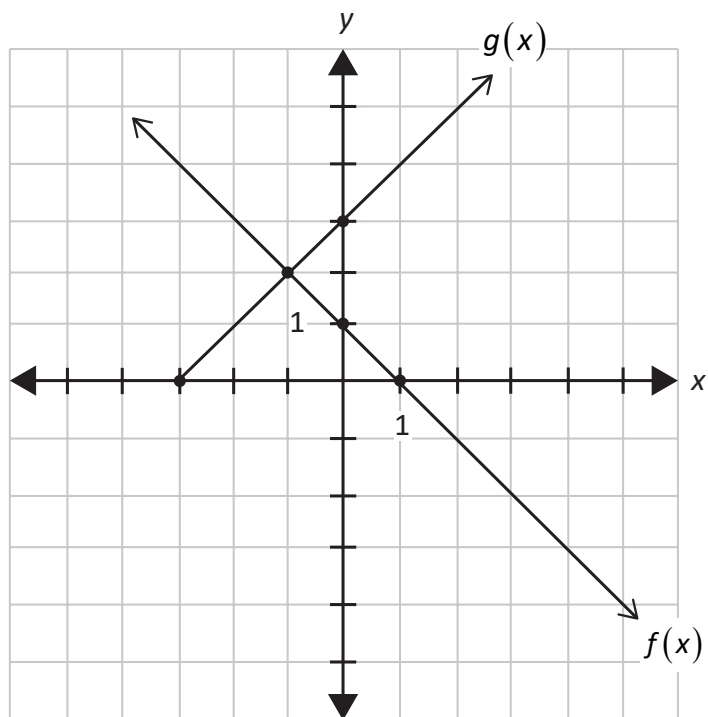
Justify that the graph of $f(x) = e^{x+1} + 5$ does not have an x -intercept.

Question 14

2 marks

115

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) \cdot g(x)$.



The graph of $f(x)$ and $g(x)$ have already been drawn for your reference.

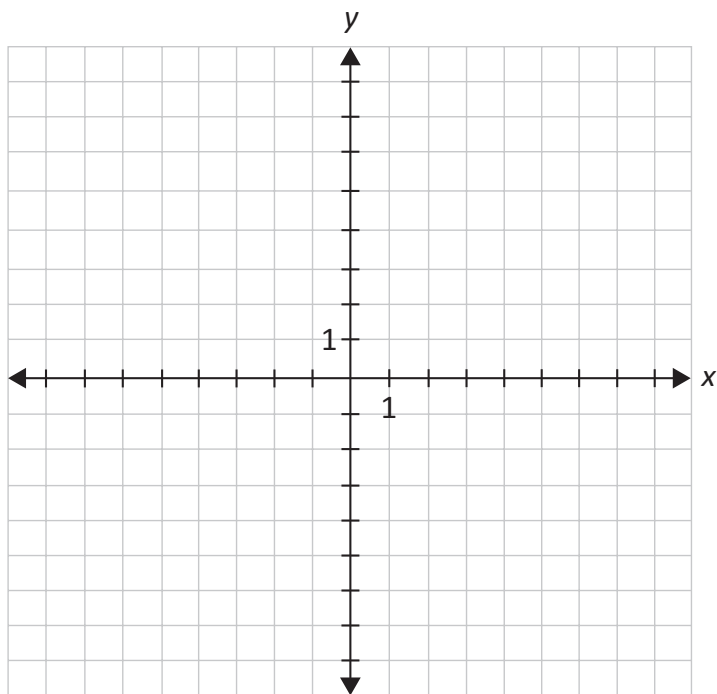
No marks will be awarded for the graph of $f(x)$ and $g(x)$.

Question 15

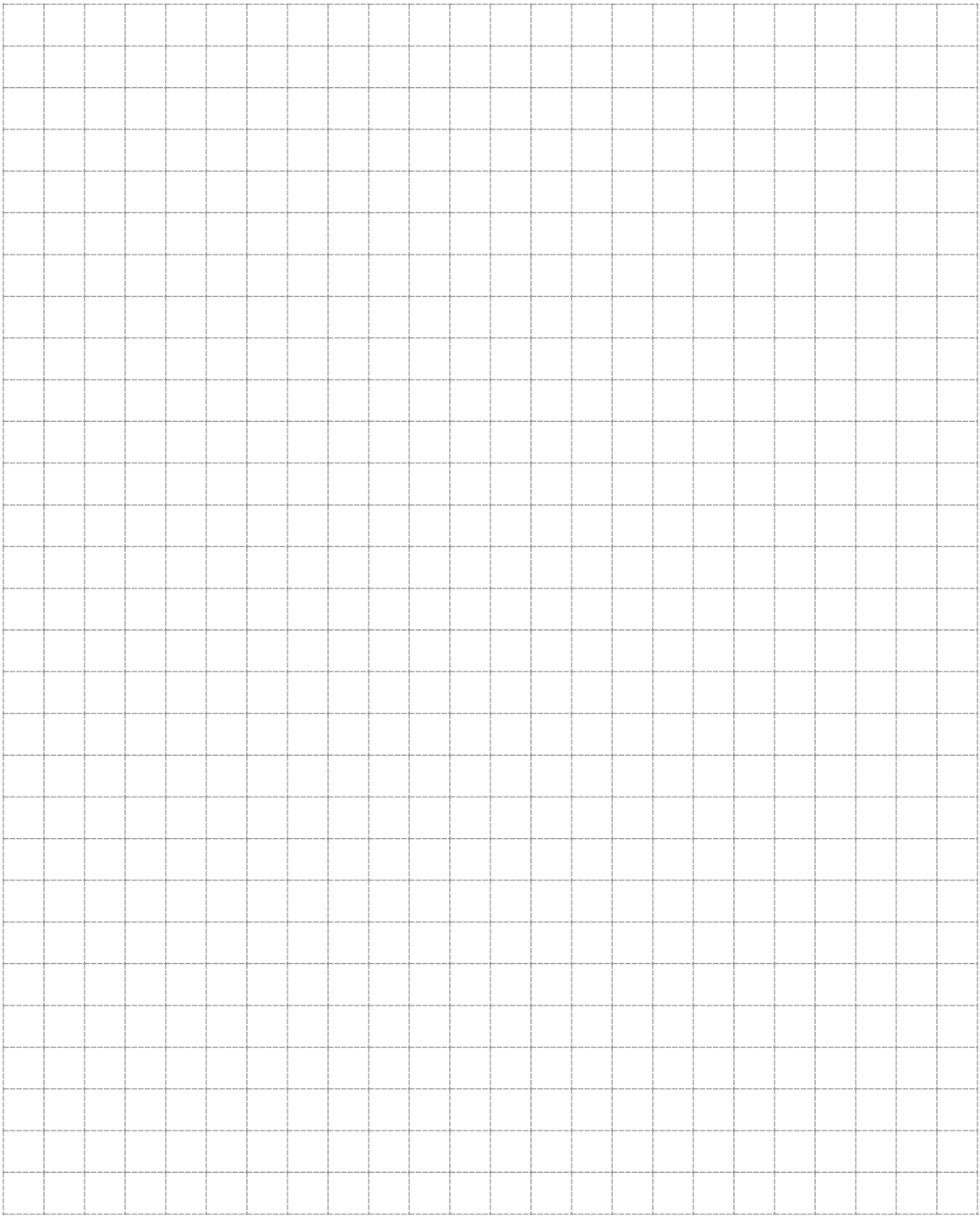
4 marks

116

Sketch the graph of $f(x) = \frac{5}{2}\sqrt{-(x-4)}$.



NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2025

Grade 12 pre-calculus mathematics achievement test.
Booklet 2. June 2025

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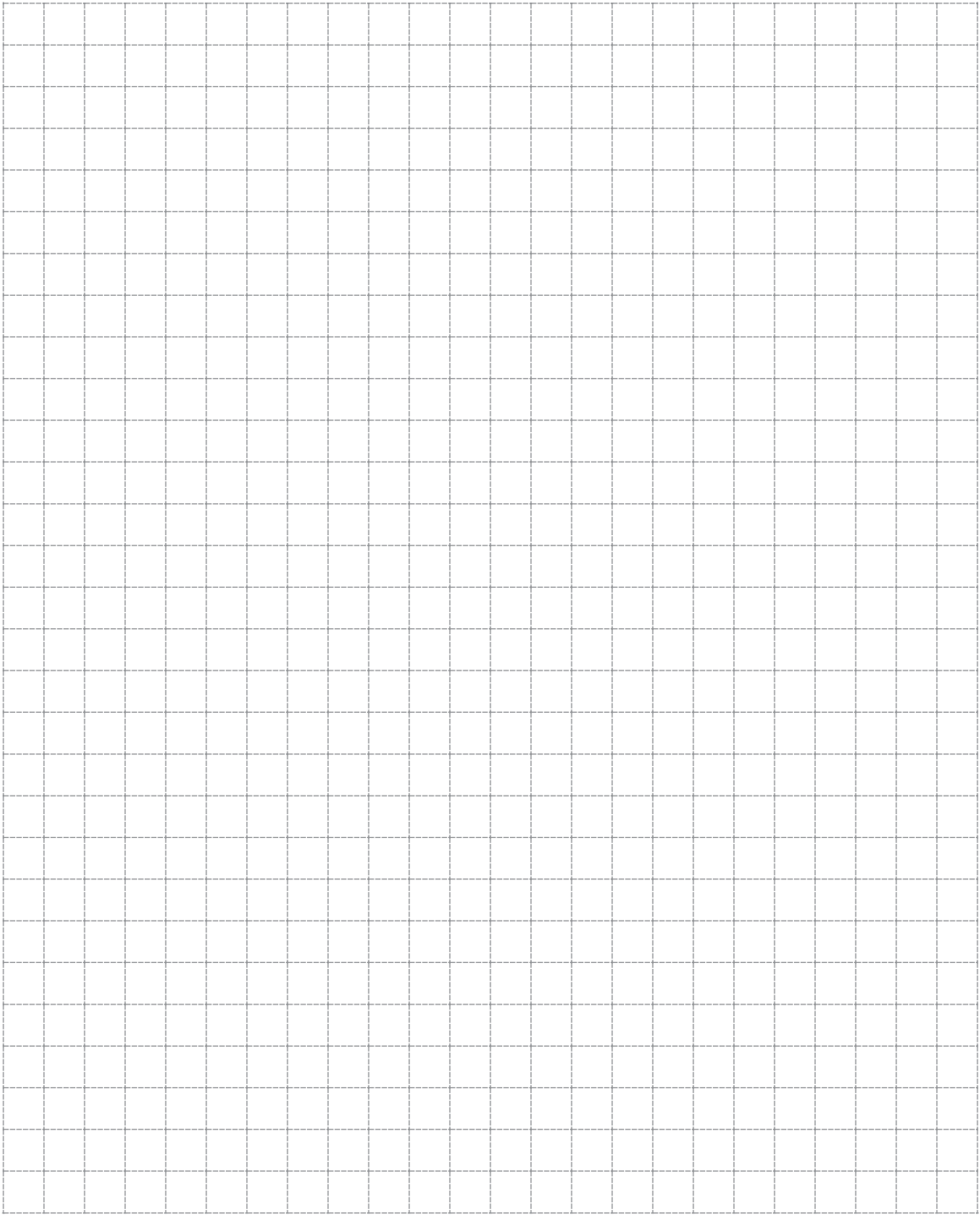
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NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 16**1 mark**

Identify the number of x -intercepts on the graph of the polynomial function,

$$p(x) = 3x(x - 8)(x^2 + 5).$$

- a. 1
- b. 2
- c. 3
- d. 4

Question 17**1 mark**

Identify the equation that represents the graph of $y = x^2 - 3$ after a reflection over the x -axis.

- a. $y = -x^2 - 3$
- b. $y = x^2 - 3$
- c. $y = x^2 + 3$
- d. $y = -x^2 + 3$

Question 18**1 mark**

Given $2^{\log_2 3} = x$, identify the value of x .

- a. 1
- b. 3
- c. 6
- d. 8

Question 19**1 mark**

Identify the y-intercept on the graph of $y = \frac{x-3}{x^2-5x+6}$.

- a. -3
- b. -2
- c. $-\frac{1}{2}$
- d. 3

Question 20**1 mark**

Identify the total number of arrangements for 10 students and 2 teachers to sit in a row if the teachers must sit together.

- a. $10!2!$
- b. $11!$
- c. $11!2!$
- d. $12!$

Question 21**1 mark**

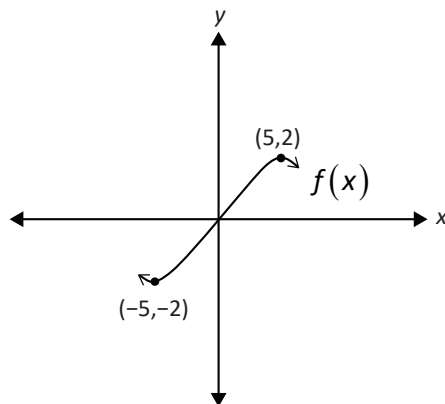
Identify which expression represents the remainder when the polynomial $P(x)$ is divided by $(x-4)$.

- a. $P(-4)$
- b. $P(4)$
- c. $P(x-4)$
- d. $P(x+4)$

Question 22**1 mark**

Identify the period of the graph of the sinusoidal function.

- a. 10π
- b. 10
- c. 20π
- d. 20

**Question 23****1 mark**

Identify an equivalent expression for $\log_3 9 + \log_3 5$.

- a. $\log_3 14$
- b. $2\log_3 5$
- c. $\log_3 45$
- d. $\log_3 \left(\frac{9}{5} \right)$

Question 24**1 mark**

Identify the coterminal angle(s) of $\theta = -\frac{11\pi}{10}$ over the interval $[-3\pi, \pi]$.

- a. $\theta = \frac{9\pi}{10}$
- b. $\theta = -\frac{31\pi}{10}, -\frac{21\pi}{10}$
- c. $\theta = -\frac{31\pi}{10}, \frac{9\pi}{10}$
- d. $\theta = -\frac{21\pi}{10}, \frac{9\pi}{10}$

Question 25**3 marks**

117

Express $p(x) = x^3 - 13x - 12$ in completely factored form.

$p(x) =$ _____

Question 26

3 marks

118

Verify, by substitution, that the equation $\frac{\cos^2\theta}{\sin^2\theta} = \frac{2\csc\theta}{\sec^2\theta}$ is true for $\theta = \frac{\pi}{6}$.

Left-Hand Side	Right-Hand Side

Question 27

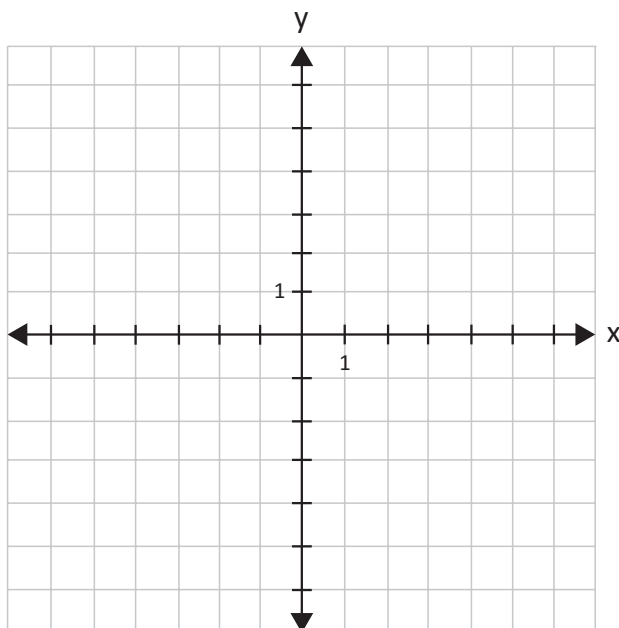
a) 1 mark b) 2 marks

119
120

Given $f(x) = \log_3 x + 2$,

a) determine the x-intercept of $f(x) = \log_3 x + 2$.

b) sketch the graph of $f(x) = \log_3 x + 2$.



Question 28

2 marks

121

Evaluate.

$$\sec^2\left(\frac{5\pi}{4}\right) \cdot \tan^2\left(-\frac{2\pi}{3}\right)$$

Question 29**1 mark**

122

Given the functions $f(x) = x - 3$ and $g(x) = \sqrt{x} + 2$, state the domain of $g(f(x))$.

Question 30**1 mark**

123

Evaluate.

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ$$

Question 31

2 marks

124

Solve, algebraically.

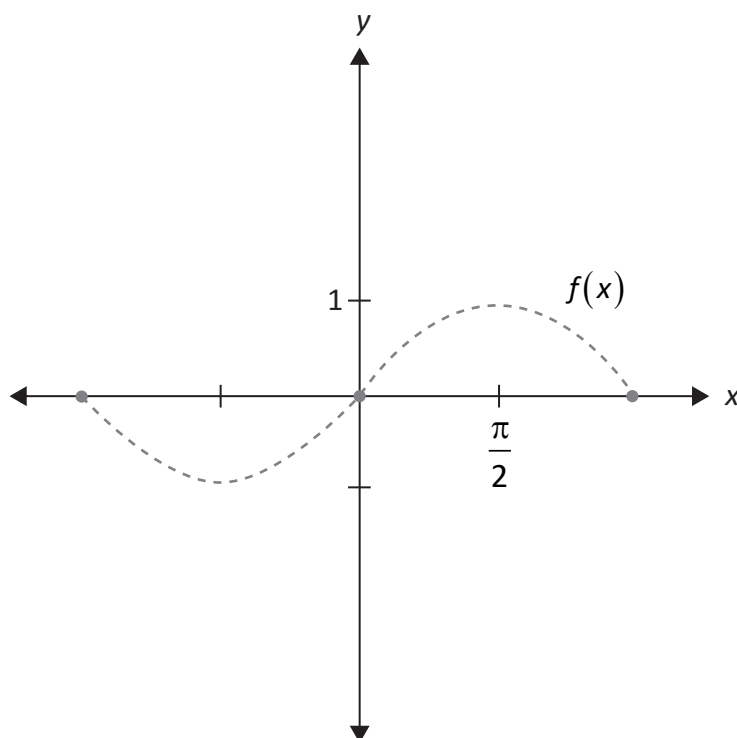
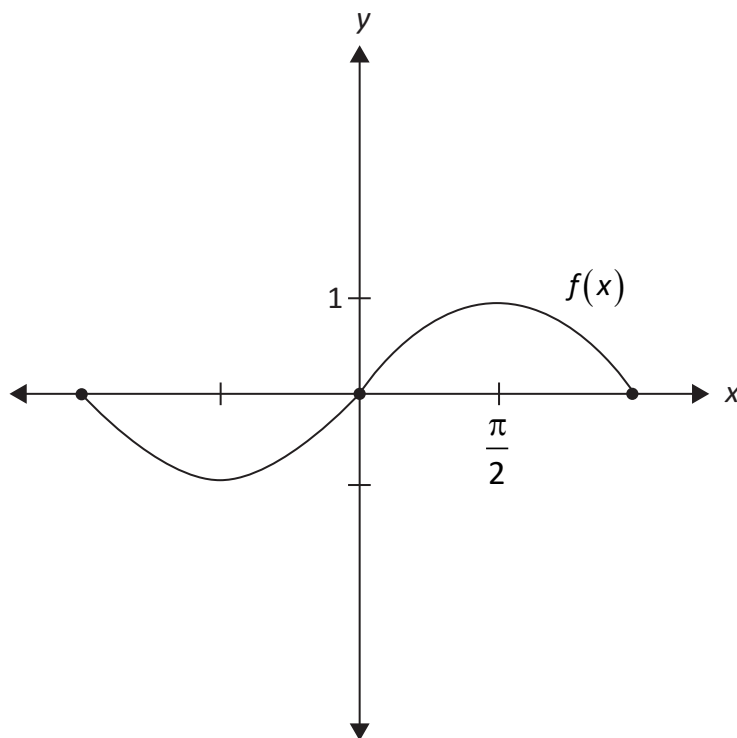
$$\log_{15}(x^2 - 1) = 1$$

Question 32

2 marks

125

Given the graph of $y = f(x)$, sketch the graph of $y = \frac{1}{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 33

a) 3 marks b) 1 mark

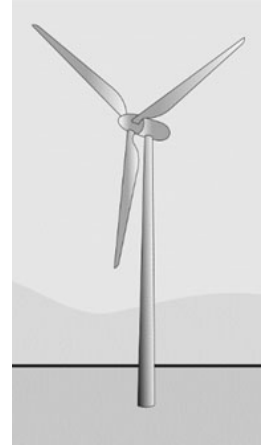
126
127

A light is attached to the end of a blade of a wind turbine. The height of the light, with respect to the ground, follows the sinusoidal equation:

$$h(t) = 80 \cos\left(\frac{\pi}{2}t\right) + 240$$

where $h(t)$ is the height of the light with respect to the ground, measured in feet and,

t is the time, measured in seconds.



- a) Sketch the graph of the height of the light with respect to the ground, over an interval of 6 seconds.



- b) If the turbine begins to rotate faster, describe the resulting effect on the period of the graph of $h(t)$.

Question 34

1 mark

128

Elodie was asked to determine the total number of arrangements of the letters in the word EXCELLENCE.

Her solution:

$$10! = 3628\ 800$$

Describe her error.

Question 35

a) 3 marks b) 1 mark

129
130

Given that $\cos \alpha = -\frac{4}{7}$ where α is in quadrant III and $\sin \beta = \frac{5}{13}$ where β is in quadrant II, determine the exact value of:

a) $\cos(\alpha + \beta)$

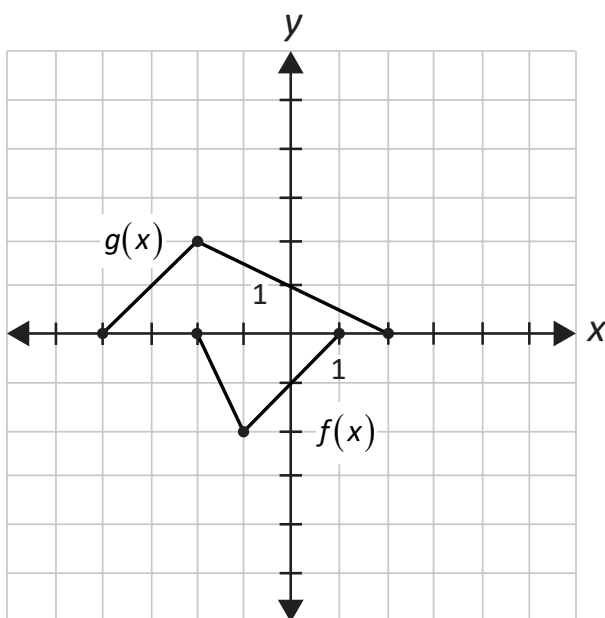
b) $\sec(\alpha + \beta)$

Question 36

2 marks

131

Determine the equation for $g(x)$, in terms of $f(x)$.



$g(x) =$ _____

Question 37

1 mark

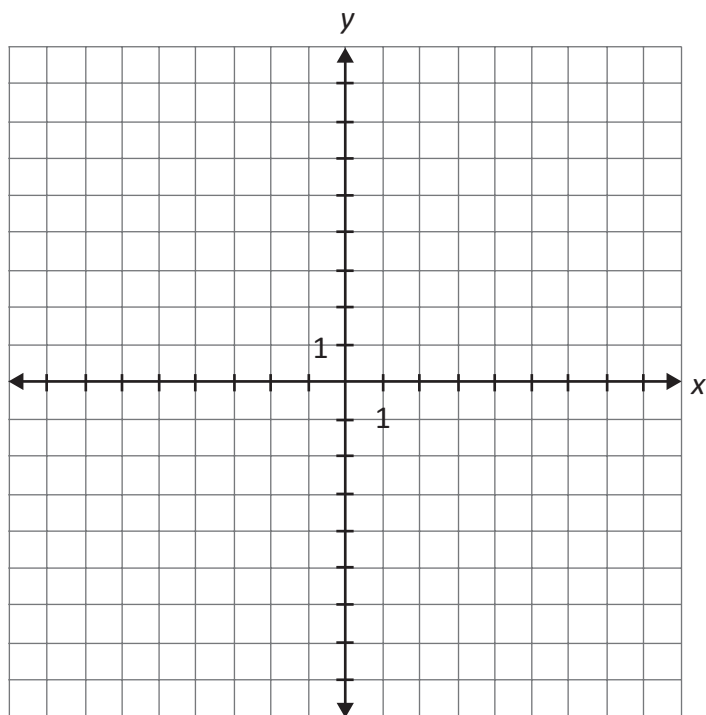
132

State possible values for m and n that satisfy the equation, $\log_m n = 5$.

Question 38**4 marks**

133

Sketch the graph of $f(x) = \frac{x-4}{x^2-7x+12}$.



Question 39

1 mark

134

Justify that the value of ${}_3P_4$ does not exist.

Question 40**1 mark**

135

State the range of the sinusoidal function that has the following characteristics:

- an amplitude of 3
- a maximum at $\left(\frac{\pi}{2}, 1\right)$

Question 41

3 marks

136

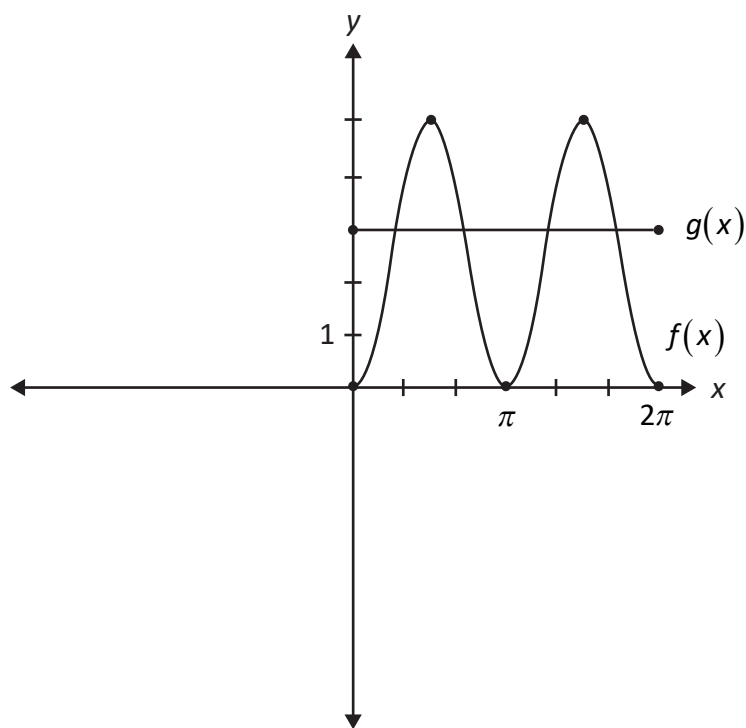
Solve, algebraically.

$${}_nC_2 = 15$$

Question 42**1 mark**

137

Describe how to use the graphs of $f(x) = 5\sin^2 x$ and $g(x) = 3$ to solve the equation, $5\sin^2 x = 3$, over the interval $[0, 2\pi]$.



Question 43**3 marks**

138

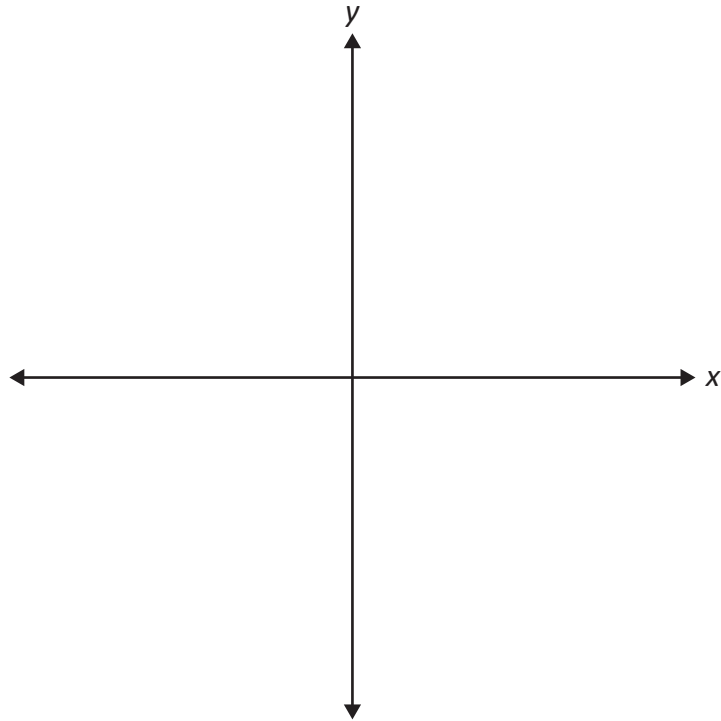
Solve, algebraically.

$$2\log_3 5 - \frac{1}{3}\log_3 125 = \log_3 a$$

Question 44**1 mark**

139

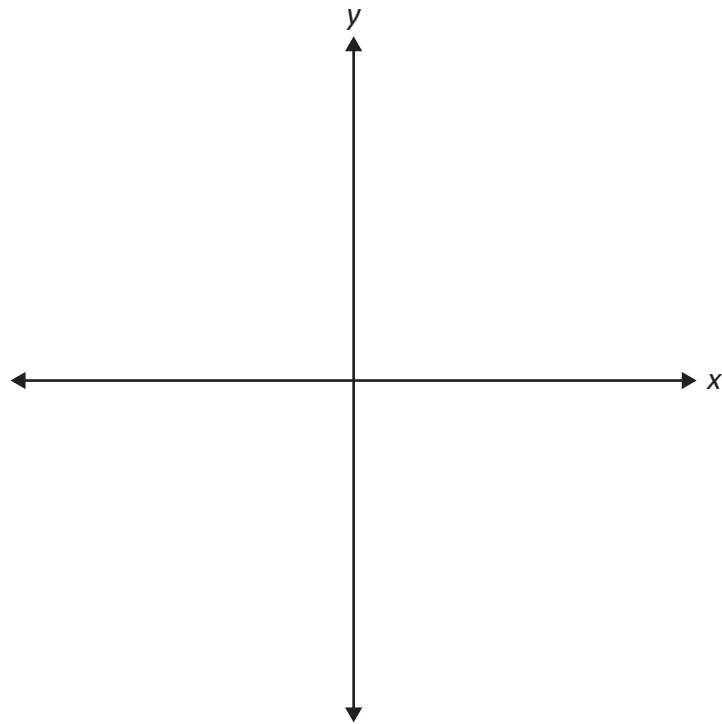
Sketch the angle, $\frac{-7\pi}{12}$, in standard position.



Question 45**3 marks**

140

Sketch the graph of $p(x) = -(x-1)(x-2)(x+4)^2$.



Question 46**2 marks**

141

The point, $P(\theta) = \left(-\frac{1}{6}, y\right)$, lies on the unit circle and is located in Quadrant III. Determine the exact value of $\csc \theta$.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

January 2026

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 1 (January 2026)

This resource is available in print and electronic formats.

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Available in alternate formats upon request.

Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Number and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	17	37
Booklet 2	8	20	53
Total	8	37	90

* The first 4 questions with the symbol  in Booklet 1 require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

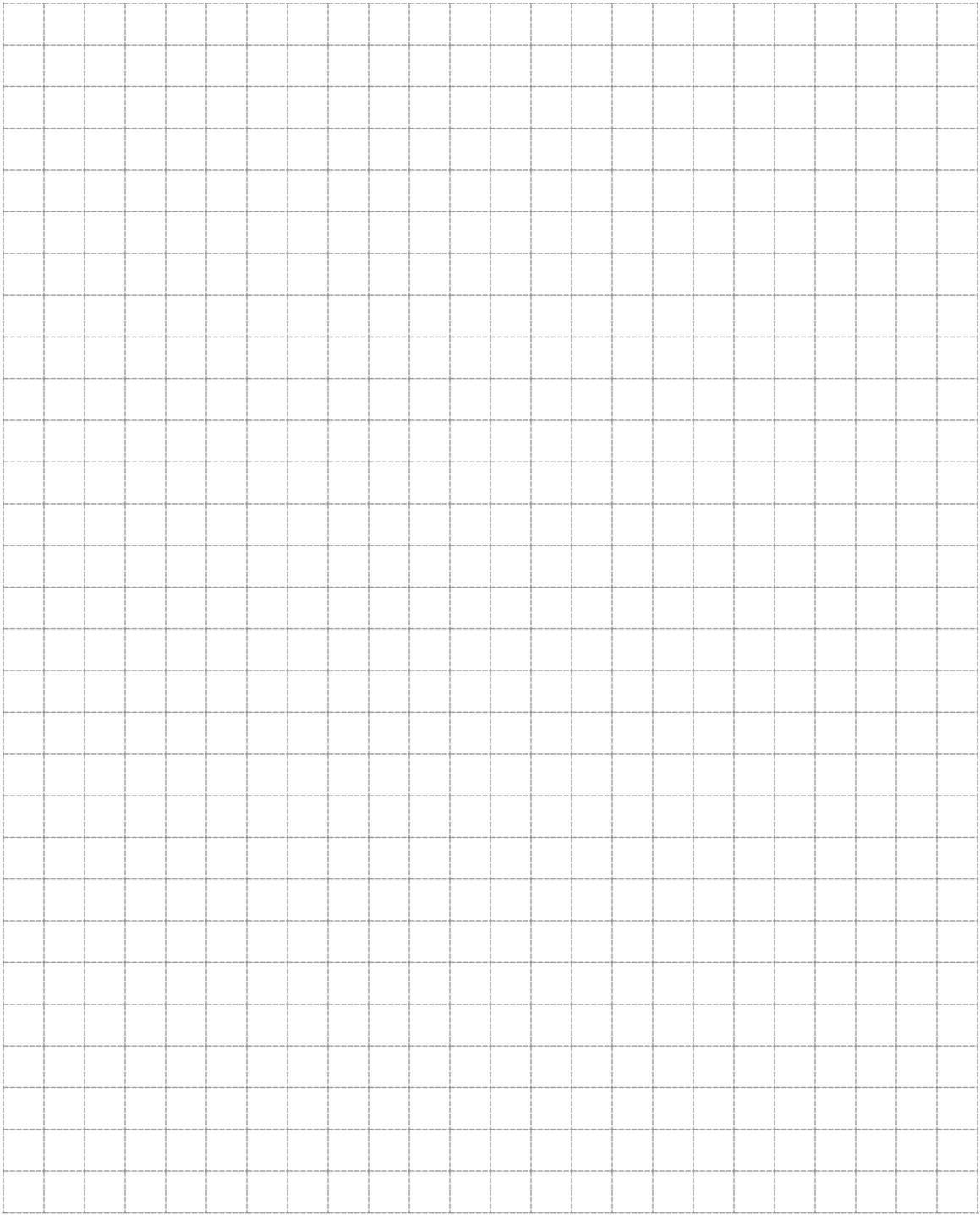
Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited. Please turn off your cell phone and all other such devices.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_nP_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_nC_k a^{n-k} b^k$$

$$\text{Given } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain, identify, and justify*. These words are defined below

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Express: Write or present a mathematical concept in a specific form, often using symbols or equations.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

Prove: Show that a mathematical statement is true using logical reasoning and/or known facts.

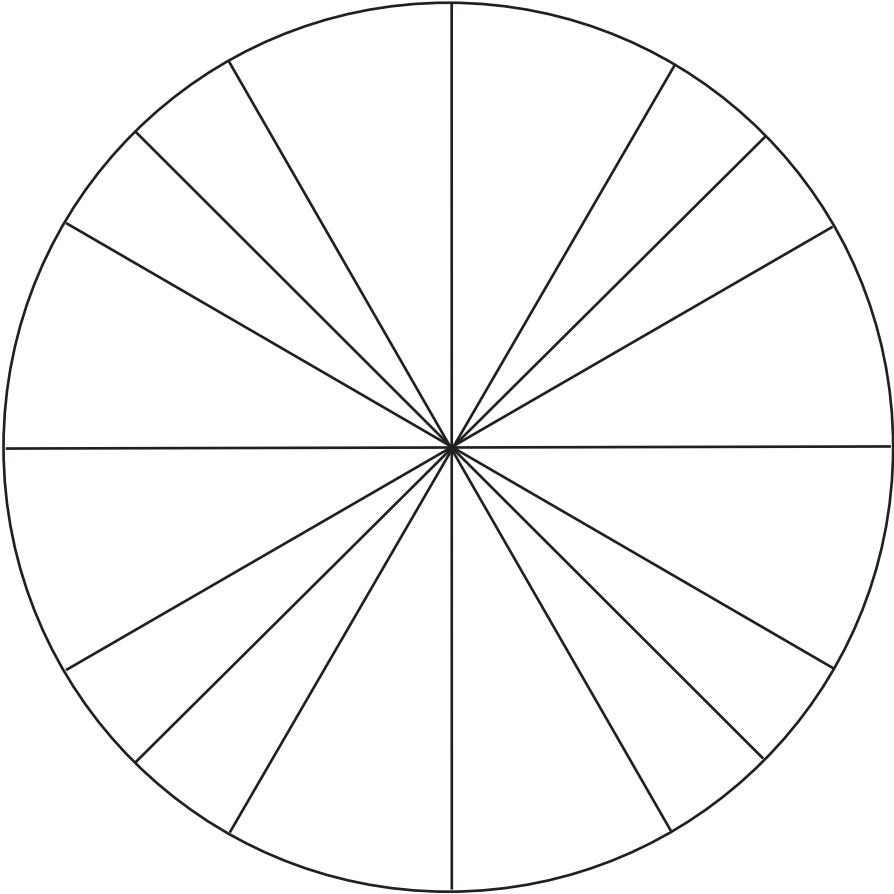
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of two coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

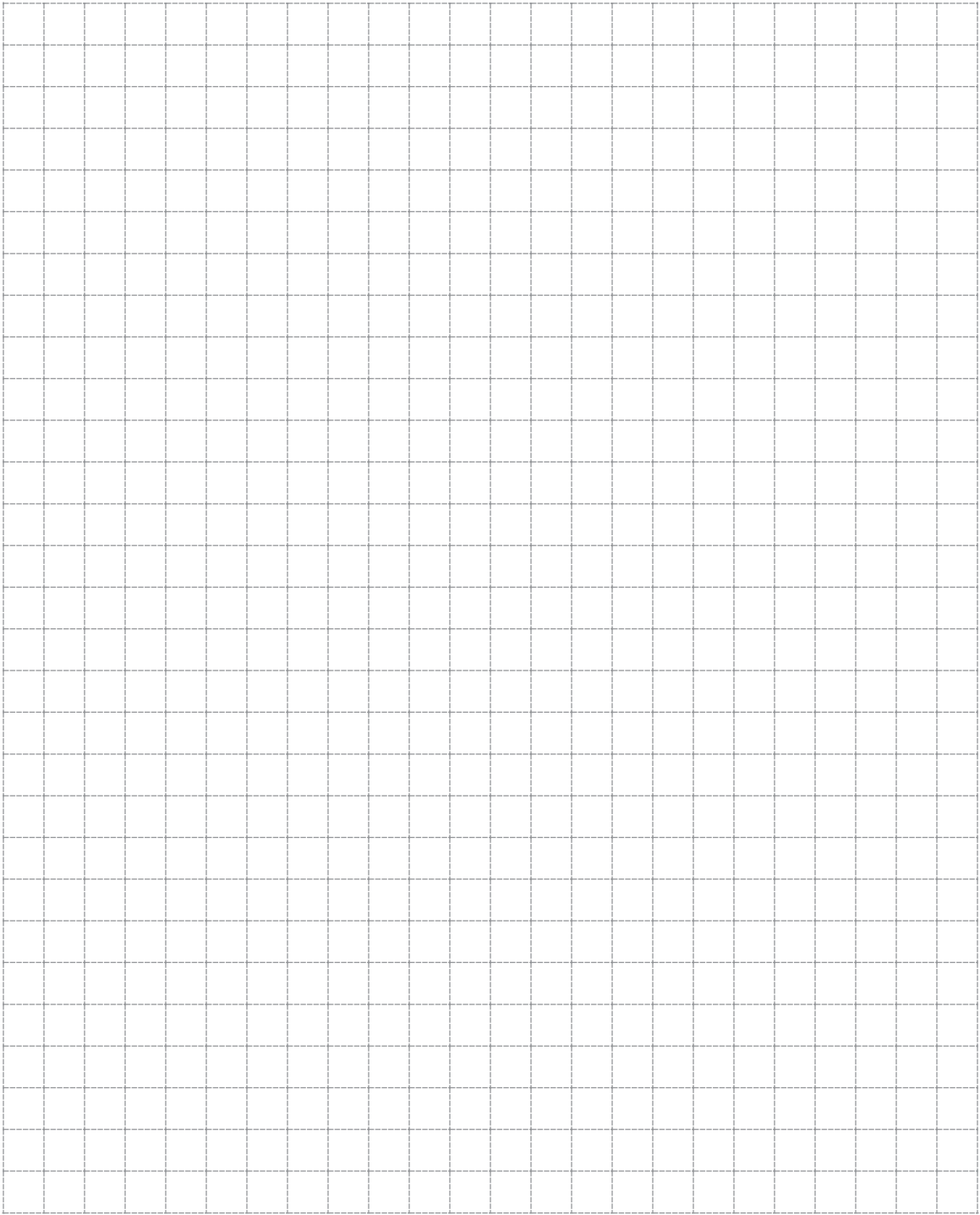
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 1 **2 marks**

101

Liam went for a ride on a Ferris wheel with a radius of 5 metres. Determine the distance he travelled if the Ferris wheel stopped after rotating 1260° .

Question 2 **4 marks**

102

Determine and simplify the term that contains x^{12} in the binomial expansion of $(2x^4 - 3)^7$.

Question 3 **a) 1 mark b) 2 marks**103
104

The normal healing of wounds can be modeled by an exponential function. The area of the wound decreases according to the formula,

$$A = A_0 e^{-0.35t}$$

where: A is the area of the wound in cm^2 , after t days,

A_0 is the initial area of the wound in cm^2 , and

t is the time, in days, after healing begins.

If a wound has an initial area of 25 cm^2 , determine:

a) the area of the wound after 3 days.

b) the amount of time it will take for the area of the wound to decrease to 4 cm^2 .

Question 4 **3 marks**

105

During a hockey practice, 4 players are wearing a green jersey and 5 players are wearing a red jersey. Determine the number of ways three players can be selected if at least two of the players must be wearing a red jersey.

Note: A calculator is not required for the remaining test questions.

Question 5

4 marks

106

Solve, algebraically, over the interval $[0, 2\pi]$.

$$2\sin^2 x + 5\cos x + 1 = 0$$

Question 6

1 mark

107

The function, $y = f(x)$, has a domain of $[-8, 4]$. The domain of the graph of $y = f(bx)$ is $[-4, 2]$.

State the value of b .

Question 7**1 mark**

108

Describe the difference between the graph of $f(x) = \frac{(x+3)(x-4)}{(x+3)}$ and the graph of $g(x) = x - 4$.

Question 8

1 mark

109

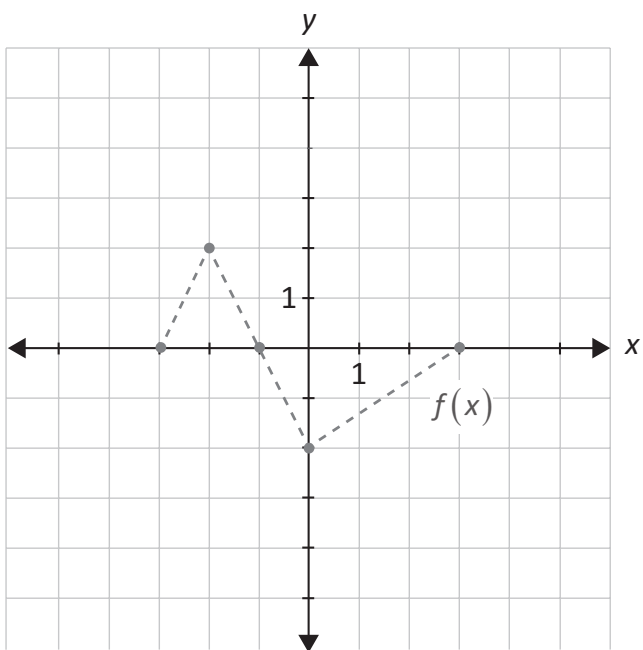
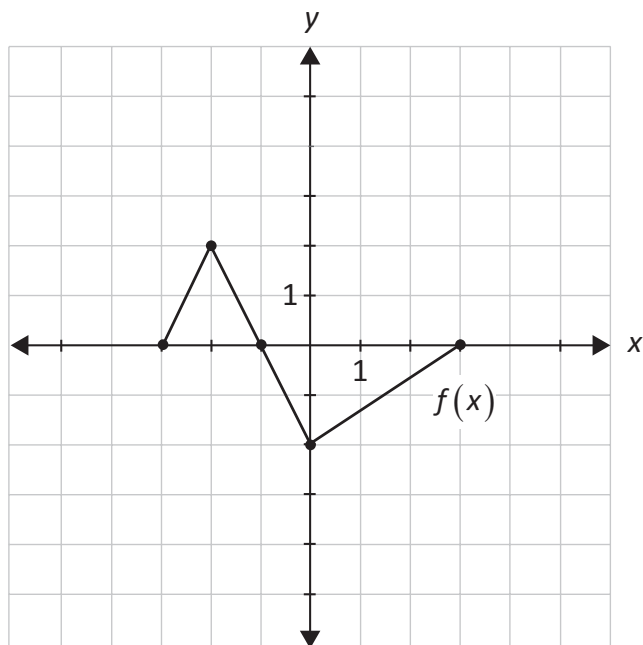
Determine if the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is on the unit circle. Justify your answer.

Question 9

3 marks

110

Given the graph of $f(x)$, sketch the graph of $g(x) = f(-(x-1)) + 3$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 10**1 mark**

111

Bonnie was asked to determine the factors of $p(x) = -3x^4 + 2x^2 - 4x + 5$.

Describe the error she made in her set up of synthetic division.

$$\begin{array}{r|rrrrr} 1 & & -3 & 2 & -4 & 5 \end{array}$$

Question 11

a) 1 mark b) 1 mark

112
113

Given the functions, $f(x) = \frac{1}{x-1}$ and $g(x) = x - 3$,

a) state the equation of $f(g(x))$.

$$f(g(x)) = \underline{\hspace{2cm}}$$

b) state the domain of $f(g(x))$.

Domain:

Question 12

3 marks

114

Prove the identity for all permissible values of x .

$$\tan 2x = \frac{2}{\cot x - \tan x}$$

Left-Hand Side	Right-Hand Side

Question 13

a) 1 mark b) 1 mark

115
116

Given $f(x) = -x^2 + 2$ and $g(x) = x^2 - 1$,

a) determine the equation of $h(x) = g(x) - f(x)$.

$h(x) =$ _____

b) state the range of $h(x)$.

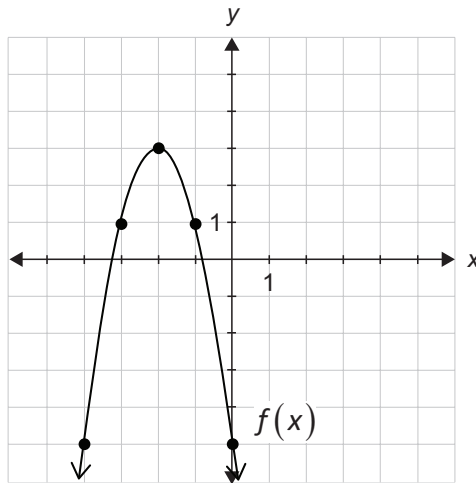
Range: _____

Question 14

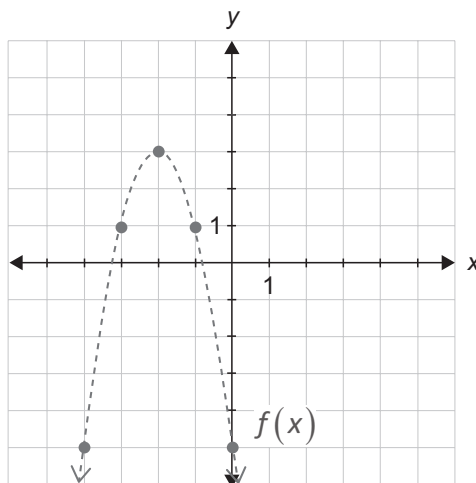
a) 1 mark b) 1 mark

117
118

Given the graph of $f(x)$,



a) sketch the graph after a reflection over the line $y = x$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

b) state a restriction on the domain of $f(x)$ in order for its inverse to be a function.

Question 15

1 mark

119

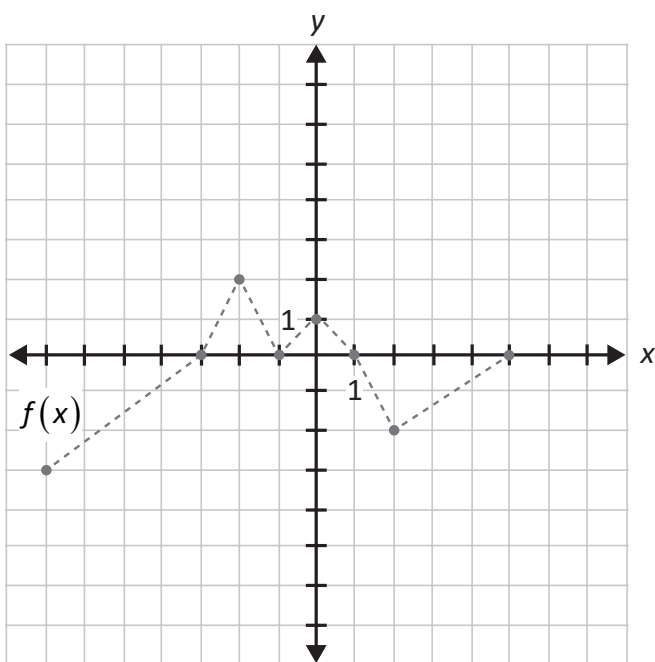
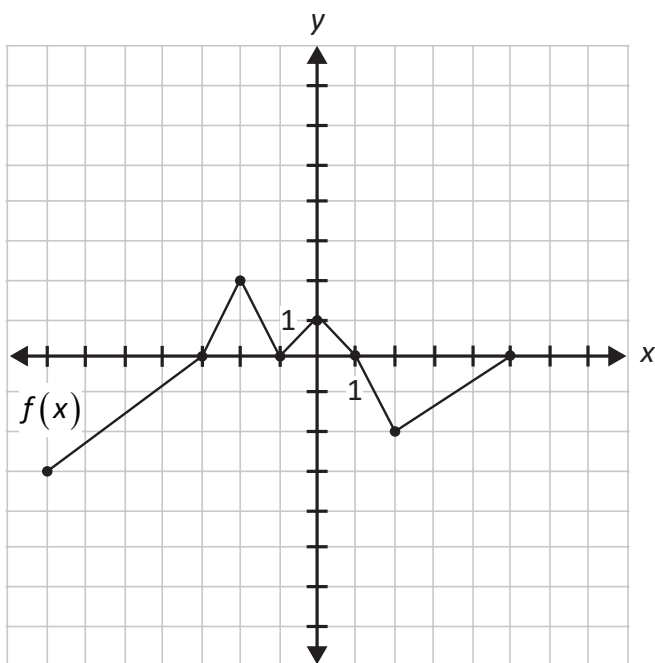
Describe a transformation of a function that would not affect its domain.

Question 16

1 mark

120

Given the graph of $f(x)$, sketch the graph of the function $y = |f(x)|$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 17

3 marks

121

Determine, algebraically, all of the zeros of the polynomial function, $p(x) = 3x^3 + x^2 - 20x + 12$.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

January 2026

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 2 (January 2026)

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Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

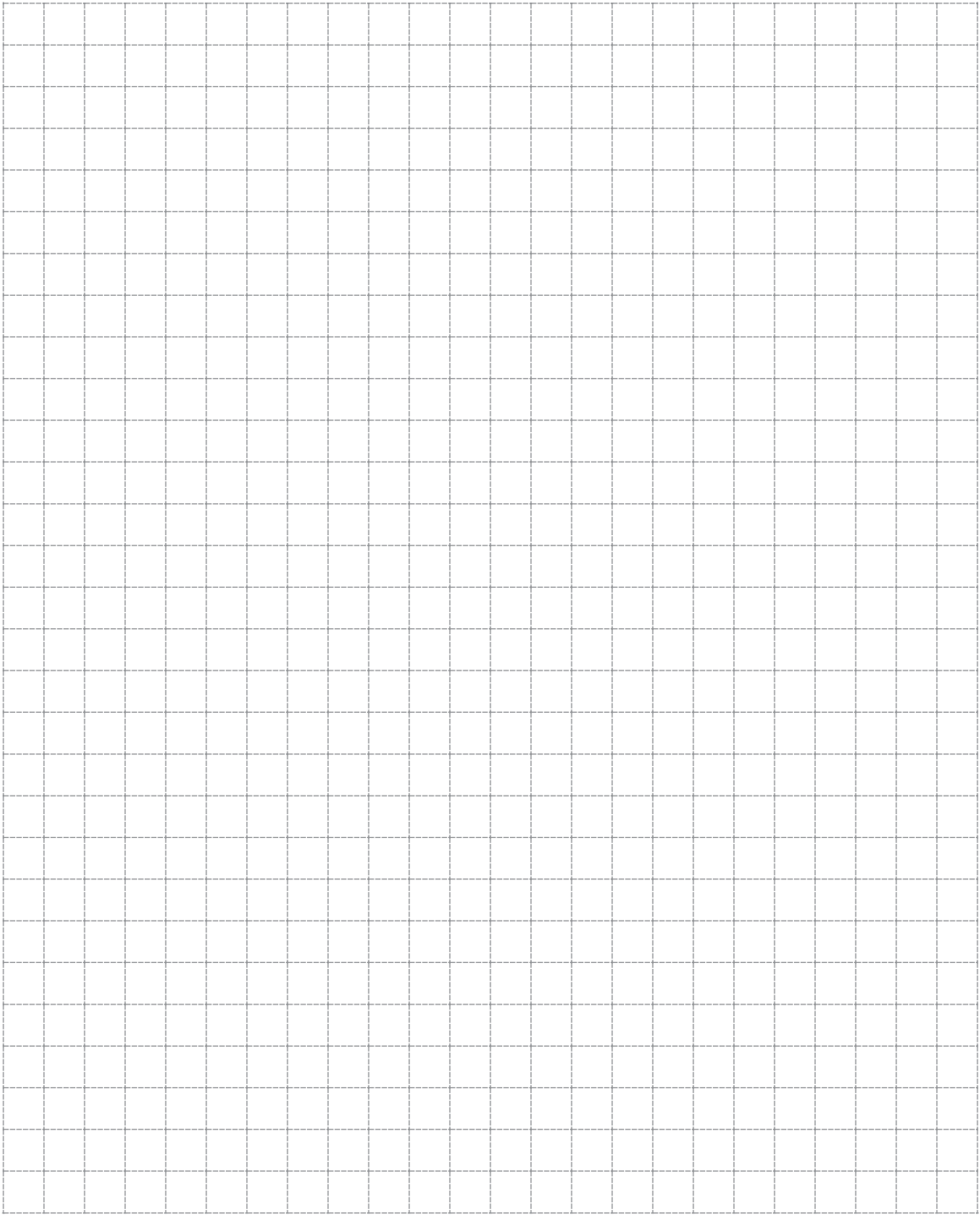
Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- Write each solution in the space provided.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 18**1 mark**

Given $\theta = 240^\circ$, identify the coordinates of the point, $P(\theta)$, on the unit circle.

- a. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- b. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
- c. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
- d. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

Question 19**1 mark**

Given the functions, $f(x) = -3x + 5$ and $g(x) = x^2 + x - 1$, identify the value of $g(f(2))$.

- a. -10
- b. -5
- c. -3
- d. -1

Question 20**1 mark**

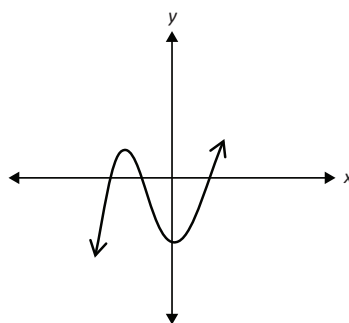
Given $f(x) = \sqrt{x} + 5$, identify the equation of the transformed graph that has the same y -intercept.

- a. $y = f(x - 3)$
- b. $y = -3f(x)$
- c. $y = f(-3x)$
- d. $y = f(x) - 3$

Question 21**1 mark**

Given the graph of $p(x) = ax^3 + bx^2 + cx + d$, identify the statement that is true.

- a. $a > 0, d < 0$
- b. $a > 0, d > 0$
- c. $a < 0, d < 0$
- d. $a < 0, d > 0$

**Question 22****1 mark**

Identify the general solution of the equation, $\csc\theta = -1$.

- a. $\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$
- b. $\theta = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$
- c. $\theta = \frac{3\pi}{2} + k\pi, k \in \mathbb{Z}$
- d. $\theta = \pi + 2k\pi, k \in \mathbb{Z}$

Question 23**1 mark**

Identify the equation of a radical function with a domain of $[-6, \infty)$ and a range of $(-\infty, 3]$.

a. $y = -\sqrt{x+6} - 3$

b. $y = -\sqrt{x+6} + 3$

c. $y = \sqrt{-(x+3)} - 6$

d. $y = \sqrt{x-6} + 3$

Question 24**1 mark**

Identify the value of n in the equation, ${}_nC_3 = {}_nC_7$.

a. 3

b. 4

c. 7

d. 10

Question 25**1 mark**

Identify the remainder when $p(x) = x^5 - 1$ is divided by $(x+1)$.

a. -2

b. -1

c. 0

d. 4

Question 26

2 marks

122

Match each function with its corresponding graph.

Write the corresponding letter in this column.

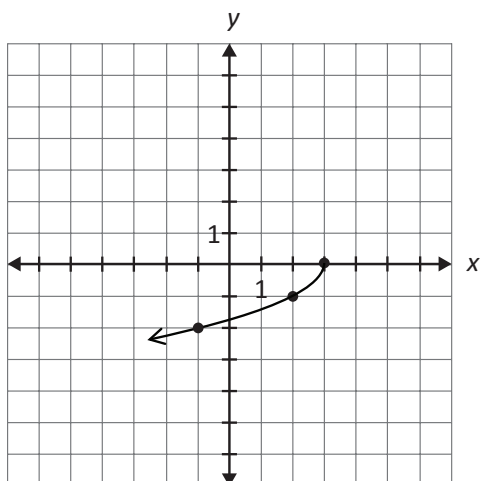
$$y = \sqrt{x} - 3$$

$$y = \sqrt{-x} + 3$$

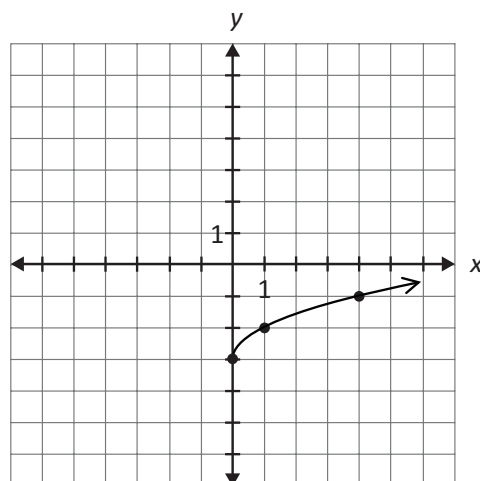
$$y = -\sqrt{x+3}$$

$$y = -\sqrt{-(x-3)}$$

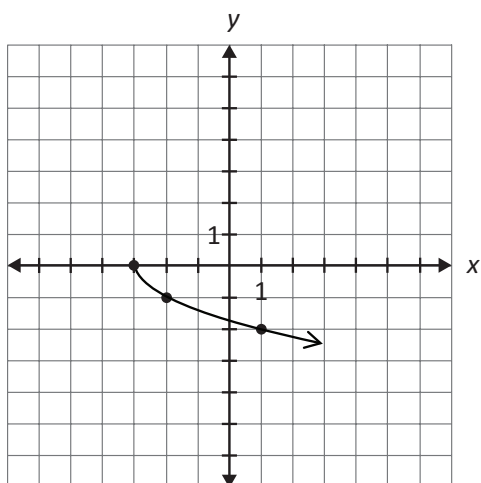
A)



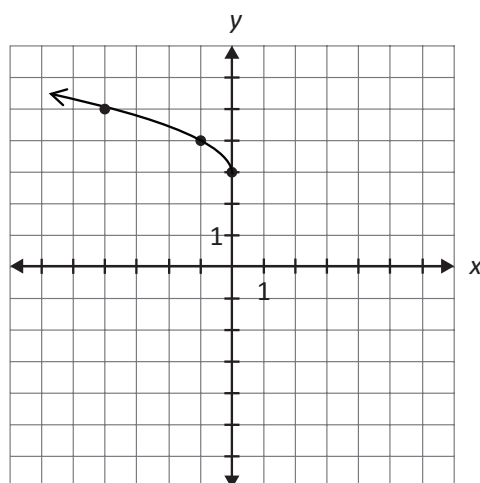
B)



C)



D)

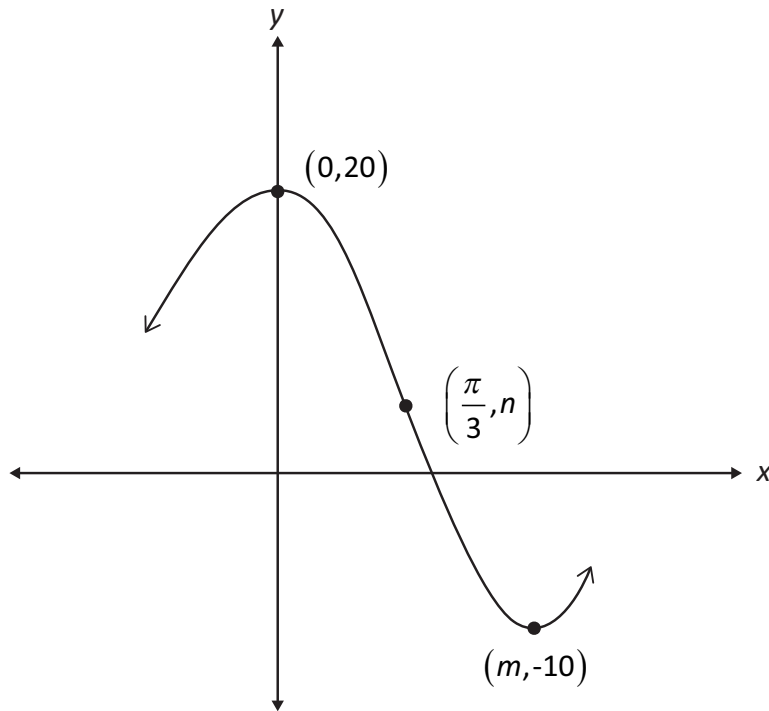


Question 27

2 marks

123

The amplitude, A , of the sinusoidal function can be determined using the equation $A = 20 - n$. State the values of m and n .



Question 28

3 marks

124

Solve, algebraically.

$${}_{n-1}P_2 = 20$$

Question 29

3 marks

125

Determine the exact value of $\sin\left(\frac{23\pi}{12}\right)$.

Question 30**1 mark**

126

Cameron was asked to solve the equation, $\log_3(x-4)=2$.

Cameron's solution:

$$x - 4 = 2^3$$

$$x - 4 = 8$$

$$x = 12$$

Describe Cameron's error.

Question 31

1 mark

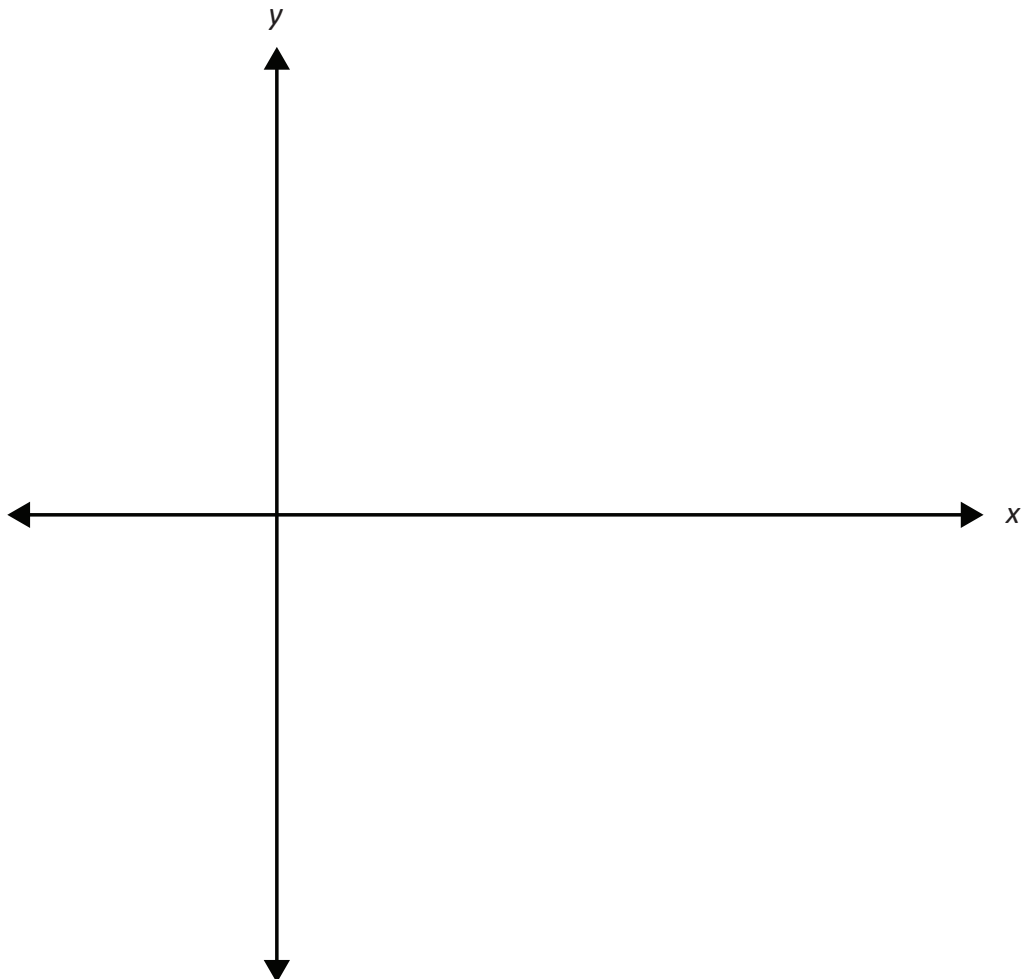
127

Given the identity $\sin^2 \theta = \frac{\tan \theta \sin \theta}{\sec \theta}$, state the non-permissible value of θ over the interval $[\pi, 2\pi]$.

Question 32**4 marks**

128

Sketch the graph of $y = -5\cos\left(\frac{\pi}{2}x\right) + 2$ over the domain $[0, 5]$.



Question 33

1 mark

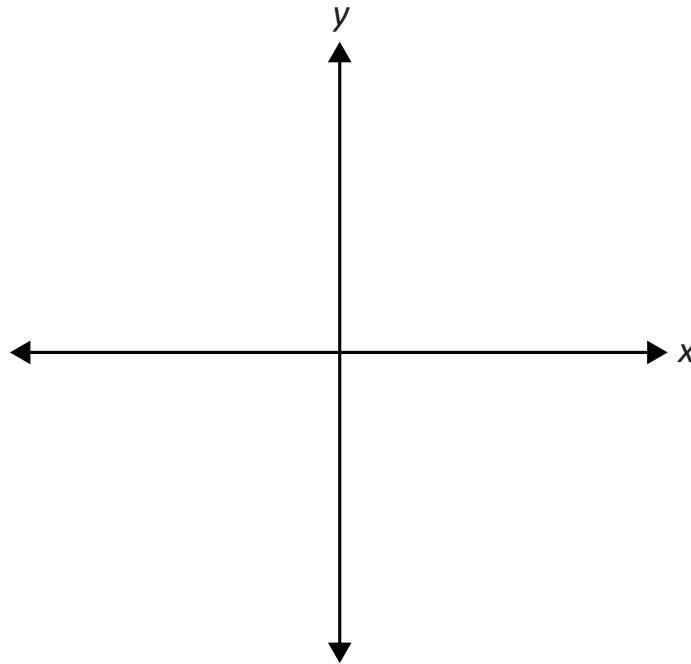
129

Describe the behaviour of the graph of the polynomial function, $p(x) = (x-3)^2(x-1)$, as it approaches the x-intercept at $x = 3$.

Question 34**1 mark**

130

Sketch the angle of 4 radians in standard position.



Question 35

4 marks

131

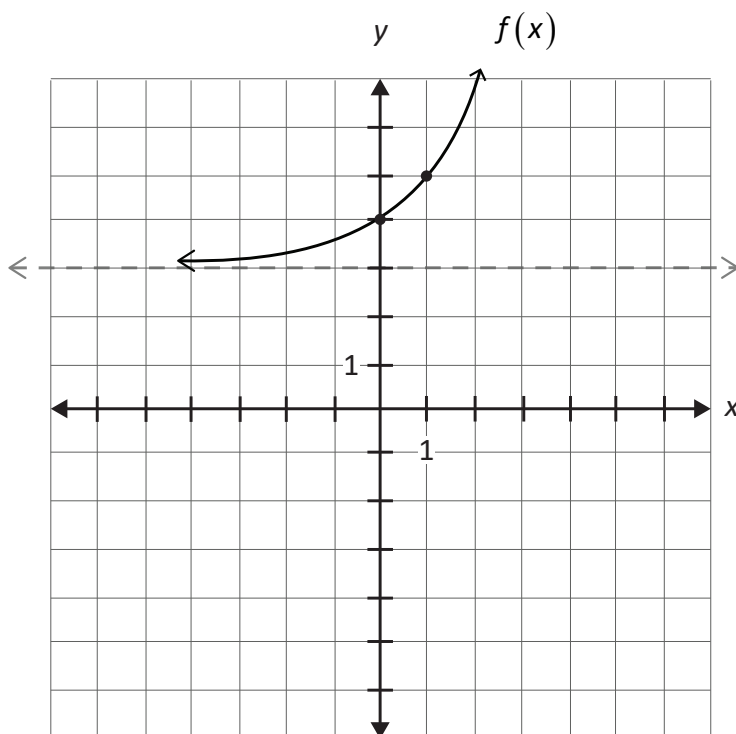
Solve, algebraically.

$$\log_5 12 + 2\log_5 x = \log_5 48$$

Question 36**2 marks**

132

State the equation of the exponential function represented by the graph of $f(x)$.



$f(x) =$ _____

Question 37

3 marks

133

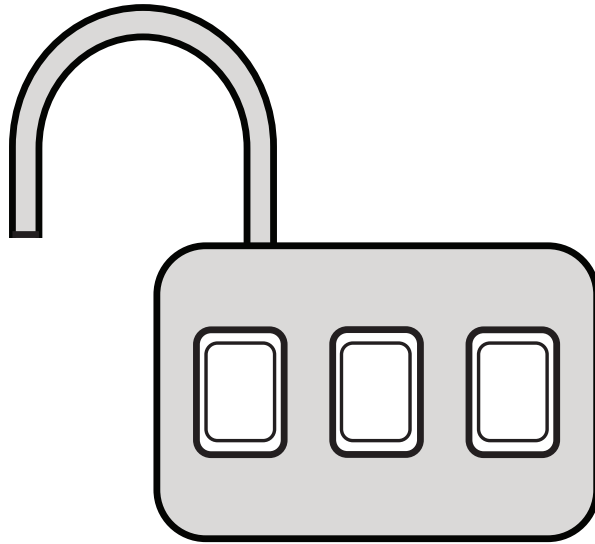
Evaluate.

$$\cos\left(\frac{11\pi}{3}\right) + \csc\left(-\frac{\pi}{3}\right) \cot\left(\frac{11\pi}{6}\right)$$

Question 38**1 mark**

134

Min Li has a 3-digit code for his lock. Determine the number of possible codes for his lock, if repetition is allowed.



Question 39**2 marks**

135

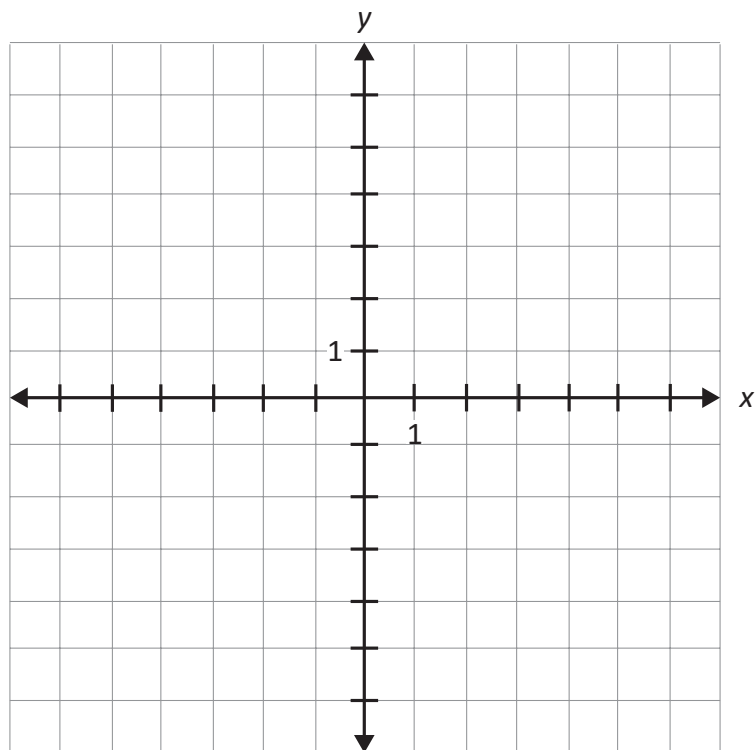
Solve, algebraically.

$$\left(\frac{1}{6}\right)^{3x+2} = 6^{2x}$$

Question 40**4 marks**

136

Sketch the graph of $f(x) = \frac{5}{(x-1)(x+3)}$ and state the y-intercept.



y-intercept: _____

Question 41**2 marks**

137

Given $\log_2 x = 5$, determine the value of $\log_4 (2x)$.

Question 42**1 mark**

138

Henry was asked to prove the identity, $\csc^2 \theta - 2\cos^2 \theta + 1 = \cot^2 \theta + 2\sin^2 \theta$, for all permissible values of θ .

His work:

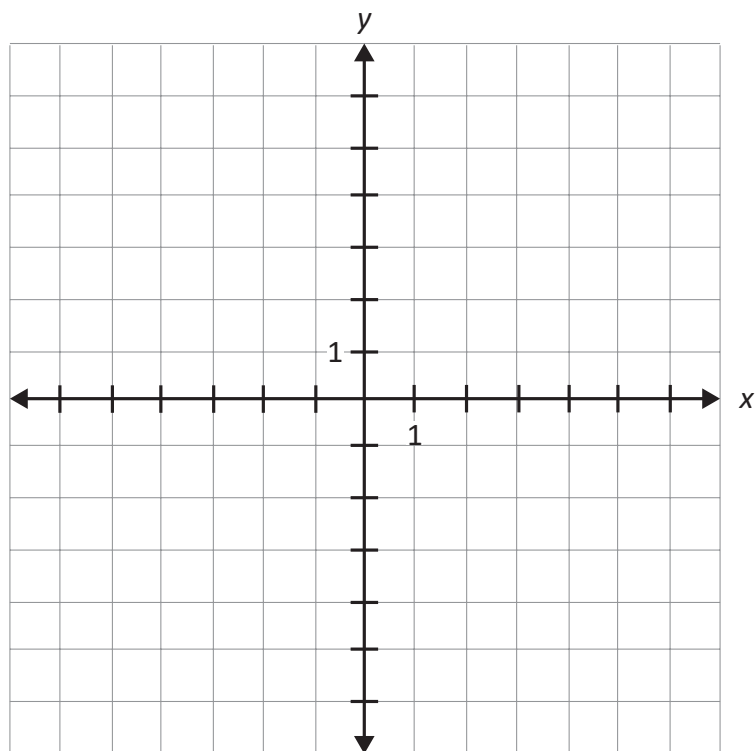
Left-Hand side	Right-Hand side
$\csc^2\left(\frac{\pi}{4}\right) - 2\cos^2\left(\frac{\pi}{4}\right) + 1$	$\cot^2\left(\frac{\pi}{4}\right) + 2\sin^2\left(\frac{\pi}{4}\right)$
$(\sqrt{2})^2 - 2\left(\frac{\sqrt{2}}{2}\right)^2 + 1$	$(1)^2 + 2\left(\frac{\sqrt{2}}{2}\right)^2$
$2 - 1 + 1$	$1 + 1$
2	2
LHS	RHS ✓

Explain why his proof is insufficient.

Question 43**4 marks**

139

Sketch the graph of $y - 2 = -\sqrt{x - 1}$.



Question 44**2 marks**

140

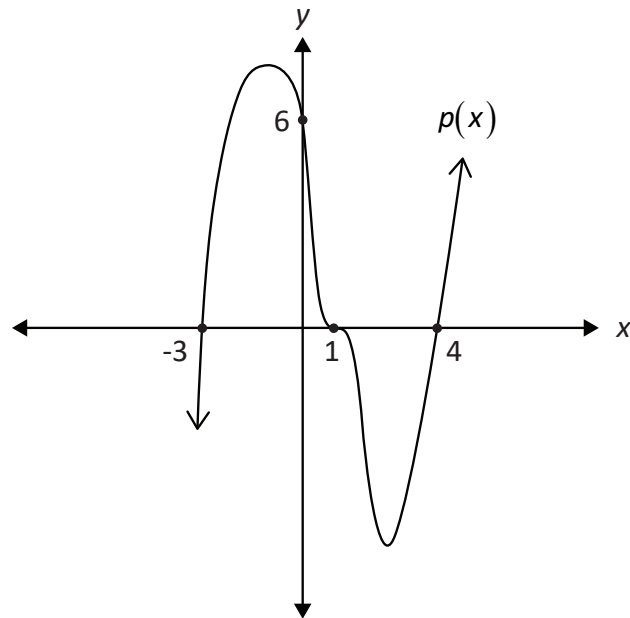
Evaluate.

$$\log_3 54 - \log_3 6$$

Question 45**2 marks**

141

Determine, algebraically, the equation of the graph of the polynomial function, $p(x)$.



$p(x) =$ _____

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.

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Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 1

June 2026

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 1 (June 2026)

This resource is available in print and electronic formats.

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Learning website at www.edu.gov.mb.ca/k12/assess/archives/index.html.

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possible, some parts of this document are not fully accessible at this time.

Available in alternate formats upon request.

Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Number and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	15	35
Booklet 2	9	20	56
Total	9	35	91

* The first five questions with the symbol  in Booklet 1 require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

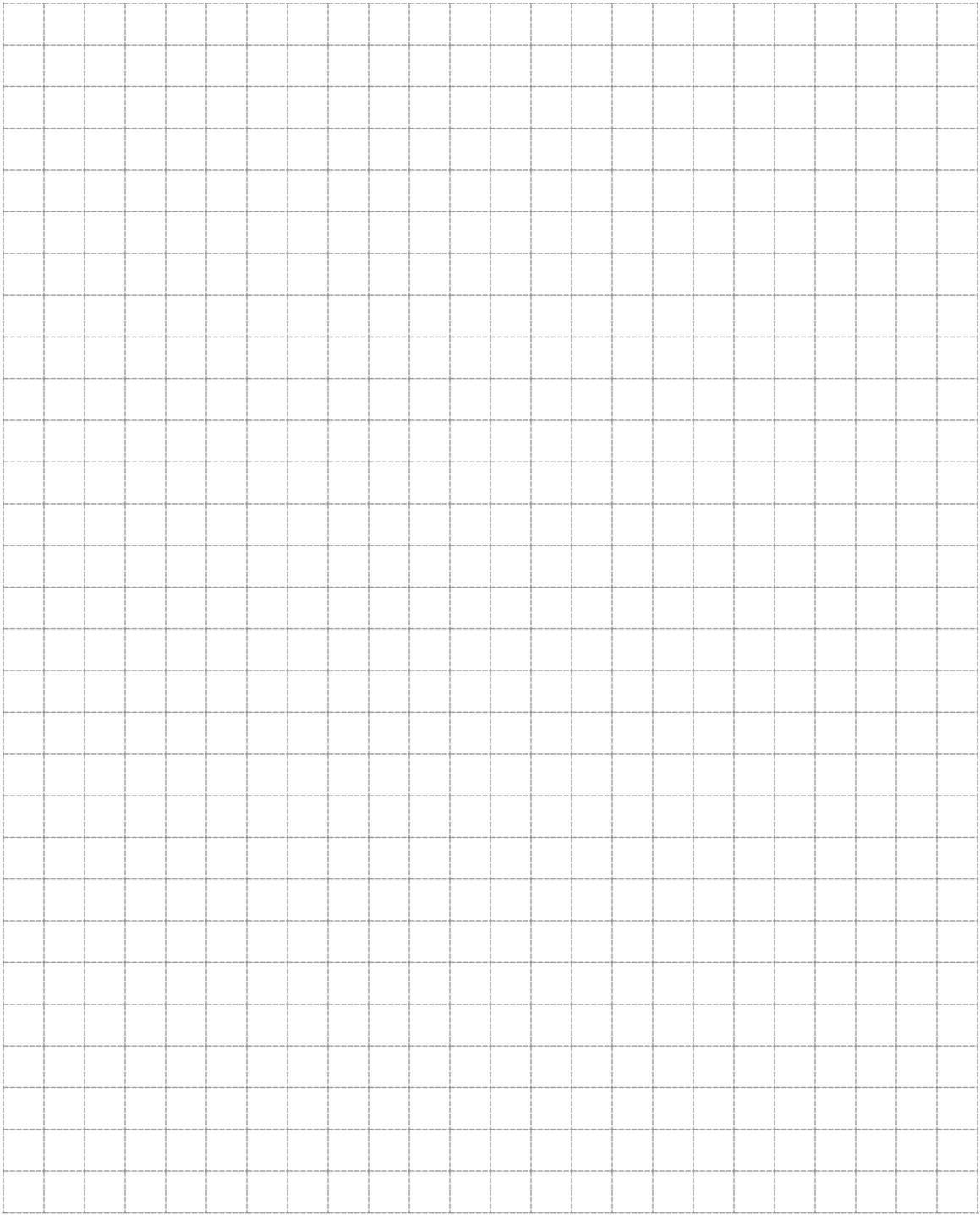
Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (three decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited. Please turn off your cell phone and all other such devices.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_nP_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_nC_k a^{n-k} b^k$$

$$\text{Given } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain, identify, and justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Express: Write or present a mathematical concept in a specific form, often using symbols or equations.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

Prove: Show that a mathematical statement is true using logical reasoning and/or known facts.

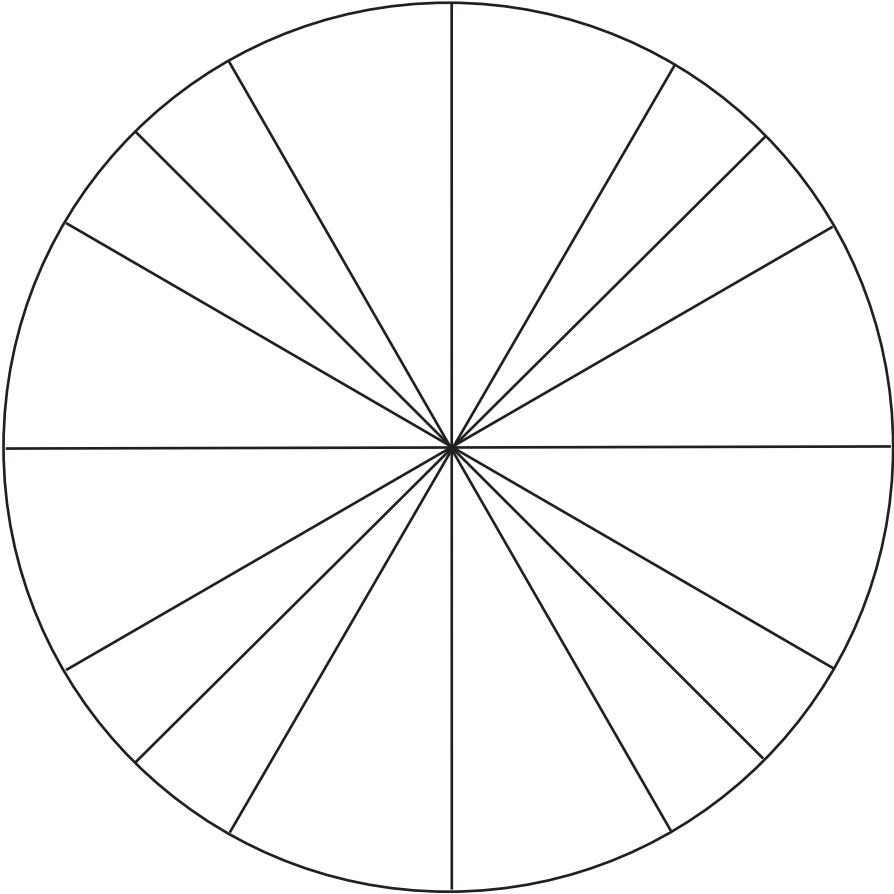
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of two coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

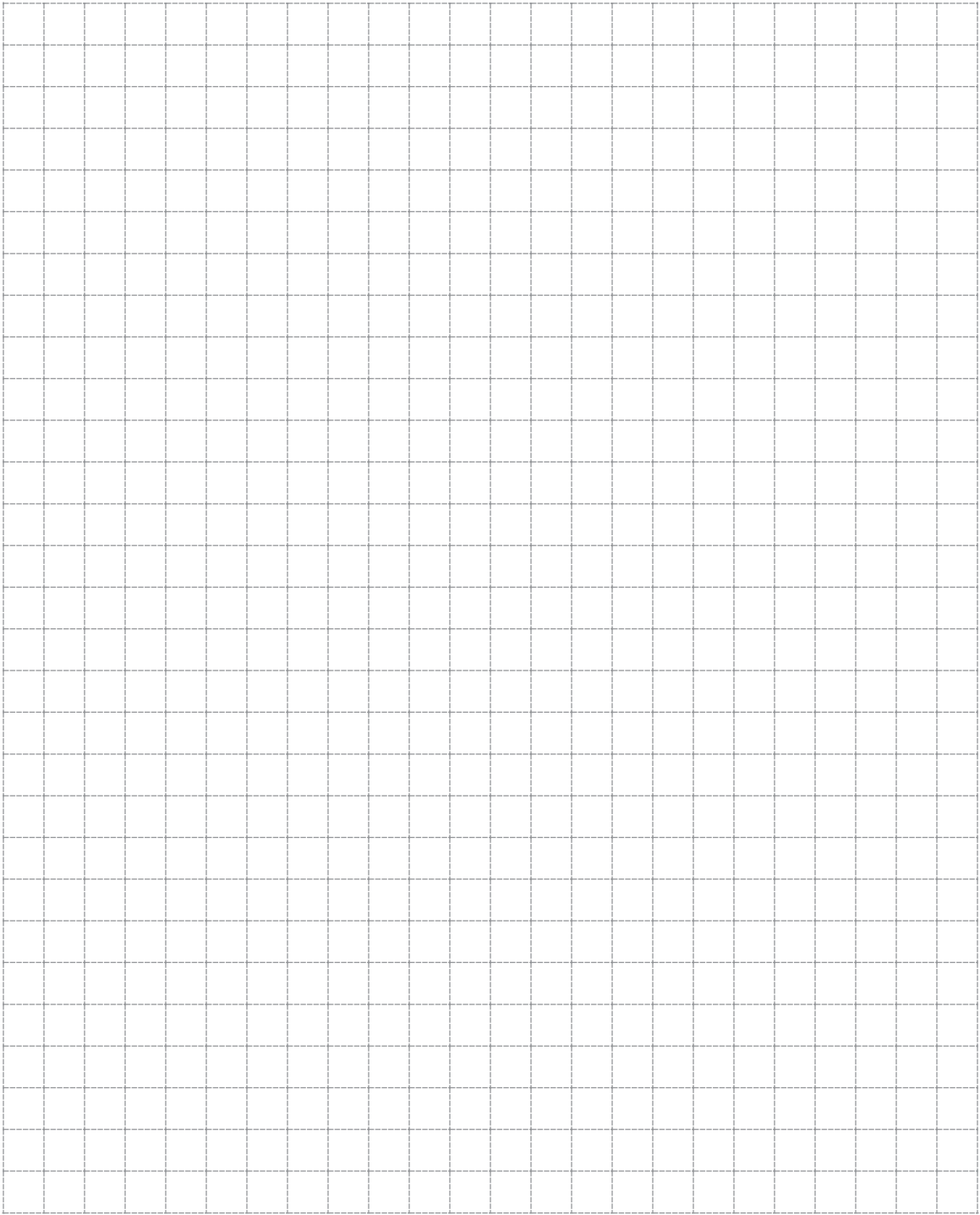
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



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Question 1 **2 marks**

101

Jack rolled a water bottle from one edge of his desk to the other. The desk measures 60 cm from edge to edge. The water bottle has a diameter of 7 cm. Determine the angle that the water bottle rotated, in degrees.

Question 2 **2 marks**

102

There are 15 dogs and 12 cats in an animal shelter. Determine the number of ways that three dogs and two cats can be selected if Scout, one of the dogs, must be selected.

Question 3 **3 marks**

103

Emily wants to save money to buy a car. She invests \$180 per month at an annual interest rate of 4.5%, compounded monthly.

Determine, algebraically, the number of monthly investments she will need to make to obtain at least \$15 000. Express the final answer as a whole number.

Use the formula: $FV = \frac{R[(1+i)^n - 1]}{i}$

where FV = the future value

R = the investment amount each period

$$i = \left[\frac{\text{the annual interest rate (as a decimal)}}{\text{the number of compounding periods per year}} \right]$$

n = the number of investments

Question 4 **3 marks**

104

Determine and simplify the 4th term in the binomial expansion of $\left(3x - \frac{2}{x^2}\right)^6$.

Question 5 **3 marks**

105

Solve, algebraically, over the interval $[0, 2\pi]$.

$$4\cos^2 x - 3\cos x - 1 = 0$$

Note: A calculator is not required for the remaining test questions.

Question 6**1 mark**

106

State an equation for a rational function, $g(x)$, whose graph has a vertical asymptote at $x = 7$.

$g(x) =$ _____

Question 7

2 marks

107

State an equation of a radical function, $f(x)$, with a domain of $x \leq 0$ and a range of $y \geq 1$.

$f(x) =$ _____

Question 8

3 marks

108

Prove the identity for all permissible values of x .

$$\frac{\cos x + \sin^2 x \sec x}{\sin x} = \sec x \csc x$$

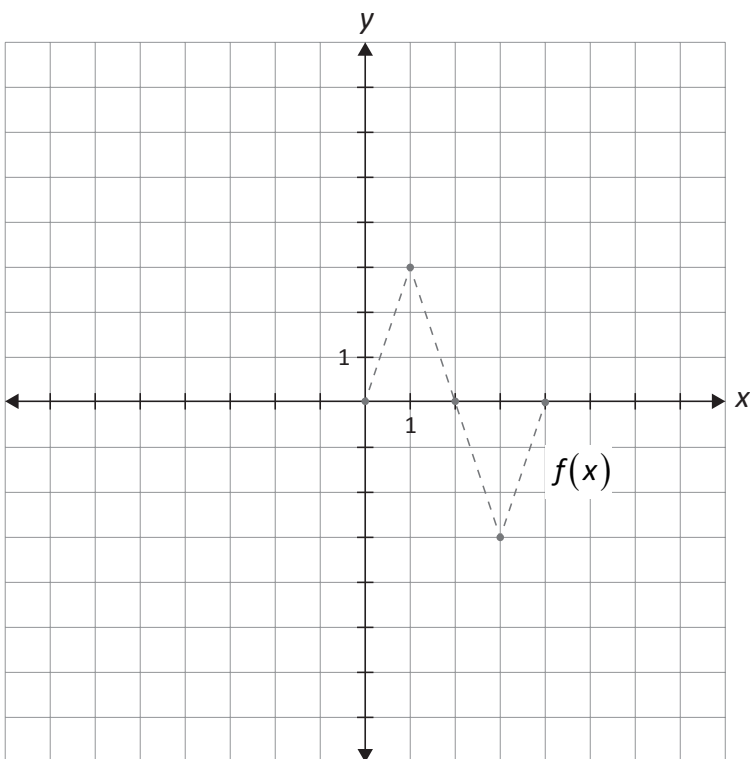
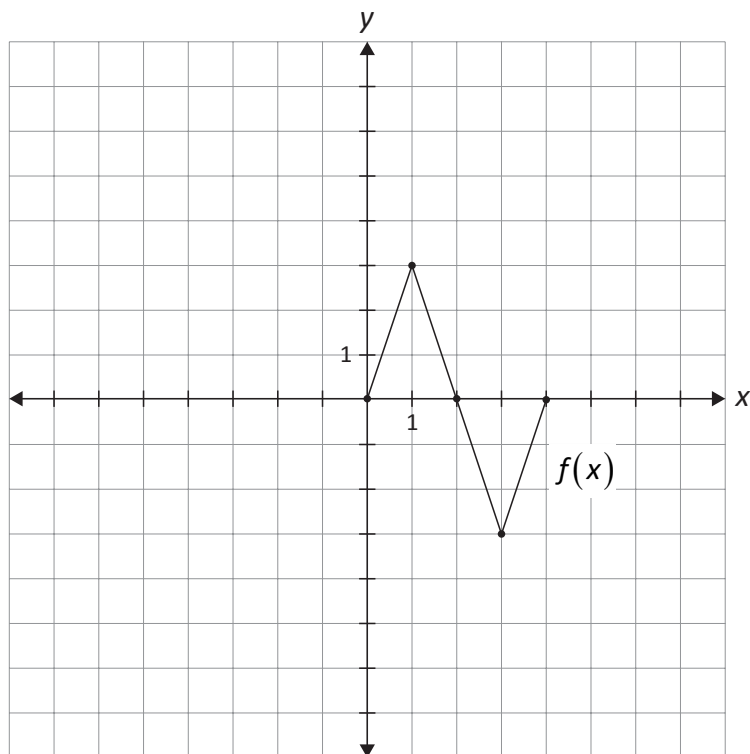
Left-Hand Side	Right-Hand Side

Question 9

3 marks

109

Given the graph of, $y = f(x)$, sketch the graph of $y = 2|f(-x)|$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 10

3 marks

110

Solve, algebraically.

$$\log(x-1) + \log(x+2) = 1$$

Question 11

1 mark

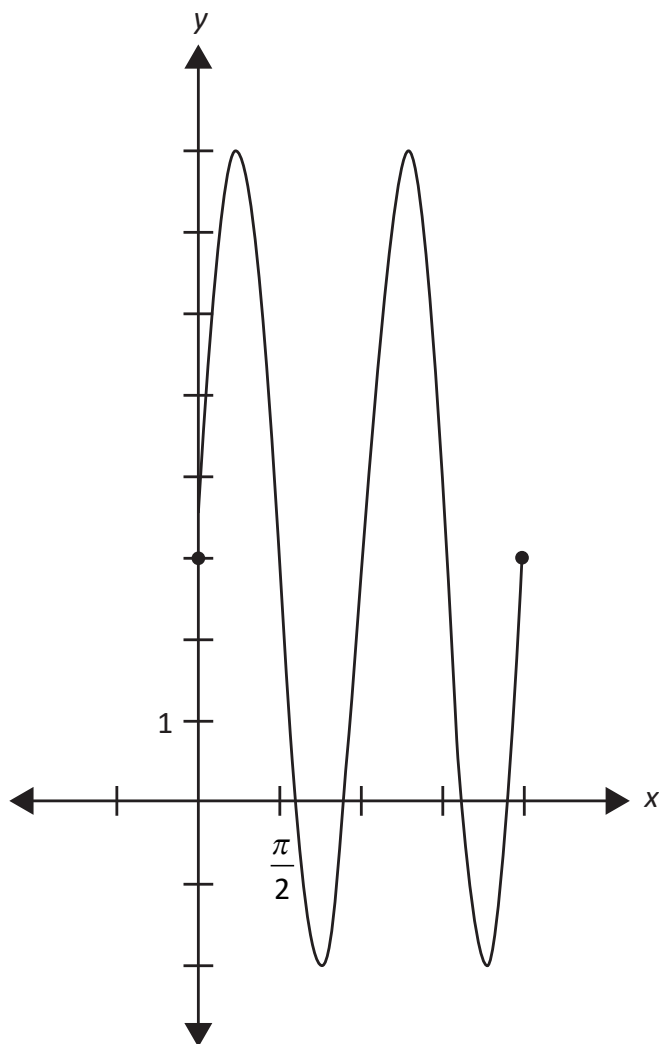
111

Justify that there are only two negative terms in the expansion of $(-2x + 5y)^4$.

Question 12**1 mark**

112

The graph of $y = 5\sin(2x) + 3$ below can be used to solve the equation $0 = 5\sin(2x) + 3$.
State how many solutions there are to the equation $0 = 5\sin(2x) + 3$ over the interval $[0, 2\pi]$.



Question 13

a) 1 mark b) 2 marks

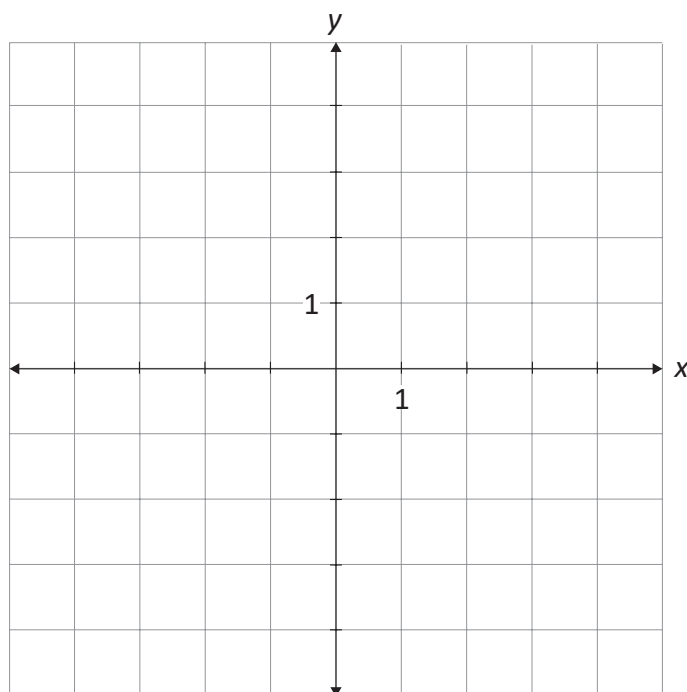
113
114

Given the functions, $f(x) = x^2 - 1$ and $g(x) = x + 1$,

a) state the equation of $h(x) = \frac{f(x)}{g(x)}$.

$h(x) =$ _____

b) sketch the graph of $h(x)$.

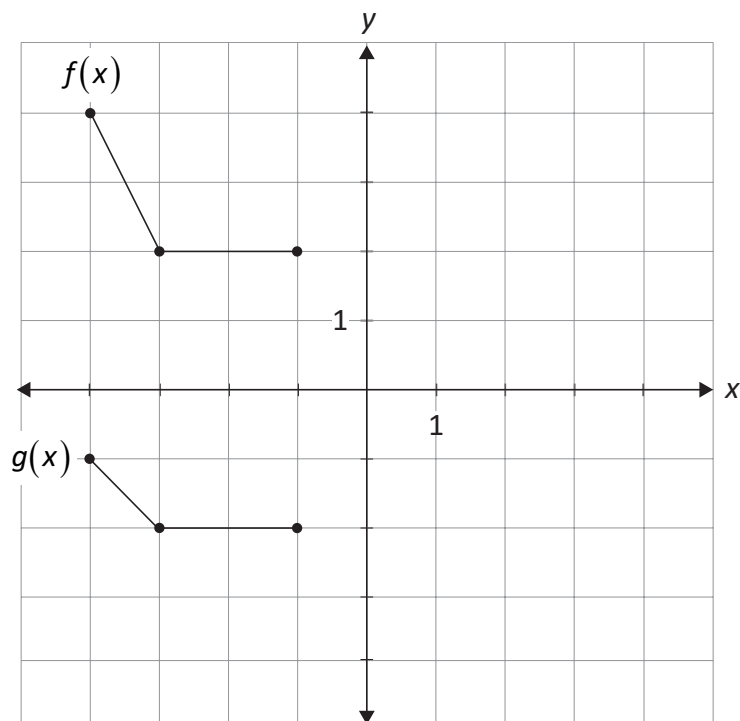


Question 14

2 marks

115

State an equation for $g(x)$, in terms of $f(x)$.



$g(x) =$ _____

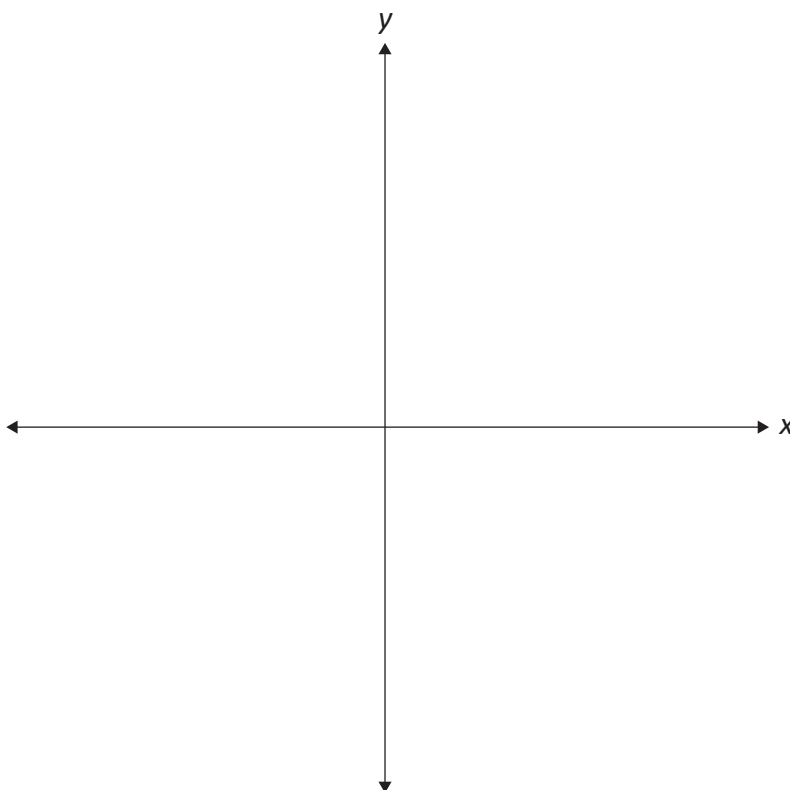
Question 15

3 marks

116

Sketch the graph of a polynomial function, $p(x)$, with the following characteristics:

- degree 5
- leading coefficient of -1
- a zero at -3 , with a multiplicity of 3
- a zero at 1 , with a multiplicity of 2



Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet 2

June 2026

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 2 (June 2026)

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Total	9	35	91

Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

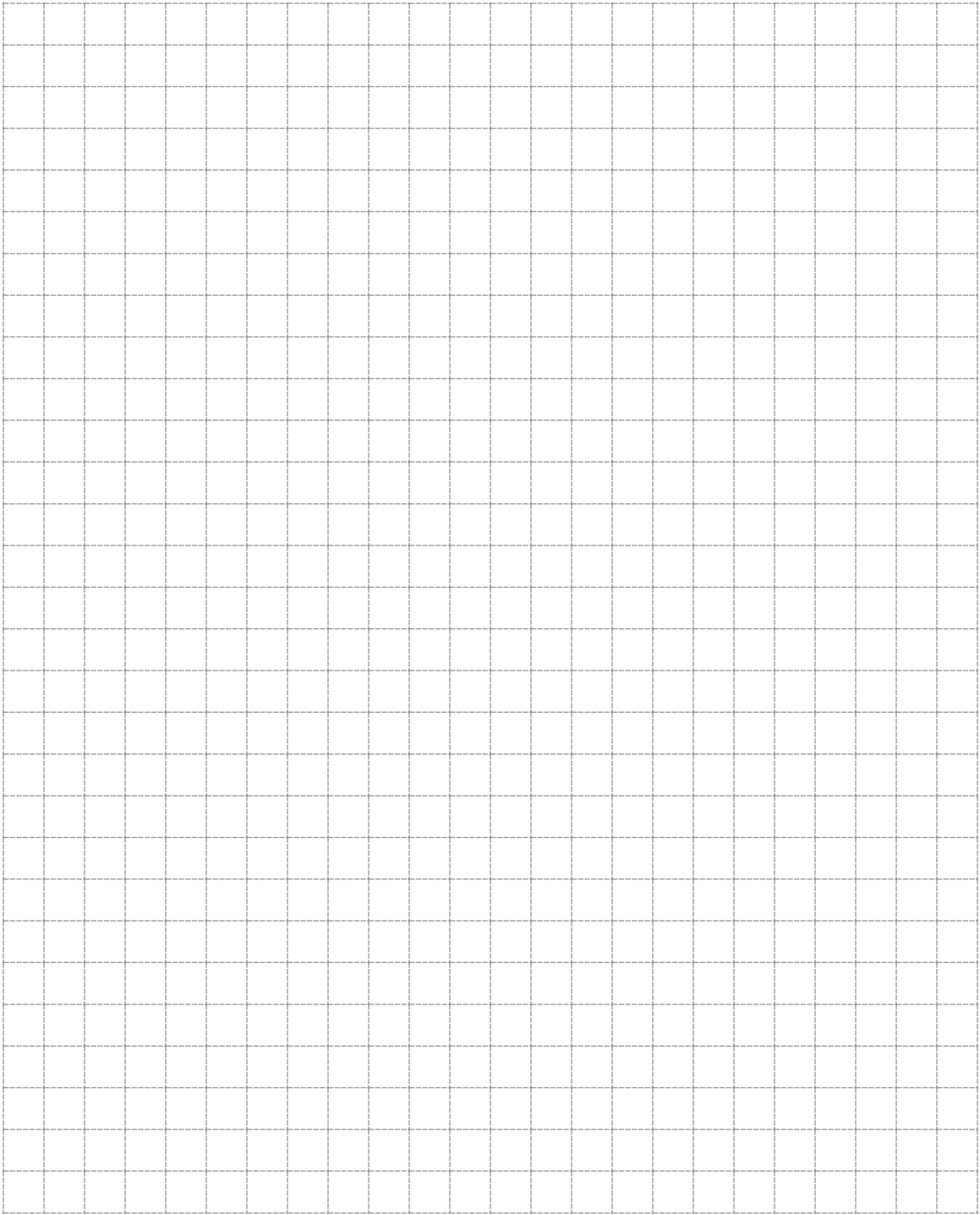
Selected Response Questions

- Calculators are **not** allowed for this part of the test.
- You may use the spaces beside each question for rough work.
- Provide only one answer per question.
- There is no penalty for guessing.
- Record your answers on the sheet provided.

Constructed Response Questions

- Calculators are **not** allowed for this part of the test.
- Write each solution in the space provided.
- For full marks, your answer must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, clear, and well organized.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Question 16**1 mark**

Identify the expression that is equivalent to $\tan\theta(\csc^2\theta - 1)$.

- a. $\cot\theta$
- b. $\tan\theta$
- c. $\sec\theta$
- d. $\sin\theta$

Question 17**1 mark**

Identify the expression that represents the number of ways five books can be arranged on a shelf if two of them must be together.

- a. $5!2!$
- b. $5! - 2!$
- c. $4!2!$
- d. $3!2!$

Question 18**1 mark**

Identify the value of x in the equation, $\log_8 x = -\frac{1}{3}$.

- a. -2
- b. $-\frac{1}{2}$
- c. $\frac{1}{2}$
- d. 2

Question 19**1 mark**

Given the functions, $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x-1}$, identify the domain of $f(g(x))$.

- a. $(-\infty, \infty)$
- b. $(-\infty, 1) \cup (1, \infty)$
- c. $[1, \infty)$
- d. $(1, \infty)$

Question 20**1 mark**

Given $f(x) = \frac{3(x+2)}{5(x+2)(x-2)}$, identify the equation of the horizontal asymptote.

- a. $y = -2$
- b. $y = 0$
- c. $y = \frac{3}{5}$
- d. $y = 2$

Question 21**1 mark**

Identify the range of the sinusoidal function, $f(x) = 2\sin x + 3$.

- a. $\{y \mid y \in \mathbb{R}\}$
- b. $\{y \mid 2 \leq y \leq 4, y \in \mathbb{R}\}$
- c. $\{y \mid -2 \leq y \leq 2, y \in \mathbb{R}\}$
- d. $\{y \mid 1 \leq y \leq 5, y \in \mathbb{R}\}$

Question 22**1 mark**

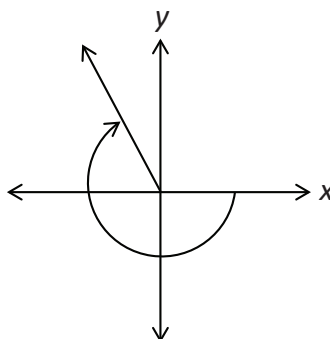
Given that $(x - 3)$ is a factor of the polynomial $P(x)$, identify which of the following is true.

- a. $P(3) = 0$
- b. $P(0) = 3$
- c. $P(0) = -3$
- d. $P(-3) = 0$

Question 23**1 mark**

Identify the value that best represents the angle in standard position.

- a. -4.4 radians
- b. -2 radians
- c. 1.3 radians
- d. 1.9 radians

**Question 24****1 mark**

Identify the expression that is equivalent to $\log_a(x + 2) + 3\log_a(x)$.

- a. $\log_a(3(x + 2))$
- b. $\log_a(x^3(x + 2))$
- c. $\log_a\left(\frac{(x + 2)}{x^3}\right)$
- d. $\log_a x + \log_a 2 + \log_a x^3$

Question 25

3 marks

117

Evaluate.

$$\tan^3\left(\frac{3\pi}{4}\right) + \csc\left(\frac{-4\pi}{3}\right) \cos\left(\frac{13\pi}{6}\right)$$

Question 26

2 marks

118

If $a = \log 4$ and $b = \log 3$, express $\log 36$ in terms of a and b .

Question 27

2 marks

119

Justify that $\tan 2x$ has a non-permissible value at $x = \frac{\pi}{4}$.

Question 28

2 marks

120

Determine the exact value of $\tan\theta$, if $\sec\theta = \frac{6}{5}$ and θ terminates in quadrant IV.

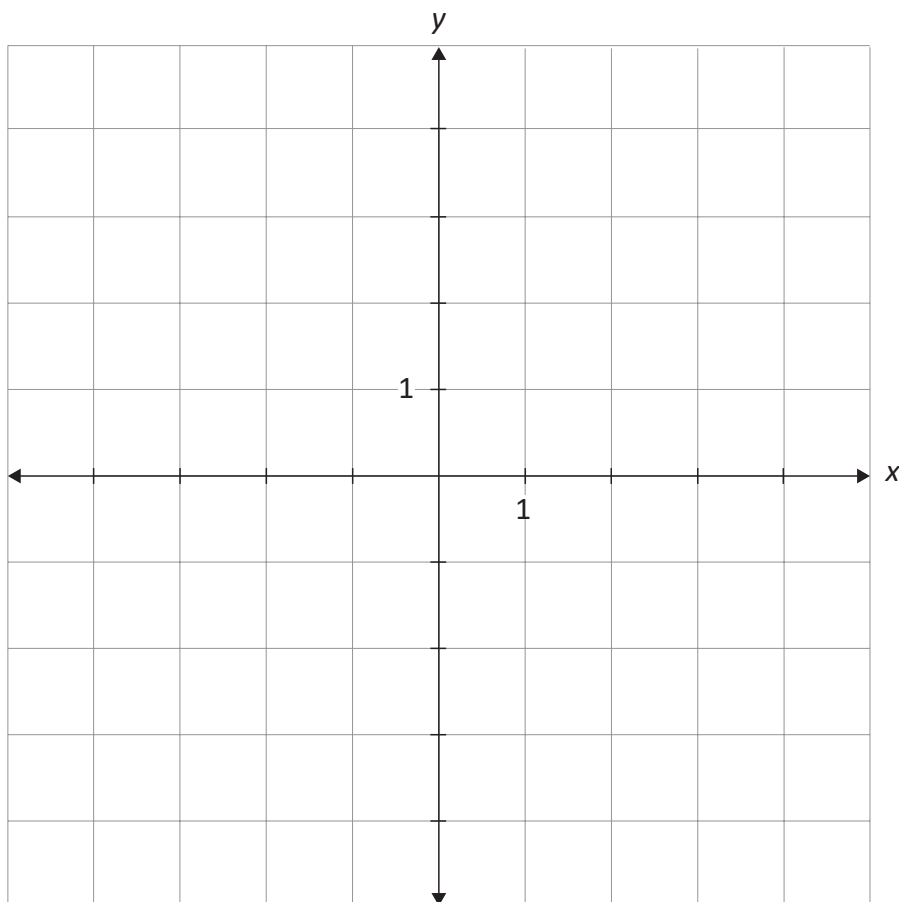
Question 29

a) 2 marks b) 1 mark

121
122

Given the function, $g(x) = \frac{4}{x^2 + 1}$,

a) sketch the graph of $g(x)$.



b) state the range of the graph of $g(x)$.

Question 30

a) 2 marks b) 1 mark

123
124

Given $y = x^2 - 4$,

a) determine the equation of the inverse.

$y =$ _____

b) state a restriction on the domain of $y = x^2 - 4$, in order for its inverse to be a function.

Question 31**3 marks**

125

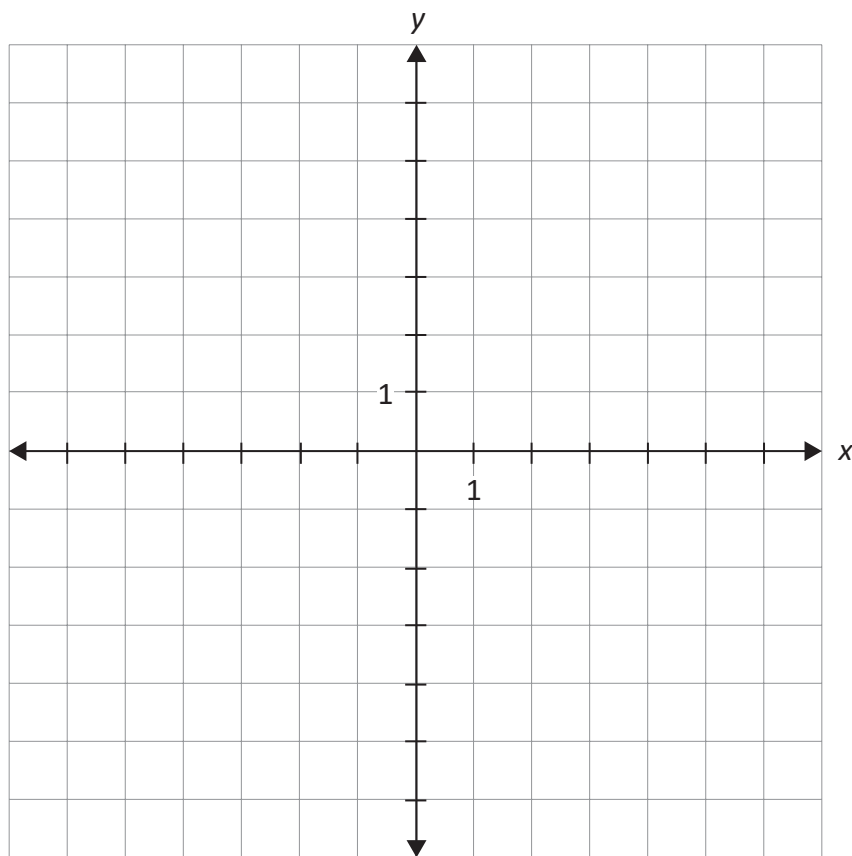
Given $\cos \alpha = -\frac{1}{3}$ and $\sin \beta = -\frac{2}{3}$ where α and β terminate in the same quadrant, determine the exact value of $\sin(\alpha - \beta)$.

Question 32

4 marks

126

Sketch the graph of $y + 2 = -\sqrt{4x}$.



Question 33**3 marks**

127

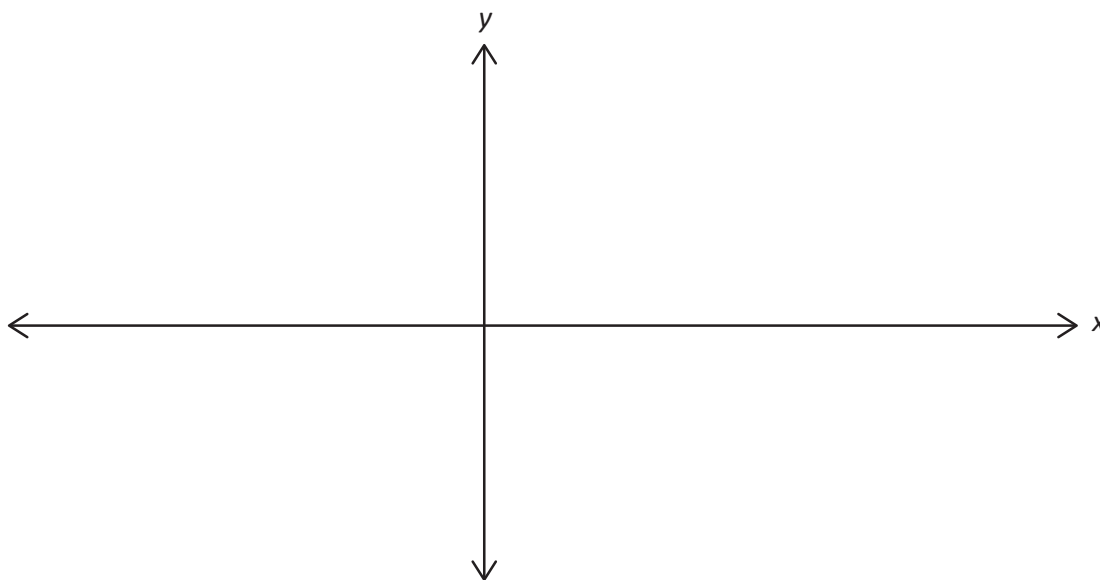
Express $p(x) = x^4 - x^3 - 6x^2 + 4x + 8$ in completely factored form, given that $(x + 1)$ is a factor of $p(x)$.

$p(x) =$ _____

Question 34**4 marks**

128

Sketch at least one period of the graph of $y = -\sin\left(2\left(x - \frac{\pi}{4}\right)\right)$.



Question 35

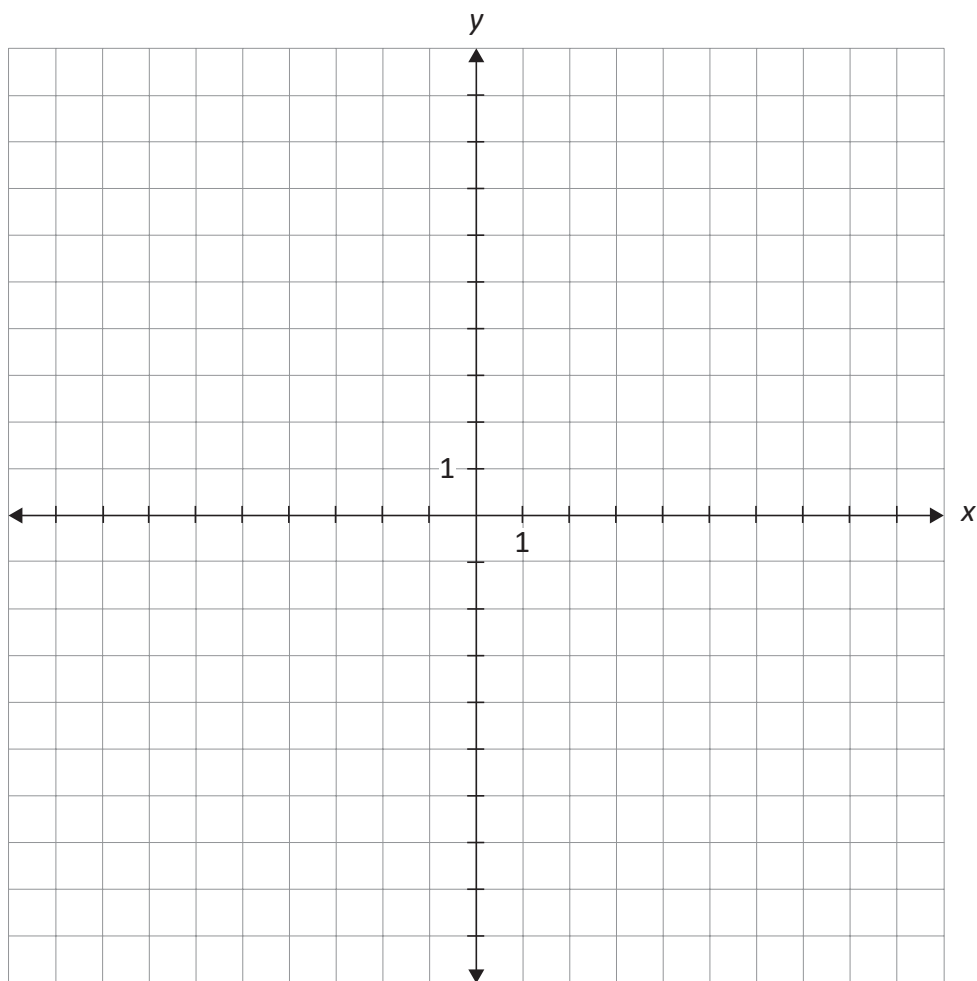
a) 1 mark b) 2 marks

129
130

Given $g(x) = \log_2(x) - 3$,

a) determine the x-intercept of the graph.

b) sketch the graph of $g(x)$.



Question 36**1 mark**

131

Modesola incorrectly evaluated the expression, $\cos^2(30^\circ) - \sin^2(30^\circ)$.

Modesola's answer:

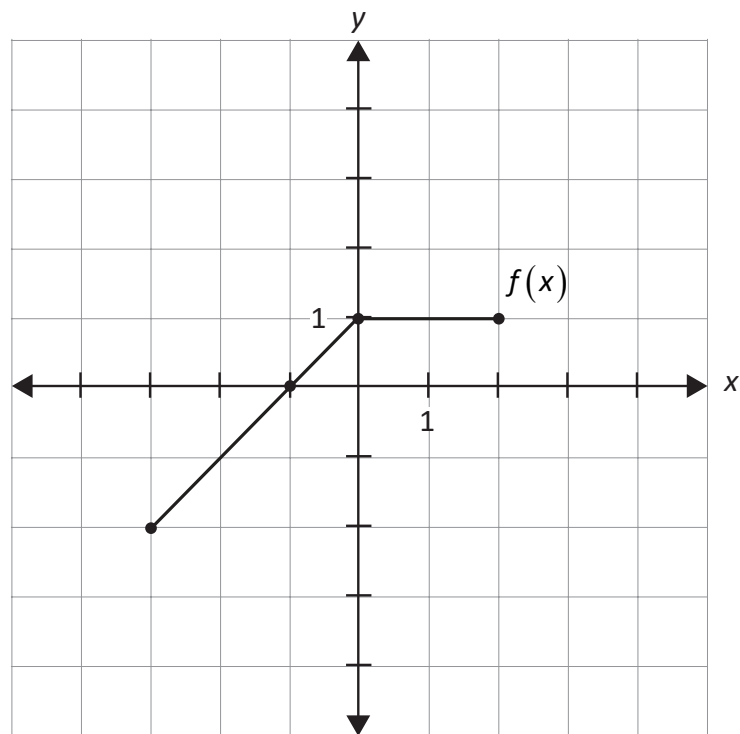
$$\begin{aligned}\cos^2(30^\circ) - \sin^2(30^\circ) \\&= 2\cos(30^\circ) \\&= 2\left(\frac{\sqrt{3}}{2}\right) \\&= \sqrt{3}\end{aligned}$$

Describe their error.

Question 37**1 mark**

132

Given the graph of $y = f(x)$, state the range of $y = \sqrt{f(x)}$.



Question 38**1 mark**

133

Determine a value of x that satisfies $\sin^2(2x - 1) + \cos^2(x + 5) = 1$.

Question 39**2 marks**

134

Solve, algebraically.

$$9^{2x+3} = \left(\frac{1}{3}\right)^{x+1}$$

Question 40

2 marks

135

Determine the total number of arrangements of the letters in the word GUIDES if the vowels and consonants must alternate.

Question 41

2 marks

136

Solve, algebraically.

$$\frac{5!(n-5)!}{3!(n-6)!} = 80$$

Question 42

1 mark

137

Justify that 2.1 is a better estimate than 2.6 for the value of $\log_5 29$.

Question 43**2 marks**

138

When $p(x) = x^3 + kx + 6$ is divided by $(x + 2)$, the remainder is 4. Determine the value of k .

Question 44

3 marks

139

Describe the transformations used to obtain the graph of the function $y = -f(7x + 14)$ from the graph of $y = f(x)$.